

Data Science for Biological, Medical and Health Research

Notes for PQHS/CRSP/MPHP 432 at CWRU

Thomas E. Love

2026-01-08

Table of contents

Introduction	24
R Setup	25
General Theme for <code>ggplot</code> work	26
1 Building the <code>nh432</code> example	27
1.1 R Setup	27
1.2 Selecting NHANES Variables	27
1.2.1 Demographics and Sample Weights	28
1.2.2 Oral Health	29
1.2.3 Hospital Utilization & Access to Care	30
1.2.4 Body Measures	30
1.2.5 Blood Pressure	31
1.2.6 Complete Blood Count	32
1.2.7 C-Reactive Protein	32
1.2.8 Alcohol Use	33
1.2.9 Dermatology	34
1.2.10 Depression Screener	34
1.2.11 Diet Behavior	35
1.2.12 Food Security	36
1.2.13 Health Insurance	37
1.2.14 Medical Conditions	38
1.2.15 Oral Health	38
1.2.16 Physical Activity	39
1.2.17 Reproductive Health	40
1.2.18 Sleep Disorders	41
1.2.19 Smoking Cigarettes	41
1.2.20 Secondhand Smoke	42
1.2.21 Weight History	43
1.3 Filtering for Inclusion	43
1.4 Merging the Data	44
1.4.1 Merging Two Data Frames at a Time	44
1.4.2 Merging Many Data Frames Together	45
1.5 The “Raw” Data	45
1.6 Cleaning Tasks	49

1.7	Our identifying variable	49
1.8	“Refused” & “Don’t Know”	50
1.9	Variables without Variation	51
1.10	Quantitative Variables	51
1.10.1	Renaming the Quantities	53
1.10.2	Sampling Weights	53
1.11	Binary Variables	54
1.11.1	Sex (RIAGENDR)	54
1.11.2	Yes/No variables	55
1.11.3	Renaming Binary Variables	57
1.12	Create PHQ-9 Scores	57
1.12.1	Forming the PHQ-9 Score	57
1.12.2	Distribution of PHQ-9 Score	60
1.12.3	Fixing the DPQ100 variable	61
1.13	Multi-Categorical Variables	62
1.13.1	Creating RACEETH from RIDRETH3	62
1.13.2	Creating EDUC from DMDEDUC2	64
1.13.3	Creating DENTAL from OHAREC	65
1.13.4	Creating SROH from HUQ010	66
1.13.5	Creating SUNSCR from DEQ034D	66
1.13.6	Creating DEPRDIFF from DPQ100	67
1.13.7	Creating DIETQUAL from DBQ700	68
1.13.8	Creating FOODSEC from FSDAD	68
1.13.9	Creating SNORE from SLQ030	69
1.13.10	Creating WTGOAL from WHQ040	70
1.14	Dropping Variables	71
1.15	Resorting Variables	71
1.16	nh432 analytic tibble	72
1.16.1	Saving the tibble as nh432.Rds	73
2	Codebook for nh432	74
2.1	R Setup	74
2.1.1	Data Load	74
2.2	Quantitative Variables in nh432	74
2.3	Two-Category (1/0) Variables in nh432	75
2.4	Factor Variables in nh432	77
2.5	Detailed Numerical Description for nh432	78
2.6	Missingness in nh432	89
3	431 Review: Comparing Means	94
3.1	R Setup	94
3.2	Data Ingest	95
3.3	Testing or Summarizing Normality?	95

3.4	Comparing Two Means using Paired Samples	95
3.5	Comparing PULSE1 to PULSE2	96
3.5.1	Distribution of Paired Differences	97
3.5.2	Using <code>t.test</code> to obtain a 90% CI for the mean pulse difference	98
3.5.3	Using linear regression to obtain a 90% CI for the mean pulse difference	99
3.5.4	Using the bootstrap to obtain a 90% CI for the mean pulse difference	100
3.5.5	Wilcoxon signed rank approach to comparing pulse rates	101
3.6	Comparing WEIGHT to ESTWT	102
3.6.1	Plotting The Paired Difference in Weight	103
3.6.2	<code>t.test</code> 90% CI for the mean weight difference	104
3.6.3	Linear Regression: 90% CI for mean weight difference	105
3.6.4	Bootstrap 90% CI for the mean weight difference	105
3.6.5	Wilcoxon signed rank approach to comparing weight estimates	106
3.7	Comparing Two Means using Independent Samples	107
3.8	Comparing White Blood Cell Count by Hospitalization Status	107
3.8.1	Exploring the Data	108
3.8.2	Pooled t test (assumes equal variances) via linear model	110
3.8.3	Pooled t test (assumes equal variances) via <code>t.test</code>	111
3.8.4	Welch t test (doesn't assume equal variance) via <code>t.test</code>	113
3.8.5	Bootstrap comparison of WBC by HOSPITAL	114
3.8.6	Transforming the WBC Counts	115
3.9	Comparing Waist Circumference by Sleep Trouble	118
3.9.1	Summarizing the Data	118
3.9.2	Pooled t test (assumes equal variances) via linear model	120
3.9.3	Pooled t test (assumes equal variances) via <code>t.test</code>	121
3.9.4	Welch t test (doesn't assume equal variance) via <code>t.test</code>	122
3.9.5	Bootstrap comparison of WAIST by SLP TROUB	124
3.9.6	Wilcoxon-Mann-Whitney Rank Sum Approach	125
3.10	Comparing 3 Means using Independent Samples: Systolic BP by Weight Goal	126
3.10.1	Summarizing SBP by WTGOAL	126
3.10.2	Fitting an ANOVA Model	128
3.10.3	Tukey HSD Pairwise Comparisons	130
3.10.4	Holm pairwise comparisons of means	132
3.11	Comparing 4 Means using Independent Samples: Weight by Food Security	133
3.11.1	Summarizing the Data	133
3.11.2	Fitting the ANOVA model	134
3.11.3	Tukey HSD Pairwise Comparisons	135
3.11.4	Kruskal-Wallis Test	137
3.11.5	Dunn Test for Pairwise Comparisons after Kruskal-Wallis Test	137
4	431 Review: Comparing Rates	139
4.1	R Setup	139
4.1.1	Data Load	139

4.2	2x2 Contingency Table: DR_LOSE and NOWLOSE	140
4.2.1	Standard Epidemiological Format	141
4.2.2	Obtaining Key Summaries with <code>twoby2()</code>	143
4.3	2x2 Table: SEDATE category and NOW_EXER	144
4.3.1	Creating a Low and High Group on SEDATE	144
4.3.2	Two-by-Two Table Summaries	146
4.3.3	Flipping Levels	147
4.4	A Larger (5x3) 2-Way Table: DIETQUAL and WTGOAL in Lighter Men . . .	148
4.4.1	What would independence look like?	149
4.4.2	Back to the DIETQUAL and WTGOAL table	150
4.5	PHQ9 Category and Race/Ethnicity	154
4.5.1	The Cochran conditions	155
4.5.2	Collapsing Categories	156
4.5.3	Pearson χ^2 Analysis	157
4.5.4	Mosaic Plot	157
4.5.5	Examining the Fit	158
5	431 Review: Fitting Linear Models	160
5.1	R Setup	160
5.1.1	Data Load	160
5.2	Modeling Weekend Sleep Hours	161
5.2.1	Should we transform our outcome?	162
5.2.2	Scatterplot Matrix	163
5.2.3	Collinearity?	164
5.2.4	Fitting and Displaying Model <code>m1</code>	164
5.2.5	Using broom functions on Model <code>m1</code>	165
5.2.6	Residual Plots for Model <code>m1</code>	167
5.2.7	Fitting and Displaying Model <code>m2</code>	172
5.2.8	Using broom functions on <code>m2</code>	172
5.2.9	Residual Plots for Model <code>m2</code>	174
5.2.10	Conclusions	176
5.3	Modeling High-Sensitivity C-Reactive Protein	176
5.3.1	Partitioning the Data	177
5.3.2	Transforming the Outcome?	177
5.3.3	Scatterplot Matrix and Collinearity	179
5.3.4	Fit Model <code>m3</code>	180
5.3.5	Residual Plots for <code>m3</code>	181
5.3.6	Fit Model <code>m4</code>	183
5.3.7	Residual Plots for <code>m4</code>	183
5.3.8	In-Sample Fit Quality Comparison (<code>m3</code> vs. <code>m4</code>)	185
5.3.9	Testing the models in new data	185
5.3.10	Conclusions	188

6	BRFSS SMART Data	189
6.1	R Setup Used Here	189
6.2	Key resources	190
6.3	Ingesting the Raw Data	190
6.4	Ingesting from our CSV file	191
6.5	What does the raw data look like?	191
6.6	Cleaning the BRFSS Data	192
6.6.1	Identifying Information	192
6.6.2	Survey Method	194
6.6.3	Health Status (1 item)	196
6.6.4	Healthy Days - Health-Related Quality of Life (3 items)	197
6.6.5	Health Care Access (4 items)	201
6.6.6	Blood Pressure (2 measures)	204
6.6.7	Cholesterol (3 items)	207
6.6.8	Chronic Health Conditions (14 items)	209
6.6.9	Arthritis Burden (4 items)	214
6.6.10	Demographics (25 items)	217
6.6.11	Tobacco Use (2 items)	236
6.6.12	E-Cigarettes (2 items)	238
6.6.13	Alcohol Consumption (6 items)	239
6.6.14	Fruits and Vegetables (8 items)	243
6.6.15	Exercise and Physical Activity (8 items)	248
6.6.16	Seatbelt Use (1 item)	258
6.6.17	Immunization (3 items)	259
6.6.18	HIV/AIDS (2 items)	261
6.7	Imputing Age and Income as Quantitative from Thin Air	262
6.7.1	age_imp: Imputing Age Data	263
6.7.2	inc_imp: Imputing Income Data	264
6.8	Clean Data in the State of Ohio	266
6.9	Clean Cleveland-Elyria Data	267
6.9.1	Cleveland - Elyria Data	267
7	Dealing with Missingness: Single Imputation	268
7.1	R Setup Used Here	268
7.1.1	Data Load	268
7.2	Selecting Some Variables from the smart_cle data	268
7.3	smart_cle1: Seeing our Missing Data	270
7.3.1	Plotting Missingness	273
7.4	Missing-data mechanisms	275
7.5	Options for Dealing with Missingness	275
7.6	Complete Case (and Available Case) analyses	276
7.7	Single Imputation	276
7.8	Multiple Imputation	276

7.9	Approach 1: Building a Complete Case Analysis: <code>smart_cle1_cc</code>	277
7.10	Approach 2: Single Imputation to create <code>smart_cle1_sh</code>	277
7.10.1	What Type of Missingness Do We Have?	278
7.10.2	Single imputation into <code>smart_cle1_sh</code>	279
7.10.3	Imputing Binary Categorical Variables	280
7.10.4	Imputing Quantitative Variables	281
7.10.5	Imputation Results	282
7.10.6	Imputing Multi-Categorical Variables	286
7.10.7	Saving the new tibbles	288
8	Summarizing <code>smart_cle1</code>	289
8.1	R Setup Used Here	289
8.1.1	Data Load	289
8.2	What's in these data?	289
8.3	General Approaches to Obtaining Numeric Summaries	290
8.3.1	<code>summary</code> for a data frame	290
8.3.2	The <code>inspect</code> function from the <code>mosaic</code> package	291
8.3.3	The <code>describe</code> function in <code>Hmisc</code>	292
8.4	Counting as exploratory data analysis	293
8.4.1	Did <code>genhealth</code> vary by smoking status?	294
8.4.2	What's the distribution of <code>physhealth</code> ?	295
8.4.3	What's the distribution of <code>bmi</code> ?	297
8.4.4	How many of the respondents have a BMI below 30?	297
8.4.5	How many of the respondents with a BMI < 30 are highly active? . . .	298
8.4.6	Is obesity associated with smoking history?	298
8.4.7	Comparing <code>drinks_wk</code> summaries by obesity status	298
8.5	Can <code>bmi</code> predict <code>physhealth</code> ?	299
8.5.1	Fitting a Simple Regression Model	300
8.5.2	Model Summary for a Simple (One-Predictor) Regression	301
8.5.3	Using the <code>broom</code> package	306
8.5.4	How does the model do? (Residuals vs. Fitted Values)	308
9	Analysis of Variance with SMART	312
9.1	R Setup Used Here	312
9.1.1	Data Load	312
9.2	A One-Factor Analysis of Variance	313
9.2.1	Can <code>activity</code> be used to predict <code>bmi</code> ?	313
9.2.2	Should we transform <code>bmi</code> ?	314
9.2.3	Building the ANOVA model	316
9.2.4	The ANOVA table	317
9.2.5	The Model Coefficients	318
9.2.6	Using <code>tidy</code> to explore the coefficients	319
9.2.7	Using <code>glance</code> to summarize the model's fit	319

9.2.8	Using augment to make predictions	321
9.3	A Two-Factor ANOVA (without Interaction)	322
9.3.1	Model Coefficients	323
9.4	A Two-Factor ANOVA (with Interaction)	326
9.4.1	Model Coefficients	326
9.4.2	Is the interaction term necessary?	329
10	Two-Way ANOVA and Interaction	334
10.1	R Setup Used Here	334
10.1.1	Data Load	334
10.2	The bonding data: A Designed Dental Experiment	334
10.3	A One-Factor Analysis of Variance	335
10.3.1	Look at the Data!	336
10.3.2	Table of Summary Statistics	337
10.4	A Two-Way ANOVA: Looking at Two Factors	340
10.5	A Means Plot (with standard deviations) to check for interaction	341
10.5.1	Summarizing the data after grouping by resin and light	343
10.6	Fitting the Two-Way ANOVA model with Interaction	344
10.6.1	The ANOVA table for our model	345
10.6.2	Is the interaction important?	345
10.6.3	Interpreting the Interaction	345
10.7	Comparing Individual Combinations of resin and light	346
10.8	The bonding model without Interaction	348
10.9	cortisol : A Hypothetical Clinical Trial	350
10.9.1	Codebook and Raw Data for cortisol	351
10.10	Creating a factor combining sex and waist	351
10.11	A Means Plot for the cortisol trial (with standard errors)	352
10.12	A Two-Way ANOVA model for cortisol with Interaction	354
10.12.1	Notes on this Question	355
10.13	A Two-Way ANOVA model for cortisol without Interaction	358
10.13.1	The Graph	358
10.13.2	The ANOVA Model	358
10.13.3	The Regression Summary	359
10.13.4	Tukey HSD Comparisons	359
11	Analysis of Covariance	361
11.1	R Setup Used Here	361
11.1.1	Data Load	361
11.2	An Emphysema Study	361
11.2.1	Codebook	361
11.3	Does sex affect the mean change in theophylline?	362
11.4	Is there an association between age and sex in this study?	363

11.5	Adding a quantitative covariate, <code>age</code> , to the model	364
11.5.1	The ANCOVA model	364
11.5.2	The ANCOVA Table	365
11.6	Rerunning the ANCOVA model after simple imputation	365
11.7	Looking at a factor-covariate interaction	367
11.8	Centering the Covariate to Facilitate ANCOVA Interpretation	368
12	Analysis of Covariance with the SMART data	370
12.1	R Setup Used Here	370
12.1.1	Data Load	370
12.2	A New Small Study: Predicting BMI	370
12.2.1	Does <code>smoke100</code> predict <code>bmi</code> well?	371
12.3	<code>mod1</code> : A simple t-test model	372
12.4	<code>mod2</code> : Adding another predictor (two-way ANOVA without interaction)	374
12.5	<code>mod3</code> : Adding the interaction term (Two-way ANOVA with interaction)	379
12.6	<code>mod4</code> : Using <code>smoke100</code> and <code>physhealth</code> in a model for <code>bmi</code>	381
12.7	Making Predictions with a Linear Regression Model	384
12.7.1	Fitting an Individual Prediction and 95% Prediction Interval	384
12.7.2	Confidence Interval for an Average Prediction	385
12.7.3	Fitting Multiple Individual Predictions to New Data	385
12.8	Centering the model	386
12.8.1	Plot of Model 4 on Centered <code>physhealth</code> : <code>mod4_c</code>	387
12.9	Rescaling an input by subtracting the mean and dividing by 2 standard deviations	388
12.9.1	Refitting model <code>mod4</code> to the rescaled data	389
12.9.2	Interpreting the model on rescaled data	389
12.9.3	Plot of model on rescaled data	390
12.10	<code>mod5</code> : What if we add more variables?	391
12.11	<code>mod6</code> : Would adding self-reported health help?	393
12.12	Key Regression Assumptions for Building Effective Prediction Models	395
12.12.1	Checking Assumptions in model <code>mod6</code>	396
13	Adding Non-linear Terms to a Linear Regression	400
13.1	R Setup Used Here	400
13.1.1	Data Load	400
13.2	The <code>pollution</code> data	400
13.3	Fitting a straight line model to predict <code>y</code> from <code>x2</code>	402
13.4	Quadratic polynomial model to predict <code>y</code> using <code>x2</code>	403
13.4.1	The raw quadratic model	403
13.4.2	Raw quadratic fit after centering <code>x2</code>	406
13.5	Orthogonal Polynomials	409
13.6	Fit a cubic polynomial to predict <code>y</code> from <code>x3</code>	412
13.7	Fitting a restricted cubic spline in a linear regression	416

13.8	“Spending” Degrees of Freedom	422
13.8.1	Overfitting and Limits on the # of Predictors	422
13.8.2	The Importance of Collinearity	423
13.8.3	Collinearity in an Explanatory Model	424
13.8.4	Collinearity in a Prediction Model	424
13.9	Spending DF on Non-Linearity: The Spearman Plot	424
13.9.1	Fitting a Big Model to the <code>pollution</code> data	426
13.9.2	Limitations of <code>lm</code> for fitting complex linear regression models	426
14	Using <code>ols</code> to fit linear models	428
14.1	R Setup Used Here	428
14.1.1	Data Load	428
14.2	Fitting a model with <code>ols</code>	429
14.2.1	The Model Likelihood Ratio Test	430
14.2.2	The <code>g</code> statistic	430
14.3	ANOVA for an <code>ols</code> model	431
14.4	Effect Estimates	431
14.4.1	Simultaneous Confidence Intervals	433
14.5	The <code>Predict</code> function for an <code>ols</code> model	433
14.6	Checking Influence via <code>dfbeta</code>	435
14.6.1	Using the <code>residuals</code> command for <code>dfbetas</code>	436
14.6.2	Using the <code>residuals</code> command for other summaries	437
14.7	Model Validation and Correcting for Optimism	438
14.8	Building a Nomogram for Our Model	440
15	A Model for Prostate Cancer	442
15.1	R Setup Used Here	442
15.1.1	Data Load	442
15.2	Data Load and Background	442
15.3	Code Book	443
15.4	Additions for Later Use	444
15.5	Fitting and Evaluating a Two-Predictor Model	445
15.5.1	Using <code>tidy</code>	446
15.5.2	Interpretation	447
15.6	Exploring Model <code>prost_A</code>	447
15.6.1	<code>summary</code> for Model <code>prost_A</code>	447
15.6.2	Adjusted R^2	448
15.6.3	Coefficient Confidence Intervals	449
15.6.4	ANOVA for Model <code>prost_A</code>	449
15.6.5	Residuals, Fitted Values and Standard Errors with <code>augment</code>	450
15.6.6	Making Predictions with <code>prost_A</code>	451
15.7	Plotting Model <code>prost_A</code>	451
15.7.1	Residual Plots of <code>prost_A</code>	453

16 Validating our Prostate Cancer Model	455
16.1 R Setup Used Here	455
16.1.1 Data Load	455
16.2 Data Cleaning	455
16.3 Fitting the <code>prostA</code> model	456
16.4 Split Validation of Model <code>prost_A</code>	456
16.5 V-fold Cross-Validation Approach for model <code>prostA</code>	458
17 Multiple Imputation and Linear Regression	461
17.1 R Setup Used Here	461
17.2 Data Load	461
17.3 Developing a <code>smart_16</code> data set	461
17.3.1 Any missing values?	463
17.4 Obtaining a Simple Imputation with <code>mice</code>	464
17.5 Linear Regression: Considering a Transformation of the Outcome	465
17.6 Linear Regression: Considering Non-Linearity in the Predictors	466
17.7 “Main Effects” Linear Regression with <code>lm</code> on the Complete Cases	467
17.7.1 Quality of Fit Statistics	468
17.7.2 Interpreting Effect Sizes	469
17.7.3 Making Predictions with the Model	470
17.8 “Augmented” Linear Regression with <code>lm</code> on the Complete Cases	472
17.8.1 Quality of Fit Statistics	473
17.8.2 ANOVA assessing the impact of the non-linear terms	473
17.8.3 Interpreting Effect Sizes	474
17.8.4 Making Predictions with the Model	477
17.9 Using <code>mice</code> to perform Multiple Imputation	478
17.10 Running the Linear Regression in <code>lm</code> with Multiple Imputation	480
17.11 Fit the Multiple Imputation Model with <code>aregImpute</code>	483
17.12 Fit Linear Regression using <code>ols</code> and <code>fit.mult.impute</code>	488
17.12.1 Summaries and Coefficients	489
17.12.2 Effect Sizes	490
17.12.3 Making Predictions with this Model	491
17.12.4 Nomogram	492
17.12.5 Validating Summary Statistics	494
18 Building Table 1	495
18.1 R Setup Used Here	495
18.2 Data Load	495
18.3 Two examples from the <i>New England Journal of Medicine</i>	495
18.3.1 A simple Table 1	495
18.3.2 A group comparison	496
18.4 The MR CLEAN trial	497
18.5 Simulated <code>fakestroke</code> data	499

18.6	Building Table 1 for fakestroke : Attempt 1	500
18.6.1	Some of this is very useful, and other parts need to be fixed.	502
18.6.2	fakestroke Cleaning Up Categorical Variables	502
18.7	fakestroke Table 1: Attempt 2	503
18.7.1	What summaries should we show?	504
18.8	Obtaining a more detailed Summary	506
18.9	Exporting the Completed Table 1 from R to Excel or Word	510
18.9.1	Approach A: Save and open in Excel	510
18.9.2	Approach B: Produce the Table so you can cut and paste it	511
18.10A	Controlled Biological Experiment - The Blood-Brain Barrier	512
18.11	The bloodbrain.csv file	512
18.12A	Table 1 for bloodbrain	513
18.12.1	Generate final Table 1 for bloodbrain	517
18.12.2A	More Finished Version (after Cleanup in Word)	520
19	Logistic Regression: The Foundations	521
19.1	R Setup Used Here	521
19.2	A First Attempt: A Linear Probability Model	521
19.3	Logistic Regression	523
19.4	The Logistic Regression Model	524
19.5	The Link Function	524
19.6	The logit or log odds	525
19.7	Interpreting the Coefficients of a Logistic Regression Model	526
19.8	The Logistic Regression has non-constant variance	526
19.9	Fitting a Logistic Regression Model to our Simulated Data	527
19.10	Plotting the Logistic Regression Model	529
20	Logistic Regression with <code>glm</code>	531
20.1	R Setup Used Here	531
20.1.1	Data Load	531
20.2	The resect data	531
20.3	Running A Simple Logistic Regression Model	533
20.3.1	Logistic Regression Can Be Harder than Linear Regression	533
20.4	Logistic Regression using glm	534
20.4.1	Interpreting the Coefficients of a Logistic Regression Model	535
20.4.2	Using predict to describe the model's fits	536
20.4.3	Odds Ratio interpretation of Coefficients	536
20.4.4	Interpreting the rest of the model output from glm	538
20.4.5	Deviance and Comparing Our Model to the Null Model	538
20.4.6	Using glance with a logistic regression model	539
20.5	Interpreting the Model Summary	540
20.5.1	Wald Z tests for Coefficients in a Logistic Regression	541
20.5.2	Confidence Intervals for the Coefficients	542

20.5.3	Deviance Residuals	542
20.5.4	Dispersion Parameter	543
20.5.5	Fisher Scoring iterations	543
20.6	Plotting a Simple Logistic Regression Model	543
20.6.1	Using <code>augment</code> to capture the fitted probabilities	543
20.6.2	Plotting a Logistic Regression Model's Fitted Values	544
20.6.3	Plotting a Simple Logistic Model using <code>binomial_smooth</code>	545
20.7	How well does Model A classify subjects?	546
20.8	The Confusion Matrix	547
20.9	Using the <code>confusionMatrix</code> tool from the <code>caret</code> package	548
20.10	Receiver Operating Characteristic Curve Analysis	549
20.10.1	Interpreting the Area under the ROC curve	550
20.11	The ROC Plot for <code>res_modA</code>	556
20.11.1	Another way to plot the ROC Curve	558
20.12	Assessing Residual Plots from Model A	559
20.13	Model B: A “Kitchen Sink” Logistic Regression Model	560
20.13.1	Comparing Model A to Model B	561
20.13.2	Interpreting Model B	562
20.14	Plotting Model B	563
20.14.1	Using <code>augment</code> to capture the fitted probabilities	563
20.14.2	Plotting Model B Fits by Observed Mortality	563
20.14.3	Confusion Matrix for Model B	564
20.14.4	The ROC curve for Model B	565
20.14.5	Residuals, Leverage and Influence	566
21	Logistic Regression with <code>lrm</code>	568
21.1	R Setup Used Here	568
21.1.1	Data Load	568
21.2	Logistic Regression using <code>lrm</code>	568
21.2.1	Interpreting Nagelkerke R^2	569
21.2.2	Interpreting the C statistic and Plotting the ROC Curve	570
21.2.3	The C statistic and Somers' D	572
21.2.4	Validating the Logistic Regression Model Summary Statistics	572
21.2.5	Plotting the Summary of the <code>lrm</code> approach	573
21.2.6	Plot In-Sample Predictions for Model C	574
21.2.7	ANOVA from the <code>lrm</code> approach	576
21.2.8	Are any points particularly influential?	576
21.2.9	A Nomogram for Model C	577
21.3	Model D: An Augmented Kitchen Sink Model	578
21.3.1	Spearman ρ^2 Plot	578
21.3.2	Fitting Model D using <code>lrm</code>	579
21.3.3	Assessing Model D using <code>lrm</code> 's tools	579
21.3.4	ANOVA and Wald Tests for Model D	580

21.3.5	Effect Sizes in Model D	580
21.3.6	Plot In-Sample Predictions for Model D	582
21.3.7	Plotting the ROC curve for Model D	583
21.3.8	Validation of Model D summaries	585
21.4	Model E: Fitting a Reduced Model in light of Model D	586
21.4.1	A Plot comparing the two intubation groups	587
21.4.2	Nomogram for Model E	588
21.4.3	Effect Sizes from Model E	589
21.4.4	Plot In-Sample Predictions for Model E	590
21.4.5	ANOVA for Model E	592
21.4.6	Validation of Model E	592
21.4.7	Do any points seem particularly influential?	593
21.4.8	Fitting Model E using <code>glm</code> to get plots about influence	593
21.5	Concordance: Comparing Model C, D and E's predictions	594
21.6	Conclusions	597
22	Estimating and Interpreting Effect Sizes	598
22.1	R Setup Used Here	598
22.1.1	Data Load	598
22.2	Available Variables	598
22.3	Effect Interpretation in A Linear Regression Model	599
22.4	Making a prediction and building a prediction interval with an <code>lm</code> fit	602
22.5	What if we include a Spline or an Interaction?	603
22.6	Making a prediction and building a prediction interval with an <code>ols</code> fit	605
22.7	Effect Estimates in A Logistic Regression fit with <code>glm</code>	606
22.8	Estimates in The Same Logistic Regression fit with <code>lrm</code>	609
22.9	Estimates in A New Logistic Regression fit with <code>lrm</code>	611
23	Colorectal Cancer Screening and Some Special Cases	615
23.1	R Setup Used Here	615
23.2	Data Load	615
23.3	Logistic Regression for Aggregated Data	616
23.3.1	Colorectal Cancer Screening Data	616
23.3.2	Fitting a Logistic Regression Model to Proportion Data	618
23.3.3	Fitting a Logistic Regression Model to Counts of Successes and Failures	619
23.3.4	How does one address this problem in <code>rms</code> ?	620
23.4	Probit Regression	621
23.4.1	Colorectal Cancer Screening Data on Individuals	621
23.4.2	A logistic regression model	623
23.4.3	Predicting status for Harry and Sally	624
23.4.4	A probit regression model	625
23.4.5	Interpreting the Probit Model's Coefficients	626
23.4.6	What about Harry and Sally?	627

24 Modeling a Count Outcome	628
24.1 R Setup Used Here	628
24.2 Data Load	628
24.3 Creating A Useful Analytic Subset, <code>ohioA</code>	629
24.3.1 Is age group associated with <code>physhealth</code> ?	629
24.4 Exploratory Data Analysis (in the 18-49 group)	632
24.4.1 Build a subset of those ages 18-49	632
24.4.2 Centering <code>bmi</code>	633
24.4.3 Distribution of the Outcome	633
24.4.4 Scatterplot Matrix	634
24.4.5 Summary of the final subset of data	635
24.5 Modeling Strategies Explored Here	635
24.5.1 What Will We Demonstrate?	636
24.5.2 Extra Data File for Harry and Sally	636
24.6 The OLS Approach	637
24.6.1 Interpreting the Coefficients	638
24.6.2 Store fitted values and residuals	638
24.6.3 Specify the R^2 and $\log(\text{likelihood})$ values	638
24.6.4 Check model assumptions	639
24.6.5 Predictions for Harry and Sally	641
24.6.6 Notes	641
24.7 OLS model on $\log(\text{physhealth} + 1)$ days	642
24.7.1 Interpreting the Coefficients	643
24.7.2 Store fitted values and residuals	643
24.7.3 Specify the R^2 and $\log(\text{likelihood})$ values	644
24.7.4 Getting R^2 on the scale of <code>physhealth</code>	645
24.7.5 Check model assumptions	645
24.7.6 Predictions for Harry and Sally	646
24.8 A Poisson Regression Model	647
24.8.1 The Fitted Equation	649
24.8.2 Interpreting the Coefficients	649
24.8.3 Testing the Predictors	650
24.8.4 Correcting for Overdispersion with <code>coeftest/coefci</code>	651
24.8.5 Store fitted values and residuals	652
24.8.6 Rootogram: see the fit of a count regression model	652
24.8.7 Specify the R^2 and $\log(\text{likelihood})$ values	653
24.8.8 Check model assumptions	654
24.8.9 Using <code>glm.diag.plots</code> from the <code>boot</code> package	655
24.8.10 Predictions for Harry and Sally	657
24.9 Overdispersion in a Poisson Model	657
24.9.1 Testing for Overdispersion?	658
24.10 Fitting the Quasi-Poisson Model	659
24.10.1 The Fitted Equation	660

24.10.2	Interpreting the Coefficients	660
24.10.3	Testing the Predictors	661
24.10.4	Store fitted values and residuals	661
24.10.5	Specify the R^2 and log(likelihood) values	662
24.10.6	Check model assumptions	663
24.10.7	Predictions for Harry and Sally	664
24.11	Poisson and Quasi-Poisson models using <code>Glm</code> from the <code>rms</code> package	665
24.11.1	Refitting the original Poisson regression with <code>Glm</code>	665
24.11.2	Refitting the overdispersed Poisson regression with <code>Glm</code>	665
24.11.3	ANOVA on a <code>Glm</code> fit	666
24.11.4	ggplots from <code>Glm</code> fit	667
24.11.5	Summary of a <code>Glm</code> fit	667
24.11.6	Plot of the Summary	668
24.11.7	Nomogram of a <code>Glm</code> fit	668
25	Negative Binomial Models for Count Data	670
25.1	R Setup Used Here	670
25.2	Data Load and Subset Creation	670
25.3	Setup for this Chapter	671
25.3.1	What Will We Demonstrate?	671
25.3.2	Extra Data File for Harry and Sally	672
25.3.3	Our Poisson Model (for comparison)	672
25.4	Negative Binomial Model	672
25.4.1	The Fitted Equation	674
25.4.2	Comparison with the (raw) Poisson model	674
25.4.3	Interpreting the Coefficients	675
25.4.4	Interpretation of Coefficients in terms of IRRs	675
25.4.5	Testing the Predictors	676
25.4.6	Store fitted values and residuals	677
25.4.7	Rootogram for Negative Binomial model	678
25.4.8	Simulating what the Negative Binomial model predicts	679
25.4.9	Specify the R^2 and log(likelihood) values	680
25.4.10	Check model assumptions	681
25.4.11	Predictions for Harry and Sally	683
25.5	The Problem: Too Few Zeros	684
26	Zero-Inflated Models for Count Data	686
26.1	R Setup Used Here	686
26.2	Data Load and Subset Creation	686
26.3	Setup for this Chapter	687
26.3.1	What Will We Demonstrate?	687
26.3.2	Extra Data File for Harry and Sally	688
26.3.3	Previous Models (for comparison)	688

26.4	The Zero-Inflated Poisson Regression Model	689
26.4.1	Comparison to a null model	690
26.4.2	Comparison to a Poisson Model with the Vuong test	691
26.4.3	The Fitted Equation	691
26.4.4	Interpreting the Coefficients	692
26.4.5	Testing the Predictors	692
26.4.6	Store fitted values and residuals	693
26.4.7	Modeled Number of Zero Counts	694
26.4.8	Rootogram for ZIP model	694
26.4.9	Specify the R^2 and log (likelihood) values	695
26.4.10	Check model assumptions	696
26.4.11	Predictions for Harry and Sally	696
26.5	The Zero-Inflated Negative Binomial Regression Model	697
26.5.1	Comparison to a null model	698
26.5.2	Comparison to a Negative Binomial Model: Vuong test	699
26.5.3	The Fitted Equation	699
26.5.4	Interpreting the Coefficients	700
26.5.5	Testing the Predictors	700
26.5.6	Store fitted values and residuals	701
26.5.7	Modeled Number of Zero Counts	701
26.5.8	Rootogram for Zero-Inflated Negative Binomial model	702
26.5.9	Specify the R^2 and log (likelihood) values	703
26.5.10	Check model assumptions	703
26.5.11	Predictions for Harry and Sally	704
26.6	A “hurdle” model (with Poisson)	704
26.6.1	Comparison to a null model	706
26.6.2	Comparison to a Poisson Model: Vuong test	707
26.6.3	Comparison to a Zero-Inflated Poisson Model: Vuong test	708
26.6.4	The Fitted Equation	708
26.6.5	Interpreting the Coefficients	708
26.6.6	Testing the Predictors	709
26.6.7	Store fitted values and residuals	710
26.6.8	Modeled Number of Zero Counts	710
26.6.9	Rootogram for Hurdle Model	711
26.6.10	Understanding the Modeled Counts in Detail	711
26.6.11	Specify the R^2 and log (likelihood) values	713
26.6.12	Check model assumptions	713
26.6.13	Predictions for Harry and Sally	714
26.7	A “hurdle” model (with negative binomial for overdispersion)	714
26.7.1	Comparison to a null model	716
26.7.2	Comparison to a Negative Binomial Model: Vuong test	717
26.7.3	Comparison to a Zero-Inflated NB Model: Vuong test	717
26.7.4	Comparing the Hurdle Models with AIC and BIC	717

26.7.5	The Fitted Equation	718
26.7.6	Interpreting the Coefficients	718
26.7.7	Testing the Predictors	719
26.7.8	Store fitted values and residuals	719
26.7.9	Rootogram for NB Hurdle Model	720
26.7.10	Specify the R^2 and log (likelihood) values	721
26.7.11	Check model assumptions	721
26.7.12	Predictions for Harry and Sally	722
26.7.13	Note: Fitting a Different Hurdle Model for Counts and Pr(zero)	722
26.7.14	Hanging Rootogram for this new Hurdle Model	723
26.8	A Tobit (Censored) Regression Model	724
26.8.1	The Fitted Equation	726
26.8.2	Interpreting the Coefficients	726
26.8.3	Testing the Predictors	726
26.8.4	Store fitted values and residuals	727
26.8.5	Building Something Like a Rootogram	728
26.8.6	Tables of the Observed and Fitted <code>physhealth</code> from Tobit	728
26.8.7	Specify the R^2 and log (likelihood) values	729
26.8.8	Check model assumptions	729
26.8.9	Predictions for Harry and Sally	731
27	Modeling an Ordinal Categorical Outcome	732
27.1	R Setup Used Here	732
27.1.1	Data Load	732
27.2	A subset of the Ohio SMART data	732
27.2.1	Several Ways of Storing Multi-Categorical data	733
27.3	Building Cross-Tabulations	734
27.3.1	Using base <code>table</code> functions	734
27.3.2	Using <code>xtabs</code>	736
27.3.3	Storing a table in a tibble	736
27.3.4	Using <code>CrossTable</code> from the <code>gmodels</code> package	737
27.4	Graphing Categorical Data	740
27.4.1	A Bar Chart for a Single Variable	740
27.4.2	A Counts Chart for a 2-Way Cross-Tabulation	741
27.5	Building a Model for <code>genh</code> using <code>veg_day</code>	742
27.5.1	A little EDA	742
27.5.2	Describing the Proportional-Odds Cumulative Logit Model	743
27.5.3	Fitting a Proportional Odds Logistic Regression with <code>polr</code>	745
27.6	Interpreting Model <code>m1</code>	746
27.6.1	Looking at Predictions	746
27.6.2	Making Predictions for Harry (and Sally) with <code>predict</code>	747
27.6.3	Predicting the actual classification of <code>genh</code>	748
27.6.4	A Cross-Tabulation of Predictions?	748

27.6.5	The Fitted Model Equations	748
27.6.6	Interpreting the <code>veg_day</code> coefficient	750
27.6.7	Exponentiating the Slope Coefficient to facilitate Interpretation	750
27.6.8	Comparison to a Null Model	751
27.7	The Assumption of Proportional Odds	751
27.7.1	Testing the Proportional Odds Assumption	753
27.8	Can model <code>m1</code> be fit using <code>rms</code> tools?	754
27.9	Building a Three-Predictor Model	755
27.9.1	Scatterplot Matrix	756
27.9.2	Our Three-Predictor Model, <code>m2</code>	756
27.9.3	Does the three-predictor model outperform <code>m1</code> ?	757
27.9.4	Wald tests for individual predictors	758
27.9.5	A Cross-Tabulation of Predictions?	759
27.9.6	Interpreting the Effect Sizes	759
27.9.7	Quality of the Model Fit	761
27.9.8	Validating the Summary Statistics in <code>m2_lrm</code>	761
27.9.9	Testing the Proportional Odds Assumption	762
27.9.10	Plotting the Fitted Model	766
27.10A	Larger Model, including income group	768
27.10.1	Cross-Tabulation of Predicted/Observed Classifications	770
27.10.2	Nomogram	770
27.10.3	Using Predict and showing mean prediction on 1-5 scale	771
27.10.4	Validating the Summary Statistics in <code>m3_lrm</code>	772
27.11	References for this Chapter	773
28	Multinomial Logistic Regression	774
28.1	R Setup Used Here	774
28.1.1	Data Load	774
28.2	The Authorship Example	774
28.3	Focus on 11 key words	775
28.3.1	Side by Side Boxplots	776
28.4	A Multinomial Logistic Regression Model	778
28.4.1	Testing Model 1	779
28.5	Model 2	780
28.5.1	Comparing Model 2 to Model 1	781
28.5.2	Testing Model 2	782
28.5.3	A little history	782
28.6	Classification Table	782
28.7	Probability Curves based on a Single Predictor	783
28.7.1	Produce the Plot of Estimated Probabilities based on “been” counts	786
28.7.2	Boxplot of “been” counts	786
28.7.3	Quote Sources	787

29 Time To Event / Survival Data	788
29.1 R Setup Used Here	788
29.1.1 Data Load	789
29.2 An Outline of Key Topics Discussed in these Notes	789
29.3 Foundations of Survival Analysis	789
29.3.1 The Survival Function, $S(t)$	790
29.3.2 Kaplan-Meier Estimator of the Survival Function	790
29.3.3 Creating a Survival Object in R	791
29.4 A First Example: Recurrent Lobar Intracerebral Hemorrhage	791
29.5 Building a Survival Object	792
29.6 Kaplan-Meier Estimate of the Survival Function	793
29.6.1 The Kaplan-Meier Plot, using Base R	795
29.6.2 Using <code>survminer</code> to draw survival curves	795
29.6.3 A “Fancy” K-M Plot with a number at risk table	796
29.7 Comparing Survival Across the Two Genotypes	798
29.7.1 Kaplan-Meier Survival Function Estimates, by Genotype	798
29.8 Testing the difference between two survival curves	800
29.8.1 Alternative log rank tests	800
29.9 A “Fancy” K-M Plot with a number at risk table	801
29.9.1 Customizing the Kaplan-Meier Plot Presentation Further	802
29.10 The Hazard Function	805
29.10.1 The Inverse Kaplan-Meier Estimator of $H(t)$	805
29.10.2 Cumulative Hazard Function from Inverse K-M	806
29.10.3 The Nelson-Aalen Estimator of $H(t)$	807
29.10.4 Convert Wide Data to Long	808
29.10.5 Plot Comparison of Hazard Estimates	808
29.11 NEW!! Checking Assumptions Using a log (-log) plot	810
30 Cox Regression Models, Part 1	812
30.1 R Setup Used Here	812
30.1.1 Data Load	812
30.2 Sources used in building this material	813
30.3 Fitting a Cox Model in R with <code>coxph</code>	813
30.3.1 Summarizing the Fit	814
30.3.2 Glancing at the model?	816
30.3.3 Plot the baseline survival function	816
30.3.4 Plot the genotype effect	817
30.3.5 Testing the Key Assumption: Proportional Hazards	818
30.3.6 Plotting the <code>cox.zph</code> results for the <code>cfit</code> model	819
30.4 Fitting a Cox Model using <code>cph</code> from the <code>rms</code> package	820
30.4.1 The Main <code>cph</code> results	820
30.4.2 Using <code>anova</code> with <code>cph</code>	821
30.4.3 Effect Sizes after <code>cph</code> fit	821

30.4.4	Validating <code>cph</code> summaries	822
30.4.5	Plotting Survival Functions for each Genotype	822
30.4.6	Genotype's effect on log relative hazard	824
30.4.7	Nomogram of our simple <code>hem</code> model	825
30.4.8	Assessing the Proportional Hazards Assumption	826
30.4.9	Plot to Check PH Assumption	826
31	Cox Regression Models, Part 2	828
31.1	R Setup Used Here	828
31.1.1	Data Load	828
31.2	A Second Example: The <code>leukem</code> data	828
31.2.1	Creating our response: A survival time object	829
31.2.2	Models We'll Fit	830
31.3	Model A: <code>coxph</code> Model for Survival Time using <code>age</code> at diagnosis	830
31.3.1	Plotting the Survival Curve implied by Model A	832
31.3.2	Testing the Proportional Hazards Assumption	833
31.4	Building Model A with <code>cph</code> for the <code>leukem</code> data	834
31.4.1	Plotting the <code>age</code> effect implied by our model.	835
31.4.2	Survival Plots (Kaplan-Meier) of the <code>age</code> effect	836
31.4.3	ANOVA test for the <code>cph</code> -built model for <code>leukem</code>	838
31.4.4	Summarizing the Effect Sizes from <code>modA_cph</code>	838
31.4.5	Validating the Cox Model Summary Statistics	839
31.4.6	Looking for Influential Points	840
31.4.7	Checking the Proportional Hazards Assumption	841
31.5	Model B: Fitting a 5-Predictor Model with <code>coxph</code>	842
31.5.1	Plotting the Survival Curve implied by Model B	843
31.5.2	Testing the Proportional Hazards Assumption	844
31.5.3	Assessing Collinearity	845
31.6	Model B2: A Stepwise Reduction of Model B	846
31.6.1	The Survival Curve implied by Model B2	847
31.6.2	Checking Proportional Hazards for Model B2	848
31.7	Model C: Using a Spearman Plot to pick a model	849
31.7.1	Fitting Model C	850
31.7.2	ANOVA for Model C	851
31.7.3	Summarizing Model C Effect Sizes	851
31.7.4	Plotting the diagnosis <code>age</code> effect in Model C	852
31.7.5	Survival Plot associated with Model C	853
31.7.6	Checking the Proportional Hazards Assumption	854
31.7.7	Model C Nomogram	855
31.7.8	Validating Model C's Summary Statistics	856
31.7.9	Calibration of Model C (12-month survival estimates)	857

32 NEW!! A Few LASSO Ideas	859
32.1 R Setup Used Here	859
32.1.1 Data Load	859
32.2 The <code>pollution</code> data	859
32.3 Should We Rescale any Predictors?	861
32.4 A Kitchen Sink Model	862
32.4.1 Considering an Outcome Transformation	862
32.4.2 How much collinearity are we dealing with?	864
32.5 Using the LASSO to suggest a smaller model	865
32.6 Would the 9-predictor model be a big improvement?	866
32.7 Using Stepwise Regression to suggest a smaller model	869
33 NEW!! Bayes and a Linear Model	876
33.1 R Setup Used Here	876
33.2 Return to the <code>smalldat</code> Example	876
33.3 The Distribution of Total Cholesterol	877
33.4 Fitting a Linear Model with <code>lm()</code> for Total Cholesterol	877
33.5 Fitting a Bayesian Linear Model	878
33.5.1 Extracting the Posterior	880
33.5.2 Describing Uncertainty	883
33.5.3 Visualizing the Coefficients and Credible Intervals	884
33.6 Summarizing the Posterior Distribution	884
33.6.1 Summarizing the Parameter values	885
33.6.2 Probability of Direction (<code>pd</code>) estimates.	885
33.6.3 The ROPE estimates	886
33.6.4 Convergence Diagnostics	887
33.7 Summarizing the Priors Used	888
33.8 Graphical Posterior Predictive Checks	888
34 NEW!! Bayes and a Logistic Model	891
34.1 R Setup Used Here	891
34.2 Return to the <code>smalldat</code> Example	892
34.3 The Distribution of Smoking Status	892
34.4 Fitting a Logistic Regression Model with <code>glm()</code>	892
34.5 Fitting a Bayesian Logistic Regression	893
34.5.1 Extracting the Posterior	896
34.5.2 Describing Uncertainty	899
34.5.3 Visualizing the Coefficients and Credible Intervals	900
34.6 Summarizing the Posterior Distribution	900
34.6.1 Summarizing the Parameter values	901
34.6.2 Probability of Direction (<code>pd</code>) estimates.	901
34.6.3 The ROPE estimates	902
34.6.4 Summarizing the Coefficients as Odds Ratios	902

34.7 Summarizing the Priors Used	903
34.8 Graphical Posterior Predictive Checks	903
References	906

Introduction

These Notes provide a series of examples using R to work through issues that are likely to come up in PQHS/CRSP/MPHP 432.

While these Notes share some of the features of a textbook, they are neither comprehensive nor completely original. The main purpose is to give students in 432 a set of common materials on which to draw during the course. In class, we will sometimes:

- reiterate points made in this document,
- amplify what is here,
- simplify the presentation of things done here,
- use new examples to show some of the same techniques,
- refer to issues not mentioned in this document,

but what we don't (always) do is follow these notes very precisely. We assume instead that you will read the materials and try to learn from them, just as you will attend classes and try to learn from them. We welcome feedback of all kinds on this document or anything else via email to **431-help at case dot edu**.

What you will mostly find are brief explanations of a key idea or summary, accompanied (most of the time) by R code and a demonstration of the results of applying that code.

Everything you see here is available to you in this HTML document. You will also have access to the Quarto files, which contain the code which generates everything in the document, including all of the R results. We will demonstrate the use of Quarto and R Studio (the “program” which we use to interface with the R language) in class. At the end of the semester, I hope to be able to get the PDF version of this document posted, but that may be a challenge.

To download the data and R code related to these notes, visit the [432-data page](#).

R Setup

Here, we'll show all of the R packages used in these notes. The list of R Packages we will use in 432 is even more extensive, and is [available on our course website](#).

```
knitr::opts_chunk$set(comment = NA) ## Dr. Love always includes this

library(arm)
library(boot)
library(car)
library(caret)
library(countreg) # install via R-Forge
library(DescTools)
library(distributions3)
library(Epi)
library(GGally)
library(ggribes)
library(glue)
library(gmodels)
library(gt)
library(gtsummary)
library(Hmisc)
library(janitor)
library(knitr)
library(lmboot)
library(lmtest)
library(MASS)
library(mice)
library(MKinfer)
library(mosaic)
library(naniar)
library(nhanesA)
library(nnet)
library(patchwork)
library(pROC)
library(psc1)
library(rms)
```

```
library(ROCR)
library(rstatix)
library(sandwich)
library(simputation)
library(survival)
library(survminer)
library(tableone)
library(topmodels) # install via R-Forge
library(vcd)
library(VGAM)
library(visdat)

library(tidymodels)
library(tidyverse)
```

General Theme for ggplot work

Dr. Love prefers `theme_bw()` to the default choice.

```
theme_set(theme_bw())
```

1 Building the nh432 example

In this Chapter, we will extract and tidy a data set using pre-pandemic (2017 - March 2020) data from the National Health and Nutrition Examination Survey (NHANES). Then we'll save this data set (which I'll call **nh432** once it's built) so that we can use it again in subsequent work. Details on the data are available at [this link to NHANES at CDC](#).

1.1 R Setup

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(gt)
library(gtsummary)
library(Hmisc)
library(naniar)
library(nhanesA)

library(tidyverse)

theme_set(theme_bw())
```

1.2 Selecting NHANES Variables

We'll focus on NHANES data describing

- participating adults ages 30-59 years who
- completed both an NHANES interview and medical examination, and who
- also completed an oral health examination, and who
- also reported their overall health as either Excellent, Very Good, Good, Fair, or Poor

We will pull the following NHANES data elements using the **nhanesA** package:

1.2.1 Demographics and Sample Weights

From the [Demographic Variables and Sample Weights](#) database (P_DEMO), we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
RIDSTATR	Interview and MEC exam status	1 = Interviewed only 2 = Interviewed & MEC examined
RIDAGEYR	Age at screening (years) We will require ages 30-59.	Top coded at 80 ¹
RIDRETH3	Race/Hispanic origin	1 = Mexican American 2 = Other Hispanic 3 = Non-Hispanic White 4 = Non-Hispanic Black 6 = Non-Hispanic Asian 7 = Other Race, including Multi-Racial
DMDE-DUC2	Education Level	1 = Less than 9th grade 2 = 9-11th grade 3 = High school graduate 4 = Some college or AA 5 = College graduate or above 7 = Refused (<i>will treat as NA</i>) 9 = Don't Know (<i>will treat as NA</i>)
RIA-GENDR	Sex	1 = Male, 2 = Female
WTINT-PRP	Full sample interview weight	Sampling weight
WT-MECPRP	Full sample MEC examination weight	Sampling weight

Note As part of our inclusion criteria, we will require that all participants in our analytic data have RIDAGEYR values of 30-59 and thus drop the participants whose responses to that item are missing or outside that range.

Here's my code to select these variables from the P_DEMO data.

```
p_demo <- nhanes('P_DEMO', translate = FALSE) |>
  select(SEQN, RIDSTATR, RIDAGEYR, RIDRETH3, DMDEDUC2, RIAGENDR,
         WTINTPRP, WTMECPRP)

dim(p_demo) # gives number of rows (participants) and columns (variables)
```

¹Those over 80 years of age are coded 80.


```
[1] 15560      8
```

Do we have any duplicate SEQN values?

```
p_demo |> get_dupes(SEQN)
```

No duplicate combinations found of: SEQN

```
[1] SEQN      dupe_count RIDSTATR  RIDAGEYR  RIDRETH3  DMDEDUC2  RIAGENDR
[8] WTINTPRP  WTMECPRP
<0 rows> (or 0-length row.names)
```

Good.

1.2.2 Oral Health

From the [Oral Health - Recommendation of Care](#) (P_OHXREF), we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
OHDEXSTS	Overall Oral Health Exam Status	1 = Complete 2 = Partial 3 = Not Done
OHAREC	Overall Recommendation for Dental Care	1 = See a dentist immediately 2 = See a dentist within the next 2 weeks 3 = See a dentist at your earliest convenience 4 = Continue your regular routine care

Note In addition to requiring that all participants in our analytic data have OHDEXSTS = 1, we will (later) collapse values 1 and 2 in OHAREC because there are only a few participants with code 1 in OHAREC.

Here's my code to select these variables from the P_OHXREF data.

```
p_ohxref <- nhanes('P_OHXREF', translate = FALSE) |>
  select(SEQN, OHDEXSTS, OHAREC)

dim(p_ohxref)
```

```
[1] 13772      3
```

1.2.3 Hospital Utilization & Access to Care

From the [Questionnaire on Hospital Utilization & Access to Care](#) (P_HUQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
HUQ010	General health condition we require a 1-5 response	1 = Excellent 2 = Very Good 3 = Good 4 = Fair 5 = Poor 7 = Refused (<i>to be dropped</i>) 9 = Don't Know (<i>to be dropped</i>)
HUQ071	Overnight hospital patient in past 12 months	1 = Yes, 2 = No 7 = Refused (<i>will treat as NA</i>) 9 = Don't Know (<i>will treat as NA</i>)
HUQ090	Seen mental health professional in past 12 months	1 = Yes, 2 = No 7 = Refused (<i>will treat as NA</i>) 9 = Don't Know (<i>will treat as NA</i>)

Note As part of our inclusion criteria, we will require that all participants in our analytic data have HUQ010 values of 1-5 and drop participants whose responses are missing, Refused or Don't Know for that item.

Here's my code to select these variables from the P_HUQ data.

```
p_huq <- nhanes('P_HUQ', translate = FALSE) |>
  select(SEQN, HUQ010, HUQ071, HUQ090)

dim(p_huq)
```

```
[1] 15560      4
```

1.2.4 Body Measures

From the [Body Measures](#) database (P_BMX), we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
BMXWT	Body weight (kg)	Measured in examination
BMXHT	Standing height (cm)	Measured in examination

Variable	Description	Values
BMXWAIST	Waist Circumference (cm)	Measured in examination

Here's my code to select these variables from the P_BMX data.

```
p_bmx <- nhanes('P_BMX', translate = FALSE) |>
  select(SEQN, BMXWT, BMXHT, BMXWAIST)

dim(p_bmx)
```

```
[1] 14300      4
```

1.2.5 Blood Pressure

From the [Blood Pressure - Oscillometric Measurement](#) (P_BPXO), we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
BPXOSY2	Systolic BP (2nd reading, in mm Hg)	Measured in examination
BPXODI2	Diastolic BP (2nd reading, in mm Hg)	Measured in examination
BPXOPLS1	Pulse (1st reading, beats/minute)	Measured in examination
BPXOPLS2	Pulse (2nd reading, beats/minute)	Measured in examination

- A “normal” blood pressure for most adults is < 120 systolic and < 80 diastolic.
- A “normal” resting pulse rate for most adults is between 60 and 100 beats/minute.

Here's my code to select these variables from the P_BPXO data.

```
p_bpxo <- nhanes('P_BPXO', translate = FALSE) |>
  select(SEQN, BPXOSY2, BPXODI2, BPXOPLS1, BPXOPLS2)

dim(p_bpxo)
```

```
[1] 11656      5
```

1.2.6 Complete Blood Count

From the [Complete Blood Count with 5-Part Differential in Whole Blood](#) (P_CBC), we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
LBXW-BCSI	White blood cell count in 1000 cells/uL	normal range: 4.5 - 11
LBXPLTSI	Platelet count in 1000 cells/uL	normal range: 150-450

Here's my code to select these variables from the P_CBC data.

```
p_cbc <- nhanes('P_CBC', translate = FALSE) |>
  select(SEQN, LBXWBCSI, LBXPLTSI)

dim(p_cbc)
```

```
[1] 13772      3
```

1.2.7 C-Reactive Protein

From the [High-Sensitivity C-Reactive Protein](#) (P_HSCRP) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
LBXH-SCRP	High-Sensitivity C-Reactive Protein (mg/L)	normal range: 1.0 - 3.0

Here's my code to select these variables from the P_HSCRP data.

```
p_hscrp <- nhanes('P_HSCRP', translate = FALSE) |>
  select(SEQN, LBXHSCRP)

dim(p_hscrp)
```

```
[1] 13772      2
```

1.2.8 Alcohol Use

From the [Questionnaire on Alcohol Use](#) (P_ALQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
ALQ111	Ever had a drink of alcohol?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
ALQ130	Average drinks per day in past 12 months (Top coded at 15 ²)	count (set to 0 if ALQ111 is No) 777 = Refused (<i>treat as NA</i>) 999 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_ALQ data.

```
p_alq <- nhanes('P_ALQ', translate = FALSE) |>
  select(SEQN, ALQ111, ALQ130)

dim(p_alq)
```

```
[1] 8965    3
```

As noted above, we set the value of ALQ130 to be 0 if the response to ALQ111 is 2 (No).

```
p_alq <- p_alq |>
  mutate(ALQ130 = ifelse(ALQ111 == 2, 0, ALQ130))

p_alq |> count(ALQ130, ALQ111)
```

	ALQ130	ALQ111	n
1	0	2	867
2	1	1	2126
3	2	1	1769
4	3	1	843
5	4	1	431
6	5	1	231
7	6	1	201
8	7	1	44

²Those who responded with 15 drinks or more are coded 15.

9	8	1	65
10	9	1	13
11	10	1	42
12	11	1	3
13	12	1	53
14	13	1	3
15	15	1	29
16	777	1	1
17	999	1	9
18	NA	1	1640
19	NA	NA	595

1.2.9 Dermatology

From the [Questionnaire on Dermatology](#) (P_DEQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
DEQ034D	Use sunscreen if outside on very sunny day?	1 = Always 2 = Most of the time 3 = Sometimes 4 = Rarely 5 = Never 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_DEQ data.

```
p_deq <- nhanes('P_DEQ', translate = FALSE) |>
  select(SEQN, DEQ034D)

dim(p_deq)
```

```
[1] 5810    2
```

1.2.10 Depression Screener

From the [Questionnaire on Mental Health - Depression Screener](#) (P_DPQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
DPQ010	Have little interest in doing things	0-3: codes below
DPQ020	Feeling down, depressed or hopeless	0-3: codes below
DPQ030	Trouble sleeping or sleeping too much	0-3: codes below
DPQ040	Feeling tired or having little energy	0-3: codes below
DPQ050	Poor appetite or overeating	0-3: codes below
DPQ060	Feeling bad about yourself	0-3: codes below
DPQ070	Trouble concentrating on things	0-3: codes below
DPQ080	Moving or speaking slowly or too fast	0-3: codes below
DPQ090	Thoughts you would be better off dead	0-3: codes below
DPQ100	Difficulty these problems have caused	0-3: codes below

- For DPQ010 - DPQ090, the codes are 0 = Not at all, 1 = Several days in the past two weeks, 2 = More than half the days in the past two weeks, 3 = Nearly every day in the past two weeks, with 7 = *Refused* and 9 = *Don't Know which we will treat as NA*.
- For DPQ100, the codes are 0 = Not at all difficult, 1 = Somewhat difficult, 2 = Very difficult, 3 = Extremely difficult, with 7 = *Refused* and 9 = *Don't Know which we will treat as NA*. Also, **the DPQ100 score should be 0 if the scores on DPQ010 through DPQ090 are all zero.**

Later, we will sum the scores in DPQ010 - DPQ090 to produce a PHQ-9 score for each participant.

Here's my code to select these variables from the P_DPQ data.

```
p_dpq <- nhanes('P_DPQ', translate = FALSE)
# we're actually pulling all available variables

dim(p_dpq)
```

```
[1] 8965    11
```

1.2.11 Diet Behavior

From the [Questionnaire on Diet Behavior and Nutrition](#) (P_DBQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
DBQ700	How healthy is your diet?	1 = Excellent 2 = Very Good 3 = Good 4 = Fair 5 = Poor 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_DBQ data.

```
p_dbq <- nhanes('P_DBQ', translate = FALSE) |>
  select(SEQN, DBQ700)

dim(p_dbq)
```

```
[1] 15560      2
```

1.2.12 Food Security

From the [Questionnaire on Food Security](#) (P_FSQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
FSDAD	Adult food security category for last 12m	1 = Full food security 2 = Marginal 3 = Low 4 = Very low

Here's my code to select these variables from the P_FSQ data.

```
p_fsq <- nhanes('P_FSQ', translate = FALSE) |>
  select(SEQN, FSDAD)

dim(p_fsq)
```

```
[1] 15560      2
```


1.2.13 Health Insurance

From the [Questionnaire on Health Insurance](#) (P_HIQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
HIQ011	Covered by health insurance now?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
HIQ210	Time when no insurance in past year?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>) (set to Yes if HIQ011 is No.)

Here's my code to select these variables from the P_HIQ data.

```
p_hiq <- nhanes('P_HIQ', translate = FALSE) |>
  select(SEQN, HIQ011, HIQ210)

dim(p_hiq)
```

```
[1] 15560      3
```

As noted above, we set the value of HIQ210 to be 1 (Yes) if HIQ011 is 2 (No).

```
p_hiq <- p_hiq |>
  mutate(HIQ210 = ifelse(HIQ011 == 2, 1, HIQ210))

p_hiq |> count(HIQ210, HIQ011)
```

	HIQ210	HIQ011	n
1	1	1	960
2	1	2	1852
3	2	1	12682
4	7	1	2
5	9	1	25
6	NA	1	2
7	NA	7	8
8	NA	9	29

1.2.14 Medical Conditions

From the [Questionnaire on Medical Conditions](#) (P_MCQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
MCQ366A	Doctor told you to control/lose weight in the past 12 months?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
MCQ366B	Doctor told you to exercise in the past 12 months?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
MCQ371A	Are you now controlling or losing weight?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
MCQ371B	Are you now increasing exercise?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_MCQ data.

```
p_mcq <- nhanes('P_MCQ', translate = FALSE) |>
  select(SEQN, MCQ366A, MCQ366B, MCQ371A, MCQ371B)

dim(p_mcq)
```

```
[1] 14986      5
```

1.2.15 Oral Health

From the [Questionnaire on Oral Health](#) (P_OHQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
OHQ870	Days using dental floss (in the last week)	count (0-7) 9 = Unknown (<i>treat as NA</i>) 99 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_OHQ data.

```
p_ohq <- nhanes('P_OHQ', translate = FALSE) |>
  select(SEQN, OHQ870)

dim(p_ohq)
```

```
[1] 14986      2
```

1.2.16 Physical Activity

From the [Questionnaire on Physical Activity](#) (P_PAQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
PAQ605	Vigorous work activity for 10 min/week?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
PAQ610	# of days of vigorous work activity in past week	count (1-7) 77 = Refused (<i>treat as NA</i>) 99 = Don't Know (<i>treat as NA</i>) (set to 0 if PAQ605 is No.)
PAQ650	Vigorous recreational activity for 10 min/week?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
PAQ655	# of days of vigorous recreational activity in past week	count (1-7) 77 = Refused (<i>treat as NA</i>) 99 = Don't Know (<i>treat as NA</i>) (set to 0 if PAQ650 is No.)
PAD680	Minutes of sedentary activity (min/day)	excludes sleeping 7777 = Refused (<i>treat as NA</i>) 9999 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_PAQ data.

```
p_paq <- nhanes('P_PAQ', translate = FALSE) |>
  select(SEQN, PAQ605, PAQ610, PAQ650, PAQ655, PAD680)

dim(p_paq)
```

```
[1] 9693      6
```

Now, let's set the value of PAQ610 to be 0 if PAQ605 is 2 (No).

```
p_paq <- p_paq |>
  mutate(PAQ610 = ifelse(PAQ605 == 2, 0, PAQ610))
```

Finally, we set the value of PAQ655 to be 0 if PAQ650 is 2 (No).

```
p_paq <- p_paq |>
  mutate(PAQ655 = ifelse(PAQ650 == 2, 0, PAQ655))
```

1.2.17 Reproductive Health

From the [Questionnaire on Reproductive Health](#) (P_RHQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
RHQ131	Ever been pregnant?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
RHQ160	How many times have you been pregnant?	count (1-11) 77 = Refused (<i>treat as NA</i>) 99 = Don't Know (<i>treat as NA</i>) (set to 0 if RHQ131 is No.)

Here's my code to select these variables from the P_RHQ data.

```
p_rhq <- nhanes('P_RHQ', translate = FALSE) |>
  select(SEQN, RHQ131, RHQ160)

dim(p_rhq)
```

```
[1] 5314    3
```

Now, let's set the value of RHQ160 to be 0 if RHQ131 is 2 (No).

```
p_rhq <- p_rhq |>
  mutate(RHQ160 = ifelse(RHQ131 == 2, 0, RHQ160))
```

1.2.18 Sleep Disorders

From the [Questionnaire on Sleep Disorders](#) (P_SLQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
SLD012	Usual hours of sleep (weekdays)	hours <i>limited to 2-14</i>
SLD013	Usual hours of sleep (weekends)	hours <i>limited to 2-14</i>
SLQ030	How often do you snore in the past 12 months?	0 = Never 1 = Rarely (1-2 nights/week) 2 = Occasionally (3-4 nights/week) 3 = Frequently (5+ nights/week) 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
SLQ050	Have you ever told a doctor you had trouble sleeping?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_SLQ data.

```
p_slq <- nhanes('P_SLQ', translate = FALSE) |>
  select(SEQN, SLD012, SLD013, SLQ030, SLQ050)

dim(p_slq)
```

```
[1] 10195      5
```

1.2.19 Smoking Cigarettes

From the [Questionnaire on Smoking - Cigarette Use](#) (P_SMQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
SMQ020	Smoked at least 100 cigarettes in your life?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
SMD641	Days (in past 30) when you smoked a cigarette?	count (0-30) 77 = Refused (<i>treat as NA</i>) 99 = Don't Know (<i>treat as NA</i>) (set to 0 if SMQ020 is No.)

Here's my code to select these variables from the P_SMQ data.

```
p_smq <- nhanes('P_SMQ', translate = FALSE) |>
  select(SEQN, SMQ020, SMD641)

dim(p_smq)
```

```
[1] 11137      3
```

Now, let's set the value of SMD641 to be 0 if SMQ020 is 2 (No).

```
p_smq <- p_smq |>
  mutate(SMD641 = ifelse(SMQ020 == 2, 0, SMD641))
```

1.2.20 Secondhand Smoke

From the [Questionnaire on Smoking - Secondhand Smoke Exposure](#) (P_SMQSHS) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
SMQ856	Last 7 days worked at a job not at home?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
SMQ860	Last 7 days spent time in a restaurant?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)
SMQ866	Last 7 days spent time in a bar?	1 = Yes, 2 = No 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_SMQSHS data.

```
p_smqshs <- nhanes('P_SMQSHS', translate = FALSE) |>
  select(SEQN, SMQ856, SMQ860, SMQ866)

dim(p_smqshs)
```

```
[1] 15560      4
```

1.2.21 Weight History

From the [Questionnaire on Weight History](#) (P_WHQ) we collect the following variables.

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
WHD010	Current self-reported height (in inches)	49 to 82 7777 = Refused (<i>treat as NA</i>) 9999 = Don't Know (<i>treat as NA</i>)
WHD020	Current self-reported weight (in pounds)	67 to 578 7777 = Refused (<i>treat as NA</i>) 9999 = Don't Know (<i>treat as NA</i>)
WHQ040	Like to weigh more, less, or same	1 = More 2 = Less 3 = Stay about the same 7 = Refused (<i>treat as NA</i>) 9 = Don't Know (<i>treat as NA</i>)

Here's my code to select these variables from the P_WHQ data.

```
p_whq <- nhanes('P_WHQ', translate = FALSE) |>
  select(SEQN, WHD010, WHD020, WHQ040)

dim(p_whq)
```

```
[1] 10195      4
```

1.3 Filtering for Inclusion

First, I'll filter the demographic data (p_demo) to the participants with known ages (RIDAGEYR here) between 30 and 59 years (inclusive), and to those who were both interviewed and examined (so RIDSTATR is 2) to match our inclusion criteria.

```
p_demo <- p_demo |>
  filter(RIDAGEYR >= 30 & RIDAGEYR <= 59,
         RIDSTATR == 2)

dim(p_demo)
```

```
[1] 4133      8
```

Second, I'll restrict the `p_ohxref` sample to the participants who had a complete oral health exam (so `OHDEXSTS` is 1) which is also part of our inclusion criteria.

```
p_ohxref <- p_ohxref |>
  filter(OHDEXSTS == 1)

dim(p_ohxref)
```

```
[1] 13271      3
```

Third, I'll restrict the `p_huq` sample to the participants who gave one of our five available responses (codes 1-5) to the general health condition question in `HUQ010`, which is the final element of our inclusion criteria.

```
p_huq <- p_huq |>
  filter(HUQ010 <= 5)

dim(p_huq)
```

```
[1] 15550      4
```

Subjects that meet all of these requirements will be included in our analytic data. To achieve that end, we'll begin merging the individual data bases.

1.4 Merging the Data

1.4.1 Merging Two Data Frames at a Time

We have two ways to merge our data. We can merge data sets two at a time. In this case, we'll use `inner_join()` from the `dplyr` package to include only those participants with data in each of the two data frames we're merging. For example, we'll create `temp01` to include data from both `p_demo` and `p_ohxref` for all participants (identified by their `SEQN`) that appear in each of those two data frames. Then, we'll merge the resulting `temp01` with `p_huq` to create `temp02` in a similar way.

```
temp01 <- inner_join(p_demo, p_ohxref, by = "SEQN")
temp02 <- inner_join(temp01, p_huq, by = "SEQN")

dim(temp02)
```



```
[1] 3931    13
```

Note that we now have 3931 participants in our data, and this should be the case after we merge in all of the other data sets, too. Rather than using `inner_join()` we will switch now to using `left_join()` many more times so that we always add new information only on those subjects who meet our inclusion criteria (as identified in `temp02`. For more on the various types of joins we can use from the `dplyr` package, visit <https://dplyr.tidyverse.org/reference/mutate-joins.html>. The problem is that that approach would force us to create lots of new temporary files as we add in each new variable.

1.4.2 Merging Many Data Frames Together

A better approach is to use the `reduce()` function in the `purrr` package³, which will let us join this `temp02` data frame with our remaining 17 data frames using `left_join()` in a much more streamlined way. We'll also ensure that the final result (which we'll call `nh_raw`) is a tibble, rather than just a data frame.

```
df_list <- list(temp02, p_bmx, p_bpxo, p_cbc, p_hscrp,
               p_alq, p_deq, p_dpq, p_dbq, p_fsq,
               p_hiq, p_mcq, p_ohq, p_paq, p_rhq,
               p_slq, p_smqshs, p_smq, p_whq)

nh_raw <- df_list |>
  reduce(left_join, by = 'SEQN') |>
  as_tibble()

dim(nh_raw)
```

```
[1] 3931    64
```

1.5 The “Raw” Data

What does the data in `nh_raw` look like? Normally, I wouldn't include this sort of intermediate description in a published bit of work, but it may be helpful to compare this description to the one we'll generate at the end of the cleaning process in this case.

```
summary(nh_raw)
```

³For more on the `purrr` package, visit <https://purrr.tidyverse.org/>

SEQN	RIDSTATR	RIDAGEYR	RIDRETH3	DMDEDUC2
Min. :109271	Min. :2	Min. :30.00	Min. :1.000	Min. :1.00
1st Qu.:113103	1st Qu.:2	1st Qu.:37.00	1st Qu.:3.000	1st Qu.:3.00
Median :117059	Median :2	Median :45.00	Median :3.000	Median :4.00
Mean :117074	Mean :2	Mean :44.79	Mean :3.561	Mean :3.64
3rd Qu.:121040	3rd Qu.:2	3rd Qu.:53.00	3rd Qu.:4.000	3rd Qu.:5.00
Max. :124818	Max. :2	Max. :59.00	Max. :7.000	Max. :7.00

RIAGENDR	WTINTPRP	WTMECPRP	OHDEXSTS	OHAREC
Min. :1.000	Min. : 2467	Min. : 2589	Min. :1	Min. :1.000
1st Qu.:1.000	1st Qu.: 10615	1st Qu.: 11365	1st Qu.:1	1st Qu.:3.000
Median :2.000	Median : 17358	Median : 18422	Median :1	Median :4.000
Mean :1.533	Mean : 28434	Mean : 30353	Mean :1	Mean :3.455
3rd Qu.:2.000	3rd Qu.: 31476	3rd Qu.: 33155	3rd Qu.:1	3rd Qu.:4.000
Max. :2.000	Max. :311265	Max. :321574	Max. :1	Max. :4.000

HUQ010	HUQ071	HUQ090	BMXWT
Min. :1.000	Min. :1.000	Min. :1.000	Min. : 36.90
1st Qu.:2.000	1st Qu.:2.000	1st Qu.:2.000	1st Qu.: 69.30
Median :3.000	Median :2.000	Median :2.000	Median : 82.10
Mean :2.741	Mean :1.913	Mean :1.883	Mean : 86.31
3rd Qu.:3.000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.: 99.10
Max. :5.000	Max. :2.000	Max. :9.000	Max. :254.30

NA's :28

BMXHT	BMXWAIST	BPXOSY2	BPXODI2
Min. :135.3	Min. : 57.9	Min. : 69.0	Min. : 31.00
1st Qu.:160.0	1st Qu.: 89.1	1st Qu.:110.0	1st Qu.: 69.00
Median :166.9	Median : 99.2	Median :120.0	Median : 76.00
Mean :167.4	Mean :101.5	Mean :121.5	Mean : 77.03
3rd Qu.:174.7	3rd Qu.:111.7	3rd Qu.:131.0	3rd Qu.: 84.00
Max. :198.7	Max. :178.0	Max. :222.0	Max. :136.00
NA's :30	NA's :149	NA's :346	NA's :346

BPXOPLS1	BPXOPLS2	LBXWBCSI	LBXPLTSI
Min. : 38.0	Min. : 37.00	Min. : 2.300	Min. : 47.0
1st Qu.: 62.0	1st Qu.: 63.00	1st Qu.: 5.700	1st Qu.:210.0
Median : 69.0	Median : 70.00	Median : 6.900	Median :246.0
Mean : 70.3	Mean : 70.96	Mean : 7.254	Mean :253.3
3rd Qu.: 77.0	3rd Qu.: 78.00	3rd Qu.: 8.400	3rd Qu.:290.0
Max. :126.0	Max. :121.00	Max. :22.800	Max. :818.0
NA's :615	NA's :617	NA's :176	NA's :176

LBXHSCRCP	ALQ111	ALQ130	DEQ034D
Min. : 0.110	Min. :1.000	Min. : 0.000	Min. :1.000
1st Qu.: 0.890	1st Qu.:1.000	1st Qu.: 1.000	1st Qu.:3.000

Median : 2.090	Median :1.000	Median : 2.000	Median :4.000
Mean : 4.326	Mean :1.089	Mean : 2.345	Mean :3.675
3rd Qu.: 4.740	3rd Qu.:1.000	3rd Qu.: 3.000	3rd Qu.:5.000
Max. :182.820	Max. :2.000	Max. :15.000	Max. :5.000
NA's :267	NA's :205	NA's :789	NA's :19
DPQ010	DPQ020	DPQ030	DPQ040
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :1.0000
Mean :0.3907	Mean :0.3732	Mean :0.6529	Mean :0.7612
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :9.0000	Max. :7.0000	Max. :9.0000	Max. :9.0000
NA's :212	NA's :212	NA's :212	NA's :212
DPQ050	DPQ060	DPQ070	DPQ080
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.4117	Mean :0.2504	Mean :0.2924	Mean :0.1622
3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :9.0000	Max. :3.0000	Max. :3.0000	Max. :3.0000
NA's :212	NA's :213	NA's :213	NA's :213
DPQ090	DPQ100	DBQ700	FSDAD
Min. :0.000	Min. :0.0000	Min. :1.00	Min. :1.000
1st Qu.:0.000	1st Qu.:0.0000	1st Qu.:2.00	1st Qu.:1.000
Median :0.000	Median :0.0000	Median :3.00	Median :1.000
Mean :0.053	Mean :0.3447	Mean :3.11	Mean :1.737
3rd Qu.:0.000	3rd Qu.:1.0000	3rd Qu.:4.00	3rd Qu.:2.000
Max. :3.000	Max. :3.0000	Max. :9.00	Max. :4.000
NA's :214	NA's :1448		NA's :231
HIQ011	HIQ210	MCQ366A	MCQ366B
Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000
Median :1.000	Median :2.000	Median :2.000	Median :2.000
Mean :1.211	Mean :1.742	Mean :1.699	Mean :1.574
3rd Qu.:1.000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:2.000
Max. :9.000	Max. :9.000	Max. :9.000	Max. :9.000
	NA's :12		
MCQ371A	MCQ371B	OHQ870	PAQ605
Min. :1.000	Min. :1.000	Min. : 0.000	Min. :1.000
1st Qu.:1.000	1st Qu.:1.000	1st Qu.: 0.000	1st Qu.:1.000
Median :1.000	Median :1.000	Median : 3.000	Median :2.000
Mean :1.371	Mean :1.399	Mean : 3.503	Mean :1.723
3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.: 7.000	3rd Qu.:2.000

Max. :9.000	Max. :9.000	Max. :99.000	Max. :9.000	
		NA's :1		
PAQ610	PAQ650	PAQ655	PAD680	
Min. : 0.000	Min. :1.000	Min. : 0.0000	Min. : 2.0	
1st Qu.: 0.000	1st Qu.:1.000	1st Qu.: 0.0000	1st Qu.: 180.0	
Median : 0.000	Median :2.000	Median : 0.0000	Median : 300.0	
Mean : 1.244	Mean :1.731	Mean : 0.9201	Mean : 363.7	
3rd Qu.: 2.000	3rd Qu.:2.000	3rd Qu.: 1.0000	3rd Qu.: 480.0	
Max. :99.000	Max. :2.000	Max. :99.0000	Max. :9999.0	
NA's :4			NA's :11	
RHQ131	RHQ160	SLD012	SLD013	
Min. :1.000	Min. : 0.000	Min. : 2.000	Min. : 2.000	
1st Qu.:1.000	1st Qu.: 2.000	1st Qu.: 6.500	1st Qu.: 7.000	
Median :1.000	Median : 3.000	Median : 7.500	Median : 8.000	
Mean :1.114	Mean : 3.159	Mean : 7.359	Mean : 8.231	
3rd Qu.:1.000	3rd Qu.: 4.000	3rd Qu.: 8.000	3rd Qu.: 9.000	
Max. :7.000	Max. :77.000	Max. :14.000	Max. :14.000	
NA's :1970	NA's :1972	NA's :34	NA's :34	
SLQ030	SLQ050	SMQ856	SMQ860	
Min. :0.000	Min. :1.000	Min. :1.000	Min. :1.000	
1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	
Median :2.000	Median :2.000	Median :1.000	Median :1.000	
Mean :2.015	Mean :1.711	Mean :1.323	Mean :1.423	
3rd Qu.:3.000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:2.000	
Max. :9.000	Max. :9.000	Max. :2.000	Max. :9.000	
SMQ866	SMQ020	SMD641	WHD010	WHD020
Min. :1.000	Min. :1.0	Min. : 0.000	Min. : 50	Min. : 86.0
1st Qu.:2.000	1st Qu.:1.0	1st Qu.: 0.000	1st Qu.: 63	1st Qu.: 152.0
Median :2.000	Median :2.0	Median : 0.000	Median : 66	Median : 180.0
Mean :1.846	Mean :1.6	Mean : 6.859	Mean : 247	Mean : 353.9
3rd Qu.:2.000	3rd Qu.:2.0	3rd Qu.: 5.000	3rd Qu.: 70	3rd Qu.: 220.0
Max. :2.000	Max. :7.0	Max. :99.000	Max. :9999	Max. :9999.0
		NA's :724	NA's :24	
WHQ040				
Min. :1.00				
1st Qu.:2.00				
Median :2.00				
Mean :2.18				
3rd Qu.:2.00				
Max. :9.00				

1.6 Cleaning Tasks

We now have a tibble called `nh_raw` containing 3931 NHANES participants in the rows and 64 variables in the columns. What must we do to clean up the data?

1. Check that every identifier (here, `SEQN`) is unique.
2. Ensure that all coded values for “Refused”, “Don’t Know” or “Missing” are interpreted as missing values by R.
3. Be sure all quantitative variables have plausible minimum and maximum values.
4. Replace the `RIAGENDR` variable with a new factor variable called `SEX` with levels Male and Female.
5. Convert all binary Yes/No variables to 1/0 numeric variables where 1 = Yes, 0 = No.
6. Create the PHQ-9 score from the nine relevant items in the depression screener (`P_DPQ`).
7. Use meaningful level names for all multi-categorical variables, and be sure R uses factors to represent them.
8. Clean and adjust the names of the variables to something more useful, as desired. (Usually, I will do this first, but in this case, I’ve decided to do it last.)

Once we’ve accomplished these cleaning tasks, we’ll save the resulting tibble as an R data set we can use later, and we’ll summarize our final analytic variables in a proper codebook.

1.7 Our identifying variable

`SEQN` is meant to identify the rows (participants) in these data, with one row per `SEQN`. Is every `SEQN` unique?

```
nrow(nh_raw)
```

```
[1] 3931
```

```
n_distinct(nh_raw$SEQN)
```

```
[1] 3931
```

It looks like the number of rows in our tibble matches the number of unique (distinct) `SEQN` values, so we’re OK. I prefer to specify that the `SEQN` be maintained by R as a character variable, which reduces the chance of my accidentally including it in a model as if it were something meaningful.

```
nh_fixing <- nh_raw |> mutate(SEQN = as.character(SEQN))
```

1.8 “Refused” & “Don’t Know”

Some of our variables have “hidden” missing values coded as “Refused” or “Don’t Know”. We must ensure that R sees these values as missing.

- The following variables use code 7 for Refused and 9 for Don’t Know:
 - DMDEDUC2, HUQ071, HUQ090, ALQ111, DEQ034D,
 - DPQ010, DPQ020, DPQ030, DPQ040, DPQ050,
 - DPQ060, DPQ070, DPQ080, DPQ090, DPQ100,
 - DBQ700, HIQ011, HIQ210, MCQ366A, MCQ366B,
 - MCQ371A, MCQ371B, PAQ605, PAQ650, RHQ131,
 - SLQ030, SLQ050, SMQ020, SMQ856, SMQ860,
 - SMQ866, WHQ040
- The following variables use code 9 for Unknown and 99 for Don’t Know:
 - OHQ870
- The following variables use code 77 for Refused and 99 for Unknown:
 - PAQ610, PAQ655, RHQ160, SMD641
- The following variables use code 777 for Refused and 999 for Don’t Know:
 - ALQ130
- The following variables use code 7777 for Refused and 9999 for Don’t Know:
 - PAD680, WHD010, WHD020

The `replace_with_na()` set of functions from the `naniar` package can be very helpful here⁴.

```
nh_fixing <- nh_fixing %>%  
  replace_with_na_at(.vars = c("DMDEDUC2", "HUQ071", "HUQ090", "ALQ111",  
                                "DEQ034D", "DPQ010", "DPQ020", "DPQ030",  
                                "DPQ040", "DPQ050", "DPQ060", "DPQ070",  
                                "DPQ080", "DPQ090", "DPQ100", "DBQ700",  
                                "HIQ011", "HIQ210", "MCQ366A", "MCQ366B",  
                                "MCQ371A", "MCQ371B", "PAQ605", "PAQ650",  
                                "RHQ131", "SLQ030", "SLQ050", "SMQ020",  
                                "SMQ856", "SMQ860", "SMQ866", "WHQ040"),
```

⁴For more on `replace_with_na()`, visit <https://naniar.njtierney.com/articles/replace-with-na.html>

```

      condition = ~.x %in% c(7, 9)) %>%
replace_with_na_at(.vars = c("OHQ870"),
      condition = ~.x %in% c(9, 99)) %>%
replace_with_na_at(.vars = c("PAQ610", "PAQ655", "RHQ160", "SMD641"),
      condition = ~.x %in% c(77, 99)) %>%
replace_with_na_at(.vars = c("ALQ130"),
      condition = ~.x %in% c(777, 999)) %>%
replace_with_na_at(.vars = c("PAD680", "WHD010", "WHD020"),
      condition = ~.x %in% c(7777, 9999))

```

1.9 Variables without Variation

Note first that we have two variables which now have the same value for all participants.

Variable	Description	Codes
RIDSTATR	Interview and examination status	2 = Both
OHDEXSTS	Complete oral health exam?	1 = Yes

```
nh_fixing |> count(RIDSTATR, OHDEXSTS)
```

```

# A tibble: 1 x 3
  RIDSTATR OHDEXSTS     n
  <dbl>    <dbl> <int>
1         2         1 3931

```

We won't use these variables in our analyses, now that we've verified them.

1.10 Quantitative Variables

Here are our quantitative variables, and some key information about the values we observe, along with their units of measurement. Our job here is to check the ranges of these variables, and be sure we have no unreasonable values. It's also helpful to keep an eye on how much missingness we might have to deal with.

We're also going to rename each of these variables, as indicated.

NHANES					
New Name	Name	Description	Units	# NA	Range
AGE	RIDAGEYR	Age at screening	years	0	30, 59
WEIGHT	BMXWT	Body weight	kg	28	36.9, 254.3
HEIGHT	BMXHT	Standing height	cm	30	135.3, 198.7
WAIST	BMXWAIST	Waist Circumference	cm	149	57.9, 178
SBP	BPXOSY2	Systolic BP (2nd reading)	mm Hg	346	69, 222
DBP	BPXODI2	Diastolic BP (2nd reading)	mm Hg	346	31, 136
PULSE1	BPXO-PLS1	Pulse (1st reading)	$\frac{beats}{minute}$	615	38, 126
PULSE2	BPXO-PLS2	Pulse (2nd reading)	$\frac{beats}{minute}$	617	37, 121
WBC	LBXW-BCSI	White blood cell count	$\frac{1000 \text{ cells}}{uL}$	176	2.3, 22.8
PLATELET	LBXPLTSI	Platelet count	$\frac{1000 \text{ cells}}{uL}$	176	47, 818
HSCRIP	LBXH-SCRIP	High-Sensitivity C-Reactive Protein	mg/L	267	0.11, 182.82
DRINKS	ALQ130	Average daily Alcoholic drinks	drinks	789	0, 15
FLOSS	OHQ870	Days using dental floss in past week	days	4	0, 7
VIGWK_D	PAQ610	Average days per week with Vigorous work	days	5	0, 7
VI-GREC_D	PAQ655	Average days per week with Vigorous recreation	days	1	0, 7
SEDATE	PAD680	Average daily Sedentary activity	minutes	24	2, 1320
PREG-NANT	RHQ160	Pregnancies	times	1975	0, 11
SLPWK-DAY	SLD012	Usual sleep (weekdays)	hours	34	2, 14
SLPWK-END	SLD013	Usual sleep (weekends)	hours	34	2, 14
SMOKE30	SMD641	Days in past 30 when you smoked a cigarette	days	726	0, 30
ESTHT	WHD010	Self-reported height	inches	95	50, 81
ESTWT	WHD020	Self-reported weight	pounds	68	86, 578

To insert the number of missing values and range (minimum, maximum among non-missing values) into the table, I used inline R code like this:

```
n_miss(nh_fixing$BMXWT)
```

```
[1] 28
```

```
range(nh_fixing$BMXWT, na.rm = TRUE)
```

```
[1] 36.9 254.3
```

1.10.1 Renaming the Quantities

Here's the renaming code:

```
nh_fixing <- nh_fixing |>
  rename(AGE = RIDAGEYR, WEIGHT = BMXWT, HEIGHT = BMXHT,
         WAIST = BMXWAIST, SBP = BPXOSY2, DBP = BPXODI2,
         PULSE1 = BPXOPLS1, PULSE2 = BPXOPLS2, WBC = LBXWBCSI,
         PLATELET = LBXPLTSI, HSCRP = LBXHSCRP, DRINKS = ALQ130,
         FLOSS = OHQ870, VIGWK_D = PAQ610, VIGREC_D = PAQ655,
         SEDATE = PAD680, PREGS = RHQ160, SLPWKDAY = SLD012,
         SLPWKEND = SLD013, SMOKE30 = SMD641,
         ESTHT = WHD010, ESTWT = WHD020)
```

1.10.2 Sampling Weights

Here are the sampling weights, which I think of as unitless, typically, though they represent people.

Variable	Description	# NA	Range
WTINTPRP	Sampling Weight (interviews)	0	2467.1, 311265.2
WTMECPRP	Sampling Weight (examinations)	0	2589.2, 321573.5

Note that to obtain these ranges formatted like this, I had to use some additional code in the table:

```
format(round_half_up(range(nh_fixing$WTINTPRP, na.rm = TRUE),1), scientific = FALSE)
```

```
[1] " 2467.1" "311265.2"
```

1.11 Binary Variables

1.11.1 Sex (RIAGENDR)

To start, let's do something about the variable describing the participant's biological sex (so we'll rename it to a more useful name), and then we'll recode the values of the **SEX** variable to more useful choices.

New Name	NHANES Name	Description
SEX	RIAGENDR	Sex

Note that we have more female than male subjects, and no missing values, as it turns out.

```
nh_fixing |> count(RIAGENDR)
```

```
# A tibble: 2 x 2
  RIAGENDR     n
  <dbl> <int>
1         1 1837
2         2 2094
```

Now, let's convert the information in **RIAGENDR** to **SEX**.

```
nh_fixing <- nh_fixing |>
  rename(SEX = RIAGENDR) |>
  mutate(SEX = factor(ifelse(SEX == 1, "Male", "Female")))
```

And we'll run a little “sanity check” here to ensure that we've recoded this variable properly⁵.

```
nh_fixing |> count(SEX)
```

⁵Sanity checks are a great idea while in the midst of coding, but can confuse the reader and thus should not be included in 432 submitted work.

```
# A tibble: 2 x 2
  SEX      n
  <fct> <int>
1 Female 2094
2 Male   1837
```

1.11.2 Yes/No variables

Now, let's tackle the variables with code 1 = Yes, and 2 = No, and (potentially) some missing values. I'll summarize each with the percentage of "Yes" responses (out of those with code 1 or 2) and the number of missing values.

New NAME	NHANES NAME	Description	% Yes	# NA
HOSPITAL	HUQ071	Overnight hospital patient in past 12m?	8.7	0
MENTALH	HUQ090	Seen mental health professional past 12m?	12.1	2
EVER-ALC	ALQ111	Ever had a drink of alcohol?	91.1	205
IN-SURNOW	HIQ011	Covered by health insurance now?	80.8	10
NOIN-SUR	HIQ210	Time when no insurance in past year?	26.6	16
DR_LOSE	MCQ366A	Doctor told you to control/lose weight in the past 12 months?	30.3	1
DR_EXER	MCQ366B	Doctor told you to exercise in the past 12 months?	42.7	1
NOW_LOSE	MCQ371A	Are you now controlling or losing weight?	63.2	2
NOW_EXER	MCQ371B	Are you now increasing exercise?	60.3	1
WORK_V	PAQ605	Vigorous work activity for 10 min/week?	28.4	4
REC_V	PAQ650	Vigorous recreational activity for 10 min/week?	26.9	0
EVER-PREG	RHQ131	Ever been pregnant?	89.2	1972
SLP-TROUB	SLQ050	Ever told a doctor you had trouble sleeping?	29.5	3
CIG100	SMQ020	Smoked at least 100 cigarettes in your life?	40.1	1
AWAY-WORK	SMQ856	Last 7 days worked at a job not at home?	67.7	0
AWAYREST	SMQ860	Last 7 days spent time in a restaurant?	58.1	2
AWAY-BAR	SMQ866	Last 7 days spent time in a bar?	15.4	0

The inline code I used in the tables was, for example:

```
round_half_up(100 * sum(nh_fixing$HUQ090 == "1", na.rm = TRUE) /  
               sum(nh_fixing$HUQ090 %in% c("1","2"), na.rm = TRUE), 1)
```

```
[1] 12.1
```

```
n_miss(nh_fixing$HUQ090)
```

```
[1] 2
```

To clean these (1 = Yes, 2 = No) variables, I'll subtract the values from 2, to obtain variables where 1 = Yes and 0 = No. I'll use the `across()` function within my `mutate()` statement so as to avoid having to type out each change individually⁶.

```
nh_fixing <- nh_fixing |>  
  mutate(across(c(HUQ071, HUQ090, ALQ111, HIQ011, HIQ210,  
                  MCQ366A, MCQ366B, MCQ371A, MCQ371B, PAQ605,  
                  PAQ650, RHQ131, SLQ050, SMQ020, SMQ856,  
                  SMQ860, SMQ866),  
           ~ 2 - .x))
```

Let's do just one of the relevant sanity checks here. In addition to verifying that our new variable has the values 0 and 1 (instead of 2 and 1), we want to be certain that we've maintained any missing values.

```
nh_fixing |> count(SLQ050)
```

```
# A tibble: 3 x 2  
  SLQ050      n  
  <dbl> <int>  
1      0  2769  
2      1  1159  
3     NA      3
```

⁶For more on the `across()` function and how it can be used, visit <https://dplyr.tidyverse.org/reference/across.html>.

1.11.3 Renaming Binary Variables

Here's the renaming code.

```
nh_fixing <- nh_fixing |>
  rename(HOSPITAL = HUQ071, MENTALH = HUQ090, EVERALC = ALQ111,
         INSURNOW = HIQ011, NOINSUR = HIQ210, DR_LOSE = MCQ366A,
         DR_EXER = MCQ366B, NOW_LOSE = MCQ371A, NOW_EXER = MCQ371B,
         WORK_V = PAQ605, REC_V = PAQ650, EVERPREG = RHQ131,
         SLPTROUB = SLQ050, CIG100 = SMQ020, AWAYWORK = SMQ856,
         AWAYREST = SMQ860, AWAYBAR = SMQ866)
```

1.12 Create PHQ-9 Scores

The questions below are asked to assess depression severity, following the prompt “Over the last two weeks, how often have you been bothered by any of the following problems?”

Variable	Description	Values
SEQN	Respondent Sequence Number (participant ID)	link across databases
DPQ010	Have little interest in doing things	0-3: see below
DPQ020	Feeling down, depressed or hopeless	0-3: see below
DPQ030	Trouble sleeping or sleeping too much	0-3: see below
DPQ040	Feeling tired or having little energy	0-3: see below
DPQ050	Poor appetite or overeating	0-3: see below
DPQ060	Feeling bad about yourself	0-3: see below
DPQ070	Trouble concentrating on things	0-3: see below
DPQ080	Moving or speaking slowly or too fast	0-3: see below
DPQ090	Thoughts you would be better off dead	0-3: see below

- For DPQ010 - DPQ090, the codes are 0 = Not at all, 1 = Several days in the past two weeks, 2 = More than half the days in the past two weeks, 3 = Nearly every day in the past two weeks.

1.12.1 Forming the PHQ-9 Score

One way to use this information is to sum the scores from items DPQ010 through DPQ090 to obtain a result on a scale from 0 - 27. Cutoffs of 5, 10, 15, and 20 then represent mild,

moderate, moderately severe, and severe levels of depressive symptoms, respectively⁷. If we had no missing values in our responses, then this would be relatively straightforward.

```
temp <- nh_fixing |>
  mutate(PHQ9 = DPQ010 + DPQ020 + DPQ030 + DPQ040 + DPQ050 +
          DPQ060 + DPQ070 + DPQ080 + DPQ090)

temp |> count(PHQ9) |> tail()
```

```
# A tibble: 6 x 2
  PHQ9      n
  <dbl> <int>
1    22     9
2    23     5
3    24     2
4    25     2
5    26     3
6    NA    221
```

It turns out that this formulation of PHQ9 regards as missing the result for any participant who failed to answer all 9 questions. A common approach to dealing with missing data in creating PHQ-9 scores⁸ is to score all questionnaires with up to two missing values, replacing any missing values with the average score of the completed items.

So how many of our subjects are missing only one or two of the 9 items?

```
temp2 <- temp |>
  select(SEQN, DPQ010, DPQ020, DPQ030, DPQ040, DPQ050, DPQ060,
          DPQ070, DPQ080, DPQ090)

miss_case_table(temp2)
```

```
# A tibble: 5 x 3
  n_miss_in_case n_cases pct_cases
  <int>    <int>    <dbl>
1         0    3710    94.4
2         1         5    0.127
3         2         3    0.0763
4         4         1    0.0254
5         9     212    5.39
```

⁷See Kroenke et al. 2010

⁸Again, see Kroenke et al. (2010) at <https://pubmed.ncbi.nlm.nih.gov/20633738/>.

With a little googling I found [an R script online](#) that will perform this task, and create three new variables:

- `nvalid_phq9` = Number of Valid Responses (out of 9) to the PHQ-9 items
- `PHQ9` = PHQ-9 score (0-27 scale, higher values indicate more depression)
- `PHQ9_CAT` = factor describing PHQ-9 score
 - `PHQ9 > 20` means `PHQ9_CAT` is “severe”,
 - 15-19 = “moderately severe”,
 - 10-14 = “moderate”
 - 5-9 = “mild”
 - 0-4 = “minimal”

```
scoring_phq9 <- function(data, items.phq9) {
  data %>%
    mutate(nvalid_phq9 = rowSums(!is.na(select(., items.phq9))),
           nvalid_phq9 = as.integer(nvalid_phq9),
           mean.temp = rowSums(select(., items.phq9), na.rm = TRUE)/nvalid_phq9,
           phq.01.temp = as.integer(unlist(data[items.phq9[1]])),
           phq.02.temp = as.integer(unlist(data[items.phq9[2]])),
           phq.03.temp = as.integer(unlist(data[items.phq9[3]])),
           phq.04.temp = as.integer(unlist(data[items.phq9[4]])),
           phq.05.temp = as.integer(unlist(data[items.phq9[5]])),
           phq.06.temp = as.integer(unlist(data[items.phq9[6]])),
           phq.07.temp = as.integer(unlist(data[items.phq9[7]])),
           phq.08.temp = as.integer(unlist(data[items.phq9[8]])),
           phq.09.temp = as.integer(unlist(data[items.phq9[9]]))) %>%
    mutate_at(vars(phq.01.temp:phq.09.temp),
              funs(ifelse(is.na(.), round(mean.temp), .))) %>%
    mutate(score.temp = rowSums(select(., phq.01.temp:phq.09.temp), na.rm = TRUE),
           PHQ9 = ifelse(nvalid_phq9 >= 7, as.integer(round(score.temp)), NA),
           PHQ9_CAT = case_when(
             PHQ9 >= 20 ~ 'severe',
             PHQ9 >= 15 ~ 'moderately severe',
             PHQ9 >= 10 ~ 'moderate',
             PHQ9 >= 5 ~ 'mild',
             PHQ9 < 5 ~ 'minimal'),
           PHQ9_CAT = factor(PHQ9_CAT, levels = c('minimal', 'mild',
                                                  'moderate', 'moderately severe',
                                                  'severe')) %>%
    select(-ends_with("temp"))
}
```

Applying this script to our `nh_fixing` data, our result is:

```
items.phq9 <- c("DPQ010", "DPQ020", "DPQ030", "DPQ040", "DPQ050",
               "DPQ060", "DPQ070", "DPQ080", "DPQ090")
nh_fixing <- nh_fixing %>% scoring_phq9(., all_of(items.phq9))

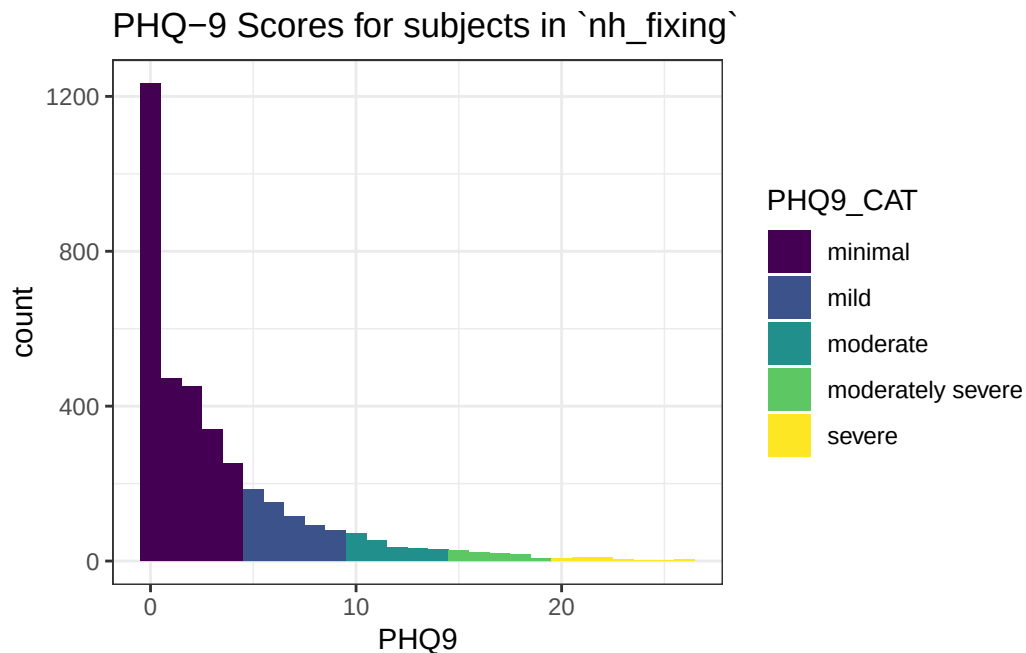
nh_fixing |> count(nvalid_phq9, PHQ9, PHQ9_CAT)
```

```
# A tibble: 36 x 4
  nvalid_phq9 PHQ9 PHQ9_CAT      n
    <int> <int> <fct>    <int>
1         0    NA <NA>      212
2         5    NA <NA>         1
3         7     1 minimal         1
4         7     2 minimal         1
5         7     7 mild           1
6         8     1 minimal         2
7         8     2 minimal         1
8         8     3 minimal         1
9         8     8 mild           1
10        9     0 minimal    1233
# i 26 more rows
```

1.12.2 Distribution of PHQ-9 Score

Here's a quick look at the distribution of PHQ-9 scores in our `nh_fixing` data.

```
nh_fixing |> filter(complete.cases(PHQ9, PHQ9_CAT)) %>%
  ggplot(., aes(x = PHQ9, fill = PHQ9_CAT)) +
  geom_histogram(binwidth = 1) +
  scale_fill_viridis_d() +
  labs(title = "PHQ-9 Scores for subjects in `nh_fixing`")
```

1.12.3 Fixing the DPQ100 variable

The DPQ100 variable should be 0 (Not at all difficult) if the PHQ-9 score is zero. We need to fix this, because NHANES participants who answered 0 (Not at all) to each of the nine elements contained in the PHQ-9 were not asked the DPQ100 question. So, we set the value of DPQ100 to be 0 if PHQ9 is 0.

```
nh_fixing <- nh_fixing |>
  mutate(DPQ100 = ifelse(PHQ9 == 0, 0, DPQ100))
```

This will eliminate the “automatic missing” values in DPQ100.

```
nh_fixing |> tabyl(DPQ100)
```

DPQ100	n	percent	valid_percent
0	3039	0.77308573	0.81781485
1	541	0.13762401	0.14558665
2	93	0.02365810	0.02502691
3	43	0.01093869	0.01157158
NA	215	0.05469346	NA

RIDRETH3	n	percent	Code
1	500	12.7%	Mexican American
2	403	10.3%	Other Hispanic
3	1192	30.3%	Non-Hispanic White
4	1049	26.7%	Non-Hispanic Black
6	588	15.0%	Non-Hispanic Asian
7	199	5.1%	Other Race

1.13 Multi-Categorical Variables

Our remaining categorical variables with more than two levels are:

Variable	Description	Codes	# NAs
RIDRETH3	Race/Hispanic origin	1, 2, 3, 4, 6, 7	0
DMDEDUC2	Education Level	1, 2, 3, 4, 5	1
OHAREC	Overall Recommendation for Care	1, 2, 3, 4	0
HUQ010	General health condition	1, 2, 3, 4, 5	0
DEQ034D	Use sunscreen if outside on very sunny day?	1, 2, 3, 4, 5	19
DPQ100	Difficulty depression problems have caused	0, 1, 2, 3	215
DBQ700	How healthy is your diet?	1, 2, 3, 4, 5	1
FSDAD	Adult food security in last 12m	1, 2, 3, 4	231
SLQ030	How often do you snore?	0, 1, 2, 3	219
WHQ040	Like to weigh more, less, or same?	1, 2, 3	3

1.13.1 Creating RACEETH from RIDRETH3

At the moment, our RIDRETH3 data look like this:

```
t_ridreth3 <- nh_fixing |> tabyl(RIDRETH3) |> adorn_pct_formatting() |>
  mutate(Code = c("Mexican American", "Other Hispanic", "Non-Hispanic White",
                  "Non-Hispanic Black", "Non-Hispanic Asian", "Other Race"))
gt(t_ridreth3)
```

Now, we'll turn this RIDRETH3 variable into a new **factor** called RACEETH with meaningful levels, and then sort those levels by their frequency in the data. We'll also collapse together

the Mexican American and Other Hispanic levels, not because the distinction is irrelevant, but more to demonstrate how this might be done.

```
nh_fixing <- nh_fixing |>
  mutate(RACEETH =
    fct_recode(
      factor(RIDRETH3),
      "Hispanic" = "1",
      "Hispanic" = "2",
      "Non-H White" = "3",
      "Non-H Black" = "4",
      "Non-H Asian" = "6",
      "Other Race" = "7")) |>
  mutate(RACEETH = fct_infreq(RACEETH))
```

I'm using `fct_infreq()` here to sort the (nominal) Race and Ethnicity data so that the most common column appears first, and will thus be treated as the “baseline” level in models. Here is the resulting order.

```
nh_fixing |> count(RACEETH)
```

```
# A tibble: 5 x 2
  RACEETH      n
  <fct>      <int>
1 Non-H White 1192
2 Non-H Black 1049
3 Hispanic    903
4 Non-H Asian 588
5 Other Race  199
```

Now, let's check⁹ to see if RACEETH and RIDRETH3 include the same information (after collapsing the Mexican American and Other Hispanic categories.)

```
nh_fixing |> count(RACEETH, RIDRETH3)
```

```
# A tibble: 6 x 3
  RACEETH      RIDRETH3      n
  <fct>      <dbl> <int>
1 Non-H White      3  1192
```

⁹Again, I wouldn't usually include this sort of coding check in published work, like either of your 432 Projects.

DMDEDUC2	n	percent	valid_percent	Code
1	272	6.9%	6.9%	Less than 9th grade
2	424	10.8%	10.8%	9th-11th grade
3	850	21.6%	21.6%	High School Grad
4	1287	32.7%	32.7%	Some College / AA
5	1097	27.9%	27.9%	College Grad
NA	1	0.0%	-	Missing

2 Non-H Black	4	1049
3 Hispanic	1	500
4 Hispanic	2	403
5 Non-H Asian	6	588
6 Other Race	7	199

1.13.2 Creating EDUC from DMDEDUC2

At the moment, our DMDEDUC2 data look like this:

```
t_dmddeduc2 <- nh_fixing |> tabyl(DMDEDUC2) |> adorn_pct_formatting() |>
  mutate(Code = c("Less than 9th grade", "9th-11th grade", "High School Grad",
                  "Some College / AA", "College Grad", "Missing"))

gt(t_dmddeduc2)
```

Now, we'll turn this DMDEDUC2 variable into a new **factor** called EDUC with meaningful levels.

```
nh_fixing <- nh_fixing |>
  mutate(EDUC =
    fct_recode(
      factor(DMDEDUC2),
      "Less than 9th Grade" = "1",
      "9th - 11th Grade" = "2",
      "High School Grad" = "3",
      "Some College / AA" = "4",
      "College Grad" = "5"))
```

Once again, checking our work...

EDUC	1	2	3	4	5	NA_
Less than 9th Grade	272	0	0	0	0	0
9th - 11th Grade	0	424	0	0	0	0
High School Grad	0	0	850	0	0	0
Some College / AA	0	0	0	1287	0	0
College Grad	0	0	0	0	1097	0
NA	0	0	0	0	0	1

OHAREC	n	percent	Code
1	4	0.1%	See a dentist immediately
2	230	5.9%	See a dentist within the next 2 weeks
3	1671	42.5%	See a dentist at your earliest convenience
4	2026	51.5%	Continue your regular routine care

```
nh_fixing |> tabyl(EDUC, DMDEDUC2) |> gt()
```

1.13.3 Creating DENTAL from OHAREC

```
t_oharec <- nh_fixing |> tabyl(OHAREC) |> adorn_pct_formatting() |>
  mutate(Code = c("See a dentist immediately", "See a dentist within the next 2 weeks", "See
gt(t_oharec)
```

We'll collapse categories 1 and 2 together since they are quite small.

```
nh_fixing <- nh_fixing |>
  mutate(DENTAL =
    fct_recode(
      factor(OHAREC),
      "See dentist urgently" = "1",
      "See dentist urgently" = "2",
      "See dentist soon" = "3",
      "Regular Routine" = "4"))
```

Once again, checking our work...

DENTAL	1	2	3	4
See dentist urgently	4	230	0	0
See dentist soon	0	0	1671	0
Regular Routine	0	0	0	2026

SROH	1	2	3	4	5
Excellent	495	0	0	0	0
Very Good	0	1071	0	0	0
Good	0	0	1462	0	0
Fair	0	0	0	765	0
Poor	0	0	0	0	138

```
nh_fixing |> tabyl(DENTAL, OHAREC) |> gt()
```

1.13.4 Creating SROH from HUQ010

Variable	Description	Codes	# NAs
HUQ010	General health condition	1 = Excellent 2 = Very Good 3 = Good 4 = Fair 5 = Poor	0

```
nh_fixing <- nh_fixing |>
  mutate(SROH =
    fct_recode(
      factor(HUQ010),
      "Excellent" = "1",
      "Very Good" = "2",
      "Good" = "3",
      "Fair" = "4",
      "Poor" = "5"))
```

Checking our work...

```
nh_fixing |> tabyl(SROH, HUQ010) |> gt()
```

1.13.5 Creating SUNSCR from DEQ034D

SUNSCR	1	2	3	4	5	NA_
Always	351	0	0	0	0	0
Most of the time	0	485	0	0	0	0
Sometimes	0	0	831	0	0	0
Rarely	0	0	0	662	0	0
Never	0	0	0	0	1583	0
NA	0	0	0	0	0	19

Variable	Description	Codes	# NAs
DEQ034D	Use sunscreen if outside on very sunny day?	1 = Always 2 = Most of the time 3 = Sometimes 4 = Rarely 5 = Never	19

```
nh_fixing <- nh_fixing |>
  mutate(SUNSCR =
    fct_recode(
      factor(DEQ034D),
      "Always" = "1",
      "Most of the time" = "2",
      "Sometimes" = "3",
      "Rarely" = "4",
      "Never" = "5"))
```

```
nh_fixing |> tabyl(SUNSCR, DEQ034D) |> gt()
```

1.13.6 Creating DEPRDIFF from DPQ100

Variable	Description	Codes	# NAs
DPQ100	Difficulty depression problems have caused	0 = Not at all difficult 1 = Somewhat difficult 2 = Very difficult 3 = Extremely difficult	215

```
nh_fixing <- nh_fixing |>
  mutate(DEPRDIFF =
    fct_recode(
```

DEPRDIFF	0	1	2	3	NA__
Not at all	3039	0	0	0	0
Somewhat	0	541	0	0	0
Very	0	0	93	0	0
Extremely	0	0	0	43	0
NA	0	0	0	0	215

DIETQUAL	1	2	3	4	5	NA__
Excellent	260	0	0	0	0	0
Very Good	0	758	0	0	0	0
Good	0	0	1519	0	0	0
Fair	0	0	0	1082	0	0
Poor	0	0	0	0	311	0
NA	0	0	0	0	0	1

```
factor(DPQ100),
  "Not at all" = "0",
  "Somewhat" = "1",
  "Very" = "2",
  "Extremely" = "3"))
```

```
nh_fixing |> tabyl(DEPRDIFF, DPQ100) |> gt()
```

1.13.7 Creating DIETQUAL from DBQ700

Variable	Description	Codes	# NAs
DBQ700	How healthy is your diet?	1 = Excellent 2 = Very Good 3 = Good 4 = Fair 5 = Poor	1

```
nh_fixing <- nh_fixing |>
  mutate(DIETQUAL =
    fct_recode(
      factor(DBQ700),
      "Excellent" = "1",
      "Very Good" = "2",
      "Good" = "3",
      "Fair" = "4",
      "Poor" = "5"))
```

```
nh_fixing |> tabyl(DIETQUAL, DBQ700) |> gt()
```

1.13.8 Creating FOODSEC from FSDAD

FOODSEC	1	2	3	4	NA_
Full	2247	0	0	0	0
Marginal	0	565	0	0	0
Low	0	0	503	0	0
Very Low	0	0	0	385	0
NA	0	0	0	0	231

Variable	Description	Codes	# NAs
FSDAD	Adult food security category for last 12m	1 = Full food security 2 = Marginal food security 3 = Low food security 4 = Very low food security	231

```
nh_fixing <- nh_fixing |>
  mutate(FOODSEC =
    fct_recode(
      factor(FSDAD),
      "Full" = "1",
      "Marginal" = "2",
      "Low" = "3",
      "Very Low" = "4"))
```

```
nh_fixing |> tabyl(FOODSEC, FSDAD) |> gt()
```

1.13.9 Creating SNORE from SLQ030

Variable	Description	Codes	# NAs
SLQ030	How often do you snore?	0 = Never 1 = Rarely (1-2 nights/week) 2 = Occasionally (3-4 nights/week) 3 = Frequently (5+ nights/week)	219

```
nh_fixing <- nh_fixing |>
  mutate(SNORE =
    fct_recode(
```

SNORE	0	1	2	3	NA__
Never	855	0	0	0	0
Rarely	0	959	0	0	0
Occasionally	0	0	700	0	0
Frequently	0	0	0	1198	0
NA	0	0	0	0	219

```
factor(SLQ030),
"Never" = "0",
"Rarely" = "1",
"Occasionally" = "2",
"Frequently" = "3"))
```

```
nh_fixing |> tabyl(SNORE, SLQ030) |> gt()
```

1.13.10 Creating WTGOAL from WHQ040

Variable	Description	Codes	# NAs
WHQ040	Like to weigh more, less, or same?	1 = More 2 = Less 3 = Stay about the same	3

Since there's a natural ordering here (more then same then less) I'll adapt it using the `fct_relevel()` function from the **forcats** package¹⁰ for this variable.

```
nh_fixing <- nh_fixing |>
  mutate(WTGOAL =
    fct_recode(
      factor(WHQ040),
      "More" = "1",
      "Less" = "2",
      "Same" = "3")) |>
  mutate(WTGOAL = fct_relevel(WTGOAL, "More", "Same", "Less"))
```

¹⁰<https://forcats.tidyverse.org/> is the home of forcats, although the Factors chapter at <https://r4ds.hadley.nz/factors.html> in R for Data Science, 2nd edition is also helpful.

WTGOAL	1	2	3	NA__
More	289	0	0	0
Same	0	0	974	0
Less	0	2665	0	0
NA	0	0	0	3

```
nh_fixing |> tabyl(WTGOAL, WHQ040) |> gt()
```

1.14 Dropping Variables

We'll drop the following variables before saving an analytic tibble.

- Our two variables with no variation
 - RIDSTATR, OHDEXSTS
- Elements of the PHQ-9 we no longer need
 - nvalid_phq, DPQ010, DPQ020, DPQ030, DPQ040
 - DPQ050, DPQ060, DPQ070, DPQ080, DPQ090
- Multi-categorical variables that we renamed
 - RIDRETH3, DMDEDUC2, OHAREC, HUQ010, DEQ034D
 - DPQ100, DBQ700, FSDAD, SLQ030, WHQ040

```
nh_fixing <- nh_fixing |>
  select(-c(RIDSTATR, OHDEXSTS, nvalid_phq9, DPQ010,
            DPQ020, DPQ030, DPQ040, DPQ050, DPQ060,
            DPQ070, DPQ080, DPQ090, RIDRETH3, DMDEDUC2,
            OHAREC, HUQ010, DEQ034D, DPQ100, DBQ700,
            FSDAD, SLQ030, WHQ040))
```

1.15 Resorting Variables

I'd like to have the variables in the following order:

```
nh432 <- nh_fixing |>
  select(SEQN, AGE, RACEETH, EDUC, SEX, INSURNOW,
         NOINSUR, SROH, WEIGHT, HEIGHT, WAIST,
         SBP, DBP, PULSE1, PULSE2, WBC, PLATELET, HSCRP,
         DR_LOSE, DR_EXER, NOW_LOSE, NOW_EXER,
         ESTHT, ESTWT, WTGOAL, DIETQUAL, FOODSEC,
         WORK_V, VIGWK_D, REC_V, VIGREC_D, SEDATE,
         PHQ9, PHQ9_CAT, DEPRDIFF, MENTALH,
         SLPWKDAY, SLPWKEND, SLPTROUB, SNORE,
         HOSPITAL, EVERALC, DRINKS, CIG100, SMOKE30,
         AWAYWORK, AWAYREST, AWAYBAR, DENTAL, FLOSS,
         EVERPREG, PREGS, SUNSCR, WTINTPRP, WTMECPRP)
```

1.16 nh432 analytic tibble

nh432

```
# A tibble: 3,931 x 55
  SEQN    AGE RACEETH    EDUC SEX  INSURNOW NOINSUR SROH  WEIGHT HEIGHT WAIST
  <chr>  <dbl> <fct>    <fct> <fct>    <dbl>    <dbl> <fct>  <dbl>  <dbl> <dbl>
1 109271   49 Non-H Wh~ 9th ~ Male      1      0 Fair   98.8   182.  120.
2 109273   36 Non-H Wh~ Some~ Male      1      1 Good   74.3   184.  86.8
3 109284   44 Hispanic 9th ~ Fema~    0      1 Fair   91.1   153.  103.
4 109291   42 Non-H As~ Coll~ Fema~    1      0 Fair   81.4   161.   NA
5 109292   58 Hispanic High~ Male      1      0 Very~   86     168.  108.
6 109293   44 Non-H Wh~ High~ Male      1      0 Good   99.4   182.  107
7 109295   54 Hispanic Less~ Fema~    1      0 Good   61.7   157.  90.5
8 109297   30 Non-H As~ Some~ Fema~    1      0 Very~   55.4   155.  73.2
9 109300   54 Non-H As~ Coll~ Fema~    1      0 Exce~   62     145.  84.8
10 109305   55 Non-H As~ Coll~ Male      1      0 Good   64     175.  82.5
# i 3,921 more rows
# i 44 more variables: SBP <dbl>, DBP <dbl>, PULSE1 <dbl>, PULSE2 <dbl>,
#   WBC <dbl>, PLATELET <dbl>, HSCRP <dbl>, DR_LOSE <dbl>, DR_EXER <dbl>,
#   NOW_LOSE <dbl>, NOW_EXER <dbl>, ESTHT <dbl>, ESTWT <dbl>, WTGOAL <fct>,
#   DIETQUAL <fct>, FOODSEC <fct>, WORK_V <dbl>, VIGWK_D <dbl>, REC_V <dbl>,
#   VIGREC_D <dbl>, SEDATE <dbl>, PHQ9 <int>, PHQ9_CAT <fct>, DEPRDIFF <fct>,
#   MENTALH <dbl>, SLPWKDAY <dbl>, SLPWKEND <dbl>, SLPTROUB <dbl>, ...
```

1.16.1 Saving the tibble as nh432.Rds

```
write_rds(nh432, "data/nh432.Rds")
```

2 Codebook for nh432

2.1 R Setup

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(gt)
library(gtsummary)
library(Hmisc)
library(mosaic)
library(naniar)
library(tidyverse)

theme_set(theme_bw())
```

2.1.1 Data Load

```
nh432 <- read_rds("data/nh432.Rds")
```

2.2 Quantitative Variables in nh432

```
t1_quantitative <- df_stats(~ AGE + WEIGHT + HEIGHT + WAIST + SBP + DBP +
  PULSE1 + PULSE2 + WBC + PLATELET + HSCRP +
  ESTHT + ESTWT + VIGWK_D + VIGREC_D + SEDATE + PHQ9 +
  SLPWKDAY + SLPWKEND + DRINKS + SMOKE30 +
  FLOSS + PREGS, data = nh432) |>
  mutate(across(.cols = -c(response, n, missing),
    round_half_up, digits = 1)) |>
  rename(med = median, "NA" = missing)
```

```

t1_quantitative |>
  mutate(description =
    c("Age (years)", "Weight (kg)", "Height (cm)",
      "Waist circumference (cm)", "Systolic BP (mm Hg)",
      "Diastolic BP (mm Hg)", "1st Pulse (beats/min)",
      "2nd Pulse (beats/min)", "White Blood Cell Count (1000 cells/uL)",
      "Platelets (1000 cells/uL)",
      "High-Sensitivity C-Reactive Protein (mg/L)",
      "Self Estimate: Height (in)", "Self-Estimate: Weight (lb)",
      "Vigorous Work per week (days)",
      "Vigorous Recreation per week (days)",
      "Sedentary Activity per day (minutes)",
      "PHQ-9 Depression Screener Score (points)",
      "Average weekday sleep (hours)", "Average weekend sleep (hours)",
      "Average Alcohol per day (drinks)",
      "Days smoked cigarette in last 30",
      "Days Flossed in last 7", "Pregnancies")) |>
  select(response, description, everything()) |>
  gt() |>
  tab_header(title = "Quantitative Variables in nh432")

```

2.3 Two-Category (1/0) Variables in nh432

```

nh_dich_vars <- nh432 |>
  select(HOSPITAL, MENTALH, EVERALC, INSURNOW, NOINSUR, DR_LOSE,
    DR_EXER, NOW_LOSE, NOW_EXER, WORK_V, REC_V, EVERPREG,
    SLPTRoub, CIG100, AWAYWORK, AWAYREST, AWAYBAR)

temp1 <- nh_dich_vars |> summarise(across(.cols = everything(),
  ~ sum(.x, na.rm = TRUE)))

temp2 <- nh_dich_vars |> summarise(across(.cols = everything(),
  ~ round_half_up(100*mean(.x, na.rm = TRUE), 1)))

temp3 <- nh_dich_vars |> summarise(across(.cols = everything(),
  ~ n_miss(.x)))

nh_dichotomous_summary <- bind_rows(temp1, temp2, temp3) |>

```

Quantitative Variables in nh432

response	description	min	Q1	med	Q3	max	mean
AGE	Age (years)	30.0	37.0	45.0	53.0	59.0	42.0
WEIGHT	Weight (kg)	36.9	69.3	82.1	99.1	254.3	82.1
HEIGHT	Height (cm)	135.3	160.0	166.9	174.7	198.7	166.9
WAIST	Waist circumference (cm)	57.9	89.1	99.2	111.7	178.0	100.0
SBP	Systolic BP (mm Hg)	69.0	110.0	120.0	131.0	222.0	120.0
DBP	Diastolic BP (mm Hg)	31.0	69.0	76.0	84.0	136.0	76.0
PULSE1	1st Pulse (beats/min)	38.0	62.0	69.0	77.0	126.0	77.0
PULSE2	2nd Pulse (beats/min)	37.0	63.0	70.0	78.0	121.0	77.0
WBC	White Blood Cell Count (1000 cells/uL)	2.3	5.7	6.9	8.4	22.8	6.9
PLATELET	Platelets (1000 cells/uL)	47.0	210.0	246.0	290.0	818.0	250.0
HSCRP	High-Sensitivity C-Reactive Protein (mg/L)	0.1	0.9	2.1	4.7	182.8	2.1
ESTHT	Self Estimate: Height (in)	50.0	63.0	66.0	69.0	81.0	66.0
ESTWT	Self-Estimate: Weight (lb)	86.0	150.0	180.0	216.0	578.0	180.0
VIGWK_D	Vigorous Work per week (days)	0.0	0.0	0.0	2.0	7.0	0.0
VIGREC_D	Vigorous Recreation per week (days)	0.0	0.0	0.0	1.0	7.0	0.0
SEDATE	Sedentary Activity per day (minutes)	2.0	180.0	300.0	480.0	1320.0	330.0
PHQ9	PHQ-9 Depression Screener Score (points)	0.0	0.0	2.0	5.0	26.0	2.0
SLPWKDAY	Average weekday sleep (hours)	2.0	6.5	7.5	8.0	14.0	7.5
SLPWKEND	Average weekend sleep (hours)	2.0	7.0	8.0	9.0	14.0	8.0
DRINKS	Average Alcohol per day (drinks)	0.0	1.0	2.0	3.0	15.0	2.0
SMOKE30	Days smoked cigarette in last 30	0.0	0.0	0.0	5.0	30.0	0.0
FLOSS	Days Flossed in last 7	0.0	0.0	3.0	7.0	7.0	3.0
PREGS	Pregnancies	0.0	2.0	3.0	4.0	11.0	3.0


```

mutate(summary = c("Yes", "% Yes", "# NA")) |>
relocate(summary) |>
pivot_longer(!summary, names_to = "variable") |>
pivot_wider(names_from = summary) |>
mutate(Description =
  c("Overnight hospital patient in past 12m?",
    "Seen mental health professional past 12m?",
    "Ever had a drink of alcohol?",
    "Covered by health insurance now?",
    "Time when no insurance in past year?",
    "Doctor said to control/lose weight past 12m?",
    "Doctor said to exercise in past 12m?",
    "Are you now controlling or losing weight?",
    "Are you now increasing exercise?",
    "Vigorous work activity for 10 min/week?",
    "Vigorous recreational activity for 10 min/week?",
    "Ever been pregnant?",
    "Ever told a doctor you had trouble sleeping?",
    "Smoked at least 100 cigarettes in your life?",
    "Last 7 days worked at a job not at home?",
    "Last 7 days spent time in a restaurant?",
    "Last 7 days spent time in a bar?"))

nh_dichotomous_summary |>
gt()

```

2.4 Factor Variables in nh432

```

nh_factor_vars <- nh432 |>
  select(where(~ is.factor(.x)))

tbl_summary(nh_factor_vars,
  label = list(RACEETH = "RACEETH: Race/Ethnicity",
    EDUC = "EDUC: Educational Attainment",
    SROH = "SROH: Self-reported Overall Health",
    WTGOAL = "WTGOAL: Like to weigh more/less/the same?",
    DIETQUAL = "DIETQUAL: How healthy is your diet?",
    FOODSEC = "FOODSEC: Adult food security (last 12m)",
    PHQ9_CAT = "PHQ9_CAT: Depression Screen Category",

```

variable	Yes	% Yes	# NA	Description
HOSPITAL	343	8.7	0	Overnight hospital patient in past 12m?
MENTALH	475	12.1	2	Seen mental health professional past 12m?
EVERALC	3393	91.1	205	Ever had a drink of alcohol?
INSURNOW	3169	80.8	10	Covered by health insurance now?
NOINSUR	1041	26.6	16	Time when no insurance in past year?
DR_LOSE	1189	30.3	1	Doctor said to control/lose weight past 12m?
DR_EXER	1680	42.7	1	Doctor said to exercise in past 12m?
NOW_LOSE	2485	63.2	2	Are you now controlling or losing weight?
NOW_EXER	2369	60.3	1	Are you now increasing exercise?
WORK_V	1116	28.4	4	Vigorous work activity for 10 min/week?
REC_V	1056	26.9	0	Vigorous recreational activity for 10 min/week?
EVERPREG	1747	89.2	1972	Ever been pregnant?
SLPTROUB	1159	29.5	3	Ever told a doctor you had trouble sleeping?
CIG100	1576	40.1	1	Smoked at least 100 cigarettes in your life?
AWAYWORK	2663	67.7	0	Last 7 days worked at a job not at home?
AWAYREST	2283	58.1	2	Last 7 days spent time in a restaurant?
AWAYBAR	605	15.4	0	Last 7 days spent time in a bar?

```

DEPRDIFF = "DEPRDIFF: Difficulty with Depression?",
SNORE = "SNORE: How often do you snore?",
DENTAL = "DENTAL: Recommendation for Dental Care?",
SUNSCR = "SUNSCR: Use sunscreen on very sunny day?"),
missing_text = "(# NA)"

```

2.5 Detailed Numerical Description for nh432

```
describe(nh432)
```

nh432

55 Variables 3931 Observations

SEQN

n	missing	distinct
3931	0	3931

Characteristic	N = 3,931 ¹
RACEETH: Race/Ethnicity	
Non-H White	1,192 (30%)
Non-H Black	1,049 (27%)
Hispanic	903 (23%)
Non-H Asian	588 (15%)
Other Race	199 (5.1%)
EDUC: Educational Attainment	
Less than 9th Grade	272 (6.9%)
9th - 11th Grade	424 (11%)
High School Grad	850 (22%)
Some College / AA	1,287 (33%)
College Grad	1,097 (28%)
(# NA)	1
SEX	
Female	2,094 (53%)
Male	1,837 (47%)
SROH: Self-reported Overall Health	
Excellent	495 (13%)
Very Good	1,071 (27%)
Good	1,462 (37%)
Fair	765 (19%)
Poor	138 (3.5%)
WTGOAL: Like to weigh more/less/the same?	
More	289 (7.4%)
Same	974 (25%)
Less	2,665 (68%)
(# NA)	3
DIETQUAL: How healthy is your diet?	
Excellent	260 (6.6%)
Very Good	758 (19%)
Good	1,519 (39%)
Fair	1,082 (28%)
Poor	311 (7.9%)
(# NA)	1
FOODSEC: Adult food security (last 12m)	
Full	2,247 (61%)
Marginal	565 (15%)
Low	503 (14%)
Very Low	385 (10%)
(# NA)	231
PHQ9_CAT: Depression Screen Category	
minimal	2,748 (74%)
mild	621 (17%)
moderate	220 (5.9%)
moderately severe	91 (2.4%)
severe	38 (1.0%)
(# NA)	213
DEPRDIFF: Difficulty with Depression?	
Not at all	3,039 (82%)
Somewhat	541 (15%)

lowest : 109271 109273 109284 109291 109292, highest: 124807 124810 124813 124815 124818

AGE

	n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3931	0	30	0.999	44.79	45	10.09	31	
.10	.25	.50	.75	.90	.95			
33	37	45	53	57	58			

lowest : 30 31 32 33 34, highest: 55 56 57 58 59

RACEETH

	n	missing	distinct
3931	0	5	

Value	Non-H White	Non-H Black	Hispanic	Non-H Asian	Other Race
Frequency	1192	1049	903	588	199
Proportion	0.303	0.267	0.230	0.150	0.051

EDUC

	n	missing	distinct
3930	1	5	

Value	Less than 9th Grade	9th - 11th Grade	High School Grad
Frequency	272	424	850
Proportion	0.069	0.108	0.216

Value	Some College / AA	College Grad
Frequency	1287	1097
Proportion	0.327	0.279

SEX

	n	missing	distinct
3931	0	2	

Value	Female	Male
Frequency	2094	1837
Proportion	0.533	0.467

INSURNOW

	n	missing	distinct	Info	Sum	Mean
3921	10	2	0.465	3169	0.8082	

NOINSUR

n	missing	distinct	Info	Sum	Mean
3915	16	2	0.586	1041	0.2659

SROH

n	missing	distinct
3931	0	5

Value	Excellent	Very Good	Good	Fair	Poor
Frequency	495	1071	1462	765	138
Proportion	0.126	0.272	0.372	0.195	0.035

WEIGHT

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3903	28	969	1	86.31	84.1	26.59	54.20
.10	.25	.50	.75	.90	.95		
58.82	69.30	82.10	99.10	119.30	131.49		

lowest : 36.9 39.4 39.6 39.8 39.9 , highest: 204.4 204.6 210.8 242.6 254.3

HEIGHT

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3901	30	484	1	167.4	167.2	11.45	152.0
.10	.25	.50	.75	.90	.95		
154.8	160.0	166.9	174.7	180.8	184.6		

lowest : 135.3 138.3 139.7 141.4 141.9 , highest: 195.8 195.9 196.6 198.3 198.7

WAIST

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3782	149	781	1	101.5	100.4	19.68	75.9
.10	.25	.50	.75	.90	.95		
80.4	89.1	99.2	111.7	125.4	134.5		

lowest : 57.9 62.7 63.2 64.5 64.9 , highest: 166 167.1 170.8 173.1 178

SBP

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3585	346	116	1	121.5	120.5	18.61	98
.10	.25	.50	.75	.90	.95		
102	110	120	131	143	152		

lowest : 69 72 77 79 80, highest: 199 200 211 219 222

DBP

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3585	346	81	0.999	77.03	76.5	13.01	59.2
.10	.25	.50	.75	.90	.95		
63.0	69.0	76.0	84.0	92.0	97.0		

lowest : 31 44 45 46 47, highest: 121 122 126 127 136

PULSE1

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3316	615	76	0.999	70.3	69.5	12.92	53
.10	.25	.50	.75	.90	.95		
57	62	69	77	86	91		

lowest : 38 40 41 42 44, highest: 114 115 120 121 126

PULSE2

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3314	617	80	0.999	70.96	70.5	12.92	54
.10	.25	.50	.75	.90	.95		
57	63	70	78	86	91		

lowest : 37 39 40 41 42, highest: 117 118 119 120 121

WBC

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3755	176	136	1	7.254	7.1	2.387	4.3
.10	.25	.50	.75	.90	.95		
4.8	5.7	6.9	8.4	10.1	11.3		

lowest : 2.3 2.5 2.6 2.7 2.8 , highest: 17.2 17.4 17.6 20.6 22.8

PLATELET

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3755	176	372	1	253.3	249.5	72.04	159
.10	.25	.50	.75	.90	.95		
179	210	246	290	337	371		

lowest : 47 48 54 57 61, highest: 583 602 638 662 818

HSCRIP

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3664	267	1065	1	4.326	2.77	5.271	0.350
.10	.25	.50	.75	.90	.95		
0.470	0.890	2.090	4.740	9.217	13.630		

lowest : 0.11 0.16 0.17 0.18 0.19 , highest: 102.94 104.48 109.81 138.81 182.82

DR_LOSE

n	missing	distinct	Info	Sum	Mean
3930	1	2	0.633	1189	0.3025

DR_EXER

n	missing	distinct	Info	Sum	Mean
3930	1	2	0.734	1680	0.4275

NOW_LOSE

n	missing	distinct	Info	Sum	Mean
3929	2	2	0.697	2485	0.6325

NOW_EXER

n	missing	distinct	Info	Sum	Mean
3930	1	2	0.718	2369	0.6028

ESTHT

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3836	95	29	0.995	66.46	66.5	4.722	60
.10	.25	.50	.75	.90	.95		
61	63	66	69	72	74		

lowest : 50 53 54 55 56, highest: 76 77 78 79 81

ESTWT

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3863	68	255	1	188.1	183.5	56.51	120.0
.10	.25	.50	.75	.90	.95		
130.0	150.0	180.0	216.0	258.0	281.9		

lowest : 86 88 90 93 95, highest: 416 434 450 457 578

WTGOAL

	n	missing	distinct
	3928	3	3

Value	More	Same	Less
Frequency	289	974	2665
Proportion	0.074	0.248	0.678

DIETQUAL

	n	missing	distinct
	3930	1	5

Value	Excellent	Very Good	Good	Fair	Poor
Frequency	260	758	1519	1082	311
Proportion	0.066	0.193	0.387	0.275	0.079

FOODSEC

	n	missing	distinct
	3700	231	4

Value	Full	Marginal	Low	Very Low
Frequency	2247	565	503	385
Proportion	0.607	0.153	0.136	0.104

WORK_V

	n	missing	distinct	Info	Sum	Mean
	3927	4	2	0.61	1116	0.2842

VIGWK_D

	n	missing	distinct	Info	Mean	pMedian	Gmd
	3926	5	8	0.632	1.22	0	1.904

Value	0	1	2	3	4	5	6	7
Frequency	2811	79	127	184	112	352	132	129
Proportion	0.716	0.020	0.032	0.047	0.029	0.090	0.034	0.033

REC_V

	n	missing	distinct	Info	Sum	Mean
	3931	0	2	0.589	1056	0.2686

VIGREC_D

	n	missing	distinct	Info	Mean	pMedian	Gmd
	3930	1	8	0.608	0.8952	0	1.436

Value	0	1	2	3	4	5	6	7
Frequency	2875	126	211	301	166	154	46	51
Proportion	0.732	0.032	0.054	0.077	0.042	0.039	0.012	0.013

SEDATE

	n	missing	distinct	Info	Mean	pMedian	Gmd	.05
	3907	24	44	0.99	332.7	330	232.2	60
	.10	.25	.50	.75	.90	.95		
	120	180	300	480	600	720		

lowest : 2 3 5 8 9, highest: 960 1020 1080 1200 1320

PHQ9

	n	missing	distinct	Info	Mean	pMedian	Gmd	.05
	3718	213	27	0.958	3.324	2.5	4.201	0
	.10	.25	.50	.75	.90	.95		
	0	0	2	5	9	13		

lowest : 0 1 2 3 4, highest: 22 23 24 25 26

PHQ9_CAT

	n	missing	distinct
	3718	213	5

Value	minimal	mild	moderate
Frequency	2748	621	220
Proportion	0.739	0.167	0.059

Value	moderately severe	severe
Frequency	91	38
Proportion	0.024	0.010

DEPRDIFF

	n	missing	distinct
	3716	215	4

Value	Not at all	Somewhat	Very	Extremely
Frequency	3039	541	93	43
Proportion	0.818	0.146	0.025	0.012

MENTALH

n	missing	distinct	Info	Sum	Mean
3929	2	2	0.319	475	0.1209

SLPWKDAY

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3897	34	22	0.984	7.359	7.5	1.735	5.0
.10	.25	.50	.75	.90	.95		
5.5	6.5	7.5	8.0	9.0	10.0		

lowest : 2 3 3.5 4 4.5 , highest: 11 11.5 12 13 14

SLPWKEND

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3897	34	24	0.983	8.231	8.25	1.928	5
.10	.25	.50	.75	.90	.95		
6	7	8	9	10	11		

lowest : 2 3 3.5 4 4.5 , highest: 12 12.5 13 13.5 14

SLPTROUB

n	missing	distinct	Info	Sum	Mean
3928	3	2	0.624	1159	0.2951

SNORE

n	missing	distinct
3712	219	4

Value	Never	Rarely	Occasionally	Frequently
Frequency	855	959	700	1198
Proportion	0.230	0.258	0.189	0.323

HOSPITAL

n	missing	distinct	Info	Sum	Mean
3931	0	2	0.239	343	0.08726

EVERALC

n	missing	distinct	Info	Sum	Mean
3726	205	2	0.244	3393	0.9106

DRINKS

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3142	789	15	0.948	2.345	2	2.051	0
.10	.25	.50	.75	.90	.95		
0	1	2	3	5	6		

Value	0	1	2	3	4	5	6	7	8	9	10
Frequency	333	912	903	412	228	123	105	21	37	4	19
Proportion	0.106	0.290	0.287	0.131	0.073	0.039	0.033	0.007	0.012	0.001	0.006

Value	11	12	13	15
Frequency	1	26	2	16
Proportion	0.000	0.008	0.001	0.005

CIG100

n	missing	distinct	Info	Sum	Mean
3930	1	2	0.721	1576	0.401

SMOKE30

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3205	726	26	0.594	6.808	0	10.5	0
.10	.25	.50	.75	.90	.95		
0	0	0	5	30	30		

lowest : 0 1 2 3 4, highest: 26 27 28 29 30

AWAYWORK

n	missing	distinct	Info	Sum	Mean
3931	0	2	0.656	2663	0.6774

AWAYREST

n	missing	distinct	Info	Sum	Mean
3929	2	2	0.73	2283	0.5811

AWAYBAR

n	missing	distinct	Info	Sum	Mean
3931	0	2	0.391	605	0.1539

DENTAL

n	missing	distinct
3931	0	3

Value	See dentist urgently	See dentist soon	Regular Routine
Frequency	234	1671	2026
Proportion	0.060	0.425	0.515

FLOSS

n	missing	distinct	Info	Mean	pMedian	Gmd
3927	4	8	0.934	3.476	3.5	3.248

Value	0	1	2	3	4	5	6	7
Frequency	1104	288	395	329	230	179	44	1358
Proportion	0.281	0.073	0.101	0.084	0.059	0.046	0.011	0.346

EVERPREG

n	missing	distinct	Info	Sum	Mean
1959	1972	2	0.29	1747	0.8918

PREGS

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
1956	1975	12	0.973	3.046	3	2.24	0
.10	.25	.50	.75	.90	.95		
0	2	3	4	6	7		

Value	0	1	2	3	4	5	6	7	8	9	10
Frequency	212	205	421	420	297	191	95	58	22	10	9
Proportion	0.108	0.105	0.215	0.215	0.152	0.098	0.049	0.030	0.011	0.005	0.005

Value	11
Frequency	16
Proportion	0.008

SUNSCR

n	missing	distinct
3912	19	5

Value	Always	Most of the time	Sometimes	Rarely
Frequency	351	485	831	662
Proportion	0.090	0.124	0.212	0.169

Value	Never
Frequency	1583
Proportion	0.405

WTINTPRP

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3931	0	3677	1	28434	20563	27437	5911
.10	.25	.50	.75	.90	.95		
7199	10615	17358	31476	65098	94422		

lowest : 2467.05 2779.46 2833.29 2917.41 2967.27
highest: 246250 248091 264719 282884 311265

WTMECPRP

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
3931	0	3701	1	30353	21829	29409	6217
.10	.25	.50	.75	.90	.95		
7634	11365	18422	33155	68569	102038		

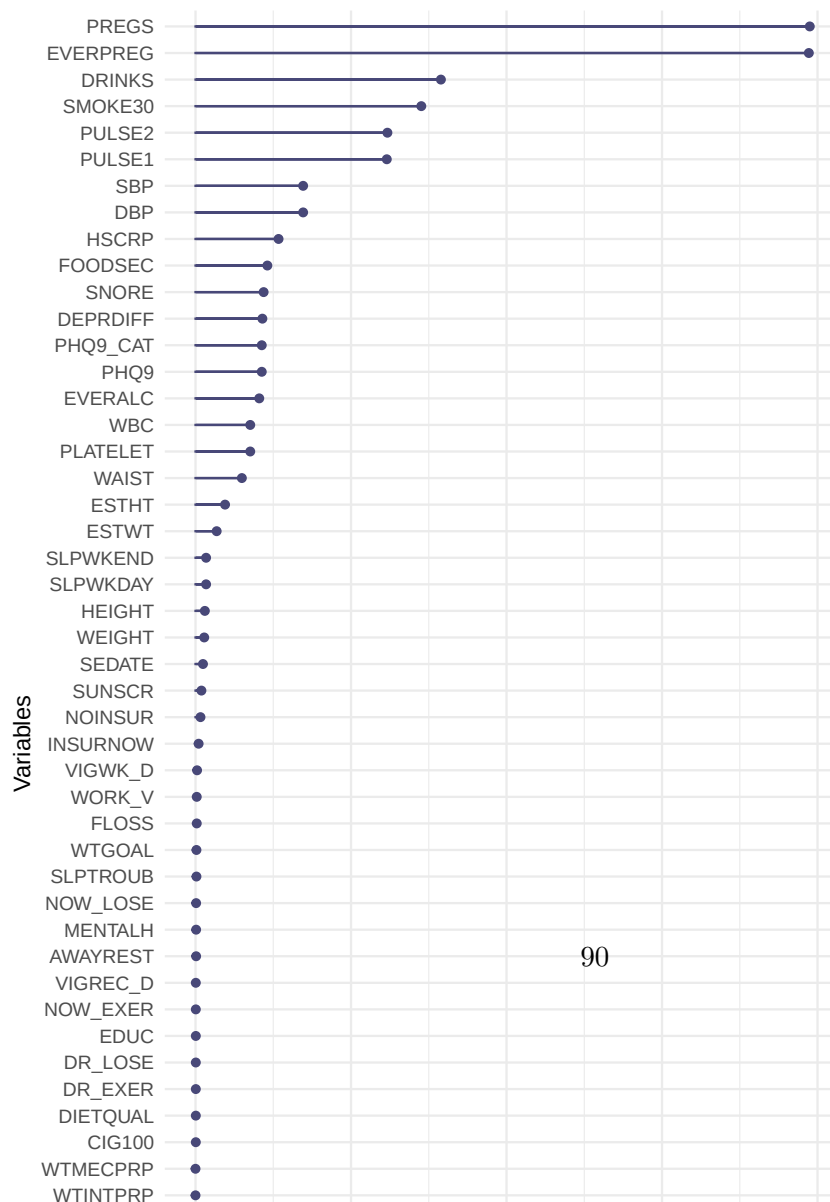
lowest : 2589.17 2782.74 3003.52 3009.53 3016.64
highest: 267064 268879 273958 308015 321574

2.6 Missingness in nh432

```
miss_case_table(nh432) |> gt()
```

```
gg_miss_var(nh432)
```

n_miss_in_case	n_cases	pct_cases
0	907	23.0730094
1	533	13.5588909
2	1030	26.2019842
3	591	15.0343424
4	307	7.8097176
5	161	4.0956500
6	87	2.2131773
7	106	2.6965149
8	68	1.7298397
9	18	0.4578988
10	20	0.5087764
11	39	0.9921140
12	27	0.6868481
13	14	0.3561435
14	9	0.2289494
15	14	0.3561435



```
miss_var_summary(nh432) |> gt()
```

```
miss_var_table(nh432) |> gt()
```

variable	n_miss	pct_miss
PREGS	1975	50.2
EVERPREG	1972	50.2
DRINKS	789	20.1
SMOKE30	726	18.5
PULSE2	617	15.7
PULSE1	615	15.6
SBP	346	8.80
DBP	346	8.80
HSCRП	267	6.79
FOODSEC	231	5.88
SNORE	219	5.57
DEPRDIFF	215	5.47
PHQ9	213	5.42
PHQ9_CAT	213	5.42
EVERALC	205	5.21
WBC	176	4.48
PLATELET	176	4.48
WAIST	149	3.79
ESTHT	95	2.42
ESTWT	68	1.73
SLPWKDAY	34	0.865
SLPWKEND	34	0.865
HEIGHT	30	0.763
WEIGHT	28	0.712
SEDATE	24	0.611
SUNSCR	19	0.483
NOINSUR	16	0.407
INSURNOW	10	0.254
VIGWK_D	5	0.127
WORK_V	4	0.102
FLOSS	4	0.102
WTGOAL	3	0.0763
SLPTROUB	3	0.0763
NOW_LOSE	2	0.0509
MENTALH	2	0.0509
AWAYREST	2	0.0509
EDUC	1	0.0254
DR_LOSE	1	0.0254
DR_EXER	1	0.0254
NOW_EXER	1	0.0254
DIETQUAL	1	0.0254
VIGREC_D	1	0.0254
CIG100	1	0.0254
SEQN	92 0	0
AGE	0	0
RACEETH	0	0
SEX	0	0
SROH	0	0
REC_V	0	0
HOSPITAL	0	0

n_miss_in_var	n_vars	pct_vars
0	12	21.818182
1	7	12.727273
2	3	5.454545
3	2	3.636364
4	2	3.636364
5	1	1.818182
10	1	1.818182
16	1	1.818182
19	1	1.818182
24	1	1.818182
28	1	1.818182
30	1	1.818182
34	2	3.636364
68	1	1.818182
95	1	1.818182
149	1	1.818182
176	2	3.636364
205	1	1.818182
213	2	3.636364
215	1	1.818182
219	1	1.818182
231	1	1.818182
267	1	1.818182
346	2	3.636364
615	1	1.818182
617	1	1.818182
726	1	1.818182
789	1	1.818182
1972	1	1.818182
1975	1	1.818182

3 431 Review: Comparing Means

In this Chapter, we will review some key issues about comparing means, mostly drawn from the 431 course. This and the other Chapters labeled “431 Review” show elements of the two studies involved in a “[431 Project B](#)” using the pre-pandemic (2017 - March 2020) data from the National Health and Nutrition Examination Survey (NHANES) called **nh432** that we developed in Chapter 1 and then summarized in Chapter 2. Note that we’ll ignore the issue of sampling weights and assume a “missing completely at random” (MCAR) mechanism in these “431 Review” chapters, so that we can work with complete cases.

The 431 course book is found at <https://thomaselove.github.io/431-book/> and will remain there until June 1.

3.1 R Setup

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(car)
library(glue)
library(gt)
library(gtsummary)
library(lmboot)
library(MKinfer)
library(mosaic)
library(naniar)
library(patchwork)
library(rstatix)

library(tidyverse)

theme_set(theme_bw())
```

3.2 Data Ingest

Since we've already got the **nh432** file formatted as an R data set, we'll use that.

```
nh432 <- read_rds("data/nh432.Rds")
```

3.3 Testing or Summarizing Normality?

As we'll see, the two most useful strategies for dealing with problematic non-Normality when comparing means are (1) transformation of the outcome to make the assumption of Normality more tenable, and (2) alternate inference approaches (for example, using a bootstrap or rank-based procedure instead of a *t* test.)

While it is possible to obtain numerical summaries of deviations from Normality, perhaps a measure of skewness (asymmetry) or kurtosis (heavy-tailed behavior), in practical work, I **never** use such summaries to overrule my assessment of the plots. It's critical instead to focus on the **pictures** of a distribution, most especially Normal Q-Q plots.

Perhaps the simplest skewness summary is $skew_1 = (\text{mean} - \text{median}) / (\text{standard deviation})$, where values below -0.2 are meant to indicate (meaningful) left skew, and values above +0.2 indicate (meaningful) right skew. Unfortunately, this approach works poorly with many distributions (for example, multimodal distributions) and so do many other (more sophisticated) measures¹.

It is also possible to develop hypothesis tests of whether a particular batch of data follows a Normal distribution, for example, the Kolmogorov-Smirnov test², or the Shapiro-Wilk test³, but again, I find these to be without value in practical work and cannot recommend their use.

3.4 Comparing Two Means using Paired Samples

Now, we'll demonstrate some approaches to comparing two means coming from paired samples. This will include:

- a paired *t* test (one-sample *t* test on the paired differences), which we can obtain from a linear model, or from `t.test()`

¹The `e1071` package in R provides several such measures of skewness and kurtosis if you really want to see them.

²The Kolmogorov-Smirnov test is available in R: see `ks.test()`

³See `shapiro.test()` in R

These procedures based on the t distribution for paired samples require that the distribution of the sample paired differences is well-approximated by a Normal model. As an alternative without that requirement, we'll focus primarily on a bootstrap comparison (not assuming Normality) from `boot.t.test()`, which comes from the `MKinfer` package. It is also possible to generate rank-based inference, such as using the Wilcoxon signed rank approach, but this introduces the major weakness of not estimating the population mean (or even the population median.)

We'll assume a Missing Completely at Random (MCAR) mechanism for missing data, so that a complete case analysis makes sense, and we'll also use functions from the `broom` package to tidy our output, and from the `gt` package to help present it in an attractive table.

3.5 Comparing PULSE1 to PULSE2

We have two measurements of pulse rate (in beats per minute) in `nh432` for each participant. Let's compare the two for all participants with two PULSE readings. Since we have a value of PULSE1 and PULSE2 for each participant, it makes sense to treat these as paired samples, and study the paired differences in pulse rate.

```
dat1 <- nh432 |> select(SEQN, PULSE1, PULSE2) |>
  drop_na() |>
  mutate(PULSEDIFF = PULSE2 - PULSE1)

summary(dat1 |> select(-SEQN))
```

PULSE1		PULSE2		PULSEDIFF	
Min.	: 38.0	Min.	: 37.00	Min.	:-22.0000
1st Qu.:	62.0	1st Qu.:	63.00	1st Qu.:	-1.0000
Median :	69.0	Median :	70.00	Median :	1.0000
Mean :	70.3	Mean :	70.96	Mean :	0.6533
3rd Qu.:	77.0	3rd Qu.:	78.00	3rd Qu.:	2.0000
Max.	:126.0	Max.	:121.00	Max.	: 26.0000

```
df_stats(~ PULSE1 + PULSE2 + PULSEDIFF, data = dat1) |>
  mutate(across(.cols = c(mean, sd), \(x) num(x, digits = 2))) |>
  gt()
```

response	min	Q1	median	Q3	max	mean	sd	n	missing
PULSE1	38	62	69	77	126	70.30	11.61	3314	0
PULSE2	37	63	70	78	121	70.96	11.57	3314	0
PULSEDIFF	-22	-1	1	2	26	0.65	3.43	3314	0

3.5.1 Distribution of Paired Differences

```

p1 <- ggplot(dat1, aes(sample = PULSEDIFF)) +
  geom_qq(col = "dodgerblue") + geom_qq_line(col = "navy") +
  theme(aspect.ratio = 1) +
  labs(title = "Normal Q-Q plot")

p2 <- ggplot(dat1, aes(x = PULSEDIFF)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 25, fill = "dodgerblue", col = "white") +
  stat_function(fun = dnorm,
    args = list(mean = mean(dat1$PULSEDIFF),
      sd = sd(dat1$PULSEDIFF)),
    col = "navy", linewidth = 1.5) +
  labs(title = "Histogram and Normal Density")

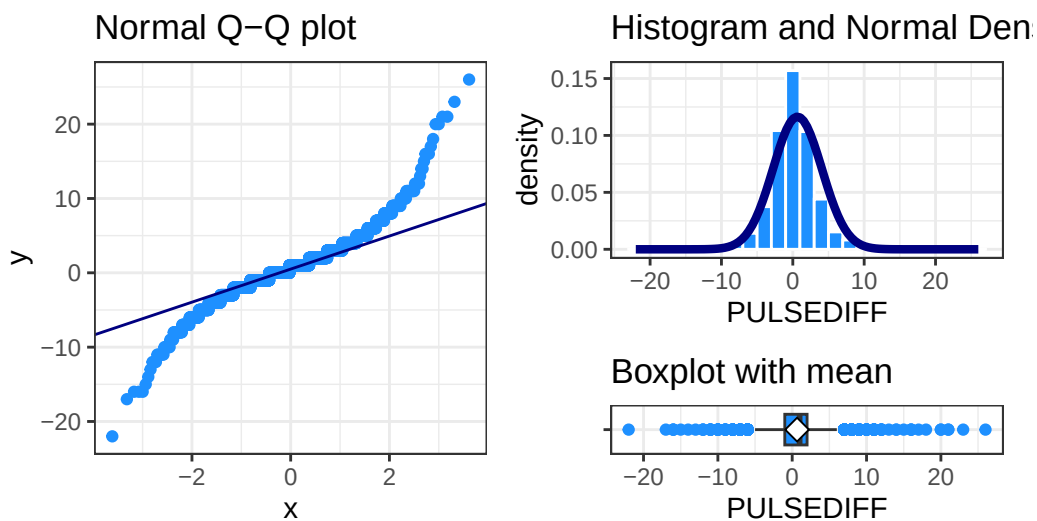
p3 <- ggplot(dat1, aes(x = PULSEDIFF, y = "")) +
  geom_boxplot(fill = "dodgerblue", outlier.color = "dodgerblue") +
  stat_summary(fun = "mean", geom = "point",
    shape = 23, size = 3, fill = "white") +
  labs(title = "Boxplot with mean", y = "")

p1 + (p2 / p3 + plot_layout(heights = c(4,1))) +
  plot_annotation(title = "Pulse 2 - Pulse 1 difference in nh432",
    subtitle = glue(nrow(dat1), " NHANES Participants ages 30-59 in nh432 data")

```

Pulse 2 – Pulse 1 difference in nh432

3314 NHANES Participants ages 30–59 in nh432 data



These data appear to come from a distribution that is essentially symmetric, but extremely heavy-tailed, with many outlier candidates on both the low and high end of the distribution. It seems unwise to assume a Normal distribution for these differences in pulse rate.

3.5.2 Using `t.test` to obtain a 90% CI for the mean pulse difference

Note that I use 90% as my confidence level here, mostly to make sure that we don't always simply default to 95% without engaging our brains.

```
tt1 <- t.test(dat1$PULSEDIFF, conf.level = 0.90)
tt1
```

One Sample t-test

```
data: dat1$PULSEDIFF
t = 10.98, df = 3313, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
90 percent confidence interval:
 0.5553989 0.7511792
sample estimates:
```

Pulse 2 - Pulse 1 Difference Estimate with 90% T-based CI

3314 NHANES Participants ages 30-59 in nh432 data

estimate	low.90	hi.90	Paired t	p.value	method
0.653	0.555	0.751	10.980	0.000	One Sample t-test

mean of x
0.6532891

```
tidy(tt1, conf.int = TRUE, conf = 0.90) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Paired t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "Pulse 2 - Pulse 1 Difference Estimate with 90% T-based CI",
             subtitle = glue(nrow(dat1), " NHANES Participants ages 30-59 in nh432 data"))
```

3.5.3 Using linear regression to obtain a 90% CI for the mean pulse difference

A linear regression model predicting the paired differences with an intercept alone produces the same result as the paired t test.

```
lm1 <- lm(PULSEDIFF ~ 1, data = dat1)

summary(lm1)
```

Call:

```
lm(formula = PULSEDIFF ~ 1, data = dat1)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.6533	-1.6533	0.3467	1.3467	25.3467

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.6533	0.0595	10.98	<2e-16 ***

Pulse 2 - Pulse 1 Difference Estimate with 90% T-based CI

3314 NHANES Participants ages 30-59 in nh432 data

estimate	low.90	hi.90	Paired t	p.value	method
0.653	0.555	0.751	10.980	0.000	Linear Model

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.425 on 3313 degrees of freedom

```
tidy(lm1, conf.int = TRUE, conf = 0.90) |>
  mutate(method = c("Linear Model")) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Paired t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "Pulse 2 - Pulse 1 Difference Estimate with 90% T-based CI",
             subtitle = glue(nrow(dat1), " NHANES Participants ages 30-59 in nh432 data"))
```

3.5.4 Using the bootstrap to obtain a 90% CI for the mean pulse difference

This is a better choice than the t test if the distribution of the paired differences veer far away from a Normal distribution, but you are still interested in making inferences about the population mean. This is a different approach to obtaining a bootstrap than I have used in the past, but I prefer it because it works well with the `tidy()` function in the `broom` package.

```
set.seed(4321)
bs1 <- boot.t.test(dat1$PULSEDIFF, conf.level = 0.90,
                  boot = TRUE, R = 999)

bs1
```

Bootstrap One Sample t-test

data: dat1\$PULSEDIFF
number of bootstrap samples: 999

Pulse 2 - Pulse 1 Difference Estimate with 90% Bootstrap CI

3314 NHANES Participants ages 30-59 in nh432 data

estimate	low.90	hi.90	Bootstrap t	p.value	method
0.653	0.555	0.751	10.980	0.000	Bootstrap One Sample t-test

```
bootstrap p-value < 0.001001
bootstrap mean of x (SE) = 0.6554904 (0.05954107)
90 percent bootstrap percentile confidence interval:
  0.5554617 0.7526554
```

```
Results without bootstrap:
t = 10.98, df = 3313, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
90 percent confidence interval:
  0.5553989 0.7511792
sample estimates:
mean of x
0.6532891
```

```
tidy(bs1, conf.int = TRUE, conf = 0.90) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Bootstrap t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "Pulse 2 - Pulse 1 Difference Estimate with 90% Bootstrap CI",
             subtitle = glue(nrow(dat1), " NHANES Participants ages 30-59 in nh432 data"))
```

Given our large sample size, it is perhaps not overly surprising that even a small difference in mean pulse rate (0.653 beats per minute) turns out to have a 90% confidence interval well above the value (0) that would occur if there were no difference at all between the groups.

3.5.5 Wilcoxon signed rank approach to comparing pulse rates

We can obtain a 90% confidence interval for the pseudo-median of our paired differences in pulse rate with the Wilcoxon signed rank approach.

```
wt1 <- wilcox.test(dat1$PULSEDIFF, conf.int = TRUE, conf.level = 0.90)

wt1
```

Wilcoxon signed rank test with continuity correction

```
data: dat1$PULSEDIFF
V = 2449203, p-value < 2.2e-16
alternative hypothesis: true location is not equal to 0
90 percent confidence interval:
 0.5000466 0.9999290
sample estimates:
(pseudo)median
 0.9999809
```

But this is of limited value, even though it doesn't assume Normality of the distribution of paired differences, because the summary statistic is a pseudo-median⁴, which isn't straightforward to interpret, unless the true distribution of the paired differences is **symmetric**, in which case the pseudo-median and the median have the same value.

Let's consider another example using two paired samples to compare means, this time with a somewhat smaller sample size.

3.6 Comparing WEIGHT to ESTWT

We have two assessments of each participant's weight in **nh432**: their **WEIGHT** (as measured using a scale, in kilograms) and their **ESTWT** (self-reported weight via questionnaire, in pounds.) First, let's create a data set containing those values, and converting pounds to kilograms for the **ESTWT** results so that we can compare the two assessments fairly. To shrink the sample size a bit, let's only look at people whose age is 43, and who describe their overall health as either **Good** or **Fair**.

```
dat2 <- nh432 |> select(SEQN, AGE, SROH, WEIGHT, ESTWT) |>
  filter(AGE == 43, SROH %in% c("Good", "Fair")) |>
  drop_na() |>
  mutate(ESTWTKG = ESTWT*0.45359,
         WTDIFF = WEIGHT - ESTWTKG)
```

⁴The pseudomedian of a distribution F is the median of the distribution of $(u + v)/2$, where u and v are independent and each have distribution F .

response	min	Q1	median	Q3	max	mean	sd	n	missing
WEIGHT	36.900	74.525	89.400	106.175	204.600	93.040	29.211	70	0
ESTWTKG	45.359	74.842	89.130	103.759	204.115	92.532	27.869	70	0
WTDIFF	-9.871	-2.256	-0.028	1.923	15.067	0.508	4.671	70	0

```
glimpse(dat2)
```

Rows: 70

Columns: 7

```
$ SEQN    <chr> "109342", "109602", "109805", "110286", "110645", "111149", "1~
$ AGE     <dbl> 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43~
$ SROH    <fct> Good, Good, Good, Fair, Good, Good, Good, Good, Good, Fair, Go~
$ WEIGHT  <dbl> 92.1, 76.5, 133.0, 86.8, 119.3, 74.1, 75.8, 106.8, 102.1, 77.0~
$ ESTWT   <dbl> 200, 167, 260, 198, 230, 145, 167, 240, 223, 172, 150, 265, 22~
$ ESTWTKG <dbl> 90.71800, 75.74953, 117.93340, 89.81082, 104.32570, 65.77055, ~
$ WTDIFF  <dbl> 1.38200, 0.75047, 15.06660, -3.01082, 14.97430, 8.32945, 0.050~
```

```
df_stats(~ WEIGHT + ESTWTKG + WTDIFF, data = dat2) |>
  mutate(across(.cols = -c(response, n, missing), \(x) num(x , digits = 3))) |>
  gt()
```

3.6.1 Plotting The Paired Difference in Weight

```
p1 <- ggplot(dat2, aes(sample = WTDIFF)) +
  geom_qq(col = "seagreen") + geom_qq_line(col = "deeppink") +
  theme(aspect.ratio = 1) +
  labs(title = "Normal Q-Q plot")

p2 <- ggplot(dat2, aes(x = WTDIFF)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 15, fill = "seagreen", col = "white") +
  stat_function(fun = dnorm,
    args = list(mean = mean(dat2$WTDIFF),
      sd = sd(dat2$WTDIFF)),
    col = "deeppink", linewidth = 1.5) +
  labs(title = "Histogram and Normal Density")
```

```

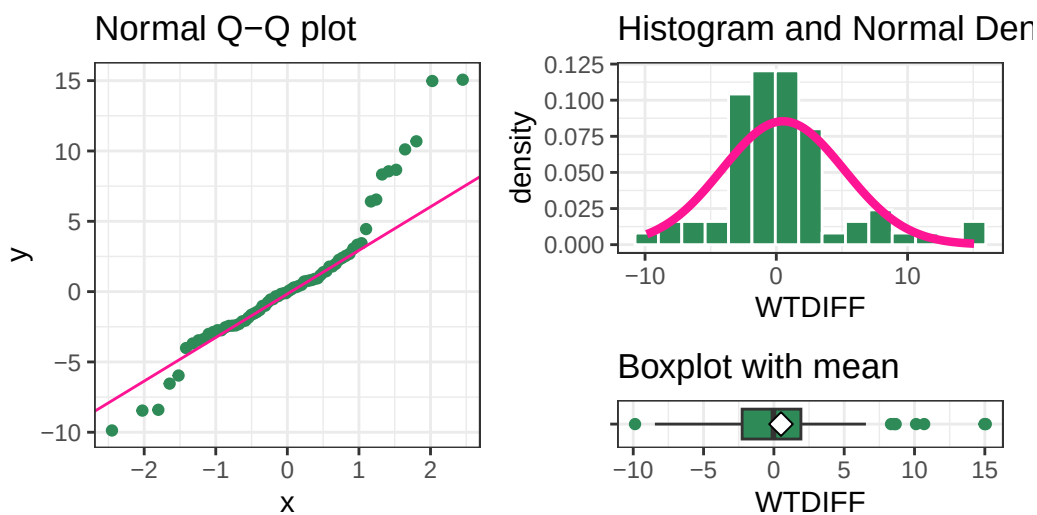
p3 <- ggplot(dat2, aes(x = WTDIFF, y = "")) +
  geom_boxplot(fill = "seagreen", outlier.color = "seagreen") +
  stat_summary(fun = "mean", geom = "point",
    shape = 23, size = 3, fill = "white") +
  labs(title = "Boxplot with mean", y = "")

p1 + (p2 / p3 + plot_layout(heights = c(4,1))) +
  plot_annotation(title = "Measured - Self-reported Weight (in kilograms)",
    subtitle = glue(nrow(dat2), " participants in Good or Fair Health aged 43 :

```

Measured – Self-reported Weight (in kilograms)

70 participants in Good or Fair Health aged 43 in nh432



As we saw with the differences in pulse rate, the differences in weight for this sample appear to come from a distribution that might be symmetric, but that still has several outlier candidates, especially on the high end of the distribution. We may want to consider whether the assumption of a t-based confidence interval is reasonable here, and whether we might be better off using a bootstrap approach.

3.6.2 t.test 90% CI for the mean weight difference

```

tt2 <- t.test(dat2$WTDIFF, conf.level = 0.90)

tidy(tt2, conf.int = TRUE, conf = 0.90) |>

```

Measured - Self-reported Weight with 90% T-based CI 70 NHANES Participants aged 43 in Good or Fair Health in nh432 data

estimate	low.90	hi.90	Paired t	p.value	method
0.508	-0.423	1.438	0.909	0.366	One Sample t-test

Measured - Self-reported Weight with 90% T-based CI 70 NHANES Participants aged 43 in Good or Fair Health in nh432 data

estimate	low.90	hi.90	Paired t	p.value	method
0.508	-0.423	1.438	0.909	0.366	Linear Model

```
select(estimate, low.90 = conf.low, hi.90 = conf.high,
       "Paired t" = statistic, p.value, method) |>
mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
gt() |>
tab_header(title = "Measured - Self-reported Weight with 90% T-based CI",
           subtitle = glue(nrow(dat2), " NHANES Participants aged 43 in Good or Fair Health"))
```

3.6.3 Linear Regression: 90% CI for mean weight difference

```
lm2 <- lm(WTDIFF ~ 1, data = dat2)

tidy(lm2, conf.int = TRUE, conf = 0.90) |>
mutate(method = c("Linear Model")) |>
select(estimate, low.90 = conf.low, hi.90 = conf.high,
       "Paired t" = statistic, p.value, method) |>
mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
gt() |>
tab_header(title = "Measured - Self-reported Weight with 90% T-based CI",
           subtitle = glue(nrow(dat2), " NHANES Participants aged 43 in Good or Fair Health"))
```

3.6.4 Bootstrap 90% CI for the mean weight difference

Measured - Self-reported Weight with 90% Bootstrap CI

estimate	low.90	hi.90	Bootstrap t	p.value	method
0.508	-0.423	1.438	0.909	0.366	Bootstrap One Sample t-test

3314 NHANES Participants age 43 in Good or Fair Health in nh432 data

```
set.seed(4322)
bs2 <- boot.t.test(dat2$WTDIFF, conf.level = 0.90,
                   boot = TRUE, R = 999)

tidy(bs2, conf.int = TRUE, conf = 0.90) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Bootstrap t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "Measured - Self-reported Weight",
             subtitle = "with 90% Bootstrap CI") |>
  tab_footnote(footnote = glue(nrow(dat1), " NHANES Participants age 43 in Good or Fair Health in nh432 data"))
```

In light of the clear issue with outliers in the plots of the weight differences, I think I would choose the bootstrap confidence interval, which clearly includes both negative and positive values as plausible estimates of the population mean difference.

3.6.5 Wilcoxon signed rank approach to comparing weight estimates

We can obtain a 90% confidence interval for the pseudo-median of our paired differences in weight with the Wilcoxon signed rank approach.

```
wt2 <- wilcox.test(dat2$WTDIFF, conf.int = TRUE, conf.level = 0.90)

wt2
```

Wilcoxon signed rank test with continuity correction

```
data:  dat2$WTDIFF
V = 1262, p-value = 0.9115
```

```
alternative hypothesis: true location is not equal to 0
90 percent confidence interval:
 -0.6347756  0.7941699
sample estimates:
(pseudo)median
 0.06155979
```

But this is of limited value, even though it doesn't assume Normality of the distribution of paired differences, because the summary statistic is a pseudo-median⁵, which isn't straightforward to interpret, unless the true distribution of the paired differences is **symmetric**, in which case the pseudo-median and the median have the same value.

3.7 Comparing Two Means using Independent Samples

Now, we'll demonstrate some approaches to comparing two means coming from independent samples. This will include:

- a pooled t test (t test assuming equal population variances), which we can obtain from a linear model, or from `t.test()`
- a Welch t test (t test not assuming equal population variances), from `t.test()`

Each of these t tests requires the distribution of each of our two independent samples to be well-approximated by a Normal model. As an alternative without that requirement, we'll focus on a bootstrap comparison (not assuming equal variances or Normality) from `boot.t.test()` (again from the `MKinfer` package.) Once more, it is also possible to generate rank-based inference, such as using the Wilcoxon-Mann-Whitney rank sum approach, but again this does not provide us with estimates of either the difference in population means or medians, which limits its utility.

3.8 Comparing White Blood Cell Count by Hospitalization Status

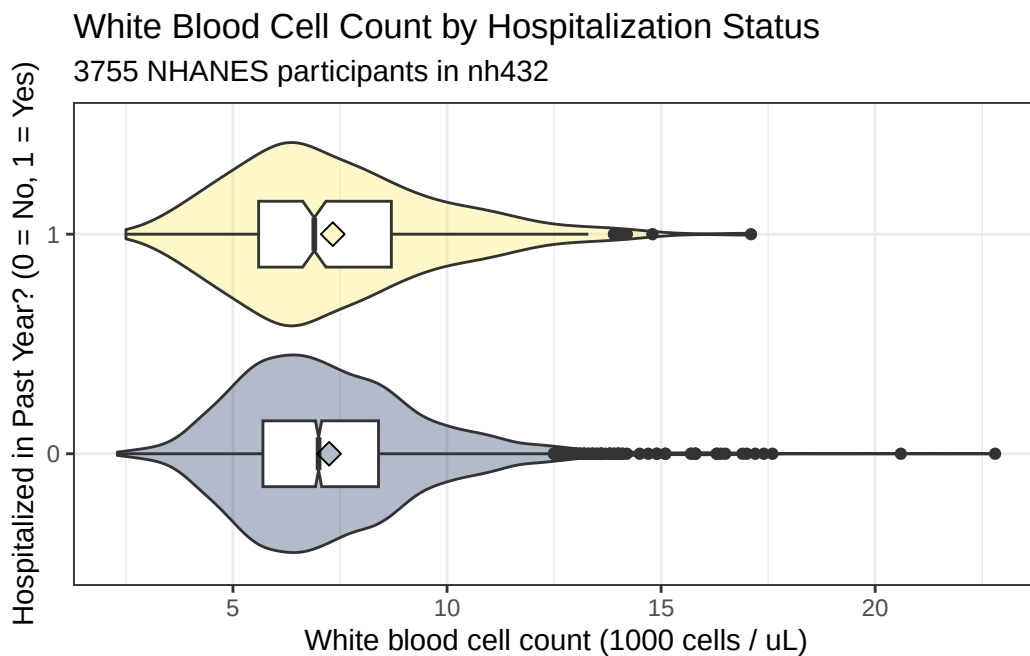
Now, we'll use independent samples to compare subjects who were hospitalized in the past year to those who were not, in terms of their white blood cell count. The normal range of WBCs in the blood is 4.5 to 11 on the scale (1000 cells per microliter) our data is available.

⁵The pseudomedian of a distribution F is the median of the distribution of $(u + v)/2$, where u and v are independent and each have distribution F .

3.8.1 Exploring the Data

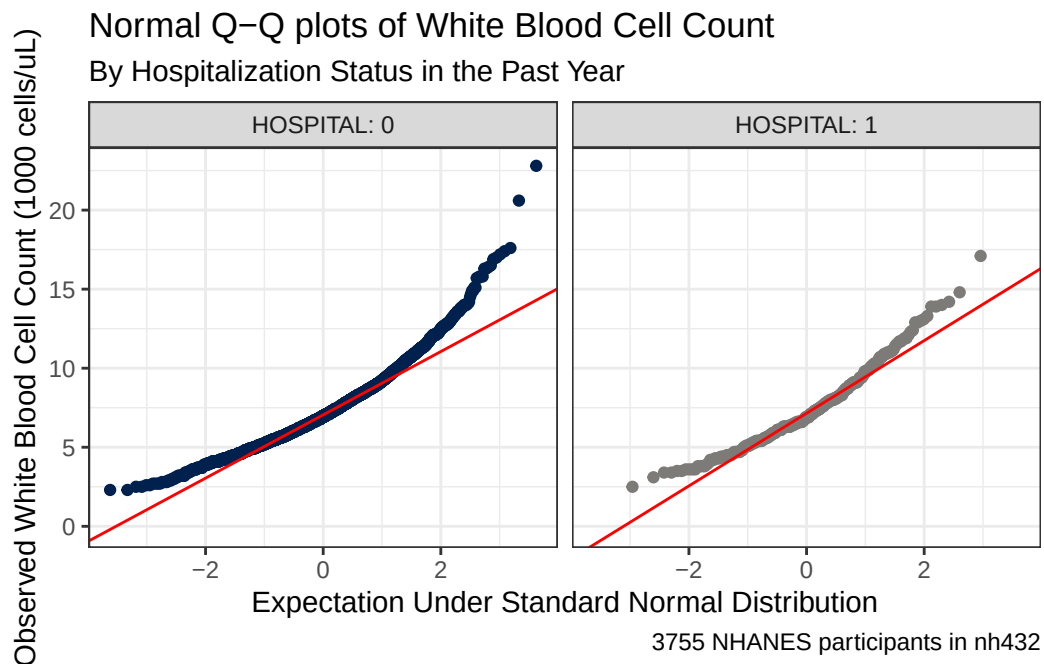
```
dat3 <- nh432 |>
  select(SEQN, HOSPITAL, WBC) |>
  drop_na()

ggplot(dat3, aes(x = factor(HOSPITAL), y = WBC)) +
  geom_violin(aes(fill = factor(HOSPITAL))) +
  geom_boxplot(width = 0.3, notch = TRUE) +
  stat_summary(aes(fill = factor(HOSPITAL)), fun = "mean", geom = "point",
    shape = 23, size = 3) +
  guides(fill = "none", col = "none") +
  scale_fill_viridis_d(option = "cividis", alpha = 0.3) +
  coord_flip() +
  labs(x = "Hospitalized in Past Year? (0 = No, 1 = Yes)",
    y = "White blood cell count (1000 cells / uL)",
    title = "White Blood Cell Count by Hospitalization Status",
    subtitle = glue(nrow(dat3), " NHANES participants in nh432"))
```



Each of these distributions shows some signs of right skew, or at least more than a few outlier candidates on the upper end of the white blood cell count's distribution, according to the boxplot. A pair of Normal Q-Q plots should help clarify issues for us.


```
ggplot(dat3, aes(sample = WBC)) +
  geom_qq(aes(col = factor(HOSPITAL))) + geom_qq_line(col = "red") +
  facet_wrap(~ HOSPITAL, labeller = "label_both") +
  guides(col = "none") +
  scale_color_viridis_d(option = "cividis", end = 0.5) +
  labs(x = "Expectation Under Standard Normal Distribution",
       y = "Observed White Blood Cell Count (1000 cells/uL)",
       title = "Normal Q-Q plots of White Blood Cell Count",
       subtitle = "By Hospitalization Status in the Past Year",
       caption = glue(nrow(dat3), " NHANES participants in nh432"))
```



It seems unreasonable to assume that each of these samples comes from a distribution that is well-approximated by the Normal. There's just too much skew here. Here are some key numerical summaries of the data in each sample.

```
favstats(WBC ~ HOSPITAL, data = dat3) |>
  mutate(across(.cols = c(mean, sd), \(x) num(x, digits = 3))) |>
  gt()
```

HOSPITAL	min	Q1	median	Q3	max	mean	sd	n	missing
0	2.3	5.7	7.0	8.4	22.8	7.246	2.170	3430	0
1	2.5	5.6	6.9	8.7	17.1	7.332	2.392	325	0

3.8.2 Pooled t test (assumes equal variances) via linear model

The pooled t test for comparison of two population means using independent samples assumes:

- that the WBC (outcome) in each of the two HOSPITAL (exposure) groups follows a Normal distribution, and
- that the population variances are equal in the two groups

The “equal population variances” assumption can be relaxed and a pooled t test used if we have a balanced design, with the same number of subjects in each exposure group.

In our setting, we shouldn’t be particularly comfortable with the assumption of Normality, as mentioned above. Were we able to get past that, though, we can see that the two distributions have fairly similar sample variances (remember this is just the square of the standard deviation.) The sample sizes are wildly different, with many more non-hospitalized subjects than hospitalized ones.

For completeness, though, we’ll start by running the pooled t test.

```
lm3 <- lm(WBC ~ HOSPITAL, data = dat3)
summary(lm3)
```

Call:

```
lm(formula = WBC ~ HOSPITAL, data = dat3)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-4.9463 -1.5463 -0.3463  1.1537 15.5537
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.24633     0.03740 193.760  <2e-16 ***
HOSPITAL      0.08567     0.12712   0.674    0.5
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

White Blood Cell Count (Hospitalized - Non-Hospitalized)

with 90% Pooled t-based CI via Linear Model

estimate	low.90	hi.90	Pooled t	p.value	method
0.086	-0.123	0.295	0.674	0.500	Pooled t

3755 NHANES Participants ages 30-59 in nh432 data

Residual standard error: 2.19 on 3753 degrees of freedom

Multiple R-squared: 0.000121, Adjusted R-squared: -0.0001454

F-statistic: 0.4542 on 1 and 3753 DF, p-value: 0.5004

```
confint(lm3, level = 0.90)
```

```

              5 %      95 %
(Intercept) 7.1847964 7.3078567
HOSPITAL    -0.1234734 0.2948204

```

```

tidy(lm3, conf.int = TRUE, conf.level = 0.90) |>
  filter(term == "HOSPITAL") |>
  mutate(method = "Pooled t") |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Pooled t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "White Blood Cell Count (Hospitalized - Non-Hospitalized)",
             subtitle = "with 90% Pooled t-based CI via Linear Model") |>
  tab_footnote(footnote = glue(nrow(dat3), " NHANES Participants ages 30-59 in nh432 data"))

```

3.8.3 Pooled t test (assumes equal variances) via t.test

Note that this approach estimates the difference with Not Hospitalized - Hospitalized, as opposed to the approach used in the linear model. Be careful to check the sample estimates provided in your output against the original summary of the sample data to avoid making a mistake.

White Blood Cell Count (Non-Hospitalized minus Hospitalized)

with 90% Pooled t-based Confidence Interval

estimate	low.90	hi.90	Pooled t	p.value	method
-0.086	-0.295	0.123	-0.674	0.500	Two Sample t-test

3755 NHANES Participants ages 30-59 in nh432 data

```
tt3p <- t.test(WBC ~ HOSPITAL, data = dat3, var.equal = TRUE, conf.level = 0.90)
```

```
tt3p
```

Two Sample t-test

data: WBC by HOSPITAL

t = -0.67395, df = 3753, p-value = 0.5004

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
90 percent confidence interval:

-0.2948204 0.1234734

sample estimates:

mean in group 0 mean in group 1

7.246327 7.332000

```
tidy(tt3p, conf.int = TRUE, conf.level = 0.90) |>  
  select(estimate, low.90 = conf.low, hi.90 = conf.high,  
         "Pooled t" = statistic, p.value, method) |>  
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>  
  gt() |>  
  tab_header(title = "White Blood Cell Count (Non-Hospitalized minus Hospitalized)",  
             subtitle = "with 90% Pooled t-based Confidence Interval") |>  
  tab_footnote(footnote = glue(nrow(dat3), " NHANES Participants ages 30-59 in nh432 data"))
```

Again, note that the `t.test()` approach estimates Non-Hospitalized - Hospitalized (so that the sample mean is negative.)

White Blood Cell Count (Non-Hospitalized minus Hospitalized)

with 90% Welch t-based Confidence Interval

estimate	low.90	hi.90	Welch t	p.value	method
-0.086	-0.313	0.142	-0.622	0.534	Welch Two Sample t-test

3755 NHANES Participants ages 30-59 in nh432 data

3.8.4 Welch t test (doesn't assume equal variance) via `t.test`

The Welch t test (which is actually the default `t.test` in R) assumes that the two groups each follow a Normal distribution, but does not require that those distributions have the same population variance.

```
tt3w <- t.test(WBC ~ HOSPITAL, data = dat3, conf.level = 0.90)

tt3w
```

Welch Two Sample t-test

data: WBC by HOSPITAL

t = -0.6218, df = 376.28, p-value = 0.5345

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
90 percent confidence interval:

-0.3128672 0.1415202

sample estimates:

mean in group 0 mean in group 1
7.246327 7.332000

```
tidy(tt3w, conf.int = TRUE, conf.level = 0.90) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Welch t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "White Blood Cell Count (Non-Hospitalized minus Hospitalized)",
             subtitle = "with 90% Welch t-based Confidence Interval") |>
  tab_footnote(footnote = glue(nrow(dat3), " NHANES Participants ages 30-59 in nh432 data"))
```

3.8.5 Bootstrap comparison of WBC by HOSPITAL

The bootstrap approach is appealing in part because it neither assumes Normality or equal population variances.

```
set.seed(4323)
bs3 <- boot.t.test(WBC ~ HOSPITAL, data = dat3,
                  R = 999, conf.level = 0.90)

bs3
```

Bootstrap Welch Two Sample t-test

```
data: WBC by HOSPITAL
number of bootstrap samples: 999
bootstrap p-value = 0.5325
bootstrap difference of means (SE) = -0.08734076 (0.1376246)
90 percent bootstrap percentile confidence interval:
-0.3137606 0.1414317
```

Results without bootstrap:

```
t = -0.6218, df = 376.28, p-value = 0.5345
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
-0.3128672 0.1415202
sample estimates:
mean in group 0 mean in group 1
7.246327 7.332000
```

```
tidy(bs3, conf.int = TRUE, conf.level = 0.90) |>
  mutate(diff = estimate1 - estimate2) |>
  select(est1 = estimate1, est2 = estimate2, diff,
         low.90 = conf.low, hi.90 = conf.high,
         p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "White Blood Cell Count (Non-Hospitalized minus Hospitalized)",
             subtitle = "with 90% Bootstrap Confidence Interval") |>
  tab_footnote(footnote = glue(nrow(dat3), " NHANES Participants ages 30-59 in nh432 data"))
```

White Blood Cell Count (Non-Hospitalized minus Hospitalized)

with 90% Bootstrap Confidence Interval

est1	est2	diff	low.90	hi.90	p.value	method
7.246	7.332	-0.086	-0.313	0.142	0.534	Bootstrap Welch Two Sample t-test

3755 NHANES Participants ages 30-59 in nh432 data

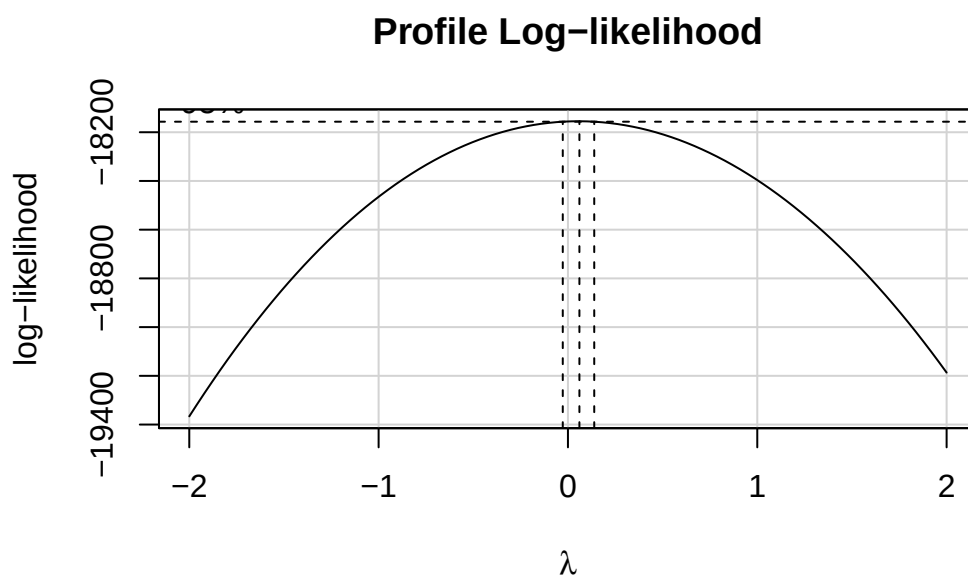
In any case, though, we come to the same basic conclusion - both positive and negative differences in WBC count are plausible.

Given the huge imbalance between the two groups in terms of sample size, and the apparent skew in the distribution of each sample, I would probably be most comfortable with the bootstrap approach here than the t-based intervals.

3.8.6 Transforming the WBC Counts

Since the White Blood Cell counts are far from Normally distributed, and in fact appear to be substantially skewed (asymmetric) we might want to consider a transformation of the data. The Box-Cox approach can be used to suggest potential transformations even in a simple case like this. We can use the `boxCox()` function from the **car** package, for example.

```
m3 <- lm(WBC ~ HOSPITAL, data = dat3)
boxCox(m3)
```



The estimated power (λ) shown in the plot is close to 0. The **ladder of power transformations** looks like this:

\$	Transformation	Formula
-2	inverse square	$1/y^2$
-1	inverse	$1/y$
-0.5	inverse square root	$1/\sqrt{y}$
0	logarithm	$\log y$
0.5	square root	$\sqrt{y}$
1	no transformation	y
2	square	y^2
3	cube	y^3

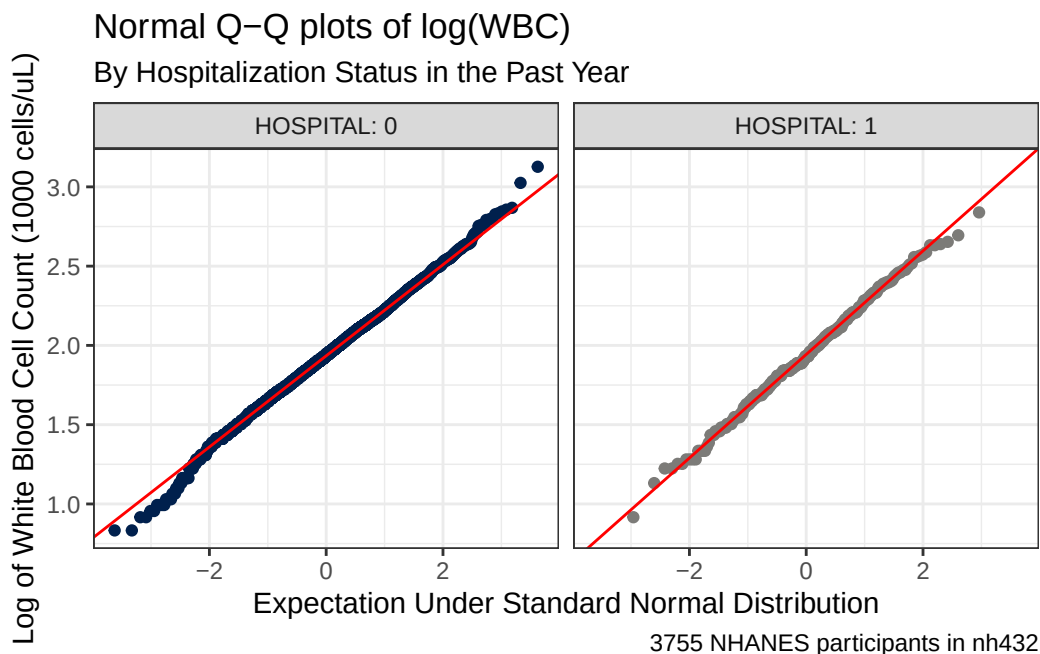
So in this case, the Box-Cox approach is suggesting we try the logarithm (we use the natural logarithm, with base e, here) of WBC.

Let's redraw our Normal Q-Q plots with this transformation applied.

```
ggplot(dat3, aes(sample = log(WBC))) +
  geom_qq(aes(col = factor(HOSPITAL))) + geom_qq_line(col = "red") +
  facet_wrap(~ HOSPITAL, labeller = "label_both") +
  guides(col = "none") +
  scale_color_viridis_d(option = "cividis", end = 0.5) +
```



```
labs(x = "Expectation Under Standard Normal Distribution",
     y = "Log of White Blood Cell Count (1000 cells/uL)",
     title = "Normal Q-Q plots of log(WBC)",
     subtitle = "By Hospitalization Status in the Past Year",
     caption = glue(nrow(dat3), " NHANES participants in nh432"))
```



The assumption of Normality now looks much more plausible for each of our samples. So we might try building a 90% confidence interval for the mean of log(WBC), as follows:

```
favstats(log(WBC) ~ HOSPITAL, data = dat3) |>
  mutate(across(.cols = -c(HOSPITAL, n, missing), \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "log(WBC) by Hospitalization Status",
             subtitle = "NHANES participants in nh432")
```

We see that there's essentially no difference at all in the means of the log(WBC) values across the two levels of hospitalization status.

```
tt3log <- t.test(log(WBC) ~ HOSPITAL, data = dat3, var.equal = TRUE, conf.level = 0.90)

tidy(tt3log, conf.int = TRUE, conf.level = 0.90) |>
  select(estimate1, estimate2, estimate, conf.low, conf.high, p.value)
```

log(WBC) by Hospitalization Status

NHANES participants in nh432

HOSPITAL	min	Q1	median	Q3	max	mean	sd	n	missing
0	0.833	1.740	1.946	2.128	3.127	1.938	0.293	3430	0
1	0.916	1.723	1.932	2.163	2.839	1.941	0.322	325	0

```
# A tibble: 1 x 6
  estimate1 estimate2 estimate conf.low conf.high p.value
    <dbl>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1      1.94      1.94 -0.00320  -0.0314    0.0250    0.852
```

Let's consider a second example for comparing means from independent samples.

3.9 Comparing Waist Circumference by Sleep Trouble

Now, we'll restrict ourselves to NHANES participants who rated their overall health as "Fair", and we'll compare the mean waist circumference (WAIST, in cm) of people in that group who responded Yes (vs. No) to the question of whether they had told a doctor that they had trouble sleeping (gathered in the SLPTROUB variable.)

3.9.1 Summarizing the Data

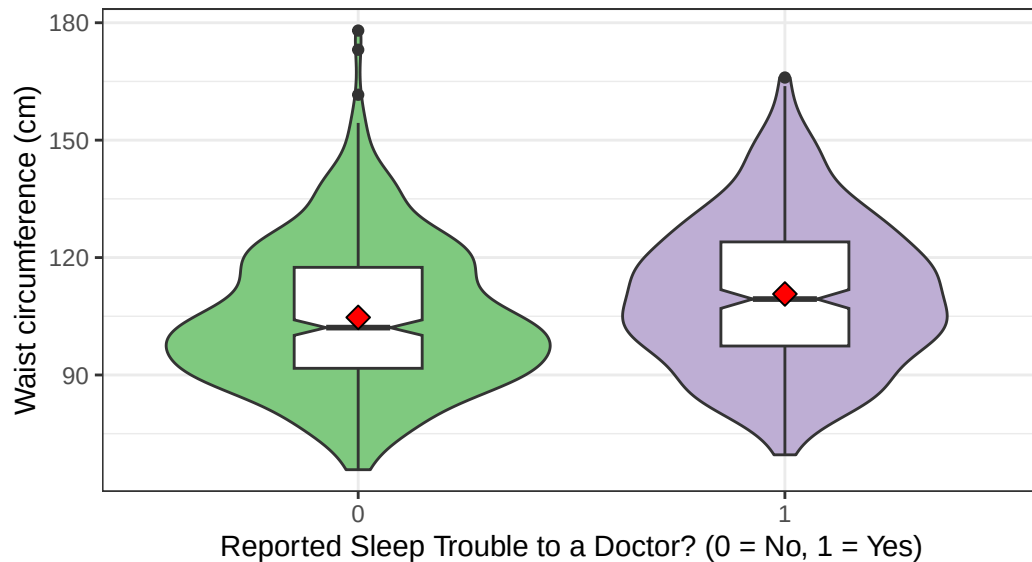
```
dat4 <- nh432 |>
  select(SEQN, SROH, SLPTROUB, WAIST) |>
  filter(SROH == "Fair") |>
  drop_na()

ggplot(dat4, aes(x = factor(SLPTROUB), y = WAIST)) +
  geom_violin(aes(fill = factor(SLPTROUB))) +
  geom_boxplot(width = 0.3, notch = TRUE) +
  stat_summary(fill = "red", fun = "mean", geom = "point",
               shape = 23, size = 3) +
  guides(fill = "none", col = "none") +
  scale_fill_brewer(palette = "Accent") +
  labs(x = "Reported Sleep Trouble to a Doctor? (0 = No, 1 = Yes)",
       y = "Waist circumference (cm)",
```

```
title = "Waist Circumference by Sleep Trouble",
subtitle = glue(nrow(dat4), " NHANES participants in Fair health in nh432"))
```

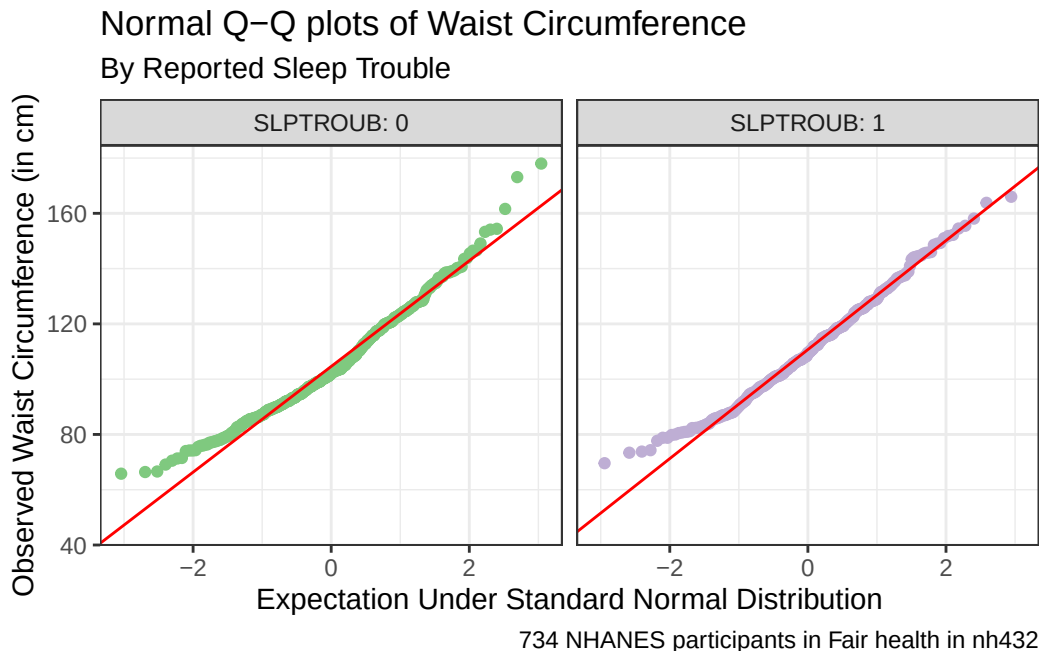
Waist Circumference by Sleep Trouble

734 NHANES participants in Fair health in nh432



```
ggplot(dat4, aes(sample = WAIST)) +
  geom_qq(aes(col = factor(SLPTRoub))) + geom_qq_line(col = "red") +
  facet_wrap(~ SLPTRoub, labeller = "label_both") +
  guides(col = "none") +
  scale_color_brewer(palette = "Accent") +
  labs(x = "Expectation Under Standard Normal Distribution",
       y = "Observed Waist Circumference (in cm)",
       title = "Normal Q-Q plots of Waist Circumference",
       subtitle = "By Reported Sleep Trouble",
       caption = glue(nrow(dat4), " NHANES participants in Fair health in nh432"))
```

SLPTROUB	min	Q1	median	Q3	max	mean	sd	n	missing
0	65.8	91.7	102.1	117.5	178	104.71	18.35	425	0
1	69.6	97.4	109.4	124.0	166	110.69	18.86	309	0



Here's a situation where we might be willing to consider a t test, since a Normal distribution is a much better fit for the data in each of our two samples. Let's look at some brief numerical summaries, too.

```
favstats(WAIST ~ SLPTROUB, data = dat4) |>
  mutate(across(.cols = c(mean, sd), \(x) num(x, digits = 2))) |>
  gt()
```

3.9.2 Pooled t test (assumes equal variances) via linear model

Here's the pooled t test via linear model.

```
lm4 <- lm(WAIST ~ SLPTROUB, data = dat4)
summary(lm4)
```

```
Call:
lm(formula = WAIST ~ SLPTRoub, data = dat4)

Residuals:
    Min       1Q   Median       3Q      Max
-41.093 -13.293  -2.293   12.790   73.290

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 104.7099     0.9006  116.27  < 2e-16 ***
SLPTRoub      5.9830     1.3881    4.31 1.85e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.57 on 732 degrees of freedom
Multiple R-squared:  0.02475,    Adjusted R-squared:  0.02342
F-statistic: 18.58 on 1 and 732 DF,  p-value: 1.853e-05
```

```
confint(lm4, level = 0.90)
```

```
              5 %      95 %
(Intercept) 103.226621 106.193144
SLPTRoub      3.696944   8.269052
```

```
tidy(lm4, conf.int = TRUE, conf.level = 0.90) |>
  filter(term == "SLPTRoub") |>
  mutate(method = "Pooled t") |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Pooled t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 2))) |>
  gt() |>
  tab_header(title = "Waist Circumference by Sleep Trouble",
             subtitle = "with 90% Pooled t-based CI via Linear Model") |>
  tab_footnote(footnote = glue(nrow(dat4), " NHANES Participants in Fair Health in nh432 data"))
```

3.9.3 Pooled t test (assumes equal variances) via t.test

Waist Circumference by Sleep Trouble

with 90% Pooled t-based CI via Linear Model

estimate	low.90	hi.90	Pooled t	p.value	method
5.98	3.70	8.27	4.31	0.00	Pooled t

734 NHANES Participants in Fair Health in nh432 data

```
tt4p <- t.test(WAIST ~ SLPTROUB, data = dat4, var.equal = TRUE, conf.level = 0.90)

tt4p
```

Two Sample t-test

data: WAIST by SLPTROUB

t = -4.3103, df = 732, p-value = 1.853e-05

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to

90 percent confidence interval:

-8.269052 -3.696944

sample estimates:

mean in group 0 mean in group 1

104.7099 110.6929

```
tidy(tt4p, conf.int = TRUE, conf.level = 0.90) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Pooled t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 2))) |>
  gt() |>
  tab_header(title = "Waist Circumference by Sleep Trouble",
             subtitle = "with 90% Pooled t-based Confidence Interval") |>
  tab_footnote(footnote = glue(nrow(dat4), " NHANES Participants in Fair Health in nh432 data")
```

3.9.4 Welch t test (doesn't assume equal variance) via t.test

```
tt4w <- t.test(WAIST ~ SLPTROUB, data = dat4, conf.level = 0.90)

tt4w
```

Waist Circumference by Sleep Trouble with 90% Pooled t-based Confidence Interval

estimate	low.90	hi.90	Pooled t	p.value	method
-5.98	-8.27	-3.70	-4.31	0.00	Two Sample t-test

734 NHANES Participants in Fair Health in nh432 data

Waist Circumference by Sleep Trouble with 90% Welch t-based Confidence Interval

estimate	low.90	hi.90	Welch t	p.value	method
-5.98	-8.28	-3.69	-4.29	0.00	Welch Two Sample t-test

734 NHANES Participants in Fair Health in nh432 data

Welch Two Sample t-test

data: WAIST by SLPTRoub

t = -4.2919, df = 653.24, p-value = 2.04e-05

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to

90 percent confidence interval:

-8.279237 -3.686759

sample estimates:

mean in group 0 mean in group 1

104.7099 110.6929

```
tidy(tt4w, conf.int = TRUE, conf.level = 0.90) |>
  select(estimate, low.90 = conf.low, hi.90 = conf.high,
         "Welch t" = statistic, p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 2))) |>
  gt() |>
  tab_header(title = "Waist Circumference by Sleep Trouble",
             subtitle = "with 90% Welch t-based Confidence Interval") |>
  tab_footnote(footnote = glue(nrow(dat4), " NHANES Participants in Fair Health in nh432 data")
```

3.9.5 Bootstrap comparison of WAIST by SLPTROUB

```
set.seed(4324)
bs4 <- boot.t.test(WAIST ~ SLPTROUB, data = dat4,
                   R = 999, conf.level = 0.90)

bs4
```

Bootstrap Welch Two Sample t-test

data: WAIST by SLPTROUB
number of bootstrap samples: 999
bootstrap p-value < 0.001001
bootstrap difference of means (SE) = -5.991814 (1.389937)
90 percent bootstrap percentile confidence interval:
-8.207833 -3.747965

Results without bootstrap:

t = -4.2919, df = 653.24, p-value = 2.04e-05
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
-8.279237 -3.686759
sample estimates:
mean in group 0 mean in group 1
104.7099 110.6929

```
tidy(bs4, conf.int = TRUE, conf.level = 0.90) |>
  mutate(diff = estimate1 - estimate2) |>
  select(est1 = estimate1, est2 = estimate2, diff,
         low.90 = conf.low, hi.90 = conf.high,
         p.value, method) |>
  mutate(across(.cols = -method, \(x) num(x, digits = 2))) |>
  gt() |>
  tab_header(title = "Waist Circumference by Sleep Trouble",
             subtitle = "with 90% Bootstrap Confidence Interval") |>
  tab_footnote(footnote = glue(nrow(dat4), " NHANES Participants in Fair Health in nh432 data"))
```


Waist Circumference by Sleep Trouble

with 90% Bootstrap Confidence Interval

est1	est2	diff	low.90	hi.90	p.value	method
104.71	110.69	-5.98	-8.28	-3.69	0.00	Bootstrap Welch Two Sample t-test

734 NHANES Participants in Fair Health in nh432 data

3.9.6 Wilcoxon-Mann-Whitney Rank Sum Approach

The Wilcoxon-Mann-Whitney rank sum approach also allows us (like the bootstrap) to avoid the assumptions of Normality and equal population variances, but at the cost of no longer yielding direct inference about the population mean.

```
wt4 <- wilcox.test(WAIST ~ SLPTRoub, data = dat4,  
                   conf.int = TRUE, conf.level = 0.90)  
wt4
```

Wilcoxon rank sum test with continuity correction

```
data:  WAIST by SLPTRoub  
W = 53322, p-value = 1.355e-05  
alternative hypothesis: true location shift is not equal to 0  
90 percent confidence interval:  
 -8.599997 -3.900026  
sample estimates:  
difference in location  
 -6.299963
```

Note that the estimated “difference in location” here is **not** the difference in the medians across the two groups, but instead the median of the difference between a sample from the SLPTRoub = Yes group and a sample from the SLPTRoub = No group.

Just to prove my point, here are the sample median WAIST results in the two SLPTRoub groups. You can see that the difference between these medians does not match the “difference in location” estimate from the Wilcoxon-Mann-Whitney rank sum output.

```
dat4 |> group_by(SLPTRoub) |> summarise(median(WAIST))
```

```
# A tibble: 2 x 2
  SLPTRoub `median(WAIST)`
    <dbl>         <dbl>
1      0         102.
2      1         109.
```

In conclusion, the confidence intervals (from any of these approaches) suggest that plausible means of waist circumference are around 3-8 centimeters larger in the “told Dr. about sleep problems” group, which I suppose isn’t especially surprising, at least in terms of its direction.

3.10 Comparing 3 Means using Independent Samples: Systolic BP by Weight Goal

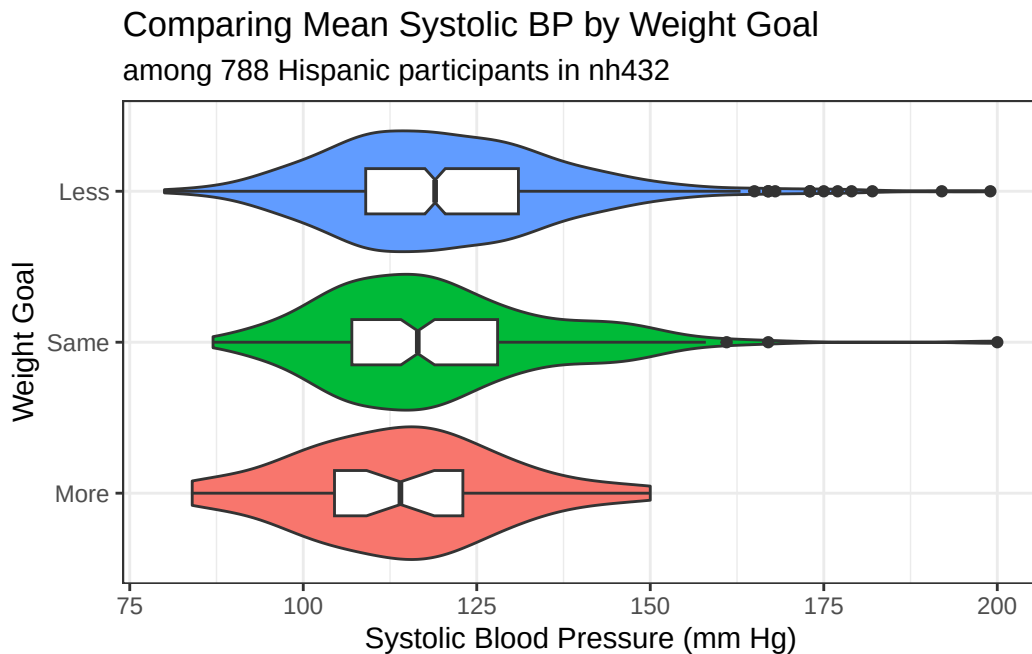
We’ll compare systolic blood pressure means across the three samples defined by WTGOAL (goal is to weigh more, less or stay about the same), restricting to our participants of Hispanic or Latinx ethnicity in **nh432**.

```
dat5 <- nh432 |>
  select(SEQN, RACEETH, SBP, WTGOAL) |>
  filter(RACEETH == "Hispanic") |>
  drop_na()
```

3.10.1 Summarizing SBP by WTGOAL

```
ggplot(dat5, aes(x = SBP, y = WTGOAL)) +
  geom_violin(aes(fill = WTGOAL)) +
  geom_boxplot(width = 0.3, notch = TRUE) +
  guides(fill = "none") +
  labs(title = "Comparing Mean Systolic BP by Weight Goal",
       subtitle = glue("among ", nrow(dat5), " Hispanic participants in nh432"),
       x = "Systolic Blood Pressure (mm Hg)", y = "Weight Goal")
```

WTGOAL	min	Q1	median	Q3	max	mean	sd	n	missing
More	84	104.5	114.0	123	150	114.08	14.76	36	0
Same	87	107.0	116.5	128	200	119.26	16.49	188	0
Less	80	109.0	119.0	131	199	120.70	17.17	564	0



```
favstats(SBP ~ WTGOAL, data = dat5) |>
  as_tibble() |>
  mutate(across(.cols = c("mean", "sd"), \(x) num(x, digits = 2))) |>
  gt()
```

The analysis of variance is our primary tool for comparing more than two means (this is the extension of the pooled t test, with similar assumptions.) So the assumptions we might want to think about here are:

- SBP in each Weight Goal group is assumed to follow a Normal distribution
- SBP in each Weight Goal group is assumed to have the same population variance

The ANOVA, however, is far more robust to minor violations of these assumptions than is the pooled t test. So we might go ahead and fit the ANOVA model anyway, despite the apparent right skew in the “Less” group.

3.10.2 Fitting an ANOVA Model

```
m5 <- lm(SBP ~ WTGOAL, data = dat5)

anova(m5)
```

Analysis of Variance Table

Response: SBP

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
WTGOAL	2	1639	819.45	2.8656	0.05754
Residuals	785	224477	285.96		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

A small p value (remember we are using 90% confidence in our 432 work) like this isn't really very important - usually it simply steers us towards trying to identify confidence intervals for differences between pairs of SBP means defined by WTGOAL.

3.10.2.1 ANOVA without assuming Equal Variances?

R will also fit an ANOVA-style model and produce a p value without the assumption of equal population SBP variance across the three groups of WTGOAL.

```
oneway.test(SBP ~ WTGOAL, data = dat5)
```

One-way analysis of means (not assuming equal variances)

data: SBP and WTGOAL

F = 3.5255, num df = 2.000, denom df = 93.753, p-value = 0.0334

I don't use this approach much, as ANOVA is pretty robust to the assumption of equal variance. The huge differences in sample size in this study (many more participants are in the Less group than the More, for instance) are most of the cause of the difference we see here.

3.10.2.2 Testing for Equal Population Variance?

Some people like to perform tests for equal population variance to help choose between ANOVA and the `oneway.test()` approach, but I do not. If I'm happy with the assumption of Normality, I virtually always just use ANOVA. There are many such tests of "equal variance", including:

- Bartlett's test
- Levene's test (which in R comes from the **car** package)
- Fligner-Killeen test

Bartlett's test is the least reliable of these when the data in at least one sample appear to be poorly described by the Normal distribution. Either Levene or Fligner-Killeen is a better choice in that setting, but again, I don't use any of these in my work.

```
bartlett.test(SBP ~ WTGOAL, data = dat5)
```

Bartlett test of homogeneity of variances

data: SBP by WTGOAL

Bartlett's K-squared = 1.6722, df = 2, p-value = 0.4334

```
leveneTest(SBP ~ WTGOAL, data = dat5)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	0.6557	0.5193
	785		

```
fligner.test(SBP ~ WTGOAL, data = dat5)
```

Fligner-Killeen test of homogeneity of variances

data: SBP by WTGOAL

Fligner-Killeen:med chi-squared = 1.26, df = 2, p-value = 0.5326

3.10.2.3 Is there a bootstrap one-way ANOVA approach?

If all you are looking for is a p value for the ANOVA model, then yes, there is a bootstrap approach available to perform one-way ANOVA testing. But I don't actually use it, again usually preferring the usual ANOVA if the data seem reasonably likely to have been drawn from a Normal distribution, and the Kruskal-Wallis rank-based test otherwise. If you are willing to install the `lmboot` package, and use its `ANOVA.boot()` function, you can do so, like this.

```
bs5 <- ANOVA.boot(SBP ~ WTGOAL, B = 1000, seed = 4325, data = dat5)
bs5$`p-value`
```

```
[1] 0.052
```

In this case, it doesn't seem that we have a wildly different result than we got from the original ANOVA. That is often the case, and I have never actually used `ANOVA.boot()` in practical work.

3.10.3 Tukey HSD Pairwise Comparisons

When pairwise comparisons are pre-planned, especially when the design is close to balanced, my favorite choice for generating adjusted inferences about the means is Tukey's Honestly Significant Differences (HSD) approach.

Here, we generate confidence intervals for the pairwise differences in the SBP means by WTGOAL group with a 90% family-wise confidence level.

```
th5 <- TukeyHSD(aov(SBP ~ WTGOAL, data = dat5),
               ordered = TRUE, conf.level = 0.90)

tidy(th5) |>
  select(-c(term, null.value)) |>
  mutate(across(.cols = -contrast, \(x) num(x, digits = 3))) |>
  arrange(desc(estimate)) |>
  gt() |>
  tab_header(title = "Comparing Mean Systolic BP across pairs of WTGOAL groups",
             subtitle = "90% Tukey HSD Confidence Intervals") |>
  tab_footnote(footnote = glue(nrow(dat5), " Hispanic participants in nh432"))
```

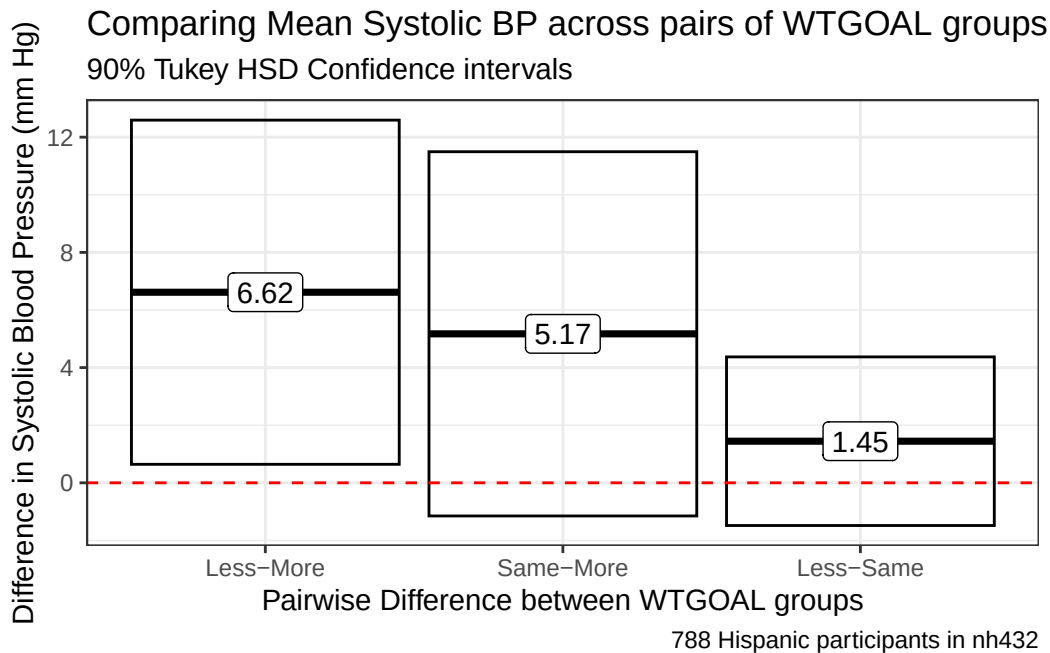
Comparing Mean Systolic BP across pairs of WTGOAL groups

90% Tukey HSD Confidence Intervals

contrast	estimate	conf.low	conf.high	adj.p.value
Less-More	6.617	0.642	12.592	0.060
Same-More	5.172	-1.151	11.495	0.213
Less-Same	1.445	-1.482	4.372	0.568

788 Hispanic participants in nh432

```
tidy(th5) |>
  mutate(contrast = fct_reorder(contrast, estimate, .desc = TRUE)) %>%
  ggplot(., aes(x = contrast, y = estimate, ymin = conf.low, ymax = conf.high)) +
  geom_crossbar() +
  geom_hline(yintercept = 0, col = "red", linetype = "dashed") +
  geom_label(aes(label = round_half_up(estimate, 2))) +
  labs(title = "Comparing Mean Systolic BP across pairs of WTGOAL groups",
        subtitle = "90% Tukey HSD Confidence intervals",
        caption = glue(nrow(dat5), " Hispanic participants in nh432"),
        x = "Pairwise Difference between WTGOAL groups",
        y = "Difference in Systolic Blood Pressure (mm Hg)")
```



The main problems here are that:

- the sample sizes in the various levels of WTGOAL are very different from one another, and
- the SBP data are not especially well-described by a Normal distribution, at least in the “Less” group.

3.10.4 Holm pairwise comparisons of means

Another approach to developing pairwise inferences would be to use either Bonferroni or (my preference) Holm-adjusted p values for the relevant t tests. First, we’ll run the appropriate Holm comparison of means assuming equal population variances of SBP across all three WTGOAL groups.

```
ht5 <- pairwise.t.test(dat5$SBP, dat5$WTGOAL, pool.sd = TRUE, p.adjust.method = "holm")
tidy(ht5) |>
  gt()
```

The results are merely p-values, and not confidence intervals. There’s nothing being estimated here of interest. We can also perform these Holm comparisons without assuming equal population variances, as shown below.

group1	group2	p.value
Same	More	0.18625400
Less	More	0.06929224
Less	Same	0.31056211

group1	group2	p.value
Same	More	0.12866665
Less	More	0.04046259
Less	Same	0.30396220

```
ht5un <- pairwise.t.test(dat5$SBP, dat5$WTGOAL, pool.sd = FALSE,
                          p.adjust.method = "holm")
tidy(ht5un) |>
  gt()
```

Again, the problem with this approach is that it's only producing a p value, which tempts us into talking about useless things like “statistical significance.” This is part of the reason I prefer Tukey HSD approaches when appropriate.

3.11 Comparing 4 Means using Independent Samples: Weight by Food Security

```
dat6 <- nh432 |>
  select(SEQN, WEIGHT, FOODSEC) |>
  drop_na()
```

3.11.1 Summarizing the Data

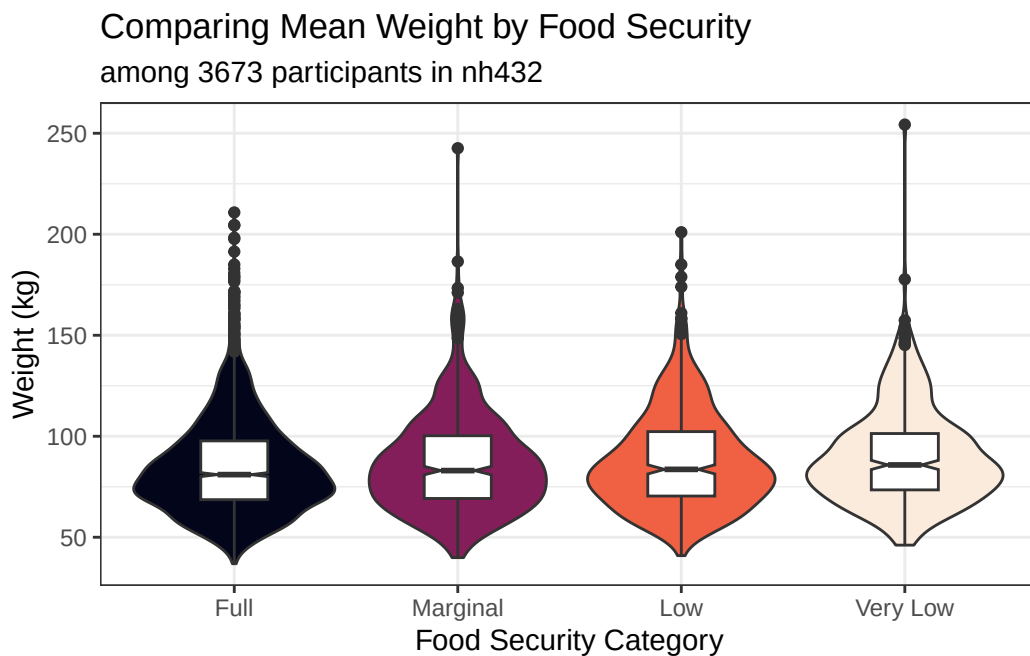
```
ggplot(dat6, aes(x = FOODSEC, y = WEIGHT)) +
  geom_violin(aes(fill = FOODSEC)) +
  geom_boxplot(width = 0.3, notch = TRUE) +
  guides(fill = "none") +
  scale_fill_viridis_d(option = "rocket") +
  labs(title = "Comparing Mean Weight by Food Security",
```

FOODSEC	min	Q1	median	Q3	max	mean	sd	n	missing
Full	36.9	68.6	81.0	97.7	210.8	85.2	24.5	2233	0
Marginal	39.9	69.2	83.0	100.2	242.6	87.0	25.0	560	0
Low	40.9	70.4	83.6	102.3	201.0	88.2	24.6	501	0
Very Low	46.1	73.4	85.8	101.3	254.3	90.1	24.4	379	0

```

  subtitle = glue("among ", nrow(dat6), " participants in nh432"),
  x = "Food Security Category", y = "Weight (kg)"

```



```

favstats(WEIGHT ~ FOODSEC, data = dat6) |>
  as_tibble() |>
  mutate(across(.cols = -c("FOODSEC", "n", "missing"), \(x) num(x, digits = 1))) |>
  gt()

```

3.11.2 Fitting the ANOVA model

```

m6 <- lm(WEIGHT ~ FOODSEC, data = dat6)

anova(m6)

```

Analysis of Variance Table

Response: WEIGHT

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
FOODSEC	3	10021	3340.3	5.525	0.0008786 ***
Residuals	3669	2218162	604.6		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Does the bootstrap ANOVA give a meaningfully different result? No.

```
bs6 <- ANOVA.boot(WEIGHT ~ FOODSEC, B = 5000, seed = 4326, data = dat6)
bs6$`p-value`
```

```
[1] 8e-04
```

3.11.3 Tukey HSD Pairwise Comparisons

```
th6 <- TukeyHSD(aov(WEIGHT ~ FOODSEC, data = dat6),
               ordered = TRUE, conf.level = 0.90)

tidy(th6) |>
  select(-c(term, null.value)) |>
  mutate(across(.cols = -contrast, \(x) num(x, digits = 3))) |>
  arrange(desc(estimate)) |>
  gt() |>
  tab_header(title = "Comparing Mean Weight across pairs of Food Security groups",
             subtitle = "90% Tukey HSD Confidence Intervals") |>
  tab_footnote(footnote = glue(nrow(dat6), " participants in nh432"))
```

```
tidy(th6) |>
  mutate(contrast = fct_reorder(contrast, estimate, .desc = TRUE)) %>%
  ggplot(., aes(x = contrast, y = estimate, ymin = conf.low, ymax = conf.high)) +
  geom_pointrange() +
  geom_hline(yintercept = 0, col = "blue", linetype = "dashed") +
  geom_label(aes(label = round_half_up(estimate, 2))) +
  labs(title = "Comparing Mean Weight across pairs of Food Security groups",
       subtitle = "90% Tukey HSD Confidence intervals",
       caption = glue(nrow(dat5), " non-Hispanic Black participants in nh432"),
```

Comparing Mean Weight across pairs of Food Security groups

90% Tukey HSD Confidence Intervals

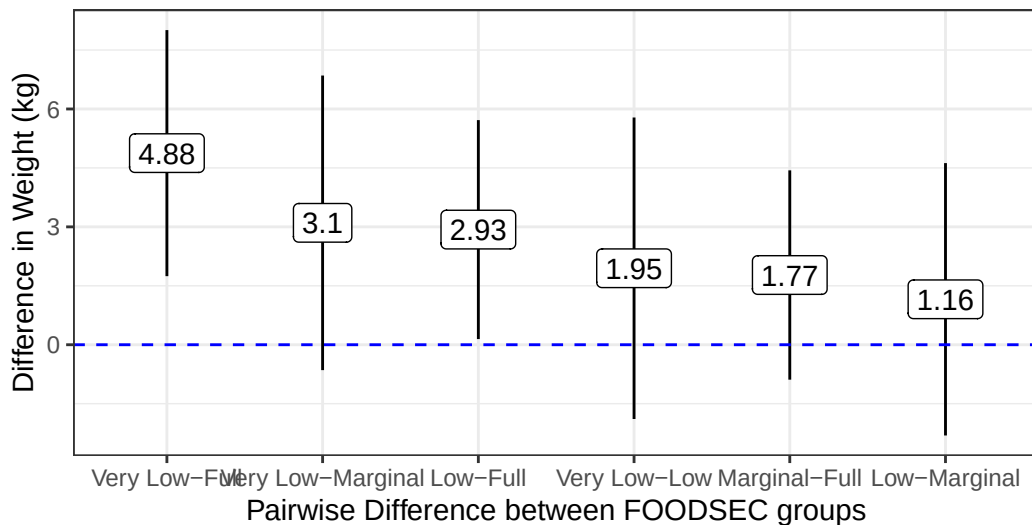
contrast	estimate	conf.low	conf.high	adj.p.value
Very Low-Full	4.877	1.746	8.008	0.002
Very Low-Marginal	3.103	-0.646	6.851	0.229
Low-Full	2.930	0.144	5.717	0.075
Very Low-Low	1.946	-1.891	5.783	0.650
Marginal-Full	1.774	-0.889	4.438	0.422
Low-Marginal	1.156	-2.310	4.622	0.870

3673 participants in nh432

```
x = "Pairwise Difference between FOODSEC groups",
y = "Difference in Weight (kg)"
```

Comparing Mean Weight across pairs of Food Security groups

90% Tukey HSD Confidence intervals



788 non-Hispanic Black participants in nh432

3.11.4 Kruskal-Wallis Test

When the assumption of Normality is really unreasonable, many people (including me) will instead use a rank-based method, called the Kruskal-Wallis test to compare the locations of WEIGHT across levels of FOODSEC.

```
kruskal.test(WEIGHT ~ FOODSEC, data = dat6)
```

Kruskal-Wallis rank sum test

data: WEIGHT by FOODSEC

Kruskal-Wallis chi-squared = 21.102, df = 3, p-value = 0.0001003

3.11.5 Dunn Test for Pairwise Comparisons after Kruskal-Wallis Test

Should you develop a Kruskal-Wallis test result which implies that running a set of pairwise comparisons is important, I would suggest the use of the Dunn test, available in the `dunn_test()` function from the **rstatix** package.

```
dunn_test(data = dat6, WEIGHT ~ FOODSEC,
          p.adjust.method = "holm", detailed = TRUE) |>
  select(group1, group2, p.adj, n1, n2, estimate1, estimate2, estimate) |>
  mutate(across(.cols = -c(group1, group2, n1, n2), \(x) num(x, digits = 3))) |>
  gt() |>
  tab_header(title = "Dunn Tests comparing WEIGHT by FOODSEC",
             subtitle = "Pairwise Comparisons after Kruskal-Wallis test") |>
  tab_footnote(footnote = glue(nrow(dat6), " participants in nh432"))
```

Again, a problem with this approach is that all it provides is a set of adjusted p values for these comparisons, but if we're not willing to assume even very approximate Normality (and thus use an ANOVA approach) this is what we'll have to cope with.

Dunn Tests comparing WEIGHT by FOODSEC

Pairwise Comparisons after Kruskal-Wallis test

group1	group2	p.adj	n1	n2	estimate1	estimate2	estimate
Full	Marginal	0.269	2233	560	1780.595	1865.621	85.026
Full	Low	0.049	2233	501	1780.595	1916.147	135.551
Full	Very Low	0.000	2233	379	1780.595	2022.412	241.816
Marginal	Low	0.438	560	501	1865.621	1916.147	50.525
Marginal	Very Low	0.105	560	379	1865.621	2022.412	156.790
Low	Very Low	0.282	501	379	1916.147	2022.412	106.265

3673 participants in nh432

4 431 Review: Comparing Rates

In this Chapter, we will review some key issues about comparing proportions or rates, mostly drawn from the 431 course. This and the other Chapters labeled “431 Review” show elements of the two studies involved in a “[431 Project B](#)” using the pre-pandemic (2017 - March 2020) data from the National Health and Nutrition Examination Survey (NHANES) called **nh432** that we developed in Chapter 1 and then summarized in Chapter 2. Note that we’ll ignore the issue of sampling weights and assume a “missing completely at random” (MCAR) mechanism in these “431 Review” chapters, so that we can work with complete cases.

The 431 course book is found at <https://thomaselove.github.io/431-book/> and will remain there until June 1.

4.1 R Setup

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(DescTools)
library(Epi)
library(gt)
library(Hmisc)
library(vcd)

library(tidyverse)

theme_set(theme_bw())
```

4.1.1 Data Load

```
nh432 <- read_rds("data/nh432.Rds")
```

4.2 2x2 Contingency Table: DR_LOSE and NOWLOSE

Let's compare the probability that NOWLOSE is 1 (The subject is currently working on losing or controlling their body weight) between NHANES participants who have (vs. who have not) been told by a doctor to lose or control their weight in the past 12 months (DR_LOSE). Each of these (DR_LOSE and NOWLOSE) is stored in R as a numeric variable with non-missing values equal to 0 or 1.

```
temp <- nh432 |>
  select(SEQN, DR_LOSE, NOW_LOSE) |>
  drop_na()
```

As with any categorical variable, we start by counting, and the natural way to display the counts of these two variables (DR_LOSE and NOW_LOSE) is in a table, rather than a graph, I think.

```
temp |>
  tabyl(DR_LOSE, NOW_LOSE) |>
  adorn_totals(where = c("row", "col")) |>
  adorn_title()
```

	NOW_LOSE		
DR_LOSE	0	1	Total
0	1198	1541	2739
1	246	943	1189
Total	1444	2484	3928

Now that we have a 2x2 table, we could consider obtaining some more detailed summary statistics, with a tool like the `twoby2()` function in the **Epi** package. There is a problem with this, though.

```
twoby2(temp$DR_LOSE, temp$NOW_LOSE)
```

2 by 2 table analysis:

```
-----
Outcome      : 0
Comparing    : 0 vs. 1

      0      1      P(0) 95% conf. interval
0 1198 1541  0.4374    0.4189    0.4560
```



```

1  246  943  0.2069    0.1848    0.2309

                                95% conf. interval
      Relative Risk: 2.1140      1.8766    2.3815
    Sample Odds Ratio: 2.9801    2.5412    3.4949
Conditional MLE Odds Ratio: 2.9793    2.5350    3.5096
    Probability difference: 0.2305    0.2002    0.2594

      Exact P-value: 0.0000
    Asymptotic P-value: 0.0000
-----

```

The code runs fine, but the table isn't really in a useful format. This table shows the probability that NOWLOSE = 0 ("No") comparing DR_LOSE = 0 ("No") to DR_LOSE = 1 ("Yes"), and that's just confusing.

It would be much better if we did two things:

1. used factors with meaningful labels to represent the 1/0 variables for this table
2. set up the table in standard epidemiological format, and then made a better choice as to what combination should be in the top left of the 2x2 table.

So let's do that.

4.2.1 Standard Epidemiological Format

Standard Epidemiological Format for a 2x2 table places the exposure in the rows, and the outcome in the columns, with the top left representing the combination of interest when we obtain things like an odds ratio or probability difference. Typically this means we want to put the "Yes" and "Yes" combination in the top left.

First, let's create factor versions (with more meaningful labels than 1 and 0) out of the two variables of interest: DR_LOSE and NOW_LOSE.

```

dat1 <- nh432 |>
  select(SEQN, DR_LOSE, NOW_LOSE) |>
  drop_na() |>
  mutate(DR_LOSE_f = fct_recode(factor(DR_LOSE), "Dr_said_Lose_Wt" = "1", No = "0"),
         DR_LOSE_f = fct_relevel(DR_LOSE_f, "Dr_said_Lose_Wt", "No"),
         NOW_LOSE_f = fct_recode(factor(NOW_LOSE), "Now_losing_Wt" = "1", No = "0"),
         NOW_LOSE_f = fct_relevel(NOW_LOSE_f, "Now_losing_Wt", "No"))

```

DR_LOSE vs. NOW_LOSE

Standard Epidemiological Format

Dr said Lose Weight?	Currently Losing Weight?		Total
	Now_losing_Wt	No	
Dr_said_Lose_Wt	943	246	1189
No	1541	1198	2739
Total	2484	1444	3928

Note that after recoding the levels to more meaningful labels, we also re-leveled the factors so that the “Yes” result comes first rather than last.

This produces the following table, which is now in standard epidemiological format, where we are using the DR_LOSE_f information to predict NOW_LOSE_f.

```
dat1 |>
  tabyl(DR_LOSE_f, NOW_LOSE_f) |>
  adorn_totals(where = c("row", "col")) |>
  adorn_title()
```

	NOW_LOSE_f		
DR_LOSE_f	Now_losing_Wt	No	Total
Dr_said_Lose_Wt	943	246	1189
No	1541	1198	2739
Total	2484	1444	3928

We could, I suppose, make the table even prettier.

```
tab1 <- dat1 |>
  tabyl(DR_LOSE_f, NOW_LOSE_f) |>
  adorn_totals(where = c("row", "col"))

gt(tab1, rowname_col = "DR_LOSE_f") |>
  tab_header(title = "DR_LOSE vs. NOW_LOSE",
             subtitle = "Standard Epidemiological Format") |>
  tab_stubhead(label = "Dr said Lose Weight?") |>
  tab_spanner(label = "Currently Losing Weight?",
             columns = c(Now_losing_Wt, No))
```

4.2.2 Obtaining Key Summaries with twoby2()

And, finally, we can obtain necessary summaries (including estimates and confidence intervals) using the `twoby2()` function.

```
twoby2(dat1$DR_LOSE_f, dat1$NOW_LOSE_f, conf.level = 0.90)
```

2 by 2 table analysis:

```
-----
Outcome      : Now_losing_Wt
Comparing     : Dr_said_Lose_Wt vs. No

              Now_losing_Wt   No      P(Now_losing_Wt) 90% conf. interval
Dr_said_Lose_Wt          943   246          0.7931      0.7731   0.8118
No                      1541 1198          0.5626      0.5470   0.5781

                                90% conf. interval
              Relative Risk: 1.4097      1.3586   1.4627
              Sample Odds Ratio: 2.9801      2.6071   3.4065
Conditional MLE Odds Ratio: 2.9793      2.5998   3.4195
              Probability difference: 0.2305      0.2052   0.2548

              Exact P-value: 0.0000
              Asymptotic P-value: 0.0000
-----
```

Some brief descriptions of these results:

- The probability that a participant is now losing weight (NOW_LOSE is 1) is estimated to be 0.79 (with 90% CI 0.77, 0.81) if the participant has been told to lose weight by a doctor in the past 12 months (DR_LOSE = 1), but only 0.56 (with 90% CI 0.55, 0.58) if the participant has not been told this.
- The relative risk of a participant now losing weight is estimated to be $\frac{0.7931}{0.5626} = 1.41$ (with 90% CI 1.36, 1.46) for a participant who has been told to lose weight vs. a participant who has not.
- The odds of a participant now losing weight are $\frac{0.7931(1-0.5626)}{0.5626(1-0.7931)} = 2.98$ times as high for a participant who has been told to lose weight than for one who has not, with 90% CI (2.61, 3.41).
- The difference in probability is estimated to be $0.7931 - 0.5626 = 0.2305$ (90% CI: 0.21, 0.25), indicating again that the true probability of now losing weight is higher in participants who have been told to lose weight than in those who have not.

The “exact” p-value listed comes from the Fisher exact test, while the “asymptotic” p-value comes from a Pearson χ^2 (chi-squared) test. I would focus on the meaningful estimates (those with confidence intervals) in making comparisons, rather than on trying to determine “statistical significance” with the p-values.

4.3 2x2 Table: SEDATE category and NOW_EXER

Let’s now look at another example, where we compare the probability that a participant is “now exercising” (NOW_EXER = 1) on the basis of their level of sedentary activity in a typical day (collected in the SEDATE variable, in minutes.)

```
dat2 <- nh432 |>
  select(SEQN, SEDATE, NOW_EXER) |>
  drop_na()

summary(dat2 |> select(-SEQN))
```

SEDATE		NOW_EXER	
Min.	: 2.0	Min.	:0.0000
1st Qu.	: 180.0	1st Qu.	:0.0000
Median	: 300.0	Median	:1.0000
Mean	: 332.8	Mean	:0.6019
3rd Qu.	: 480.0	3rd Qu.	:1.0000
Max.	:1320.0	Max.	:1.0000

As you can see above, the information in SEDATE is quantitative, and suppose we want to compare a High SEDATE group vs. a Low SEDATE group.

4.3.1 Creating a Low and High Group on SEDATE

We can use the `cut2()` function from the **Hmisc** package to partition the data by the SEDATE variable into three groups of equal sample size. At the same time, we’ll make NOW_EXER into a more useful (for tabulation) factor with more meaningful level descriptions.

```
dat2 <- dat2 |>
  mutate(SED_f = cut2(SEDATE, g = 3),
         NOW_EXER_f = fct_recode(factor(NOW_EXER), "Now_exercising" = "1", No = "0"),
         NOW_EXER_f = fct_relevel(NOW_EXER_f, "Now_exercising", "No"))
```

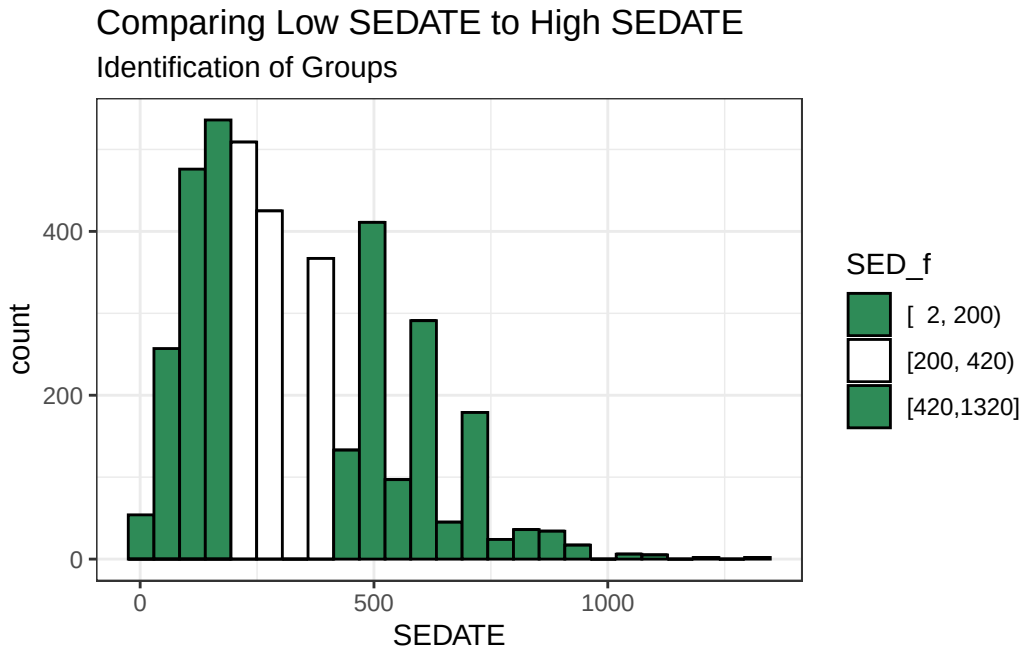
As you can see, we now have three groups defined by their SEDATE values, of roughly equal sample sizes.

```
dat2 |> tabyl(SED_f)
```

	SED_f	n	percent
[	2, 200)	1323	0.3387097
[	200, 420)	1301	0.3330773
[	420,1320]	1282	0.3282130

The group labeled [2, 200) contains the 1323 subjects who had SEDATE values ranging from 2 up to (but not including) 200 minutes, for example.

```
ggplot(dat2, aes(x = SEDATE)) +  
  geom_histogram(aes(fill = SED_f), col = "black", bins = 25) +  
  scale_fill_manual(values = c("seagreen", "white", "seagreen")) +  
  labs(title = "Comparing Low SEDATE to High SEDATE",  
       subtitle = "Identification of Groups")
```



Now, we want to compare the Lowest SEDATE group ($SED_f = [2, 200)$) to the Highest SEDATE group ($SED_F = [420, 1320]$). To do that, we'll drop the middle group, and then look at the cross-tabulation of our two remaining SEDATE groups with our outcome: NOW_EXER (in factor form.)

```
dat2 <- dat2 |>
  filter(SED_f != "[200, 420)") |>
  mutate(SED_f = fct_drop(SED_f))

dat2 |> tabyl(SED_f, NOW_EXER_f)
```

```
      SED_f Now_exercising No
[ 2, 200)           776 547
[420,1320]           789 493
```

4.3.2 Two-by-Two Table Summaries

Let's look at the analytic results for this table.

```
twoby2(dat2$SED_f, dat2$NOW_EXER_f)
```

2 by 2 table analysis:

```
-----
Outcome   : Now_exercising
Comparing : [ 2, 200) vs. [420,1320]
```

	Now_exercising	No	P(Now_exercising)	95% conf. interval	
[2, 200)	776	547	0.5865	0.5598	0.6128
[420,1320]	789	493	0.6154	0.5885	0.6417

		95% conf. interval	
Relative Risk:	0.9530	0.8952	1.0146
Sample Odds Ratio:	0.8864	0.7577	1.0371
Conditional MLE Odds Ratio:	0.8865	0.7553	1.0403
Probability difference:	-0.0289	-0.0664	0.0087

```
Exact P-value: 0.1388
Asymptotic P-value: 0.1322
-----
```

Uh, oh. There's a bit of a problem here now. We have the right rows and the right columns, but they're not in the best possible order, since the estimated probability of Now Exercising for the group on top (SED = [2, 200)) is smaller than it is for the people in the high group in terms of sedentary activity. As a result of this problem with ordering, our relative risk and odds ratio estimates are less than 1, and our probability difference is negative.

4.3.3 Flipping Levels

Since which exposure goes at the top is an arbitrary decision, let's switch the factor levels in `SED_f`, so that the people with high sedentary activity and who are now exercising are shown in the top left cell of the table. This should flip the point estimates of the relative risk and odds ratio above 1, and the estimated probability difference to a positive number. Note the use of the `fct_rev()` function from the **forcats** package to accomplish this.

```
dat2 <- dat2 |>
  mutate(SED_f = fct_rev(SED_f))

dat2 |> tabyl(SED_f, NOW_EXER_f) |>
  adorn_totals(where = c("row", "col")) |>
  adorn_title()
```

	NOW_EXER_f		
SED_f	Now_exercising	No	Total
[420,1320]	789	493	1282
[2, 200)	776	547	1323
Total	1565	1040	2605

```
twoby2(dat2$SED_f, dat2$NOW_EXER_f, conf.level = 0.90)
```

2 by 2 table analysis:

Outcome : Now_exercising
Comparing : [420,1320] vs. [2, 200)

	Now_exercising	No	P(Now_exercising)	90% conf. interval	
[420,1320]	789	493	0.6154	0.5929	0.6375
[2, 200)	776	547	0.5865	0.5641	0.6086

	90% conf. interval		
Relative Risk:	1.0493	0.9956	1.1059
Sample Odds Ratio:	1.1281	0.9889	1.2869
Conditional MLE Odds Ratio:	1.1281	0.9858	1.2910
Probability difference:	0.0289	-0.0027	0.0604

Exact P-value: 0.1388
Asymptotic P-value: 0.1322

We conclude now that the participants who were in the high SEDATION group (as compared to those in the low SEDATION group) had:

- a relative risk of 1.05 (90% CI: 0.995, 1.106) for Now exercising,
- a sample odds ratio of 1.13 (90% CI: 0.989, 1.287) for Now exercising,
- and probability for Now exercising that was 0.029 higher (-0.003, 0.060) than for those in the low SEDATION group.

4.4 A Larger (5x3) 2-Way Table: DIETQUAL and WTGOAL in Lighter Men

Here, we'll look at Male participants who weighed less than 100 kg (approximately 220 pounds) and ask whether their DIETQUAL (diet quality: self-rated as Excellent to poor in 5 categories) response is associated with their response to WTGOAL (would you like to weigh more, about the same, or less than you do now: 3 categories.)

The resulting two-way contingency table includes 5 rows and 3 columns. We are interested in evaluating the relationship between the rows and the columns. It's called a two-way table because there are two categorical variables (DIETQUAL and WTGOAL) under study.

If the rows and columns were found to be *independent* of one another, this would mean that the probabilities of falling in each column do not change, regardless of what row of the table we look at.

If the rows and columns are *associated*, then the probabilities of falling in each column do depend on which row we're looking at.

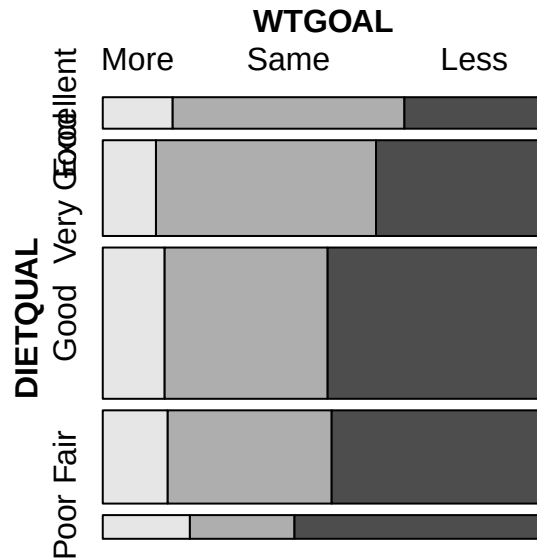
```
dat3 <- nh432 |>
  select(SEQN, DIETQUAL, WTGOAL, WEIGHT, SEX) |>
  filter(WEIGHT < 100 & SEX == "Male") |>
  drop_na()

dat3 |>
  tabyl(DIETQUAL, WTGOAL)
```

DIETQUAL	More	Same	Less
Excellent	16	53	32
Very Good	37	153	117
Good	68	179	238
Fair	44	111	144
Poor	15	18	43

If we want a graphical representation of a two-way table, the most common choice is probably a **mosaic plot**.

```
vcd::mosaic(~ DIETQUAL + WTGOAL, data = dat3,
             highlighting = "WTGOAL")
```

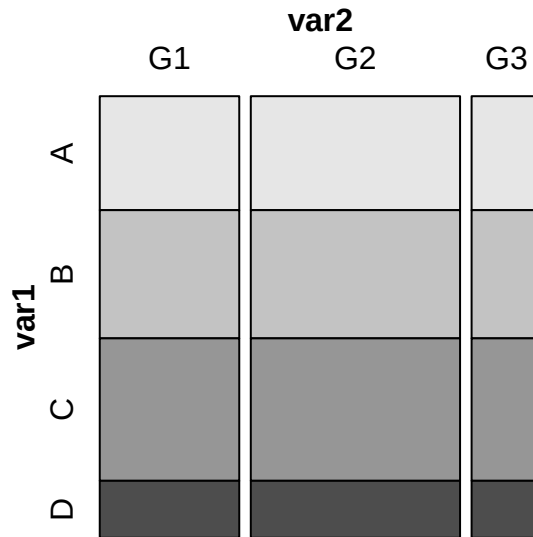


Larger observed frequencies in the contingency table show up with larger tile areas in the in the mosaic plot. So, for instance, we see the larger proportion of “less” WTGOAL in the “Poor” DIETQUAL category, as compared to most of the other DIETQUAL categories.

4.4.1 What would independence look like?

A mosaic plot displaying perfect independence (using simulated data) might look something like this:

```
var1 <- c(rep("A", 48), rep("B", 54), rep("C", 60), rep("D", 24) )
var2 <- c( rep(c("G1", "G1", "G2", "G2", "G2", "G3"), 31) )
temp_tab <- tibble(var1, var2); rm(var1, var2)
vcd::mosaic(~ var1 + var2, data = temp_tab, highlighting = "var1")
```



Here's the table for our simulated data, where independence holds perfectly.

```
xtabs(~ var1 + var2, data = temp_tab)
```

```

      var2
var1 G1 G2 G3
A   16 24  8
B   18 27  9
C   20 30 10
D    8 12  4

```

Note that in these simulated data, we have the same fraction of people in each of the four var1 categories (A, B, C, and D) regardless of which of the three var2 categories (G1, G2 and G3) we are in, and vice versa. That's what it means for rows and columns to be independent.

4.4.2 Back to the DIETQUAL and WTGOAL table

Now, returning to our problem, to obtain detailed results from the Pearson χ^2 test, I use the `xtabs()` function and then the `chisq.test()` function, like this:

```
chi3 <- chisq.test(xtabs(~ DIETQUAL + WTGOAL, data = dat3))

chi3
```

Pearson's Chi-squared test

```
data:  xtabs(~DIETQUAL + WTGOAL, data = dat3)
X-squared = 32.603, df = 8, p-value = 7.261e-05
```

The null hypothesis being tested here is that DIETQUAL and WTGOAL are independent of each other. A small p value like this is indicative of an association between the two variables.

The `chi3` object we have created also contains:

- the observed frequencies in each cell, as well as
- the expected frequencies under the hypothesis of independence of the rows and the columns¹, and
- the Pearson residuals $(\text{observed} - \text{expected})/\sqrt{\text{expected}}$ for each cell, among other things.

```
chi3$observed
```

DIETQUAL	WTGOAL		
	More	Same	Less
Excellent	16	53	32
Very Good	37	153	117
Good	68	179	238
Fair	44	111	144
Poor	15	18	43

```
chi3$expected
```

DIETQUAL	WTGOAL		
	More	Same	Less
Excellent	14.33754	40.94164	45.72082
Very Good	43.58044	124.44637	138.97319
Good	68.84858	196.60095	219.55047
Fair	42.44479	121.20347	135.35174
Poor	10.78864	30.80757	34.40379

¹The expected frequency for a cell under independence is the total for the cell's row multiplied by the total for the cell's column, divided by the grand total for the whole table.

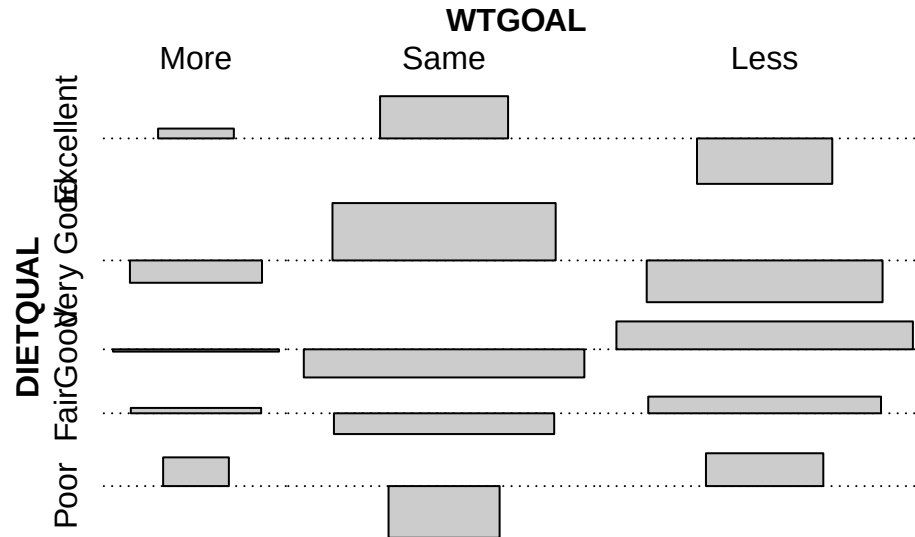
```
chi3$residuals # Pearson residuals
```

	WTGOAL		
DIETQUAL	More	Same	Less
Excellent	0.4390501	1.8845411	-2.0291917
Very Good	-0.9968028	2.5595886	-1.8639211
Good	-0.1022694	-1.2552875	1.2451396
Fair	0.2387127	-0.9268093	0.7433564
Poor	1.2821492	-2.3074805	1.4655618

An **association plot** presents a graphical description of the Pearson residuals, with the area of each box shown proportional to the difference between the observed and expected frequencies.

- If the observed frequency of a cell is greater than the expectation under the hypothesis of independence, then the box rises above the baseline.
 - An example here is the (DIETQUAL = Very Good, WTGOAL = Same) which had an observed frequency of 153 but an expected frequency of 124.4, yielding the largest positive Pearson residual at 2.56.
- Boxes shown below the baseline indicate that the observed frequency was less than the expectation under the independence hypothesis.
 - The largest negative Pearson residual is the (DIETQUAL = Poor, WTGOAL = Same) cell, where we observed 18 observations but the independence model would predict 30.8, yielding a Pearson residual of -2.31.

```
vcd::assoc(~ DIETQUAL + WTGOAL, data = dat3)
```



Some people also like to calculate a correlation between categorical variables. If each of your categorical variables is ordinal (as in this case) then Kendall's tau (version b) is probably the best choice. As with a Pearson correlation for quantities, the value for this measure ranges from -1 to 1, with -1 indicating a strong negative correlation, and +1 a strong positive correlation, with 0 indicating no correlation.

To use this approach, though, we first have to be willing to treat our multi-categorical variables as if they were numeric, which may or may not be reasonable.

```
dat3 <- dat3 |>
  mutate(DIETQUAL_num = as.numeric(DIETQUAL))

dat3 |> tabyl(DIETQUAL_num, DIETQUAL)
```

DIETQUAL_num	Excellent	Very Good	Good	Fair	Poor
1	101	0	0	0	0
2	0	307	0	0	0
3	0	0	485	0	0
4	0	0	0	299	0
5	0	0	0	0	76

```
dat3 <- dat3 |>
  mutate(WTGOAL_num = as.numeric(WTGOAL))

dat3 |> tabyl(WTGOAL_num, WTGOAL)
```

WTGOAL_num	More	Same	Less
1	180	0	0
2	0	514	0
3	0	0	574

```
cor(dat3$DIETQUAL_num, dat3$WTGOAL_num, method = "kendall")
```

```
[1] 0.07193663
```

If you want to obtain a confidence interval for this correlation coefficient, then you would need to use the `KendallTauB()` function from the **DescTools** package.

```
KendallTauB(dat3$DIETQUAL_num, dat3$WTGOAL_num, conf.level = 0.90)
```

tau_b	lwr.ci	upr.ci
0.07193663	0.03147130	0.11240196

Again, it's just a number, and not especially valuable.

4.5 PHQ9 Category and Race/Ethnicity

Let's look next at the association of race-ethnicity (`RACEETH`, which has 5 levels) and the depression category (minimal, mild, moderate, moderately severe, or severe) available in `PHQ9_CAT`, which we derived from the PHQ-9 depression screener score. We'll restrict this small analysis to NHANES participants who did not receive care from a mental health provider (so `MENTALH` is 0) in the last 12 months.

```
temp <- nh432 |>
  select(SEQN, RACEETH, PHQ9_CAT, MENTALH) |>
  filter(MENTALH == 0) |>
  drop_na()
```

So here's our first attempt at a 5x5 table describing this association.

```
temp |>
  tabyl(RACEETH, PHQ9_CAT)
```

	RACEETH	minimal	mild	moderate	moderately	severe	severe
Non-H	White	770	151	41		20	8
Non-H	Black	668	143	34		13	1
	Hispanic	581	133	36		12	5
Non-H	Asian	428	49	11		1	1
	Other Race	105	34	12		5	1

We note some very small observed frequencies, especially in the bottom right of the table. Should we try to run a Pearson χ^2 test on these results, we will generate a warning that the Chi-square approximation may be incorrect.

```
xtabs( ~ RACEETH + PHQ9_CAT, data = temp ) |>
  chisq.test()
```

```
Warning in stats::chisq.test(x, y, ...): Chi-squared approximation may be
incorrect
```

Pearson's Chi-squared test

```
data:  xtabs(~RACEETH + PHQ9_CAT, data = temp)
X-squared = 49.288, df = 16, p-value = 2.974e-05
```

4.5.1 The Cochran conditions

R sets off this warning when the “Cochran conditions” are not met. The Cochran conditions require that we have:

- no cells with 0 counts
- at least 80% of the cells in our table with counts of 5 or higher
- expected counts in each cell of the table should be 5 or more

In our table, we have four cells with observed counts below 5 (all have count 1) and two more with observed counts of exactly 5. If we look at the expected frequencies under the hypothesis of independence, what do we see?

```
temp_chi <- xtabs( ~ RACEETH + PHQ9_CAT, data = temp ) |>
  chisq.test()
```

Warning in stats::chisq.test(x, y, ...): Chi-squared approximation may be incorrect

```
temp_chi$expected
```

		PHQ9_CAT				
RACEETH		minimal	mild	moderate	moderately severe	severe
Non-H White		774.2813	154.73491	40.655838	15.473491	4.8544284
Non-H Black		671.8259	134.25988	35.276126	13.425988	4.2120748
Hispanic		599.8725	119.88048	31.498008	11.988048	3.7609562
Non-H Asian		383.2302	76.58596	20.122587	7.658596	2.4026969
Other Race		122.7901	24.53877	6.447441	2.453877	0.7698437

Every cell in the “severe” category has an expected frequency below 5, and we also have some generally small counts, in the Non-Hispanic Asian and Other Race categories, as well as the “moderately severe” category.

4.5.2 Collapsing Categories

So what might we do about this?

Let us consider two approaches that we’ll use simultaneously:

1. drop two of the RACEETH groups, and just use the top 3 (Non-H White, Non-H Black and Hispanic) using `filter()`
2. collapse together the two right-most levels of PHQ9_CAT (moderately severe and severe) into a new level which I’ll call “More Severe”, using `fct_lump_n()`

```
dat5 <- nh432 |>
  select(SEQN, RACEETH, PHQ9_CAT, MENTALH) |>
  filter(MENTALH == 0) |>
  filter(RACEETH %in% c("Non-H White", "Non-H Black", "Hispanic")) |>
  drop_na() |>
  mutate(RACEETH = fct_drop(RACEETH),
         PHQ9_CAT = fct_lump_n(PHQ9_CAT, 3,
                               other_level = "More Severe"))

dat5 |>
  tabyl(RACEETH, PHQ9_CAT)
```


	RACEETH	minimal	mild	moderate	More	Severe
Non-H	White	770	151	41		28
Non-H	Black	668	143	34		14
	Hispanic	581	133	36		17

Now, we have at least 14 participants in every cell of the table.

4.5.3 Pearson χ^2 Analysis

Now, let's consider what the Pearson χ^2 test suggests.

```
tab5 <- xtabs(~ RACEETH + PHQ9_CAT, data = dat5)
tab5
```

		PHQ9_CAT				
RACEETH		minimal	mild	moderate	More	Severe
Non-H	White	770	151	41		28
Non-H	Black	668	143	34		14
	Hispanic	581	133	36		17

```
chisq.test(tab5)
```

Pearson's Chi-squared test

```
data:  tab5
X-squared = 5.0157, df = 6, p-value = 0.5418
```

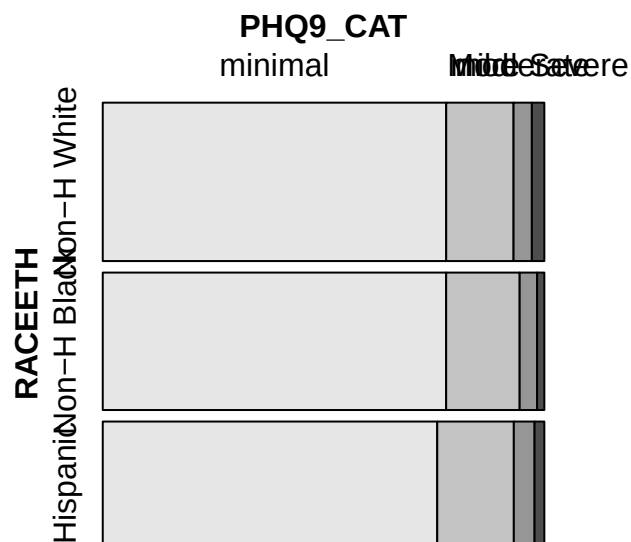
Now we have no warning, and notice also how large a change this has meant in terms of the p-value, as compared to our original χ^2 result.

4.5.4 Mosaic Plot

Here's a mosaic plot² of the table.

²The **ggmosaic** package has a `geom_mosaic()` tool for building such plots but its maintenance has been spotty in recent weeks.

```
vcd::mosaic(tab5, highlighting = "PHQ9_CAT")
```



4.5.5 Examining the Fit

We'll finish up with a look at the expected frequencies, and a table and association plot of the Pearson residuals.

```
chisq.test(tab5)$observed
```

	PHQ9_CAT			
RACEETH	minimal	mild	moderate	More Severe
Non-H White	770	151	41	28
Non-H Black	668	143	34	14
Hispanic	581	133	36	17

```
chisq.test(tab5)$expected
```

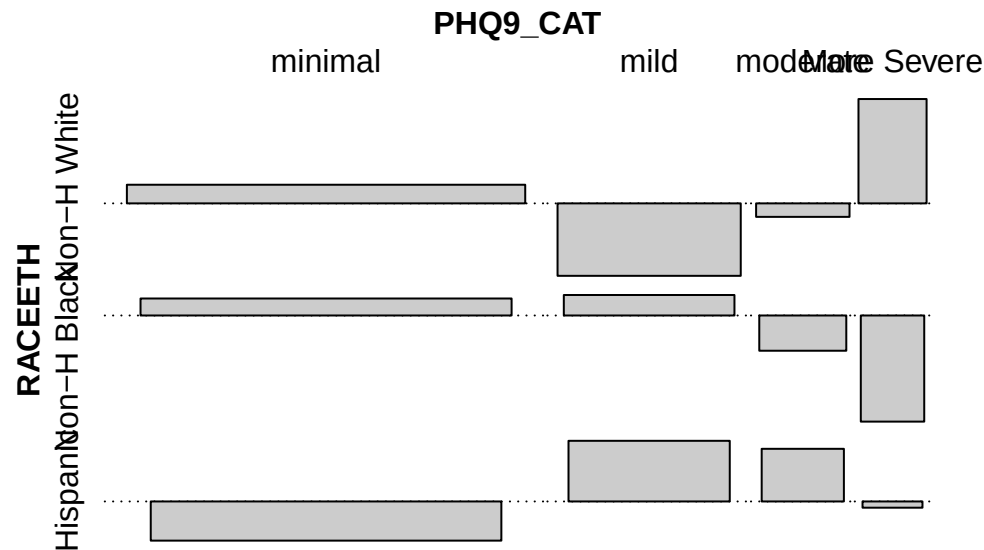
	PHQ9_CAT			
RACEETH	minimal	mild	moderate	More Severe
Non-H White	764.0711	161.5940	42.00688	22.32798

Non-H Black	662.9667	140.2114	36.44839	19.37347
Hispanic	591.9622	125.1946	32.54472	17.29855

```
chisq.test(tab5)$residuals
```

RACEETH	PHQ9_CAT			
	minimal	mild	moderate	More Severe
Non-H White	0.21449006	-0.83339100	-0.15535235	1.20036381
Non-H Black	0.19548040	0.23550271	-0.40554793	-1.22081874
Hispanic	-0.45055624	0.69759600	0.60567876	-0.07178083

```
assoc(tab5)
```



5 431 Review: Fitting Linear Models

In this Chapter, we will review some key issues about comparing proportions or rates, mostly drawn from the 431 course. This and the other Chapters labeled “431 Review” show elements of the two studies involved in a “[431 Project B](#)” using the pre-pandemic (2017 - March 2020) data from the National Health and Nutrition Examination Survey (NHANES) called **nh432** that we developed in Chapter 1 and then summarized in Chapter 2. Note that we’ll ignore the issue of sampling weights and assume a “missing completely at random” (MCAR) mechanism in these “431 Review” chapters, so that we can work with complete cases.

The 431 course book is found at <https://thomaselove.github.io/431-book/> and will remain there until June 1.

5.1 R Setup

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(car)
library(GGally)
library(gt)
library(Hmisc)
library(knitr)
library(patchwork)

library(tidyverse)

theme_set(theme_bw())
```

5.1.1 Data Load

```
nh432 <- read_rds("data/nh432.Rds")
```

5.2 Modeling Weekend Sleep Hours

In this example, we'll try to build an effective model to predict our outcome: average weekend hours of sleep (SLPWKEND) on the basis of four predictors:

- average weekday hours of sleep (SLPWKDAY)
- systolic blood pressure (SBP)
- PHQ-9 depression screener score (PHQ9), and
- whether or not the participant has mentioned trouble sleeping to a physician (SLPTROUB)

We'll compare a model using all four of these predictors to a model using just the two directly related to sleep (SLPWKDAY and SLPTROUB), and we'll restrict our analysis to those participants whose self-reported overall health (SROH) was "Good".

```
dat1 <- nh432 |>
  select(SEQN, SLPWKEND, SLPWKDAY, SBP, PHQ9, SLPTROUB, SROH) |>
  filter(SROH == "Good") |>
  drop_na()

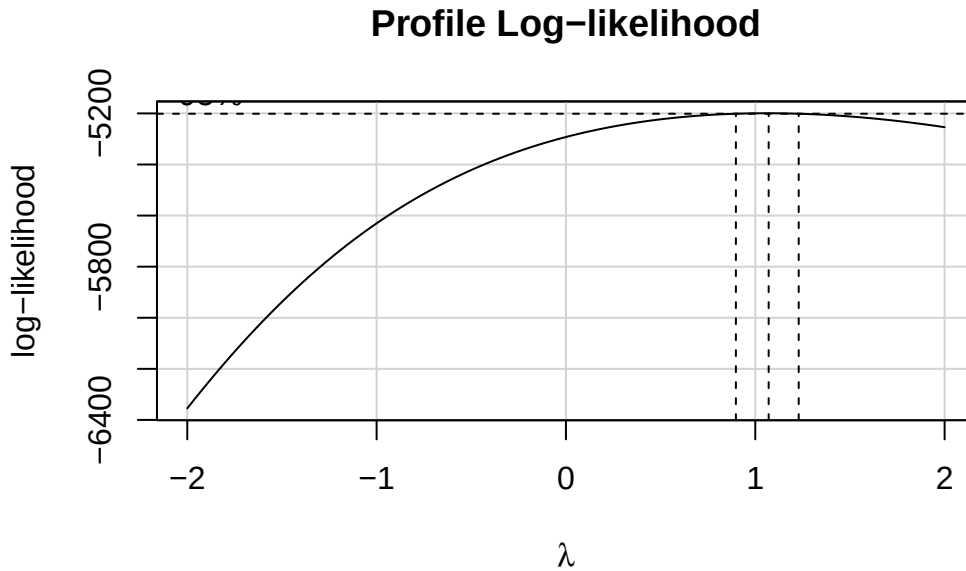
dat1
```

```
# A tibble: 1,293 x 7
   SEQN  SLPWKEND SLPWKDAY  SBP  PHQ9 SLPTROUB SROH
   <chr>    <dbl>    <dbl> <dbl> <int>    <dbl> <fct>
1 109273      8      6.5  110   15      1 Good
2 109293     6.5     7.5  130    3      0 Good
3 109295      7      7    161    0      1 Good
4 109305     6.5      6    125    0      0 Good
5 109307     11     7.5  114    0      0 Good
6 109315      5      5    123    1      0 Good
7 109336      8      4    148    1      1 Good
8 109342      8     6.5   107   16      1 Good
9 109365     9.5     9.5   133    7      0 Good
10 109378      9      9    133    0      0 Good
# i 1,283 more rows
```

5.2.1 Should we transform our outcome?

We can develop a Box-Cox plot to help us choose between potential transformations of our outcome, so as to improve the adherence to regression assumptions. To do so, we first fit our larger model.

```
m1 <- lm(SLPWKEND ~ SLPWKDAY + SBP + PHQ9 + SLPTROUB, data = dat1)
boxCox(m1)
```



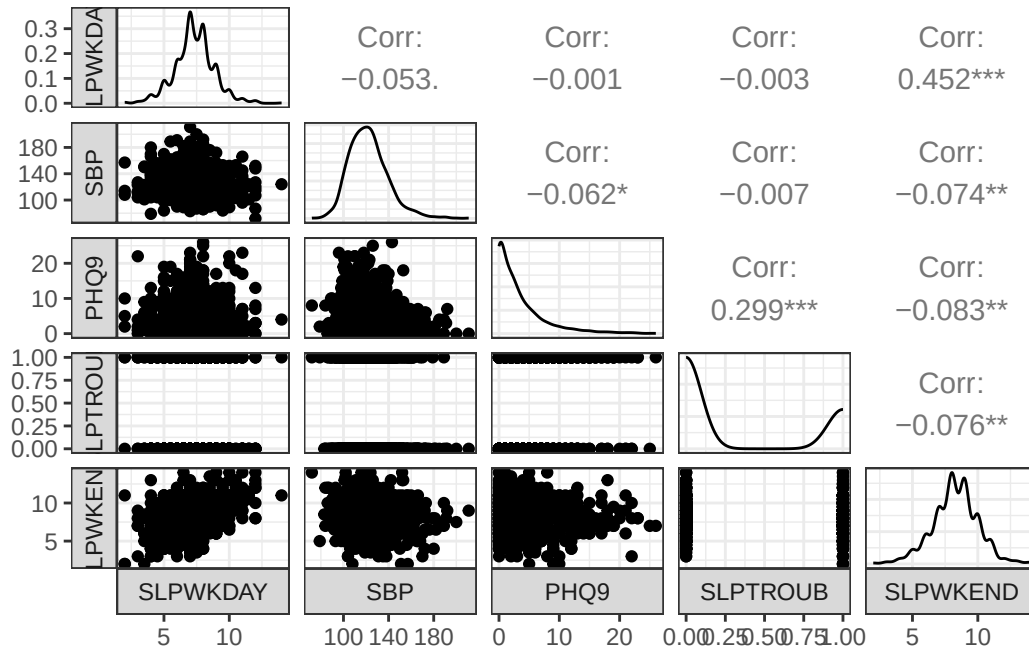
The Box-Cox suggested set of transformations centers near $\lambda = 1$. As we saw back in Chapter 3, The **ladder of power transformations** looks like this:

λ	Transformation	Formula
-2	inverse square	$1/y^2$
-1	inverse	$1/y$
-0.5	inverse square root	$1/\sqrt{y}$
0	logarithm	$\log y$
0.5	square root	$\sqrt{y}$
1	no transformation	y
2	square	y^2
3	cube	y^3

So, in this case, the Box-Cox approach (again, with λ near 1) suggests that we leave the existing SLPWKEND outcome alone.

5.2.2 Scatterplot Matrix

```
ggpairs(dat1, columns = c(3:6, 2), switch = "both",
        lower=list(combo=wrap("facethist", bins=25)))
```



- The reason I included column 2 (our outcome: SLPWKEND) last in this plot is so that the bottom row would include each of our predictors plotted on the X (horizontal) axis against the outcome on the Y (vertical) axis, next to a density plot of the outcome.
- I also switched the locations of the facet labels on both the x and y axis from their defaults, so that the labels are to the left and below the plots, since I find that a bit easier to work with.
- The `lower` business is to avoid getting a warning about binwidths.
- The binary variable (SLPTROUB) is included here as a 1-0 numeric variable, rather than a factor, which is why the scatterplot matrix looks as it does, rather than creating a series of boxplots (as we'll see when we work with a factor later.)

5.2.3 Collinearity?

In any multiple regression setting, two or more of the predictors might be highly correlated with one another, and this is referred to as multicollinearity or just collinearity. If we have a serious problem with collinearity, this can cause several problems, including difficulty fitting and interpreting the resulting model.

Is collinearity a serious concern in our situation? Looking at the scatterplot matrix, we see that the largest observed correlation between two predictors is between PHQ9 and SLPTROUB. Does that rise to the level of a problem?

I usually use the `vif()` function from the **car** package to help make this decision. The *variance inflation factor* (or VIF) measures how much the variance of a regression coefficient is inflated due to collinearity in the model. The smallest possible VIF value is 1, and VIFs near 1, as we'll see here, indicate no problems with collinearity worth worrying about.

```
vif(m1)
```

```
SLPWKDAY      SBP      PHQ9 SLPTROUB
1.002807 1.006778 1.102341 1.098187
```

Should we see a VIF (or generalized VIF, which is produced by the `vif()` function when we have factor variables in the model) above, say, 5, that would be an indication that the model would be improved by not including the variable that exhibits collinearity. Here, we have no such issues, and will proceed to fit the model including all of these predictors.

5.2.4 Fitting and Displaying Model m1

Here are the coefficients obtained from fitting the model `m1`.

```
m1 <- lm(SLPWKEND ~ SLPWKDAY + SBP + PHQ9 + SLPTROUB, data = dat1)
m1
```

Call:

```
lm(formula = SLPWKEND ~ SLPWKDAY + SBP + PHQ9 + SLPTROUB, data = dat1)
```

Coefficients:

(Intercept)	SLPWKDAY	SBP	PHQ9	SLPTROUB
5.26266	0.52661	-0.00560	-0.02945	-0.20813

If Harry and Sally have the same values of SLPWKDAY, SBP and SLPTRoub, but Harry's PHQ9 is one point higher than Sally's, then model `m1` predicts that Harry will sleep 0.029 hours less than Sally on the weekends.

A summary of the regression model `m1` provides lots of useful information about the parameters (including their standard errors) and the quality of fit (at least as measured by R^2 and adjusted R^2 .)

```
summary(m1)
```

Call:

```
lm(formula = SLPWKEND ~ SLPWKDAY + SBP + PHQ9 + SLPTRoub, data = dat1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.1173	-0.9609	-0.1005	0.9248	6.3659

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.262657	0.388427	13.549	< 2e-16 ***
SLPWKDAY	0.526612	0.028950	18.190	< 2e-16 ***
SBP	-0.005600	0.002515	-2.227	0.02613 *
PHQ9	-0.029450	0.010955	-2.688	0.00727 **
SLPTRoub	-0.208129	0.098758	-2.107	0.03527 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.554 on 1288 degrees of freedom

Multiple R-squared: 0.2171, Adjusted R-squared: 0.2147

F-statistic: 89.3 on 4 and 1288 DF, p-value: < 2.2e-16

5.2.5 Using broom functions on Model `m1`

If we want to actually use the information in the model summary elsewhere, we use the `tidy()` and `glance()` functions from the **broom** package to help us.

```
tidy(m1, conf.int = TRUE, conf.level = 0.90)
```

A tibble: 5 x 7

term	estimate	std.error	statistic	p.value	conf.low	conf.high
------	----------	-----------	-----------	---------	----------	-----------

	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	5.26	0.388	13.5	3.41e-39	4.62	5.90
2	SLPWKDAY	0.527	0.0290	18.2	5.52e-66	0.479	0.574
3	SBP	-0.00560	0.00251	-2.23	2.61e- 2	-0.00974	-0.00146
4	PHQ9	-0.0294	0.0110	-2.69	7.27e- 3	-0.0475	-0.0114
5	SLPTROUB	-0.208	0.0988	-2.11	3.53e- 2	-0.371	-0.0456

We can produce a neater version of the `tidy()` output, produced using the `gt()` function, but I will just show the code here, since it causes trouble when I build a PDF of these Notes.

```
tidy(m1, conf.int = TRUE, conf.level = 0.90) |>
  mutate(across(.cols = -c(term), \(x) round_half_up(x, digits = 3))) |>
  gt()
```

Note that none of the 90% confidence intervals here cross zero. This just means that we have a pretty good handle on the direction of effects - for example, our estimate for the slope of SLPWKDAY is positive, suggesting that people who sleep more during the week also sleep more on the weekend, after accounting for SBP, PHQ9 and SLPPTROUB.

```
glance(m1)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
1   0.217        0.215  1.55     89.3 4.93e-67     4 -2402. 4817. 4848.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Model `m1` shows an R^2 value of 0.217, which means that 21.7% of the variation in our outcome SLPWKEND is accounted for by the model using SLPWKDAY, SBP, PHQ9 and SLPPTROUB.

The adjusted R^2 value isn't a percentage or proportion of anything, but it is a handy index when comparing two models fit to the same outcome for the same observations. It penalizes the raw R^2 value for models that require more coefficients to be fit. If the raw R^2 is much larger than the adjusted R^2 value, this is also an indication that the model may be "overfit" - capitalizing on noise in the data more than we'd like, so that the amount of signal in the predictors may be overstated by raw R^2 .

Here's a cleaner presentation of some of the more important elements in the `glance()` output:

```
glance(m1) |>
  select(r2 = r.squared, adjr2 = adj.r.squared,
         AIC, BIC, sigma, nobs, df, df.res = df.residual) |>
  kable(digits = c(3, 3, 1, 1, 2, 0, 0, 0))
```

r2	adjr2	AIC	BIC	sigma	nobs	df	df.res
0.217	0.215	4816.6	4847.6	1.55	1293	4	1288

- AIC and BIC are measures used to compare models for the same outcome using the same data, so we'll see those again when we fit a second model to these data. In those comparisons, smaller values of AIC and BIC indicate better fitting models.
- nobs is the number of observations used to actually fit our model `m1`,
- df indicates the number of degrees of freedom used by the model, and represents the number of estimated coefficients fit, while
- $\text{df.res} = \text{nobs} - \text{df} - 1 = \text{residual degrees of freedom}$.

5.2.6 Residual Plots for Model `m1`

The key assumptions for a linear regression model include:

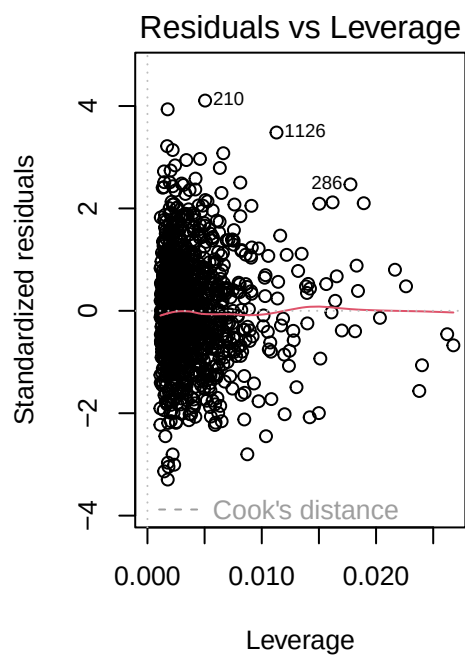
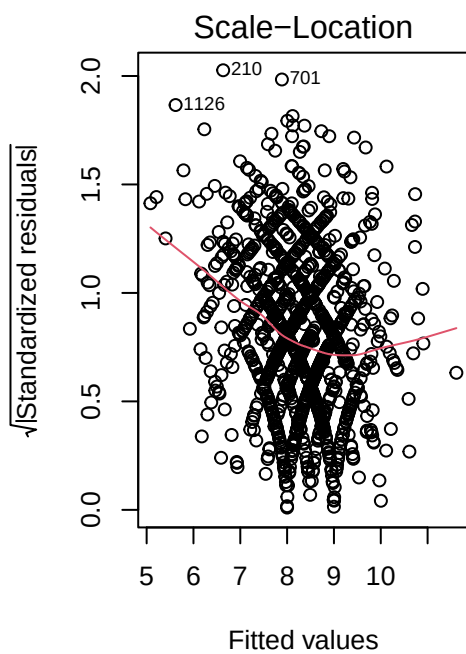
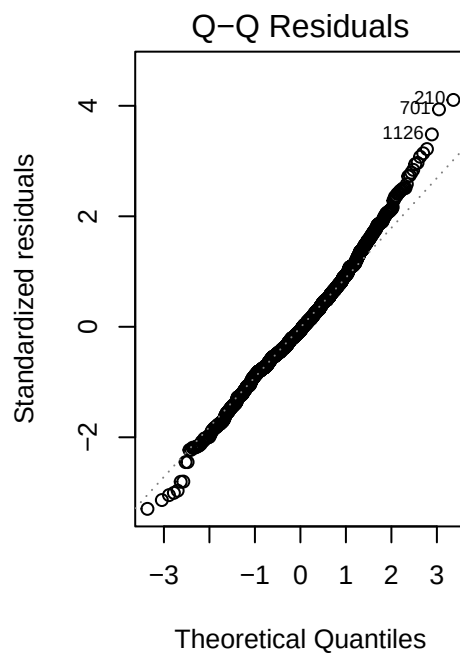
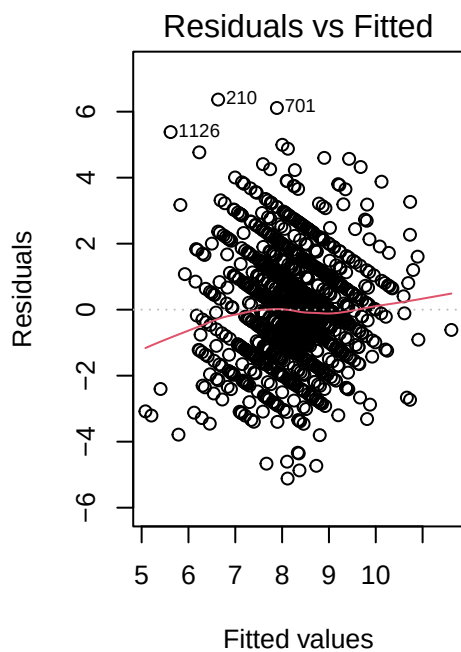
1. Linearity of the association under study
2. Normality of the residuals
3. Constant Variance (Homoscedasticity)
4. Independence (not an issue with cross-sectional data like this)

A residual for a point in a regression model is just the observed value of our outcome (here, `SLPWKEND`) minus the value predicted by the model based on the predictor values (also called the fitted value.)

The four key plots that R will generate for you to help assess these results are shown below for model `m1`.

```
## I used
## #/ fig.height: 8
## at the top of this code chunk
## to make the plots tall enough to see well

par(mfrow = c(2,2)); plot(m1); par(mfrow = c(1,1))
```



5.2.6.1 Residuals vs. Fitted values

The top left plot (Residuals vs. Fitted Values) helps us to assess the linearity and constant variance assumptions.

- We want to see a “fuzzy football” shape.
- A clear curve is indicative of a problem with linearity, and suggests that perhaps a transformation of the outcome (or perhaps one or more predictors) may be in order
- A fan shape, with much more variation at one end of the fitted values (left or right) than the other indicates a problem with the constant variance assumption, and again a transformation may be needed.

The diagonal lines we see in the Residuals vs. Fitted plot are the result of the fact that both the outcome (SLPWKEND) and a key predictor (SLPWKDAY5) aren’t really continuous in the data, as most of the responses to those questions were either integers, or used 0.5 as the fraction. So those two variables are more discrete than we might have expected.

5.2.6.2 Normal Q-Q plot of standardized residuals

The top right plot (Normal Q-Q) is a Normal Q-Q plot of the standardized regression residuals for our model `m1`. Substantial issues with skew (a curve in the plot) or a major problem with outliers (as indicated by a reverse S-shape) indicate potential concerns with the Normality assumption. Since the y-axis here shows standardized residuals, we can also assess whether what we’re seeing is especially surprising relative to our expectations for any standardized values (for example, we should see values above +3 or below -3 approximately 3 times in 1000).

- Remember that this plot represents `nobs = 1293` residuals, so a few values near 3 in absolute value aren’t surprising.
- We’re looking for big deviations from Normality here.
- The `plot()` function in R will always identify three of the cases, by default, in these four plots.

Suppose we wanted to look at the data for case 210, identified by these plots as a potential outlier, or at least a poorly fit point.

```
dat1_aug <- augment(m1, data = dat1)

dat1_aug |> slice(210) |>
  select(SEQN, SLPWKEND, .fitted, .resid, .std.resid, everything())
```

```
# A tibble: 1 x 13
  SEQN   SLPWKEND .fitted .resid .std.resid SLPWKDAY   SBP  PHQ9 SLPTROUB SROH
  <chr>    <dbl>   <dbl> <dbl>    <dbl>   <dbl> <dbl> <int>    <dbl> <fct>
1 111688     13    6.63  6.37     4.11     4   126     1      0 Good
# i 3 more variables: .hat <dbl>, .sigma <dbl>, .cooksad <dbl>
```

We see that row 210 contains SEQN 111688, whose

- observed SLPWKEND is 13
- fitted SLPWKEND is 6.63
- yielding a residual of 6.37,
- or a standardized residual of 4.11

We can use the `outlierTest()` function in the **car** package to help assess whether this value is unusual enough to merit more careful consideration. This function actually works with the studentized residual, which is similar to the standardized residual we saw above. Here, this point (SEQN 111688) is fit poorly enough to be flagged by the Bonferroni outlier test as a mean-shift outlier.

```
outlierTest(m1)
```

```
      rstudent unadjusted p-value Bonferroni p
210 4.131972      3.8289e-05    0.049508
```

Having seen that, though, I’m going to essentially ignore it for the moment, and press on to the rest of our residual analysis.

5.2.6.3 Scale-Location plot

The bottom left plot in our set of four residual plots is the Scale-Location plot, which presents the square root of the standardized residuals against the fitted values. This plot provides another check on the “equal variance” assumption - if the plot shows a clear trend either up or down as we move left to right, then that indicates an issue with constant variance. While a loess smooth is provided (red curve) to help guide our thinking, it’s important not to get too excited about small changes or changes associated with small numbers of observations.

You’ll also note the presence of curves (in particular, little “V” shapes) formed by the points of the plot. Again, this is caused by the discrete nature of the outcome (and one of the key predictors) and wouldn’t be evident if our outcome was more continuous.

Despite the drop in the red loess smooth as fitted values move from 5 to about 8, I don’t see much of a pattern here to indicate trouble with non-constant variance.

5.2.6.4 Residuals vs. Leverage plot

The bottom-left plot is a plot of residuals vs. leverage, with influence contours.

Highly leveraged points have unusual combinations of predictor values.

Highly influential points have a big impact on the model, in that the model's coefficients or quality of fit would change markedly were those points to be removed from the model. To measure influence, we combine leverage and residuals together, with a measure like Cook's distance.

- To look for points with substantial **leverage** on the model by virtue of having unusual values of the predictors - look for points whose leverage is at least 3 times as large as the average leverage value.
- The average leverage is always k/n , where k is the number of coefficients fit by the model (including the slopes and intercept), and n is the number of observations in the model.
- To obtain the leverage values, the `augment()` function stores them in `.hat`.
- To look for points with substantial **influence** on the model, that is, removing them from the model would change it substantially, consider the Cook's distance, plotted in contours here.
- Any Cook's distance point > 1 will likely have a substantial impact on the model.
- Any points with Cook's distance > 0.5 , if any, will be indicated in the bottom-right (Residuals vs. Leverage) plot, and are worthy of investigation.
- In model `m1`, we have no points with values of Cook's distance > 0.5 . To obtain the Cook's distance values for each point, use the `augment()` function, which stores them in `.cooks`.

Here, for example, we identify the points with largest leverage and with largest Cook's distance, across the points used to fit `m1`.

```
dat1_aug <- augment(m1, data = dat1)
```

```
dat1_aug |> slice_max(.hat) |>  
  select(SEQN, .hat, .resid, .fitted, .cooks, everything())
```

```
# A tibble: 1 x 13
```

	SEQN	.hat	.resid	.fitted	.cooks	SLPWKEND	SLPWKDAY	SBP	PHQ9	SLPTROUB
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>
1	123474	0.0268	-1.03	8.03	0.00250	7	8	126	25	0

```
# i 3 more variables: SROH <fct>, .sigma <dbl>, .std.resid <dbl>
```

```
dat1_aug |> slice_max(.cooks) |>  
  select(SEQN, .hat, .resid, .fitted, .cooks, everything())
```

```
# A tibble: 1 x 13
  SEQN      .hat .resid .fitted .cooksd SLPWKEND SLPWKDAY   SBP  PHQ9 SLPTROUB
  <chr>   <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <dbl> <int>   <dbl>
1 122894 0.0113  5.38    5.62  0.0277     11       2  114    2      0
# i 3 more variables: SROH <fct>, .sigma <dbl>, .std.resid <dbl>
```

It turns out that SEQN 123474 has the largest value of leverage (`.hat`) and SEQN 122894 has the largest value of influence (`.cooksd`) in our model. We will worry about `.cooksd` values above 0.5, but the largest value in this model is much smaller than that, so I think we're OK for now.

5.2.7 Fitting and Displaying Model `m2`

We will now move on to compare the results of this model (`m1`) to a smaller model.

Our second model, `m2` is a subset of `m1`, including only the two predictors directly related to sleep, `SLPWKDAY` and `SLPTROUB`.

```
m2 <- lm(SLPWKEND ~ SLPWKDAY + SLPTROUB, data = dat1)

m2
```

Call:

```
lm(formula = SLPWKEND ~ SLPWKDAY + SLPTROUB, data = dat1)
```

Coefficients:

(Intercept)	SLPWKDAY	SLPTROUB
4.4813	0.5301	-0.2862

Note that the slopes of both `SLPWKDAY` and `SLPTROUB` have changed from model `m1` (although not very much), and that the intercept has changed more substantially.

5.2.8 Using broom functions on `m2`

```
tidy(m2, conf.int = TRUE, conf.level = 0.90) |>
  kable(digits = 3)
```


term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	4.481	0.219	20.475	0.000	4.121	4.842
SLPWKDAY	0.530	0.029	18.268	0.000	0.482	0.578
SLPTROUB	-0.286	0.095	-3.025	0.003	-0.442	-0.130

```
glance(m2) |>
  select(r2 = r.squared, adjr2 = adj.r.squared,
         AIC, BIC, sigma, nobs, df, df.res = df.residual) |>
  kable(digits = c(3, 3, 1, 1, 2, 0, 0, 0))
```

r2	adjr2	AIC	BIC	sigma	nobs	df	df.res
0.21	0.209	4824.1	4844.8	1.56	1293	2	1290

Since we want to compare the fit of `m1` to that of `m2`, we probably want to do so in a single table, like this:

```
temp1 <- glance(m1) |>
  select(r2 = r.squared, adjr2 = adj.r.squared,
         AIC, BIC, sigma, nobs, df, df.res = df.residual) |>
  mutate(model = "m1") |>
  relocate(model)

temp2 <- glance(m2) |>
  select(r2 = r.squared, adjr2 = adj.r.squared,
         AIC, BIC, sigma, nobs, df, df.res = df.residual) |>
  mutate(model = "m2") |>
  relocate(model)

bind_rows(temp1, temp2) |>
  kable(digits = c(0, 3, 3, 1, 1, 2, 1, 0, 0, 0))
```

model	r2	adjr2	AIC	BIC	sigma	nobs	df	df.res
m1	0.217	0.215	4816.6	4847.6	1.55	1293	4	1288
m2	0.210	0.209	4824.1	4844.8	1.56	1293	2	1290

Each model uses the same number of observations to predict the same outcome (SLPWKEND). So we can compare them directly. As compared to model `m2`, model `m1` has:

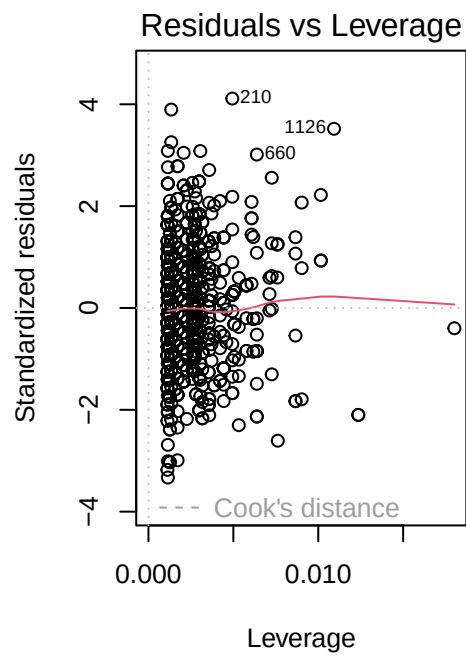
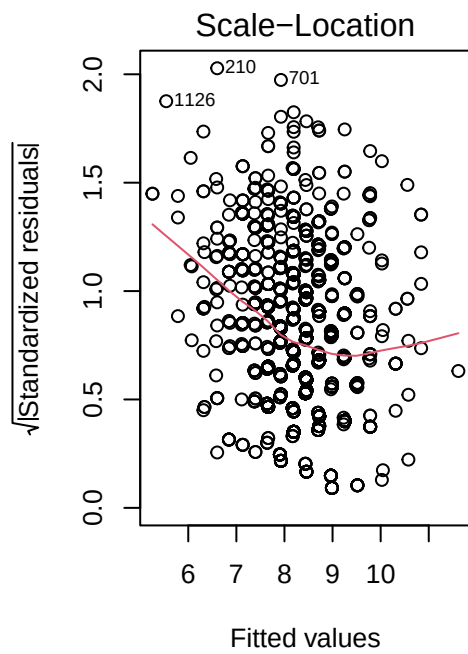
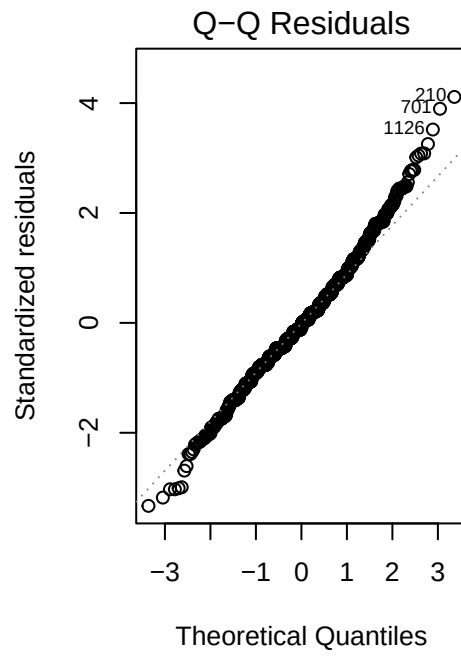
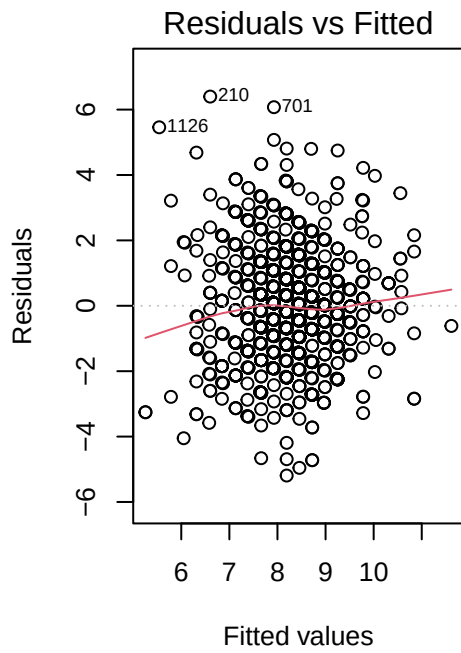
- the larger R^2 (as it must, since model `m2` includes a subset of the predictors in model `m1`),
- the larger adjusted R^2 ,
- the smaller AIC (Akaike Information Criterion: smaller values are better),
- the larger BIC (Bayes Information Criterion: again, smaller values are better),
- and the smaller residual standard error (σ) (smaller values are better.)

The key realizations for these data are that the AIC, adjusted R^2 and σ results favor model `m1` while the BIC favors model `m2`.

5.2.9 Residual Plots for Model `m2`

```
## I used #/ fig.height: 8 in this code chunk
## to make the plots tall enough to see well

par(mfrow = c(2,2)); plot(m2); par(mfrow = c(1,1))
```



The residual plots here show (even more starkly than in model `m1`) the discrete nature of our outcome and the two variables we're using to predict it. I see no especially serious problems with the assumptions of linearity or constant variance here, and while there are still some fairly poorly fit values, there are no highly influential points, so I'll accept these residual plots as indicative of a fairly reasonable model on the whole.

5.2.10 Conclusions

Three of our four in-sample measures of fit quality (AIC, σ and adjusted R^2) favor the larger model `m1` over `m2`, but there's not a lot to choose from here. Neither model showed important problems with regression assumptions, so I would probably wind up choosing `m1` based on the analyses we've done in this Chapter.

However, a more appropriate strategy for prediction assessment would be to partition the data into separate samples for model training (the development or building sample) and model testing. We adopt such a model validation strategy in our next little study.

5.3 Modeling High-Sensitivity C-Reactive Protein

In this, our second linear modeling example, we will try to predict High-Sensitivity C-Reactive Protein levels (HSCRCP) on the basis of these three predictor variables:

- the participant's mean pulse rate, specifically the mean of the two gathered pulse rates, `PULSE1` and `PULSE2`
- the participant's self-reported overall health (`SROH`, which is an ordinal factor with levels Excellent, Very Good, Good, Fair and Poor)
- `HOSPITAL`, a 1-0 binary variable indicating whether or not the participant was hospitalized in the past year.

In this case, we'll use all NHANES participants with complete data on the relevant variables to fit the three-predictor model, and then a second model using mean pulse rate alone.

```
dat2 <- nh432 |>
  select(SEQN, HSCRCP, PULSE1, PULSE2, SROH, HOSPITAL) |>
  drop_na() |>
  mutate(MEANPULSE = 0.5*(PULSE1 + PULSE2))

glimpse(dat2)
```

```

Rows: 3,117
Columns: 7
$ SEQN      <chr> "109271", "109273", "109291", "109292", "109293", "109295", ~
$ HSCRIP    <dbl> 28.68, 0.98, 5.31, 3.08, 15.10, 6.28, 0.56, 1.45, 0.32, 0.86~
$ PULSE1    <dbl> 73, 71, 77, 93, 62, 93, 74, 59, 66, 83, 64, 55, 54, 63, 68, ~
$ PULSE2    <dbl> 71, 70, 76, 91, 64, 93, 74, 58, 64, 87, 68, 55, 54, 63, 70, ~
$ SROH      <fct> Fair, Good, Fair, Very Good, Good, Good, Very Good, Excellen~
$ HOSPITAL   <dbl> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ MEANPULSE <dbl> 72.0, 70.5, 76.5, 92.0, 63.0, 93.0, 74.0, 58.5, 65.0, 85.0, ~

```

5.3.1 Partitioning the Data

Before partitioning, it's always a good idea to be sure that the number of rows in the tibble matches the number of distinct (unique) values in the identifier column.

```
identical(nrow(dat2), n_distinct(dat2 |> select(SEQN)))
```

```
[1] TRUE
```

OK. Now, be sure to set a seed so that we can replicate the selection. We'll put 70% of the data in our training sample, setting aside the remaining 30% for the test sample.

```

set.seed(432005)

dat2_train <- slice_sample(dat2, prop = 0.70)

dat2_test <- anti_join(dat2, dat2_train, by = "SEQN")

c(nrow(dat2), nrow(dat2_train), nrow(dat2_test))

```

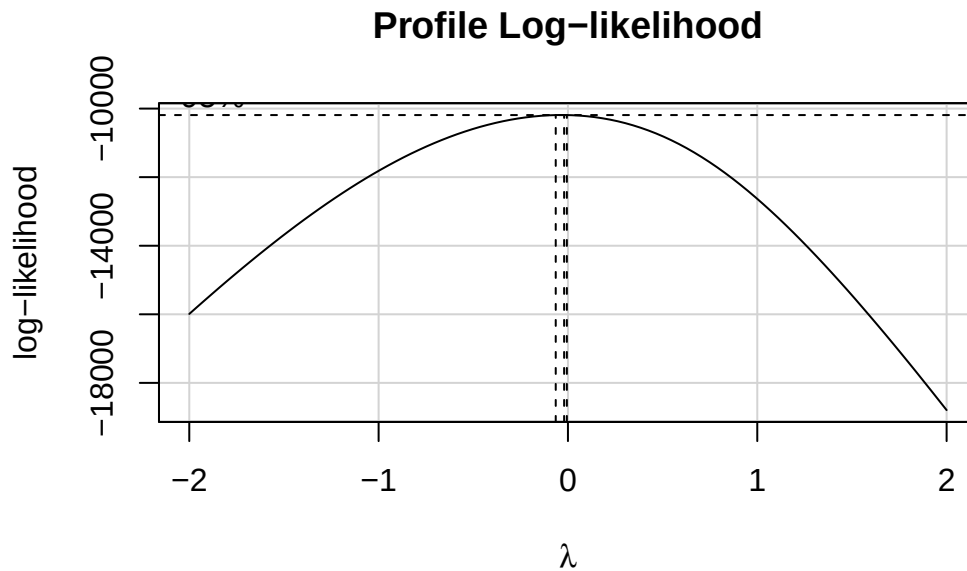
```
[1] 3117 2181 936
```

In what follows, we'll work with the `dat2_train` sample, and set aside the `dat2_test` sample for a while.

5.3.2 Transforming the Outcome?

Let's use the Box-Cox approach to help us think about which potential transformations of our outcome might be helpful, within our training sample.

```
m_temp <- lm(HSCRIP ~ MEANPULSE + SROH + HOSPITAL, data = dat2_train)
boxCox(m_temp)
```

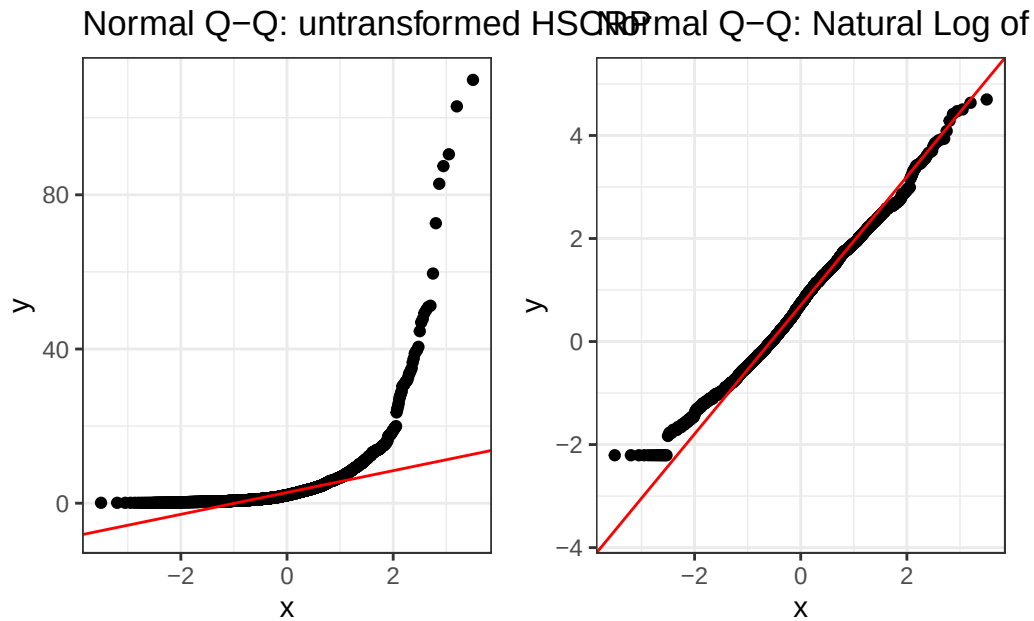


The estimated λ value is very close to 0, which according to the ladder of power transformations, suggests we take the logarithm of our outcome, so as to improve the residual plots for the model. This will also, as it turns out, lead to a much less right-skewed outcome variable.

```
p1 <- ggplot(dat2_train, aes(sample = HSCRIP)) +
  geom_qq() + geom_qq_line(col = "red") +
  labs(title = "Normal Q-Q: untransformed HSCRIP")

p2 <- ggplot(dat2_train, aes(sample = log(HSCRIP))) +
  geom_qq() + geom_qq_line(col = "red") +
  labs(title = "Normal Q-Q: Natural Log of HSCRIP")

p1 + p2
```



Clearly, one benefit of the transformation is some improvement in the Normality of our outcome's distribution.

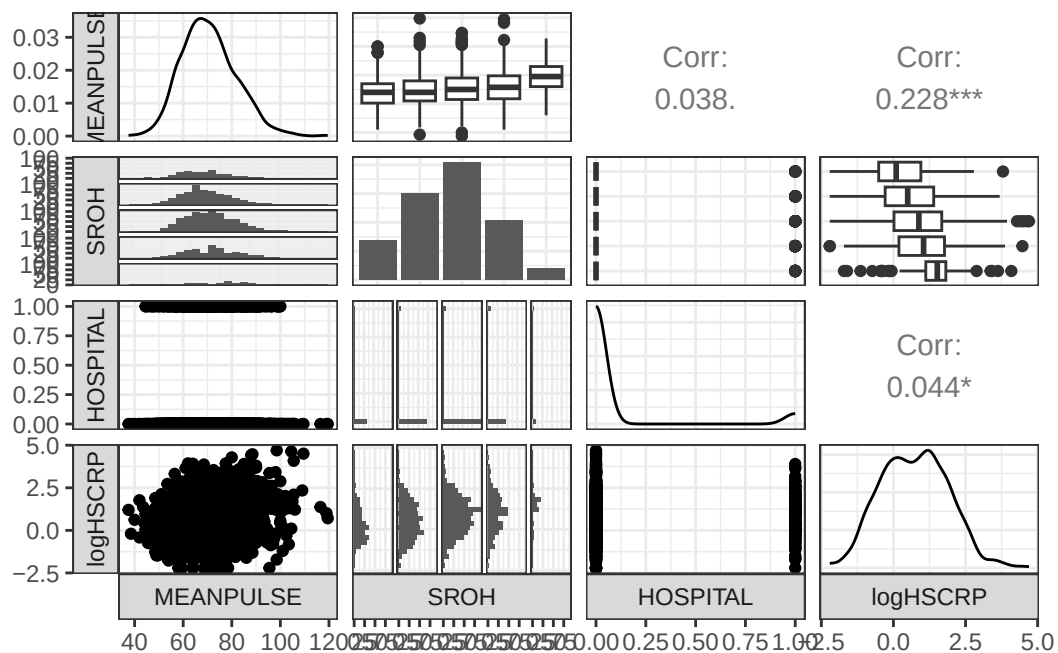
5.3.3 Scatterplot Matrix and Collinearity

To build the relevant scatterplot matrix with our transformed outcome, I'll create a variable containing the result of the transformation within our training sample.

```
dat2_train <- dat2_train |>
  mutate(logHSCRP = log(HSCRP))
names(dat2_train)
```

```
[1] "SEQN"      "HSCRP"      "PULSE1"     "PULSE2"     "SROH"      "HOSPITAL"
[7] "MEANPULSE" "logHSCRP"
```

```
ggpairs(dat2_train, columns = c(7,5,6,8), switch = "both",
  lower=list(combo=wrap("facethist", bins=25)))
```



As a collinearity check, we'll run `vif()` from the **car** package here.

```
m3 <- lm(log(HSCRIP) ~ MEANPULSE + SROH + HOSPITAL,
         data = dat2_train)

vif(m3)
```

	GVIF	Df	GVIF^(1/(2*Df))
MEANPULSE	1.036382	1	1.018028
SROH	1.063545	4	1.007731
HOSPITAL	1.027860	1	1.013834

Again, no signs of meaningful collinearity. Note the presentation of the factor variable `SROH` in the scatterplot matrix, and in the generalized VIF output.

5.3.4 Fit Model `m3`

```
m3 <- lm(log(HSCRIP) ~ MEANPULSE + SROH + HOSPITAL,
         data = dat2_train)
```



```
tidy(m3, conf.int = TRUE, conf.level = 0.90) |>
  kable(digits = 3)
```

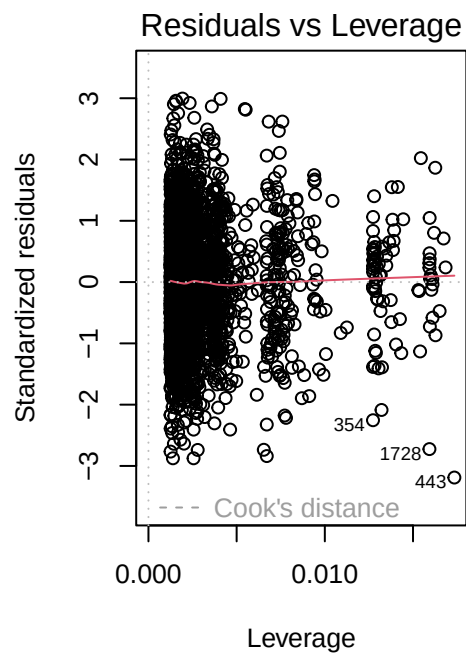
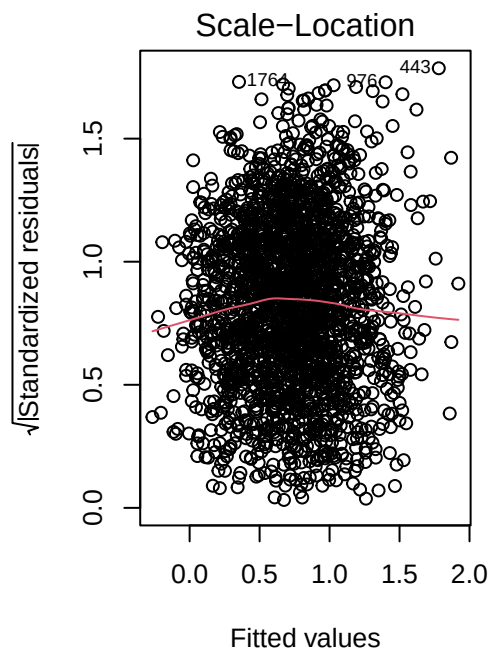
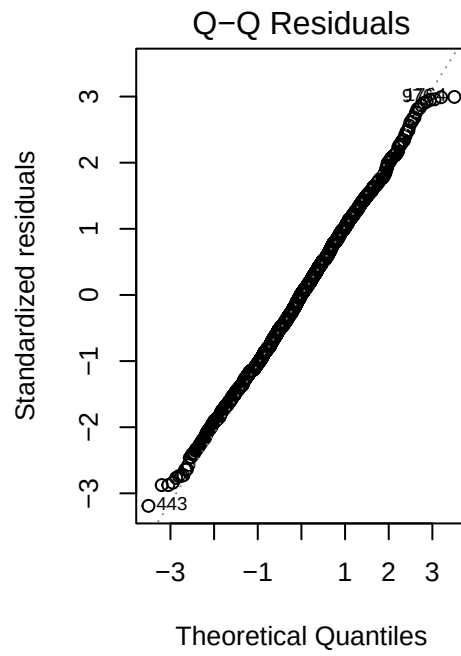
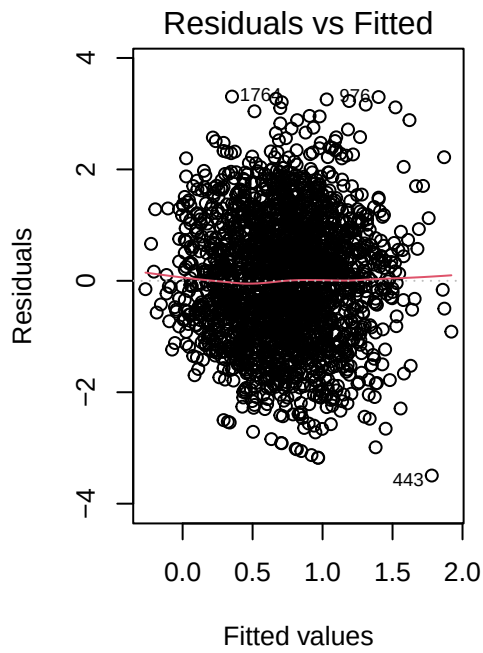
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-1.108	0.160	-6.914	0.000	-1.372	-0.845
MEANPULSE	0.020	0.002	9.359	0.000	0.017	0.024
SROHVery Good	0.237	0.081	2.939	0.003	0.104	0.370
SROHGood	0.532	0.078	6.857	0.000	0.404	0.660
SROHFair	0.641	0.087	7.353	0.000	0.497	0.784
SROHPoor	0.860	0.143	6.020	0.000	0.625	1.095
HOSPITAL	0.052	0.088	0.594	0.552	-0.092	0.197

- If Harry and Sally have the same values of HOSPITAL and MEANPULSE, but Harry's SROH is "Very Good" while Sally's is "Excellent", then model `m3` predicts that Harry will have a $\log(\text{HSCR})$ that is 0.237 (90% CI: 0.104, 0.370) larger than Sally's $\log(\text{HSCR})$.
- On the other hand, if Harry and Gary have the same values of HOSPITAL and MEANPULSE, but Harry's SROH remains "Very Good" while Gary's is only "Good", then model `m3` predicts that *Gary* will have a $\log(\text{HSCR})$ that is $(0.532 - 0.237 = 0.295)$ larger than Harry's $\log(\text{HSCR})$.

5.3.5 Residual Plots for `m3`

```
## don't forget to use #/ fig.height: 8
## to make the residual plots taller

par(mfrow = c(2,2)); plot(m3); par(mfrow = c(1,1))
```



I see no serious concerns with regression assumptions here. The residuals vs. fitted plot shows no signs of meaningful non-linearity or heteroscedasticity. The standardized residuals in the Normal Q-Q plot follow the reference line closely. There is no clear trend in the scale-location plot, and the residuals vs. leverage plot reveals no particularly influential points.

5.3.6 Fit Model m4

Let's now fit the simple regression model, `m4`, with only `MEANPULSE` as a predictor of the log of `HSCRCP`.

```
m4 <- lm(log(HSCRCP) ~ MEANPULSE,
          data = dat2_train)
```

Now, let's look at the tidied coefficients.

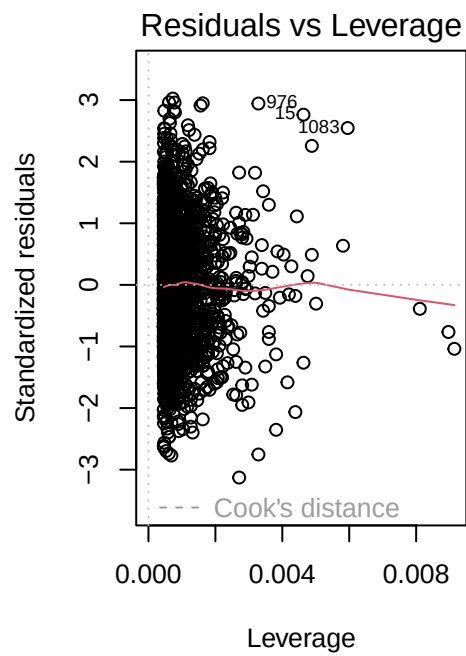
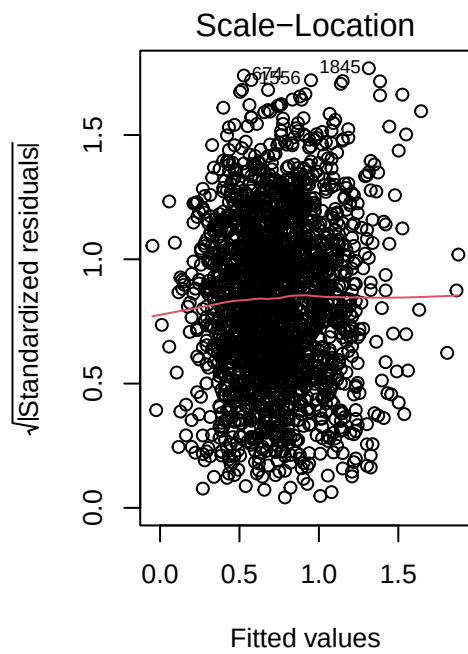
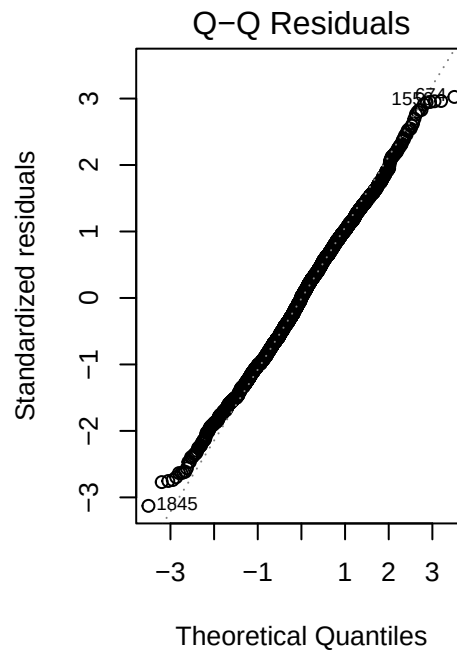
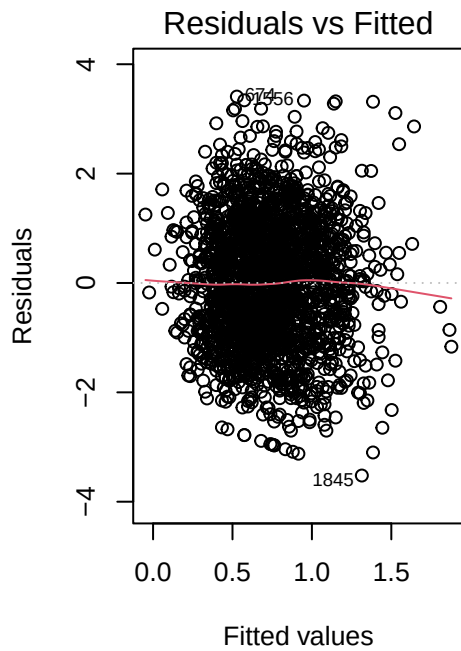
```
tidy(m4, conf.int = TRUE, conf.level = 0.90) |>
  kable(digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.929	0.154	-6.048	0	-1.181	-0.676
MEAN- PULSE	0.023	0.002	10.934	0	0.020	0.027

- If Harry's mean pulse rate is one beat per minute higher than Sally's, then model `m4` predicts that the logarithm of Harry's `HSCRCP` will be 0.023 higher than Sally's, with 90% CI (0.020, 0.027).
- Note that if Harry's mean pulse rate is *ten* beats per minute higher than Sally's, then model `m4` predicts that the logarithm of Harry's `HSCRCP` will be 0.23 higher than Sally's, with 90% CI (0.20, 0.27).

5.3.7 Residual Plots for m4

```
par(mfrow = c(2,2)); plot(m4); par(mfrow = c(1,1))
```



5.3.8 In-Sample Fit Quality Comparison (m3 vs. m4)

```
g3 <- glance(m3) |>
  select(r2 = r.squared, adjr2 = adj.r.squared,
         AIC, BIC, sigma, nobs, df, df.res = df.residual) |>
  mutate(model = "m3") |>
  relocate(model)

g4 <- glance(m4) |>
  select(r2 = r.squared, adjr2 = adj.r.squared,
         AIC, BIC, sigma, nobs, df, df.res = df.residual) |>
  mutate(model = "m4") |>
  relocate(model)

bind_rows(g3, g4) |> kable()
```

model	r2	adjr2	AIC	BIC	sigma	nobs	df	df.res
m3	0.0907079	0.0881984	6636.024	6681.525	1.105532	2181	6	2174
m4	0.0520082	0.0515731	6716.928	6733.990	1.127517	2181	1	2179

The larger model (model **m3**) has better results than model **m4** in the sense that it produces a larger adjusted R^2 , and smaller values for AIC, BIC and σ . Based on this comparison within the training sample, we clearly prefer **m3**, since each model shows reasonable adherence to the assumptions of a linear regression model.

5.3.9 Testing the models in new data

At last we return to the `dat2_test` sample which was not used in fitting models **m3** and **m4** to investigate which of these models has better predictive results in new data. When doing this sort of testing, I recommend a look at the following 4 summaries, each of which is based on the fitted (predicted) and observed values of our outcome in our new data, using the models we want to compare:

- squared correlation of predicted with observed values (validated R^2 ; higher values are better)
- mean absolute prediction error (MAPE; smaller values indicate smaller errors, hence better prediction)
- square root of the mean squared prediction error (RMSPE; again, smaller values indicate better predictions)

- maximum (in absolute value) prediction error (Max Error)

To obtain observed, predicted, and error (observed - predicted) values for each new data point when we apply model `m3`, we first use the `augment()` function from the **broom** package to obtain our `.fitted` values.

```
m3_test_aug <- augment(m3, newdata = dat2_test)
head(m3_test_aug)
```

```
# A tibble: 6 x 9
  SEQN   HSCR PULSE1 PULSE2 SROH   HOSPITAL MEANPULSE .fitted .resid
  <chr> <dbl> <dbl> <dbl> <fct>    <dbl>    <dbl>    <dbl> <dbl>
1 109273 0.98    71    70 Good      0      70.5    0.839 -0.859
2 109293 15.1    62    64 Good      0      63     0.688  2.03
3 109312 0.86    83    87 Very Good  0      85     0.835 -0.985
4 109332 2.29    63    63 Excellent 0      63     0.156  0.673
5 109340 4.64    78    78 Fair      0      78     1.10   0.437
6 109342 5.51    72    70 Good      0      71     0.849  0.858
```

Remember, however, that our models `m3` and `m4` do not predict HSCR, but rather the logarithm of HSCR, so we need to exponentiate the `.fitted` values to get what we want.

```
m3_test_aug <- augment(m3, newdata = dat2_test) |>
  mutate(fits = exp(.fitted),
         resid = HSCR - fits) |>
  select(SEQN, HSCR, fits, resid, everything())

head(m3_test_aug)
```

```
# A tibble: 6 x 11
  SEQN   HSCR fits resid PULSE1 PULSE2 SROH   HOSPITAL MEANPULSE .fitted .resid
  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <fct>    <dbl>    <dbl>    <dbl> <dbl>
1 109273 0.98  2.31 -1.33    71    70 Good      0      70.5    0.839 -
0.859
2 109293 15.1  1.99 13.1    62    64 Good      0      63     0.688  2.03
3 109312 0.86  2.30 -1.44    83    87 Very~    0      85     0.835 -
0.985
4 109332 2.29  1.17  1.12    63    63 Exce~    0      63     0.156  0.673
5 109340 4.64  3.00  1.64    78    78 Fair     0      78     1.10   0.437
6 109342 5.51  2.34  3.17    72    70 Good     0      71     0.849  0.858
```

Now, we can obtain our summaries, as follows.

```
m3_test_results <- m3_test_aug |>
  summarize(validated_R_sq = cor(HSCR_P, fits)^2,
            MAPE = mean(abs(resid)),
            RMSPE = sqrt(mean(resid^2)),
            max_Error = max(abs(resid)))

m3_test_results
```

```
# A tibble: 1 x 4
  validated_R_sq MAPE RMSPE max_Error
      <dbl> <dbl> <dbl>      <dbl>
1      0.114  3.45  10.7      177.
```

For model m4, we have:

```
m4_test_aug <- augment(m4, newdata = dat2_test) |>
  mutate(fits = exp(.fitted),
         resid = HSCR_P - fits) |>
  select(SEQN, HSCR_P, fits, resid, everything())

m4_test_results <- m4_test_aug |>
  summarize(validated_R_sq = cor(HSCR_P, fits)^2,
            MAPE = mean(abs(resid)),
            RMSPE = sqrt(mean(resid^2)),
            max_Error = max(abs(resid)))

m4_test_results
```

```
# A tibble: 1 x 4
  validated_R_sq MAPE RMSPE max_Error
      <dbl> <dbl> <dbl>      <dbl>
1      0.102  3.50  10.8      177.
```

And we can put the two sets of results together into a nice table.

```
bind_rows(m3_test_results, m4_test_results) |>
  mutate(model = c("m3", "m4")) |>
  relocate(model) |>
  kable()
```

model	validated_R_sq	MAPE	RMSPE	max_Error
m3	0.1135858	3.451295	10.72894	176.9855
m4	0.1015280	3.496673	10.79613	177.0716

Based on these out-of-sample validation results, it seems that Model **m3** has the better results across each of these four summaries than Model 4 does.

5.3.10 Conclusions

We fit two models to predict HSCRP, a larger model (**m3**) containing three predictors (MEANPULSE, SROH and HOSPITAL), and a smaller model (**m4**) containing only the MEANPULSE as a predictor.

- Both models (after transforming to $\log(\text{HSCRP})$ for our outcome) seem to generally meet the assumptions of linear regression
- Model **m3** had a raw R^2 value of 0.091, so it accounted for about 9.1% of the variation in $\log(\text{HSCRP})$ within our training sample. Model **m4** accounted for 5.2%.
- In our in-sample checks, Model **m3** had better results in terms of adjusted R^2 , AIC, BIC and σ .
- In a validation (test) sample, our Model **m3** also showed superior predictive performance, including better results in terms of MAPE, RMSPE and maximum absolute error, as well as a validated R^2 of 11.4%, higher than model **m4**'s result of 10.2%.

Overall, model **m3** seems like the meaningfully better choice.

6 BRFSS SMART Data

The Centers for Disease Control analyzes Behavioral Risk Factor Surveillance System (BRFSS) survey data for specific metropolitan and micropolitan statistical areas (MMSAs) in a program called the Selected Metropolitan/Micropolitan Area Risk Trends of BRFSS (SMART BRFSS.)

In this work, we will focus on data from the 2017 SMART, and in particular on data from the state of Ohio, and from the Cleveland-Elyria, OH, Metropolitan Statistical Area. The purpose of this survey is to provide localized health information that can help public health practitioners identify local emerging health problems, plan and evaluate local responses, and efficiently allocate resources to specific needs.

In this chapter, I describe some cleaning of the BRFSS SMART data, and break it out into national, statewide, and local samples.

The data files produced by this chapter include:

- `smart_ohio.Rds` which includes data on approximately 100 variables for over 7000 subjects in six MMSAs that are at least partially located in the state of Ohio.
- `smart_cle.Rds` which includes data on those same variables for a little over 1000 subjects in the Cleveland-Elyria-Lorain OH MMSA.

6.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(Hmisc)
library(patchwork)
library(tidyverse)

theme_set(theme_bw())
```

6.2 Key resources

- the “raw” data, in the form of the 2017 SMART BRFSS MMSA Data, found in a zipped [SAS Transport Format](#) file. The data were released in October 2018.
- the [MMSA Variable Layout](#) which simply lists the variables included in the data file
- the [Calculated Variables PDF](#) which describes the risk factors by data variable names - there is also an [online summary matrix of these calculated variables](#).
- the lengthy [2017 Survey Questions PDF](#) which lists all questions asked as part of the BRFSS in 2017
- the enormous [Codebook for the 2017 BRFSS Survey PDF](#) which identifies the variables by name for us.

Also, for each subject, we are also provided with a sampling weight, in `_MMSAWT`, which will help us incorporate the sampling design later. These weights are at the MMSA level, and are used for generating MMSA-level estimates for variables in the data set. [Details on the weighting methodology are available at this PDF](#).

6.3 Ingesting the Raw Data

To create the data files we’ll use, I used the `read_xpt` function from the `haven` package to bring in the SAS XPT data file that is provided by CDC. The codes I used (but won’t use in these Notes) were:

```
smart_raw <- read_xpt("MMSA2017/MMSA2017.xpt")
```

This gives the nationwide data, which has 230,875 rows and 177 columns.

But for the purposes of putting these Notes online, I needed to crank down the sample size enormously. To that end, I created a new data file, which I developed by

- importing the MMSA2017.xpt file as above
- filtering away all observations except those from MMSAs which include Ohio in their name, and
- saving the result, which now has 7,412 rows and 177 columns.

The code (again, not run here) that I used to filter to the OH-based MMSAs was:

```
smart_ohio_raw <- smart_raw |>
  filter(str_detect(MMSANAME, "OH"))

write_csv(smart_ohio_raw, "data/smart_ohio_raw.csv")
```

So, for purposes of these notes, our complete data set is actually coming from `smart_ohio_raw.csv` and consists only of the 7,412 observations associated with the six MMSAs that include Ohio in their names.

6.4 Ingesting from our CSV file

Note that the `smart_ohio_raw.csv` and other data files we're developing in this Chapter are available on our [432-Data website](#)

```
smart_ohio_raw <- read_csv("data/smart_ohio_raw.csv", show_col_types = FALSE)

dim(smart_ohio_raw)
```

```
[1] 7412 177
```

6.5 What does the raw data look like?

Here is a list of all variable names included in this file. We're not going to use all of those variables, but this will give you a sense of what is available.

```
names(smart_ohio_raw)
```

```
[1] "DISPCODE" "STATERE1" "SAFETIME" "HHADULT" "GENHLTH" "PHYSHLTH"
[7] "MENTHLTH" "POORHLTH" "HLTHPLN1" "PERSDOC2" "MEDCOST" "CHECKUP1"
[13] "BPHIGH4" "BPMEDS" "CHOLCHK1" "TOLDHI2" "CHOLMED1" "CVDINFR4"
[19] "CVDCRHD4" "CVDSTRK3" "ASTHMA3" "ASTHNOW" "CHCSCNCR" "CHCOCNCR"
[25] "CHCCOPD1" "HAVARTH3" "ADDEPEV2" "CHCKIDNY" "DIABETE3" "DIABAGE2"
[31] "LMTJOIN3" "ARTHDIS2" "ARTHSOCL" "JOINPAI1" "SEX" "MARITAL"
[37] "EDUCA" "RENTHOM1" "NUMMHOL2" "NUMPHON2" "CPDEMO1A" "VETERAN3"
[43] "EMPLOY1" "CHILDREN" "INCOME2" "INTERNET" "WEIGHT2" "HEIGHT3"
[49] "PREGNANT" "DEAF" "BLIND" "DECIDE" "DIFFWALK" "DIFFDRES"
[55] "DIFFALON" "SMOKE100" "SMOKDAY2" "STOPSMK2" "LASTSMK2" "USENOW3"
[61] "ECIGARET" "ECIGNOW" "ALCDAY5" "AVEDRNK2" "DRNK3GE5" "MAXDRNKS"
[67] "FRUIT2" "FRUITJU2" "FVGREEN1" "FRENCHF1" "POTATOE1" "VEGETAB2"
[73] "EXERANY2" "EXTRACT11" "EXEROFT1" "EXERHMM1" "EXTRACT21" "EXEROFT2"
[79] "EXERHMM2" "STRENGTH" "SEATBELT" "FLUSHOT6" "FLSHTMY2" "PNEUVAC3"
[85] "SHINGLE2" "HIVTST6" "HIVTSTD3" "HIVRISK5" "CASTHDX2" "CASTHNO2"
[91] "CALLBCKZ" "WDUSENOW" "WDINFTRK" "WDHOWOFT" "WDSHARE" "NAMTRIBE"
[97] "NAMOTHR" "_URBNRRL" "_STSTR" "_IMPSEX" "_RFHLTH" "_PHYS14D"
```

```

[103] "_MENT14D" "_HCVU651" "_RFHYPE5" "_CHOLCH1" "_RFCHOL1" "_MICH1"
[109] "_LTASTH1" "_CASTHM1" "_ASTHMS1" "_DRDXAR1" "_LMTACT1" "_LMTWRK1"
[115] "_LMTSCL1" "_PRACE1" "_MRACE1" "_HISPANC" "_RACE" "_RACEG21"
[121] "_RACEGR3" "_AGEG5YR" "_AGE65YR" "_AGE80" "_AGE_G" "WTKG3"
[127] "_BMI5" "_BMI5CAT" "_RFBMI5" "_EDUCAG" "_INCOMG" "_SMOKER3"
[133] "_RFSMOK3" "_ECIGSTS" "_CURECIG" "DRNKANY5" "_RFBING5" "_DRNKWEK"
[139] "_RFRHRV5" "FTJUDA2_" "FRUTDA2_" "GREND1_" "FRNCHDA_" "POTADA1_"
[145] "VEGEDA2_" "_MISFRT1" "_MISVEG1" "_FRTRES1" "_VEGRES1" "_FRUTSU1"
[151] "_VEGESU1" "_FRTL1A" "_VEGLT1A" "_FRT16A" "_VEG23A" "_FRUTE1"
[157] "_VEGETE1" "_TOTINDA" "_MINAC11" "_MINAC21" "_PACAT1" "_PAINDX1"
[163] "_PA15OR2" "_PA30OR2" "_PA30O21" "_PASTRNG" "_PAREC1" "_PASTAE1"
[169] "_RFSEAT2" "_RFSEAT3" "_FLSHOT6" "_PNEUMO2" "_AIDTST3" "_MMSA"
[175] "_MMSAWT" "SEQNO" "MMSANAME"

```

6.6 Cleaning the BRFSS Data

6.6.1 Identifying Information

The identifying variables for each subject are gathered in `SEQNO`, which I'll leave alone.

- Each statistical (geographic) area is identified by a `_MMSA` variable, which I'll rename `mmsa_code`, and by an `MMSANAME` which I'll rename as `mmsa_name`
- For each subject, we are also provided with a sampling weight, in `_MMSAWT`, which will help us incorporate the sampling design later in the semester. We'll rename this as `mmsa_wt`. Details on the weighting methodology are available at https://www.cdc.gov/brfss/annual_data/2017/pdf/2017_SMART_BRFSS_MMSA_Methodology-508.pdf

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(mmsa_code = `_MMSA`,
         mmsa_name = `MMSANAME`,
         mmsa_wt = `_MMSAWT`)

smart_ohio_raw |> count(mmsa_code, mmsa_name)

```

A tibble: 6 x 3

	mmsa_code	mmsa_name	n
	<dbl>	<chr>	<int>
1	17140	Cincinnati, OH-KY-IN, Metropolitan Statistical Area	1737
2	17460	Cleveland-Elyria, OH, Metropolitan Statistical Area	1133
3	18140	Columbus, OH, Metropolitan Statistical Area	2033

4	19380 Dayton, OH, Metropolitan Statistical Area	587
5	26580 Huntington-Ashland, WV-KY-OH, Metropolitan Statistical Area	1156
6	45780 Toledo, OH, Metropolitan Statistical Area	766

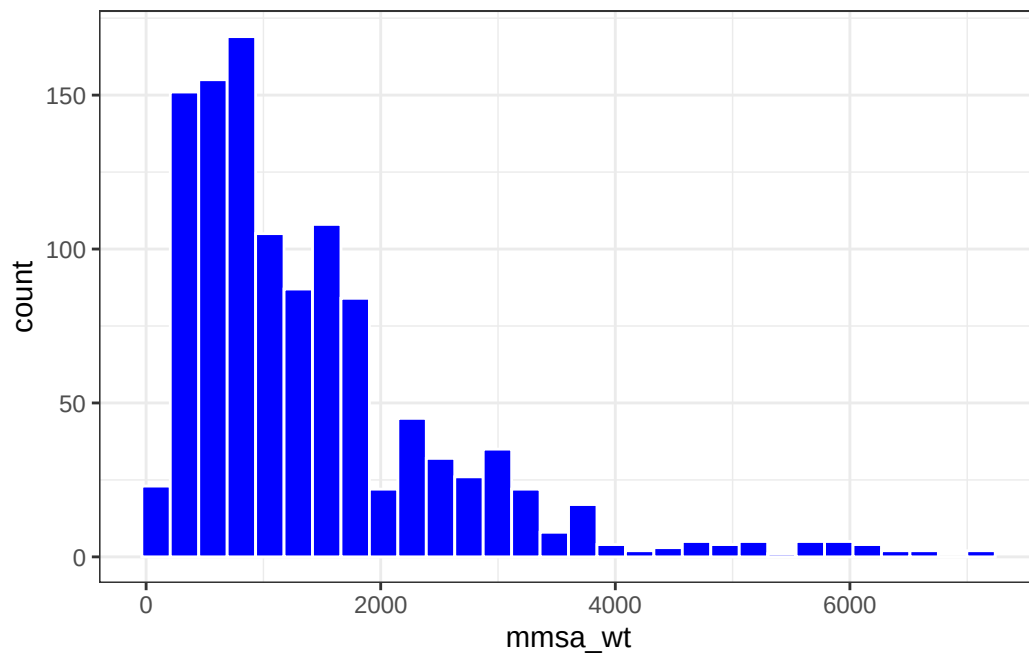
Those names are very long. I'll build some shorter ones, by dropping everything after the comma.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(mmsa = str_replace_all(string = mmsa_name, pattern="\\,.*$", replacement=" "))
smart_ohio_raw |> count(mmsa, mmsa_name)
```

```
# A tibble: 6 x 3
  mmsa          mmsa_name          n
<chr>         <chr>         <int>
1 "Cincinnati " Cincinnati, OH-KY-IN, Metropolitan Statistical Ar~ 1737
2 "Cleveland-Elyria " Cleveland-Elyria, OH, Metropolitan Statistical Ar~ 1133
3 "Columbus " Columbus, OH, Metropolitan Statistical Area 2033
4 "Dayton " Dayton, OH, Metropolitan Statistical Area 587
5 "Huntington-Ashland " Huntington-Ashland, WV-KY-OH, Metropolitan Statis~ 1156
6 "Toledo " Toledo, OH, Metropolitan Statistical Area 766
```

And here are the sampling weights for the subjects in the Cleveland-Elyria MSA.

```
smart_ohio_raw |>
  filter(mmsa_code == 17460) %>%
  ggplot(., aes(x = mmsa_wt)) +
  geom_histogram(bins = 30, fill = "blue", col = "white")
```



6.6.2 Survey Method

6.6.2.1 DISPCODE and its cleanup to completed

DISPCODE which is 1100 if the subject completed the interview, and 1200 if they partially completed the interview. We'll create a variable called `completed` that indicates (1 = complete, 0 = not) whether the subject completed the interview.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(completed = 12 - (DISPCODE/100))

smart_ohio_raw |> count(DISPCODE, completed)
```

```
# A tibble: 2 x 3
  DISPCODE completed     n
  <dbl>      <dbl> <int>
1    1100         1  6277
2    1200         0  1135
```

6.6.2.2 STATERE1 and SAFETIME and their reduction to landline

BRFSSS is conducted by telephone. The next two variables help us understand whether the subject was contacted via land line or via cellular phone.

- **STATERE1** is 1 if the subject is a resident of the state (only asked of people in the land line version of the survey).
- **SAFETIME** is 1 if this is a safe time to talk (only asked of people in the cell phone version of the survey).
- We'll use **STATERE1** and **SAFETIME** to create an indicator variable **landline** that specifies how the respondent was surveyed (1 = land line, 0 = cell phone), as follows...

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(landline = replace_na(STATERE1, 0))

smart_ohio_raw |> count(STATERE1, SAFETIME, landline)
```

```
# A tibble: 2 x 4
  STATERE1 SAFETIME landline     n
    <dbl>    <dbl>    <dbl> <int>
1         1      NA         1  3649
2        NA         1         0  3763
```

6.6.2.3 HHADULT and its cleanup to hhadults

- **HHADULT** is the response to “How many members of your household, including yourself, are 18 years of age or older?”
 - The permitted responses range from 1-76, with special values 77 for Don't Know/Not Sure and 99 for refused, with BLANK for missing or not asked.
 - So we should change all numerical values above 76 to NA for our analyses (the blanks are already regarded as NAs by R in the ingestion process.)

```
smart_ohio_raw |> tabyl(HHADULT)
```

HHADULT	n	percent	valid_percent
1	274	0.0369670804	0.236206897
2	603	0.0813545602	0.519827586
3	170	0.0229357798	0.146551724
4	73	0.0098488937	0.062931034
5	28	0.0037776579	0.024137931

6	4	0.0005396654	0.003448276
7	3	0.0004047491	0.002586207
8	1	0.0001349164	0.000862069
10	1	0.0001349164	0.000862069
11	1	0.0001349164	0.000862069
99	2	0.0002698327	0.001724138
NA	6252	0.8434970318	NA

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(hhadults = HHADULT,
         hhadults = replace(hhadults, hhadults > 76, NA))

smart_ohio_raw |> count(HHADULT, hhadults) |> tail()
```

```
# A tibble: 6 x 3
  HHADULT hhadults      n
  <dbl>    <dbl> <int>
1      7         7     3
2      8         8     1
3     10        10     1
4     11        11     1
5     99        NA     2
6     NA        NA  6252
```

6.6.3 Health Status (1 item)

The next variable describes relate to the subject's health status.

6.6.3.1 GENHLTH and its cleanup to genhealth

- GENHLTH, the General Health variable, which is the response to “Would you say that in general your health is ...”
 - 1 = Excellent
 - 2 = Very good
 - 3 = Good
 - 4 = Fair
 - 5 = Poor
 - 7 = Don't know/Not sure
 - 9 = Refused
 - BLANK = Not asked or missing

To clean up the `GENHLTH` data into a new variable called `genhealth` we'll need to - convince R that the 7 and 9 values are in fact best interpreted as `NA`, - and perhaps change the variable to a factor and incorporate the names into the levels.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(genhealth = fct_recode(factor(GENHLTH),
    "1_Excellent" = "1",
    "2_VeryGood" = "2",
    "3_Good" = "3",
    "4_Fair" = "4",
    "5_Poor" = "5",
    NULL = "7",
    NULL = "9"))

smart_ohio_raw |> count(GENHLTH, genhealth)
```

```
# A tibble: 7 x 3
  GENHLTH genhealth      n
  <dbl> <fct>      <int>
1      1 1_Excellent 1057
2      2 2_VeryGood 2406
3      3 3_Good     2367
4      4 4_Fair     1139
5      5 5_Poor      428
6      7 <NA>        10
7      9 <NA>         5
```

6.6.4 Healthy Days - Health-Related Quality of Life (3 items)

The next three variables describe the subject's health-related quality of life.

6.6.4.1 PHYSHLTH and its cleanup to physhealth

`PHYSHLTH`, the Number of Days Physical Health Not Good variable, which is the response to "Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?"

- Values of 1-30 are numeric and reasonable.
- A value of 88 indicates "none" and should be recoded to 0.
- 77 is the code for Don't know/Not sure
- 99 is the code for Refused

- BLANK indicates Not asked or missing, and R recognizes this as NA properly.

To clean up PHYSHLTH to a new variable called **physhealth**, we'll need: - to convince R that the 77 and 99 values are in fact best interpreted as NA, and - to convince R that the 88 should be interpreted as 0.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(physhealth = PHYSHLTH,
         physhealth = replace(physhealth, physhealth %in% c(77, 99), NA),
         physhealth = replace(physhealth, physhealth == 88, 0))

smart_ohio_raw |> count(PHYSHLTH, physhealth) |> tail()
```

```
# A tibble: 6 x 3
  PHYSHLTH physhealth     n
  <dbl>      <dbl> <int>
1      28         28     12
2      29         29     14
3      30         30    677
4      77         NA    123
5      88          0   4380
6      99         NA     15
```

Note that we present the **tail** of the counts in this case so we can see what happens to the key values (77, 88, 99) of our original variable PHYSHLTH.

6.6.4.2 MENTHLTH and its cleanup to menthealth

MENTHLTH⁴, the Number of Days Mental Health Not Good variable, which is the response to “Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?”

- This is coded just like the PHYSHLTH variable, so we need to do the same cleaning we did there.

To clean up MENTHLTH to a new variable called **menthealth**, we'll need: - to convince R that the 77 and 99 values are in fact best interpreted as NA, and - to convince R that the 88 should be interpreted as 0.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(menthealth = MENTHLTH,
         menthealth = replace(menthealth, menthealth %in% c(77, 99), NA),
         menthealth = replace(menthealth, menthealth == 88, 0))

smart_ohio_raw |> count(MENTHLTH, menthealth) |> tail()
```

```
# A tibble: 6 x 3
  MENTHLTH menthealth     n
  <dbl>      <dbl> <int>
1      28         28     7
2      29         29    10
3      30         30   475
4      77         NA    86
5      88          0  4823
6      99         NA    28
```

6.6.4.3 POORHLTH and its cleanup to poorhealth

POORHLTH, the Poor Physical or Mental Health variable, which is the response to “During the past 30 days, for about how many days did poor physical or mental health keep you from doing your usual activities, such as self-care, work, or recreation?”

- Again, we recode just like the PHYSHLTH variable.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(poorhealth = POORHLTH,
         poorhealth = replace(poorhealth, poorhealth %in% c(77, 99), NA),
         poorhealth = replace(poorhealth, poorhealth == 88, 0))

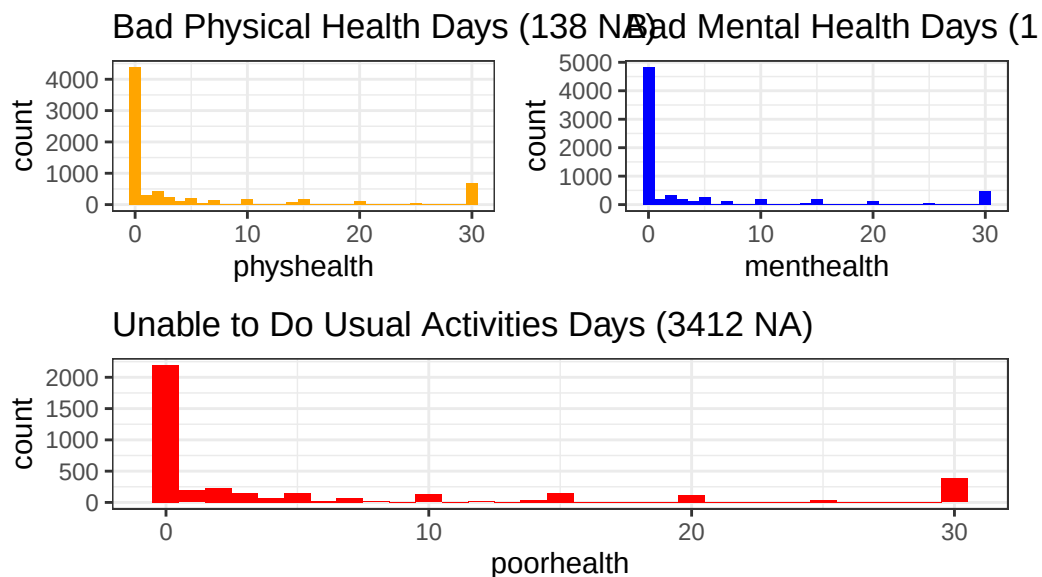
smart_ohio_raw |> count(POORHLTH, poorhealth) |> tail()
```

```
# A tibble: 6 x 3
  POORHLTH poorhealth     n
  <dbl>      <dbl> <int>
1      29         29     4
2      30         30   382
3      77         NA    64
4      88          0  2194
5      99         NA    11
6      NA         NA  3337
```

There's a lot more missingness in the `poorhealth` counts than in the other health-related quality of life measures. There's also a strong mode at 0, and a smaller mode at 30 in each variable.

```
p1 <- ggplot(smart_ohio_raw, aes(x = physhealth)) +  
  geom_histogram(binwidth = 1, fill = "orange") +  
  labs(title = paste0("Bad Physical Health Days (",  
                      sum(is.na(smart_ohio_raw$physhealth)),  
                      " NA)"))  
  
p2 <- ggplot(smart_ohio_raw, aes(x = menthealth)) +  
  geom_histogram(binwidth = 1, fill = "blue") +  
  labs(title = paste0("Bad Mental Health Days (",  
                      sum(is.na(smart_ohio_raw$menthealth)),  
                      " NA)"))  
  
p3 <- ggplot(smart_ohio_raw, aes(x = poorhealth)) +  
  geom_histogram(binwidth = 1, fill = "red") +  
  labs(title = paste0("Unable to Do Usual Activities Days (",  
                      sum(is.na(smart_ohio_raw$poorhealth)),  
                      " NA)"))  
  
(p1 + p2) / p3 +  
  plot_annotation(title = "Health Related Quality of Life Measures in BRFSS/SMART (Ohio MM
```

Health Related Quality of Life Measures in BRFSS/SMART (Ohio M



6.6.5 Health Care Access (4 items)

The next four variables relate to the subject's health care access.

6.6.5.1 HLTHPLN1 and its cleanup to healthplan

HLTHPLN1, the Have any health care coverage variable, is the response to “Do you have any kind of health care coverage, including health insurance, prepaid plans such as HMOs, or government plans such as Medicare, or Indian Health Service?”

- 1 = Yes
- 2 = No
- 7 = Don't know/Not sure
- 9 = Refused

To clean up the HLTHPLN1 data into a new variable called `healthplan` we'll

- convince R that the 7 and 9 values are in fact best interpreted as NA, - and turn it into an indicator variable, e.g., we will leave the variable as numeric, but change the values to 1 = Yes and 0 = No.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(healthplan = HLTHPLN1,
         healthplan = replace(healthplan, healthplan %in% c(7, 9), NA),
         healthplan = replace(healthplan, healthplan == 2, 0))

smart_ohio_raw |> count(HLTHPLN1, healthplan)
```

```
# A tibble: 4 x 3
  HLTHPLN1 healthplan      n
  <dbl>      <dbl> <int>
1       1         1  6994
2       2         0   398
3       7        NA    10
4       9        NA    10
```

6.6.5.2 PERSDOC2 and its cleanup to hasdoc and to numdocs2

PERSDOC2, the Multiple Health Care Professionals variable, is the response to “Do you have one person you think of as your personal doctor or health care provider?” where if the response is “No”, the survey then asks “Is there more than one or is there no person who you think of as your personal doctor or health care provider?”

- 1 = Yes, only one
- 2 = More than one
- 3 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

To clean up the PERSDOC2 data into a new variable called `hasdoc` we’ll - convince R that the 7 and 9 values are in fact best interpreted as `NA`, - and turn it into an indicator variable, e.g., we will leave the variable as numeric, but change the values to 1 = Yes and 0 = No, so that the original 1 and 2 become 1, and the original 3 becomes 0.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(hasdoc = PERSDOC2,
         hasdoc = replace(hasdoc, hasdoc %in% c(7, 9), NA),
         hasdoc = replace(hasdoc, hasdoc %in% c(1, 2), 1),
         hasdoc = replace(hasdoc, hasdoc == 3, 0))

smart_ohio_raw |> count(PERSDOC2, hasdoc)
```

```
# A tibble: 5 x 3
  PERSDOC2 hasdoc      n
    <dbl>   <dbl> <int>
1         1     1  5784
2         2     1   623
3         3     0   990
4         7    NA    14
5         9    NA     1
```

6.6.5.3 MEDCOST and its cleanup to costprob

MEDCOST, the Could Not See Doctor Because of Cost variable, is the response to “Was there a time in the past 12 months when you needed to see a doctor but could not because of cost?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

This is just like HLTHPLAN.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(costprob = MEDCOST,
         costprob = replace(costprob, costprob %in% c(7, 9), NA),
         costprob = replace(costprob, costprob == 2, 0))

smart_ohio_raw |> count(MEDCOST, costprob)
```

```
# A tibble: 4 x 3
  MEDCOST costprob      n
    <dbl>   <dbl> <int>
1         1     1   714
2         2     0  6680
3         7    NA    14
4         9    NA     4
```

6.6.5.4 CHECKUP1 and its cleanup to t_checkup

CHECKUP1, the Length of time since last routine checkup variable, is the response to “About how long has it been since you last visited a doctor for a routine checkup? [A routine checkup is a general physical exam, not an exam for a specific injury, illness, or condition.]”

- 1 = Within past year (anytime less than 12 months ago)
- 2 = Within past 2 years (1 year but less than 2 years ago)
- 3 = Within past 5 years (2 years but less than 5 years ago)
- 4 = 5 or more years ago
- 7 = Don't know/Not sure
- 8 = Never
- 9 = Refused
- BLANK = Not asked or missing

To clean up the `CHECKUP1` data into a new variable called `t_checkup` we'll - convince R that the 7 and 9 values are in fact best interpreted as `NA`, - relabel options 1, 2, 3, 4 and 8 while turning the variable into a factor.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(t_checkup = fct_recode(factor(CHECKUP1),
    "1_In-past-year" = "1",
    "2_1-to-2-years" = "2",
    "3_2-to-5-years" = "3",
    "4_5_plus_years" = "4",
    "8_Never" = "8",
    NULL = "7",
    NULL = "9"))

smart_ohio_raw |> count(CHECKUP1, t_checkup)
```

```
# A tibble: 7 x 3
  CHECKUP1 t_checkup      n
  <dbl>   <fct>    <int>
1       1 1_In-past-year 5803
2       2 2_1-to-2-years  714
3       3 3_2-to-5-years  413
4       4 4_5_plus_years  376
5       7 <NA>          68
6       8 8_Never       32
7       9 <NA>          6
```

6.6.6 Blood Pressure (2 measures)

6.6.6.1 BPHIGH4 and its cleanup to `bp_high`

BPHIGH4 is asking about awareness of a hypertension diagnosis. It's the response to the question: "Have you EVER been told by a doctor, nurse or other health professional that you have

high blood pressure?” In addition, if the answer was “Yes” and the respondent is female, they were then asked “Was this only when you were pregnant?”

The available codes are:

- 1 = Yes
- 2 = Yes, but female told only during pregnancy
- 3 = No
- 4 = Told borderline high or pre-hypertensive
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

To clean up the BPHIGH4 data into a new variable called `bp_high` we’ll - convince R that the 7 and 9 values are in fact best interpreted as NA, - relabel (and re-order) options 1, 2, 3, 4 while turning the variable into a factor.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(bp_high = fct_recode(factor(BPHIGH4),
                                "0_No" = "3",
                                "1_Yes" = "1",
                                "2_Only_while_pregnant" = "2",
                                "4_Borderline" = "4",
                                NULL = "7",
                                NULL = "9"),
         bp_high = fct_relevel(bp_high,
                                "0_No", "1_Yes",
                                "2_Only_while_pregnant",
                                "4_Borderline"))

smart_ohio_raw |> count(BPHIGH4, bp_high)
```

```
# A tibble: 6 x 3
  BPHIGH4 bp_high      n
  <dbl> <fct>    <int>
1     1 1_Yes      3161
2     2 2_Only_while_pregnant    67
3     3 0_No      4114
4     4 4_Borderline         49
5     7 <NA>         19
6     9 <NA>          2
```

6.6.6.2 BPMEDS and its cleanup to bp_meds

BPMEDS is the response to the question “Are you currently taking medicine for your high blood pressure?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

To clean up the BPMEDS data into a new variable called `bp_meds` we’ll treat it just as we did with `HLTHPLN1` and - convince R that the 7 and 9 values are in fact best interpreted as `NA`, - and turn it into an indicator variable, e.g., we will leave the variable as numeric, but change the values to 1 = Yes and 0 = No.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(bp_meds = BPMEDS,
         bp_meds = replace(bp_meds, bp_meds %in% c(7, 9), NA),
         bp_meds = replace(bp_meds, bp_meds == 2, 0))

smart_ohio_raw |> count(BPMEDS, bp_meds)
```

```
# A tibble: 5 x 3
  BPMEDS bp_meds      n
  <dbl>   <dbl> <int>
1      1       1  2675
2      2       0   481
3      7      NA     4
4      9      NA     1
5     NA      NA  4251
```

What is the relationship between our two blood pressure variables? Only the people with `bp_meds = “1_Yes”` were asked the `bp_meds` question.

```
smart_ohio_raw |> tabyl(bp_high, bp_meds)
```

```
      bp_high  0    1  NA_
0_No         0    0  4114
1_Yes      481 2675     5
2_Only_while_pregnant 0    0   67
4_Borderline  0    0   49
<NA>         0    0   21
```

6.6.7 Cholesterol (3 items)

6.6.7.1 CHOLCHK1 and its cleanup to t_chol

CHOLCHK1, the Length of time since cholesterol was checked, is the response to “Blood cholesterol is a fatty substance found in the blood. About how long has it been since you last had your blood cholesterol checked?”

- 1 = Never
- 2 = Within past year (anytime less than 12 months ago)
- 3 = Within past 2 years (1 year but less than 2 years ago)
- 4 = Within past 5 years (2 years but less than 5 years ago)
- 5 = 5 or more years ago
- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

To clean up the CHOLCHK1 data into a new variable called `t_chol` we'll - convince R that the 7 and 9 values are in fact best interpreted as NA, - relabel options 1, 2, 3, 4 and 8 while turning the variable into a factor.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(t_chol = fct_recode(factor(CHOLCHK1),
                                "1_Never" = "1",
                                "2_In-past-year" = "2",
                                "3_1-to-2-years" = "3",
                                "4_2-to-5-years" = "4",
                                "5_5-plus-years" = "5",
                                NULL = "7",
                                NULL = "9"))

smart_ohio_raw |> count(CHOLCHK1, t_chol)
```

```
# A tibble: 8 x 3
  CHOLCHK1 t_chol      n
  <dbl> <fct>    <int>
1      1 1_Never    424
2      2 2_In-past-year 5483
3      3 3_1-to-2-years  559
4      4 4_2-to-5-years  289
5      5 5_5-plus-years  272
6      7 <NA>       376
7      9 <NA>        8
```

The next two measures are not gathered from the people who answered “Never” to this question.

6.6.7.2 TOLDHI2 and its cleanup to chol_high

TOLDHI2 is asking about awareness of a diagnosis of high cholesterol. It’s the response to the question: “Have you EVER been told by a doctor, nurse or other health professional that your blood cholesterol is high?”

The available codes are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

To clean up the TOLDHI2 data into a new variable called `chol_high` we’ll treat it like `BPMEDS` and `HLTHPLN1` - convince R that the 7 and 9 values are in fact best interpreted as `NA`, - and turn it into an indicator variable, e.g., we will leave the variable as numeric, but change the values to 1 = Yes and 0 = No.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(chol_high = TOLDHI2,
         chol_high = replace(chol_high, chol_high %in% c(7, 9), NA),
         chol_high = replace(chol_high, chol_high == 2, 0))

smart_ohio_raw |> count(TOLDHI2, chol_high)
```

```
# A tibble: 5 x 3
  TOLDHI2 chol_high     n
  <dbl>    <dbl> <int>
1      1         1  2612
2      2         0  4286
3      7        NA    70
4      9        NA     4
5     NA        NA   440
```

6.6.7.3 CHOLMED1 and its cleanup to chol_meds

CHOLMED1 is the response to the question “Are you currently taking medicine prescribed by a doctor or other health professional for your blood cholesterol?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

To clean up the CHOLMED1 data into a new variable called `chol_meds` we’ll treat it just as we did with `HLTHPLN1` and - convince R that the 7 and 9 values are in fact best interpreted as `NA`, - and turn it into an indicator variable, e.g., we will leave the variable as numeric, but change the values to 1 = Yes and 0 = No.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(chol_meds = CHOLMED1,
         chol_meds = replace(chol_meds, chol_meds %in% c(7, 9), NA),
         chol_meds = replace(chol_meds, chol_meds == 2, 0))

smart_ohio_raw |> count(CHOLMED1, chol_meds)
```

```
# A tibble: 4 x 3
  CHOLMED1 chol_meds      n
  <dbl>    <dbl> <int>
1         1         1  1781
2         2         0   826
3         7        NA     5
4        NA        NA  4800
```

6.6.8 Chronic Health Conditions (14 items)

6.6.8.1 Self-reported diagnosis history (11 items)

The next few variables describe whether or not the subject meets a particular standard, and are all coded in the raw data the same way:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

- BLANK = Not asked or missing

and we'll recode them all to 1 = Yes, 0 = No, otherwise NA, as we've done previously.

The questions are all started with "Has a doctor, nurse, or other health professional ever told you that you had any of the following? For each, tell me Yes, No, or you're Not sure."

Original	Revised	Details
CVDINFR4	hx_mi	(Ever told) you had a heart attack, also called a myocardial infarction?
CVDCRHD4	hx_chd	(Ever told) you had angina or coronary heart disease?
CVDSTRK3	hx_stroke	(Ever told) you had a stroke?
ASTHMA3	hx_asthma	(Ever told) you had asthma?
ASTHNOW	now_asthma	Do you still have asthma? (only asked of those with Yes in ASTHMA3)
CHCSCNCR	hx_skinc	(Ever told) you had skin cancer?
CHCOCNCR	hx_otherc	(Ever told) you had any other types of cancer?
CHCCOPD1	hx_copd	(Ever told) you have Chronic Obstructive Pulmonary Disease or COPD, emphysema or chronic bronchitis?
HAVARTH3	hx_arthr	(Ever told) you have some form of arthritis, rheumatoid arthritis, gout, lupus, or fibromyalgia? (Arthritis diagnoses include: rheumatism, polymyalgia rheumatica; osteoarthritis (not osteoporosis); tendonitis, bursitis, bunion, tennis elbow; carpal tunnel syndrome, tarsal tunnel syndrome; joint infection, etc.)
ADDEPEV2	hx_depress	(Ever told) you that you have a depressive disorder, including depression, major depression, dysthymia, or minor depression?
CHCKIDNY	hx_kidney	(Ever told) you have kidney disease? Do NOT include kidney stones, bladder infection or incontinence.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(hx_mi = CVDINFR4,
         hx_mi = replace(hx_mi, hx_mi %in% c(7, 9), NA),
         hx_mi = replace(hx_mi, hx_mi == 2, 0),
         hx_chd = CVDCRHD4,
         hx_chd = replace(hx_chd, hx_chd %in% c(7, 9), NA),
         hx_chd = replace(hx_chd, hx_chd == 2, 0),
         hx_stroke = CVDSTRK3,
         hx_stroke = replace(hx_stroke, hx_stroke %in% c(7, 9), NA),
         hx_stroke = replace(hx_stroke, hx_stroke == 2, 0),
         hx_asthma = ASTHMA3,
         hx_asthma = replace(hx_asthma, hx_asthma %in% c(7, 9), NA),
```

```

hx_asthma = replace(hx_asthma, hx_asthma == 2, 0),
now_asthma = ASTHNOW,
now_asthma = replace(now_asthma, now_asthma %in% c(7, 9), NA),
now_asthma = replace(now_asthma, now_asthma == 2, 0),
hx_skinc = CHCSCNCR,
hx_skinc = replace(hx_skinc, hx_skinc %in% c(7, 9), NA),
hx_skinc = replace(hx_skinc, hx_skinc == 2, 0),
hx_otherc = CHCOCNCR,
hx_otherc = replace(hx_otherc, hx_otherc %in% c(7, 9), NA),
hx_otherc = replace(hx_otherc, hx_otherc == 2, 0),
hx_copd = CHCCOPD1,
hx_copd = replace(hx_copd, hx_copd %in% c(7, 9), NA),
hx_copd = replace(hx_copd, hx_copd == 2, 0),
hx_arthr = HAVARTH3,
hx_arthr = replace(hx_arthr, hx_arthr %in% c(7, 9), NA),
hx_arthr = replace(hx_arthr, hx_arthr == 2, 0),
hx_depress = ADDEPEV2,
hx_depress = replace(hx_depress, hx_depress %in% c(7, 9), NA),
hx_depress = replace(hx_depress, hx_depress == 2, 0),
hx_kidney = CHCKIDNY,
hx_kidney = replace(hx_kidney, hx_kidney %in% c(7, 9), NA),
hx_kidney = replace(hx_kidney, hx_kidney == 2, 0))

```

We definitely should have written a function to do that, of course.

6.6.8.2 _ASTHMS1 and its cleanup to asthma

_ASTHMS1 categorizes subjects by asthma status as:

- 1 = Current
- 2 = Former
- 3 = Never
- 9 = Don't Know / Not Sure / Refused / Missing

We'll turn this into a factor with appropriate levels and NA information.

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(asthma = fct_recode(
    factor(`_ASTHMS1`),
    "Current" = "1",
    "Former" = "2",

```

```

    "Never" = "3",
    NULL = "9"))

smart_ohio_raw |> count(`_ASTHMS1`, asthma)

```

```

# A tibble: 4 x 3
  `_ASTHMS1` asthma      n
    <dbl> <fct>    <int>
1         1 Current    734
2         2 Former    248
3         3 Never     6376
4         9 <NA>        54

```

6.6.8.3 DIABETE3 and its cleanup to hx_diabetes and dm_status

DIABETE3, the (Ever told) you have diabetes variable, is the response to “(Ever told) you have diabetes (If Yes and respondent is female, ask Was this only when you were pregnant?. If Respondent says pre-diabetes or borderline diabetes, use response code 4.)”

- 1 = Yes
- 2 = Yes, but female told only during pregnancy
- 3 = No
- 4 = No, pre-diabetes or borderline diabetes
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

I’ll create one variable called `hx_diabetes` which is 1 if `DIABETE3` = 1, and 0 otherwise, with appropriate NAs, like our other variables. Then I’ll create `dm_status` to include all of this information in a factor, but again recode the missing values properly.

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(hx_diabetes = DIABETE3,
         hx_diabetes = replace(hx_diabetes, hx_diabetes %in% c(7, 9), NA),
         hx_diabetes = replace(hx_diabetes, hx_diabetes %in% 2:4, 0),
         dm_status = fct_recode(factor(DIABETE3),
                                "Diabetes" = "1",
                                "Pregnancy-Induced" = "2",
                                "No-Diabetes" = "3",
                                "Pre-Diabetes" = "4",
                                NULL = "7",

```



```

NULL = "9"),
  dm_status = fct_relevel(dm_status,
    "No-Diabetes",
    "Pre-Diabetes",
    "Pregnancy-Induced",
    "Diabetes"))

smart_ohio_raw |> count(DIABETE3, hx_diabetes, dm_status)

```

```

# A tibble: 6 x 4
  DIABETE3 hx_diabetes dm_status      n
  <dbl>    <dbl> <fct>    <int>
1       1         1 1 Diabetes    1098
2       2         0 0 Pregnancy-Induced    67
3       3         0 0 No-Diabetes    6100
4       4         0 0 Pre-Diabetes    133
5       7         NA <NA>      12
6       9         NA <NA>       2

```

6.6.8.4 DIABAGE2 and its cleanup to dm_age

DIABAGE2, the Age When Told Diabetic variable, is the response to “How old were you when you were told you have diabetes?” It is asked only of people with DIABETE3 = 1 (Yes).

- The response is 1-97, with special values 98 for Don’t Know/Not Sure and 99 for refused, with BLANK for missing or not asked. People 97 years of age and above were listed as 97.

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(dm_age = DIABAGE2,
    dm_age = replace(dm_age, dm_age > 97, NA))

smart_ohio_raw |> count(DIABAGE2, dm_age) |> tail()

```

```

# A tibble: 6 x 3
  DIABAGE2 dm_age      n
  <dbl>    <dbl> <int>
1      84     84      1
2      85     85      2
3      90     90      1
4      98     NA     61

```

5	99	NA	4
6	NA	NA	6314

6.6.9 Arthritis Burden (4 items)

The first two measures are only asked of people with `hx_arthr = 1`, and are coded as:

- 1 = Yes
- 2 = No
- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

and we'll recode them to 1 = Yes, 0 = No, otherwise NA, as we've done previously.

6.6.9.1 LMTJOIN3 (Limited because of joint symptoms), and its cleanup to `arth_lims`

This is the response to “Are you now limited in any way in any of your usual activities because of arthritis or joint symptoms?”

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(arth_lims = LMTJOIN3,
         arth_lims = replace(arth_lims, arth_lims %in% c(7, 9), NA),
         arth_lims = replace(arth_lims, arth_lims == 2, 0))

smart_ohio_raw |> count(hx_arthr, LMTJOIN3, arth_lims)
```

```
# A tibble: 6 x 4
  hx_arthr LMTJOIN3 arth_lims      n
  <dbl>    <dbl>    <dbl> <int>
1       0      NA      NA  4587
2       1       1       1  1378
3       1       2       0  1388
4       1       7      NA   17
5       1       9      NA    2
6      NA      NA      NA   40
```

6.6.9.2 ARTHDIS2 (Does Arthritis Affect Whether You Work), and its cleanup to arth_work

This is the response to “Do arthritis or joint symptoms now affect whether you work, the type of work you do or the amount of work you do?”

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(arth_work = ARTHDIS2,
         arth_work = replace(arth_work, arth_work %in% c(7, 9), NA),
         arth_work = replace(arth_work, arth_work == 2, 0))

smart_ohio_raw |> count(ARTHDIS2, arth_work)
```

```
# A tibble: 5 x 3
  ARTHDIS2 arth_work      n
    <dbl>     <dbl> <int>
1         1         1  925
2         2         0 1808
3         7        NA   42
4         9        NA   10
5        NA        NA 4627
```

6.6.9.3 ARTHSOCL (Social Activities Limited Because of Joint Symptoms) and its cleanup to arth_soc

This is the response to “During the past 30 days, to what extent has your arthritis or joint symptoms interfered with your normal social activities, such as going shopping, to the movies, or to religious or social gatherings?”

The responses are:

- 1 = A lot
- 2 = A little
- 3 = Not at all
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(arth_soc = fct_recode(factor(ARTHSOCL),
                                "A lot" = "1",
                                "A little" = "2",
```

```

"Not at all" = "3",
NULL = "7",
NULL = "9"))

smart_ohio_raw |> count(ARTHSOCL, arth_soc)

```

```

# A tibble: 6 x 3
  ARTHSOCL arth_soc      n
  <dbl> <fct>      <int>
1       1 A lot      606
2       2 A little    734
3       3 Not at all 1427
4       7 <NA>        15
5       9 <NA>         3
6      NA <NA>     4627

```

6.6.9.4 JOINPAI1 (How Bad Was Joint Pain - scale of 0-10) and its cleanup to joint_pain

This is the response to the following question: “Please think about the past 30 days, keeping in mind all of your joint pain or aching and whether or not you have taken medication. On a scale of 0 to 10 where 0 is no pain or aching and 10 is pain or aching as bad as it can be, DURING THE PAST 30 DAYS, how bad was your joint pain ON AVERAGE?”

The available values are 0-10, plus codes 77 (Don’t Know / Not Sure), 99 (Refused) and BLANK.

To clean up JOINPAI1 to a new variable called `joint_pain`, we’ll need to convince R that the 77 and 99 values are, like BLANK, in fact best interpreted as NA.

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(joint_pain = JOINPAI1,
         joint_pain = replace(joint_pain, joint_pain %in% c(77, 99), NA))

smart_ohio_raw |> count(JOINPAI1, joint_pain) |> tail()

```

```

# A tibble: 6 x 3
  JOINPAI1 joint_pain      n
  <dbl>      <dbl> <int>
1       8         8    277
2       9         9     72

```

3	10	10	158
4	77	NA	28
5	99	NA	5
6	NA	NA	4627

6.6.10 Demographics (25 items)

6.6.10.1 _AGEG5YR, which we'll edit into agegroup

The `_AGEG5YR` variable is a calculated variable (by CDC) obtained from the subject's age. Since the `age` data are not available, we instead get these groupings, which we'll rearrange into the `agegroup` factor.

<code>_AGEG5YR</code>	Age range	<code>agegroup</code>
1	18 <= AGE <= 24	18-24
2	25 <= AGE <= 29	25-29
3	30 <= AGE <= 34	30-34
4	35 <= AGE <= 39	35-39
5	40 <= AGE <= 44	40-44
6	45 <= AGE <= 49	45-49
7	50 <= AGE <= 54	50-54
8	55 <= AGE <= 59	55-59
9	60 <= AGE <= 64	60-64
10	65 <= AGE <= 69	65-69
11	70 <= AGE <= 74	70-74
12	75 <= AGE <= 79	75-79
13	AGE >= 80	80plus
14	Don't Know, Refused or Missing	NA

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(agegroup = fct_recode(factor(`_AGEG5YR`),
    "18-24" = "1",
    "25-29" = "2",
    "30-34" = "3",
    "35-39" = "4",
    "40-44" = "5",
    "45-49" = "6",
    "50-54" = "7",
    "55-59" = "8",
    "60-64" = "9",
    "65-69" = "10",
```

```

"70-74" = "11",
"75-79" = "12",
"80-96" = "13",
NULL = "14"))

smart_ohio_raw |> count(`_AGEG5YR`, agegroup)

```

```

# A tibble: 14 x 3
  `_AGEG5YR` agegroup      n
    <dbl> <fct>    <int>
1         1 18-24      448
2         2 25-29      327
3         3 30-34      375
4         4 35-39      446
5         5 40-44      426
6         6 45-49      509
7         7 50-54      604
8         8 55-59      786
9         9 60-64      837
10        10 65-69      810
11        11 70-74      685
12        12 75-79      499
13        13 80-96      592
14        14 <NA>        68

```

6.6.10.2 _MRACE1 recoded to race

We'll create three variables describing race/ethnicity. The first comes from the `_MRACE1` variable categorized by CDC, and the available responses are:

- 1 = White only
- 2 = Black or African-American only
- 3 = American Indian or Alaskan Native only
- 4 = Asian only
- 5 = Native Hawaiian or Pacific Islander only
- 6 = Other race only
- 7 = Multiracial
- 77 = Don't know / Not Sure
- 99 = Refused
- BLANK = Missing

We'll create a factor out of this information, with appropriate level names.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(race = fct_recode(factor(`_MRACE1`),
                                "White" = "1",
                                "Black or African A" = "2",
                                "Amer Indian or Alaskan" = "3",
                                "Asian" = "4",
                                "Hawaiian or Pac Island" = "5",
                                "Other Race" = "6",
                                "Multiracial" = "7",
                                NULL = "77",
                                NULL = "99"))

smart_ohio_raw |> count(`_MRACE1`, race)
```

```
# A tibble: 9 x 3
  `_MRACE1` race      n
    <dbl> <fct>      <int>
1         1 White      6177
2         2 Black or African A    739
3         3 Amer Indian or Alaskan    66
4         4 Asian      115
5         5 Hawaiian or Pac Island     5
6         6 Other Race     43
7         7 Multiracial    153
8        77 <NA>        14
9         99 <NA>       100
```

6.6.10.3 _HISPANC recoded to hispanic

The `_HISPANC` variable specifies whether or not the respondent is of Hispanic or Latinx origin. The available responses are:

- 1 = Hispanic, Latinx or Spanish origin
- 2 = Not of Hispanic, Latinx or Spanish origin
- 9 = Don't Know, Refused, or Missing

We'll turn the 9s into NA, and create an indicator variable (1 = Hispanic or Latinx, 0 = not)

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(hispanic = 2 - `_HISPANC`,
         hispanic = replace(hispanic, hispanic < 0, NA))

smart_ohio_raw |> count(`_HISPANC`, hispanic)
```

```
# A tibble: 3 x 3
  `_HISPANC` hispanic      n
    <dbl>      <dbl> <int>
1         1         1   146
2         2         0  7217
3         9        NA    49
```

6.6.10.4 _RACEGR3 recoded to race_eth

The `_RACEGR3` variable is a five-level combination of race and ethnicity. The responses are:

- 1 = White non-Hispanic
- 2 = Black non-Hispanic
- 3 = Other race non-Hispanic
- 4 = Multiracial non-Hispanic
- 5 = Hispanic
- 9 = Don't Know / Not Sure / Refused

We'll create a factor out of this information, with appropriate level names.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(race_eth = fct_recode(
    factor(`_RACEGR3`),
    "White non-Hispanic" = "1",
    "Black non-Hispanic" = "2",
    "Other race non-Hispanic" = "3",
    "Multiracial non-Hispanic" = "4",
    "Hispanic" = "5",
    NULL = "9"))

smart_ohio_raw |> count(`_RACEGR3`, race_eth)
```

```
# A tibble: 6 x 3
  `_RACEGR3` race_eth      n
    <dbl> <fct>      <int>
1         1     1         146
2         2     2      7217
3         3     3         49
4         4     4         49
5         5     5         49
6         9     9         49
```


1	1 White non-Hispanic	6086
2	2 Black non-Hispanic	725
3	3 Other race non-Hispanic	193
4	4 Multiracial non-Hispanic	143
5	5 Hispanic	146
6	9 <NA>	119

6.6.10.5 SEX recoded to female

The available levels of SEX are:

- 1 = Male
- 2 = Female
- 9 = Refused

We'll recode that to `female = 1` for Female, 0 Male, otherwise NA. Note the trick here is to subtract one from the coded SEX to get the desired `female`, but this requires that we move 8 to NA, rather than 9.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(female = SEX - 1,
         female = replace(female, female == 8, NA))

smart_ohio_raw |> count(SEX, female)
```

```
# A tibble: 2 x 3
  SEX female     n
  <dbl> <dbl> <int>
1     1     0  3136
2     2     1  4276
```

6.6.10.6 MARITAL status, revised to marital

The available levels of MARITAL are:

- 1 = Married
- 2 = Divorced
- 3 = Widowed
- 4 = Separated
- 5 = Never married
- 6 = A member of an unmarried couple
- 9 = Refused

- BLANK = Not asked or missing

We'll just turn this into a factor, and move 9 to NA.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(marital = fct_recode(factor(MARITAL),
    "Married" = "1",
    "Divorced" = "2",
    "Widowed" = "3",
    "Separated" = "4",
    "Never_Married" = "5",
    "Unmarried_Couple" = "6",
    NULL = "9"))

smart_ohio_raw |> count(MARITAL, marital)
```

```
# A tibble: 7 x 3
  MARITAL marital      n
  <dbl> <fct>      <int>
1      1 Married    3668
2      2 Divorced   1110
3      3 Widowed    978
4      4 Separated   142
5      5 Never_Married 1248
6      6 Unmarried_Couple 208
7      9 <NA>        58
```

6.6.10.7 EDUCA recoded to educgroup

The available levels of EDUCA (Education Level) are responses to: “What is the highest grade or year of school you completed?”

- 1 = Never attended school or only kindergarten
- 2 = Grades 1 through 8 (Elementary)
- 3 = Grades 9 through 11 (Some high school)
- 4 = Grade 12 or GED (High school graduate)
- 5 = College 1 year to 3 years (Some college or technical school)
- 6 = College 4 years or more (College graduate)
- 9 = Refused
- BLANK = Not asked or missing

We'll just turn this into a factor, and move 9 to NA.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(educgroup = fct_recode(factor(EDUCA),
                                "Kindergarten" = "1",
                                "Elementary" = "2",
                                "Some_HS" = "3",
                                "HS_Grad" = "4",
                                "Some_College" = "5",
                                "College_Grad" = "6",
                                NULL = "9"))

smart_ohio_raw |> count(EDUCA, educgroup)
```

```
# A tibble: 7 x 3
  EDUCA educgroup      n
  <dbl> <fct>      <int>
1     1 Kindergarten    3
2     2 Elementary   117
3     3 Some_HS      332
4     4 HS_Grad     2209
5     5 Some_College 2079
6     6 College_Grad 2646
7     9 <NA>         26
```

6.6.10.8 RENTHOM1 recoded to home_own

The available levels of RENTHOM1 (Own or Rent Home) are responses to: “Do you own or rent your home? (Home is defined as the place where you live most of the time/the majority of the year.)”

- 1 = Own
- 2 = Rent
- 3 = Other Arrangement
- 7 = Don’t know/Not Sure
- 9 = Refused
- BLANK = Not asked or missing

We’ll recode as `home_own` = 1 if they own their home, and 0 otherwise, and dealing with missingness properly.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(home_own = RENTHOM1,
         home_own = replace(home_own, home_own %in% c(7,9), NA),
         home_own = replace(home_own, home_own %in% c(2,3), 0))

smart_ohio_raw |> count(RENTHOM1, home_own)
```

```
# A tibble: 5 x 3
  RENTHOM1 home_own      n
    <dbl>    <dbl> <int>
1         1         1  5216
2         2         0  1793
3         3         0   348
4         7        NA    28
5         9        NA    27
```

6.6.10.9 CPDEM01A and its cleanup to cell_own

CPDEM01A is the response to “Including phones for business and personal use, do you have a cell phone for personal use?”

Available responses are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

and we’ll recode them to 1 = Yes, 0 = No, otherwise NA, as we’ve done previously.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(cell_own = 2 - CPDEM01A,
         cell_own = replace(cell_own, cell_own < 0, NA))

smart_ohio_raw |> count(CPDEM01A, cell_own)
```

```
# A tibble: 5 x 3
  CPDEM01A cell_own      n
    <dbl>    <dbl> <int>
1         1         1  2930
```

2	2	0	698
3	7	NA	2
4	9	NA	19
5	NA	NA	3763

6.6.10.10 VETERAN3 and its cleanup to veteran

VETERAN3, the Are You A Veteran variable, is the response to “Have you ever served on active duty in the United States Armed Forces, either in the regular military or in a National Guard or military reserve unit? (Active duty does not include training for the Reserves or National Guard, but DOES include activation, for example, for the Persian Gulf War.)”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(veteran = VETERAN3,
         veteran = replace(veteran, veteran %in% c(7, 9), NA),
         veteran = replace(veteran, veteran == 2, 0))

smart_ohio_raw |> count(VETERAN3, veteran)
```

```
# A tibble: 3 x 3
  VETERAN3 veteran      n
    <dbl>    <dbl> <int>
1         1         1  927
2         2         0 6479
3         9        NA    6
```

6.6.10.11 EMPLOY1 and its cleanup to employment

EMPLOY1, the Employment Status variable, is the response to “Are you currently ... ?”

- 1 = Employed for wages
- 2 = Self-employed
- 3 = Out of work for 1 year or more
- 4 = Out of work for less than 1 year
- 5 = A homemaker

- 6 = A student
- 7 = Retired
- 8 = Unable to work
- 9 = Refused
- BLANK = Not asked or missing

We'll just turn this into a factor, and move 9 to NA.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(employment = fct_recode(factor(EMPLOY1),
    "Employed_for_wages" = "1",
    "Self-employed" = "2",
    "Outofwork_1yearormore" = "3",
    "Outofwork_lt1year" = "4",
    "Homemaker" = "5",
    "Student" = "6",
    "Retired" = "7",
    "Unable_to_work" = "8",
    NULL = "9"))

smart_ohio_raw |> count(EMPLOY1, employment)
```

```
# A tibble: 9 x 3
  EMPLOY1 employment      n
  <dbl> <fct>          <int>
1      1 Employed_for_wages 3119
2      2 Self-employed      466
3      3 Outofwork_1yearormore 254
4      4 Outofwork_lt1year   134
5      5 Homemaker          411
6      6 Student            190
7      7 Retired           2202
8      8 Unable_to_work     603
9      9 <NA>              33
```

6.6.10.12 CHILDREN and its cleanup to kids

CHILDREN, the Number of Children in Household variable, is the response to “How many children less than 18 years of age live in your household?”

- 1-87 = legitimate responses
- 88 = None

- 99 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(kids = CHILDREN,
         kids = replace(kids, kids == 99, NA),
         kids = replace(kids, kids == 88, 0))

smart_ohio_raw |> count(CHILDREN, kids) |> tail()
```

```
# A tibble: 6 x 3
  CHILDREN kids      n
    <dbl> <dbl> <int>
1         6     6     7
2         7     7     5
3         8     8     2
4        12    12     1
5        88     0  5449
6        99    NA    43
```

6.6.10.13 INCOME2 to incomegroup

The available levels of INCOME2 (Income Level) are responses to: “Is your annual household income from all sources ...”

- 1 = Less than \$10,000
- 2 = \$10,000 to less than \$15,000
- 3 = \$15,000 to less than \$20,000
- 4 = \$20,000 to less than \$25,000
- 5 = \$25,000 to less than \$35,000
- 6 = \$35,000 to less than \$50,000
- 7 = \$50,000 to less than \$75,000
- 8 = \$75,000 or more
- 77 = Don’t know/Not sure
- 99 = Refused
- BLANK = Not asked or missing

We’ll just turn this into a factor, and move 77 and 99 to NA.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(incomegroup = fct_recode(factor(`INCOME2`),
    "0-9K" = "1",
    "10-14K" = "2",
    "15-19K" = "3",
    "20-24K" = "4",
    "25-34K" = "5",
    "35-49K" = "6",
    "50-74K" = "7",
    "75K+" = "8",
    NULL = "77",
    NULL = "99"))

smart_ohio_raw |> count(`INCOME2`, incomegroup)
```

```
# A tibble: 11 x 3
  INCOME2 incomegroup    n
  <dbl> <fct>      <int>
1      1 0-9K         285
2      2 10-14K        306
3      3 15-19K        477
4      4 20-24K        589
5      5 25-34K        685
6      6 35-49K        922
7      7 50-74K        928
8      8 75K+       1910
9     77 <NA>         610
10     99 <NA>         678
11     NA <NA>          22
```

6.6.10.14 INTERNET and its cleanup to internet30

INTERNET, the Internet use in the past 30 days variable, is the response to “Have you used the internet in the past 30 days?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing


```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(internet30 = INTERNET,
         internet30 = replace(internet30, internet30 %in% c(7, 9), NA),
         internet30 = replace(internet30, internet30 == 2, 0))

smart_ohio_raw |> count(INTERNET, internet30)
```

```
# A tibble: 5 x 3
  INTERNET internet30     n
  <dbl>      <dbl> <int>
1       1         1  6020
2       2         0  1335
3       7        NA    10
4       9        NA    10
5      NA        NA    37
```

6.6.10.15 WTKG3 is weight_kg

WTKG3 is computed by CDC, as the respondent's weight in kilograms with two implied decimal places. We calculate the actual weight in kg, with the following:

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(weight_kg = WTKG3/100)

smart_ohio_raw |> count(WTKG3, weight_kg) |> tail()
```

```
# A tibble: 6 x 3
  WTKG3 weight_kg     n
  <dbl>      <dbl> <int>
1 19051      191.     1
2 19278      193.     1
3 19504      195.     1
4 20412      204.     2
5 20865      209.     1
6    NA        NA    462
```

6.6.10.16 HEIGHT3 is replaced with height_m

HEIGHT3 is strangely gathered to allow people to specify their height in either feet and inches or in meters and centimeters.

- 200-711 indicates height in feet (first digit) and inches (second two digits)
- 9000 - 9998 indicates height in meters (second digit) and centimeters (last two digits)
- 7777 = Don't know/Not sure
- 9999 = Refused

Note that there is one impossible value of 575 in the data set. We'll make that an NA, and we'll also make NA any heights below 3 feet, or above 2.24 meters. Specifically, we calculate the actual height in meters, with the following:

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(height_m = case_when(
    HEIGHT3 >= 300 & HEIGHT3 <= 511 ~ round((12*floor(HEIGHT3/100) + (HEIGHT3 - 100*floor(HEIGHT3/100))/12)/3.048),
    HEIGHT3 >= 600 & HEIGHT3 <= 711 ~ round((12*floor(HEIGHT3/100) + (HEIGHT3 - 100*floor(HEIGHT3/100))/12)/3.048),
    HEIGHT3 >= 9000 & HEIGHT3 <= 9224 ~ ((HEIGHT3 - 9000)/100)))

smart_ohio_raw |> count(HEIGHT3, height_m) |> tail()
```

```
# A tibble: 6 x 3
  HEIGHT3 height_m     n
  <dbl>     <dbl> <int>
1     607      2.01     2
2     608      2.03     6
3     609      2.06     1
4    7777      NA     27
5    9999      NA     86
6      NA      NA     67
```

6.6.10.17 bmi is calculated from height_m and weight_kg

We'll calculate body-mass index from height and weight.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(bmi = round(weight_kg/(height_m)^2,2))

smart_ohio_raw |> count(height_m, weight_kg, bmi)# |> tail()
```

```
# A tibble: 1,806 x 4
  height_m weight_kg   bmi     n
  <dbl>     <dbl> <dbl> <int>
1     1.35      39.0  21.4     1
2     1.35      52.2  28.6     1
```

```

3      1.4      89.8  45.8    1
4      1.42     31.8  15.8    1
5      1.42     45.4  22.5    1
6      1.42     55.8  27.7    1
7      1.42     58.5  29.0    1
8      1.42     59.9  29.7    1
9      1.42     60.8  30.1    1
10     1.42     71.2  35.3    1
# i 1,796 more rows

```

6.6.10.18 bmigroup is calculated from bmi

We'll then divide the respondents into adult BMI categories, in the usual way.

- BMI < 18.5 indicates underweight
- BMI from 18.5 up to 25 indicates normal weight
- BMI from 25 up to 30 indicates overweight
- BMI of 30 and higher indicates obesity

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(bmigroup = factor(cut2(as.numeric(bmi),
                                cuts = c(18.5, 25.0, 30.0))))

smart_ohio_raw |> count(bmigroup)

```

```

# A tibble: 5 x 2
  bmigroup      n
  <fct>      <int>
1 [13.3,18.5)  119
2 [18.5,25.0) 2017
3 [25.0,30.0) 2445
4 [30.0,75.5] 2338
5 <NA>         493

```

6.6.10.19 PREGNANT and its cleanup to pregnant

PREGNANT, the Pregnancy Status variable, is the response to “To your knowledge, are you now pregnant?”

- 1 = Yes
- 2 = No

- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing (includes SEX = male)

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(pregnant = PREGNANT,
         pregnant = replace(pregnant, pregnant %in% c(7, 9), NA),
         pregnant = replace(pregnant, pregnant == 2, 0))

smart_ohio_raw |> count(PREGNANT, pregnant)
```

```
# A tibble: 5 x 3
  PREGNANT pregnant    n
  <dbl>     <dbl> <int>
1      1         1    41
2      2         0  1329
3      7        NA     3
4      9        NA     3
5     NA        NA  6036
```

6.6.10.20 DEAF and its cleanup to deaf

DEAF, the Are you deaf or do you have serious difficulty hearing variable, is the response to “Are you deaf or do you have serious difficulty hearing?”

- 1 = Yes
- 2 = No
- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(deaf = DEAF,
         deaf = replace(deaf, deaf %in% c(7, 9), NA),
         deaf = replace(deaf, deaf == 2, 0))

smart_ohio_raw |> count(DEAF, deaf)
```

```
# A tibble: 5 x 3
  DEAF deaf    n
  <dbl> <dbl> <int>
```

1	1	1	708
2	2	0	6551
3	7	NA	15
4	9	NA	4
5	NA	NA	134

6.6.10.21 BLIND and its cleanup to blind

BLIND, the Blind or Difficulty seeing variable, is the response to “Are you blind or do you have serious difficulty seeing, even when wearing glasses?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(blind = BLIND,
         blind = replace(blind, blind %in% c(7, 9), NA),
         blind = replace(blind, blind == 2, 0))

smart_ohio_raw |> count(BLIND, blind)
```

```
# A tibble: 5 x 3
  BLIND blind    n
  <dbl> <dbl> <int>
1     1     1  415
2     2     0 6834
3     7    NA   14
4     9    NA    1
5    NA    NA  148
```

6.6.10.22 DECIDE and its cleanup to decide

DECIDE, the Difficulty Concentrating or Remembering variable, is the response to “Because of a physical, mental, or emotional condition, do you have serious difficulty concentrating, remembering, or making decisions?”

- 1 = Yes
- 2 = No

- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(decide = DECIDE,
         decide = replace(decide, decide %in% c(7, 9), NA),
         decide = replace(decide, decide == 2, 0))

smart_ohio_raw |> count(DECIDE, decide)
```

```
# A tibble: 5 x 3
  DECIDE decide      n
  <dbl>   <dbl> <int>
1     1     1    870
2     2     0   6348
3     7    NA    30
4     9    NA     2
5    NA    NA   162
```

6.6.10.23 DIFFWALK and its cleanup to diffwalk

DIFFWALK, the Difficulty Walking or Climbing Stairs variable, is the response to “Do you have serious difficulty walking or climbing stairs?”

- 1 = Yes
- 2 = No
- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(diffwalk = DIFFWALK,
         diffwalk = replace(diffwalk, diffwalk %in% c(7, 9), NA),
         diffwalk = replace(diffwalk, diffwalk == 2, 0))

smart_ohio_raw |> count(DIFFWALK, diffwalk)
```

```
# A tibble: 5 x 3
  DIFFWALK diffwalk      n
  <dbl>     <dbl> <int>
1     1     1    870
2     2     0   6348
3     7    NA    30
4     9    NA     2
5    NA    NA   162
```

1	1	1	1482
2	2	0	5738
3	7	NA	19
4	9	NA	2
5	NA	NA	171

6.6.10.24 DIFFDRES and its cleanup to diffdress

DIFFDRES, the Difficulty Dressing or Bathing variable, is the response to “Do you have difficulty dressing or bathing?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(diffdress = DIFFDRES,
         diffdress = replace(diffdress, diffdress %in% c(7, 9), NA),
         diffdress = replace(diffdress, diffdress == 2, 0))

smart_ohio_raw |> count(DIFFDRES, diffdress)
```

```
# A tibble: 5 x 3
  DIFFDRES diffdress      n
    <dbl>     <dbl> <int>
1       1         1   352
2       2         0  6868
3       7        NA    12
4       9        NA     1
5      NA        NA   179
```

6.6.10.25 DIFFALON and its cleanup to diffalone

DIFFALON, the Difficulty Doing Errands Alone variable, is the response to “Because of a physical, mental, or emotional condition, do you have difficulty doing errands alone such as visiting a doctor’s office or shopping?”

- 1 = Yes
- 2 = No

- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(diffalone = DIFFALON,
         diffalone = replace(diffalone, diffalone %in% c(7, 9), NA),
         diffalone = replace(diffalone, diffalone == 2, 0))

smart_ohio_raw |> count(DIFFALON, diffalone)
```

```
# A tibble: 5 x 3
  DIFFALON diffalone     n
    <dbl>     <dbl> <int>
1         1         1  636
2         2         0 6560
3         7        NA   15
4         9        NA    4
5        NA        NA  197
```

6.6.11 Tobacco Use (2 items)

6.6.11.1 SMOKE100 and its cleanup to smoke100

SMOKE100, the Smoked at Least 100 Cigarettes variable, is the response to “Have you smoked at least 100 cigarettes in your entire life? [Note: 5 packs = 100 cigarettes]”

- 1 = Yes
- 2 = No
- 7 = Don't know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(smoke100 = SMOKE100,
         smoke100 = replace(smoke100, smoke100 %in% c(7, 9), NA),
         smoke100 = replace(smoke100, smoke100 == 2, 0))

smart_ohio_raw |> count(SMOKE100, smoke100)
```



```
# A tibble: 5 x 3
  SMOKE100 smoke100     n
    <dbl>    <dbl> <int>
1         1         1 3294
2         2         0 3881
3         7        NA   31
4         9        NA    4
5        NA        NA  202
```

6.6.11.2 `_SMOKER3` and its cleanup to `smoker`

`_SMOKER3`, is a calculated variable which categorizes subjects by their smoking status:

- 1 = Current smoker who smokes daily
- 2 = Current smoker but not every day
- 3 = Former smoker
- 4 = Never smoked
- 9 = Don't Know / Refused / Missing

We'll reclassify this as a factor with appropriate labels and NAs.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(smoker = fct_recode(factor(`_SMOKER3`),
                                "Current_daily" = "1",
                                "Current_not_daily" = "2",
                                "Former" = "3",
                                "Never" = "4",
                                NULL = "9"))

smart_ohio_raw |> count(`_SMOKER3`, smoker)
```

```
# A tibble: 5 x 3
  `_SMOKER3` smoker           n
    <dbl> <fct>          <int>
1         1 Current_daily     990
2         2 Current_not_daily  300
3         3 Former          1999
4         4 Never           3881
5         9 <NA>             242
```

6.6.12 E-Cigarettes (2 items)

6.6.12.1 ECIGARET and its cleanup to ecig_ever

ECIGARET, the Ever used an e-cigarette variable, is the response to “Have you ever used an e-cigarette or other electronic vaping product, even just one time, in your entire life?”

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(ecig_ever = ECIGARET,
         ecig_ever = replace(ecig_ever, ecig_ever %in% c(7, 9), NA),
         ecig_ever = replace(ecig_ever, ecig_ever == 2, 0))

smart_ohio_raw |> count(ECIGARET, ecig_ever)
```

```
# A tibble: 5 x 3
  ECIGARET ecig_ever      n
  <dbl>     <dbl> <int>
1       1         1  1354
2       2         0  5799
3       7        NA     9
4       9        NA     3
5      NA        NA    247
```

6.6.12.2 _ECIGSTS and its cleanup to ecigs

_ECIGSTS, is a calculated variable which categorizes subjects by their smoking status:

- 1 = Current and uses daily
- 2 = Current user but not every day
- 3 = Former user
- 4 = Never used e-cigarettes
- 9 = Don’t Know / Refused / Missing

We’ll reclassify this as a factor with appropriate labels and NAs.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(ecigs = fct_recode(factor(`_ECIGSTS`),
    "Current_daily" = "1",
    "Current_not_daily" = "2",
    "Former" = "3",
    "Never" = "4",
    NULL = "9"))

smart_ohio_raw |> count(`_ECIGSTS`, ecigs)
```

```
# A tibble: 5 x 3
  `_ECIGSTS` ecigs      n
    <dbl> <fct>    <int>
1         1 Current_daily  102
2         2 Current_not_daily 165
3         3 Former      1085
4         4 Never       5799
5         9 <NA>        261
```

6.6.13 Alcohol Consumption (6 items)

6.6.13.1 ALCDAY5 and its cleanup to alcdays

ALCDAY5, the Days in past 30 had alcoholic beverage variable, is the response to “During the past 30 days, how many days per week or per month did you have at least one drink of any alcoholic beverage such as beer, wine, a malt beverage or liquor?”

- 101-107 = # of days per week (101 = 1 day per week, 107 = 7 days per week)
- 201-230 = # of days in past 30 days (201 = 1 day in last 30, 230 = 30 days in last 30)
- 777 = Don’t know/Not sure
- 888 = No drinks in past 30 days
- 999 = Refused
- BLANK = Not asked or Missing

We’re going to convert this to a single numeric value. Answers in days per week (in the past 7 days) will be converted (after rounding) to days in the past 30. This is a little bit of a mess, really, but we can do it.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(alcdays = as.numeric(ALCDAY5)) |>
  mutate(alcdays = replace(alcdays, alcdays == 888, 0),
```

```

      alcdays = replace(alcdays, alcdays %in% c(777, 999), NA)) |>
mutate(alcdays = case_when(ALCDAY5 > 199 & ALCDAY5 < 231 ~ ALCDAY5 - 200,
      ALCDAY5 > 100 & ALCDAY5 < 108 ~ round((ALCDAY5 - 100)*30/7,0),
      TRUE ~ alcdays))

smart_ohio_raw |> count(ALCDAY5, alcdays)

```

```

# A tibble: 39 x 3
  ALCDAY5 alcdays      n
    <dbl>   <dbl> <int>
1     101       4   263
2     102       9   197
3     103      13   142
4     104      17    76
5     105      21    53
6     106      26    18
7     107      30   114
8     201       1   621
9     202       2   448
10    203       3   233
# i 29 more rows

```

6.6.13.2 AVEDRNK2 and its cleanup to avgdrinks

AVEDRNK2, the Avg alcoholic drinks per day in past 30 variable, is the response to “One drink is equivalent to a 12-ounce beer, a 5-ounce glass of wine, or a drink with one shot of liquor. During the past 30 days, on the days when you drank, about how many drinks did you drink on the average? (A 40 ounce beer would count as 3 drinks, or a cocktail drink with 2 shots would count as 2 drinks.)”

- 1-76 = # of drinks per day
- 77 = Don’t know/Not sure
- 99 = Refused
- BLANK = Not asked or Missing (always happens when ALCDAY5 = 777, 888 or 999)

```

smart_ohio_raw <- smart_ohio_raw |>
  mutate(avgdrinks = AVEDRNK2,
         avgdrinks = replace(avgdrinks, avgdrinks > 76, NA))

smart_ohio_raw |> count(AVEDRNK2, avgdrinks) |> tail()

```

```
# A tibble: 6 x 3
  AVEDRINK2 avgdrinks      n
    <dbl>      <dbl> <int>
1      42        42     1
2      60        60     2
3      76        76     1
4      77        NA    46
5      99        NA     5
6      NA        NA  3876
```

6.6.13.3 MAXDRINKS and its cleanup to maxdrinks

MAXDRINKS, the most drinks on a single occasion in the past 30 days variable, is the response to “During the past 30 days, what is the largest number of drinks you had on any occasion?”

- 1-76 = # of drinks
- 77 = Don’t know/Not sure
- 99 = Refused
- BLANK = Not asked or Missing (always happens when ALCDAY5 = 777, 888 or 999)

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(maxdrinks = MAXDRINKS,
         maxdrinks = replace(maxdrinks, maxdrinks > 76, NA))

smart_ohio_raw |> count(MAXDRINKS, maxdrinks) |> tail()
```

```
# A tibble: 6 x 3
  MAXDRINKS maxdrinks      n
    <dbl>      <dbl> <int>
1      42        42     1
2      48        48     1
3      76        76     2
4      77        NA    94
5      99        NA    11
6      NA        NA  3899
```

6.6.13.4 _RFBING5 and its cleanup to binge

_RFBING5 identifies binge drinkers (males having five or more drinks on one occasion, females having four or more drinks on one occasion in the past 30 days)

The values are

- 1 = No
- 2 = Yes
- 9 = Don't Know / Refused / Missing

People who reported no `alcdays` are reported here as “No”, so we’ll adjust this into an indicator variable, and create the necessary NAs.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(binge = `_RFBING5` - 1,
         binge = replace(binge, binge > 1, NA))

smart_ohio_raw |> count(`_RFBING5`, binge)
```

```
# A tibble: 3 x 3
  `_RFBING5` binge      n
    <dbl> <dbl> <int>
1         1     0  6035
2         2     1  1000
3         9    NA   377
```

6.6.13.5 `_DRNKWEK` and its cleanup to `drinks_wk`

`_DRNKWEK` provides the computed number of alcoholic drinks per week, with two implied decimal places. The code 99900 is used for “Don’t know / Not sure / Refused / Missing” so we’ll fix that, and also divide by 100 to get an average with a decimal point.

Note: We’re also going to treat all results of 100 or more drinks per week as incorrect, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(drinks_wk = `_DRNKWEK` / 100,
         drinks_wk = replace(drinks_wk, drinks_wk > 99, NA))

smart_ohio_raw |> count(`_DRNKWEK`, drinks_wk) |> tail(12)
```

```
# A tibble: 12 x 3
  `_DRNKWEK` drinks_wk      n
    <dbl>      <dbl> <int>
1     9333      93.3     2
2    10000      NA     1
3    10500      NA     2
```

4	11667	NA	1
5	14000	NA	2
6	16800	NA	2
7	17500	NA	1
8	18200	NA	1
9	28000	NA	1
10	29400	NA	1
11	53200	NA	1
12	99900	NA	379

6.6.13.6 `_RFDRHV5` and its cleanup to `drink_heavy`

`_RFDRHV5` identifies heavy drinkers (males having 14 or more drinks per week, females having 7 or more drinks per week)

The values are

- 1 = No
- 2 = Yes
- 9 = Don't Know / Refused / Missing

People who reported no `alcdays` are reported here as “No”, so we’ll adjust this into an indicator variable, and create the necessary NAs.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(drink_heavy = `_RFDRHV5` - 1,
         drink_heavy = replace(drink_heavy, drink_heavy > 1, NA))

smart_ohio_raw |> count(`_RFDRHV5`, drink_heavy)
```

```
# A tibble: 3 x 3
  `_RFDRHV5` drink_heavy      n
    <dbl>         <dbl> <int>
1         1             0  6607
2         2             1   426
3         9            NA   379
```

6.6.14 Fruits and Vegetables (8 items)

6.6.14.1 `_FRUTSU1` and its cleanup to `fruit_day`

`_FRUTSU1` provides the computed number of fruit servings consumed per day, with two implied decimal places. We’ll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here, following some CDC procedures.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(fruit_day = `_FRUTSU1` / 100,
         fruit_day = replace(fruit_day, fruit_day > 16, NA))

smart_ohio_raw |> count(`_FRUTSU1`, fruit_day) |> tail()
```

```
# A tibble: 6 x 3
  `_FRUTSU1` fruit_day      n
    <dbl>      <dbl> <int>
1      913      9.13      1
2     1000     10      4
3     1400     14      1
4     3000     NA      1
5     7600     NA      1
6        NA     NA     555
```

6.6.14.2 _VEGESU1 and its cleanup to veg_day

_VEGESU1 provides the computed number of vegetable servings consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 23 servings per day as implausible, and thus indicate them as missing data here, following some CDC procedures.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(veg_day = `_VEGESU1` / 100,
         veg_day = replace(veg_day, veg_day > 23, NA))

smart_ohio_raw |> count(`_VEGESU1`, veg_day) |> tail()
```

```
# A tibble: 6 x 3
  `_VEGESU1` veg_day      n
    <dbl>      <dbl> <int>
1     1414     14.1      1
2     1603     16.0      1
3     1891     18.9      1
4     2167     21.7      1
5     3150     NA      1
6        NA     NA     666
```


6.6.14.3 FTJUDA2_ and its cleanup to eat_juice

FTJUDA2_ provides the servings of fruit juice consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(eat_juice = `FTJUDA2_` / 100,
         eat_juice = replace(eat_juice, eat_juice > 16, NA))

smart_ohio_raw |> count(`FTJUDA2_`, eat_juice) |> tail()
```

```
# A tibble: 6 x 3
  FTJUDA2_ eat_juice      n
  <dbl>    <dbl> <int>
1     500         5     6
2     600         6     1
3     700         7     1
4    1200        12     1
5    7500        NA     1
6      NA        NA   469
```

6.6.14.4 FRUTDA2_ and its cleanup to eat_fruit

FRUTDA2_ provides the servings of fruit consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(eat_fruit = `FRUTDA2_` / 100,
         eat_fruit = replace(eat_fruit, eat_fruit > 16, NA))

smart_ohio_raw |> count(`FRUTDA2_`, eat_fruit) |> tail()
```

```
# A tibble: 6 x 3
  FRUTDA2_ eat_fruit      n
  <dbl>    <dbl> <int>
1     700         7     5
```

2	800	8	3
3	900	9	1
4	1000	10	1
5	3000	NA	1
6	NA	NA	456

6.6.14.5 GREND1_ and its cleanup to eat_greenveg

GREND1_ provides the servings of dark green vegetables consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(eat_greenveg = `GREND1_` / 100,
         eat_greenveg = replace(eat_greenveg, eat_greenveg > 16, NA))

smart_ohio_raw |> count(`GREND1_`, eat_greenveg) |> tail()
```

```
# A tibble: 6 x 3
  GREND1_ eat_greenveg     n
  <dbl>      <dbl> <int>
1     700          7     4
2     786         7.86     1
3     800          8     2
4    2000         NA     1
5    3000         NA     1
6      NA         NA    447
```

6.6.14.6 FRNCHDA_ and its cleanup to eat_fries

FRNCHDA_ provides the servings of french fries consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(eat_fries = `FRNCHDA_` / 100,
         eat_fries = replace(eat_fries, eat_fries > 16, NA))

smart_ohio_raw |> count(`FRNCHDA_`, eat_fries) |> tail()
```

```
# A tibble: 6 x 3
  FRNCHDA_ eat_fries      n
  <dbl>    <dbl> <int>
1     300         3      9
2     314        3.14     1
3     400         4      3
4     500         5      1
5     700         7      1
6      NA      NA    453
```

6.6.14.7 POTADA1_ and its cleanup to eat_potato

POTADA1_ provides the servings of potatoes consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(eat_potato = `POTADA1_` / 100,
         eat_potato = replace(eat_potato, eat_potato > 16, NA))

smart_ohio_raw |> count(`POTADA1_`, eat_potato) |> tail()
```

```
# A tibble: 6 x 3
  POTADA1_ eat_potato      n
  <dbl>    <dbl> <int>
1     314        3.14     1
2     329        3.29     1
3     400         4      3
4     471        4.71     1
5     700         7      1
6      NA      NA    501
```

6.6.14.8 VEGEDA2_ and its cleanup to eat_otherveg

VEGEDA2_ provides the servings of other vegetables consumed per day, with two implied decimal places. We'll divide by 100 to insert the decimal point.

Note: We're also going to treat all results exceeding 16 servings per day as implausible, and thus indicate them as missing data here.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(eat_otherveg = `VEGEDA2_` / 100,
         eat_otherveg = replace(eat_otherveg, eat_otherveg > 16, NA))

smart_ohio_raw |> count(`VEGEDA2_`, eat_otherveg) |> tail()
```

```
# A tibble: 6 x 3
  VEGEDA2_ eat_otherveg     n
  <dbl>      <dbl> <int>
1     600          6     3
2     700          7    11
3     800          8     1
4    1000         10     2
5    1100         11     1
6      NA         NA   509
```

6.6.15 Exercise and Physical Activity (8 items)

6.6.15.1 _TOTINDA and its cleanup to exerany

_TOTINDA, the Exercise in Past 30 Days variable, is the response to “During the past month, other than your regular job, did you participate in any physical activities or exercises such as running, calisthenics, golf, gardening, or walking for exercise?”

- 1 = Yes
- 2 = No
- 7 = Don't know/Not sure
- 9 = Refused
- BLANK = Not asked or missing

This is just like HLTHPLAN.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(exerany = `_TOTINDA`,
         exerany = replace(exerany, exerany %in% c(7, 9), NA),
         exerany = replace(exerany, exerany == 2, 0))

smart_ohio_raw |> count(`_TOTINDA`, exerany)
```

```
# A tibble: 3 x 3
  `_TOTINDA` exerany      n
    <dbl>    <dbl> <int>
1         1         1  4828
2         2         0  2137
3         9        NA   447
```

6.6.15.2 `\_PACAT1` and its cleanup to activity

`\_PACAT1` contains physical activity categories, estimated from responses to the BRFSS. The categories are:

- 1 = Highly Active
- 2 = Active
- 3 = Insufficiently Active
- 4 = Inactive
- 9 = Don't Know / Not Sure / Refused / Missing

So we'll create a factor.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(activity = factor(`_PACAT1`),
         activity = fct_recode(activity,
                                "Highly_Active" = "1",
                                "Active" = "2",
                                "Insufficiently_Active" = "3",
                                "Inactive" = "4",
                                NULL = "9"))

smart_ohio_raw |> count(`_PACAT1`, activity)
```

```
# A tibble: 5 x 3
  `_PACAT1` activity      n
    <dbl> <fct>    <int>
1         1 Highly_Active  4828
2         2 Active        2137
3         3 Insufficiently_Active  447
4         4 Inactive
5         9 NULL
```

1	1 Highly_Active	2053
2	2 Active	1132
3	3 Insufficiently_Active	1293
4	4 Inactive	2211
5	9 <NA>	723

6.6.15.3 _PAINDX1 and its cleanup to rec_aerobic

_PAINDX1 indicates whether the respondent's stated levels of physical activity meet recommendations for aerobic activity. The responses are:

- 1 = Yes
- 2 = No
- 9 = Don't know/Not sure/Refused/Missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(rec_aerobic = 2 - `_PAINDX1`,
         rec_aerobic = replace(rec_aerobic, rec_aerobic < 0, NA))

smart_ohio_raw |> count(`_PAINDX1`, rec_aerobic)
```

```
# A tibble: 3 x 3
  `_PAINDX1` rec_aerobic      n
    <dbl>         <dbl> <int>
1         1             1 3228
2         2             0 3504
3         9            NA  680
```

6.6.15.4 _PASTRNG and its cleanup to rec_strength

_PASTRNG indicates whether the respondent's stated levels of physical activity meet recommendations for strength-building activity. The responses are:

- 1 = Yes
- 2 = No
- 9 = Don't know/Not sure/Refused/Missing

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(rec_strength = 2 - `_PASTRNG`,
         rec_strength = replace(rec_strength, rec_strength < 0, NA))

smart_ohio_raw |> count(`_PASTRNG`, rec_strength)
```

```
# A tibble: 3 x 3
  `_PASTRNG` rec_strength      n
    <dbl>         <dbl> <int>
1       1         1     1852
2       2         0     5004
3       9        NA      556
```

6.6.15.5 EXTRACT11 and its cleanup to exer1_type

Respondents are asked “What type of physical activity or exercise did you spend the most time doing during the past month?” and these responses are gathered into a set of 76 named categories, including an “other” category. Codes 77 (Don’t Know / Not Sure) and 99 (Refused) are dropped into NA in my code below, and Code 98 (“Other type of activity”) remains. Then I went through the tedious work of converting the factor levels from numbers to names, following the value labels provided by BRFSS.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(exer1_type = factor(EXTRACT11),
         exer1_type = fct_recode(
           exer1_type,
           "Active Gaming Devices" = "1",
           "Aerobics video or class" = "2",
           "Backpacking" = "3",
           "Badminton" = "4",
           "Basketball" = "5",
           "Bicycling machine" = "6",
           "Bicycling" = "7",
           "Boating" = "8",
           "Bowling" = "9",
           "Boxing" = "10",
           "Calisthenics" = "11",
           "Canoeing" = "12",
           "Carpentry" = "13",
           "Dancing" = "14",
           "Elliptical machine" = "15",
           "Fishing" = "16",
           "Frisbee" = "17",
           "Gardening" = "18",
           "Golf with cart" = "19",
           "Golf without cart" = "20",
           "Handball" = "21",
           "Hiking" = "22",
```

```
"Hockey" = "23",
"Horseback riding" = "24",
"Hunting large game" = "25",
"Hunting small game" = "26",
"Inline skating" = "27",
"Jogging" = "28",
"Lacrosse" = "29",
"Mountain climbing" = "30",
"Mowing lawn" = "31",
"Paddleball" = "32",
"Painting house" = "33",
"Pilates" = "34",
"Racquetball" = "35",
"Raking lawn" = "36",
"Running" = "37",
"Rock climbing" = "38",
"Rope skipping" = "39",
"Rowing machine" = "40",
"Rugby" = "41",
"Scuba diving" = "42",
"Skateboarding" = "43",
"Skating" = "44",
"Sledding" = "45",
"Snorkeling" = "46",
"Snow blowing" = "47",
"Snow shoveling" = "48",
"Snow skiing" = "49",
"Snowshoeing" = "50",
"Soccer" = "51",
"Softball/Baseball" = "52",
"Squash" = "53",
"Stair Climbing" = "54",
"Stream fishing" = "55",
"Surfing" = "56",
"Swimming" = "57",
"Swimming in laps" = "58",
"Table tennis" = "59",
"Tai Chi" = "60",
"Tennis" = "61",
"Touch football" = "62",
"Volleyball" = "63",
"Walking" = "64",
```



```

    "Waterskiing" = "66",
    "Weight lifting" = "67",
    "Wrestling" = "68",
    "Yoga" = "69",
    "Child Care" = "71",
    "Farm Work" = "72",
    "Household Activities" = "73",
    "Martial Arts" = "74",
    "Upper Body Cycle" = "75",
    "Yard Work" = "76",
    "Other Activities" = "98",
    NULL = "77",
    NULL = "99")
)

```

Warning: There was 1 warning in `mutate()`.
 i In argument: `exer1\_type = fct\_recode(...)`.
 Caused by warning:
 ! Unknown levels in `f`: 3, 17, 21, 32, 36, 41, 42, 45, 47, 53, 55, 56, 59

The warning generated here is caused by the fact that some of the available types of exercise were not mentioned by people in our sample. Looking at the last few results, we can see how many people fell into several categories.

```
smart_ohio_raw |> count(EXTRACT11, exer1_type) |> tail()
```

```
# A tibble: 6 x 3
  EXTRACT11 exer1_type      n
    <dbl> <fct>      <int>
1      75 Upper Body Cycle      6
2      76 Yard Work           78
3      77 <NA>                10
4      98 Other Activities    276
5      99 <NA>                 4
6       NA <NA>            2588
```

The most common activities are:

```
smart_ohio_raw |> count(exer1_type, sort = TRUE) |> head(10)
```

```
# A tibble: 10 x 2
  exer1_type      n
  <fct>          <int>
1 Walking      2605
2 <NA>         2602
3 Running       324
4 Other Activities 276
5 Gardening     242
6 Weight lifting 189
7 Aerobics video or class 103
8 Bicycling machine 103
9 Bicycling      96
10 Golf with cart  90
```

6.6.15.6 EXTRACT21 and its cleanup to exer2_type

As a follow-up, respondents are asked “What other type of physical activity gave you the next most exercise during the past month?” and these responses are also gathered into the same set of 76 named categories, including an “other” category, but now also adding a “No Other Activity” category (code 88). Codes 77 (Don’t Know / Not Sure) and 99 (Refused) are dropped into NA in my code below, and Code 98 (“Other type of activity”) remains. Then I went through the tedious work of converting the factor levels from numbers to names, following the value labels provided by BRFSS. I’m sure there’s a better way to do this.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(exer2_type = factor(EXTRACT21),
         exer2_type = fct_recode(
           exer2_type,
           "Active Gaming Devices" = "1",
           "Aerobics video or class" = "2",
           "Backpacking" = "3",
           "Badminton" = "4",
           "Basketball" = "5",
           "Bicycling machine" = "6",
           "Bicycling" = "7",
           "Boating" = "8",
           "Bowling" = "9",
           "Boxing" = "10",
           "Calisthenics" = "11",
           "Canoeing" = "12",
           "Carpentry" = "13",
           "Dancing" = "14",
```

```
"Elliptical machine" = "15",
"Fishing" = "16",
"Frisbee" = "17",
"Gardening" = "18",
"Golf with cart" = "19",
"Golf without cart" = "20",
"Handball" = "21",
"Hiking" = "22",
"Hockey" = "23",
"Horseback riding" = "24",
"Hunting large game" = "25",
"Hunting small game" = "26",
"Inline skating" = "27",
"Jogging" = "28",
"Lacrosse" = "29",
"Mountain climbing" = "30",
"Mowing lawn" = "31",
"Paddleball" = "32",
"Painting house" = "33",
"Pilates" = "34",
"Racquetball" = "35",
"Raking lawn" = "36",
"Running" = "37",
"Rock climbing" = "38",
"Rope skipping" = "39",
"Rowing machine" = "40",
"Rugby" = "41",
"Scuba diving" = "42",
"Skateboarding" = "43",
"Skating" = "44",
"Sledding" = "45",
"Snorkeling" = "46",
"Snow blowing" = "47",
"Snow shoveling" = "48",
"Snow skiing" = "49",
"Snowshoeing" = "50",
"Soccer" = "51",
"Softball/Baseball" = "52",
"Squash" = "53",
"Stair Climbing" = "54",
"Stream fishing" = "55",
"Surfing" = "56",
```

```

        "Swimming" = "57",
        "Swimming in laps" = "58",
        "Table tennis" = "59",
        "Tai Chi" = "60",
        "Tennis" = "61",
        "Touch football" = "62",
        "Volleyball" = "63",
        "Walking" = "64",
        "Waterskiing" = "66",
        "Weight lifting" = "67",
        "Wrestling" = "68",
        "Yoga" = "69",
        "Child Care" = "71",
        "Farm Work" = "72",
        "Household Activities" = "73",
        "Martial Arts" = "74",
        "Upper Body Cycle" = "75",
        "Yard Work" = "76",
        "No Other Activity" = "88",
        "Other Activities" = "98",
        NULL = "77",
        NULL = "99")
)

```

Warning: There was 1 warning in `mutate()`.
 i In argument: `exer2\_type = fct\_recode(...)`.
 Caused by warning:
 ! Unknown levels in `f`: 3, 21, 30, 39, 41, 46, 50, 62

```
smart_ohio_raw |> count(EXTRACT21, exer2_type) |> tail()
```

```

# A tibble: 6 x 3
  EXTRACT21 exer2_type      n
  <dbl> <fct>          <int>
1      76 Yard Work      153
2      77 <NA>           26
3      88 No Other Activity 1854
4      98 Other Activities   246
5      99 <NA>            19
6      NA <NA>          2627

```

The most common activity types in this group are:

```
smart_ohio_raw |> count(exer2_type, sort = TRUE) |> head(10)
```

```
# A tibble: 10 x 2
  exer2_type      n
  <fct>         <int>
1 <NA>         2672
2 No Other Activity 1854
3 Walking        629
4 Weight lifting   272
5 Other Activities 246
6 Gardening       202
7 Household Activities 169
8 Yard Work       153
9 Running         148
10 Bicycling      118
```

6.6.15.7 _MINAC11 and its cleanup to exer1_min

_MINAC11 is minutes of physical activity per week for the first activity (listed as `exer1_type` above.) Since there are only about 10,080 minutes in a typical week, we'll treat as implausible any values larger than 4200 minutes (which would indicate 70 hours per week.)

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(exer1_min = `_MINAC11`,
         exer1_min = replace(exer1_min, exer1_min > 4200, NA))

smart_ohio_raw |> count(`_MINAC11`, exer1_min) |> tail()
```

```
# A tibble: 6 x 3
  `_MINAC11` exer1_min      n
  <dbl>      <dbl> <int>
1    3780    3780      8
2    3959    3959      1
3    3960    3960      1
4    4193    4193      6
5   27000     NA       1
6      NA     NA    2760
```

6.6.15.8 _MINAC21 and its cleanup to exer2_min

_MINAC21 is minutes of physical activity per week for the second activity (listed as `exer2_type` above.) Again, we'll treat as implausible any values larger than 4200 minutes (which would indicate 70 hours per week.)

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(exer2_min = `_MINAC21`,
         exer2_min = replace(exer2_min, exer2_min > 4200, NA))

smart_ohio_raw |> count(`_MINAC21`, exer2_min) |> tail()
```

```
# A tibble: 6 x 3
  `_MINAC21` exer2_min      n
    <dbl>      <dbl> <int>
1     3360      3360     3
2     3780      3780     7
3     4193      4193     3
4     6120        NA     1
5     8400        NA     1
6        NA        NA    2770
```

6.6.16 Seatbelt Use (1 item)

6.6.16.1 SEATBELT and its cleanup to seatbelt

This question asks “How often do you use seat belts when you drive or ride in a car?” Possible responses are:

- 1 = Always
- 2 = Nearly always
- 3 = Sometimes
- 4 = Seldom
- 5 = Never
- 7 = Don't know / Not sure
- 8 = Never drive or ride in a car
- 9 = Refused

We'll treat codes 7, 8 and 9 as NA, and turn this into a factor.

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(seatbelt = fct_recode(factor(SEATBELT),
                                "Always" = "1",
                                "Nearly_always" = "2",
                                "Sometimes" = "3",
                                "Seldom" = "4",
                                "Never" = "5",
                                NULL = "7",
                                NULL = "8",
                                NULL = "9"))

smart_ohio_raw |> count(SEATBELT, seatbelt)
```

```
# A tibble: 9 x 3
  SEATBELT seatbelt      n
  <dbl> <fct>      <int>
1     1 Always      6047
2     2 Nearly_always 409
3     3 Sometimes    191
4     4 Seldom       81
5     5 Never       148
6     7 <NA>         7
7     8 <NA>        21
8     9 <NA>         2
9    NA <NA>       506
```

6.6.17 Immunization (3 items)

6.6.17.1 FLUSHOT6 and its cleanup to vax_flu

FLUSHOT6 gives the response to “During the past 12 months, have you had either a flu shot or a flu vaccine that was sprayed in your nose?” The responses are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(vax_flu = 2 - FLUSHOT6,
         vax_flu = replace(vax_flu, vax_flu < 0, NA))

smart_ohio_raw |> count(FLUSHOT6, vax_flu)
```

```
# A tibble: 5 x 3
  FLUSHOT6 vax_flu      n
    <dbl>   <dbl> <int>
1         1         1 3453
2         2         0 3410
3         7        NA   26
4         9        NA    3
5        NA        NA  520
```

6.6.17.2 PNEUVAC3 and its cleanup to vax_pneumo

PNEUVAC3 gives the response to “A pneumonia shot or pneumococcal vaccine is usually given only once or twice in a person’s lifetime and is different from the flu shot. Have you ever had a pneumonia shot?” The responses are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(vax_pneumo = 2 - PNEUVAC3,
         vax_pneumo = replace(vax_pneumo, vax_pneumo < 0, NA))

smart_ohio_raw |> count(PNEUVAC3, vax_pneumo)
```

```
# A tibble: 5 x 3
  PNEUVAC3 vax_pneumo      n
    <dbl>   <dbl> <int>
1         1         1 3112
2         2         0 3262
3         7        NA   509
4         9        NA    3
5        NA        NA  526
```


6.6.17.3 SHINGLE2 and its cleanup to vax_shingles

SHINGLE2 gives the response to “Have you ever had the shingles or zoster vaccine?” The responses are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(vax_shingles = 2 - SHINGLE2,
         vax_shingles = replace(vax_shingles, vax_shingles < 0, NA))

smart_ohio_raw |> count(SHINGLE2, vax_shingles)
```

```
# A tibble: 4 x 3
  SHINGLE2 vax_shingles      n
    <dbl>      <dbl> <int>
1       1          1  1503
2       2          0  2979
3       7         NA    78
4      NA         NA  2852
```

6.6.18 HIV/AIDS (2 items)

6.6.18.1 HIVTST6 and its cleanup to hiv_test

HIVTST6 gives the response to “Have you ever been tested for HIV? Do not count tests you may have had as part of a blood donation. Include testing fluid from your mouth.” The responses are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(hiv_test = 2 - HIVTST6,
         hiv_test = replace(hiv_test, hiv_test < 0, NA))

smart_ohio_raw |> count(HIVTST6, hiv_test)
```

```
# A tibble: 5 x 3
  HIVTST6 hiv_test      n
  <dbl>    <dbl> <int>
1      1      1    2017
2      2      0   4565
3      7     NA    260
4      9     NA     14
5     NA     NA    556
```

6.6.18.2 HIVRISK5 and its cleanup to hiv_risk

HIVRISK5 gives the response to “I am going to read you a list. When I am done, please tell me if any of the situations apply to you. You do not need to tell me which one. You have injected any drug other than those prescribed for you in the past year. You have been treated for a sexually transmitted disease or STD in the past year. You have given or received money or drugs in exchange for sex in the past year.” The responses are:

- 1 = Yes
- 2 = No
- 7 = Don’t know/Not sure
- 9 = Refused

```
smart_ohio_raw <- smart_ohio_raw |>
  mutate(hiv_risk = 2 - HIVRISK5,
         hiv_risk = replace(hiv_risk, hiv_risk < 0, NA))

smart_ohio_raw |> count(HIVRISK5, hiv_risk)
```

```
# A tibble: 5 x 3
  HIVRISK5 hiv_risk      n
  <dbl>    <dbl> <int>
1      1      1    277
2      2      0   6537
3      7     NA      2
4      9     NA     17
5     NA     NA    579
```

6.7 Imputing Age and Income as Quantitative from Thin Air

This section is purely for teaching purposes. I would never use the variables created in this section for research work.

6.7.1 age_imp: Imputing Age Data

I want a quantitative age variable, so I'm going to create an imputed `age_imp` value for each subject based on their `agegroup`. For each age group, I will assume that each of the ages represented by a value in that age group will be equally likely, and will draw from the relevant uniform distribution to impute age.

```
set.seed(2020432002)

smart_ohio_raw <- smart_ohio_raw |>
  mutate(age_low = as.numeric(str_sub(as.character(agegroup), 1, 2))) |>
  mutate(age_high = as.numeric(str_sub(as.character(agegroup), 4, 5))) |>
  rowwise() |>
  mutate(age_imp = ifelse(!is.na(agegroup),
                          round(runif(1, min = age_low, max = age_high), 0),
                          NA))

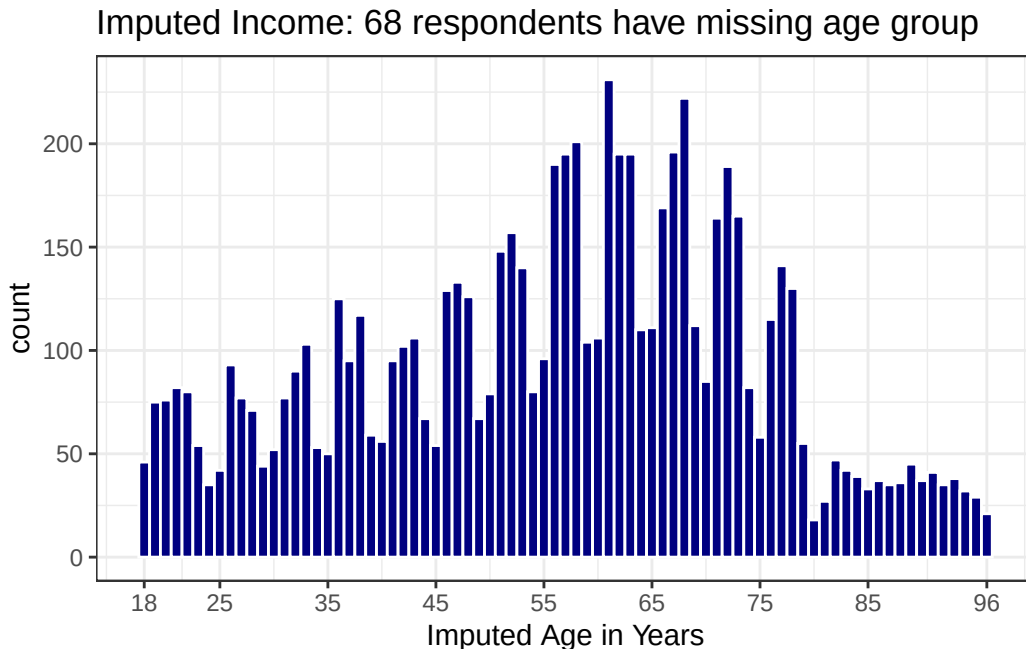
smart_ohio_raw |> count(agegroup, age_imp) #|> tail()
```

```
# A tibble: 80 x 3
# Rowwise:
  agegroup age_imp    n
  <fct>      <dbl> <int>
1 18-24      18     46
2 18-24      19     75
3 18-24      20     76
4 18-24      21     82
5 18-24      22     80
6 18-24      23     54
7 18-24      24     35
8 25-29      25     42
9 25-29      26     93
10 25-29      27     77
# i 70 more rows
```

Here is a histogram of the `age_imp` variable.

```
ggplot(smart_ohio_raw, aes(x = age_imp)) +
  geom_histogram(fill = "navy", col = "white",
                 binwidth = 1) +
  scale_x_continuous(breaks = c(18, 25, 35, 45, 55, 65, 75, 85, 96)) +
```

```
labs(x = "Imputed Age in Years",
     title = paste0("Imputed Income: ",
                     sum(is.na(smart_ohio_raw$age_imp)),
                     " respondents have missing age group"))
```



6.7.2 inc_imp: Imputing Income Data

I want a quantitative income variable, so I'm going to create an imputed `inc_imp` value for each subject based on their `incomegroup`. For most income groups, I will assume that each of the incomes represented by a value in that income group will be equally likely, and will draw from the relevant uniform distribution to impute income. The exception is the highest income group, where I will impute a value drawn from a distribution that places all values at \$75,000 or more, but has a substantial right skew and long tail.

```
set.seed(2020432001)

smart_ohio_raw <- smart_ohio_raw |>
  mutate(inc_imp = case_when(
    incomegroup == "0-9K" ~ round(runif(1, min = 100, max = 9999)),
    incomegroup == "10-14K" ~ round(runif(1, min = 10000, max = 14999)),
    incomegroup == "15-19K" ~ round(runif(1, min = 15000, max = 19999)),
```

```

incomegroup == "20-24K" ~ round(runif(1, min = 20000, max = 24999)),
incomegroup == "25-34K" ~ round(runif(1, min = 25000, max = 34999)),
incomegroup == "35-49K" ~ round(runif(1, min = 35000, max = 49999)),
incomegroup == "50-74K" ~ round(runif(1, min = 50000, max = 74999)),
incomegroup == "75K+" ~ round((rnorm(n = 1, mean = 0, sd = 300)^2) + 74999)))

smart_ohio_raw |> count(incomegroup, inc_imp) |> tail()

```

```

# A tibble: 6 x 3
# Rowwise:
  incomegroup inc_imp      n
  <fct>        <dbl> <int>
1 75K+         774009      1
2 75K+         798174      1
3 75K+         806161      1
4 75K+         847758      1
5 75K+        1085111      1
6 <NA>          NA     1310

```

Here are density plots of the `inc_imp` variable. The top picture shows the results on a linear scale, and the bottom shows them on a log (base 10) scale.

```

p1 <- ggplot(smart_ohio_raw, aes(x = inc_imp/1000)) +
  geom_density(fill = "darkgreen", col = "white") +
  labs(x = "Imputed Income in Thousands of Dollars",
       title = "Imputed Income on the Linear scale") +
  scale_x_continuous(breaks = c(25, 75, 250, 1000))

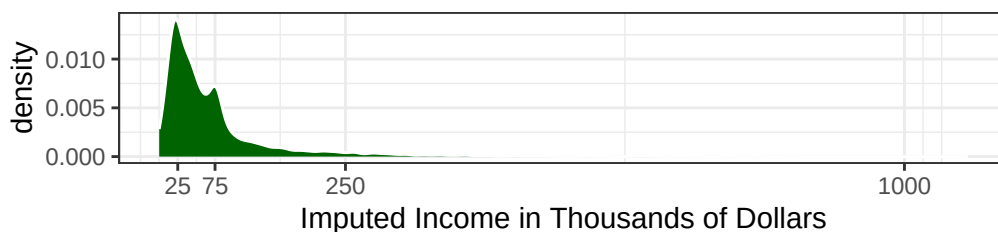
p2 <- ggplot(smart_ohio_raw, aes(x = inc_imp/1000)) +
  geom_density(fill = "darkgreen", col = "white") +
  labs(x = "Imputed Income in Thousands of Dollars",
       title = "Imputed Income on the Log (base 10) scale") +
  scale_x_log10(breaks = c(0.1, 1, 5, 25, 75, 250, 1000))

p1 / p2 +
  plot_annotation(title =
    paste0("Imputed Income: ", sum(is.na(smart_ohio_raw$inc_imp)), " res

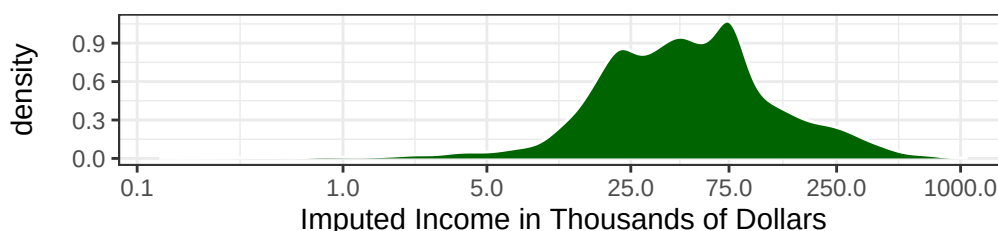
```

Imputed Income: 1310 respondents have missing income group

Imputed Income on the Linear scale



Imputed Income on the Log (base 10) scale



6.8 Clean Data in the State of Ohio

There are six MMSAs associated with the state of Ohio. We're going to create a `smart_ohio` that includes each of them. First, I'll ungroup the data that I created earlier, so I get a clean tibble.

```
smart_ohio_raw <- smart_ohio_raw |> ungroup()
```

Next, I'll select the variables I want to retain (they are the ones I created, plus `SEQNO`.)

```
smart_ohio <- smart_ohio_raw |>
  select(SEQNO, mmsa, mmsa_code, mmsa_name, mmsa_wt, completed,
         landline, hhadults,
         genhealth, physhealth, menthealth, poorhealth,
         agegroup, age_imp, race, hispanic, race_eth,
         female, marital, kids, educgroup, home_own,
         veteran, employment, incomegroup, inc_imp,
         cell_own, internet30,
         weight_kg, height_m, bmi, bmigroup,
         pregnant, deaf, blind, decide,
         diffwalk, diffdress, diffalone,
```

```

smoke100, smoker, ecig_ever, ecigs,
healthplan, hasdoc, costprob, t_checkup,
bp_high, bp_meds,
t_chol, chol_high, chol_meds,
asthma, hx_asthma, now_asthma,
hx_mi, hx_chd, hx_stroke, hx_skinc, hx_otherc,
hx_copd, hx_depress, hx_kidney,
hx_diabetes, dm_status, dm_age,
hx_arthr, arth_lims, arth_work, arth_soc,
joint_pain, alcdays, avgdrinks, maxdrinks,
binge, drinks_wk, drink_heavy,
fruit_day, veg_day, eat_juice, eat_fruit,
eat_greenveg, eat_fries, eat_potato,
eat_otherveg, exerany, activity, rec_aerobic,
rec_strength, exer1_type, exer2_type,
exer1_min, exer2_min, seatbelt,
vax_flu, vax_pneumo, vax_shingles,
hiv_test, hiv_risk)

saveRDS(smart_ohio, "data/smart_ohio.Rds")

write_csv(smart_ohio, "data/smart_ohio.csv")

```

The `smart_ohio` file should contain 99 variables, describing 7412 respondents.

6.9 Clean Cleveland-Elyria Data

6.9.1 Cleveland - Elyria Data

The `mmsa_name` variable is probably the simplest way for us to filter our data down to the MMSA we are interested in. Here, I'm using the `str_detect` function to identify the values of `mmsa_name` that contain the text "Cleveland".

```

smart_cle <- smart_ohio |>
  filter(str_detect(mmsa_name, 'Cleveland'))

saveRDS(smart_cle, "data/smart_cle.Rds")

```

In the Cleveland-Elyria MSA, we have 1133 observations on the same 99 variables.

We'll build a variety of smaller subsets from these data, eventually.

7 Dealing with Missingness: Single Imputation

7.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(ggribes)
library(knitr)
library(naniar)
library(simputation)
library(Hmisc)
library(mosaic)
library(patchwork)
library(visdat)
library(tidyverse)

theme_set(theme_bw())
```

7.1.1 Data Load

```
smart_cle <- readRDS("data/smart_cle.Rds")
```

7.2 Selecting Some Variables from the smart_cle data

```
smart_cle1 <- smart_cle |>
  select(SEQNO, physhealth, genhealth, bmi,
         age_imp, female, race_eth, internet30,
         smoke100, activity, drinks_wk, veg_day)
```


The `smart_cle.Rds` data file available on the Data and Code page of [our website](#) describes information on 99 variables for 1133 respondents to the BRFSS 2017, who live in the Cleveland-Elyria, OH, Metropolitan Statistical Area. The variables in the `smart_cle1.csv` file are listed below, along with the items that generate these responses.

Variable	Description
<code>SEQNO</code>	respondent identification number (all begin with 2016)
<code>physhealth</code>	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?
<code>genhealth</code>	Would you say that in general, your health is ... (five categories: Excellent, Very Good, Good, Fair or Poor)
<code>bmi</code>	Body mass index, in kg/m ²
<code>age_imp</code>	Age, imputed, in years
<code>female</code>	Sex, 1 = female, 0 = male
<code>race_eth</code>	Race and Ethnicity, in five categories
<code>internet30</code>	Have you used the internet in the past 30 days? (1 = yes, 0 = no)
<code>smoke100</code>	Have you smoked at least 100 cigarettes in your life? (1 = yes, 0 = no)
<code>activity</code>	Physical activity (Highly Active, Active, Insufficiently Active, Inactive)
<code>drinks_wk</code>	On average, how many drinks of alcohol do you consume in a week?
<code>veg_day</code>	How many servings of vegetables do you consume per day, on average?

```
str(smart_cle1)
```

```
tibble [1,133 x 12] (S3: tbl_df/tbl/data.frame)
 $ SEQNO      : num [1:1133] 2.02e+09 2.02e+09 2.02e+09 2.02e+09 2.02e+09 ...
 $ physhealth: num [1:1133] 4 0 0 0 0 2 2 0 0 0 ...
 $ genhealth  : Factor w/ 5 levels "1_Excellent",...: 1 1 3 3 3 2 3 2 4 1 ...
 $ bmi        : num [1:1133] NA 23.1 26.9 26.5 24.2 ...
 $ age_imp    : num [1:1133] 51 28 37 36 88 43 23 34 58 54 ...
 $ female     : num [1:1133] 1 1 1 1 0 0 0 0 0 1 ...
 $ race_eth   : Factor w/ 5 levels "White non-Hispanic",...: 1 1 3 1 1 1 1 3 2 1 ...
 $ internet30: num [1:1133] 1 1 0 1 1 1 1 1 1 1 ...
 $ smoke100   : num [1:1133] 1 0 0 1 1 1 0 0 0 1 ...
 $ activity   : Factor w/ 4 levels "Highly_Active",...: 4 4 3 1 1 NA 1 1 1 1 ...
 $ drinks_wk : num [1:1133] 0.7 0 0 4.67 0.93 0 2 0 0 0.47 ...
 $ veg_day    : num [1:1133] NA 3 4.06 2.07 1.31 NA 1.57 0.83 0.49 1.72 ...
```

7.3 smart_cle1: Seeing our Missing Data

The `naniar` package provides several useful functions for summarizing missingness in our data set. Like all tidy data sets, our `smart_cle1` tibble contains rows which describe observations, sometimes called *cases*, and also contains columns which describe variables.

Overall, there are 1133 cases, and 1133 observations in our `smart_cle1` tibble.

- We can obtain a count of the number of missing cells in the entire tibble.

```
smart_cle1 |> n_miss()
```

```
[1] 479
```

- We can use the `miss_var_summary` function to get a sorted table of each variable by number missing.

```
miss_var_summary(smart_cle1) |> kable()
```

variable	n_miss	pct_miss
activity	109	9.62
veg_day	101	8.91
bmi	91	8.03
drinks_wk	66	5.83
smoke100	40	3.53
race_eth	26	2.29
physhealth	24	2.12
age_imp	11	0.971
internet30	7	0.618
genhealth	4	0.353
SEQNO	0	0
female	0	0

- Or we can use the `miss_var_table` function to tabulate the number of variables that have each observed level of missingness.

```
miss_var_table(smart_cle1)
```

```
# A tibble: 11 x 3
  n_miss_in_var n_vars pct_vars
      <int>   <int>   <dbl>
1         0     2    16.7
2         4     1     8.33
3         7     1     8.33
4        11     1     8.33
5        24     1     8.33
6        26     1     8.33
7        40     1     8.33
8        66     1     8.33
9        91     1     8.33
10       101     1     8.33
11       109     1     8.33
```

- Or we can get a count for a specific variable, like `activity`:

```
smart_cle1 |> select(activity) |> n_miss()
```

```
[1] 109
```

- We can also use `prop_miss_case` or `pct_miss_case` to specify the proportion (or percentage) of missing observations across an entire data set, or within a specific variable.

```
prop_miss_case(smart_cle1)
```

```
[1] 0.2127096
```

```
smart_cle1 |> select(activity) |> pct_miss_case()
```

```
[1] 9.620477
```

- We can also use `prop_miss_var` or `pct_miss_var` to specify the proportion (or percentage) of variables with missing observations across an entire data set.

```
prop_miss_var(smart_cle1)
```

```
[1] 0.8333333
```

```
pct_miss_var(smart_cle1)
```

```
[1] 83.33333
```

- We use `miss_case_table` to identify the number of missing values for each of the cases (rows) in our tibble.

```
miss_case_table(smart_cle1)
```

```
# A tibble: 7 x 3
  n_miss_in_case n_cases pct_cases
      <int>      <int>      <dbl>
1         0       892      78.7
2         1       129     11.4
3         2        51      4.50
4         3        22      1.94
5         4        21      1.85
6         5        10      0.883
7         6         8      0.706
```

- Use `miss_case_summary` to specify individual observations and count their missing values.

```
miss_case_summary(smart_cle1)
```

```
# A tibble: 1,133 x 3
  case n_miss pct_miss
   <int>  <int>    <dbl>
1     17      6      50
2     42      6      50
3    254      6      50
4    425      6      50
5    521      6      50
6    729      6      50
7    757      6      50
8   1051      6      50
9     89      5     41.7
10    94      5     41.7
# i 1,123 more rows
```

The case numbers identified here are row numbers. Extract the data for case 17, for instance, with the `slice` function.

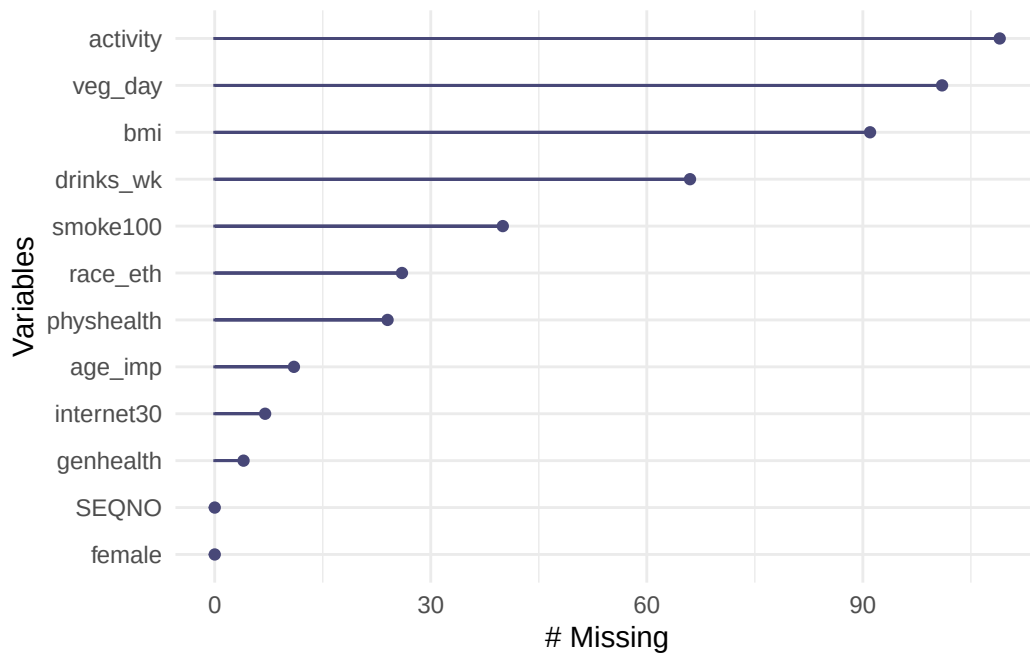
```
smart_cle1 |> slice(17)
```

```
# A tibble: 1 x 12
  SEQNO physhealth genhealth  bmi age_imp female race_eth internet30 smoke100
  <dbl>   <dbl> <fct>      <dbl> <dbl> <dbl> <fct>      <dbl>   <dbl>
1  2.02e9     0 1_Excellent NA     50     0 White n~     NA     NA
# i 3 more variables: activity <fct>, drinks_wk <dbl>, veg_day <dbl>
```

7.3.1 Plotting Missingness

The `gg_miss_var` function plots the number of missing observations in each variable in our data set.

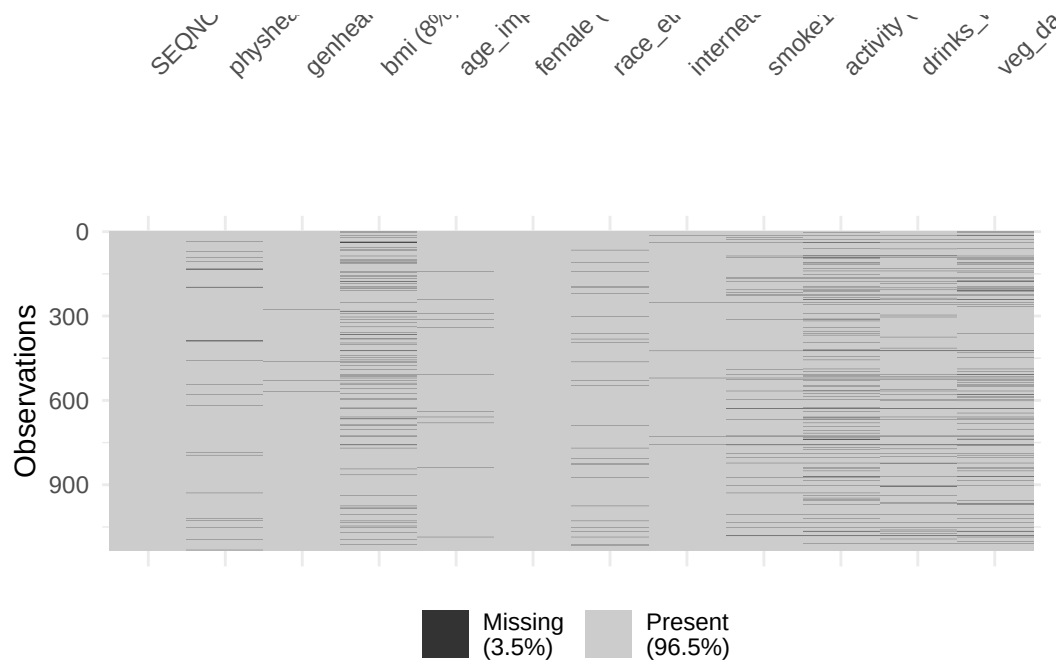
```
gg_miss_var(smart_cle1)
```



So the most commonly missing variable is `activity`.

To get a general sense of the missingness in our data, we might use either the `vis_dat` or the `vis_miss` function from the `visdat` package.

```
vis_miss(smart_cle1)
```



```
vis_dat(smart_cle1)
```



7.4 Missing-data mechanisms

My source for this description of mechanisms is Chapter 25 of @GelmanHill2007, and that chapter is [available at this link](#).

1. **MCAR = Missingness completely at random.** A variable is missing completely at random if the probability of missingness is the same for all units, for example, if for each subject, we decide whether to collect the **diabetes** status by rolling a die and refusing to answer if a “6” shows up. If data are missing completely at random, then throwing out cases with missing data does not bias your inferences.
2. **Missingness that depends only on observed predictors.** A more general assumption, called **missing at random** or **MAR**, is that the probability a variable is missing depends only on available information. Here, we would have to be willing to assume that the probability of nonresponse to **diabetes** depends only on the other, fully recorded variables in the data. It is often reasonable to model this process as a logistic regression, where the outcome variable equals 1 for observed cases and 0 for missing. When an outcome variable is missing at random, it is acceptable to exclude the missing cases (that is, to treat them as NA), as long as the regression controls for all the variables that affect the probability of missingness.
3. **Missingness that depends on unobserved predictors.** Missingness is no longer “at random” if it depends on information that has not been recorded and this information also predicts the missing values. If a particular treatment causes discomfort, a patient is more likely to drop out of the study. This missingness is not at random (unless “discomfort” is measured and observed for all patients). If missingness is not at random, it must be explicitly modeled, or else you must accept some bias in your inferences.
4. **Missingness that depends on the missing value itself.** Finally, a particularly difficult situation arises when the probability of missingness depends on the (potentially missing) variable itself. For example, suppose that people with higher earnings are less likely to reveal them.

Essentially, situations 3 and 4 are referred to collectively as **non-random missingness**, and cause more trouble for us than 1 and 2.

7.5 Options for Dealing with Missingness

There are several available methods for dealing with missing data that are MCAR or MAR, but they basically boil down to:

- Complete Case (or Available Case) analyses
- Single Imputation
- Multiple Imputation

7.6 Complete Case (and Available Case) analyses

In **Complete Case** analyses, rows containing NA values are omitted from the data before analyses commence. This is the default approach for many statistical software packages, and may introduce unpredictable bias and fail to include some useful, often hard-won information.

- A complete case analysis can be appropriate when the number of missing observations is not large, and the missing pattern is either MCAR (missing completely at random) or MAR (missing at random.)
- Two problems arise with complete-case analysis:
 1. If the units with missing values differ systematically from the completely observed cases, this could bias the complete-case analysis.
 2. If many variables are included in a model, there may be very few complete cases, so that most of the data would be discarded for the sake of a straightforward analysis.
- A related approach is *available-case* analysis where different aspects of a problem are studied with different subsets of the data, perhaps identified on the basis of what is missing in them.

7.7 Single Imputation

In **single imputation** analyses, NA values are estimated/replaced *one time* with *one particular data value* for the purpose of obtaining more complete samples, at the expense of creating some potential bias in the eventual conclusions or obtaining slightly *less* accurate estimates than would be available if there were no missing values in the data.

- A single imputation can be just a replacement with the mean or median (for a quantity) or the mode (for a categorical variable.) However, such an approach, though easy to understand, underestimates variance and ignores the relationship of missing values to other variables.
- Single imputation can also be done using a variety of models to try to capture information about the NA values that are available in other variables within the data set.
- The **simputation** package can help us execute single imputations using a wide variety of techniques, within the pipe approach used by the **tidyverse**. Another approach I have used in the past is the **mice** package, which can also perform single imputations.

7.8 Multiple Imputation

Multiple imputation, where NA values are repeatedly estimated/replaced with multiple data values, for the purpose of obtaining more complete samples *and* capturing details of the

variation inherent in the fact that the data have missingness, so as to obtain *more* accurate estimates than are possible with single imputation.

- We'll postpone the discussion of multiple imputation in these Notes for a while.

7.9 Approach 1: Building a Complete Case Analysis:

`smart_cle1_cc`

In the 431 course, we usually dealt with missing data by restricting our analyses to respondents with complete data on all variables. Let's start by doing that here. We'll create a new tibble called `smart_cle1_cc` which includes all respondents with complete data on all of these variables.

```
smart_cle1_cc <- smart_cle1 |>
  drop_na()

dim(smart_cle1_cc)
```

```
[1] 892  12
```

Our `smart_cle1_cc` tibble now has many fewer observations than its predecessors, but all of the variables in this complete cases tibble have no missing observations.

Data Set	Rows	Columns	Missingness?
<code>smart_cle</code>	1133	99	Quite a bit.
<code>smart_cle1</code>	1133	12	Quite a bit.
<code>smart_cle1_cc</code>	892	12	None.

7.10 Approach 2: Single Imputation to create `smart_cle1_sh`

Next, we'll create a data set which has all of the rows in the original `smart_cle1` tibble, but deals with missingness by imputing (estimating / filling in) new values for each of the missing values. To do this, we'll make heavy use of the `simputation` package in R.

The `simputation` package is designed for single imputation work. Note that we'll eventually adopt a **multiple imputation** strategy in some of our modeling work, and we'll use some specialized tools to facilitate that later.

To begin, we'll create a "shadow" in our tibble to track what we'll need to impute.

```
smart_cle1_sh <- bind_shadow(smart_cle1)
```

```
names(smart_cle1_sh)
```

```
[1] "SEQNO"      "physhealth"  "genhealth"   "bmi"
[5] "age_imp"    "female"      "race_eth"    "internet30"
[9] "smoke100"   "activity"     "drinks_wk"   "veg_day"
[13] "SEQNO_NA"   "physhealth_NA" "genhealth_NA" "bmi_NA"
[17] "age_imp_NA" "female_NA"    "race_eth_NA"  "internet30_NA"
[21] "smoke100_NA" "activity_NA"  "drinks_wk_NA" "veg_day_NA"
```

Note that the `bind_shadow()` function doubles the number of variables in our tibble, specifically by creating a new variable for each that takes the value `!NA` or `NA`. For example, consider

```
smart_cle1_sh |> count(activity, activity_NA)
```

```
# A tibble: 5 x 3
  activity      activity_NA      n
  <fct>         <fct>    <int>
1 Highly_Active !NA       338
2 Active        !NA       173
3 Insufficiently_Active !NA       201
4 Inactive      !NA       312
5 <NA>          NA        109
```

The `activity_NA` variable takes the value `!NA` (meaning not missing) when the value of the `activity` variable is known, and takes the value `NA` for observations where the `activity` variable is missing. This background tracking will be helpful to us when we try to assess the impact of imputation on some of our summaries.

7.10.1 What Type of Missingness Do We Have?

There are three types of missingness that we might assume in any given setting: missing completely at random (MCAR), missing at random (MAR) and missing not at random (MNAR). Together, MCAR and MAR are sometimes called *ignorable* non-response, which essentially means that imputation provides a way to useful estimates. MNAR or missing NOT at random is sometimes called non-ignorable missingness, implying that even high-quality imputation may not be sufficient to provide useful information to us.

Missing Completely at Random means that the missing data points are a random subset of the data. Essentially, there is nothing that makes some data more likely to be missing than others. If the data truly match the standard for MCAR, then a complete-case analysis will be about as good as an analysis after single or multiple imputation.

Missing at Random means that there is a systematic relationship between the observed data and the missingness mechanism. Another way to say this is that the missing value is not related to the reason why it is missing, but is related to the other variables collected in the study. The implication is that the missingness can be accounted for by studying the variables with complete information. Imputation strategies can be very helpful here, incorporating what we know (or think we know) about the relationships between the results that are missing and the results that we see.

- Wikipedia provides a nice example. If men are less likely to fill in a depression survey, but this has nothing to do with their level of depression after accounting for the fact that they are male, then the missingness can be assumed MAR.
- Determining whether missingness is MAR or MNAR can be tricky. We'll spend more time discussing this later.

Missing NOT at Random means that the missing value is related to the reason why it is missing.

- Continuing the Wikipedia example, if men failed to fill in a depression survey because of their level of depression, then this would be MNAR.
- Single imputation is most helpful in the MAR situation, although it is also appropriate when we assume MCAR.
- Multiple imputation will, similarly, be more helpful in MCAR and MAR situations than when data are missing NOT at random.

It's worth noting that many people are unwilling to impute values for outcomes or key predictors in a modeling setting, but are happy to impute for less important covariates. For now, we'll assume MCAR or MAR for all of the missingness in our `smart_cle1` data, which will allow us to adopt a single imputation strategy.

7.10.2 Single imputation into `smart_cle1_sh`

Which variables in `smart_cle1_sh` contain missing data?

```
miss_var_summary(smart_cle1_sh)
```

```
# A tibble: 24 x 3
  variable    n_miss pct_miss
  <chr>      <int>   <num>
1 activity     109    9.62
2 veg_day     101    8.91
3 bmi          91    8.03
4 drinks_wk    66    5.83
5 smoke100     40    3.53
6 race_eth     26    2.29
7 physhealth    24    2.12
8 age_imp      11    0.971
9 internet30     7    0.618
10 genhealth     4    0.353
# i 14 more rows
```

We will impute these variables using several different strategies, all supported nicely by the `imputation` package.

These include imputation methods based solely on the distribution of the complete cases of the variable being imputed.

- `impute_median`: impute the median value of all non-missing observations into the missing values for the variable
- `impute_rhd`: random “hot deck” imputation involves drawing at random from the complete cases for that variable

Also available are imputation strategies that impute predicted values from models using other variables in the data set besides the one being imputed.

- `impute_pmm`: imputation using predictive mean matching
- `impute_rlm`: imputation using robust linear models
- `impute_cart`: imputation using classification and regression trees
- `impute_knn`: imputation using k-nearest neighbors methods

7.10.3 Imputing Binary Categorical Variables

Here, we’ll arbitrarily impute our 1/0 variables as follows:

- For `internet30` we’ll use the `impute_rhd` approach to draw a random observation from the existing set of 1s and 0s in the complete `internet30` data.
- For `smoke100` we’ll use a method called predictive mean matching (`impute_pmm`) which takes the result from a model based on the (imputed) `internet30` value and whether or not the subject is `female`, and converts it to the nearest value in the observed `smoke100` data. This is a good approach for imputing discrete variables.

These are completely arbitrary choices, for demonstration purposes.

```
set.seed(2020001)

smart_cle1_sh <- smart_cle1_sh |> data.frame() |>
  impute_rhd(internet30 ~ 1) |>
  impute_pmm(smoke100 ~ internet30 + female) |>
  as_tibble()

smart_cle1_sh |> count(smoke100, smoke100_NA)
```

```
# A tibble: 4 x 3
  smoke100 smoke100_NA      n
  <dbl> <fct>      <int>
1         0 !NA         579
2         0 NA           7
3         1 !NA        514
4         1 NA          33
```

```
smart_cle1_sh |> count(internet30, internet30_NA)
```

```
# A tibble: 4 x 3
  internet30 internet30_NA      n
  <dbl> <fct>      <int>
1         0 !NA        207
2         0 NA           1
3         1 !NA        919
4         1 NA           6
```

Other approaches that may be used with 1/0 variables include `impute_knn` and `impute_pmm`.

7.10.4 Imputing Quantitative Variables

We'll demonstrate a different approach for imputing each of the quantitative variables with missing observations. Again, we're making purely arbitrary decisions here about what to include in each imputation. In practical work, we'd want to be a bit more thoughtful about this.

Note that I'm choosing to use `impute_pmm` with the `physhealth` and `age_imp` variables. This is (in part) because I want my imputations to be integers, as the other observations are for those variables. `impute_rhd` would also accomplish this.

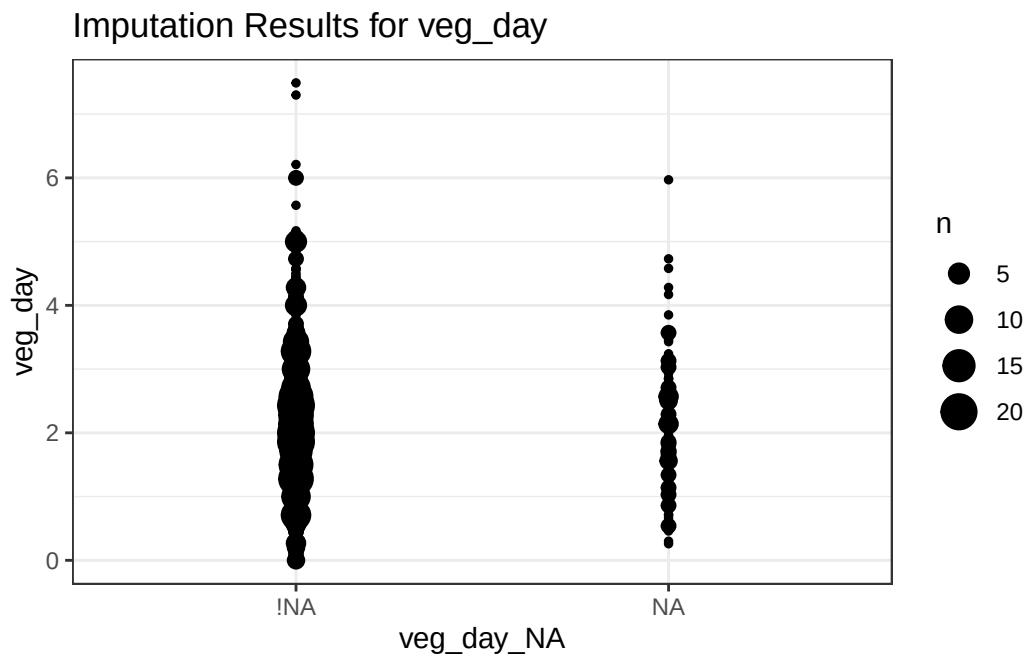
```
set.seed(2020001)
smart_cle1_sh <- smart_cle1_sh |> data.frame() |>
  impute_rhd(veg_day ~ 1) |>
  impute_median(drinks_wk ~ 1) |>
  impute_pmm(physhealth ~ drinks_wk + female + smoke100) |>
  impute_pmm(age_imp ~ drinks_wk + physhealth) |>
  impute_rlm(bmi ~ physhealth + smoke100) |>
  as_tibble()
```

7.10.5 Imputation Results

Let's plot a few of these results, so we can see what imputation has done to the distribution of these quantities.

1. veg_day

```
ggplot(smart_cle1_sh, aes(x = veg_day_NA, y = veg_day)) +
  geom_count() +
  labs(title = "Imputation Results for veg_day")
```

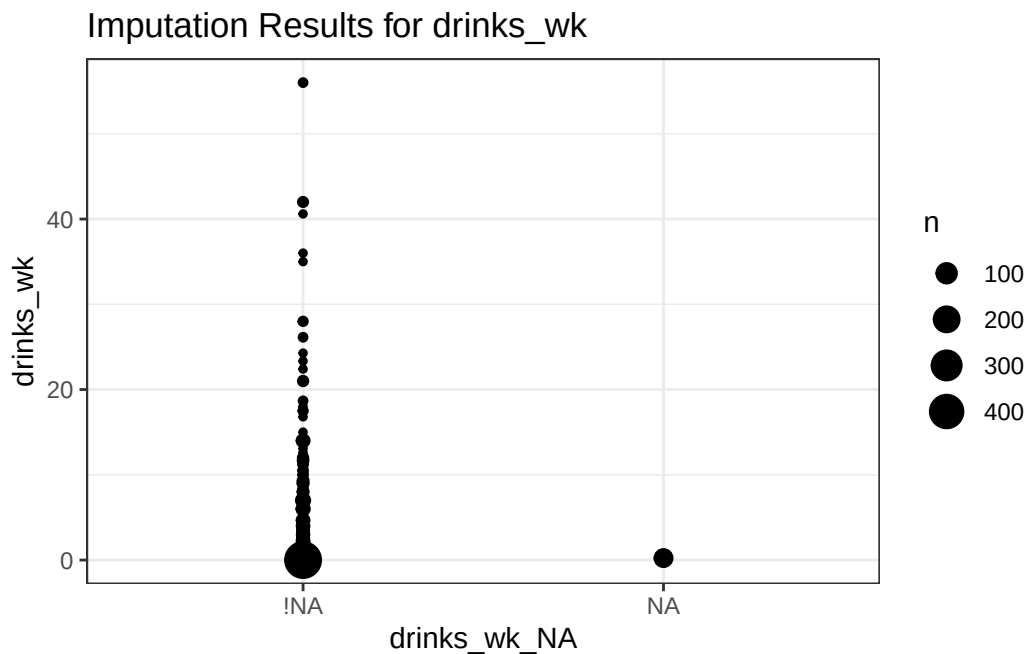


```
favstats(veg_day ~ veg_day_NA, data = smart_cle1_sh)
```

	veg_day_NA	min	Q1	median	Q3	max	mean	sd	n	missing
1	!NA	0.00	1.2675	1.72	2.42	7.49	1.912548	1.038403	1032	0
2	NA	0.26	1.3400	1.86	2.72	5.97	2.085050	1.062316	101	0

2. drinks_wk for which we imputed the median value...

```
ggplot(smart_cle1_sh, aes(x = drinks_wk_NA, y = drinks_wk)) +  
  geom_count() +  
  labs(title = "Imputation Results for drinks_wk")
```



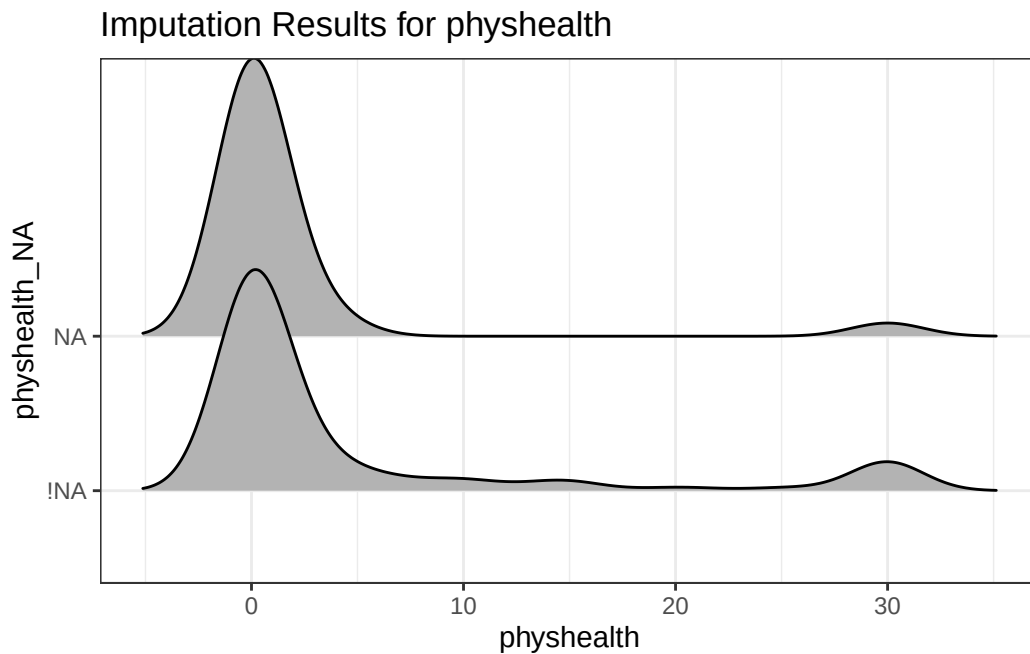
```
smart_cle1_sh |> filter(drinks_wk_NA == "NA") |>  
  tabyl(drinks_wk)
```

drinks_wk	n	percent
0.23	66	1

3. physhealth, a count between 0 and 30...

```
ggplot(smart_cle1_sh,
       aes(x = physhealth, y = physhealth_NA)) +
  geom_density_ridges() +
  labs(title = "Imputation Results for physhealth")
```

Picking joint bandwidth of 1.71

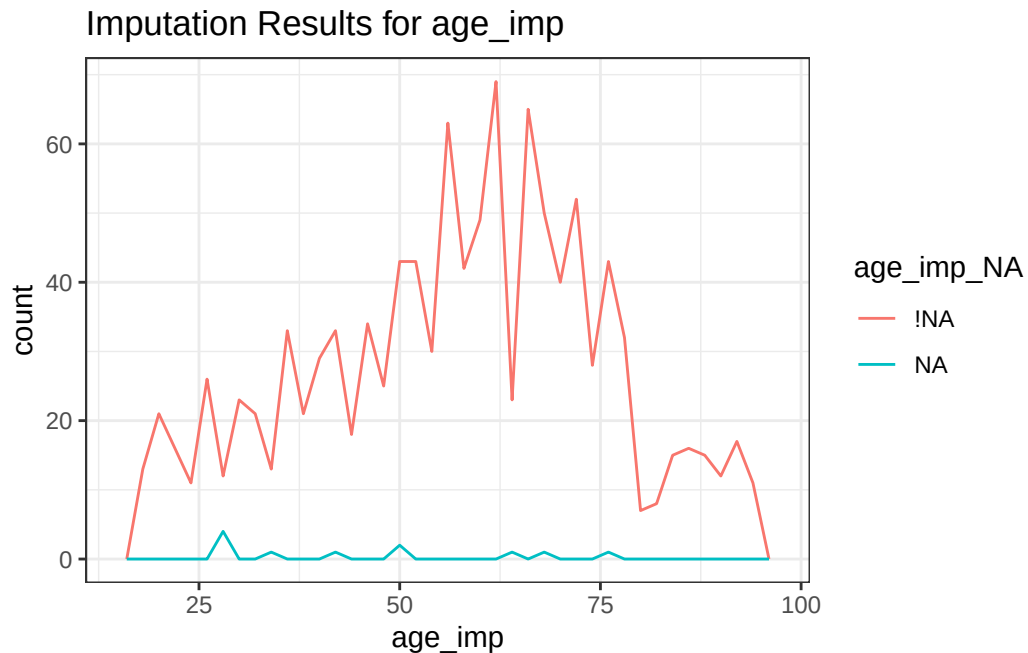


```
smart_cle1_sh |> filter(physhealth_NA == "NA") |>
  tabyl(physhealth)
```

physhealth	n	percent
0	20	0.83333333
2	2	0.08333333
4	1	0.04166667
30	1	0.04166667

4. age_imp, in (integer) years

```
ggplot(smart_cle1_sh,
       aes(x = age_imp, color = age_imp_NA)) +
  geom_freqpoly(binwidth = 2) +
  labs(title = "Imputation Results for age_imp")
```

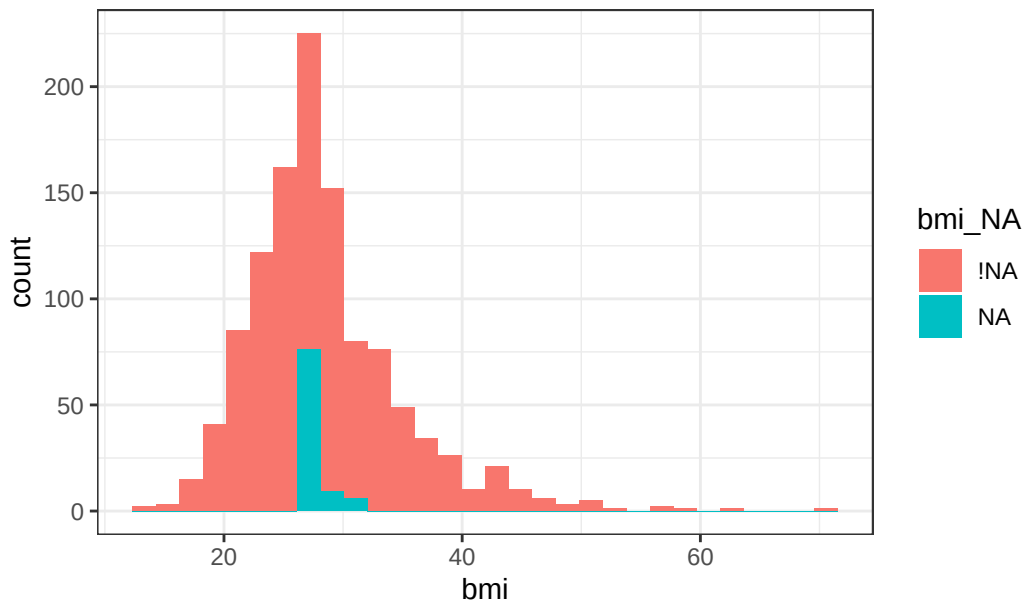
```
smart_cle1_sh |> filter(age_imp_NA == "NA") |>
  tabyl(age_imp)
```

age_imp	n	percent
28	4	0.36363636
35	1	0.09090909
43	1	0.09090909
50	2	0.18181818
65	1	0.09090909
69	1	0.09090909
76	1	0.09090909

5. bmi or body mass index

```
ggplot(smart_cle1_sh, aes(x = bmi, fill = bmi_NA)) +
  geom_histogram(bins = 30) +
  labs(title = "Histogram of BMI and imputed BMI")
```

Histogram of BMI and imputed BMI



```
favstats(bmi ~ bmi_NA, data = smart_cle1_sh)
```

	bmi_NA	min	Q1	median	Q3	max	mean	sd	n
1	!NA	13.30000	24.11000	27.30000	31.68000	70.56000	28.40947	6.6289286	1042
2	NA	27.07078	27.07078	27.52932	27.59447	30.67146	27.66006	0.8695896	91
	missing								
1		0							
2		0							

7.10.6 Imputing Multi-Categorical Variables

The three multi-categorical variables we have left to impute are `activity`, `race_eth` and `genhealth`, and each is presented as a factor in R, rather than as a character variable.

We'll arbitrarily decide to impute

- `activity` and `genhealth` with a classification tree using `physhealth`, `bmi` and `smoke100`,
- and then impute `race_eth` with a random draw from the distribution of complete cases.

```
set.seed(2020001)
smart_cle1_sh <- smart_cle1_sh |>
  data.frame() |>
  impute_cart(activity + genhealth ~ physhealth + bmi + smoke100) |>
  impute_rhd(race_eth ~ 1) |>
  as_tibble()
```

Let's check our results.

```
smart_cle1_sh |> count(activity_NA, activity)
```

```
# A tibble: 6 x 3
  activity_NA activity          n
  <fct>      <fct>          <int>
1 !NA       Highly_Active      338
2 !NA       Active          173
3 !NA       Insufficiently_Active 201
4 !NA       Inactive           312
5 NA        Highly_Active      90
6 NA        Inactive           19
```

```
smart_cle1_sh |> count(race_eth_NA, race_eth)
```

```
# A tibble: 9 x 3
  race_eth_NA race_eth          n
  <fct>      <fct>          <int>
1 !NA       White non-Hispanic      805
2 !NA       Black non-Hispanic     222
3 !NA       Other race non-Hispanic  24
4 !NA       Multiracial non-Hispanic  22
5 !NA       Hispanic             34
6 NA        White non-Hispanic     19
7 NA        Black non-Hispanic       4
8 NA        Multiracial non-Hispanic  2
9 NA        Hispanic              1
```

```
smart_cle1_sh |> count(genhealth_NA, genhealth)
```

```
# A tibble: 7 x 3
```

	genhealth_NA	genhealth	n
	<fct>	<fct>	<int>
1	!NA	1_Excellent	164
2	!NA	2_VeryGood	383
3	!NA	3_Good	364
4	!NA	4_Fair	158
5	!NA	5_Poor	60
6	NA	2_VeryGood	3
7	NA	3_Good	1

And now, we should have no missing values in the data, at all.

```
miss_case_table(smart_cle1_sh)
```

```
# A tibble: 1 x 3
  n_miss_in_case n_cases pct_cases
      <int>      <int>      <dbl>
1           0      1133         100
```

7.10.7 Saving the new tibbles

```
saveRDS(smart_cle1_cc, ("data/smart_cle1_cc.Rds"))
saveRDS(smart_cle1_sh, ("data/smart_cle1_sh.Rds"))
```

8 Summarizing smart_cle1

In this chapter, we'll work with the two data files we built in Chapter 7.

8.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(Hmisc)
library(mosaic)
library(knitr)
library(tidyverse)

theme_set(theme_bw())
```

8.1.1 Data Load

```
smart_cle1_sh <- read_rds("data/smart_cle1_sh.Rds")
smart_cle1_cc <- read_rds("data/smart_cle1_cc.Rds")
```

8.2 What's in these data?

Those files (`_sh` contains single imputations, and a shadow set of variables which have `_NA` at the end of their names, while `_cc` contains only the complete cases) describe information on the following variables from the BRFSS 2017, who live in the Cleveland-Elyria, OH, Metropolitan Statistical Area.

Variable	Description
SEQNO	respondent identification number (all begin with 2016)
physhealth	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?
genhealth	Would you say that in general, your health is ... (five categories: Excellent, Very Good, Good, Fair or Poor)
bmi	Body mass index, in kg/m ²
age_imp	Age, imputed, in years
female	Sex, 1 = female, 0 = male
race_eth	Race and Ethnicity, in five categories
internet30	Have you used the internet in the past 30 days? (1 = yes, 0 = no)
smoke100	Have you smoked at least 100 cigarettes in your life? (1 = yes, 0 = no)
activity	Physical activity (Highly Active, Active, Insufficiently Active, Inactive)
drinks_wk	On average, how many drinks of alcohol do you consume in a week?
veg_day	How many servings of vegetables do you consume per day, on average?

8.3 General Approaches to Obtaining Numeric Summaries

8.3.1 summary for a data frame

Of course, we can use the usual `summary` to get some basic information about the data.

```
summary(smart_cle1_cc)
```

```

      SEQNO      physhealth      genhealth      bmi
Min.   :2.017e+09  Min.    : 0.000  1_Excellent:134  Min.    :13.30
1st Qu.:2.017e+09  1st Qu.: 0.000  2_VeryGood :314  1st Qu.:24.23
Median :2.017e+09  Median : 0.000  3_Good    :284  Median :27.48
Mean   :2.017e+09  Mean   : 4.361  4_Fair    :114  Mean   :28.51
3rd Qu.:2.017e+09  3rd Qu.: 3.000  5_Poor    : 46  3rd Qu.:31.82
Max.   :2.017e+09  Max.   :30.000                Max.   :63.00

      age_imp      female      race_eth
Min.   :18.00  Min.    :0.0000  White non-Hispanic    :661
1st Qu.:43.00  1st Qu.:0.0000  Black non-Hispanic    :168
Median :58.00  Median :1.0000  Other race non-Hispanic : 20
Mean   :56.46  Mean   :0.5807  Multiracial non-Hispanic: 17
3rd Qu.:69.00  3rd Qu.:1.0000  Hispanic              : 26
Max.   :95.00  Max.   :1.0000

```

internet30	smoke100	activity	drinks_wk
Min. :0.000	Min. :0.0000	Highly_Active :308	Min. : 0.000
1st Qu.:1.000	1st Qu.:0.0000	Active :155	1st Qu.: 0.000
Median :1.000	Median :0.0000	Insufficiently_Active:167	Median : 0.470
Mean :0.833	Mean :0.4787	Inactive :262	Mean : 2.727
3rd Qu.:1.000	3rd Qu.:1.0000		3rd Qu.: 2.850
Max. :1.000	Max. :1.0000		Max. :56.000

veg_day
Min. :0.000
1st Qu.:1.280
Median :1.730
Mean :1.919
3rd Qu.:2.422
Max. :7.300

8.3.2 The inspect function from the mosaic package

```
inspect(smart_cle1_cc)
```

categorical variables:

	name	class	levels	n	missing
1	genhealth	factor	5	892	0
2	race_eth	factor	5	892	0
3	activity	factor	4	892	0

distribution

1 2_VeryGood (35.2%), 3_Good (31.8%) ...

2 White non-Hispanic (74.1%) ...

3 Highly_Active (34.5%) ...

quantitative variables:

	name	class	min	Q1	median	Q3
1	SEQNO	numeric	2.017e+09	2.0170e+09	2.017001e+09	2.017001e+09
2	physhealth	numeric	0.000e+00	0.0000e+00	0.000000e+00	3.000000e+00
3	bmi	numeric	1.330e+01	2.4235e+01	2.747500e+01	3.181500e+01
4	age_imp	numeric	1.800e+01	4.3000e+01	5.800000e+01	6.900000e+01
5	female	numeric	0.000e+00	0.0000e+00	1.000000e+00	1.000000e+00
6	internet30	numeric	0.000e+00	1.0000e+00	1.000000e+00	1.000000e+00
7	smoke100	numeric	0.000e+00	0.0000e+00	0.000000e+00	1.000000e+00
8	drinks_wk	numeric	0.000e+00	0.0000e+00	4.700000e-01	2.850000e+00

```

9    veg_day numeric 0.000e+00 1.2800e+00 1.730000e+00 2.422500e+00
      max      mean      sd    n missing
1 2017001133.0 2.017001e+09 326.8928344 892      0
2      30.0 4.360987e+00  8.8153373 892      0
3      63.0 2.850905e+01  6.5057975 892      0
4      95.0 5.645516e+01 18.0333027 892      0
5       1.0 5.807175e-01  0.4937185 892      0
6       1.0 8.329596e-01  0.3732212 892      0
7       1.0 4.786996e-01  0.4998263 892      0
8      56.0 2.726525e+00  5.7172011 892      0
9       7.3 1.918643e+00  1.0262415 892      0

```

8.3.3 The describe function in Hmisc

This provides some useful additional summaries, including a list of the lowest and highest values (which is very helpful when checking data.)

```

smart_cle1_cc |>
  select(bmi, genhealth, female) |>
  describe()

```

```
select(smart_cle1_cc, bmi, genhealth, female)
```

```

3 Variables      892 Observations
-----
bmi
      n missing distinct      Info      Mean  pMedian      Gmd      .05
      892      0      481      1      28.51   27.88   6.952   20.09
      .10      .25      .50      .75      .90      .95
      21.40   24.23   27.48   31.81   36.78   41.09

lowest : 13.3  13.64 15.71 15.75 17.07, highest: 52.74 56.31 57.12 58.98 63
-----
genhealth
      n missing distinct
      892      0      5

Value      1_Excellent  2_VeryGood  3_Good  4_Fair  5_Poor
Frequency      134      314      284      114      46
Proportion      0.150      0.352      0.318      0.128      0.052
-----

```


n	missing	distinct	Info	Sum	Mean
892	0	2	0.73	518	0.5807

- The **Info** measure is used for quantitative and binary variables. It is a relative information measure that increases towards 1 for variables with no ties, and is smaller for variables with many ties.
- The **Gmd** is the Gini mean difference. It is a measure of spread (or dispersion), where larger values indicate greater spread in the distribution, like the standard deviation or the interquartile range. It is defined as the mean absolute difference between any pairs of observations.

See the Help file for **describe** in the **Hmisc** package for more details on these measures, and on the settings for **describe**.

8.4 Counting as exploratory data analysis

Counting and/or tabulating things can be amazingly useful. Suppose we want to understand the **genhealth** values, after our single imputation.

```
smart_cle1_sh |> count(genhealth) |>
  mutate(percent = 100*n / sum(n))
```

```
# A tibble: 5 x 3
  genhealth      n percent
  <fct>      <int>   <dbl>
1 1_Excellent   164    14.5
2 2_VeryGood   386    34.1
3 3_Good       365    32.2
4 4_Fair       158    13.9
5 5_Poor        60     5.30
```

We might use **tabyl** to do this job...

```
smart_cle1_sh |>
  tabyl(genhealth) |>
  adorn_pct_formatting(digits = 1) |>
  kable()
```

genhealth	n	percent
1_Excellent	164	14.5%
2_VeryGood	386	34.1%
3_Good	365	32.2%
4_Fair	158	13.9%
5_Poor	60	5.3%

8.4.1 Did genhealth vary by smoking status?

```
smart_cle1_sh |>
  count(genhealth, smoke100) |>
  mutate(percent = 100*n / sum(n))
```

```
# A tibble: 10 x 4
  genhealth  smoke100      n percent
  <fct>      <dbl> <int>   <dbl>
1 1_Excellent      0   102    9.00
2 1_Excellent      1    62    5.47
3 2_VeryGood       0   217   19.2
4 2_VeryGood       1   169   14.9
5 3_Good           0   178   15.7
6 3_Good           1   187   16.5
7 4_Fair           0    65    5.74
8 4_Fair           1    93    8.21
9 5_Poor           0    24    2.12
10 5_Poor          1    36    3.18
```

Suppose we want to find the percentage within each smoking status group. Here's one approach...

```
smart_cle1_sh |>
  count(smoke100, genhealth) |>
  group_by(smoke100) |>
  mutate(prob = 100*n / sum(n))
```

```
# A tibble: 10 x 4
# Groups:   smoke100 [2]
  smoke100 genhealth      n prob
  <dbl>    <fct>      <int> <dbl>
1 0       1_Excellent    102  9.00
2 0       2_VeryGood   217 19.20
3 0       3_Good       178 15.70
4 0       4_Fair        65  5.74
5 0       5_Poor        24  2.12
6 1       1_Excellent     62  5.47
7 1       2_VeryGood   169 14.90
8 1       3_Good       187 16.50
9 1       4_Fair        93  8.21
10 1       5_Poor        36  3.18
```

	<dbl>	<fct>	<int>	<dbl>
1	0	1_Excellent	102	17.4
2	0	2_VeryGood	217	37.0
3	0	3_Good	178	30.4
4	0	4_Fair	65	11.1
5	0	5_Poor	24	4.10
6	1	1_Excellent	62	11.3
7	1	2_VeryGood	169	30.9
8	1	3_Good	187	34.2
9	1	4_Fair	93	17.0
10	1	5_Poor	36	6.58

And here's another ...

```
smart_cle1_sh |>
  tabyl(smoke100, genhealth) |>
  adorn_totals(where = c("row", "col")) |>
  adorn_percentages(denominator = "row") |>
  adorn_pct_formatting(digits = 1) |>
  adorn_ns(position = "front")
```

smoke100	1_Excellent	2_VeryGood	3_Good	4_Fair	5_Poor
0	102 (17.4%)	217 (37.0%)	178 (30.4%)	65 (11.1%)	24 (4.1%)
1	62 (11.3%)	169 (30.9%)	187 (34.2%)	93 (17.0%)	36 (6.6%)
Total	164 (14.5%)	386 (34.1%)	365 (32.2%)	158 (13.9%)	60 (5.3%)
Total	586 (100.0%)				
	547 (100.0%)				
	1,133 (100.0%)				

8.4.2 What's the distribution of physhealth?

We can count quantitative variables with discrete sets of possible values, like `physhealth`, which is captured as an integer (that must fall between 0 and 30.)

```
smart_cle1_sh |> count(physhealth)
```

```
# A tibble: 21 x 2
  physhealth      n
  <dbl> <int>
```

```

1      0    710
2      1     49
3      2     63
4      3     38
5      4     16
6      5     30
7      6      5
8      7     18
9      8      5
10     10     32
# i 11 more rows

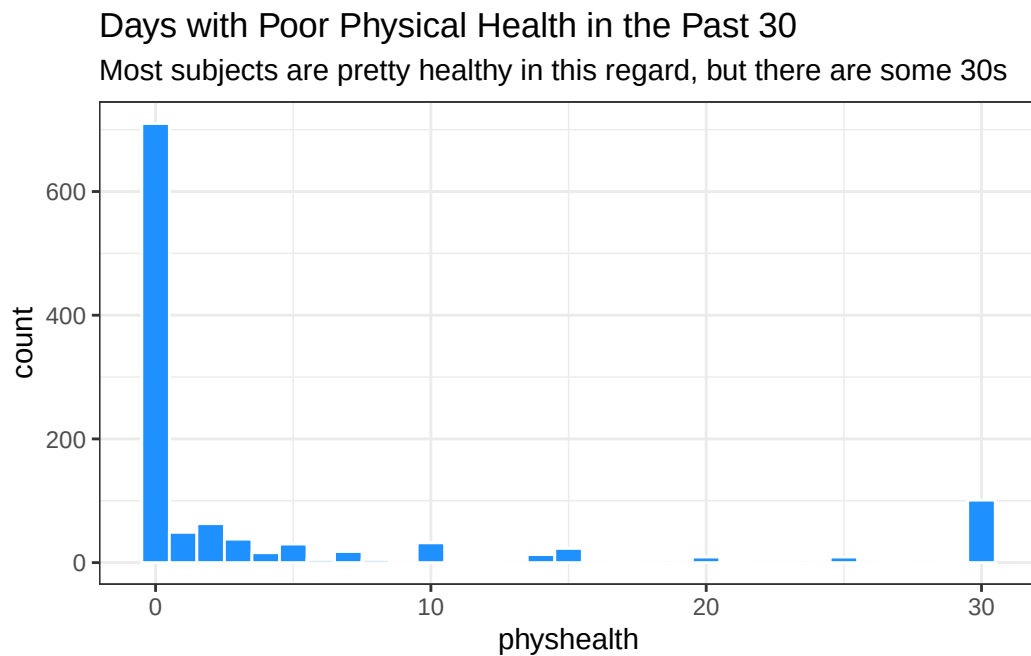
```

Of course, a natural summary of a quantitative variable like this would be graphical.

```

ggplot(smart_cle1_sh, aes(physhealth)) +
  geom_histogram(binwidth = 1,
                 fill = "dodgerblue", col = "white") +
  labs(title = "Days with Poor Physical Health in the Past 30",
       subtitle = "Most subjects are pretty healthy in this regard, but there are some 30s

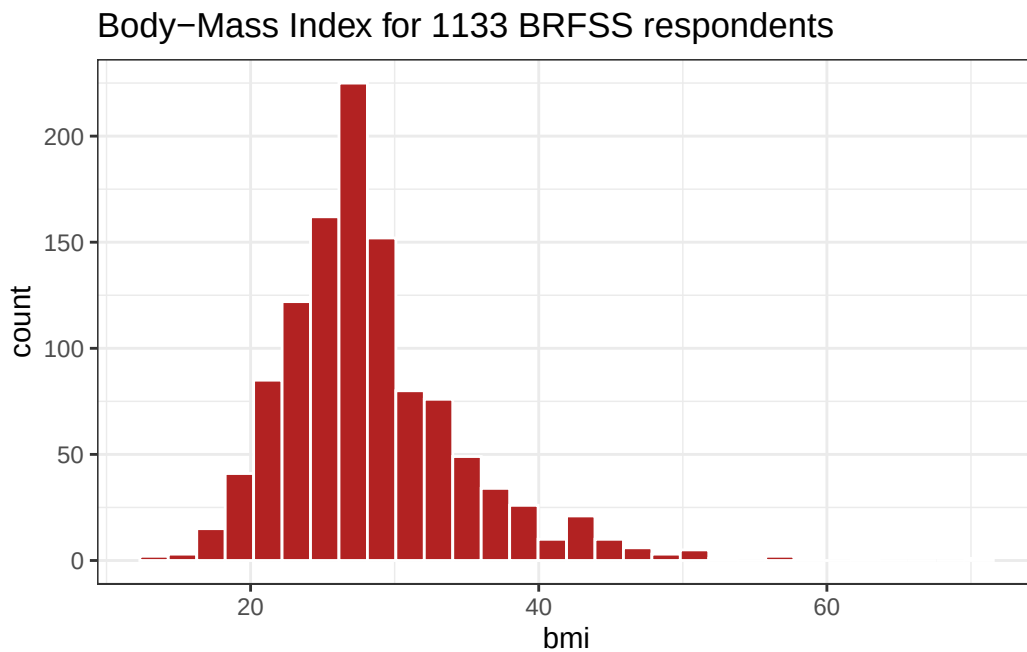
```



8.4.3 What's the distribution of bmi?

bmi is the body-mass index, an indicator of size (thickness, really.)

```
ggplot(smart_cle1_sh, aes(bmi)) +  
  geom_histogram(bins = 30,  
                 fill = "firebrick", col = "white") +  
  labs(title = paste0("Body-Mass Index for ",  
                      nrow(smart_cle1_sh),  
                      " BRFSS respondents"))
```



8.4.4 How many of the respondents have a BMI below 30?

```
smart_cle1_sh |> count(bmi < 30) |>  
  mutate(proportion = n / sum(n))
```

```
# A tibble: 2 x 3  
  `bmi < 30`      n proportion  
  <lgl>      <int>      <dbl>  
1 FALSE      330      0.291  
2 TRUE       803      0.709
```

8.4.5 How many of the respondents with a BMI < 30 are highly active?

```
smart_cle1_sh |>
  filter(bmi < 30) |>
  tabyl(activity) |>
  adorn_pct_formatting()
```

	activity	n	percent
	Highly_Active	343	42.7%
	Active	133	16.6%
	Insufficiently_Active	129	16.1%
	Inactive	198	24.7%

8.4.6 Is obesity associated with smoking history?

```
smart_cle1_sh |> count(smoke100, bmi < 30) |>
  group_by(smoke100) |>
  mutate(percent = 100*n/sum(n))
```

```
# A tibble: 4 x 4
# Groups:   smoke100 [2]
  smoke100 `bmi < 30`      n percent
  <dbl> <lgl>      <int>   <dbl>
1      0 FALSE      159    27.1
2      0 TRUE       427    72.9
3      1 FALSE      171    31.3
4      1 TRUE       376    68.7
```

8.4.7 Comparing drinks_wk summaries by obesity status

Can we compare the `drinks_wk` means, medians and 75th percentiles for respondents whose BMI is below 30 to the respondents whose BMI is not?

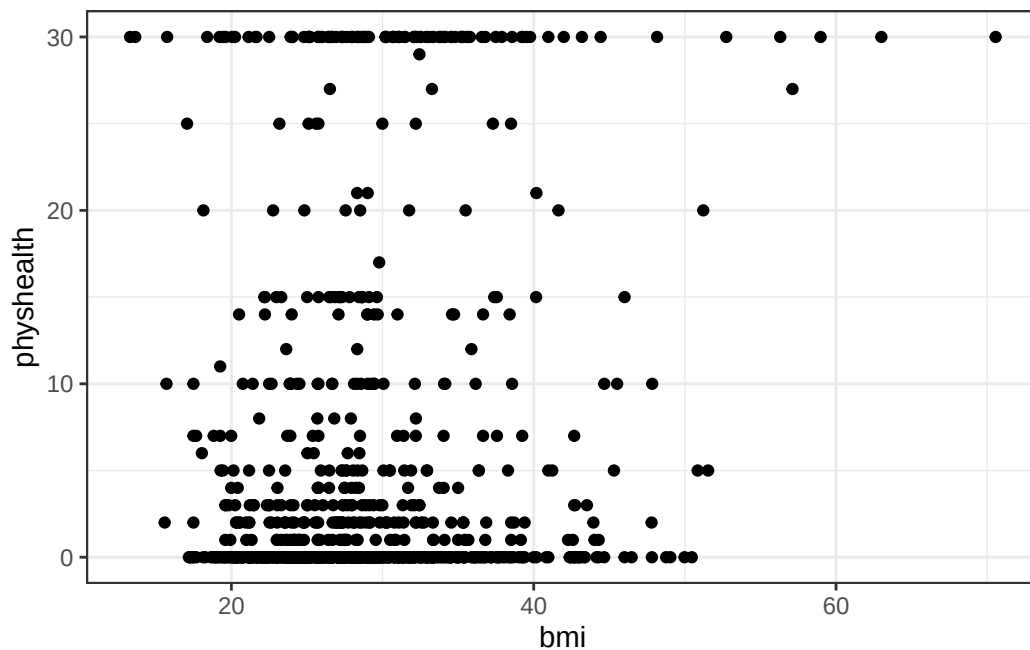
```
smart_cle1_sh |>
  group_by(bmi < 30) |>
  summarize(mean(drinks_wk), median(drinks_wk),
    q75 = quantile(drinks_wk, 0.75))
```

```
# A tibble: 2 x 4
  `bmi < 30` `mean(drinks_wk)` `median(drinks_wk)` q75
  <lgl>          <dbl>          <dbl> <dbl>
1 FALSE          1.67          0.23  1.17
2 TRUE           2.80          0.23  2.8
```

8.5 Can bmi predict physhealth?

We'll start with an effort to predict `physhealth` using `bmi`. A natural graph would be a scatterplot.

```
ggplot(data = smart_cle1_sh, aes(x = bmi, y = physhealth)) +
  geom_point()
```

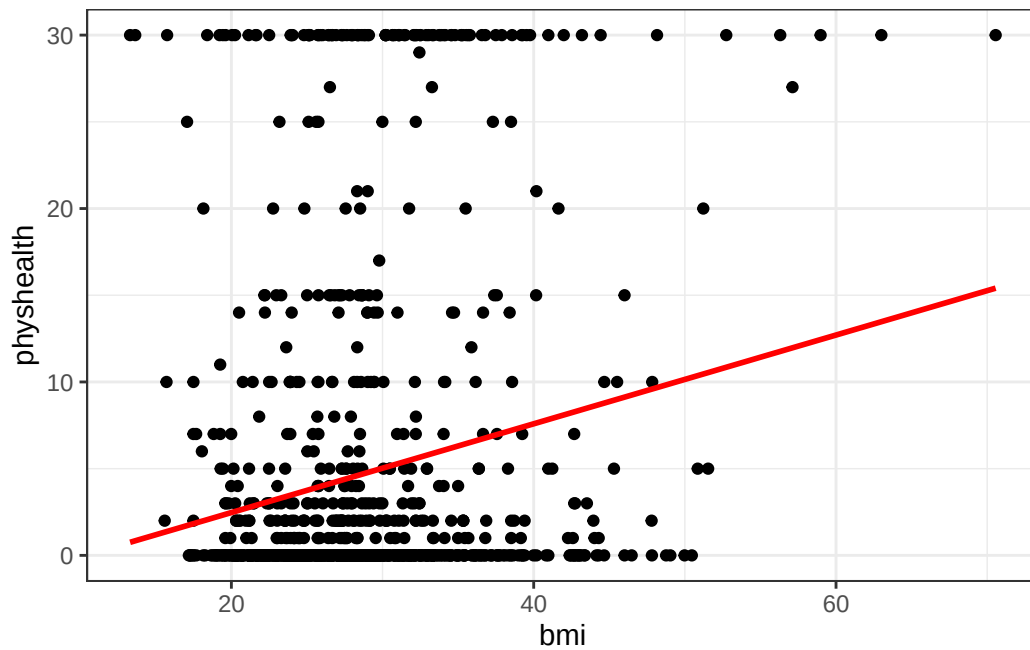


A good question to ask ourselves here might be: “In what BMI range can we make a reasonable prediction of `physhealth`?”

Now, we might take the plot above and add a simple linear model ...

```
ggplot(data = smart_cle1_sh, aes(x = bmi, y = physhealth)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, col = "red")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



which shows the same least squares regression model that we can fit with the `lm` command.

8.5.1 Fitting a Simple Regression Model

```
model_A <- lm(physhealth ~ bmi, data = smart_cle1_sh)
```

```
model_A
```

Call:

```
lm(formula = physhealth ~ bmi, data = smart_cle1_sh)
```

Coefficients:

(Intercept)	bmi
-2.6497	0.2559

```
summary(model_A)
```



```

Call:
lm(formula = physhealth ~ bmi, data = smart_cle1_sh)

Residuals:
    Min       1Q   Median       3Q      Max
-10.2641  -4.5596  -3.5897  -0.7778   29.2460

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.64970     1.22523  -2.163   0.0308 *
bmi          0.25592     0.04217   6.069 1.76e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.031 on 1131 degrees of freedom
Multiple R-squared:  0.03154,    Adjusted R-squared:  0.03068
F-statistic: 36.83 on 1 and 1131 DF,  p-value: 1.757e-09

```

```
confint(model_A, level = 0.95)
```

```

                2.5 %      97.5 %
(Intercept) -5.0536859 -0.2457115
bmi          0.1731799  0.3386617

```

The model coefficients can be obtained by printing the model object, and the `summary` function provides several useful descriptions of the model's residuals, its statistical significance, and quality of fit.

8.5.2 Model Summary for a Simple (One-Predictor) Regression

The fitted model predicts `physhealth` using a prediction equation we can read off from the model coefficient estimates. Specifically, we have:

```
coef(model_A)
```

```

(Intercept)      bmi
-2.6496987    0.2559208

```

so the equation is $\text{physhealth} = -2.82 + 0.265 \text{ bmi}$.

Each of the 1133 respondents included in the `smart_cle1_sh` data makes a contribution to this model.

8.5.2.1 Residuals

Suppose Harry is one of the people in that group, and Harry's data is `bmi = 20`, and `physhealth = 3`.

- Harry's *observed* value of `physhealth` is just the value we have in the data for them, in this case, observed `physhealth = 3` for Harry.
- Harry's *fitted* or *predicted* `physhealth` value is the result of calculating $-2.82 + 0.265 \text{ bmi}$ for Harry. So, if Harry's BMI was 20, then Harry's predicted `physhealth` value is $-2.82 + 0.265 (20) = 2.48$.
- The *residual* for Harry is then his *observed* outcome minus his *fitted* outcome, so Harry has a residual of $3 - 2.48 = 0.52$.
- Graphically, a residual represents vertical distance between the observed point and the fitted regression line.
- Points above the regression line will have positive residuals, and points below the regression line will have negative residuals. Points on the line have zero residuals.

The residuals are summarized at the top of the `summary` output for linear model.

```
summary(model_A)
```

Call:

```
lm(formula = physhealth ~ bmi, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.2641	-4.5596	-3.5897	-0.7778	29.2460

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.64970	1.22523	-2.163	0.0308 *
bmi	0.25592	0.04217	6.069	1.76e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.031 on 1131 degrees of freedom

Multiple R-squared: 0.03154, Adjusted R-squared: 0.03068

F-statistic: 36.83 on 1 and 1131 DF, p-value: 1.757e-09

- The mean residual will always be zero in an ordinary least squares model, but a five number summary of the residuals is provided by the summary, as is an estimated standard deviation of the residuals (called here the Residual standard error.)

- In the `smart_cle1_sh` data, the minimum residual was -10.53, so for one subject, the observed value was 10.53 days smaller than the predicted value. This means that the prediction was 10.53 days too large for that subject.
- Similarly, the maximum residual was 29.30 days, so for one subject the prediction was 29.30 days too small. Not a strong performance.
- In a least squares model, the residuals are assumed to follow a Normal distribution, with mean zero, and standard deviation (for the `smart_cle1_sh` data) of about 9.0 days. We know this because the residual standard error is specified as 8.968 later in the linear model output. Thus, by the definition of a Normal distribution, we'd expect
 - about 68% of the residuals to be between -9 and +9 days,
 - about 95% of the residuals to be between -18 and +18 days,
 - about all (99.7%) of the residuals to be between -27 and +27 days.

8.5.2.2 Coefficients section

The `summary` for a linear model shows Estimates, Standard Errors, t values and *p* values for each coefficient fit.

```
summary(model_A)
```

Call:

```
lm(formula = physhealth ~ bmi, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.2641	-4.5596	-3.5897	-0.7778	29.2460

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.64970	1.22523	-2.163	0.0308 *
bmi	0.25592	0.04217	6.069	1.76e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.031 on 1131 degrees of freedom

Multiple R-squared: 0.03154, Adjusted R-squared: 0.03068

F-statistic: 36.83 on 1 and 1131 DF, p-value: 1.757e-09

- The Estimates are the point estimates of the intercept and slope of `bmi` in our model.

- In this case, our estimated slope is 0.265, which implies that if Harry's BMI is 20 and Sally's BMI is 21, we predict that Sally's `physhealth` will be 0.265 days larger than Harry's.
- The Standard Errors are also provided for each estimate. We can create rough 95% uncertainty intervals for these estimated coefficients by adding and subtracting two standard errors from each coefficient, or we can get a slightly more accurate answer with the `confint` function.
- Here, the 95% uncertainty interval for the slope of `bmi` is estimated to be (0.18, 0.35). This is a good measure of the uncertainty in the slope that is captured by our model. We are 95% confident in the process of building this interval, but this doesn't mean we're 95% sure that the true slope is actually in that interval.

Also available are a t value (just the Estimate divided by the Standard Error) and the appropriate p value for testing the null hypothesis that the true value of the coefficient is 0 against a two-tailed alternative.

- If a slope coefficient is statistically detectably different from 0, this implies that 0 will not be part of the uncertainty interval obtained through `confint`.
- If the slope was zero, it would suggest that `bmi` would add no predictive value to the model. But that's unlikely here.

If the `bmi` slope coefficient is associated with a small p value, as in the case of our `model_A`, it suggests that the model including `bmi` is statistically detectably better at predicting `physhealth` than the model without `bmi`.

- Without `bmi` our `model_A` would become an *intercept-only* model, in this case, which would predict the mean `physhealth` for everyone, regardless of any other information.

8.5.2.3 Model Fit Summaries

```
summary(model_A)
```

Call:

```
lm(formula = physhealth ~ bmi, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.2641	-4.5596	-3.5897	-0.7778	29.2460

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.64970      1.22523  -2.163   0.0308 *
bmi          0.25592      0.04217   6.069 1.76e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.031 on 1131 degrees of freedom
Multiple R-squared:  0.03154,    Adjusted R-squared:  0.03068
F-statistic: 36.83 on 1 and 1131 DF,  p-value: 1.757e-09

```

The summary of a linear model also displays:

- The residual standard error and associated degrees of freedom for the residuals.
- For a simple (one-predictor) least regression like this, the residual degrees of freedom will be the sample size minus 2.
- The multiple R-squared (or coefficient of determination)
- This is interpreted as the proportion of variation in the outcome (`physhealth`) accounted for by the model, and will always fall between 0 and 1 as a result.
- Our model_A accounts for a mere 3.4% of the variation in `physhealth`.
- The Adjusted R-squared value “adjusts” for the size of our model in terms of the number of coefficients included in the model.
- The adjusted R-squared will always be smaller than the Multiple R-squared.
- We still hope to find models with relatively large adjusted R^2 values.
- In particular, we hope to find models where the adjusted R^2 isn’t substantially less than the Multiple R-squared.
- The adjusted R-squared is usually a better estimate of likely performance of our model in new data than is the Multiple R-squared.
- The adjusted R-squared result is no longer interpretable as a proportion of anything - in fact, it can fall below 0.
- We can obtain the adjusted R^2 from the raw R^2 , the number of observations N and the number of predictors p included in the model, as follows:

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1},$$

- The F statistic and p value from a global ANOVA test of the model.
 - Obtaining a statistically significant result here is usually pretty straightforward, since the comparison is between our model, and a model which simply predicts the mean value of the outcome for everyone.
 - In a simple (one-predictor) linear regression like this, the t statistic for the slope is just the square root of the F statistic, and the resulting p values for the slope’s t test and for the global F test will be identical.

- To see the complete ANOVA F test for this model, we can run `anova(model_A)`.

```
anova(model_A)
```

Analysis of Variance Table

Response: physhealth

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
bmi	1	3004	3003.58	36.83	1.757e-09 ***
Residuals	1131	92237	81.55		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

8.5.3 Using the broom package

The broom package has three functions of particular use in a linear regression model:

8.5.3.1 The tidy function

`tidy` builds a data frame/tibble containing information about the coefficients in the model, their standard errors, *t* statistics and *p* values.

```
tidy(model_A)
```

```
# A tibble: 2 x 5
  term      estimate std.error statistic    p.value
<chr>      <dbl>      <dbl>      <dbl>    <dbl>
1 (Intercept) -2.65      1.23      -2.16  0.0308
2 bmi          0.256    0.0422     6.07 0.00000000176
```

It's often useful to include other summaries in this tidying, for instance:

```
tidy(model_A, conf.int = TRUE, conf.level = 0.9) |>
  select(term, estimate, conf.low, conf.high)
```

```
# A tibble: 2 x 4
  term      estimate conf.low conf.high
<chr>      <dbl>      <dbl>      <dbl>
1 (Intercept) -2.65      -4.67     -0.633
2 bmi          0.256    0.186     0.325
```

8.5.3.2 The glance function

`glance` builds a data frame/tibble containing summary statistics about the model, including

- the (raw) multiple R^2 and adjusted R^2
- `sigma` which is the residual standard error
- the F statistic, `p.value` model `df` and `df.residual` associated with the global ANOVA test, plus
- several statistics that will be useful in comparing models down the line:
- the model's log likelihood function value, `logLik`
- the model's Akaike's Information Criterion value, `AIC`
- the model's Bayesian Information Criterion value, `BIC`
- and the model's `deviance` statistic

```
glance(model_A)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic      p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>    <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
1   0.0315      0.0307  9.03     36.8 0.00000000176     1 -4100. 8206. 8221.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

8.5.3.3 The augment function

`augment` builds a data frame/tibble which adds fitted values, residuals and other diagnostic summaries that describe each observation to the original data used to fit the model, and this includes

- `.fitted` and `.resid`, the fitted and residual values, in addition to
- `.hat`, the leverage value for this observation
- `.cooksd`, the Cook's distance measure of *influence* for this observation
- `.stdresid`, the standardized residual (think of this as a z-score - a measure of the residual divided by its associated standard deviation `.sigma`)
- and `se.fit` which will help us generate prediction intervals for the model downstream

Note that each of the new columns begins with `.` to avoid overwriting any data.

```
head(augment(model_A))
```

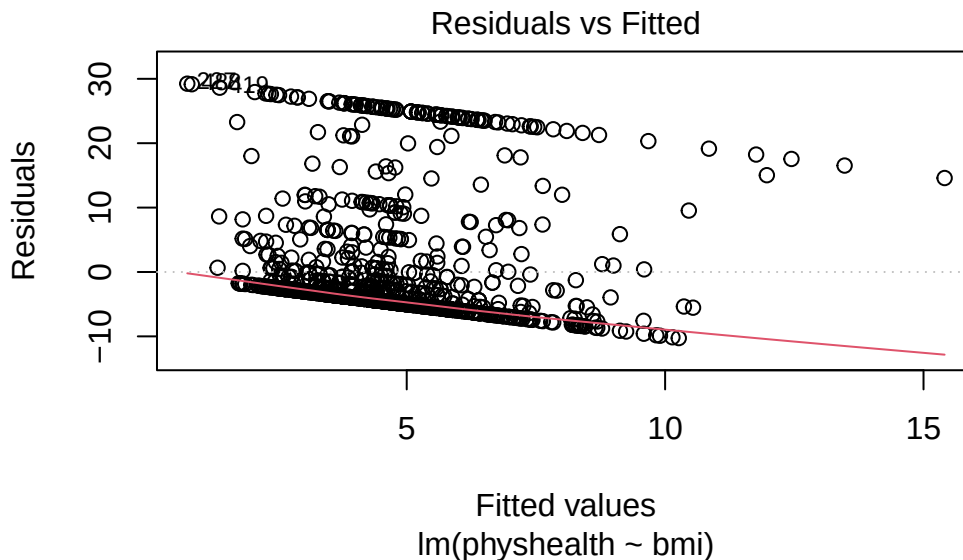
```
# A tibble: 6 x 8
  physhealth  bmi .fitted .resid    .hat .sigma    .cooksd .std.resid
    <dbl> <dbl>   <dbl> <dbl>   <dbl> <dbl>   <dbl>   <dbl>
1         4  27.9    4.50 -0.503 0.000886  9.03 0.00000138 -0.0557
2         0  23.0    3.25 -3.25  0.00149  9.03 0.0000971 -0.360
3         0  26.9    4.24 -4.24  0.000927  9.03 0.000102 -0.470
4         0  26.5    4.13 -4.13  0.000956  9.03 0.000100 -0.458
5         0  24.2    3.56 -3.56  0.00125  9.03 0.0000971 -0.394
6         2  27.7    4.45 -2.45  0.000891  9.03 0.0000328 -0.271
```

For more on the `broom` package, you may want to look at [this vignette](#).

8.5.4 How does the model do? (Residuals vs. Fitted Values)

- Remember that the R^2 value was about 3.4%.

```
plot(model_A, which = 1)
```

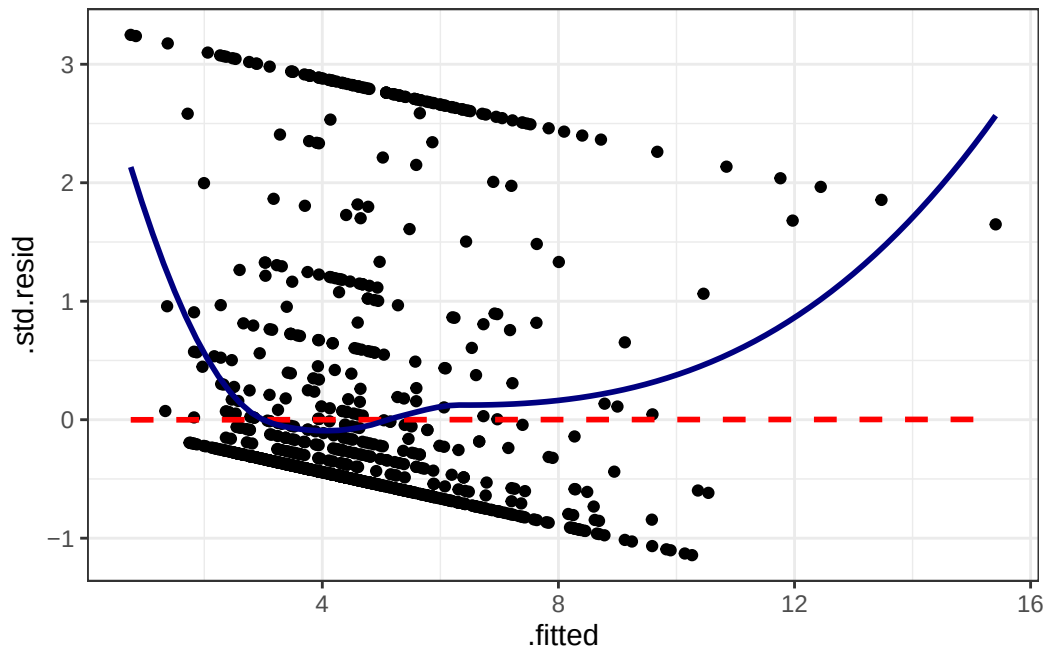


This is a plot of residuals vs. fitted values. The goal here is for this plot to look like a random scatter of points, perhaps like a “fuzzy football”, and that’s **not** what we have. Why?

If you prefer, here’s a `ggplot2` version of a similar plot, now looking at standardized residuals instead of raw residuals, and adding a loess smooth and a linear fit to the result.

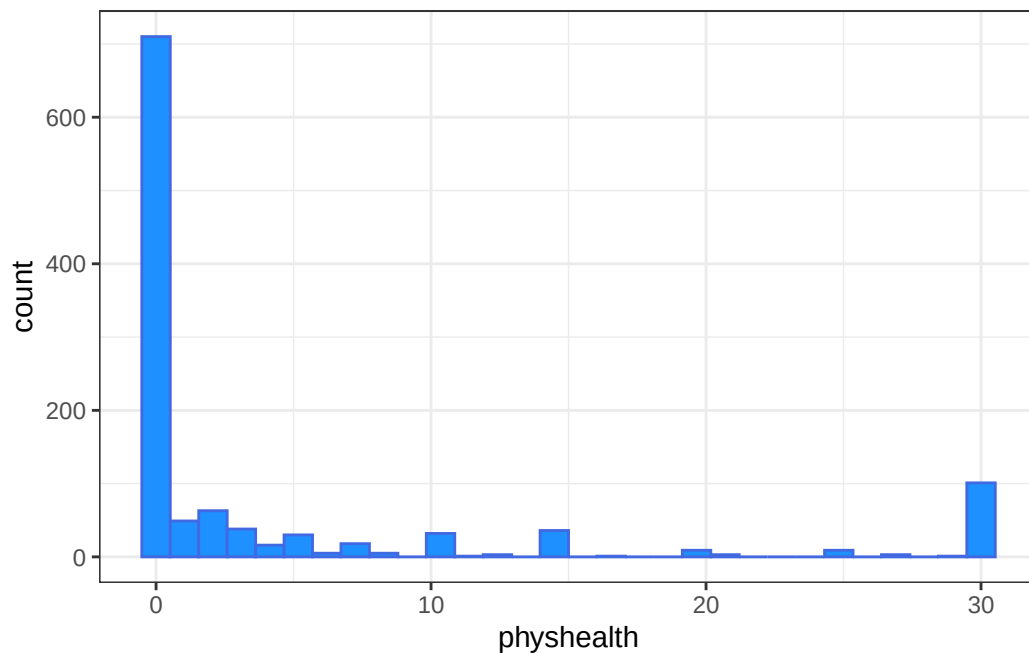

```
ggplot(augment(model_A), aes(x = .fitted, y = .std.resid)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, col = "red", linetype = "dashed") +
  geom_smooth(method = "loess", se = FALSE, col = "navy") +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



The problem we're having here becomes, I think, a little more obvious if we look at what we're predicting. Does `physhealth` look like a good candidate for a linear model?

```
ggplot(smart_cle1_sh, aes(x = physhealth)) +
  geom_histogram(bins = 30, fill = "dodgerblue",
    color = "royalblue")
```



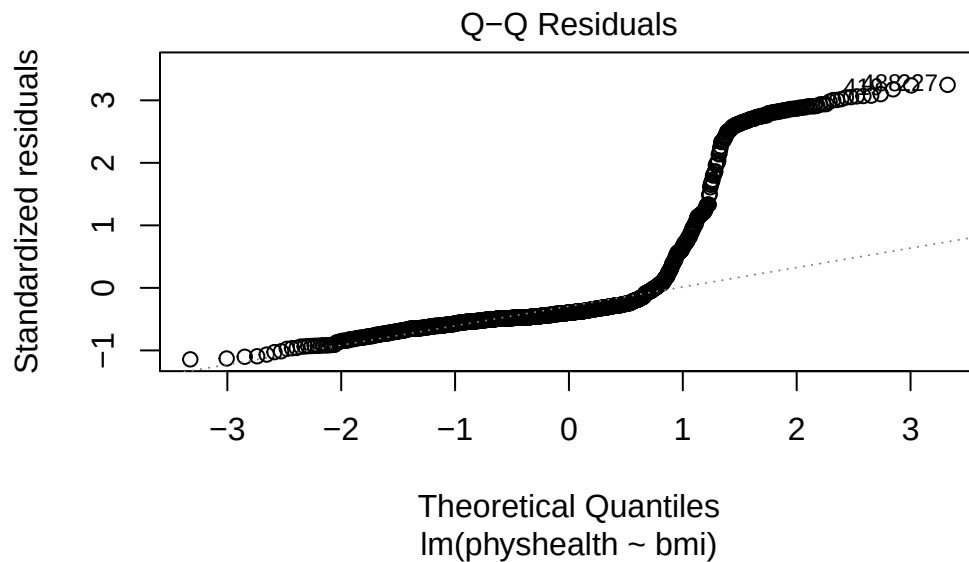
```
smart_cle1_sh |> count(physhealth == 0, physhealth == 30)
```

```
# A tibble: 3 x 3
  `physhealth == 0` `physhealth == 30`     n
  <lgl>             <lgl>             <int>
1 FALSE            FALSE             322
2 FALSE            TRUE              101
3 TRUE             FALSE             710
```

No matter what model we fit, if we are predicting **physhealth**, and most of the data are values of 0 and 30, we have limited variation in our outcome, and so our linear model will be somewhat questionable just on that basis.

A normal Q-Q plot of the standardized residuals for our `model_A` shows this problem, too.

```
plot(model_A, which = 2)
```



We're going to need a method to deal with this sort of outcome, that has both a floor and a ceiling. We'll get there eventually, but linear regression alone doesn't look promising.

All right, so that didn't go anywhere great. We'll try again, with a new outcome, in the next chapter.

9 Analysis of Variance with SMART

In this chapter, we'll work with the `smart_cle1_sh` data file again.

9.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(knitr)
library(mosaic)
library(tidyverse)

theme_set(theme_bw())
```

9.1.1 Data Load

```
smart_cle1_sh <- read_rds("data/smart_cle1_sh.Rds")
```

The variables we'll look at in this chapter are as follows.

Variable	Description
SEQNO	respondent identification number (all begin with 2016)
bmi	Body mass index, in kg/m ²
female	Sex, 1 = female, 0 = male
smoke100	Have you smoked at least 100 cigarettes in your life? (1 = yes, 0 = no)
activity	Physical activity (Highly Active, Active, Insufficiently Active, Inactive)
drinks_wk	On average, how many drinks of alcohol do you consume in a week?

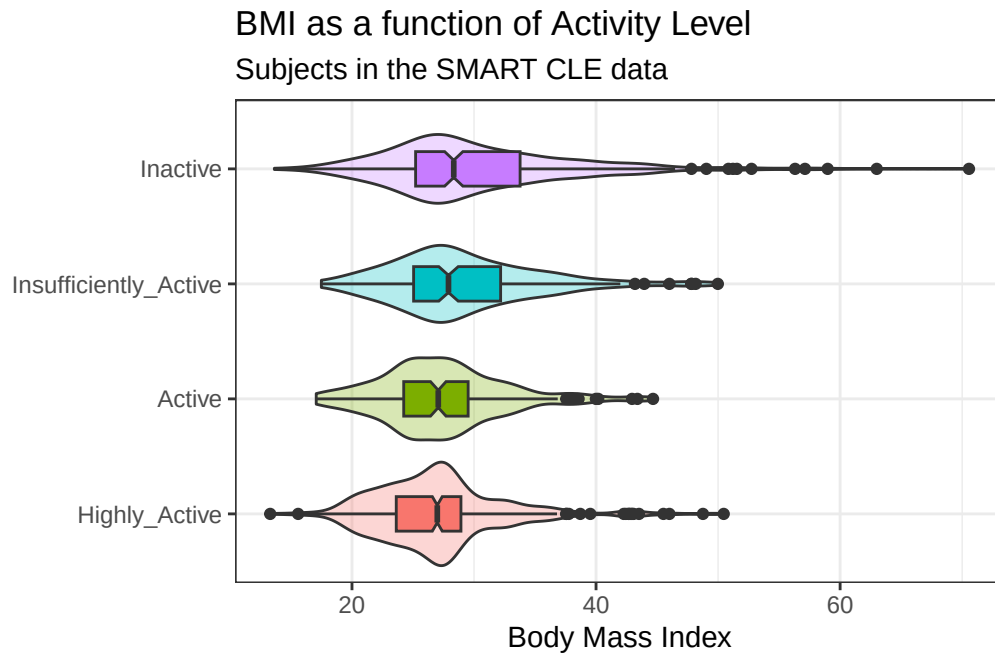
Variable	Description
physhealth	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?

9.2 A One-Factor Analysis of Variance

We'll be predicting body mass index, at first using a single factor as a predictor: the activity level.

9.2.1 Can activity be used to predict bmi?

```
ggplot(smart_cle1_sh, aes(x = activity, y = bmi,
                        fill = activity)) +
  geom_violin(alpha = 0.3) +
  geom_boxplot(width = 0.3, notch = TRUE) +
  guides(fill = "none") +
  coord_flip() +
  labs(title = "BMI as a function of Activity Level",
       subtitle = "Subjects in the SMART CLE data",
       x = "", y = "Body Mass Index")
```



Here's a numerical summary of the distributions of `bmi` within each `activity` group.

```
favstats(bmi ~ activity, data = smart_cle1_sh)
```

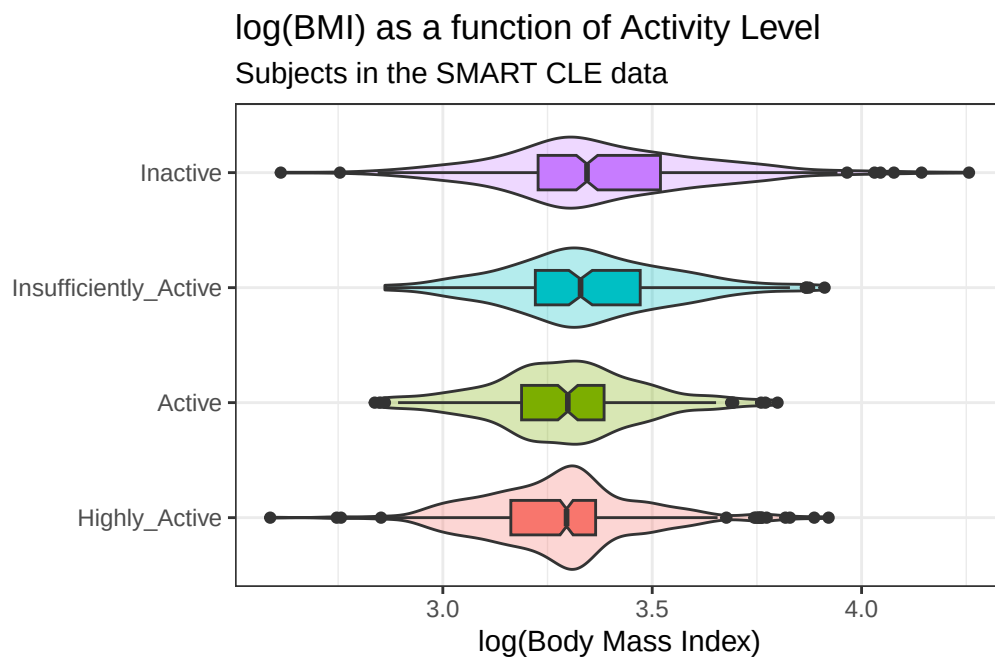
	activity	min	Q1	median	Q3	max	mean	sd
1	Highly_Active	13.30	23.6275	26.99000	28.930	50.46	27.02533	5.217758
2	Active	17.07	24.2400	27.07078	29.520	44.67	27.36435	5.151762
3	Insufficiently_Active	17.49	25.0500	27.90869	32.180	49.98	29.04259	6.051687
4	Inactive	13.64	25.2150	28.34000	33.775	70.56	30.15498	7.833659

	n	missing
1	428	0
2	173	0
3	201	0
4	331	0

9.2.2 Should we transform `bmi`?

The analysis of variance is something of a misnomer. What we're doing is using the variance to say something about population means. In light of the apparent right skew of the `bmi` results in each `activity` group, might it be a better choice to use a logarithmic transformation? We'll use the natural logarithm here, which in R, is symbolized by `log`.

```
ggplot(smart_cle1_sh, aes(x = activity, y = log(bmi),
                        fill = activity)) +
  geom_violin(alpha = 0.3) +
  geom_boxplot(width = 0.3, notch = TRUE) +
  guides(fill = "none") +
  coord_flip() +
  labs(title = "log(BMI) as a function of Activity Level",
       subtitle = "Subjects in the SMART CLE data",
       x = "", y = "log(Body Mass Index)")
```



The logarithmic transformation yields distributions that look much more symmetric in each activity group, so we'll proceed to build our regression model predicting `log(bmi)` using `activity`. Here's the numerical summary of these logged results:

```
favstats(log(bmi) ~ activity, data = smart_cle1_sh)
```

	activity	min	Q1	median	Q3	max	mean
1	Highly_Active	2.587764	3.162411	3.295466	3.364879	3.921181	3.279348
2	Active	2.837323	3.188004	3.298455	3.385068	3.799302	3.292134
3	Insufficiently_Active	2.861629	3.220874	3.328938	3.471345	3.911623	3.348360
4	Inactive	2.613007	3.227439	3.344274	3.519721	4.256463	3.376299
sd	n missing						

```

1 0.1851673 428      0
2 0.1850651 173      0
3 0.2007171 201      0
4 0.2411383 331      0

```

9.2.3 Building the ANOVA model

```

model_5a <- lm(log(bmi) ~ activity, data = smart_cle1_sh)

model_5a

```

Call:

```
lm(formula = log(bmi) ~ activity, data = smart_cle1_sh)
```

Coefficients:

	(Intercept)	activityActive
	3.27935	0.01279
activityInsufficiently_Active		activityInactive
	0.06901	0.09695

The activity data is categorical and there are four levels. The model equation is:

$$\begin{aligned}
 \log(\text{bmi}) = & 3.279 + 0.013 (\text{activity} = \text{Active}) \\
 & + 0.069 (\text{activity} = \text{Insufficiently Active}) \\
 & + 0.097 (\text{activity} = \text{Inactive})
 \end{aligned}$$

where, for example, (activity = Active) is 1 if activity is Active, and 0 otherwise. The fourth level (Highly Active) is not shown here and is used as a baseline. Thus the model above can be interpreted as follows.

activity	Predicted log(bmi)	Predicted bmi
Highly Active	3.279	$\exp(3.279) = 26.55$
Active	$3.279 + 0.013 = 3.292$	$\exp(3.292) = 26.90$
Insufficiently Active	$3.279 + 0.069 = 3.348$	$\exp(3.348) = 28.45$
Inactive	$3.279 + 0.097 = 3.376$	$\exp(3.376) = 29.25$

Those predicted log(bmi) values should look familiar. They are just the means of log(bmi) in each group, but I'm sure you'll also notice that the predicted bmi values are not exact matches for the observed means of bmi.


```
smart_cle1_sh |> group_by(activity) |>
  summarise(mean(log(bmi)), mean(bmi))
```

```
# A tibble: 4 x 3
  activity      `mean(log(bmi))` `mean(bmi)`
  <fct>          <dbl>          <dbl>
1 Highly_Active      3.28          27.0
2 Active             3.29          27.4
3 Insufficiently_Active 3.35          29.0
4 Inactive           3.38          30.2
```

9.2.4 The ANOVA table

Now, let's press on to look at the ANOVA results for this model.

```
anova(model_5a)
```

Analysis of Variance Table

```
Response: log(bmi)
          Df Sum Sq Mean Sq F value    Pr(>F)
activity    3  2.049  0.68283   16.136 2.831e-10 ***
Residuals 1129 47.778  0.04232
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The total variation in $\log(\text{bmi})$, our outcome, is captured by the sums of squares here. $SS(\text{Total}) = 2.058 + 47.770 = 49.828$
- Here, the `activity` variable (with 4 levels, so $4-1 = 3$ degrees of freedom) accounts for 4.13% ($2.058 / 49.828$) of the variation in $\log(\text{bmi})$. Another way of saying this is that the model R^2 or η^2 is 0.0413.
- The variation accounted for by the `activity` categories meets the standard for a statistically detectable result, according to the ANOVA F test, although that's not really important.
- The square root of the Mean Square(Residuals) is the residual standard error, σ , we've seen in the past. $MS(\text{Residual})$ estimates the variance (0.0423), so the residual standard error is $\sqrt{0.0423} \approx 0.206$.

9.2.5 The Model Coefficients

To address the question of effect size for the various levels of `activity` on `log(bmi)`, we could look directly at the regression model coefficients. For that, we might look at the model summary.

```
summary(model_5a)
```

Call:

```
lm(formula = log(bmi) ~ activity, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.76329	-0.12592	-0.00296	0.11045	0.88016

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.279348	0.009944	329.795	< 2e-16 ***
activityActive	0.012786	0.018533	0.690	0.49
activityInsufficiently_Active	0.069012	0.017590	3.923	9.26e-05 ***
activityInactive	0.096951	0.015057	6.439	1.78e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2057 on 1129 degrees of freedom

Multiple R-squared: 0.04111, Adjusted R-squared: 0.03857

F-statistic: 16.14 on 3 and 1129 DF, p-value: 2.831e-10

If we want to see the confidence intervals around these estimates, we could use

```
confint(model_5a, conf.level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	3.25983814	3.29885812
activityActive	-0.02357818	0.04914976
activityInsufficiently_Active	0.03449920	0.10352547
activityInactive	0.06740762	0.12649490

The model suggests, based on these 1133 subjects, that (remember that the baseline category is Highly Active)

- a 95% confidence (uncertainty) interval for the difference between Active and Highly Active subjects in log(BMI) ranges from -0.024 to 0.049
- a 95% confidence (uncertainty) interval for the difference between Insufficiently Active and Highly Active subjects in log(BMI) ranges from 0.035 to 0.104
- a 95% confidence (uncertainty) interval for the difference between Inactive and Highly Active subjects in log(BMI) ranges from 0.068 to 0.127
- the model accounts for 4.13% of the variation in log(BMI), so that knowing the respondent's activity level somewhat reduces the size of the prediction errors as compared to an intercept only model that would predict the overall mean log(BMI), regardless of activity level, for all subjects.
- from the summary of residuals, we see that one subject had a residual of 0.88 - that means they were predicted to have a log(BMI) 0.88 lower than their actual log(BMI) and one subject had a log(BMI) that is 0.76 larger than their actual log(BMI), at the extremes.

9.2.6 Using tidy to explore the coefficients

A better strategy for displaying the coefficients in any regression model is to use the `tidy` function from the `broom` package.

```
tidy(model_5a, conf.int = TRUE, conf.level = 0.95) |>
  kable(digits = 3)
```

term	esti- mate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	3.279	0.010	329.795	0.00	3.260	3.299
activityActive	0.013	0.019	0.690	0.49	-0.024	0.049
activityInsufficiently_Ac- tive	0.069	0.018	3.923	0.00	0.034	0.104
activityInactive	0.097	0.015	6.439	0.00	0.067	0.126

9.2.7 Using glance to summarize the model's fit

```
glance(model_5a) |> select(1:3) |>
  kable(digits = c(4, 4, 3))
```

r.squared	adj.r.squared	sigma
0.0411	0.0386	0.206

- The `r.squared` or R^2 value is interpreted for a linear model as the percentage of variation in the outcome (here, `log(bmi)`) that is accounted for by the model.
- The `adj.r.squared` or adjusted R^2 value incorporates a small penalty for the number of predictors included in the model. Adjusted R^2 is useful for models with more than one predictor, not simple regression models like this one. Like R^2 and most of these other summaries, its primary value comes when making comparisons between models for the same outcome.
- The `sigma` or σ is the residual standard error. Doubling this value gives us a good idea of the range of errors made by the model (approximately 95% of the time if the normal distribution assumption for the residuals holds perfectly.)

```
glance(model_5a) |> select(4:7) |>
  kable(digits = c(2, 3, 0, 2))
```

statistic	p.value	df	logLik
16.14	0	3	185.92

- The `statistic` and `p.value` shown here refer to the ANOVA F test and p value. They test the null hypothesis that the `activity` information is of no use in separating out the `bmi` data, or, equivalently, that the true R^2 is 0.
- The `df` indicates the model degrees of freedom, and in this case simply specifies the number of parameters fitted attributed to the model. Models that require more `df` for estimation require larger sample sizes.
- The `logLik` is the log likelihood for the model. This is a function of the sample size, but we can compare the fit of multiple models by comparing this value across different models for the same outcome. You want to maximize the log-likelihood.

```
glance(model_5a) |> select(8:9) |>
  kable(digits = 2)
```

AIC	BIC
-361.84	-336.68

- The `AIC` (or Akaike information criterion) and `BIC` (Bayes information criterion) are also used only to compare models. You want to minimize `AIC` and `BIC` in selecting a model. `AIC` and `BIC` are unique only up to a constant, so different packages or routines in R may give differing values, but in comparing two models - the difference in `AIC` (or `BIC`) should be consistent.

9.2.8 Using augment to make predictions

We can obtain residuals and predicted (fitted) values for the points used to fit the model with `augment` from the `broom` package.

```
augment(model_5a, se_fit = TRUE) |>
  select(1:5) |> slice(1:4) |>
  kable(digits = 3)
```

log(bmi)	activity	.fitted	.se.fit	.resid
3.330	Inactive	3.376	0.011	-0.046
3.138	Inactive	3.376	0.011	-0.239
3.293	Insufficiently_Active	3.348	0.015	-0.055
3.278	Highly_Active	3.279	0.010	-0.002

- The `.fitted` value is the predicted value of `log(bmi)` for this subject.
- The `.se.fit` value shows the standard error associated with the fitted value.
- The `.resid` is the residual value (observed - fitted `log(bmi)`)

```
augment(model_5a, se_fit = TRUE) |>
  select(1:2, 6:9) |> slice(1:4) |>
  kable(digits = 3)
```

log(bmi)	activity	.hat	.sigma	.cooksd	.std.resid
3.330	Inactive	0.003	0.206	0.000	-0.224
3.138	Inactive	0.003	0.206	0.001	-1.162
3.293	Insufficiently_Active	0.005	0.206	0.000	-0.269
3.278	Highly_Active	0.002	0.206	0.000	-0.009

- The `.hat` value shows the leverage index associated with the observation (this is a function of the predictors - higher leveraged points have more unusual predictor values)
- The `.sigma` value shows the estimate of the residual standard deviation if this observation were to be dropped from the model, and thus indexes how much of an outlier this observation's residual is.
- The `.cooksd` or Cook's distance value shows the influence that the observation has on the model - it is one of a class of leave-one-out diagnostic measures. Larger values of Cook's distance indicate more influential points.
- The `.std.resid` shows the standardized residual (which is designed to have mean 0 and standard deviation 1, facilitating comparisons across models for differing outcomes)

9.3 A Two-Factor ANOVA (without Interaction)

Let's add `race_eth` to the predictor set for `log(BMI)`.

```
model_5b <- lm(log(bmi) ~ activity + race_eth, data = smart_cle1_sh)
anova(model_5b)
```

Analysis of Variance Table

```
Response: log(bmi)
          Df Sum Sq Mean Sq F value    Pr(>F)
activity    3  2.049  0.68283  16.4184 1.904e-10 ***
race_eth    4  0.989  0.24731   5.9464 9.792e-05 ***
Residuals 1125 46.788  0.04159
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Notice that the ANOVA model assesses these variables sequentially, so the $SS(\text{activity}) = 2.058$ is accounted for before we consider the $SS(\text{race_eth}) = 0.990$. Thus, in total, the model accounts for $2.058 + 0.990 = 3.048$ of the sums of squares in `log(bmi)` in these data.

If we flip the order in the model, like this:

```
lm(log(bmi) ~ race_eth + activity, data = smart_cle1_sh) |>
  anova()
```

Analysis of Variance Table

```
Response: log(bmi)
          Df Sum Sq Mean Sq F value    Pr(>F)
race_eth    4  1.120  0.27992   6.7304 2.363e-05 ***
activity    3  1.918  0.63935  15.3729 8.314e-10 ***
Residuals 1125 46.788  0.04159
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- After flipping the order of the predictors, `race_eth` accounts for a larger Sum of Squares than it did previously, but `activity` accounts for a smaller amount, and the total between `race_eth` and `activity` remains the same, as $1.121 + 1.927$ is still 3.048 .

9.3.1 Model Coefficients

The model coefficients are unchanged regardless of the order of the variables in our two-factor ANOVA model.

```
tidy(model_5b, conf.int = TRUE, conf.level = 0.95) |>
  select(term, estimate, std.error, conf.low, conf.high) |>
  kable(digits = 3)
```

term	estimate	std.error	conf.low	conf.high
(Intercept)	3.268	0.010	3.248	3.289
activityActive	0.012	0.018	-0.024	0.048
activityInsufficiently__Active	0.073	0.018	0.039	0.108
activityInactive	0.092	0.015	0.063	0.121
race_ethBlack non-Hispanic	0.066	0.015	0.036	0.096
race_ethOther race non-Hispanic	-0.086	0.042	-0.169	-0.002
race_ethMultiracial non-Hispanic	0.020	0.042	-0.063	0.102
race_ethHispanic	0.012	0.035	-0.057	0.082

The `model_5b` equation is:

```
log(BMI) = 3.268
+ 0.012 (activity = Active)
+ 0.073 (activity = Insufficiently Active)
+ 0.092 (activity = Inactive)
+ 0.066 (race_eth = Black non-Hispanic)
- 0.086 (race_eth = Other race non-Hispanic)
+ 0.020 (race_eth = Multiracial non-Hispanic)
+ 0.012 (race_eth = Hispanic)
```

and we can make predictions by filling in appropriate 1s and 0s for the indicator variables in parentheses.

For example, the predicted `log(BMI)` for a White Highly Active person is 3.268, as White and Highly Active are the baseline categories in our two factors.

For all other combinations, we can make predictions as follows:

```

new_dat = tibble(
  race_eth = rep(c("White non-Hispanic",
                  "Black non-Hispanic",
                  "Other race non-Hispanic",
                  "Multiracial non-Hispanic",
                  "Hispanic"), 4),
  activity = c(rep("Highly_Active", 5),
               rep("Active", 5),
               rep("Insufficiently_Active", 5),
               rep("Inactive", 5))
)

augment(model_5b, newdata = new_dat)

```

```

# A tibble: 20 x 3
  race_eth          activity    .fitted
  <chr>            <chr>      <dbl>
1 White non-Hispanic Highly_Active  3.27
2 Black non-Hispanic Highly_Active  3.33
3 Other race non-Hispanic Highly_Active  3.18
4 Multiracial non-Hispanic Highly_Active  3.29
5 Hispanic          Highly_Active  3.28
6 White non-Hispanic Active        3.28
7 Black non-Hispanic Active        3.35
8 Other race non-Hispanic Active        3.19
9 Multiracial non-Hispanic Active        3.30
10 Hispanic          Active        3.29
11 White non-Hispanic Insufficiently_Active  3.34
12 Black non-Hispanic Insufficiently_Active  3.41
13 Other race non-Hispanic Insufficiently_Active  3.26
14 Multiracial non-Hispanic Insufficiently_Active  3.36
15 Hispanic          Insufficiently_Active  3.35
16 White non-Hispanic Inactive      3.36
17 Black non-Hispanic Inactive      3.43
18 Other race non-Hispanic Inactive      3.27
19 Multiracial non-Hispanic Inactive      3.38
20 Hispanic          Inactive      3.37

```

```

augment(model_5b, newdata = new_dat) |>
  mutate(race_eth = fct_relevel(factor(race_eth),
                                   "White non-Hispanic",

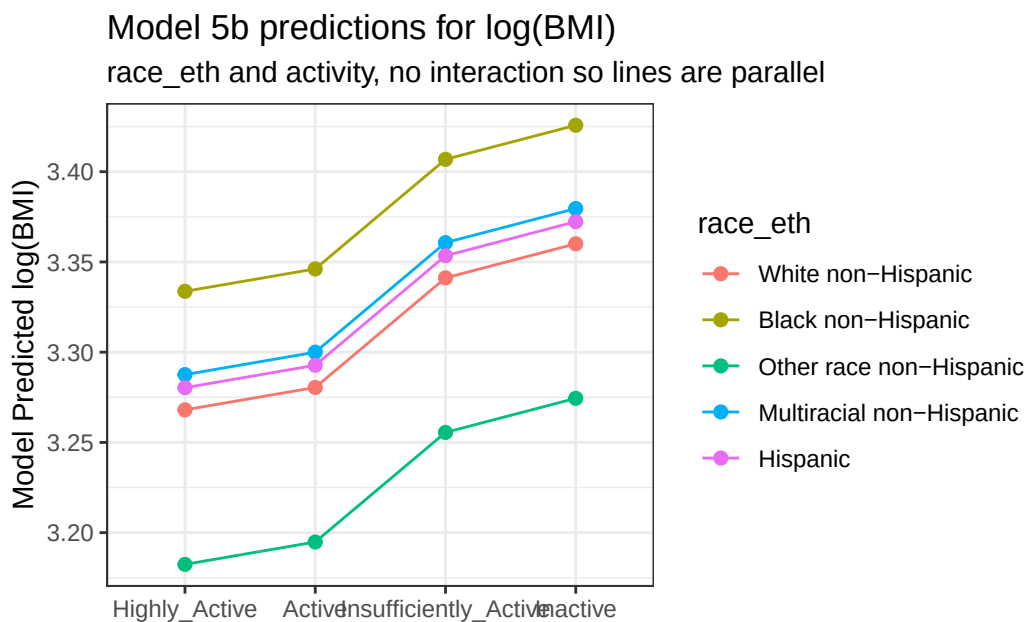
```



```

    "Black non-Hispanic",
    "Other race non-Hispanic",
    "Multiracial non-Hispanic",
    "Hispanic"),
  activity = fct_relevel(factor(activity),
    "Highly_Active",
    "Active",
    "Insufficiently_Active",
    "Inactive")) %>%
ggplot(., aes(x = activity, y = .fitted,
              col = race_eth, group = race_eth)) +
geom_point(size = 2) +
geom_line() +
labs(title = "Model 5b predictions for log(BMI)",
     subtitle = "race_eth and activity, no interaction so lines are parallel",
     y = "Model Predicted log(BMI)",
     x = "")

```



The lines joining the points for each **race_eth** category are parallel to each other. The groups always hold the same position relative to each other, regardless of their activity levels, and vice versa. There is no interaction in this model allowing the predicted effects of, say, **activity** on **log(BMI)** values to differ for the various **race_eth** groups. To do that, we'd have to fit the two-factor ANOVA model incorporating an interaction term.

9.4 A Two-Factor ANOVA (with Interaction)

Let's add the interaction of `activity` and `race_eth` (symbolized in R by `activity * race_eth`) to the model for `log(BMI)`.

```
model_5c <-  
  lm(log(bmi) ~ activity * race_eth, data = smart_cle1_sh)  
  
anova(model_5c)
```

Analysis of Variance Table

Response: log(bmi)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
activity	3	2.049	0.68283	16.3560	2.09e-10 ***
race_eth	4	0.989	0.24731	5.9238	0.0001021 ***
activity:race_eth	12	0.323	0.02688	0.6438	0.8056437
Residuals	1113	46.466	0.04175		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The ANOVA model shows that the $SS(\text{interaction}) = SS(\text{activity:race_eth})$ is 0.324, and uses 12 degrees of freedom. The model including the interaction term now accounts for $2.058 + 0.990 + 0.324 = 3.372$, which is 6.8% of the variation in `log(BMI)` overall (which is calculated as $SS(\text{Total}) = 2.058 + 0.990 + 0.324 + 46.456 = 49.828$.)

9.4.1 Model Coefficients

The model coefficients now include additional product terms that incorporate indicator variables for both `activity` and `race_eth`. For each of the product terms to take effect, both their `activity` and `race_eth` status must yield a 1 in the indicator variables.

```
tidy(model_5c, conf.int = TRUE, conf.level = 0.95) |>  
  select(term, estimate, std.error, conf.low, conf.high) |>  
  kable(digits = 3)
```

term	esti- mate	std.er- ror	conf.low	conf.high
(Intercept)	3.264	0.011	3.242	3.287

term	esti- mate	std.er- ror	conf.low	conf.high
activityActive	0.021	0.021	-0.021	0.062
activityInsufficiently__Active	0.078	0.020	0.039	0.118
activityInactive	0.097	0.018	0.062	0.132
race_ethBlack non-Hispanic	0.062	0.026	0.011	0.113
race_ethOther race non-Hispanic	-0.070	0.078	-0.223	0.083
race_ethMultiracial non-Hispanic	0.065	0.060	-0.053	0.183
race_ethHispanic	0.110	0.060	-0.008	0.228
activityActive:race_ethBlack non-Hispanic	-0.002	0.048	-0.097	0.093
activityInsufficiently__Active:race_ethBlack non-Hispanic	0.005	0.046	-0.086	0.096
activityInactive:race_ethBlack non-Hispanic	0.008	0.037	-0.064	0.080
activityActive:race_ethOther race non-Hispanic	-0.065	0.165	-0.389	0.259
activityInsufficiently__Active:race_ethOther race non-Hispanic	-0.035	0.101	-0.233	0.163
activityInactive:race_ethOther race non-Hispanic	0.033	0.129	-0.220	0.287
activityActive:race_ethMultiracial non-Hispanic	-0.206	0.134	-0.469	0.056
activityInsufficiently__Active:race_ethMultiracial non-Hispanic	-0.049	0.120	-0.284	0.186
activityInactive:race_ethMultiracial non-Hispanic	-0.054	0.110	-0.270	0.162
activityActive:race_ethHispanic	-0.104	0.096	-0.291	0.084
activityInsufficiently__Active:race_ethHispanic	-0.240	0.214	-0.660	0.179
activityInactive:race_ethHispanic	-0.169	0.082	-0.331	-0.008

The model_5c equation is:

$$\begin{aligned}
\log(\text{BMI}) = & 3.264 \\
& + 0.021 (\text{activity} = \text{Active}) \\
& + 0.079 (\text{activity} = \text{Insufficiently Active}) \\
& + 0.097 (\text{activity} = \text{Inactive}) \\
& + 0.062 (\text{race_eth} = \text{Black non-Hispanic}) \\
& - 0.070 (\text{race_eth} = \text{Other race non-Hispanic}) \\
& + 0.067 (\text{race_eth} = \text{Multiracial non-Hispanic}) \\
& + 0.110 (\text{race_eth} = \text{Hispanic}) \\
& - 0.002 (\text{activity} = \text{Active})(\text{race_eth} = \text{Black non-Hispanic}) \\
& + 0.005 (\text{Insufficiently Active})(\text{Black non-Hispanic}) \\
& + 0.008 (\text{Inactive})(\text{Black non-Hispanic}) \\
& - 0.065 (\text{Active})(\text{Other race non-Hispanic}) \\
& - 0.035 (\text{Insufficiently Active})(\text{Other race non-Hispanic}) \\
& + 0.033 (\text{Inactive})(\text{Other race non-Hispanic})
\end{aligned}$$

- 0.208 (Active)(Multiracial non-Hispanic)
- 0.050 (Insufficiently Active)(Multiracial non-Hispanic)
- 0.056 (Inactive)(Multiracial non-Hispanic)
- 0.104 (Active)(Hispanic)
- 0.240 (Insufficiently Active)(Hispanic)
- 0.169 (Inactive)(Hispanic)

and again, we can make predictions by filling in appropriate 1s and 0s for the indicator variables in parentheses.

For example, the predicted $\log(\text{BMI})$ for a White Highly Active person is 3.264, as White and Highly Active are the baseline categories in our two factors.

But the predicted $\log(\text{BMI})$ for a Hispanic Inactive person would be $3.264 + 0.097 + 0.110 - 0.169 = 3.302$.

Again, we'll plot the predicted $\log(\text{BMI})$ predictions for each possible combination.

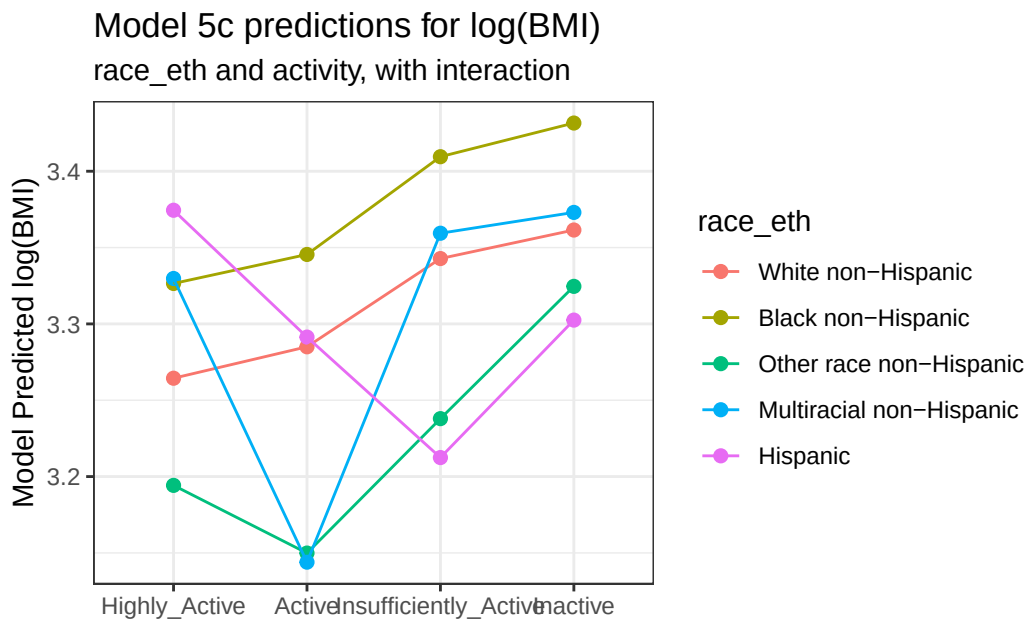
```
new_dat = tibble(
  race_eth = rep(c("White non-Hispanic",
                  "Black non-Hispanic",
                  "Other race non-Hispanic",
                  "Multiracial non-Hispanic",
                  "Hispanic"), 4),
  activity = c(rep("Highly_Active", 5),
               rep("Active", 5),
               rep("Insufficiently_Active", 5),
               rep("Inactive", 5))
)

augment(model_5c, newdata = new_dat) |>
  mutate(race_eth = fct_relevel(factor(race_eth),
                                "White non-Hispanic",
                                "Black non-Hispanic",
                                "Other race non-Hispanic",
                                "Multiracial non-Hispanic",
                                "Hispanic"),
         activity = fct_relevel(factor(activity),
                                "Highly_Active",
                                "Active",
                                "Insufficiently_Active",
                                "Inactive")) %>%
  ggplot(., aes(x = activity, y = .fitted,
```

```

    col = race_eth, group = race_eth)) +
geom_point(size = 2) +
geom_line() +
labs(title = "Model 5c predictions for log(BMI)",
     subtitle = "race_eth and activity, with interaction",
     y = "Model Predicted log(BMI)",
     x = "")

```



Note that the lines joining the points for each `race_eth` category are no longer parallel to each other. The race-ethnicity group relative positions on `log(BMI)` is now changing depending on the `activity` status.

9.4.2 Is the interaction term necessary?

We can assess this in three ways, in order of importance:

1. With an interaction plot
2. By assessing the fraction of the variation in the outcome accounted for by the interaction
3. By assessing whether the interaction accounts for statistically detectable outcome variation

9.4.2.1 The Interaction Plot

A simple interaction plot is just a plot of the unadjusted outcome means, stratified by the two factors. For example, consider this plot for our two-factor ANOVA model. To obtain this plot, we first summarize the means within each group.

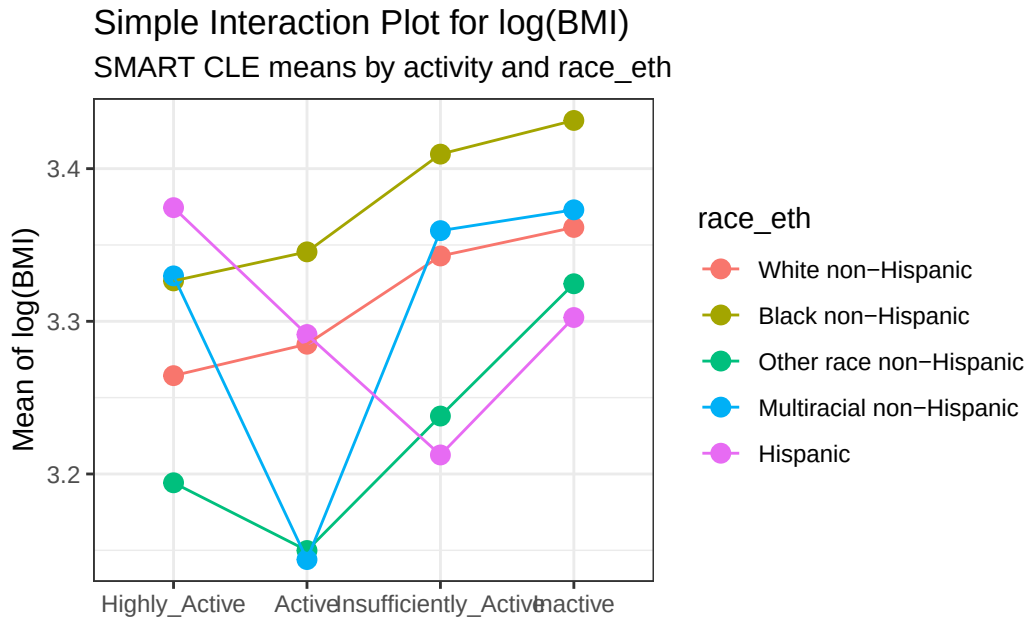
```
summaries_5 <- smart_cle1_sh |>
  group_by(activity, race_eth) |>
  summarize(n = n(), mean = mean(log(bmi)),
            sd = sd(log(bmi)))
```

`summarise()` has grouped output by 'activity'. You can override using the `groups` argument.

```
summaries_5
```

```
# A tibble: 20 x 5
# Groups:   activity [4]
  activity      race_eth      n mean      sd
  <fct>      <fct>    <int> <dbl> <dbl>
1 Highly_Active White non-Hispanic    320  3.26  0.176
2 Highly_Active Black non-Hispanic     77  3.33  0.190
3 Highly_Active Other race non-Hispanic     7  3.19  0.198
4 Highly_Active Multiracial non-Hispanic    12  3.33  0.187
5 Highly_Active Hispanic          12  3.37  0.296
6 Active      White non-Hispanic   129  3.29  0.173
7 Active      Black non-Hispanic     31  3.35  0.224
8 Active      Other race non-Hispanic     2  3.15  0.0845
9 Active      Multiracial non-Hispanic     3  3.14  0.121
10 Active      Hispanic           8  3.29  0.213
11 Insufficiently_Active White non-Hispanic   150  3.34  0.194
12 Insufficiently_Active Black non-Hispanic    35  3.41  0.213
13 Insufficiently_Active Other race non-Hispanic    11  3.24  0.137
14 Insufficiently_Active Multiracial non-Hispanic     4  3.36  0.374
15 Insufficiently_Active Hispanic           1  3.21  NA
16 Inactive    White non-Hispanic   225  3.36  0.238
17 Inactive    Black non-Hispanic    83  3.43  0.247
18 Inactive    Other race non-Hispanic     4  3.32  0.238
19 Inactive    Multiracial non-Hispanic     5  3.37  0.129
20 Inactive    Hispanic          14  3.30  0.264
```

```
ggplot(summaries_5, aes(x = activity, y = mean,
                        color = race_eth,
                        group = race_eth)) +
  geom_point(size = 3) +
  geom_line() +
  labs(title = "Simple Interaction Plot for log(BMI)",
       subtitle = "SMART CLE means by activity and race_eth",
       x = "", y = "Mean of log(BMI)")
```



The interaction plot suggests that there is a modest interaction here. The White non-Hispanic and Black non-Hispanic groups appear pretty parallel (and they are the two largest groups) and Other race non-Hispanic has a fairly similar pattern, but the other two groups (Hispanic and Multiracial non-Hispanic) bounce around quite a bit based on activity level.

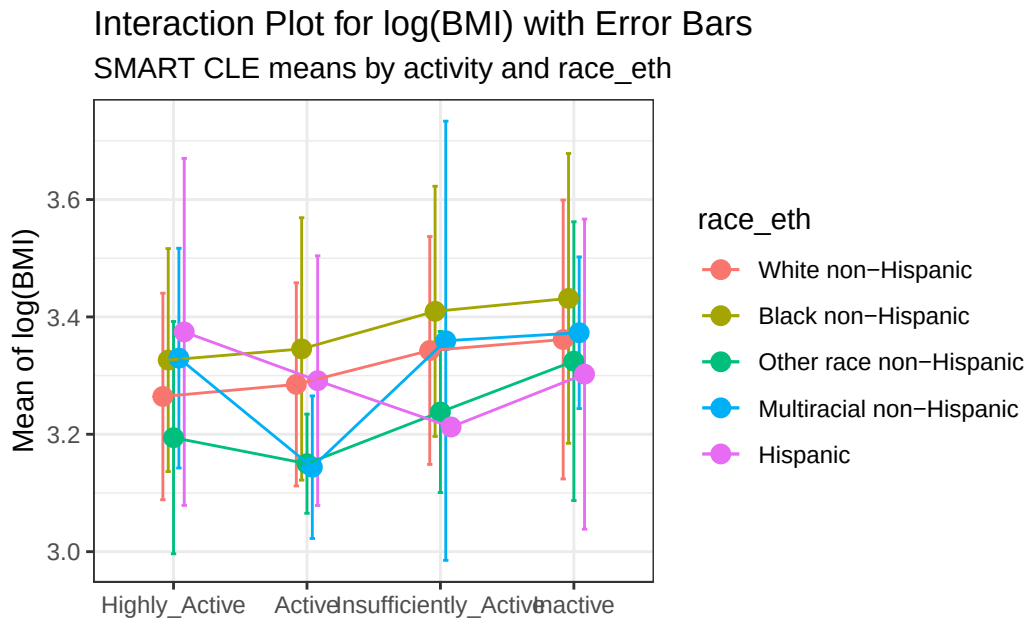
An alternative would be to include a small “dodge” for each point and include error bars (means $\pm$ standard deviation) for each combination.

```
pd = position_dodge(0.2)
ggplot(summaries_5, aes(x = activity, y = mean,
                        color = race_eth,
                        group = race_eth)) +
  geom_errorbar(aes(ymin = mean - sd,
                    ymax = mean + sd),
```

```

width = 0.2, position = pd) +
geom_point(size = 3, position = pd) +
geom_line(position = pd) +
labs(title = "Interaction Plot for log(BMI) with Error Bars",
      subtitle = "SMART CLE means by activity and race_eth",
      x = "", y = "Mean of log(BMI)")

```



Here, we see a warning flag because we have one combination (which turns out to be Insufficiently Active and Hispanic) with only one observation in it, so a standard deviation cannot be calculated. In general, I'll stick with the simpler means plot most of the time.

9.4.2.2 Does the interaction account for substantial variation?

In this case, we can look at the fraction of the overall sums of squares accounted for by the interaction.

```
anova(model_5c)
```

Analysis of Variance Table

Response: log(bmi)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
activity	3	2.049	0.68283	16.3560	2.09e-10	***
race_eth	4	0.989	0.24731	5.9238	0.0001021	***
activity:race_eth	12	0.323	0.02688	0.6438	0.8056437	
Residuals	1113	46.466	0.04175			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Here we have

$$\eta^2(Interaction) = \frac{0.324}{2.058 + 0.990 + 0.324 + 46.456} = 0.0065$$

so the interaction accounts for 0.65% of the variation in `bmi`. That looks pretty modest.

9.4.2.3 Does the interaction account for statistically detectable variation?

We can test this directly with the p value from the ANOVA table, which shows $p = 0.803$, which is far above any of our usual standards for a statistically detectable effect.

On the whole, I don't think the interaction term is especially helpful in improving this model.

In the next chapter, we'll look at two different examples of ANOVA models, now in more designed experiments. We'll also add some additional details on how the analyses might proceed.

We'll return to the SMART CLE data later in these Notes.

10 Two-Way ANOVA and Interaction

10.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(ggribbles)
library(glue)
library(gt)
library(mosaic)
library(patchwork)
library(tidyverse)

theme_set(theme_bw())
```

10.1.1 Data Load

```
bonding <- read_csv("data/bonding.csv", show_col_types = FALSE)
cortisol <- read_csv("data/cortisol.csv", show_col_types = FALSE)
```

10.2 The bonding data: A Designed Dental Experiment

The `bonding` data describe a designed experiment into the properties of four different resin types (`resin` = A, B, C, D) and two different curing light sources (`light` = Halogen, LED) as they relate to the resulting bonding strength (measured in MPa¹) on the surface of teeth. The source is @Kim2014.

¹The MPa is defined as the failure load (in Newtons) divided by the entire bonded area, in mm².

The experiment involved making measurements of bonding strength under a total of 80 experimental setups, or runs, with 10 runs completed at each of the eight combinations of a light source and a resin type. The data are gathered in the `bonding.csv` file.

```
bonding
```

```
# A tibble: 80 x 4
  runID light  resin strength
  <chr> <chr>   <chr>   <dbl>
1 R101  LED     B       12.8
2 R102  Halogen B       22.2
3 R103  Halogen B       24.6
4 R104  LED     A       17
5 R105  LED     C       32.2
6 R106  Halogen B       27.1
7 R107  LED     A       23.4
8 R108  Halogen A       23.5
9 R109  Halogen D       37.3
10 R110 Halogen A       19.7
# i 70 more rows
```

10.3 A One-Factor Analysis of Variance

Suppose we are interested in the distribution of the `strength` values for the four different types of `resin`.

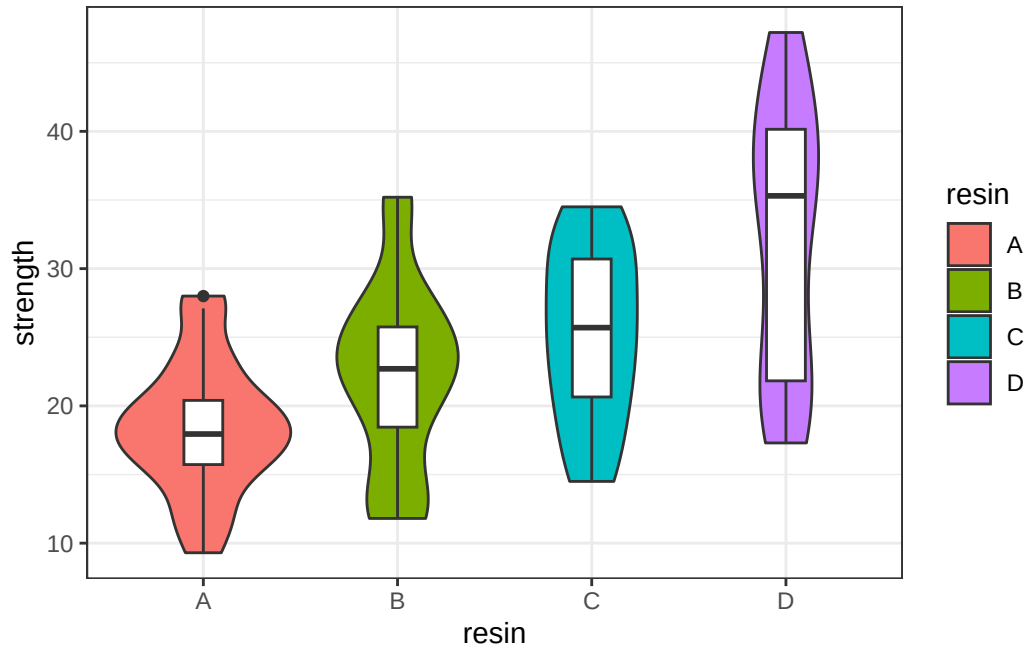
```
bonding |> group_by(resin) |>
  summarise(n = n(), mean(strength), median(strength))
```

```
# A tibble: 4 x 4
  resin      n `mean(strength)` `median(strength)`
  <chr> <int>         <dbl>         <dbl>
1 A      20      18.4      18.0
2 B      20      22.2      22.7
3 C      20      25.2      25.7
4 D      20      32.1      35.3
```

I'd begin serious work with a plot.

10.3.1 Look at the Data!

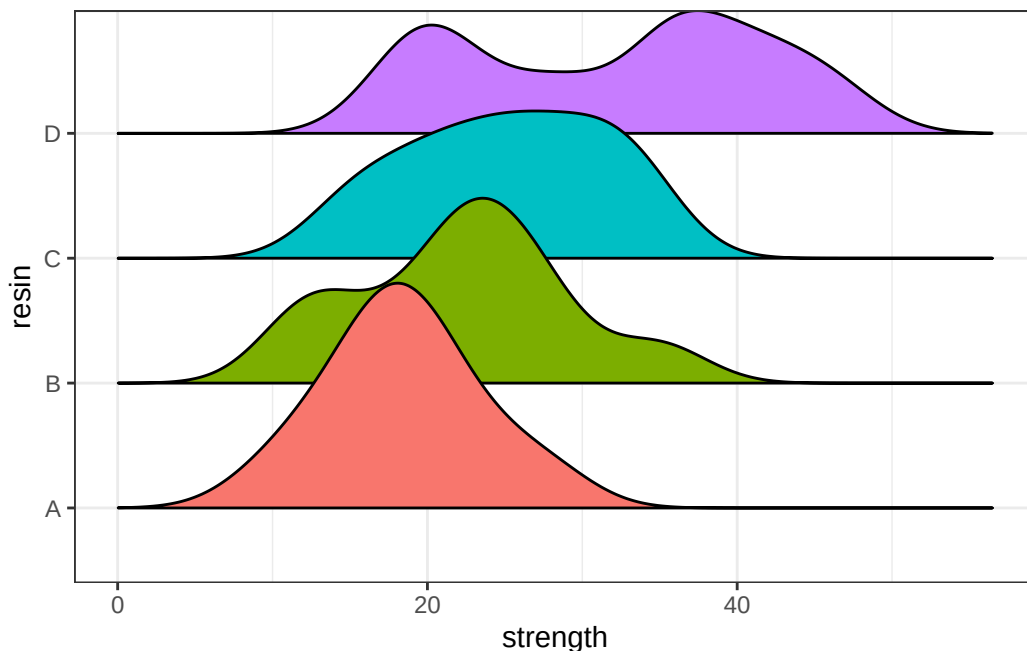
```
ggplot(bonding, aes(x = resin, y = strength)) +  
  geom_violin(aes(fill = resin)) +  
  geom_boxplot(width = 0.2)
```



Another good plot for this purpose is a ridgeline plot.

```
ggplot(bonding, aes(x = strength, y = resin, fill = resin)) +  
  geom_density_ridges2() +  
  guides(fill = "none")
```

Picking joint bandwidth of 3.09



10.3.2 Table of Summary Statistics

With the small size of this experiment ($n = 20$ for each **resin** type), graphical summaries may not perform as well as they often do. We'll also produce a quick table of summary statistics for **strength** within each **resin** type.

```
favstats(strength ~ resin, data = bonding)
```

	resin	min	Q1	median	Q3	max	mean	sd	n	missing
1	A	9.3	15.725	17.95	20.40	28.0	18.415	4.805948	20	0
2	B	11.8	18.450	22.70	25.75	35.2	22.230	6.748263	20	0
3	C	14.5	20.650	25.70	30.70	34.5	25.155	6.326425	20	0
4	D	17.3	21.825	35.30	40.15	47.2	32.075	9.735063	20	0

Since the means and medians within each group are fairly close, and the distributions (with the possible exception of **resin** D) are reasonably well approximated by the Normal, I'll fit an ANOVA model².

²If the data weren't approximately Normally distributed, we might instead consider a rank-based alternative to ANOVA, like the Kruskal-Wallis test.

```
anova(lm(strength ~ resin, data = bonding))
```

Analysis of Variance Table

Response: strength

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
resin	3	1999.7	666.57	13.107	5.52e-07 ***
Residuals	76	3865.2	50.86		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

It appears that the **resin** types have a significant association with mean **strength** of the bonds. Can we identify which **resin** types have generally higher or lower **strength**?

```
t_bond <- TukeyHSD(aov(strength ~ resin, data = bonding),
                  ordered = TRUE, conf.level = 0.90)

tidy(t_bond) |>
  select(-c(term, null.value)) |>
  mutate(across(.cols = -contrast, num, digits = 3)) |>
  arrange(desc(estimate)) |>
  gt() |>
  tab_header(title = "Comparing Mean Bond Strength across pairs of resin types",
             subtitle = "90% Tukey HSD Confidence Intervals") |>
  tab_footnote(footnote = glue(nrow(bonding), " teeth in bonding data"))
```

Warning: There was 1 warning in `mutate()`.

i In argument: `across(.cols = -contrast, num, digits = 3)`.

Caused by warning:

! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.

Supply arguments directly to `.fns` through an anonymous function instead.

Previously

```
across(a:b, mean, na.rm = TRUE)
```

Now

```
across(a:b, \(x) mean(x, na.rm = TRUE))
```

Comparing Mean Bond Strength across pairs of resin types

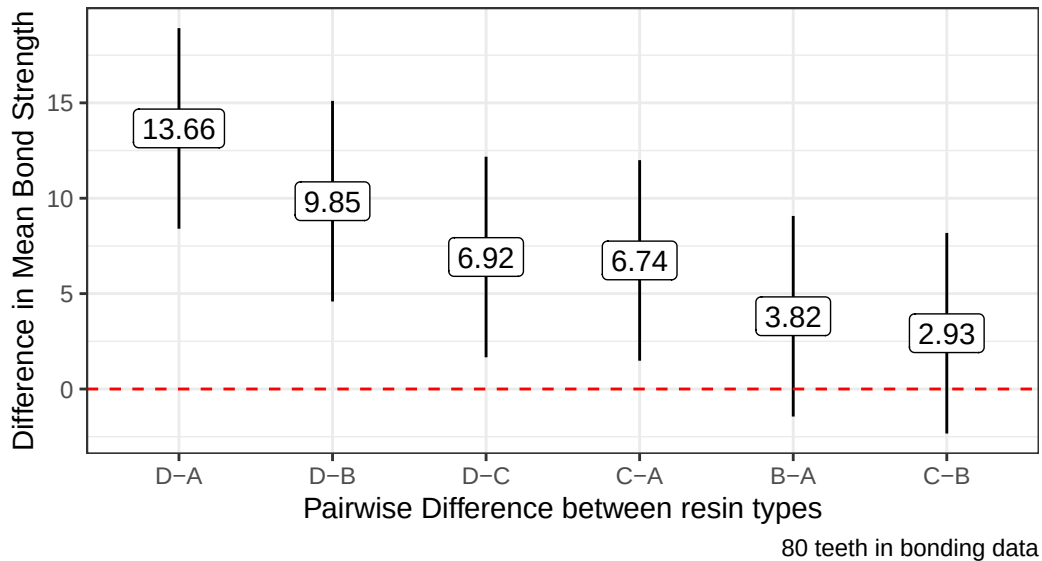
90% Tukey HSD Confidence Intervals

contrast	estimate	conf.low	conf.high	adj.p.value
D-A	13.660	8.403	18.917	0.000
D-B	9.845	4.588	15.102	0.000
D-C	6.920	1.663	12.177	0.015
C-A	6.740	1.483	11.997	0.019
B-A	3.815	-1.442	9.072	0.335
C-B	2.925	-2.332	8.182	0.568

80 teeth in bonding data

```
tidy(t_bond) |>
  mutate(contrast = fct_reorder(contrast, estimate, .desc = TRUE)) %>%
  ggplot(., aes(x = contrast, y = estimate, ymin = conf.low, ymax = conf.high)) +
  geom_pointrange() +
  geom_hline(yintercept = 0, col = "red", linetype = "dashed") +
  geom_label(aes(label = round_half_up(estimate, 2))) +
  labs(title = "Comparing Mean Bond Strength across pairs of resin types",
       subtitle = "90% Tukey HSD Confidence intervals",
       caption = glue(nrow(bonding), " teeth in bonding data"),
       x = "Pairwise Difference between resin types",
       y = "Difference in Mean Bond Strength")
```

Comparing Mean Bond Strength across pairs of resin types
90% Tukey HSD Confidence intervals

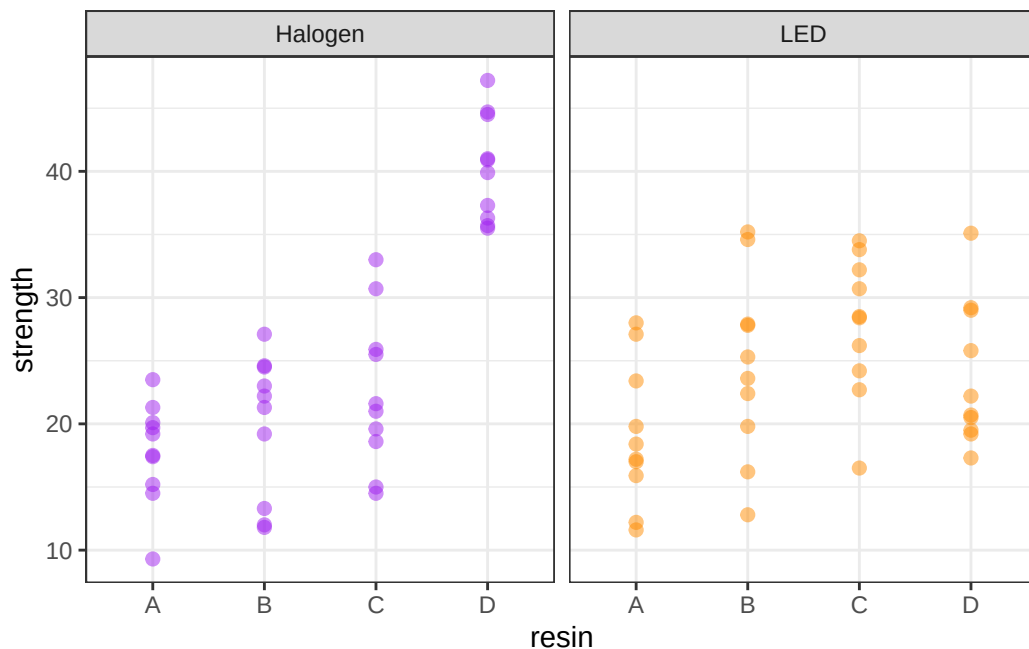


Based on these confidence intervals (which have a family-wise 90% confidence level), we see that D shows larger mean **strength** than A or B or C, and that C is also associated with larger mean **strength** than A.

10.4 A Two-Way ANOVA: Looking at Two Factors

Now, we'll now add consideration of the **light** source into our study. We can look at the distribution of the **strength** values at the combinations of both **light** and **resin**, with a plot like this one.

```
ggplot(bonding, aes(x = resin, y = strength, color = light)) +
  geom_point(size = 2, alpha = 0.5) +
  facet_wrap(~ light) +
  guides(color = "none") +
  scale_color_manual(values = c("purple", "darkorange")) +
  theme_bw()
```

10.5 A Means Plot (with standard deviations) to check for interaction

Sometimes, we'll instead look at a plot simply of the means (and, often, the standard deviations) of `strength` at each combination of `light` and `resin`. We'll start by building up a data set with the summaries we want to plot.

```
bond.sum <- bonding |>
  group_by(resin, light) |>
  summarize(mean.str = mean(strength), sd.str = sd(strength))
```

``summarise()`` has grouped output by 'resin'. You can override using the ``.groups`` argument.

```
bond.sum
```

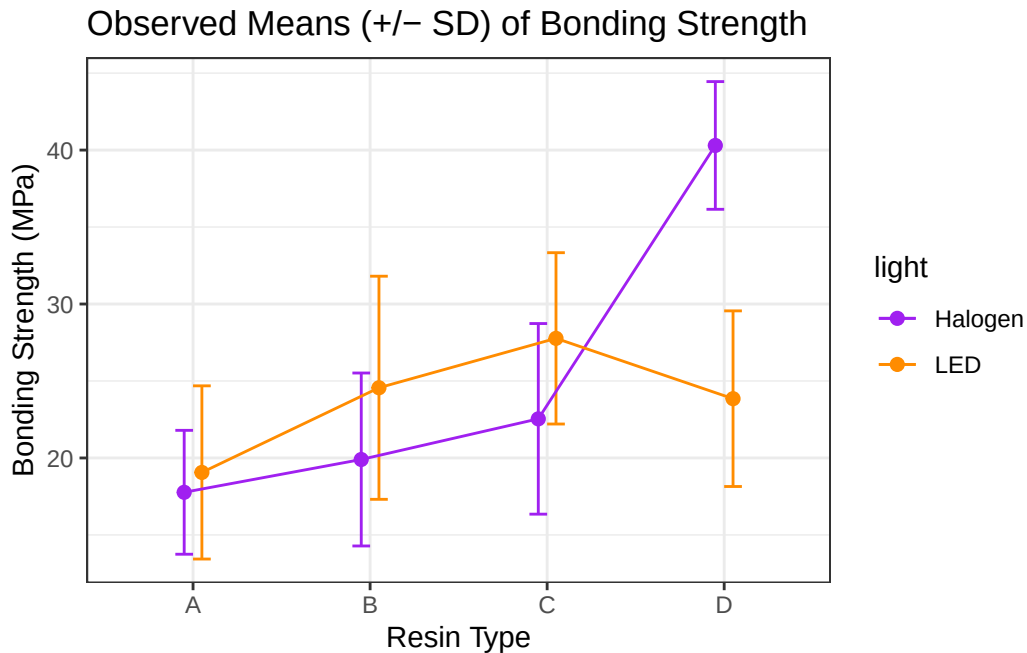
```
# A tibble: 8 x 4
# Groups:   resin [4]
  resin light mean.str sd.str
  <chr> <chr>    <dbl> <dbl>
```

1	A	Halogen	17.8	4.02
2	A	LED	19.1	5.63
3	B	Halogen	19.9	5.62
4	B	LED	24.6	7.25
5	C	Halogen	22.5	6.19
6	C	LED	27.8	5.56
7	D	Halogen	40.3	4.15
8	D	LED	23.8	5.70

Now, we'll use this new data set to plot the means and standard deviations of **strength** at each combination of **resin** and **light**.

```
## The error bars will overlap unless we adjust the position.
pd <- position_dodge(0.2) # move them .1 to the left and right

ggplot(bond.sum, aes(x = resin, y = mean.str, col = light)) +
  geom_errorbar(aes(ymin = mean.str - sd.str,
                    ymax = mean.str + sd.str),
                width = 0.2, position = pd) +
  geom_point(size = 2, position = pd) +
  geom_line(aes(group = light), position = pd) +
  scale_color_manual(values = c("purple", "darkorange")) +
  theme_bw() +
  labs(y = "Bonding Strength (MPa)", x = "Resin Type",
       title = "Observed Means (+/- SD) of Bonding Strength")
```



Is there evidence of a meaningful interaction between the resin type and the `light` source on the bonding strength in this plot?

- Sure. A meaningful interaction just means that the strength associated with different resin types depends on the `light` source.
 - With LED light, it appears that resin C leads to the strongest bonding strength.
 - With Halogen light, though, it seems that resin D is substantially stronger.
- Note that the lines we see here connecting the `light` sources aren't in parallel (as they would be if we had zero interaction between `resin` and `light`), but rather, they cross.

10.5.1 Summarizing the data after grouping by resin and light

We might want to look at a numerical summary of the `strengths` within these groups, too.

```
favstats(strength ~ resin + light, data = bonding) |>
  select(resin.light, median, mean, sd, n, missing)
```

	resin.light	median	mean	sd	n	missing
1	A.Halogen	18.35	17.77	4.024108	10	0
2	B.Halogen	21.75	19.90	5.617631	10	0
3	C.Halogen	21.30	22.54	6.191069	10	0

4	D.Halogen	40.40	40.30	4.147556	10	0
5	A.LED	17.80	19.06	5.625181	10	0
6	B.LED	24.45	24.56	7.246792	10	0
7	C.LED	28.45	27.77	5.564980	10	0
8	D.LED	21.45	23.85	5.704043	10	0

10.6 Fitting the Two-Way ANOVA model with Interaction

```
c3_m1 <- lm(strength ~ resin * light, data = bonding)
summary(c3_m1)
```

Call:

```
lm(formula = strength ~ resin * light, data = bonding)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-11.760	-3.663	-0.320	3.697	11.250

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.770	1.771	10.033	2.57e-15 ***
resinB	2.130	2.505	0.850	0.3979
resinC	4.770	2.505	1.904	0.0609 .
resinD	22.530	2.505	8.995	2.13e-13 ***
lightLED	1.290	2.505	0.515	0.6081
resinB:lightLED	3.370	3.542	0.951	0.3446
resinC:lightLED	3.940	3.542	1.112	0.2697
resinD:lightLED	-17.740	3.542	-5.008	3.78e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.601 on 72 degrees of freedom

Multiple R-squared: 0.6149, Adjusted R-squared: 0.5775

F-statistic: 16.42 on 7 and 72 DF, p-value: 9.801e-13

10.6.1 The ANOVA table for our model

In a two-way ANOVA model, we begin by assessing the interaction term. If it's important, then our best model is the model including the interaction. If it's not important, we will often move on to consider a new model, fit without an interaction.

The ANOVA table is especially helpful in this case, because it lets us look specifically at the interaction effect.

```
anova(c3_m1)
```

Analysis of Variance Table

Response: strength

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
resin	3	1999.72	666.57	21.2499	5.792e-10 ***
light	1	34.72	34.72	1.1067	0.2963
resin:light	3	1571.96	523.99	16.7043	2.457e-08 ***
Residuals	72	2258.52	31.37		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

10.6.2 Is the interaction important?

In this case, the interaction:

- is evident in the means plot, and
- is highly statistically significant, and
- accounts for a sizable fraction (27%) of the overall variation

$$\eta_{interaction}^2 = \frac{SS(\text{resin:light})}{SS(\text{Total})} = \frac{1571.96}{1999.72 + 34.72 + 1571.96 + 2258.52} = 0.268$$

If the interaction were *either* large or significant we would be inclined to keep it in the model. In this case, it's both, so there's no real reason to remove it.

10.6.3 Interpreting the Interaction

Recall the model equation, which is:

```
c3_m1
```

Call:

```
lm(formula = strength ~ resin * light, data = bonding)
```

Coefficients:

(Intercept)	resinB	resinC	resinD
17.77	2.13	4.77	22.53
lightLED	resinB:lightLED	resinC:lightLED	resinD:lightLED
1.29	3.37	3.94	-17.74

So, if `light` = Halogen, our equation is:

$$strength = 17.77 + 2.13resinB + 4.77resinC + 22.53resinD$$

And if `light` = LED, our equation is:

$$strength = 19.06 + 5.50resinB + 8.71resinC + 4.79resinD$$

Note that both the intercept and the slopes change as a result of the interaction. The model yields a different prediction for every possible combination of a `resin` type and a `light` source.

10.7 Comparing Individual Combinations of resin and light

To make comparisons between individual combinations of a `resin` type and a `light` source, using something like Tukey's HSD approach for multiple comparisons, we first refit the model using the `aov` structure, rather than `lm`.

```
c3m1_aov <- aov(strength ~ resin * light, data = bonding)
summary(c3m1_aov)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
resin	3	1999.7	666.6	21.250	5.79e-10	***
light	1	34.7	34.7	1.107	0.296	
resin:light	3	1572.0	524.0	16.704	2.46e-08	***

```
Residuals    72 2258.5    31.4
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

And now, we can obtain Tukey HSD comparisons (which will maintain an overall 90% family-wise confidence level) across the `resin` types, the `light` sources, and the combinations, with the `TukeyHSD` command. This approach is only completely appropriate if these comparisons are pre-planned, and if the design is balanced (as this is, with the same sample size for each combination of a `light` source and `resin` type.)

```
TukeyHSD(c3m1_aov, conf.level = 0.9)
```

```
Tukey multiple comparisons of means
 90% family-wise confidence level
```

```
Fit: aov(formula = strength ~ resin * light, data = bonding)
```

```
$resin
```

	diff	lwr	upr	p adj
B-A	3.815	-0.3176052	7.947605	0.1461960
C-A	6.740	2.6073948	10.872605	0.0016436
D-A	13.660	9.5273948	17.792605	0.0000000
C-B	2.925	-1.2076052	7.057605	0.3568373
D-B	9.845	5.7123948	13.977605	0.0000026
D-C	6.920	2.7873948	11.052605	0.0011731

```
$light
```

	diff	lwr	upr	p adj
LED-Halogen	-1.3175	-3.404306	0.7693065	0.2963128

```
$`resin:light`
```

	diff	lwr	upr	p adj
B:Halogen-A:Halogen	2.13	-4.9961962	9.256196	0.9893515
C:Halogen-A:Halogen	4.77	-2.3561962	11.896196	0.5525230
D:Halogen-A:Halogen	22.53	15.4038038	29.656196	0.0000000
A:LED-A:Halogen	1.29	-5.8361962	8.416196	0.9995485
B:LED-A:Halogen	6.79	-0.3361962	13.916196	0.1361092
C:LED-A:Halogen	10.00	2.8738038	17.126196	0.0037074
D:LED-A:Halogen	6.08	-1.0461962	13.206196	0.2443200
C:Halogen-B:Halogen	2.64	-4.4861962	9.766196	0.9640100
D:Halogen-B:Halogen	20.40	13.2738038	27.526196	0.0000000
A:LED-B:Halogen	-0.84	-7.9661962	6.286196	0.9999747

B:LED-B:Halogen	4.66	-2.4661962	11.786196	0.5818695
C:LED-B:Halogen	7.87	0.7438038	14.996196	0.0473914
D:LED-B:Halogen	3.95	-3.1761962	11.076196	0.7621860
D:Halogen-C:Halogen	17.76	10.6338038	24.886196	0.0000000
A:LED-C:Halogen	-3.48	-10.6061962	3.646196	0.8591455
B:LED-C:Halogen	2.02	-5.1061962	9.146196	0.9922412
C:LED-C:Halogen	5.23	-1.8961962	12.356196	0.4323859
D:LED-C:Halogen	1.31	-5.8161962	8.436196	0.9995004
A:LED-D:Halogen	-21.24	-28.3661962	-14.113804	0.0000000
B:LED-D:Halogen	-15.74	-22.8661962	-8.613804	0.0000006
C:LED-D:Halogen	-12.53	-19.6561962	-5.403804	0.0001014
D:LED-D:Halogen	-16.45	-23.5761962	-9.323804	0.0000002
B:LED-A:LED	5.50	-1.6261962	12.626196	0.3665620
C:LED-A:LED	8.71	1.5838038	15.836196	0.0185285
D:LED-A:LED	4.79	-2.3361962	11.916196	0.5471915
C:LED-B:LED	3.21	-3.9161962	10.336196	0.9027236
D:LED-B:LED	-0.71	-7.8361962	6.416196	0.9999920
D:LED-C:LED	-3.92	-11.0461962	3.206196	0.7690762

One conclusion from this is that the combination of D and Halogen appears stronger than each of the other seven combinations.

10.8 The bonding model without Interaction

It seems incorrect in this situation to fit a model without the interaction term, but we'll do so just so you can see what's involved.

```
c3_m2 <- lm(strength ~ resin + light, data = bonding)
summary(c3_m2)
```

Call:

```
lm(formula = strength ~ resin + light, data = bonding)
```

Residuals:

Min	1Q	Median	3Q	Max
-14.1162	-4.9531	0.1187	4.4612	14.4663

Coefficients:


```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)  19.074      1.787  10.676 < 2e-16 ***
resinB        3.815      2.260   1.688  0.09555 .
resinC        6.740      2.260   2.982  0.00386 **
resinD       13.660      2.260   6.044 5.39e-08 ***
lightLED      -1.317      1.598  -0.824  0.41229
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.147 on 75 degrees of freedom
Multiple R-squared:  0.3469,    Adjusted R-squared:  0.312
F-statistic: 9.958 on 4 and 75 DF,  p-value: 1.616e-06

```

In the no-interaction model, if `light` = Halogen, our equation is:

$$strength = 19.07 + 3.82resinB + 6.74resinC + 13.66resinD$$

And if `light` = LED, our equation is:

$$strength = 17.75 + 3.82resinB + 6.74resinC + 13.66resinD$$

So, in the no-interaction model, only the intercept changes.

```
anova(c3_m2)
```

Analysis of Variance Table

```

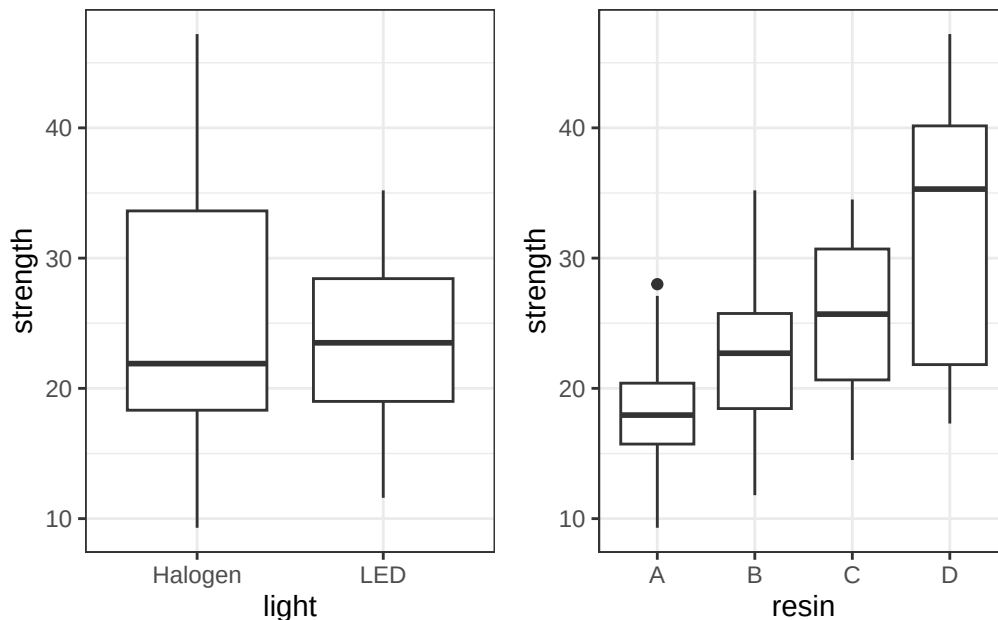
Response: strength
      Df Sum Sq Mean Sq F value    Pr(>F)
resin   3 1999.7   666.57  13.0514 6.036e-07 ***
light   1   34.7    34.72   0.6797   0.4123
Residuals 75 3830.5    51.07
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

And, it appears, if we ignore the interaction, then `resin` type has a large impact on `strength` but `light` source doesn't. This is clearer when we look at boxplots of the separated `light` and `resin` groups.

```
p1 <- ggplot(bonding, aes(x = light, y = strength)) +
  geom_boxplot()
p2 <- ggplot(bonding, aes(x = resin, y = strength)) +
  geom_boxplot()

p1 + p2
```



10.9 cortisol: A Hypothetical Clinical Trial

156 adults who complained of problems with a high-stress lifestyle were enrolled in a hypothetical clinical trial of the effectiveness of a behavioral intervention designed to help reduce stress levels, as measured by salivary cortisol.

The subjects were randomly assigned to one of three intervention groups (usual care, low dose, and high dose.) The “low dose” subjects received a one-week intervention with a follow-up at week 5. The “high dose” subjects received a more intensive three-week intervention, with follow up at week 5.

Since cortisol levels rise and fall with circadian rhythms, the cortisol measurements were taken just after rising for all subjects. These measurements were taken at baseline, and again at five weeks. The difference (baseline - week 5) in cortisol level (in micrograms / l) serves as the primary outcome.

10.9.1 Codebook and Raw Data for cortisol

The data are gathered in the `cortisol` data set. Included are:

Variable	Description
<code>subject</code>	subject identification code
<code>interv</code>	intervention group (UC = usual care, Low, High)
<code>waist</code>	waist circumference at baseline (in inches)
<code>sex</code>	male or female
<code>cort.1</code>	salivary cortisol level (microg/l) week 1
<code>cort.5</code>	salivary cortisol level (microg/l) week 5

```
cortisol
```

```
# A tibble: 156 x 6
  subject interv waist sex   cort.1 cort.5
  <chr>    <chr>  <dbl> <chr> <dbl>  <dbl>
1 S1001    UC      48.3 M     13.4   13.3
2 S1002    Low     58.3 M     17.8   16.6
3 S1003    High    43    M     14.4   12.7
4 S1004    Low     44.9 M      9     9.8
5 S1005    High    46.1 M     14.2   14.2
6 S1006    UC      41.3 M     14.8   15.1
7 S1007    Low     51    F     13.7   16
8 S1008    UC      42    F     17.3   18.7
9 S1009    Low     24.7 F     15.3   15.8
10 S1010   Low     59.4 M     12.4   11.7
# i 146 more rows
```

10.10 Creating a factor combining sex and waist

Next, we'll put the `waist` and `sex` data in the `cortisol` example together. We want to build a second categorical variable (called `fat_est`) combining this information, to indicate “healthy” vs. “unhealthy” levels of fat around the waist.

- Male subjects whose waist circumference is 40 inches or more, and
- Female subjects whose waist circumference is 35 inches or more, will fall in the “unhealthy” group.

```

cortisol <- cortisol |>
  mutate(
    fat_est = factor(case_when(
      sex == "M" & waist >= 40 ~ "unhealthy",
      sex == "F" & waist >= 35 ~ "unhealthy",
      TRUE ~ "healthy")),
    cort_diff = cort.1 - cort.5)

summary(cortisol)

```

subject	interv	waist	sex
Length:156	Length:156	Min. :20.80	Length:156
Class :character	Class :character	1st Qu.:33.27	Class :character
Mode :character	Mode :character	Median :40.35	Mode :character
		Mean :40.42	
		3rd Qu.:47.77	
		Max. :59.90	
cort.1	cort.5	fat_est	cort_diff
Min. : 6.000	Min. : 4.2	healthy : 56	Min. : -2.3000
1st Qu.: 9.675	1st Qu.: 9.6	unhealthy:100	1st Qu.: -0.5000
Median :12.400	Median :12.6		Median : 0.2000
Mean :12.686	Mean :12.4		Mean : 0.2821
3rd Qu.:16.025	3rd Qu.:15.7		3rd Qu.: 1.2000
Max. :19.000	Max. :19.7		Max. : 2.0000

10.11 A Means Plot for the cortisol trial (with standard errors)

Again, we'll start by building up a data set with the summaries we want to plot.

```

cort.sum <- cortisol |>
  group_by(interv, fat_est) |>
  summarize(mean.cort = mean(cort_diff),
            se.cort = sd(cort_diff)/sqrt(n()))

```

``summarise()`` has grouped output by 'interv'. You can override using the ``groups`` argument.

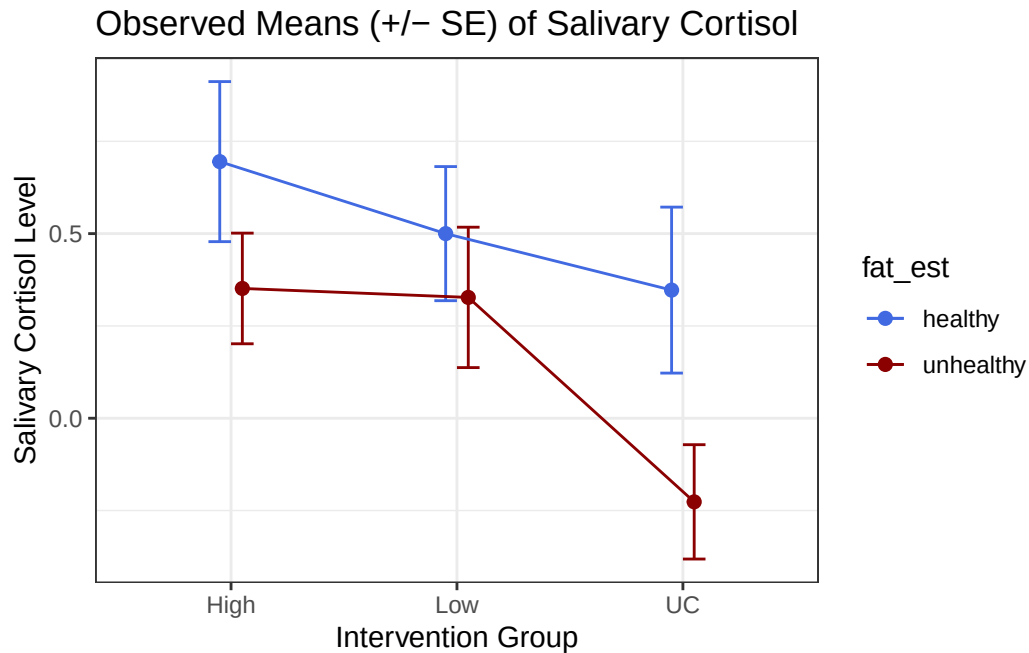
```
cort.sum
```

```
# A tibble: 6 x 4
# Groups:   interv [3]
  interv fat_est mean.cort se.cort
  <chr>   <fct>      <dbl>  <dbl>
1 High   healthy      0.695  0.217
2 High   unhealthy    0.352  0.150
3 Low    healthy      0.5    0.182
4 Low    unhealthy    0.327  0.190
5 UC     healthy      0.347  0.225
6 UC     unhealthy    -0.226  0.155
```

Now, we'll use this new data set to plot the means and standard errors.

```
## The error bars will overlap unless we adjust the position.
pd <- position_dodge(0.2) # move them .1 to the left and right

ggplot(cort.sum, aes(x = interv, y = mean.cort, col = fat_est)) +
  geom_errorbar(aes(ymin = mean.cort - se.cort,
                    ymax = mean.cort + se.cort),
               width = 0.2, position = pd) +
  geom_point(size = 2, position = pd) +
  geom_line(aes(group = fat_est), position = pd) +
  scale_color_manual(values = c("royalblue", "darkred")) +
  theme_bw() +
  labs(y = "Salivary Cortisol Level", x = "Intervention Group",
       title = "Observed Means (+/- SE) of Salivary Cortisol")
```



10.12 A Two-Way ANOVA model for cortisol with Interaction

```
c3_m3 <- lm(cort_diff ~ interv * fat_est, data = cortisol)
anova(c3_m3)
```

Analysis of Variance Table

Response: cort_diff

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
interv	2	7.847	3.9235	4.4698	0.01301 *
fat_est	1	4.614	4.6139	5.2564	0.02326 *
interv:fat_est	2	0.943	0.4715	0.5371	0.58554
Residuals	150	131.666	0.8778		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Does it seem like we need the interaction term in this case?

```
summary(c3_m3)
```

Call:

```
lm(formula = cort_diff ~ interv * fat_est, data = cortisol)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.62727	-0.75702	0.08636	0.84848	2.12647

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.6950	0.2095	3.317	0.00114 **
intervLow	-0.1950	0.3001	-0.650	0.51689
intervUC	-0.3479	0.3091	-1.126	0.26206
fat_estunhealthy	-0.3435	0.2655	-1.294	0.19774
intervLow:fat_estunhealthy	0.1708	0.3785	0.451	0.65256
intervUC:fat_estunhealthy	-0.2300	0.3846	-0.598	0.55068

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9369 on 150 degrees of freedom

Multiple R-squared: 0.0924, Adjusted R-squared: 0.06214

F-statistic: 3.054 on 5 and 150 DF, p-value: 0.01179

10.12.1 Notes on this Question

When we're evaluating a two-factor ANOVA model with an interaction, we are choosing between models with either:

1. just one factor
2. both factors, but only as main effects
3. both factors and an interaction between them

But we don't get to pick models that include any other combination of terms. For this two-way ANOVA, then, our choices are:

- a model with 'interv only
- a model with 'fat_est only
- a model with both **interv** and **fat_est** but not their interaction
- a model with **interv** and **fat_est** and their interaction

Those are the only modeling options available to us.

First, consider the ANOVA table, repeated below...

```
anova(c3_m3)
```

Analysis of Variance Table

Response: cort_diff

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
interv	2	7.847	3.9235	4.4698	0.01301 *
fat_est	1	4.614	4.6139	5.2564	0.02326 *
interv:fat_est	2	0.943	0.4715	0.5371	0.58554
Residuals	150	131.666	0.8778		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The conclusions here are as follows:

1. The interaction effect (`interv:fat_est`) has a large p value (0.58554) and assesses whether the two interaction terms (product terms) included in the model add detectable predictive value to the main effects model that includes only `interv` and `fat_est` alone. You are right to say that this ANOVA is sequential, which means that the p value for the interaction effect is looking at the additional effect of the interaction once we already have the main effects `interv` and `fat_est` included.
2. The `interv` and `fat_est` terms aren't usually evaluated with hypothesis tests or interpreted in the ANOVA for this setting, since if we intend to include the interaction term (as this model does) we also need these main effects. If we wanted to look at those terms individually in a model without the interaction, then we'd want to fit that model (without the interaction term) to do so.

Next, let's look at the summary of the `c3_m3` model, specifically the coefficients...

```
summary(c3_m3)
```

Call:

```
lm(formula = cort_diff ~ interv * fat_est, data = cortisol)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.62727	-0.75702	0.08636	0.84848	2.12647

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.6950	0.2095	3.317	0.00114 **
intervLow	-0.1950	0.3001	-0.650	0.51689
intervUC	-0.3479	0.3091	-1.126	0.26206
fat_estunhealthy	-0.3435	0.2655	-1.294	0.19774
intervLow:fat_estunhealthy	0.1708	0.3785	0.451	0.65256
intervUC:fat_estunhealthy	-0.2300	0.3846	-0.598	0.55068

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9369 on 150 degrees of freedom

Multiple R-squared: 0.0924, Adjusted R-squared: 0.06214

F-statistic: 3.054 on 5 and 150 DF, p-value: 0.01179

So here, we see two p values associated with the interaction terms (the two product terms at the bottom of the regression) but these aren't especially helpful, because we're either going to include the interaction (in which case both of these terms will be in the model) or we're not going to include the interaction (in which case neither of these terms will be in the model.)

So the p values provided here aren't very helpful - like all such p values for t tests, they are looking at the value of the term in their row as the last predictor in to the model, essentially comparing the full model to the model without that specific component, but none of those tests enable us to decide which of the 4 available model choices is our best fit.

Now, let's consider the reason why, for example, the p value for `fat_est` in the `summary()` which is looking at comparing the following models ...

- a model including `interv` (which has 2 coefficients to account for its 3 categories), `fat_est` (which has 1 coefficient to account for its 2 categories), and the `interv*fat_est` interaction terms (which are 2 terms)
- a model including `interv` and the `interv*fat_est` interaction (but somehow not the main effect of `fat_est`, which actually makes no sense: if we include the interaction we always include the main effect)

to the p value for `fat_est` in the ANOVA which is looking at comparing

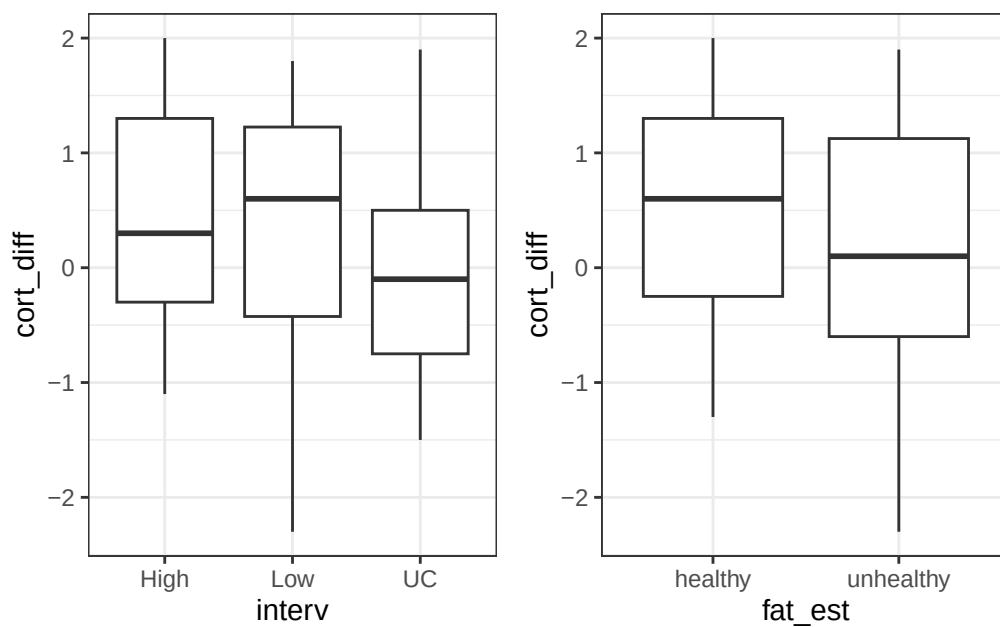
- the model with `interv` alone to
- the model with `interv` and `fat_est` as main effects, but no interaction

Only the ANOVA p value is therefore in any way useful, and it suggests that once you have the main effect of `interv`, adding `fat_est`'s main effect adds statistically detectable value (p = 0.023)

10.13 A Two-Way ANOVA model for cortisol without Interaction

10.13.1 The Graph

```
p1 <- ggplot(cortisol, aes(x = interv, y = cort_diff)) +  
  geom_boxplot()  
p2 <- ggplot(cortisol, aes(x = fat_est, y = cort_diff)) +  
  geom_boxplot()  
  
p1 + p2
```



10.13.2 The ANOVA Model

```
c3_m4 <- lm(cort_diff ~ interv + fat_est, data = cortisol)  
  
anova(c3_m4)
```

Analysis of Variance Table

```

Response: cort_diff
      Df Sum Sq Mean Sq F value Pr(>F)
interv    2   7.847   3.9235   4.4972 0.01266 *
fat_est    1   4.614   4.6139   5.2886 0.02283 *
Residuals 152 132.609   0.8724
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

How do these results compare to those we saw in the model with interaction?

10.13.3 The Regression Summary

```
summary(c3_m4)
```

```

Call:
lm(formula = cort_diff ~ interv + fat_est, data = cortisol)

Residuals:
      Min       1Q   Median       3Q      Max
-2.55929 -0.74527  0.05457  0.86456  2.05489

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.70452    0.16093   4.378 2.22e-05 ***
intervLow      -0.08645    0.18232  -0.474  0.63606
intervUC       -0.50063    0.18334  -2.731  0.00707 **
fat_estunhealthy -0.35878    0.15601  -2.300  0.02283 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.934 on 152 degrees of freedom
Multiple R-squared:  0.0859,    Adjusted R-squared:  0.06785
F-statistic: 4.761 on 3 and 152 DF,  p-value: 0.00335

```

10.13.4 Tukey HSD Comparisons

Without the interaction term, we can make direct comparisons between levels of the intervention, and between levels of the `fat_est` variable. This is probably best done here in a Tukey HSD comparison.

```
TukeyHSD(aov(cort_diff ~ interv + fat_est, data = cortisol), conf.level = 0.9)
```

Tukey multiple comparisons of means
90% family-wise confidence level

```
Fit: aov(formula = cort_diff ~ interv + fat_est, data = cortisol)
```

```
$interv
```

	diff	lwr	upr	p adj
Low-High	-0.09074746	-0.4677566	0.28626166	0.8724916
UC-High	-0.51642619	-0.8952964	-0.13755598	0.0150150
UC-Low	-0.42567873	-0.8063312	-0.04502625	0.0570728

```
$fat_est
```

	diff	lwr	upr	p adj
unhealthy-healthy	-0.3582443	-0.6162415	-0.100247	0.0229266

What conclusions can we draw here?

11 Analysis of Covariance

11.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(mosaic)
library(tidyverse)

theme_set(theme_bw())
```

11.1.1 Data Load

```
emphysema <- read_csv("data/emphysema.csv", show_col_types = FALSE)
```

11.2 An Emphysema Study

My source for this example is @Riffenburgh2006, section 18.4. Serum theophylline levels (in mg/dl) were measured in 16 patients with emphysema at baseline, then 5 days later (at the end of a course of antibiotics) and then at 10 days after baseline. Clinicians anticipate that the antibiotic will increase the theophylline level. The data are stored in the `emphysema.csv` data file, and note that the age for patient 5 is not available.

11.2.1 Codebook

Variable	Description
patient	ID code
age	patient's age in years

Variable	Description
sex	patient's sex (F or M)
st_base	patient's serum theophylline at baseline (mg/dl)
st_day5	patient's serum theophylline at day 5 (mg/dl)
st_day10	patient's serum theophylline at day 10 (mg/dl)

We're going to look at the change from baseline to day 5 as our outcome of interest, since the clinical expectation is that the antibiotic (azithromycin) will increase theophylline levels.

```
emphysema <- emphysema |>
  mutate(st_delta = st_day5 - st_base)

emphysema
```

```
# A tibble: 16 x 7
  patient age sex st_base st_day5 st_day10 st_delta
  <dbl> <dbl> <chr> <dbl> <dbl> <dbl> <dbl>
1     1     1  61 F    14.1     2.3    10.3   -11.8
2     2     2  70 F     7.2     5.4     7.3    -1.8
3     3     3  65 M    14.2    11.9    11.3    -2.3
4     4     4  65 M    10.3    10.7    13.8     0.400
5     5     5  NA M     9.9    10.7    11.7     0.800
6     6     6  76 M     5.2     6.8     4.2     1.6
7     7     7  72 M    10.4    14.6    14.1     4.2
8     8     8  69 F    10.5     7.2     5.4    -3.3
9     9     9  66 M     5      5      5.1     0
10    10    10  62 M     8.6     8.1     7.4    -0.5
11    11    11  65 F    16.6    14.9    13      -1.70
12    12    12  71 M    16.4    18.6    17.1     2.20
13    13    13  51 F    12.2    11      12.3    -1.2
14    14    14  71 M     6.6     3.7     4.5    -2.9
15    15    15  64 F    15.4    15.2    13.6    -0.200
16    16    16  50 M    10.2    10.8    11.2     0.600
```

11.3 Does sex affect the mean change in theophylline?

```
favstats(~ st_delta, data = emphysema)
```

	min	Q1	median	Q3	max	mean	sd	n	missing
	-11.8	-1.925	-0.35	0.65	4.2	-0.99375	3.484149	16	0

```
favstats(st_delta ~ sex, data = emphysema)
```

	sex	min	Q1	median	Q3	max	mean	sd	n	missing
1	F	-11.8	-2.925	-1.75	-1.325	-0.2	-3.333333	4.267864	6	0
2	M	-2.9	-0.375	0.50	1.400	4.2	0.410000	2.067446	10	0

Overall, the mean change in theophylline during the course of the antibiotic is -0.99, but this is -3.33 for female patients and 0.41 for male patients.

A one-way ANOVA model looks like this:

```
anova(lm(st_delta ~ sex, data = emphysema))
```

Analysis of Variance Table

```
Response: st_delta
          Df Sum Sq Mean Sq F value Pr(>F)
sex         1  52.547   52.547   5.6789 0.03189 *
Residuals  14 129.542    9.253
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA F test finds a fairly large difference between the mean `st_delta` among males and the mean `st_delta` among females. But is there more to the story?

11.4 Is there an association between age and sex in this study?

```
favstats(age ~ sex, data = emphysema)
```

	sex	min	Q1	median	Q3	max	mean	sd	n	missing
1	F	51	61.75	64.5	68	70	63.33333	6.889606	6	0
2	M	50	65.00	66.0	71	76	66.44444	7.568208	9	1

But we note that the male patients are also older than the female patients, on average (mean age for males is 66.4, for females 63.3)

- Does the fact that male patients are older affect change in theophylline level?
- And how should we deal with the one missing `age` value (in a male patient)?

11.5 Adding a quantitative covariate, age, to the model

We could fit an ANOVA model to predict `st_delta` using `sex` and `age` directly, but only if we categorized `age` into two or more groups. Because `age` is not categorical, we cannot include it in an ANOVA. But if age is an influence, and we don't adjust for it, it may well bias the outcome of our initial ANOVA. With a quantitative variable like `age`, we will need a method called ANCOVA, for **analysis of covariance**.

11.5.1 The ANCOVA model

ANCOVA in this case is just an ANOVA model with our outcome (`st_delta`) adjusted for a continuous covariate, called `age`. For the moment, we'll ignore the one subject with missing `age` and simply fit the regression model with `sex` and `age`.

```
summary(lm(st_delta ~ sex + age, data = emphysema))
```

Call:

```
lm(formula = st_delta ~ sex + age, data = emphysema)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-8.3352	-0.4789	0.6948	1.5580	3.5202

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-6.90266	7.92948	-0.871	0.4011
sexM	3.52466	1.75815	2.005	0.0681 .
age	0.05636	0.12343	0.457	0.6561

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.255 on 12 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.2882, Adjusted R-squared: 0.1696

F-statistic: 2.43 on 2 and 12 DF, p-value: 0.13

This model assumes that the slope of the regression line between `st_delta` and `age` is the same for both sexes.

Note that the model yields $\text{st_delta} = -6.9 + 3.52 (\text{sex} = \text{male}) + 0.056 \text{ age}$, or

- $\text{st_delta} = -6.9 + 0.056 \text{ age}$ for female patients, and
- $\text{st_delta} = (-6.9 + 3.52) + 0.056 \text{ age} = -3.38 + 0.056 \text{ age}$ for male patients.

Note that we can test this assumption of equal slopes by fitting an alternative model (with a product term between `sex` and `age`) that doesn't require the assumption, and we'll do that later.

11.5.2 The ANCOVA Table

First, though, we'll look at the ANCOVA table.

```
anova(lm(st_delta ~ sex + age, data = emphysema))
```

Analysis of Variance Table

```
Response: st_delta
      Df Sum Sq Mean Sq F value Pr(>F)
sex      1  49.284   49.284   4.6507 0.05203 .
age      1   2.209    2.209   0.2085 0.65612
Residuals 12 127.164   10.597
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

When we tested `sex` without accounting for `age`, we found a p value of 0.032, which is less than our usual cutpoint of 0.05. But when we adjusted for `age`, we find that `sex`'s p value rises, even though `age` doesn't seem to have a particularly strong influence on `st_delta` by itself, according to the ANCOVA table.

11.6 Rerunning the ANCOVA model after simple imputation

We could have *imputed* the missing `age` value for patient 5, rather than just deleting that patient. Suppose we do the simplest potentially reasonable thing to do: insert the mean `age` in where the NA value currently exists.

```
emph_imp <- replace_na(emphysema, list(age = mean(emphysema$age, na.rm = TRUE)))
emph_imp
```

```
# A tibble: 16 x 7
  patient age sex st_base st_day5 st_day10 st_delta
  <dbl> <dbl> <chr> <dbl> <dbl> <dbl> <dbl>
1     1     1 61 F      14.1     2.3    10.3 -11.8
2     2     2 70 F       7.2     5.4     7.3  -1.8
3     3     3 65 M      14.2    11.9    11.3  -2.3
4     4     4 65 M      10.3    10.7    13.8   0.400
5     5     5 65.2 M       9.9    10.7    11.7   0.800
6     6     6 76 M       5.2     6.8     4.2   1.6
7     7     7 72 M      10.4    14.6    14.1   4.2
8     8     8 69 F      10.5     7.2     5.4  -3.3
9     9     9 66 M       5       5       5.1   0
10    10    10 62 M       8.6     8.1     7.4  -0.5
11    11    11 65 F      16.6    14.9    13    -1.70
12    12    12 71 M      16.4    18.6    17.1   2.20
13    13    13 51 F      12.2    11     12.3  -1.2
14    14    14 71 M       6.6     3.7     4.5  -2.9
15    15    15 64 F      15.4    15.2    13.6  -0.200
16    16    16 50 M      10.2    10.8    11.2   0.600
```

More on simple imputation and missing data is coming soon.

For now, we can rerun the ANCOVA model on this new data set, after imputation...

```
anova(lm(st_delta ~ sex + age, data = emph_imp))
```

Analysis of Variance Table

Response: st_delta

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sex	1	52.547	52.547	5.3623	0.03755 *
age	1	2.151	2.151	0.2195	0.64721
Residuals	13	127.392	9.799		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

When we do this, we see that now the **sex** variable returns to a *p* value below 0.05. Our complete case analysis (which omitted patient 5) gives us a different result than the ANCOVA based on the data after mean imputation.

11.7 Looking at a factor-covariate interaction

Let's run a model including the interaction (product) term between **age** and **sex**, which implies that the slope of **age** on our outcome (**st_delta**) depends on the patient's sex. We'll use the imputed data again. Here is the new ANCOVA table, which suggests that the interaction of **age** and **sex** is small (because it accounts for only a small amount of the total Sum of Squares) with a *p* value of 0.91.

```
anova(lm(st_delta ~ sex * age, data = emph_imp))
```

Analysis of Variance Table

```
Response: st_delta
      Df Sum Sq Mean Sq F value    Pr(>F)
sex      1  52.547   52.547    4.9549 0.04594 *
age      1   2.151    2.151    0.2028 0.66051
sex:age   1   0.130    0.130    0.0123 0.91355
Residuals 12 127.261   10.605
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since the interaction term isn't accounting for much variation, we probably don't need it here. But let's look at its interpretation anyway, just to fix ideas. To do that, we'll need the coefficients from the underlying regression model.

```
tidy(lm(st_delta ~ sex * age, data = emph_imp))
```

```
# A tibble: 4 x 5
  term          estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)  -5.65         13.5      -0.420   0.682
2 sexM          1.72         16.8       0.102   0.920
3 age           0.0365      0.211      0.173   0.866
4 sexM:age       0.0289      0.260      0.111   0.914
```

Our ANCOVA model for **st_delta** incorporating the **age** x **sex** product term is $-5.65 + 1.72$ (**sex** = M) + 0.037 **age** + 0.029 (**sex** = M)(**age**). So that means:

- our model for females is **st_delta** = $-5.65 + 0.037$ **age**
- our model for males is **st_delta** = $(-5.65 + 1.72) + (0.037 + 0.029)$ **age**, or $-3.93 + 0.066$ **age**

but, again, our conclusion from the ANCOVA table is that this increase in complexity (letting both the slope and intercept vary by `sex`) doesn't add much in the way of predictive value for our `st_delta` outcome.

11.8 Centering the Covariate to Facilitate ANCOVA Interpretation

When developing an ANCOVA model, we will often **center** or even **center and rescale** the covariate to facilitate interpretation of the product term. In this case, let's center `age` and rescale it by dividing by two standard deviations.

```
favstats(~ age, data = emph_imp)
```

min	Q1	median	Q3	max	mean	sd	n	missing
50	63.5	65.1	70.25	76	65.2	6.978061	16	0

Note that in our imputed data, the mean `age` is 65.2 and the standard deviation of `age` is 7 years.

So we build the rescaled `age` variable that I'll call `age_z`, and then use it to refit our model.

```
emph_imp <- emph_imp |>
  mutate(age_z = (age - mean(age)) / (2 * sd(age)))

anova(lm(st_delta ~ sex * age_z, data = emph_imp))
```

Analysis of Variance Table

Response: st_delta

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sex	1	52.547	52.547	4.9549	0.04594 *
age_z	1	2.151	2.151	0.2028	0.66051
sex:age_z	1	0.130	0.130	0.0123	0.91355
Residuals	12	127.261	10.605		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
tidy(lm(st_delta ~ sex * age_z, data = emph_imp))
```

```
# A tibble: 4 x 5
  term          estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   -3.27       1.39     -2.35   0.0364
2 sexM           3.60       1.74      2.08   0.0601
3 age_z          0.510      2.95      0.173  0.866
4 sexM:age_z     0.403      3.63      0.111  0.914
```

Comparing the two models, we have:

- (unscaled): $\text{st_delta} = -5.65 + 1.72 (\text{sex} = \text{M}) + 0.037 \text{ age} + 0.029 (\text{sex} = \text{M}) \times (\text{age})$
- (rescaled): $\text{st_delta} = -3.27 + 3.60 (\text{sex} = \text{M}) + 0.510 \text{ rescaled age_z} + 0.402 (\text{sex} = \text{M}) \times (\text{rescaled age_z})$

In essence, the rescaled model on `age_z` is:

- $\text{st_delta} = -3.27 + 0.510 \text{ age_z}$ for female subjects, and
- $\text{st_delta} = (-3.27 + 3.60) + (0.510 + 0.402) \text{ age_z} = 0.33 + 0.912 \text{ age_z}$ for male subjects

Interpreting the centered, rescaled model, we have:

- no change in the ANOVA results or R-squared or residual standard deviation compared to the uncentered, unscaled model, but
- the intercept (-3.27) now represents the `st_delta` for a female of average age,
- the `sex` slope (3.60) represents the (male - female) difference in predicted `st_delta` for a person of average age,
- the `age_z` slope (0.510) represents the difference in predicted `st_delta` for a female one standard deviation older than the mean age as compared to a female one standard deviation younger than the mean age, and
- the product term's slope (0.402) represents the male - female difference in the slope of `age_z`, so that if you add the `age_z` slope (0.510) and the interaction slope (0.402) you see the difference in predicted `st_delta` for a male one standard deviation older than the mean age as compared to a male one standard deviation younger than the mean age.

12 Analysis of Covariance with the SMART data

In this chapter, we'll work with the `smart_cle1_sh` data file that we developed in Chapter 7.

12.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(mosaic)
library(tidyverse)

theme_set(theme_bw())
```

12.1.1 Data Load

```
smart_cle1_sh <- readRDS("data/smart_cle1_sh.Rds")
```

12.2 A New Small Study: Predicting BMI

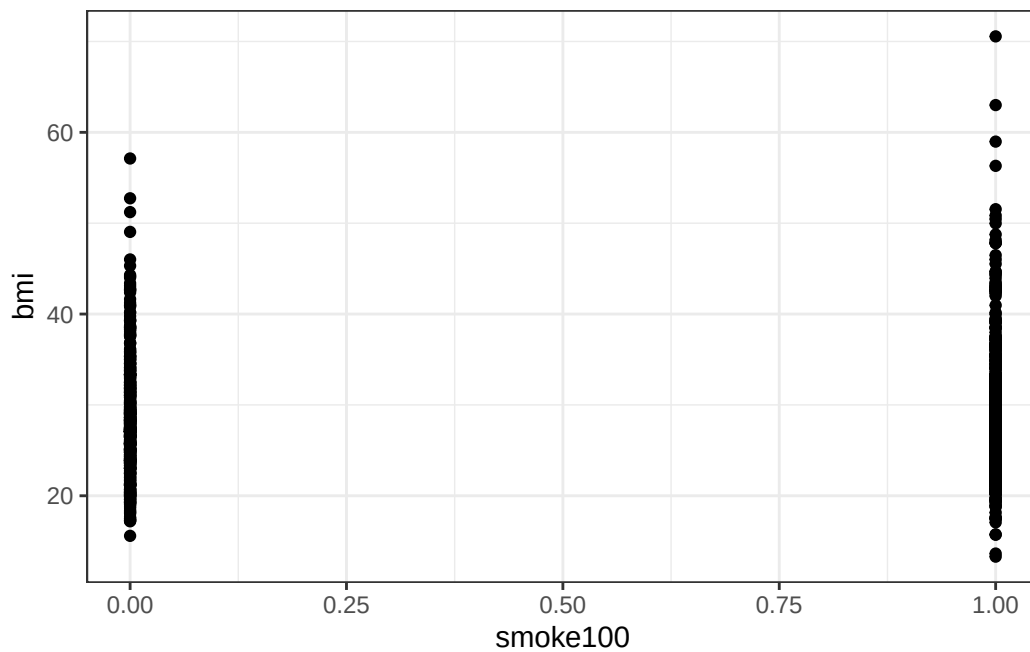
We'll begin by investigating the problem of predicting `bmi`, at first with just three regression inputs: `smoke100`, `female` and `physhealth`, in our `smart_cle1_sh` data set.

- The outcome of interest is `bmi`.
- Inputs to the regression model are:
 - `smoke100` = 1 if the subject has smoked 100 cigarettes in their lifetime
 - `female` = 1 if the subject is female, and 0 if they are male
 - `physhealth` = number of poor physical health days in past 30 (treated as quantitative)

12.2.1 Does smoke100 predict bmi well?

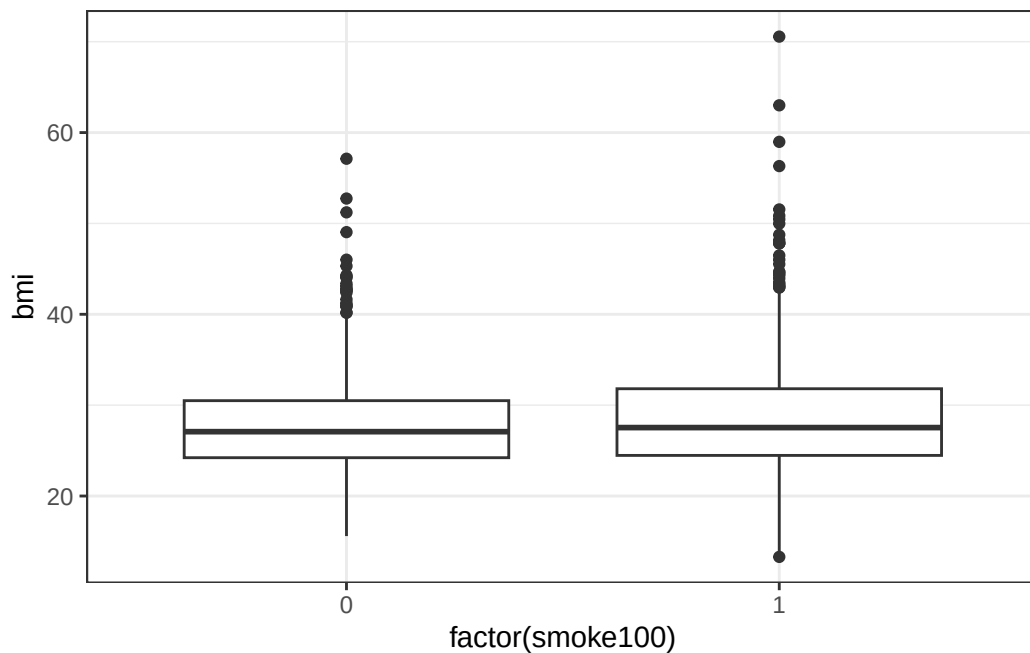
12.2.1.1 Graphical Assessment

```
ggplot(smart_cle1_sh, aes(x = smoke100, y = bmi)) +  
  geom_point()
```



Not so helpful. We should probably specify that `smoke100` is a factor, and try another plotting approach.

```
ggplot(smart_cle1_sh, aes(x = factor(smoke100), y = bmi)) +  
  geom_boxplot()
```



The median BMI looks a little higher for those who have smoked 100 cigarettes. Let's see if a model reflects that.

12.3 mod1: A simple t-test model

```
mod1 <- lm(bmi ~ smoke100, data = smart_cle1_sh)
mod1
```

Call:

```
lm(formula = bmi ~ smoke100, data = smart_cle1_sh)
```

Coefficients:

(Intercept)	smoke100
27.9150	0.8996

```
summary(mod1)
```

Call:


```
lm(formula = bmi ~ smoke100, data = smart_cle1_sh)

Residuals:
    Min       1Q   Median       3Q      Max
-15.515  -3.995  -0.844   2.865  41.745

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  27.9150     0.2624 106.387  <2e-16 ***
smoke100      0.8996     0.3776   2.382   0.0174 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.352 on 1131 degrees of freedom
Multiple R-squared:  0.004993, Adjusted R-squared:  0.004113
F-statistic: 5.675 on 1 and 1131 DF, p-value: 0.01737
```

```
confint(mod1)
```

```

                2.5 %    97.5 %
(Intercept) 27.4001291 28.429781
smoke100     0.1586812 1.640556
```

```
tidy(mod1, conf.int = TRUE, conf.level = 0.95)
```

```
# A tibble: 2 x 7
  term          estimate std.error statistic p.value conf.low conf.high
<chr>         <dbl>     <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
1 (Intercept)  27.9       0.262    106.    0       27.4    28.4
2 smoke100     0.900     0.378     2.38  0.0174   0.159    1.64
```

```
glance(mod1)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
  <dbl>      <dbl>   <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.00499    0.00411  6.35      5.68  0.0174     1 -3701. 7409. 7424.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

The model suggests, based on these 1133 subjects, that

- our best prediction for non-smokers is BMI = 27.94 kg/m², and
- our best prediction for those who have smoked 100 cigarettes is BMI = 27.94 + 0.87 = 28.81 kg/m².
- the mean difference between smokers and non-smokers is +0.87 kg/m² in BMI
- a 95% confidence (uncertainty) interval for that mean smoker - non-smoker difference in BMI ranges from 0.13 to 1.61
- the model accounts for 0.47% of the variation in BMI, so that knowing the respondent's tobacco status does very little to reduce the size of the prediction errors as compared to an intercept only model that would predict the overall mean BMI for each of our subjects.
- the model makes some enormous errors, with one subject being predicted to have a BMI 42 points lower than his/her actual BMI.

Note that this simple regression model just gives us the t-test.

```
t.test(bmi ~ smoke100, var.equal = TRUE, data = smart_cle1_sh)
```

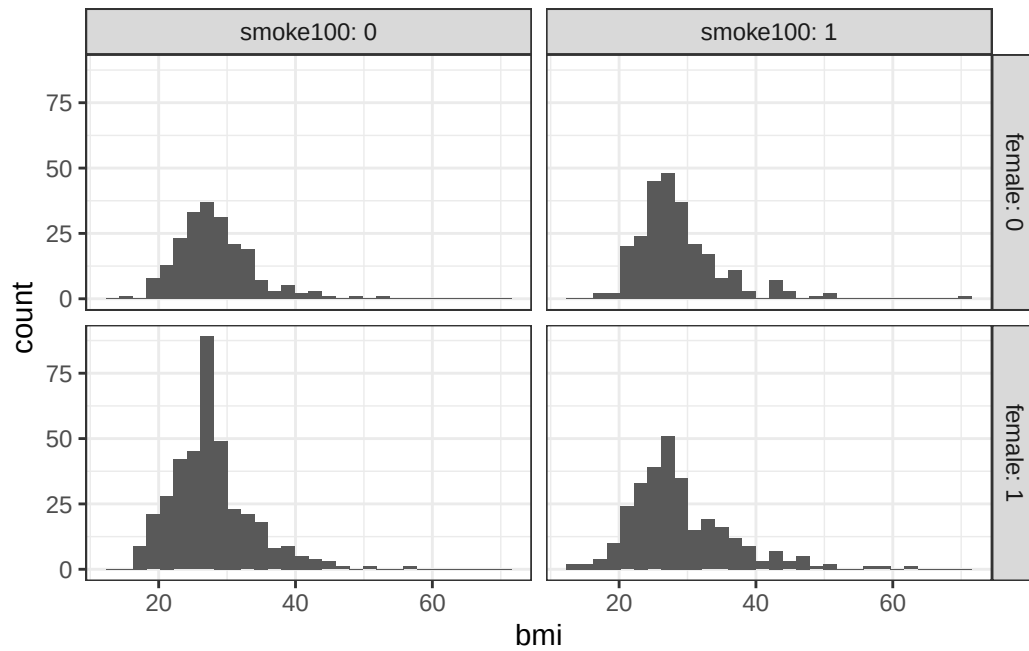
Two Sample t-test

```
data:  bmi by smoke100
t = -2.3823, df = 1131, p-value = 0.01737
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -1.6405563 -0.1586812
sample estimates:
mean in group 0 mean in group 1
    27.91495      28.81457
```

12.4 mod2: Adding another predictor (two-way ANOVA without interaction)

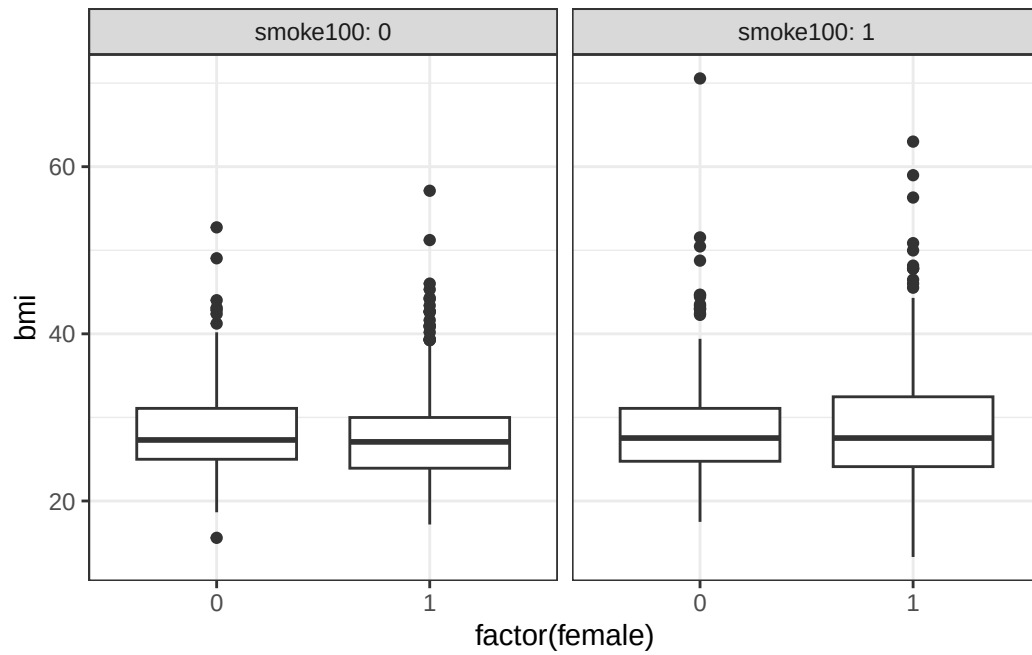
When we add in the information about `female` (sex) to our original model, we might first picture the data. We could look at separate histograms,

```
ggplot(smart_cle1_sh, aes(x = bmi)) +
  geom_histogram(bins = 30) +
  facet_grid(female ~ smoke100, labeller = label_both)
```

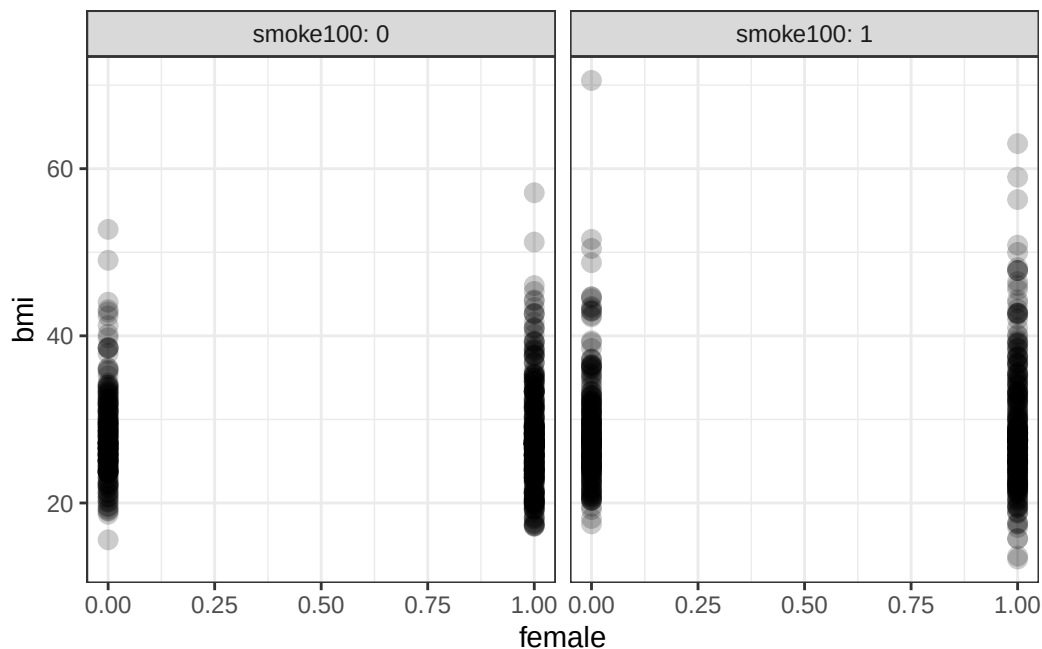


or maybe boxplots?

```
ggplot(smart_cle1_sh, aes(x = factor(female), y = bmi)) +
  geom_boxplot() +
  facet_wrap(~ smoke100, labeller = label_both)
```



```
ggplot(smart_cle1_sh, aes(x = female, y = bmi))+  
  geom_point(size = 3, alpha = 0.2) +  
  theme_bw() +  
  facet_wrap(~ smoke100, labeller = label_both)
```



OK. Let's try fitting a model.

```
mod2 <- lm(bmi ~ female + smoke100, data = smart_cle1_sh)
mod2
```

Call:

```
lm(formula = bmi ~ female + smoke100, data = smart_cle1_sh)
```

Coefficients:

(Intercept)	female	smoke100
28.0101	-0.1479	0.8842

This new model predicts only four predicted values:

- $\text{bmi} = 28.0265$ if the subject is male and has not smoked 100 cigarettes (so $\text{female} = 0$ and $\text{smoke100} = 0$)
- $\text{bmi} = 28.0265 - 0.1342 = 27.8923$ if the subject is female and has not smoked 100 cigarettes ($\text{female} = 1$ and $\text{smoke100} = 0$)
- $\text{bmi} = 28.0265 + 0.8555 = 28.8820$ if the subject is male and has smoked 100 cigarettes (so $\text{female} = 0$ and $\text{smoke100} = 1$), and, finally
- $\text{bmi} = 28.0265 - 0.1342 + 0.8555 = 28.7478$ if the subject is female and has smoked 100 cigarettes (so both female and $\text{smoke100} = 1$).

Another way to put this is that for those who have not smoked 100 cigarettes, the model is:

- $\text{bmi} = 28.0265 - 0.1342 \text{ female}$

and for those who have smoked 100 cigarettes, the model is:

- $\text{bmi} = 28.8820 - 0.1342 \text{ female}$

Only the intercept of the `bmi-female` model changes depending on `smoke100`.

```
summary(mod2)
```

Call:

```
lm(formula = bmi ~ female + smoke100, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.446	-3.960	-0.854	2.790	41.666

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.0101	0.3615	77.475	<2e-16 ***
female	-0.1479	0.3864	-0.383	0.7019
smoke100	0.8842	0.3799	2.327	0.0201 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.354 on 1130 degrees of freedom

Multiple R-squared: 0.005122, Adjusted R-squared: 0.003361

F-statistic: 2.909 on 2 and 1130 DF, p-value: 0.05495

```
confint(mod2)
```

	2.5 %	97.5 %
(Intercept)	27.3007728	28.7194992
female	-0.9061739	0.6102793
smoke100	0.1388225	1.6296303

The slopes of both `female` and `smoke100` have confidence intervals that are completely below zero, indicating that both `female` sex and `smoke100` appear to be associated with reductions in `bmi`.

The R^2 value suggests that 0.4788% of the variation in `bmi` is accounted for by this ANOVA model.

In fact, this regression (on two binary indicator variables) is simply a two-way ANOVA model without an interaction term.

```
anova(mod2)
```

Analysis of Variance Table

Response: `bmi`

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
female	1	16	16.163	0.4003	0.52705
smoke100	1	219	218.721	5.4171	0.02012 *
Residuals	1130	45625	40.376		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

12.5 mod3: Adding the interaction term (Two-way ANOVA with interaction)

Suppose we want to let the effect of `female` vary depending on the `smoke100` status. Then we need to incorporate an interaction term in our model.

```
mod3 <- lm(bmi ~ female * smoke100, data = smart_cle1_sh)
mod3
```

Call:

```
lm(formula = bmi ~ female * smoke100, data = smart_cle1_sh)
```

Coefficients:

(Intercept)	female	smoke100	female:smoke100
28.2687	-0.5499	0.4112	0.7996

So, for example, for a male who has smoked 100 cigarettes, this model predicts

- $\text{bmi} = 28.269 - 0.506 (0) + 0.412 (1) + 0.754 (0)(1) = 28.269 + 0.412 = 28.681$

And for a female who has smoked 100 cigarettes, the model predicts

- $\text{bmi} = 28.269 - 0.506 (1) + 0.412 (1) + 0.754 (1)(1) = 28.269 - 0.506 + 0.412 + 0.754 = 28.929$

For those who have not smoked 100 cigarettes, the model is:

- $\text{bmi} = 28.269 - 0.506 \text{ female}$

But for those who have smoked 100 cigarettes, the model is:

- $\text{bmi} = (28.269 + 0.412) + (-0.506 + 0.754) \text{ female}$, or ,,
- $\text{bmi} = 28.681 - 0.248 \text{ female}$

Now, both the slope and the intercept of the **bmi-female** model change depending on **smoke100**.

```
summary(mod3)
```

Call:

```
lm(formula = bmi ~ female * smoke100, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.630	-3.939	-0.879	2.760	41.880

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.2687	0.4395	64.318	<2e-16 ***
female	-0.5499	0.5480	-1.004	0.316
smoke100	0.4112	0.5945	0.692	0.489
female:smoke100	0.7996	0.7729	1.035	0.301

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.354 on 1129 degrees of freedom

Multiple R-squared: 0.006064, Adjusted R-squared: 0.003423

F-statistic: 2.296 on 3 and 1129 DF, p-value: 0.07614

```
confint(mod3)
```

	2.5 %	97.5 %
(Intercept)	27.4063786	29.1310929
female	-1.6250493	0.5252321


```
smoke100          -0.7552196  1.5775258
female:smoke100 -0.7167828  2.3160753
```

In fact, this regression (on two binary indicator variables and a product term) is simply a two-way ANOVA model with an interaction term.

```
anova(mod3)
```

Analysis of Variance Table

Response: bmi

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
female	1	16	16.163	0.4003	0.52704
smoke100	1	219	218.721	5.4175	0.02011 *
female:smoke100	1	43	43.219	1.0705	0.30106
Residuals	1129	45581	40.373		

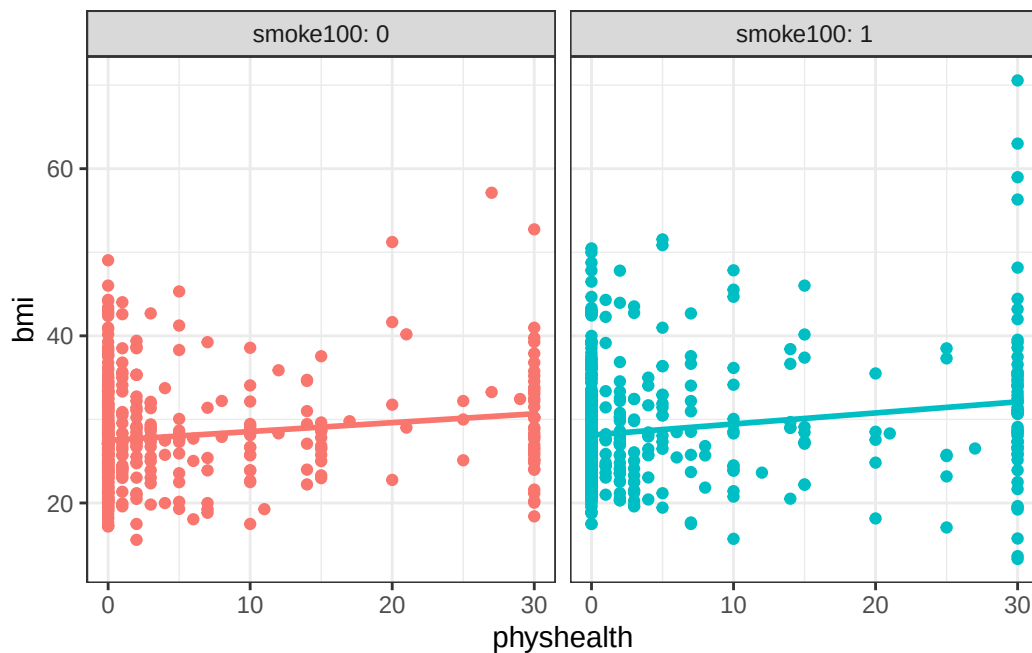
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The interaction term doesn't change very much here. Its uncertainty interval includes zero, and the overall model still accounts for less than 1% of the variation in bmi.

12.6 mod4: Using smoke100 and physhealth in a model for bmi

```
ggplot(smart_cle1_sh, aes(x = physhealth, y = bmi, color = factor(smoke100))) +
  geom_point() +
  guides(col = "none") +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ smoke100, labeller = label_both)
```

`geom\_smooth()` using formula = 'y ~ x'



Does the difference in slopes of `bmi` and `physhealth` for those who have and haven't smoked at least 100 cigarettes appear to be substantial and important?

```
mod4 <- lm(bmi ~ smoke100 * physhealth, data = smart_cle1_sh)
summary(mod4)
```

Call:

```
lm(formula = bmi ~ smoke100 * physhealth, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.800	-3.785	-0.596	2.615	38.460

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.48538	0.28609	96.073	< 2e-16 ***
smoke100	0.63964	0.41697	1.534	0.125302
physhealth	0.10617	0.03026	3.509	0.000467 ***
smoke100:physhealth	0.02631	0.04084	0.644	0.519497

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.259 on 1129 degrees of freedom
Multiple R-squared: 0.03544, Adjusted R-squared: 0.03288
F-statistic: 13.83 on 3 and 1129 DF, p-value: 7.37e-09

Does it seem as though the addition of `physhealth` has improved our model substantially over a model with `smoke100` alone (which, you recall, was `mod1`)?

Since the `mod4` model contains the `mod1` model's predictors as a subset and the outcome is the same for each model, we consider the models *nested* and have some extra tools available to compare them.

- I might start by looking at the basic summaries for each model.

```
glance(mod4)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic      p.value    df logLik   AIC   BIC
    <dbl>         <dbl> <dbl>     <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.0354         0.0329  6.26     13.8 0.00000000737     3 -3684. 7377. 7403.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
glance(mod1)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.00499         0.00411  6.35      5.68 0.0174     1 -3701. 7409. 7424.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

- The R^2 is much larger for the model with `physhealth`, but still very tiny.
- Smaller AIC and smaller BIC statistics are more desirable. Here, there's little to choose from, so `mod4` looks better, too.
- We might also consider a significance test by looking at an ANOVA model comparison. This is only appropriate because `mod1` is nested in `mod4`.

```
anova(mod4, mod1)
```

Analysis of Variance Table

```
Model 1: bmi ~ smoke100 * physhealth
Model 2: bmi ~ smoke100
      Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1      1129 44234
2      1131 45630 -2    -1396.3 17.819 2.405e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The addition of the `physhealth` term appears to be an improvement, not that that means very much.

12.7 Making Predictions with a Linear Regression Model

Recall model 4, which yields predictions for body mass index on the basis of the main effects of having smoked (`smoke100`) and days of poor physical health (`physhealth`) and their interaction.

```
mod4
```

Call:

```
lm(formula = bmi ~ smoke100 * physhealth, data = smart_cle1_sh)
```

Coefficients:

(Intercept)	smoke100	physhealth
27.48538	0.63964	0.10617
smoke100:physhealth		
0.02631		

12.7.1 Fitting an Individual Prediction and 95% Prediction Interval

What do we predict for the `bmi` of a subject who has smoked at least 100 cigarettes in their life and had 8 poor physical health days in the past 30?

```
new1 <- tibble(smoke100 = 1, physhealth = 8)
predict(mod4, newdata = new1, interval = "prediction", level = 0.95)
```

	fit	lwr	upr
1	29.18491	16.89142	41.47839

The predicted `bmi` for this new subject is shown above. The prediction interval shows the bounds of a 95% uncertainty interval for a predicted `bmi` for an individual smoker¹ who has 8 days of poor physical health out of the past 30. From the `predict` function applied to a linear model, we can get the prediction intervals for any new data points in this manner.

12.7.2 Confidence Interval for an Average Prediction

- What do we predict for the **average body mass index of a population of subjects** who are smokers and have `physhealth = 8`?

```
predict(mod4, newdata = new1, interval = "confidence", level = 0.95)
```

	fit	lwr	upr
1	29.18491	28.63866	29.73115

- How does this result compare to the prediction interval?

12.7.3 Fitting Multiple Individual Predictions to New Data

- How does our prediction change for a respondent if they instead have 7, or 9 poor physical health days? What if they have or don't have a history of smoking?

```
c8_new2 <- tibble(subjectid = 1001:1006, smoke100 = c(1, 1, 1, 0, 0, 0), physhealth = c(7, 8, 9, 7, 8, 9))
pred2 <- predict(mod4, newdata = c8_new2, interval = "prediction", level = 0.95) |> as_tibble()

result2 <- bind_cols(c8_new2, pred2)
result2
```

```
# A tibble: 6 x 6
  subjectid smoke100 physhealth  fit   lwr   upr
    <int>      <dbl>      <dbl> <dbl> <dbl> <dbl>
1     1001         1         7  29.1  16.8  41.3
2     1002         1         8  29.2  16.9  41.5
3     1003         1         9  29.3  17.0  41.6
4     1004         0         7  28.2  15.9  40.5
```

¹I'll now refer to those who have smoked at least 100 cigarettes in their life as smokers, and those who have not as non-smokers to save some space.

5	1005	0	8	28.3	16.0	40.6
6	1006	0	9	28.4	16.1	40.7

The `result2` tibble contains predictions for each scenario.

- Which has a bigger impact on these predictions and prediction intervals? A one category change in `smoke100` or a one hour change in `physhealth`?

12.8 Centering the model

Our model `mod4` has four predictors (the constant, `physhealth`, `smoke100` and their interaction) but just two inputs (`smoke100` and `physhealth`.) If we **center** the quantitative input `physhealth` before building the model, we get a more interpretable interaction term.

```
smart_cle1_sh_c <- smart_cle1_sh |>
  mutate(physhealth_c = physhealth - mean(physhealth))

mod4_c <- lm(bmi ~ smoke100 * physhealth_c, data = smart_cle1_sh_c)

summary(mod4_c)
```

Call:

```
lm(formula = bmi ~ smoke100 * physhealth_c, data = smart_cle1_sh_c)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-18.800	-3.785	-0.596	2.615	38.460

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.97435	0.25913	107.956	< 2e-16 ***
smoke100	0.76083	0.37289	2.040	0.041544 *
physhealth_c	0.10617	0.03026	3.509	0.000467 ***
smoke100:physhealth_c	0.02631	0.04084	0.644	0.519497

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.259 on 1129 degrees of freedom

Multiple R-squared: 0.03544, Adjusted R-squared: 0.03288

F-statistic: 13.83 on 3 and 1129 DF, p-value: 7.37e-09

What has changed as compared to the original `mod4`?

- Our original model was $\text{bmi} = 27.5 + 0.57 \text{ smoke100} + 0.11 \text{ physhealth} - 0.03 \text{ smoke100} \times \text{physhealth}$
- Our new model is $\text{bmi} = 28.0 + 0.73 \text{ smoke100} + 0.11 \text{ centered physhealth} + 0.03 \text{ smoke100} \times \text{centered physhealth}$.

So our new model on centered data is:

- $28.0 + 0.11 \text{ centered physhealth_c}$ for non-smokers, and
- $(28.0 + 0.73) + (0.11 - 0.03) \text{ centered physhealth_c}$, or $28.73 + 0.08 \text{ centered physhealth_c}$ for smokers.

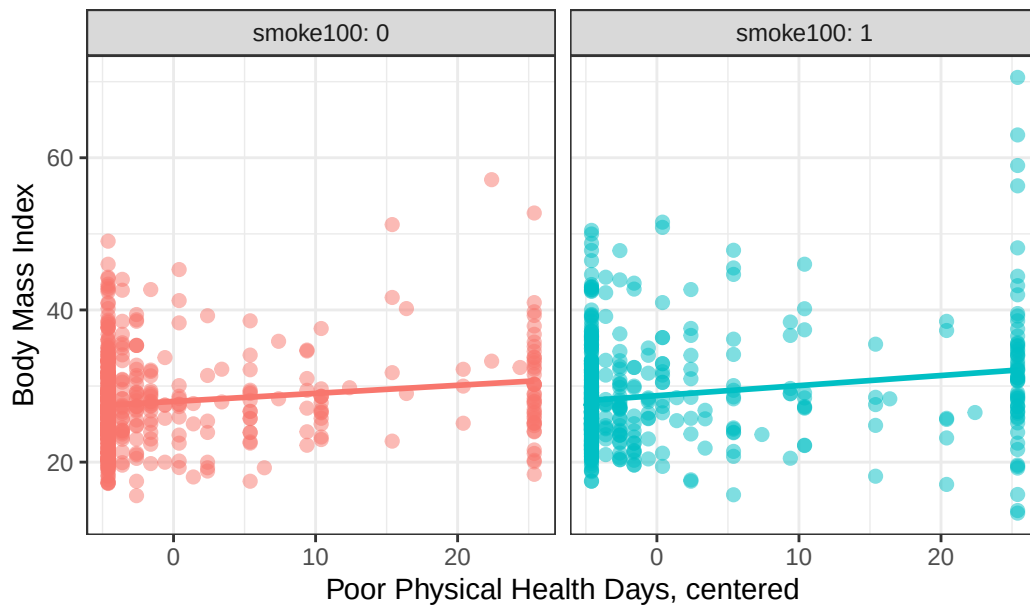
In our new (centered `physhealth_c`) model,

- the main effect of `smoke100` now corresponds to a predictive difference (smoker - non-smoker) in `bmi` with `physhealth` at its mean value, 4.68 days,
- the intercept term is now the predicted `bmi` for a non-smoker with an average `physhealth`, and
- the product term corresponds to the change in the slope of centered `physhealth_c` on `bmi` for a smoker rather than a non-smoker, while
- the residual standard deviation and the R-squared values remain unchanged from the model before centering.

12.8.1 Plot of Model 4 on Centered `physhealth`: `mod4_c`

```
ggplot(smart_cle1_sh_c, aes(x = physhealth_c, y = bmi, group = smoke100, col = factor(smoke100))) +  
  geom_point(alpha = 0.5, size = 2) +  
  geom_smooth(method = "lm", se = FALSE, formula = y ~ x) +  
  guides(color = "none") +  
  labs(x = "Poor Physical Health Days, centered", y = "Body Mass Index",  
       title = "Model `mod4` on centered data") +  
  facet_wrap(~ smoke100, labeller = label_both)
```

Model `mod4` on centered data



12.9 Rescaling an input by subtracting the mean and dividing by 2 standard deviations

Centering helped us interpret the main effects in the regression, but it still leaves a scaling problem.

- The `smoke100` coefficient estimate is much larger than that of `physhealth`, but this is misleading, considering that we are comparing the complete change in one variable (smoking or not) to a 1-day change in `physhealth`.
- @GelmanHill2007 recommend all continuous predictors be scaled by dividing by 2 standard deviations, so that:
 - a 1-unit change in the rescaled predictor corresponds to a change from 1 standard deviation below the mean, to 1 standard deviation above.
 - an unscaled binary (1/0) predictor with 50% probability of occurring will be exactly comparable to a rescaled continuous predictor done in this way.

```
smart_cle1_sh_rescale <- smart_cle1_sh |>
  mutate(physhealth_z = (physhealth - mean(physhealth))/(2*sd(physhealth)))
```


12.9.1 Refitting model mod4 to the rescaled data

```
mod4_z <- lm(bmi ~ smoke100 * physhealth_z, data = smart_cle1_sh_rescale)
summary(mod4_z)
```

Call:

```
lm(formula = bmi ~ smoke100 * physhealth_z, data = smart_cle1_sh_rescale)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.800	-3.785	-0.596	2.615	38.460

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.9743	0.2591	107.956	< 2e-16 ***
smoke100	0.7608	0.3729	2.040	0.041544 *
physhealth_z	1.9477	0.5550	3.509	0.000467 ***
smoke100:physhealth_z	0.4827	0.7492	0.644	0.519497

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.259 on 1129 degrees of freedom

Multiple R-squared: 0.03544, Adjusted R-squared: 0.03288

F-statistic: 13.83 on 3 and 1129 DF, p-value: 7.37e-09

12.9.2 Interpreting the model on rescaled data

What has changed as compared to the original mod4?

- Our original model was $\text{bmi} = 27.5 + 0.57 \text{ smoke100} + 0.11 \text{ physhealth} + 0.03 \text{ smoke100} \times \text{physhealth}$
- Our model on centered physhealth was $\text{bmi} = 28.0 + 0.73 \text{ smoke100} + 0.11 \text{ centered physhealth} + 0.03 \text{ smoke100} \times \text{centered physhealth}$.
- Our new model on rescaled physhealth is $\text{bmi} = 28.0 + 0.73 \text{ smoke100} + 1.98 \text{ rescaled physhealth} + 0.61 \text{ smoke100} \times \text{rescaled physhealth}$

So our rescaled model is:

- $28.0 + 1.98 \text{ rescaled physhealth_z}$ for non-smokers, and

- $(28.0 + 0.73) + (1.98 + 0.61)$ rescaled `physhealth_z`, or $28.73 + 2.59$ rescaled `physhealth_z` for smokers.

In this new rescaled (`physhealth_z`) model, then,

- the main effect of `smoke100`, 0.73, still corresponds to a predictive difference (smoker - non-smoker) in `bmi` with `physhealth` at its mean value, 4.68 days,
- the intercept term is still the predicted `bmi` for a non-smoker with an average `physhealth` count, and
- the residual standard deviation and the R-squared values remain unchanged,

as before, but now we also have that:

- the coefficient of `physhealth_z` indicates the predictive difference in `bmi` associated with a change in `physhealth` of 2 standard deviations (from one standard deviation below the mean of 4.68 to one standard deviation above 4.68.)
 - Since the standard deviation of `physhealth` is 9.12 (see below), this covers a massive range of potential values of `physhealth` from 0 all the way up to $4.68 + 2(9.12) = 22.92$ days.

```
favstats(~ physhealth, data = smart_cle1_sh)
```

min	Q1	median	Q3	max	mean	sd	n	missing
0	0	0	3	30	4.605472	9.172505	1133	0

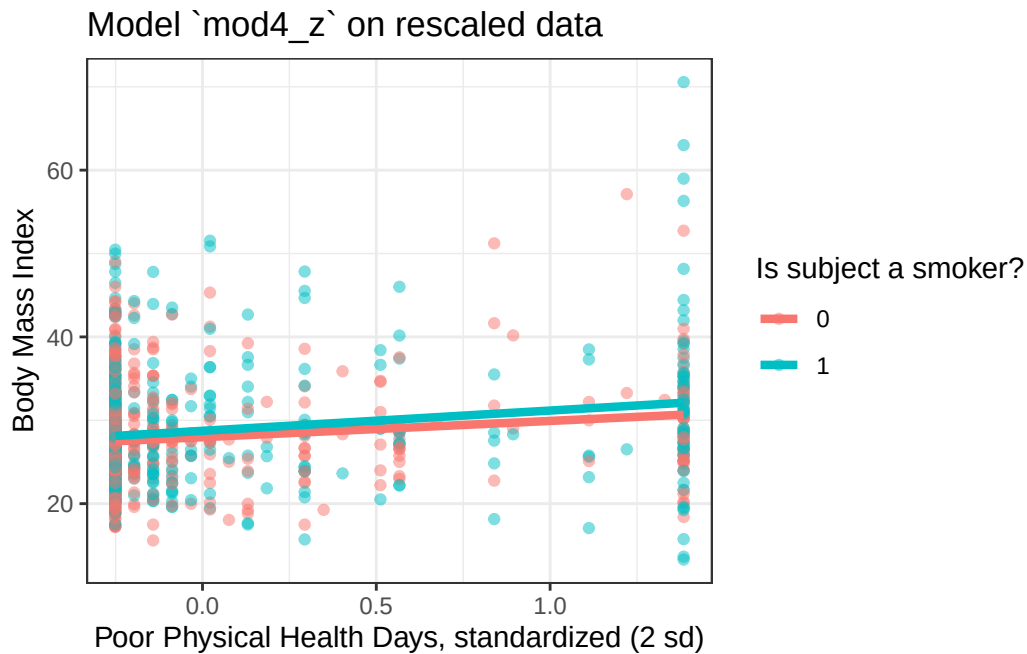
- the coefficient of the product term (0.61) corresponds to the change in the coefficient of `physhealth_z` for smokers as compared to non-smokers.

12.9.3 Plot of model on rescaled data

```
ggplot(smart_cle1_sh_rescale, aes(x = physhealth_z, y = bmi,
                                group = smoke100, col = factor(smoke100))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, size = 1.5) +
  scale_color_discrete(name = "Is subject a smoker?") +
  labs(x = "Poor Physical Health Days, standardized (2 sd)", y = "Body Mass Index",
       title = "Model `mod4_z` on rescaled data")
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

```
`geom_smooth()` using formula = 'y ~ x'
```



There's very little difference here.

12.10 mod5: What if we add more variables?

We can boost our R^2 a bit, to nearly 5%, by adding in two new variables, related to whether or not the subject (in the past 30 days) used the internet, and the average number of alcoholic drinks per week consumed by the subject.

```
mod5 <- lm(bmi ~ smoke100 + female + physhealth + internet30 + drinks_wk,
            data = smart_cle1_sh)
summary(mod5)
```

Call:

```
lm(formula = bmi ~ smoke100 + female + physhealth + internet30 +
    drinks_wk, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-18.271 -3.841 -0.723 2.654 38.187

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.59416	0.55756	49.491	< 2e-16 ***
smoke100	0.85750	0.37612	2.280	0.02280 *
female	-0.44346	0.38471	-1.153	0.24928
physhealth	0.11875	0.02060	5.765	1.05e-08 ***
internet30	0.38241	0.48782	0.784	0.43326
drinks_wk	-0.10307	0.03352	-3.074	0.00216 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.238 on 1127 degrees of freedom

Multiple R-squared: 0.04373, Adjusted R-squared: 0.03949

F-statistic: 10.31 on 5 and 1127 DF, p-value: 1.114e-09

1. Here's the ANOVA for this model. What can we study with this?

```
anova(mod5)
```

Analysis of Variance Table

Response: bmi

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
smoke100	1	229	228.97	5.8842	0.01543 *
female	1	6	5.92	0.1521	0.69663
physhealth	1	1394	1393.62	35.8147	2.912e-09 ***
internet30	1	9	9.26	0.2379	0.62586
drinks_wk	1	368	367.80	9.4521	0.00216 **
Residuals	1127	43854	38.91		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

2. Consider the revised output below. Now what can we study?

```
anova(lm(bmi ~ smoke100 + internet30 + drinks_wk + female + physhealth,  
data = smart_cle1_sh))
```

Analysis of Variance Table

Response: bmi

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
smoke100	1	229	228.97	5.8842	0.0154334	*
internet30	1	8	8.38	0.2155	0.6426003	
drinks_wk	1	440	439.87	11.3042	0.0007993	***
female	1	35	34.92	0.8973	0.3437094	
physhealth	1	1293	1293.42	33.2397	1.051e-08	***
Residuals	1127	43854	38.91			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

3. What does the output below let us conclude?

```
anova(lm(bmi ~ smoke100 + internet30 + drinks_wk + female + physhealth,
  data = smart_cle1_sh),
  lm(bmi ~ smoke100 + female + drinks_wk,
  data = smart_cle1_sh))
```

Analysis of Variance Table

Model 1: bmi ~ smoke100 + internet30 + drinks_wk + female + physhealth

Model 2: bmi ~ smoke100 + female + drinks_wk

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	1127	43854				
2	1129	45148	-2	-1293.8	16.625	7.661e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

4. What does it mean for the models to be “nested”?

12.11 mod6: Would adding self-reported health help?

And we can do even a bit better than that by adding in a multi-categorical measure: self-reported general health.

```
mod6 <- lm(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genhealth,
  data = smart_cle1_sh)
summary(mod6)
```

Call:

```
lm(formula = bmi ~ female + smoke100 + physhealth + internet30 +
    drinks_wk + genhealth, data = smart_cle1_sh)
```

Residuals:

Min	1Q	Median	3Q	Max
-19.205	-3.637	-0.750	2.667	36.839

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	25.21949	0.71005	35.518	< 2e-16 ***
female	-0.32079	0.37604	-0.853	0.3938
smoke100	0.48496	0.37062	1.309	0.1910
physhealth	0.03710	0.02475	1.499	0.1341
internet30	0.91322	0.48259	1.892	0.0587 .
drinks_wk	-0.07751	0.03293	-2.354	0.0187 *
genhealth2_VeryGood	1.21718	0.56839	2.141	0.0325 *
genhealth3_Good	3.23814	0.58002	5.583	2.97e-08 ***
genhealth4_Fair	4.19627	0.73221	5.731	1.28e-08 ***
genhealth5_Poor	6.00832	1.08770	5.524	4.12e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

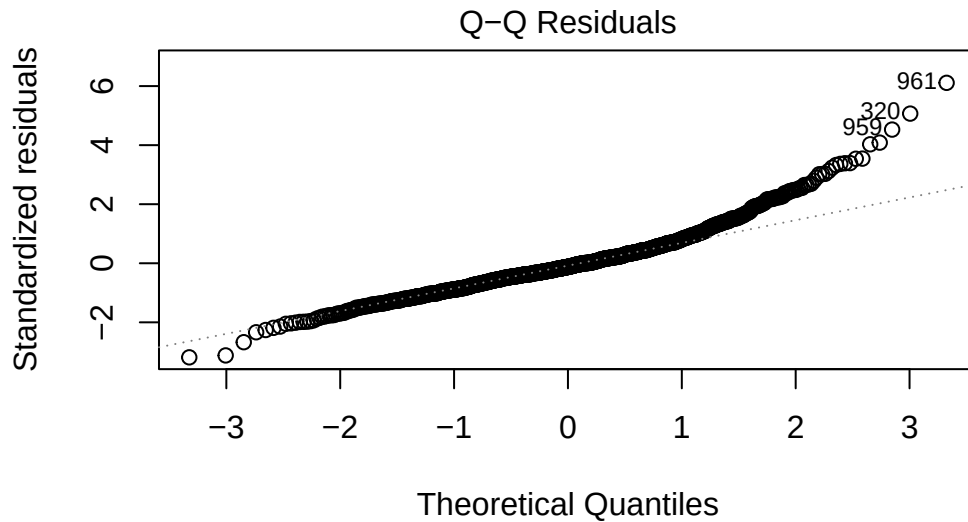
Residual standard error: 6.091 on 1123 degrees of freedom

Multiple R-squared: 0.09148, Adjusted R-squared: 0.0842

F-statistic: 12.56 on 9 and 1123 DF, p-value: < 2.2e-16

1. If Harry and Marty have the same values of `female`, `smoke100`, `physhealth`, `internet30` and `drinks_wk`, but Harry rates his health as Good, and Marty rates his as Fair, then what is the difference in the predictions? Who is predicted to have a larger BMI, and by how much?
2. What does this normal probability plot of the residuals suggest?

```
plot(mod6, which = 2)
```



m(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genh

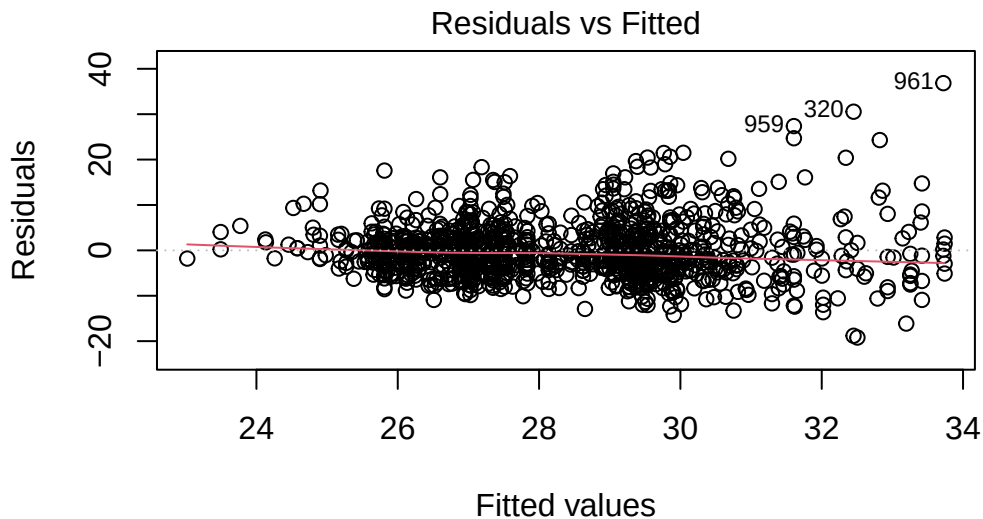
12.12 Key Regression Assumptions for Building Effective Prediction Models

1. Validity - the data you are analyzing should map to the research question you are trying to answer.
 - The outcome should accurately reflect the phenomenon of interest.
 - The model should include all relevant predictors. (It can be difficult to decide which predictors are necessary, and what to do with predictors that have large standard errors.)
 - The model should generalize to all of the cases to which it will be applied.
 - Can the available data answer our question reliably?
2. Additivity and linearity - most important assumption of a regression model is that its deterministic component is a linear function of the predictors. We often think about transformations in this setting.
3. Independence of errors - errors from the prediction line are independent of each other
4. Equal variance of errors - if this is violated, we can more efficiently estimate parameters using *weighted least squares* approaches, where each point is weighted inversely proportional to its variance, but this doesn't affect the coefficients much, if at all.
5. Normality of errors - not generally important for estimating the regression line

12.12.1 Checking Assumptions in model `mod6`

1. How does the assumption of linearity behind this model look?

```
plot(mod6, which = 1)
```

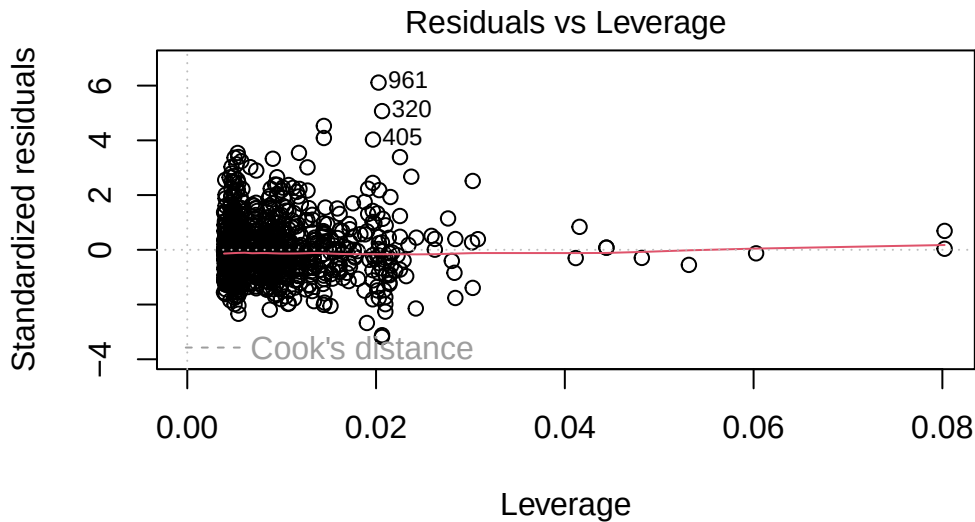


`m(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genh`

We see no strong signs of serious non-linearity here. There's no obvious curve in the plot, for example. We may have a problem with increasing variance as we move to the right.

2. What can we conclude from the plot below?

```
plot(mod6, which = 5)
```

```
m(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genh
```

This plot can help us identify points with large standardized residuals, large leverage values, and large influence on the model (as indicated by large values of Cook's distance.) In this case, I see no signs of any points used in the model with especially large influence, although there are some poorly fitted points (with especially large standardized residuals.)

We might want to identify the point listed here as 961, which appears to have an enormous standardized residual. To do so, we can use the `slice` function from `dplyr`.

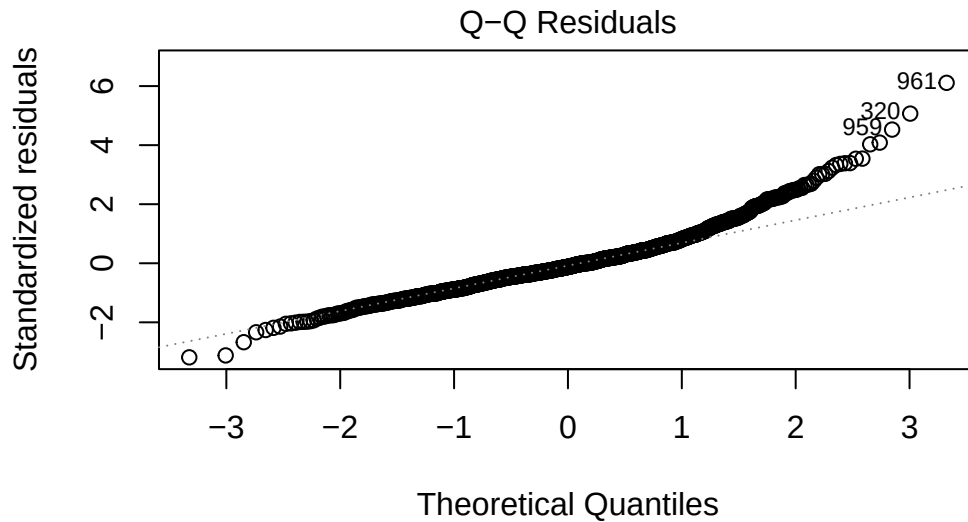
```
smart_cle1_sh |> slice(961) |> select(SEQNO)
```

```
# A tibble: 1 x 1
  SEQNO
  <dbl>
1 2017000961
```

Now we know exactly which subject we're talking about.

3. What other residual plots are available with `plot` and how do we interpret them?

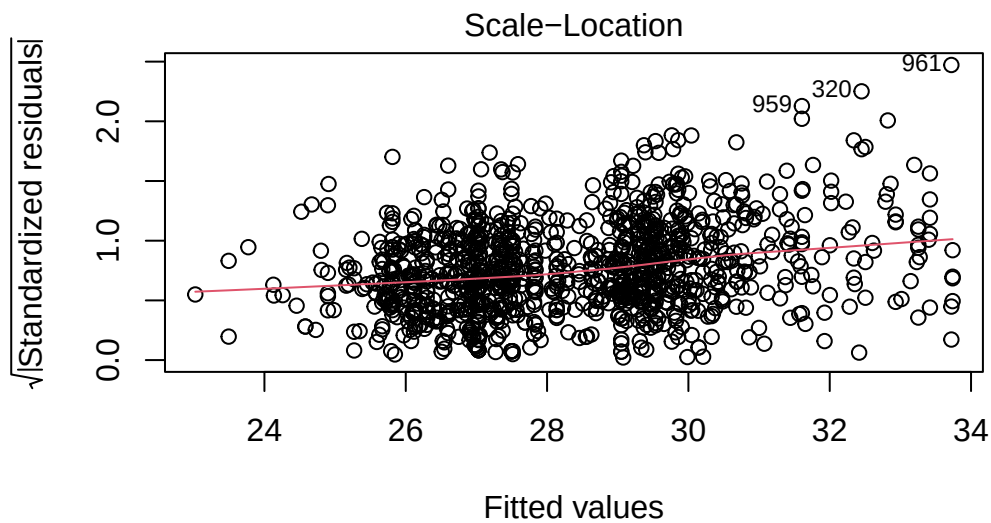
```
plot(mod6, which = 2)
```



`m(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genh`

This plot is simply a Normal Q-Q plot of the standardized residuals from our model. We're looking here for serious problems with the assumption of Normality.

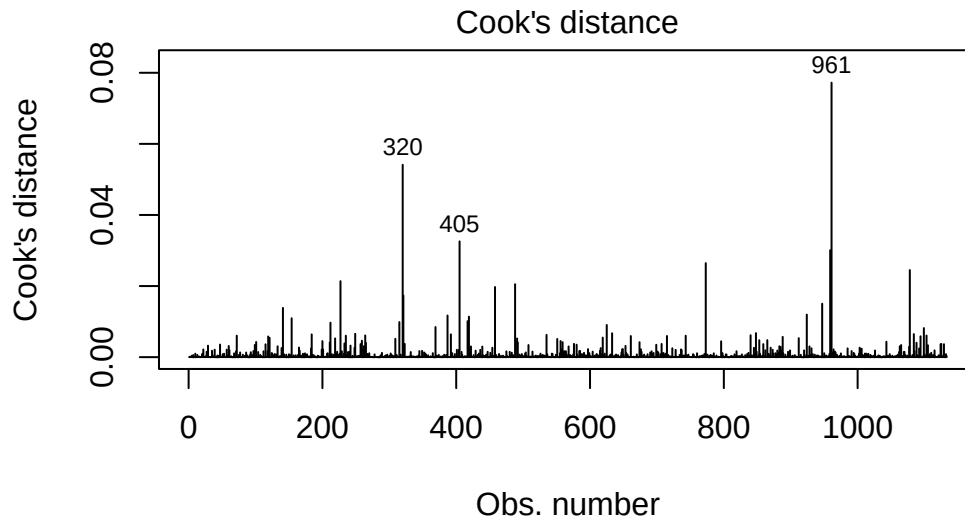
```
plot(mod6, which = 3)
```



`m(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genh`

This is a scale-location plot, designed to help us see non-constant variance in the residuals as we move across the fitted values as a linear trend, rather than as a fan shape, by plotting the square root of the residuals on the vertical axis.

```
plot(mod6, which = 4)
```



`m(bmi ~ female + smoke100 + physhealth + internet30 + drinks_wk + genh`

Finally, this is an index plot of the Cook's distance values, allowing us to identify points that are particularly large. Remember that a value of 0.5 (or perhaps even 1.0) is a reasonable boundary for a substantially influential point.

13 Adding Non-linear Terms to a Linear Regression

13.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(car)
library(Hmisc)
library(mosaic)
library(rms)
library(patchwork)
library(tidyverse)

theme_set(theme_bw())
```

13.1.1 Data Load

```
pollution <- read_csv("data/pollution.csv", show_col_types = FALSE)
```

13.2 The pollution data

Consider the pollution data set, which contain 15 independent variables and a measure of mortality, describing 60 US metropolitan areas in 1959-1961. The data come from @McDonald1973, and are available at <http://www4.stat.ncsu.edu/~boos/var.select/pollution.html> and our web site.

pollution

```
# A tibble: 60 x 16
      x1    x2    x3    x4    x5    x6    x7    x8    x9    x10    x11    x12    x13
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1    13    49    68     7   3.36  12.2  90.7 2702     3   51.9    9.7   105    32
2    28    32    81     7   3.27  12.1   81  3665    7.5   51.6   13.2     4     2
3    10    55    70    7.3   3.11  12.1  88.9 3033    5.9    51    14   144    66
4    43    32    74   10.1   3.38   9.5  79.2 3214    2.9   43.7   12     11     7
5    25    12    73    9.2   3.28  12.1  83.1 2095     2   51.9    9.8    20    11
6    35    46    85    7.1   3.22  11.8  79.9 1441   14.8   51.2   16.1     1     1
7    60    67    82   10    2.98  11.5  88.6 4657   13.5   47.3   22.4     3     1
8    11    53    68    9.2   2.99  12.1  90.6 4700    7.8   48.9   12.3   648   319
9    31    24    72     9   3.37  10.9  82.8 3226    5.1   45.2   12.3     5     3
10   15    30    73    8.2   3.15  12.2  84.2 4824    4.7   53.1   12.7    17     8
# i 50 more rows
# i 3 more variables: x14 <dbl>, x15 <dbl>, y <dbl>
```

Here's a codebook:

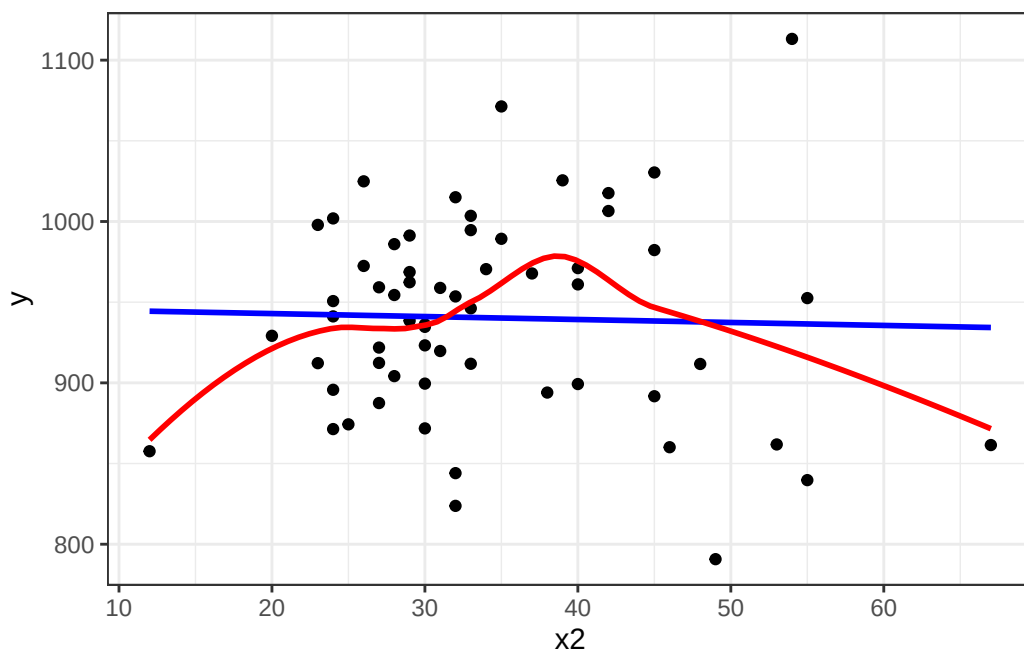
Vari- able	Description
y	Total Age Adjusted Mortality Rate
x1	Mean annual precipitation in inches
x2	Mean January temperature in degrees Fahrenheit
x3	Mean July temperature in degrees Fahrenheit
x4	Percent of 1960 SMSA population that is 65 years of age or over
x5	Population per household, 1960 SMSA
x6	Median school years completed for those over 25 in 1960 SMSA
x7	Percent of housing units that are found with facilities
x8	Population per square mile in urbanized area in 1960
x9	Percent of 1960 urbanized area population that is non-white
x10	Percent employment in white-collar occupations in 1960 urbanized area
x11	Percent of families with income under \$30,000 in 1960 urbanized area
x12	Relative population potential of hydrocarbons, HC
x13	Relative pollution potential of oxides of nitrogen, NOx
x14	Relative pollution potential of sulfur dioxide, SO2
x15	Percent relative humidity, annual average at 1 p.m.

13.3 Fitting a straight line model to predict y from x_2

Consider the relationship between y , the age-adjusted mortality rate, and x_2 , the mean January temperature, across these 60 areas. I'll include both a linear model (in blue) and a loess smooth (in red.) Does the relationship appear to be linear?

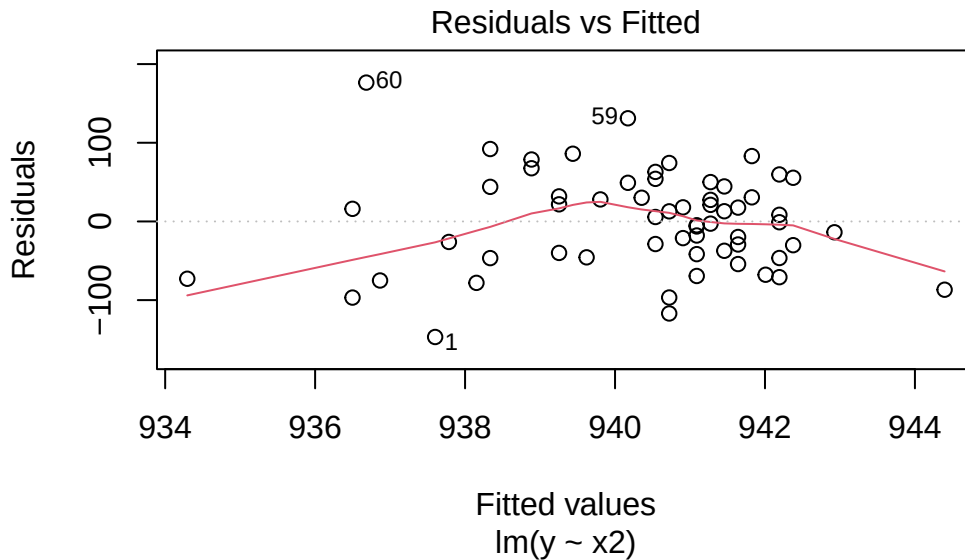
```
ggplot(pollution, aes(x = x2, y = y)) +  
  geom_point() +  
  geom_smooth(method = "lm", col = "blue", se = F) +  
  geom_smooth(method = "loess", col = "red", se = F)
```

```
`geom_smooth()` using formula = 'y ~ x'  
`geom_smooth()` using formula = 'y ~ x'
```



Suppose we plot the residuals that emerge from the linear model shown in blue, above. Do we see a curve in a plot of residuals against fitted values?

```
plot(lm(y ~ x2, data = pollution), which = 1)
```



13.4 Quadratic polynomial model to predict y using x_2

A polynomial in the variable x of degree D is a linear combination of the powers of x up to D .

For example:

- Linear: $y = \beta_0 + \beta_1 x$
- Quadratic: $y = \beta_0 + \beta_1 x + \beta_2 x^2$
- Cubic: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$
- Quartic: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4$
- Quintic: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + \beta_5 x^5$

Fitting such a model creates a **polynomial regression**.

13.4.1 The raw quadratic model

Let's look at a **quadratic model** which predicts y using x_2 and the square of x_2 , so that our model is of the form:

$$y = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \text{error}$$

There are several ways to fit this exact model.

- One approach is to calculate the square of `x2` within our `pollution` data set, and then feed both `x2` and `x2squared` to `lm`.
- Another approach uses the `I` function within our `lm` to specify the use of both `x2` and its square.
- Yet another approach uses the `poly` function within our `lm`, which can be used to specify raw models including `x2` and `x2squared`.

```
pollution <- pollution |>
  mutate(x2squared = x2^2)

mod2a <- lm(y ~ x2 + x2squared, data = pollution)
mod2b <- lm(y ~ x2 + I(x2^2), data = pollution)
mod2c <- lm(y ~ poly(x2, degree = 2, raw = TRUE), data = pollution)
```

Each of these approaches produces the same model, as they are just different ways of expressing the same idea.

```
summary(mod2a)
```

Call:

```
lm(formula = y ~ x2 + x2squared, data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-148.977	-38.651	6.889	35.312	189.346

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	785.77449	79.54086	9.879	5.87e-14	***
x2	8.87640	4.27394	2.077	0.0423	*
x2squared	-0.11704	0.05429	-2.156	0.0353	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.83 on 57 degrees of freedom

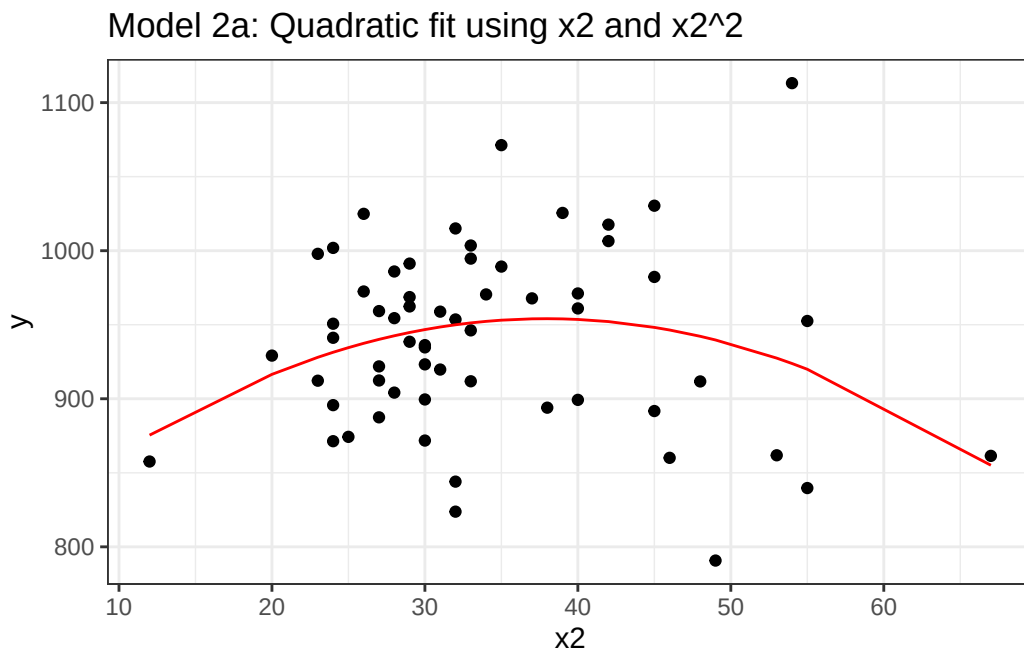
Multiple R-squared: 0.07623, Adjusted R-squared: 0.04382

F-statistic: 2.352 on 2 and 57 DF, p-value: 0.1044

And if we plot the fitted values for this mod2 using whatever approach you like, we get exactly the same result.

```
mod2a_aug <- augment(mod2a, pollution)

ggplot(mod2a_aug, aes(x = x2, y = y)) +
  geom_point() +
  geom_line(aes(x = x2, y = .fitted), col = "red") +
  labs(title = "Model 2a: Quadratic fit using x2 and x2^2")
```



```
mod2b_aug <- augment(mod2b, pollution)

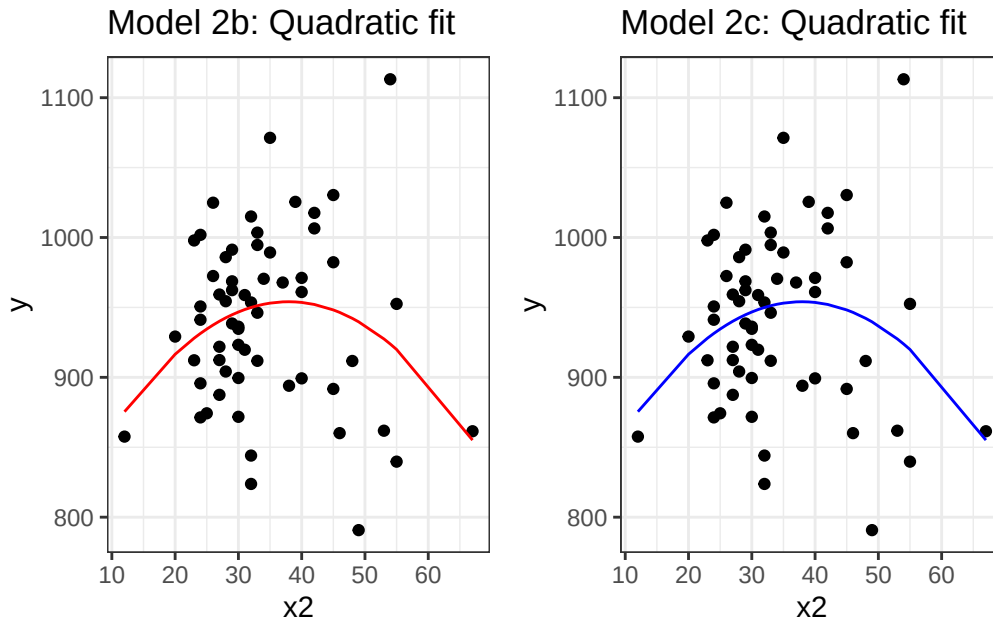
mod2c_aug <- augment(mod2c, pollution)

p1 <- ggplot(mod2b_aug, aes(x = x2, y = y)) +
  geom_point() +
  geom_line(aes(x = x2, y = .fitted), col = "red") +
  labs(title = "Model 2b: Quadratic fit")

p2 <- ggplot(mod2c_aug, aes(x = x2, y = y)) +
  geom_point() +
  geom_line(aes(x = x2, y = .fitted), col = "blue") +
```

```
labs(title = "Model 2c: Quadratic fit")
```

```
p1 + p2
```



13.4.2 Raw quadratic fit after centering x2

Sometimes, we'll center (and perhaps rescale, too) the x2 variable before including it in a quadratic fit like this.

```
pollution <- pollution |>
  mutate(x2_c = x2 - mean(x2))

mod2d <- lm(y ~ x2_c + I(x2_c^2), data = pollution)

summary(mod2d)
```

Call:

```
lm(formula = y ~ x2_c + I(x2_c^2), data = pollution)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-148.977	-38.651	6.889	35.312	189.346

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	952.25941	9.59896	99.204	<2e-16 ***
x2_c	0.92163	0.93237	0.988	0.3271
I(x2_c^2)	-0.11704	0.05429	-2.156	0.0353 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.83 on 57 degrees of freedom

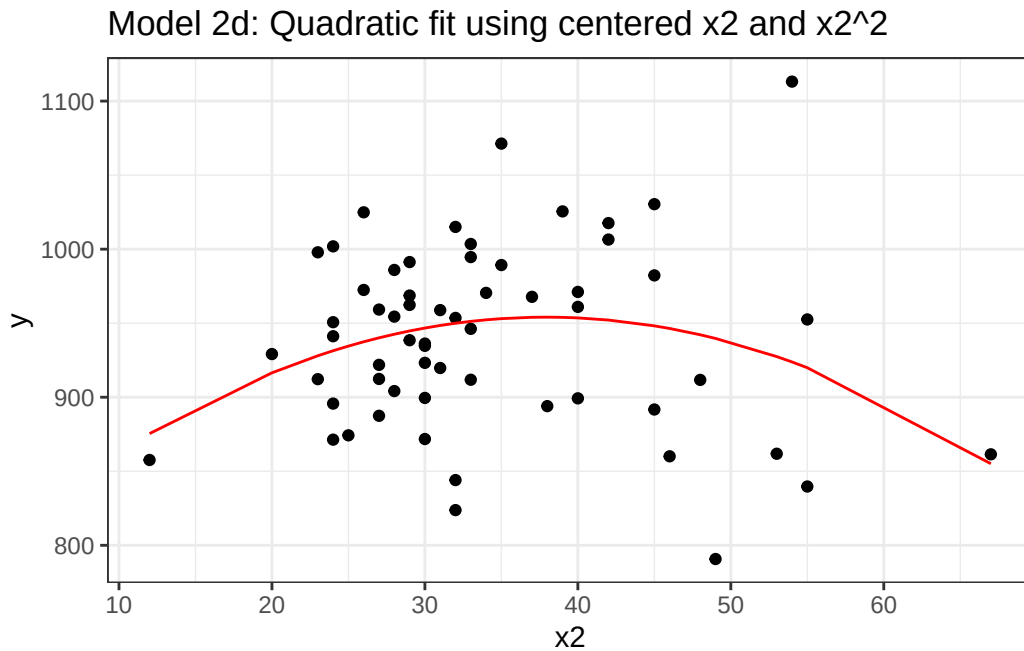
Multiple R-squared: 0.07623, Adjusted R-squared: 0.04382

F-statistic: 2.352 on 2 and 57 DF, p-value: 0.1044

Note that this model looks very different, with the exception of the second order quadratic term. But, it produces the same fitted values as the models we fit previously.

```
mod2d_aug <- augment(mod2d, pollution)

ggplot(mod2d_aug, aes(x = x2, y = y)) +
  geom_point() +
  geom_line(aes(x = x2, y = .fitted), col = "red") +
  labs(title = "Model 2d: Quadratic fit using centered x2 and x2^2")
```



Or, if you don't believe me yet, look at the four sets of fitted values another way.

```
favstats(~ .fitted, data = mod2a_aug)
```

min	Q1	median	Q3	max	mean	sd	n	missing
855.1041	936.7155	945.597	950.2883	954.073	940.3585	17.17507	60	0

```
favstats(~ .fitted, data = mod2b_aug)
```

min	Q1	median	Q3	max	mean	sd	n	missing
855.1041	936.7155	945.597	950.2883	954.073	940.3585	17.17507	60	0

```
favstats(~ .fitted, data = mod2c_aug)
```

min	Q1	median	Q3	max	mean	sd	n	missing
855.1041	936.7155	945.597	950.2883	954.073	940.3585	17.17507	60	0

```
favstats(~ .fitted, data = mod2d_aug)
```

min	Q1	median	Q3	max	mean	sd	n	missing
855.1041	936.7155	945.597	950.2883	954.073	940.3585	17.17507	60	0

13.5 Orthogonal Polynomials

Now, let's fit an orthogonal polynomial of degree 2 to predict y using x2.

```
mod2_orth <- lm(y ~ poly(x2, 2), data = pollution)

summary(mod2_orth)
```

Call:

```
lm(formula = y ~ poly(x2, 2), data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-148.977	-38.651	6.889	35.312	189.346

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	940.358	7.853	119.746	<2e-16 ***
poly(x2, 2)1	-14.345	60.829	-0.236	0.8144
poly(x2, 2)2	-131.142	60.829	-2.156	0.0353 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.83 on 57 degrees of freedom

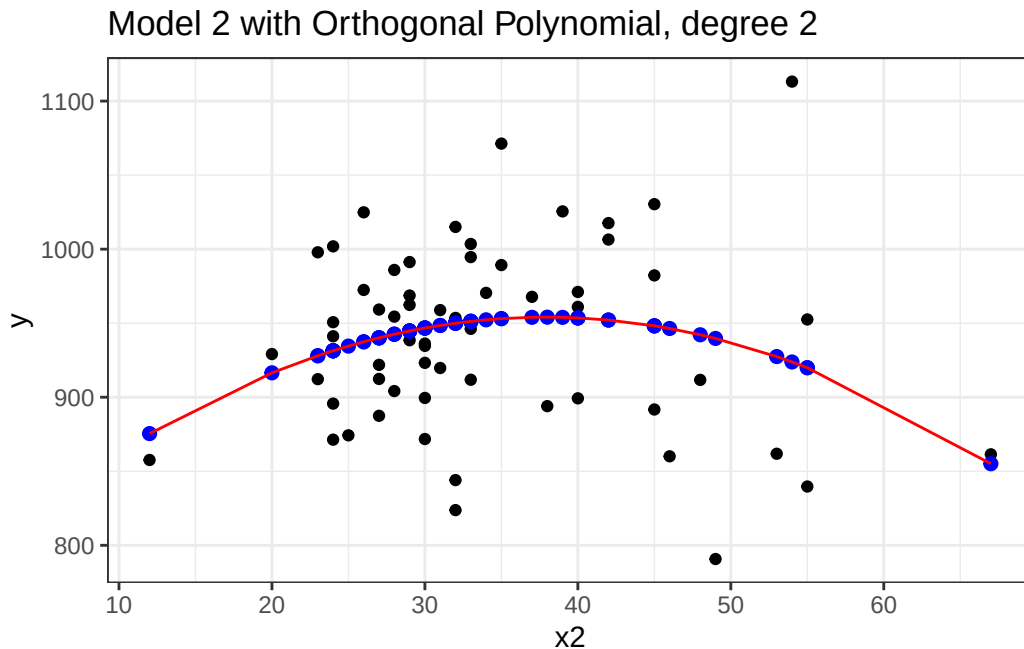
Multiple R-squared: 0.07623, Adjusted R-squared: 0.04382

F-statistic: 2.352 on 2 and 57 DF, p-value: 0.1044

Now this looks very different in the equation, but, again, we can see that this produces exactly the same fitted values as our previous models, and the same model fit summaries. Is it, in fact, the same model? Here, we'll plot the fitted Model 2a in a red line, and this new Model 2 with Orthogonal Polynomials as blue points.

```
mod2orth_aug <- augment(mod2_orth, pollution)

ggplot(mod2orth_aug, aes(x = x2, y = y)) +
  geom_point() +
  geom_point(aes(x = x2, y = .fitted),
             col = "blue", size = 2) +
  geom_line(data = mod2a_aug, aes(x = x2, y = .fitted),
           col = "red") +
  labs(title = "Model 2 with Orthogonal Polynomial, degree 2")
```



Yes, it is again the same model in terms of the predictions it makes for y .

By default, with `raw = FALSE`, the `poly()` function within a linear model computes what is called an **orthogonal polynomial**. An orthogonal polynomial sets up a model design matrix using the coding we've seen previously: `x2` and `x2^2` in our case, and then scales those columns so that each column is **orthogonal** to the previous ones. This eliminates the collinearity (correlation between predictors) and lets our t tests tell us whether the addition of any particular polynomial term improves the fit of the model over the lower orders.

Would the addition of a cubic term help us much in predicting y from $x2$?

```
mod3 <- lm(y ~ poly(x2, 3), data = pollution)
summary(mod3)
```

Call:

```
lm(formula = y ~ poly(x2, 3), data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-146.262	-39.679	5.569	35.984	191.536

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

```

(Intercept)    940.358      7.917 118.772   <2e-16 ***
poly(x2, 3)1   -14.345      61.328  -0.234   0.8159
poly(x2, 3)2  -131.142      61.328  -2.138   0.0369 *
poly(x2, 3)3    16.918      61.328   0.276   0.7837
---

```

```

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 61.33 on 56 degrees of freedom

```

```

Multiple R-squared:  0.07748,    Adjusted R-squared:  0.02806

```

```

F-statistic: 1.568 on 3 and 56 DF,  p-value: 0.2073

```

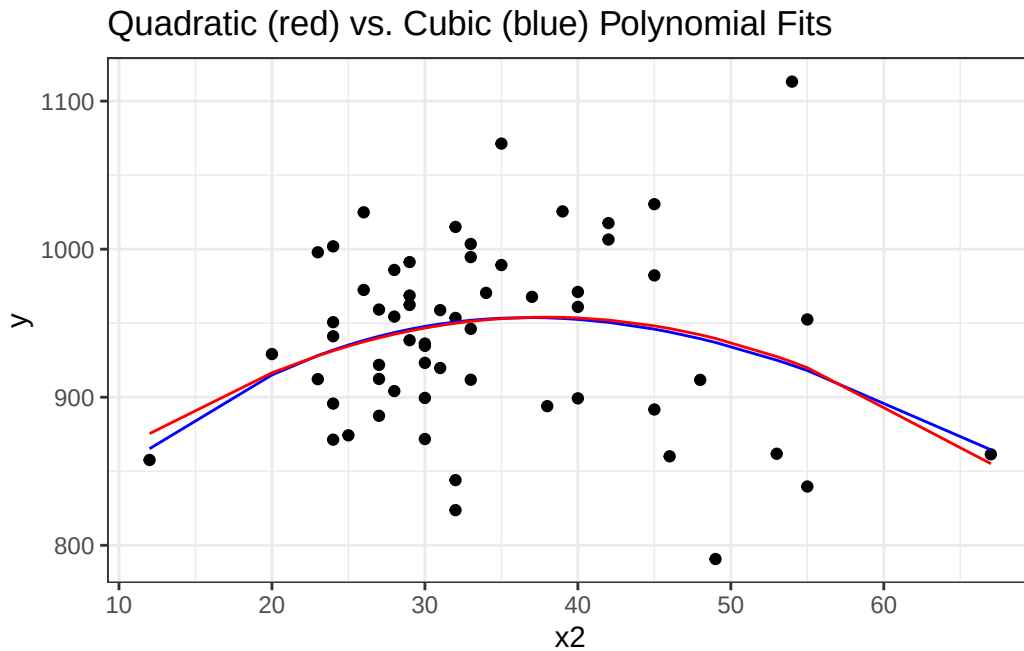
It doesn't appear that the cubic term adds much here, if anything. The p value is not significant for the third degree polynomial, the summaries of fit quality aren't much improved, and as we can see from the plot below, the predictions don't actually change all that much.

```

mod3_aug <- augment(mod3, pollution)

ggplot(mod3_aug, aes(x = x2, y = y)) +
  geom_point() +
  geom_line(aes(x = x2, y = .fitted),
            col = "blue") +
  geom_line(data = mod2orth_aug, aes(x = x2, y = .fitted),
            col = "red") +
  labs(title = "Quadratic (red) vs. Cubic (blue) Polynomial Fits")

```

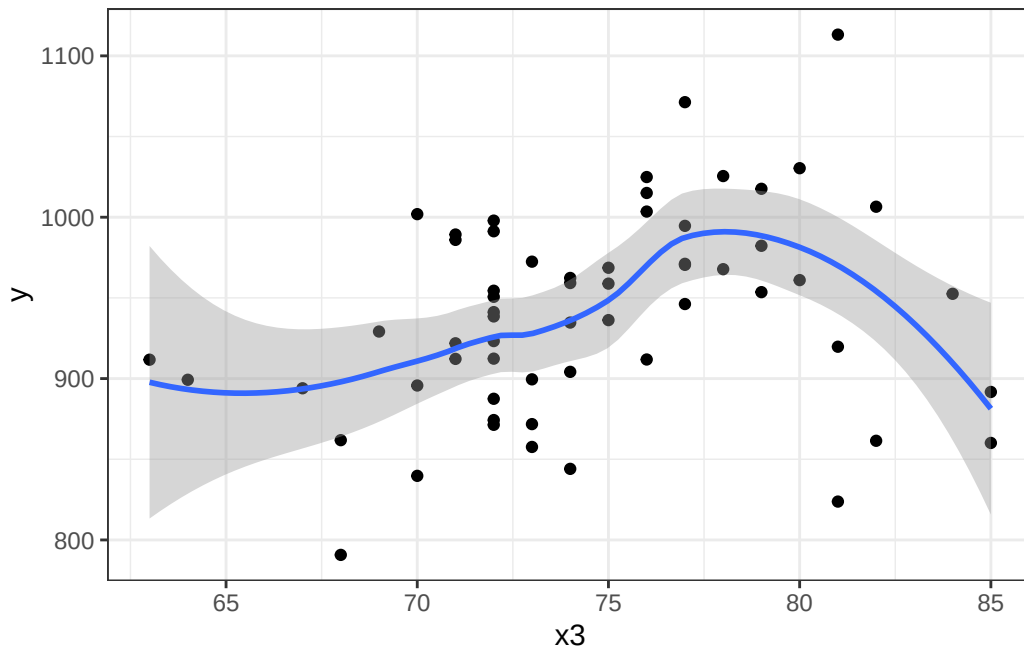


13.6 Fit a cubic polynomial to predict y from x_3

What if we consider another predictor instead? Let's look at x_3 , the Mean July temperature in degrees Fahrenheit. Here is the `loess` smooth.

```
ggplot(pollution, aes(x = x3, y = y)) +  
  geom_point() +  
  geom_smooth(method = "loess")
```

`geom_smooth()` using formula = ' $y \sim x$ '



That looks pretty curvy - perhaps we need a more complex polynomial. We'll consider a linear model (`mod4_L`), a quadratic fit (`mod4_Q`) and a polynomial of degree 3: a **cubic** fit (`mod4_C`)

```
mod4_L <- lm(y ~ x3, data = pollution)
summary(mod4_L)
```

Call:

```
lm(formula = y ~ x3, data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-139.813	-34.341	4.271	38.197	149.587

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	670.529	123.140	5.445	1.1e-06 ***
x3	3.618	1.648	2.196	0.0321 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.29 on 58 degrees of freedom

Multiple R-squared: 0.07674, Adjusted R-squared: 0.06082

F-statistic: 4.821 on 1 and 58 DF, p-value: 0.03213

```
mod4_Q <- lm(y ~ poly(x3, 2), data = pollution)
summary(mod4_Q)
```

Call:

```
lm(formula = y ~ poly(x3, 2), data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-132.004	-42.184	4.069	47.126	157.396

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	940.358	7.553	124.503	<2e-16 ***
poly(x3, 2)1	132.364	58.504	2.262	0.0275 *
poly(x3, 2)2	-125.270	58.504	-2.141	0.0365 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 58.5 on 57 degrees of freedom

Multiple R-squared: 0.1455, Adjusted R-squared: 0.1155

F-statistic: 4.852 on 2 and 57 DF, p-value: 0.01133

```
mod4_C <- lm(y ~ poly(x3, 3), data = pollution)
summary(mod4_C)
```

Call:

```
lm(formula = y ~ poly(x3, 3), data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-148.004	-29.998	1.441	34.579	141.396

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	940.358	7.065	133.095	< 2e-16 ***
poly(x3, 3)1	132.364	54.728	2.419	0.01886 *
poly(x3, 3)2	-125.270	54.728	-2.289	0.02588 *

```
poly(x3, 3)3 -165.439      54.728  -3.023  0.00377 **
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 54.73 on 56 degrees of freedom

Multiple R-squared: 0.2654, Adjusted R-squared: 0.226

F-statistic: 6.742 on 3 and 56 DF, p-value: 0.0005799

It looks like the cubic polynomial term is of some real importance here. Do the linear, quadratic and cubic model fitted values look different?

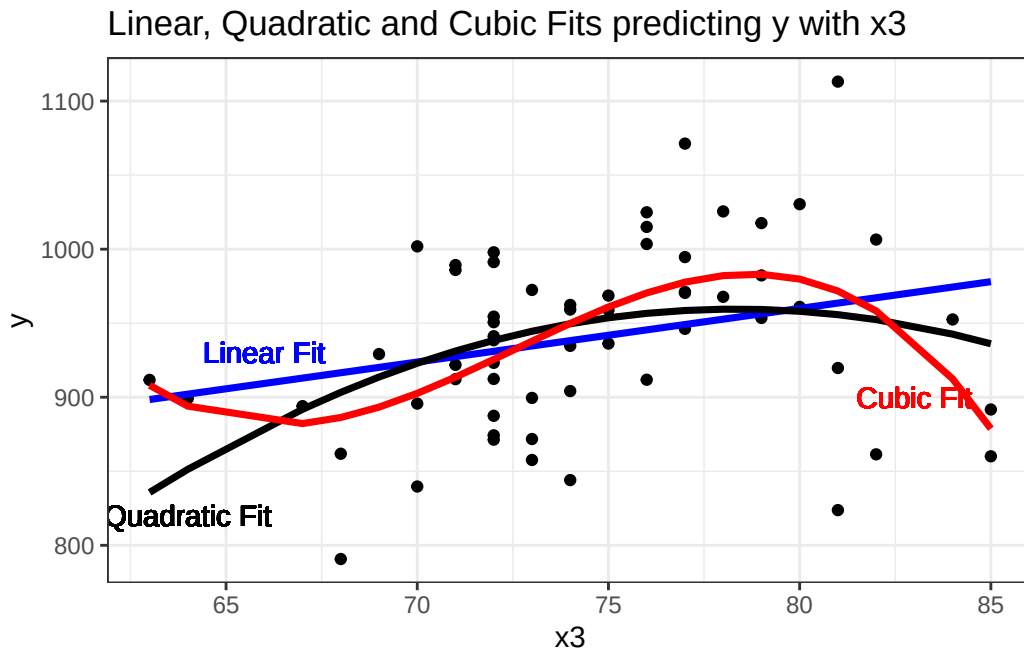
```
mod4_L_aug <- augment(mod4_L, pollution)

mod4_Q_aug <- augment(mod4_Q, pollution)

mod4_C_aug <- augment(mod4_C, pollution)

ggplot(pollution, aes(x = x3, y = y)) +
  geom_point() +
  geom_line(data = mod4_L_aug, aes(x = x3, y = .fitted),
            col = "blue", size = 1.25) +
  geom_line(data = mod4_Q_aug, aes(x = x3, y = .fitted),
            col = "black", size = 1.25) +
  geom_line(data = mod4_C_aug, aes(x = x3, y = .fitted),
            col = "red", size = 1.25) +
  geom_text(x = 66, y = 930, label = "Linear Fit", col = "blue") +
  geom_text(x = 64, y = 820, label = "Quadratic Fit", col = "black") +
  geom_text(x = 83, y = 900, label = "Cubic Fit", col = "red") +
  labs(title = "Linear, Quadratic and Cubic Fits predicting y with x3") +
  theme_bw()
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



13.7 Fitting a restricted cubic spline in a linear regression

- A **linear spline** is a continuous function formed by connecting points (called **knots** of the spline) by line segments.
- A **restricted cubic spline** is a way to build highly complicated curves into a regression equation in a fairly easily structured way.
- A restricted cubic spline is a series of polynomial functions joined together at the knots.
 - Such a spline gives us a way to flexibly account for non-linearity without over-fitting the model.
 - Restricted cubic splines can fit many different types of non-linearities.
 - Specifying the number of knots is all you need to do in R to get a reasonable result from a restricted cubic spline.

The most common choices are 3, 4, or 5 knots. Each additional knot adds to the non-linearity, and spends an additional degree of freedom:

- 3 Knots, 2 degrees of freedom, allows the curve to “bend” once.
- 4 Knots, 3 degrees of freedom, lets the curve “bend” twice.
- 5 Knots, 4 degrees of freedom, lets the curve “bend” three times.

For most applications, three to five knots strike a nice balance between complicating the model needlessly and fitting data pleasingly. Let's consider a restricted cubic spline model for our y based on x_3 again, but now with:

- in `mod5a`, 3 knots,
- in `mod5b`, 4 knots, and
- in `mod5c`, 5 knots

```
mod5a_rcs <- lm(y ~ rcs(x3, 3), data = pollution)
mod5b_rcs <- lm(y ~ rcs(x3, 4), data = pollution)
mod5c_rcs <- lm(y ~ rcs(x3, 5), data = pollution)
```

Here, for instance, is the summary of the 5-knot model:

```
summary(mod5c_rcs)
```

Call:

```
lm(formula = y ~ rcs(x3, 5), data = pollution)
```

Residuals:

Min	1Q	Median	3Q	Max
-141.522	-32.009	1.674	31.971	147.878

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	468.113	396.319	1.181	0.243
$\text{rcs}(x_3, 5)x_3$	6.447	5.749	1.121	0.267
$\text{rcs}(x_3, 5)x_3'$	-25.633	46.810	-0.548	0.586
$\text{rcs}(x_3, 5)x_3''$	323.137	293.065	1.103	0.275
$\text{rcs}(x_3, 5)x_3'''$	-612.578	396.270	-1.546	0.128

Residual standard error: 54.35 on 55 degrees of freedom

Multiple R-squared: 0.2883, Adjusted R-squared: 0.2366

F-statistic: 5.571 on 4 and 55 DF, p-value: 0.0007734

We'll begin by storing the fitted values from these three models and other summaries, for plotting.

```
mod5a_aug <- augment(mod5a_rcs, pollution)
```

```
mod5b_aug <- augment(mod5b_rcs, pollution)
```

```
mod5c_aug <- augment(mod5c_rcs, pollution)
```

```
p2 <- ggplot(pollution, aes(x = x3, y = y)) +  
  geom_point() +  
  geom_smooth(method = "loess", col = "purple", se = F) +  
  labs(title = "Loess Smooth") +  
  theme_bw()
```

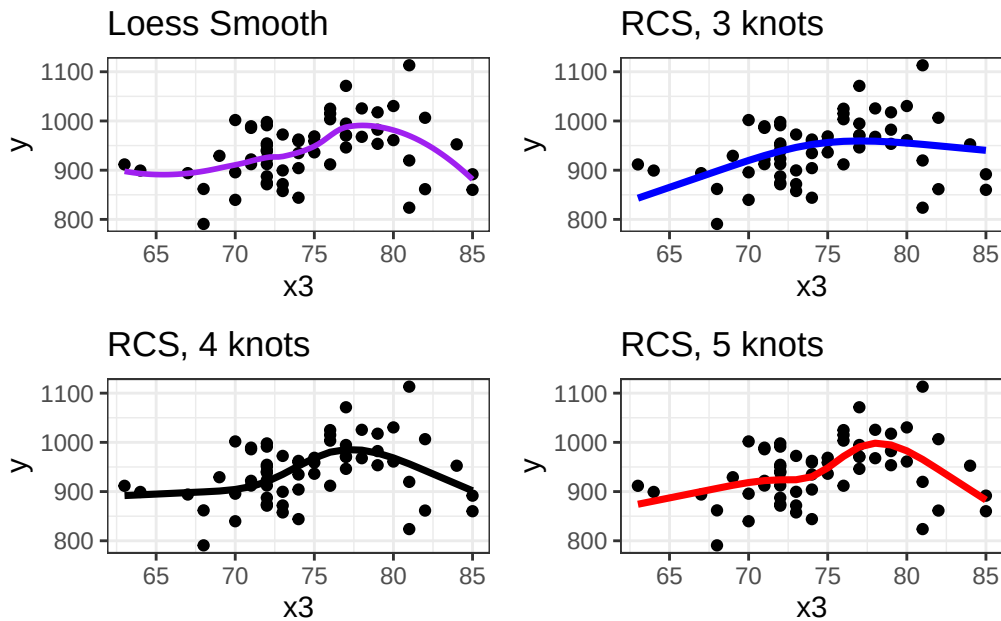
```
p3 <- ggplot(mod5a_aug, aes(x = x3, y = y)) +  
  geom_point() +  
  geom_line(aes(x = x3, y = .fitted),  
            col = "blue", size = 1.25) +  
  labs(title = "RCS, 3 knots") +  
  theme_bw()
```

```
p4 <- ggplot(mod5b_aug, aes(x = x3, y = y)) +  
  geom_point() +  
  geom_line(aes(x = x3, y = .fitted),  
            col = "black", size = 1.25) +  
  labs(title = "RCS, 4 knots") +  
  theme_bw()
```

```
p5 <- ggplot(mod5c_aug, aes(x = x3, y = y)) +  
  geom_point() +  
  geom_line(aes(x = x3, y = .fitted),  
            col = "red", size = 1.25) +  
  labs(title = "RCS, 5 knots") +  
  theme_bw()
```

```
(p2 + p3) / (p4 + p5)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Does it seem like the fit improves markedly (perhaps approaching the loess smooth result) as we increase the number of knots?

```
anova(mod5a_rcs, mod5b_rcs, mod5c_rcs)
```

Analysis of Variance Table

```
Model 1: y ~ rcs(x3, 3)
Model 2: y ~ rcs(x3, 4)
Model 3: y ~ rcs(x3, 5)
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	57	194935				
2	56	171448	1	23486.9	7.9503	0.006672 **
3	55	162481	1	8967.2	3.0354	0.087057 .

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on an ANOVA comparison, the fourth knot adds significant predictive value ($p = 0.0067$), but the fifth knot is borderline ($p = 0.0871$). From the `glance` function in the `broom` package, we can also look at some key summaries.

```
glance(mod5a_rcs)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1    0.146         0.116  58.5      4.88 0.0111     2 -328.  663.  672.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
glance(mod5b_rcs)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1    0.249         0.209  55.3      6.19 0.00104     3 -324.  658.  668.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
glance(mod5c_rcs)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1    0.288         0.237  54.4      5.57 0.000773     4 -322.  657.  669.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Model	Knots	R^2	Adj. R^2	AIC	BIC
5a	3	0.146	0.116	663.4	671.8
5b	4	0.249	0.209	657.7	668.2
5c	5	0.288	0.237	656.5	669.1

Within our sample, the five-knot RCS outperforms the 3- and 4-knot versions on adjusted R^2 and AIC (barely) and does a little worse than the 4-knot RCS on BIC.

Of course, we could also use the cross-validation methods we've developed for other linear regressions to assess predictive capacity of these models. I'll skip that for now.

To see the values of `x3` where the splines place their knots, we can use the `attributes` function.


```
attributes(rcs(pollution$x3, 5))
```

```
$dim
[1] 60  4

$dimnames
$dimnames[[1]]
NULL

$dimnames[[2]]
[1] "pollution"      "pollution'"      "pollution'"'"    "pollution'"'"

$class
[1] "rms"

$name
[1] "pollution"

$label
[1] "pollution"

$assume
[1] "rcspline"

$assume.code
[1] 4

$parms
[1] 68 72 74 77 82

$nonlinear
[1] FALSE TRUE TRUE TRUE

$colnames
[1] "pollution"      "pollution'"      "pollution'"'"    "pollution'"'"
```

The knots in this particular 5-knot spline are placed by the computer at 68, 72, 74, 77 and 82, it seems.

There are two kinds of Multivariate Regression Models

1. [Prediction] Those that are built so that we can make accurate predictions.
2. [Explanatory] Those that are built to help understand underlying phenomena.

While those two notions overlap considerably, they do imply different things about how we strategize about model-building and model assessment. Harrell's primary concern is effective use of the available data for **prediction** - this implies some things that will be different from what we've seen in the past.

Harrell refers to multivariable regression modeling strategy as the process of **spending degrees of freedom**. The main job in strategizing about multivariate modeling is to

1. Decide the number of degrees of freedom that can be spent
2. Decide where to spend them
3. Spend them, wisely.

What this means is essentially linked to making decisions about predictor complexity, both in terms of how many predictors will be included in the regression model, and about how we'll include those predictors.

13.8 “Spending” Degrees of Freedom

- “Spending” df includes
 - fitting parameter estimates in models, or
 - examining figures built using the outcome variable Y that tell you how to model the predictors.

If you use a scatterplot of Y vs. X or the residuals of the Y - X regression model vs. X to decide whether a linear model is appropriate, then how many degrees of freedom have you actually spent?

Grambsch and O'Brien conclude that if you wish to preserve the key statistical properties of the various estimation and fitting procedures used in building a model, you can't retrieve these degrees of freedom once they have been spent.

13.8.1 Overfitting and Limits on the # of Predictors

Suppose you have a total sample size of n observations, then you really shouldn't be thinking about estimating more than $n/15$ regression coefficients, at the most.

- If k is the number of parameters in a full model containing all candidate predictors for a stepwise analysis, then k should be no greater than $n/15$.
- k should include all variables screened for association with the response, including interaction terms.

- Sometimes I hold myself to a tougher standard, or $n/50$ predictors, at maximum.

So if you have 97 observations in your data, then you can probably just barely justify the use of a stepwise analysis using the main effects alone of 5 candidate variables (with one additional DF for the intercept term) using the $n/15$ limit.

@Harrell2001 also mentions that if you have a **narrowly distributed** predictor, without a lot of variation to work with, then an even larger sample size n should be required. See @Vittinghoff2012, Section 10.3 for more details.

13.8.2 The Importance of Collinearity

Collinearity denotes correlation between predictors high enough to degrade the precision of the regression coefficient estimates substantially for some or all of the correlated predictors

- @Vittinghoff2012, section 10.4.1
- Can one predictor in a model be predicted well using the other predictors in the model?
 - Strong correlations (for instance, $r \geq 0.8$) are especially troublesome.
- Effects of collinearity
 - decreases precision, in the sense of increasing the standard errors of the parameter estimates
 - decreases power
 - increases the difficulty of interpreting individual predictor effects
 - overall F test is significant, but individual t tests may not be

Suppose we want to assess whether variable X_j is collinear with the other predictors in a model. We run a regression predicting X_j using the other predictors, and obtain the R^2 . The VIF is defined as $1 / (1 - \text{this } R^2)$, and we usually interpret VIFs above 5 as indicating a serious multicollinearity problem (i.e. R^2 values for this predictor of 0.8 and above would thus concern us.)

```
vif(lm(y ~ x1 + x2 + x3 + x4 + x5 + x6, data = pollution))
```

	x1	x2	x3	x4	x5	x6
	2.238862	2.058731	2.153044	4.174448	3.447399	1.792996

Occasionally, you'll see the inverse of VIF reported, and this is called *tolerance*.

- $\text{tolerance} = 1 / \text{VIF}$

13.8.3 Collinearity in an Explanatory Model

- When we are attempting to **identify multiple independent predictors** (the explanatory model approach), then we will need to choose between collinear variables
 - options suggested by @Vittinghoff2012, p. 422, include choosing on the basis of plausibility as a causal factor,
 - choosing the variable that has higher data quality (is measured more accurately or has fewer missing values.)
 - Often, we choose to include a variable that is statistically significant as a predictor, and drop others, should we be so lucky.
- Larger effects, especially if they are associated with predictors that have minimal correlation with the other predictors under study, cause less trouble in terms of potential violation of the $n/15$ rule for what constitutes a reasonable number of predictors.

13.8.4 Collinearity in a Prediction Model

- If we are primarily building a **prediction model** for which inference on the individual predictors is not of interest, then it is totally reasonable to use both predictors in the model, if doing so reduces prediction error.
 - Collinearity doesn't affect predictions in our model development sample.
 - Collinearity doesn't affect predictions on new data so long as the new data have similar relationships between predictors.
 - If our key predictor is correlated strongly with a confounder and if the predictor remains an important part of the model after adjustment for the confounder, then this suggests a meaningful independent effect.
 - * If the effects of the predictor are clearly confounded by the adjustment variable, we again have a clear result.
 - * If neither appears to add meaningful predictive value after adjustment, the data may be inadequate.
 - If the collinearity is between adjustment variables, but doesn't involve the key predictor, then inclusion of the collinear variables is unlikely to cause substantial problems.

13.9 Spending DF on Non-Linearity: The Spearman Plot

We need a flexible approach to assessing non-linearity and fitting models with non-linear predictors. This will lead us to a measure of what @Harrell2001 calls **potential predictive punch** which hides the true form of the regression from the analyst so as to preserve statistical properties, but that lets us make sensible decisions about whether a predictor should be

included in a model, and the number of parameters (degrees of freedom, essentially) we are willing to devote to it.

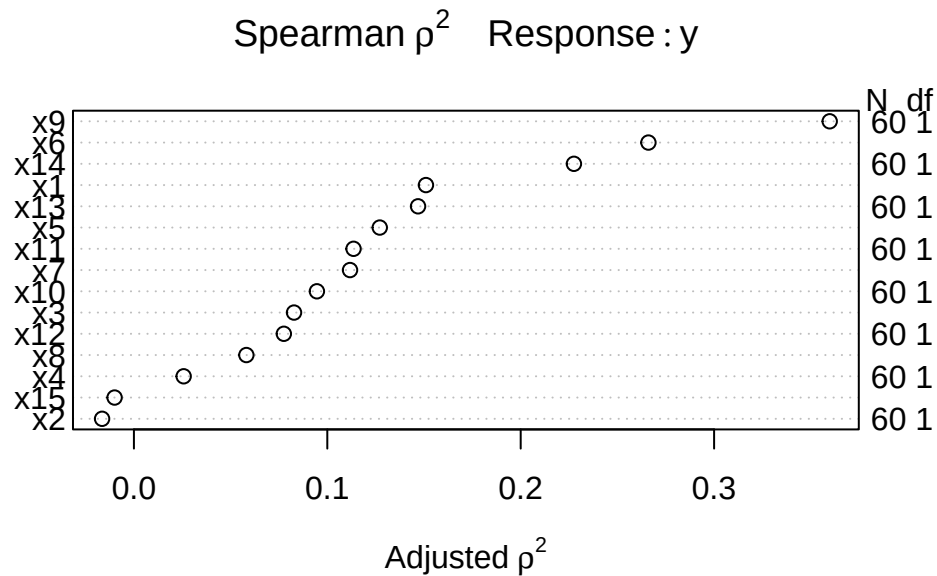
What if we want to consider where best to spend our degrees of freedom on non-linear predictor terms, like interactions, polynomial functions or curved splines to represent our input data? The approach we'll find useful in the largest variety of settings is a combination of

1. a rank correlation assessment of potential predictive punch (using a Spearman ρ^2 plot, available in the `Hmisc` package), followed by
2. the application of restricted cubic splines to fit and assess models.

Let's try such a plot for our fifteen predictors:

```
sp2 <- spearman2(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 +
                 x8 + x9 + x10 + x11 + x12 + x13 +
                 x14 + x15, data = pollution)

plot(sp2)
```



The variable with the largest adjusted squared Spearman ρ statistic in this setting is `x9`, followed by `x6` and `x14`. With only 60 observations, we might well want to restrict ourselves to a very small model. What the Spearman plot suggests is that we focus any non-linear terms on `x9` first, and then perhaps `x6` and `x14` as they have some potential predictive power. It may or may not work out that the non-linear terms are productive.

13.9.1 Fitting a Big Model to the pollution data

So, one possible model built in reaction this plot might be to fit:

- a restricted cubic spline with 5 knots on `x9`,
- a restricted cubic spline with 3 knots on `x6`,
- a quadratic polynomial on `x14`, and
- a linear fit to `x1` and `x13`

That's way more degrees of freedom (4 for `x9`, 2 for `x6`, 2 for `x14` and 1 each for `x1` and `x13` makes a total of 10 without the intercept term) than we can really justify with a sample of 60 observations. But let's see what happens.

```
mod_big <- lm(y ~ rcs(x9, 5) + rcs(x6, 3) + poly(x14, 2) +  
             x1 + x13, data = pollution)  
  
anova(mod_big)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
rcs(x9, 5)	4	100164	25040.9	17.8482	4.229e-09	***
rcs(x6, 3)	2	38306	19152.8	13.6513	1.939e-05	***
poly(x14, 2)	2	15595	7797.7	5.5579	0.006677	**
x1	1	4787	4787.3	3.4122	0.070759	.
x13	1	712	711.9	0.5074	0.479635	
Residuals	49	68747	1403.0			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

This `anova` suggests that we have at least some predictive value in each spline (`x9` and `x6`) and some additional value in `x14`, although it's not as clear that the linear terms (`x1` and `x13`) did much good.

13.9.2 Limitations of `lm` for fitting complex linear regression models

We can certainly assess this big, complex model using `lm` in comparison to other models:

- with in-sample summary statistics like adjusted R^2 , AIC and BIC,
- we can assess its assumptions with residual plots, and
- we can also compare out-of-sample predictive quality through cross-validation,

But to really delve into the details of how well this complex model works, and to help plot what is actually being fit, we'll probably want to fit the model using an alternative method for fitting linear models, called `ols`, from the `rms` package developed by Frank Harrell and colleagues. That's where we're heading, in [Chapter 14](#).

14 Using `ols` to fit linear models

Back at the end of Chapter 13, we fit a model to the `pollution` data that predicted an outcome y = Age-Adjusted Mortality Rate, using:

- a restricted cubic spline with 5 knots on `x9`
- a restricted cubic spline with 3 knots on `x6`
- a polynomial in 2 degrees on `x14`
- linear terms for `x1` and `x13`

but this model was hard to evaluate in some ways. Now, instead of using `lm` to fit this model, we'll use a new function called `ols` from the `rms` package developed by Frank Harrell and colleagues, in part to support ideas developed in @Harrell2001 for clinical prediction models.

14.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(Hmisc)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

14.1.1 Data Load

```
pollution <- read_csv("data/pollution.csv", show_col_types = FALSE)
```


14.2 Fitting a model with `ols`

We will use the `datadist` approach when fitting a linear model with `ols` from the `rms` package, so as to store additional important elements of the model fit.

```
d <- datadist(pollution)
options(datadist = "d")
```

Next, we'll fit the model using `ols` and place its results in `newmod`.

```
newmod <- ols(y ~ rcs(x9, 5) + rcs(x6, 3) + pol(x14, 2) +
             x1 + x13,
             data = pollution, x = TRUE, y = TRUE)
newmod
```

Linear Regression Model

```
ols(formula = y ~ rcs(x9, 5) + rcs(x6, 3) + pol(x14, 2) + x1 +
      x13, data = pollution, x = TRUE, y = TRUE)
```

		Model Likelihood	Discrimination
		Ratio Test	Indexes
Obs	60	LR chi2 72.02	R2 0.699
sigma	37.4566	d.f. 10	R2 adj 0.637
d.f.	49	Pr(> chi2) 0.0000	g 58.961

Residuals

Min	1Q	Median	3Q	Max
-86.189	-18.554	-1.799	18.645	104.307

	Coef	S.E.	t	Pr(> t)
Intercept	796.2658	162.3269	4.91	<0.0001
x9	-2.6328	6.3504	-0.41	0.6803
x9'	121.4651	124.4827	0.98	0.3340
x9''	-219.8025	227.6775	-0.97	0.3391
x9'''	151.5700	171.3867	0.88	0.3808
x6	7.6817	15.5230	0.49	0.6229
x6'	-29.4388	18.0531	-1.63	0.1094
x14	0.5652	0.2547	2.22	0.0311
x14^2	-0.0010	0.0010	-0.96	0.3407

x1	1.0717	0.7317	1.46	0.1494
x13	-0.1028	0.1443	-0.71	0.4796

Some of the advantages and disadvantages of fitting linear regression models with `ols` or `lm` will reveal themselves over time. For now, one advantage for `ols` is that the entire variance-covariance matrix is saved. Most of the time, there will be some value to considering both `ols` and `lm` approaches.

Most of this output should be familiar, but a few pieces are different.

14.2.1 The Model Likelihood Ratio Test

The **Model Likelihood Ratio Test** compares `newmod` to the null model with only an intercept term. It is a goodness-of-fit test that we'll use in several types of model settings this semester.

- In many settings, the logarithm of the likelihood ratio, multiplied by -2, yields a value which can be compared to a χ^2 distribution. So here, the value 72.02 is $-2(\log \text{likelihood})$, and is compared to a χ^2 distribution with 10 degrees of freedom. We reject the null hypothesis that `newmod` is no better than the null model, and conclude instead that at least one of these predictors adds some value.
 - For `ols`, interpret the model likelihood ratio test like the global (ANOVA) F test in `lm`.
 - The likelihood function is the probability of observing our data under the specified model.
 - We can compare two nested models by evaluating the difference in their likelihood ratios and degrees of freedom, then comparing the result to a χ^2 distribution.

14.2.2 The g statistic

The **g statistic** is new and is referred to as the g-index. it's based on Gini's mean difference and is purported to be a robust and highly efficient measure of variation.

- Here, $g = 58.961$, which implies that if you randomly select two of the 60 areas included in the model, the average difference in predicted y (Age-Adjusted Mortality Rate) using this model will be 58.961.
 - Technically, g is Gini's mean difference of the predicted values.

14.3 ANOVA for an `ols` model

One advantage of the `ols` approach is that when you apply an `anova` to it, it separates out the linear and non-linear components of restricted cubic splines and polynomial terms (as well as product terms, if your model includes them.)

```
anova(newmod)
```

Analysis of Variance				Response: y	
Factor	d.f.	Partial SS	MS	F	P
x9	4	35219.7647	8804.9412	6.28	0.0004
Nonlinear	3	1339.3081	446.4360	0.32	0.8121
x6	2	9367.6008	4683.8004	3.34	0.0437
Nonlinear	1	3730.7388	3730.7388	2.66	0.1094
x14	2	18679.6957	9339.8478	6.66	0.0028
Nonlinear	1	1298.7625	1298.7625	0.93	0.3407
x1	1	3009.1829	3009.1829	2.14	0.1494
x13	1	711.9108	711.9108	0.51	0.4796
TOTAL NONLINEAR	5	6656.1824	1331.2365	0.95	0.4582
REGRESSION	10	159563.8285	15956.3829	11.37	<.0001
ERROR	49	68746.8004	1402.9959		

Unlike the `anova` approach in `lm`, in `ols` ANOVA, *partial* F tests are presented - each predictor is assessed as “last predictor in” much like the usual *t* tests in `lm`. In essence, the partial sums of squares and F tests here describe the marginal impact of removing each covariate from `newmod`.

We conclude that the non-linear parts of `x9` and `x6` and `x14` combined don’t seem to add much value, but that overall, `x9`, `x6` and `x14` seem to be valuable. So it must be the linear parts of those variables within our model that are doing most of the predictive work.

14.4 Effect Estimates

A particularly useful thing to get out of the `ols` approach that is not as easily available in `lm` (without recoding or standardizing our predictors) is a summary of the effects of each predictor in an interesting scale.

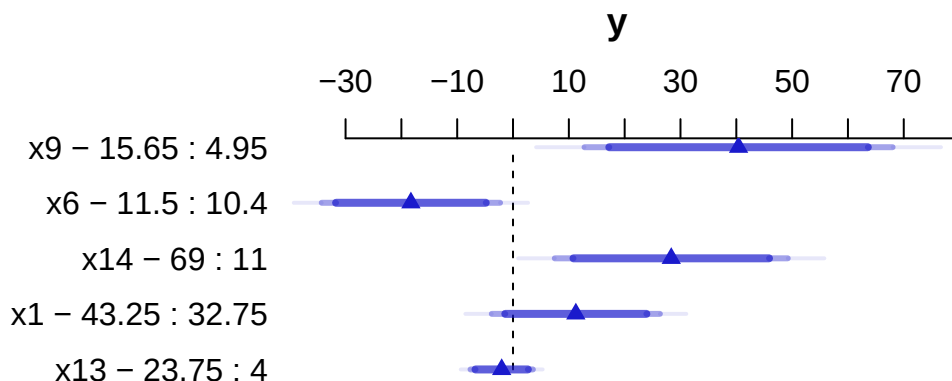
```
summary(newmod)
```

Effects			Response : y				
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
x9	4.95	15.65	10.70	40.4060	14.0790	12.1120	68.6990
x6	10.40	11.50	1.10	-18.2930	8.1499	-34.6710	-1.9153
x14	11.00	69.00	58.00	28.3480	10.6480	6.9503	49.7460
x1	32.75	43.25	10.50	11.2520	7.6833	-4.1878	26.6930
x13	4.00	23.75	19.75	-2.0303	2.8502	-7.7579	3.6973

This “effects summary” shows the effect on y of moving from the 25th to the 75th percentile of each variable (along with a standard error and 95% confidence interval) while holding the other variable at the level specified at the bottom of the output.

The most useful way to look at this sort of analysis is often a plot.

```
plot(summary(newmod))
```



For x_9 note from the `summary` above that the 25th percentile is 4.95 and the 75th is 15.65. Our conclusion is that the estimated effect of moving x_9 from 4.95 to 15.65 is an increase of 40.4 on y , with a 95% CI of (12.1, 68.7).

For a categorical variable, the low level is shown first and then the high level.

The plot shows the point estimate (arrow head) and then the 90% (narrowest bar), 95% (middle bar) and 99% (widest bar in lightest color) confidence intervals for each predictor's effect.

- It's easier to distinguish this in the `x9` plot than the one for `x13`.
- Remember that what is being compared is the first value to the second value's impact on the outcome, with other predictors held constant.

14.4.1 Simultaneous Confidence Intervals

These confidence intervals make no effort to deal with the multiple comparisons problem, but just fit individual 95% (or whatever level you choose) confidence intervals for each predictor. The natural alternative is to make an adjustment for multiple comparisons in fitting the confidence intervals, so that the set of (in this case, five - one for each predictor) confidence intervals for effect sizes has a family-wise 95% confidence level. You'll note that the effect estimates and standard errors are unchanged from those shown above, but the confidence limits are a bit wider.

```
summary(newmod, conf.type=c('simultaneous'))
```

Effects					Response : y		
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
x9	4.95	15.65	10.70	40.4060	14.0790	3.11910	77.6920
x6	10.40	11.50	1.10	-18.2930	8.1499	-39.87700	3.2903
x14	11.00	69.00	58.00	28.3480	10.6480	0.14913	56.5470
x1	32.75	43.25	10.50	11.2520	7.6833	-9.09540	31.6000
x13	4.00	23.75	19.75	-2.0303	2.8502	-9.57840	5.5178

Remember that if you're looking for the usual `lm` summary for an `ols` object, use `summary.lm`.

14.5 The Predict function for an ols model

The `Predict` function is very flexible, and can be used to produce individual or simultaneous confidence limits.

```
Predict(newmod, x9 = 12, x6 = 12, x14 = 40, x1 = 40, x13 = 20)
```

	x9	x6	x14	x1	x13	yhat	lower	upper
1	12	12	40	40	20	923.0982	893.0984	953.098

Response variable (y): y

Limits are 0.95 confidence limits

```
# individual limits
```

```
Predict(newmod, x9 = 5:15) # individual limits
```

	x9	x6	x14	x1	x13	yhat	lower	upper
1	5	11.05	30	38	9	913.7392	889.4802	937.9983
2	6	11.05	30	38	9	916.3490	892.0082	940.6897
3	7	11.05	30	38	9	921.3093	898.9657	943.6529
4	8	11.05	30	38	9	927.6464	907.0355	948.2574
5	9	11.05	30	38	9	934.3853	913.3761	955.3946
6	10	11.05	30	38	9	940.5510	917.8371	963.2648
7	11	11.05	30	38	9	945.2225	921.9971	968.4479
8	12	11.05	30	38	9	948.2885	926.4576	970.1194
9	13	11.05	30	38	9	950.2608	930.3003	970.2213
10	14	11.05	30	38	9	951.6671	932.2370	971.0971
11	15	11.05	30	38	9	953.0342	932.1662	973.9021

Response variable (y): y

Adjust to: x6=11.05 x14=30 x1=38 x13=9

Limits are 0.95 confidence limits

```
Predict(newmod, x9 = 5:15, conf.type = 'simult')
```

	x9	x6	x14	x1	x13	yhat	lower	upper
1	5	11.05	30	38	9	913.7392	882.4839	944.9946
2	6	11.05	30	38	9	916.3490	884.9884	947.7096
3	7	11.05	30	38	9	921.3093	892.5219	950.0968
4	8	11.05	30	38	9	927.6464	901.0914	954.2015
5	9	11.05	30	38	9	934.3853	907.3170	961.4536
6	10	11.05	30	38	9	940.5510	911.2865	969.8155
7	11	11.05	30	38	9	945.2225	915.2989	975.1461
8	12	11.05	30	38	9	948.2885	920.1616	976.4154

```

9  13 11.05  30 38    9 950.2608 924.5437 975.9778
10 14 11.05  30 38    9 951.6671 926.6334 976.7007
11 15 11.05  30 38    9 953.0342 926.1479 979.9204

```

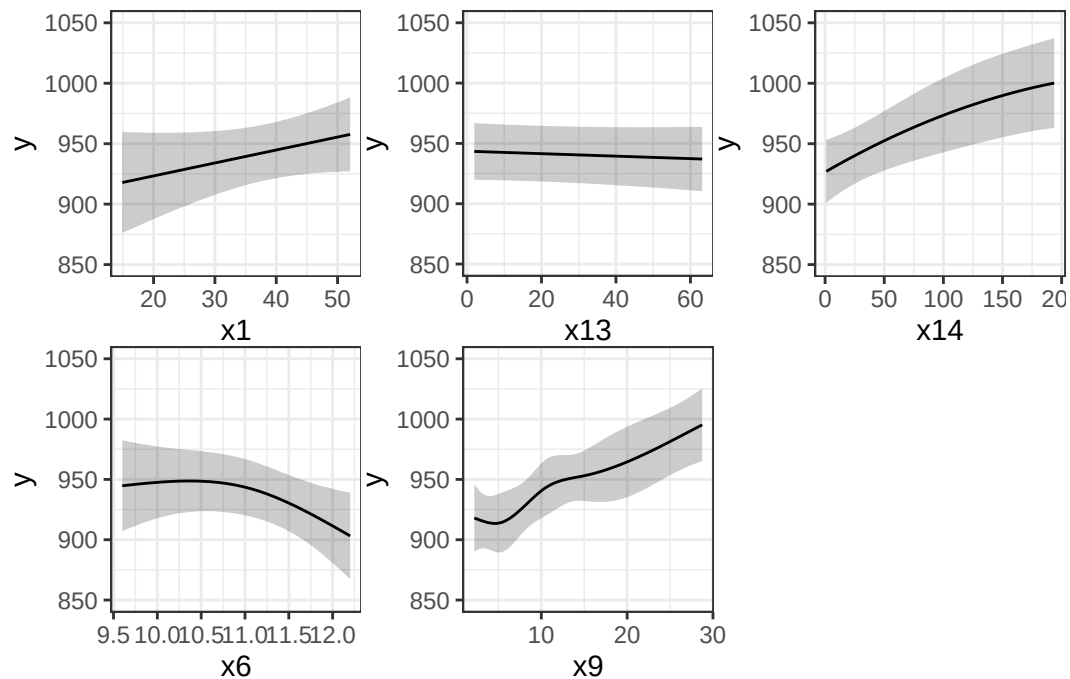
Response variable (y): y

Adjust to: x6=11.05 x14=30 x1=38 x13=9

Limits are 0.95 confidence limits

The plot below shows the individual effects in `newmod` in five subpanels, using the default approach of displaying the same range of values as are seen in the data. Note that each panel shows point and interval estimates of the effects, and spot the straight lines in `x1` and `x13`, the single bends in `x14` and `x6` and the wiggles in `x9`, corresponding to the amount of non-linearity specified in the model.

```
ggplot(Predict(newmod))
```



14.6 Checking Influence via `dfbeta`

For an `ols` object, we have several tools for looking at residuals. The most interesting to me is `which.influence` which is reliant on the notion of `dfbeta`.

- DFBETA is estimated for each observation in the data, and each coefficient in the model.
- The DFBETA is the difference in the estimated coefficient caused by deleting the observation, scaled by the coefficient's standard error estimated with the observation deleted.
- The `which.influence` command applied to an `ols` model produces a list of all of the predictors estimated by the model, including the intercept.
 - For each predictor, the command lists all observations (by row number) that, if removed from the model, would cause the estimated coefficient (the “beta”) for that predictor to change by at least some particular cutoff.
 - The default is that the DFBETA for that predictor is 0.2 or more.

```
which.influence(newmod)
```

```
$Intercept
```

```
[1] "1" "4" "7" "20" "50" "55" "60"
```

```
$x9
```

```
[1] "1" "15" "38" "39" "50" "51" "52" "53" "58"
```

```
$x6
```

```
[1] "2" "4" "7" "16" "20" "36" "50" "55" "60"
```

```
$x14
```

```
[1] "6" "7" "27" "42" "50" "56" "58" "60"
```

```
$x1
```

```
[1] "1" "7" "10" "27" "52" "60"
```

```
$x13
```

```
[1] "7" "8" "60"
```

The implication here, for instance, is that if we drop row 15 from our data frame, and refit the model, this will have a meaningful impact on the estimate of `x9` but not on the other coefficients. But if we drop, say, row 60, we will affect the estimates of the intercept, `x6`, `x14`, `x1`, and `x13`.

14.6.1 Using the `residuals` command for `dfbetas`

To see the `dfbeta` values, standardized according to the approach I used above, you can use the following code (I'll use `head` to just show the first few rows of results) to get a matrix of the results.


```
head(residuals(newmod, type = "dfbetas"))
```

	Intercept	x9	x9'	x9''	x9'''
1	-0.2142788779	0.2464314961	-0.1638633763	0.1420748132	-0.0837601846
2	-0.1609522082	-0.1651907163	0.0813708161	-0.0571478560	-0.0104199085
3	0.0002978421	0.0003616715	-0.0003743893	0.0003648207	-0.0003104531
4	-0.6689504955	0.1060165855	-0.0582613403	0.0567350766	-0.0514665792
5	-0.0185374830	0.0941987899	-0.0712140473	0.0651961744	-0.0473578767
6	-0.0733189060	0.0441321361	-0.0775996337	0.0686460863	-0.0310662575

	x6	x6'	x14	x14^2	x1
1	0.0885057227	-0.1236601786	0.1976364429	-0.1948087617	0.4602900927
2	0.1761716451	-0.2487893025	0.1331560976	-0.1425694930	0.0532802632
3	-0.0001793501	0.0001379045	-0.0002413065	0.0001584916	-0.0008536018
4	0.6480133104	-0.4398587906	0.1281557908	-0.0762194326	-0.1230959028
5	-0.0027596047	-0.0242159160	-0.0125205909	0.0030015343	0.0456581256
6	0.0598043946	-0.1228968480	0.2222504825	-0.1842672286	0.0425024727

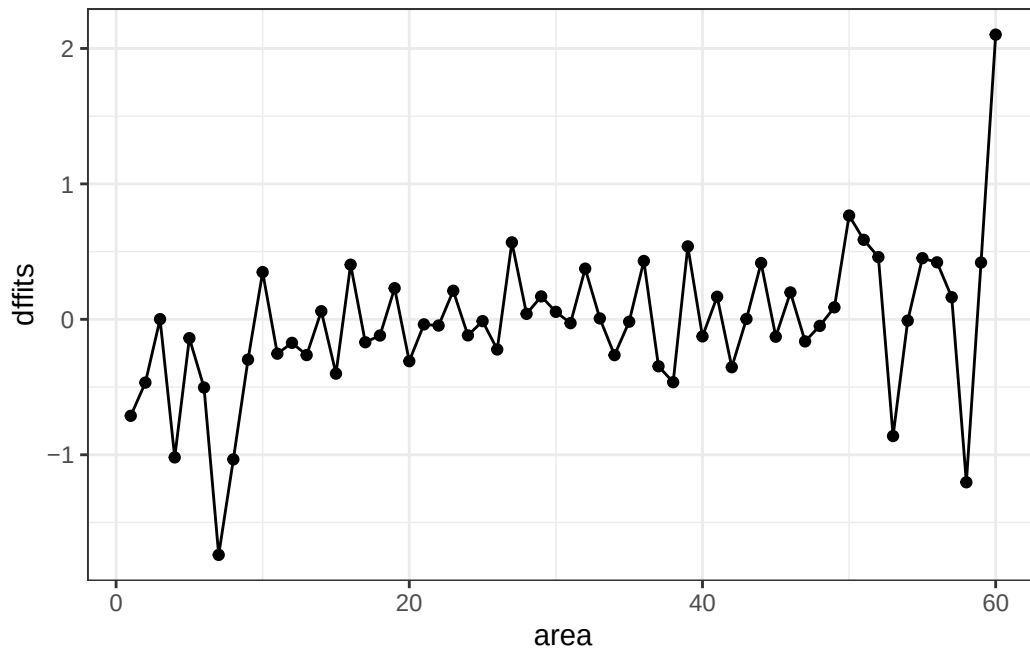
	x13
1	1.476010e-01
2	1.561944e-01
3	-8.851148e-05
4	-9.553782e-02
5	4.318623e-02
6	5.491929e-02

14.6.2 Using the residuals command for other summaries

The `residuals` command will also let you get ordinary residuals, leverage values and `dffits` values, which are the normalized differences in predicted values when observations are omitted. See `?residuals.ols` for more details.

```
temp <- data.frame(area = 1:60)
temp$residual <- residuals(newmod, type = "ordinary")
temp$leverage <- residuals(newmod, type = "hat")
temp$dfits <- residuals(newmod, type = "dffits")
temp <- as_tibble(temp)

ggplot(temp, aes(x = area, y = dfits)) +
  geom_point() +
  geom_line()
```



It appears that point 60 has the largest (positive) `dffits` value. Recall that point 60 seemed influential on several predictors and the intercept term. Point 7 has the smallest (or largest negative) `dffits`, and also appears to have been influential on several predictors and the intercept.

```
which.max(temp$dffits)
```

```
[1] 60
```

```
which.min(temp$dffits)
```

```
[1] 7
```

14.7 Model Validation and Correcting for Optimism

In 431, we learned about splitting our regression models into **training** samples and **test** samples, performing variable selection work on the training sample to identify two or three candidate models (perhaps via a stepwise approach), and then comparing the predictions made by those models in a test sample.

At the final project presentations, I mentioned (to many folks) that there was a way to automate this process a bit in 432, that would provide some ways to get the machine to split the data for you multiple times, and then average over the results, using a bootstrap approach. This is it.

The `validate` function allows us to perform cross-validation of our models for some summary statistics (and then correct those statistics for optimism in describing likely predictive accuracy) in an easy way.

`validate` develops:

- Resampling validation with or without backward elimination of variables
- Estimates of the *optimism* in measures of predictive accuracy
- Estimates of the intercept and slope of a calibration model

$$(\text{observed } y) = \text{Intercept} + \text{Slope}(\text{predicted } y)$$

with the following code...

```
set.seed(432002); validate(newmod, method = "boot", B = 40)
```

	index.orig	training	test	optimism	index.corrected	Lower
R-square	0.6989	0.7589	0.6176	0.1413	0.5576	0.3381
MSE	1145.7800	899.5893	1454.9828	-555.3935	1701.1735	1003.4277
g	58.9614	60.5120	56.0360	4.4759	54.4855	38.1532
Intercept	0.0000	0.0000	72.3650	-72.3650	72.3650	-130.9111
Slope	1.0000	1.0000	0.9217	0.0783	0.9217	0.6599
	Upper	n				
R-square	0.7351	40				
MSE	2441.6337	40				
g	69.9747	40				
Intercept	314.3001	40				
Slope	1.1392	40				

So, for **R-square** we see that our original estimate was 0.6989

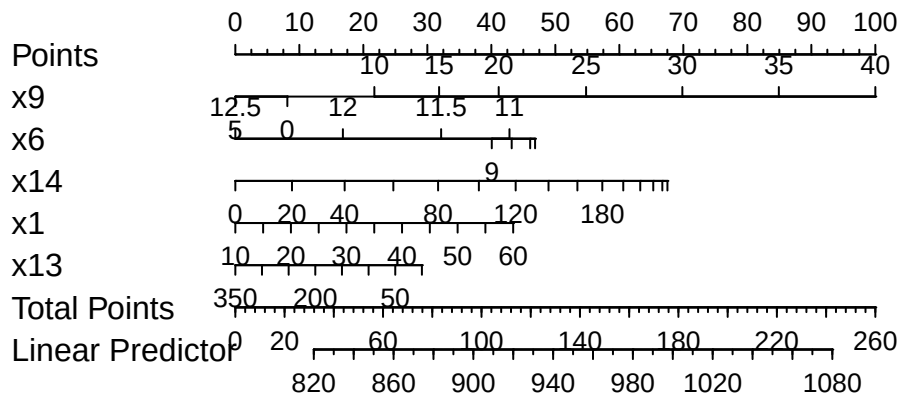
- Our estimated **R-square** across **n = 40** training samples was 0.7589, but in the resulting tests, the average **R-square** was only 0.6176
- This suggests an optimism of $0.7589 - 0.6176 = 0.1413$.
- We then apply that optimism to obtain a new estimate of R^2 corrected for overfitting, at 0.5576, which is probably a better estimate of what our results might look like in new data that were similar to (but not the same as) the data we used in building `newmod` than our initial estimate of 0.6989

We also obtain optimism-corrected estimates of the mean squared error (square of the residual standard deviation), the g index, and the intercept and slope of the calibration model. The “corrected” slope is a shrinkage factor that takes overfitting into account.

14.8 Building a Nomogram for Our Model

Another nice feature of an `ols` model object is that we can picture the model with a **nomogram** easily. Here is model `newmod`.

```
plot(nomogram(newmod))
```



For this model, we can use this plot to predict y as follows:

1. find our values of x_9 on the appropriate line
2. draw a vertical line up to the points line to count the points associated with our subject
3. repeat the process to obtain the points associated with x_6 , x_{14} , x_1 , and x_{13} . Sum the points.
4. draw a vertical line down from that number in the Total Points line to estimate y (the Linear Predictor) = Age-Adjusted Mortality Rate.

The impact of the non-linearity is seen in the x_6 results, for example, which turn around from 9-10 to 11-12. We also see non-linearity's effects in the scales of the non-linear terms in terms of points awarded.

An area with a combination of predictor values leading to a total of 100 points, for instance, would lead to a prediction of a Mortality Rate near 905. An area with a total of 140 points would have a predicted Mortality Rate of 955, roughly.

15 A Model for Prostate Cancer

15.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

15.1.1 Data Load

```
prost <- read_csv("data/prost.csv", show_col_types = FALSE)
```

15.2 Data Load and Background

The data in `prost.csv` is derived from @Stamey1989 who examined the relationship between the level of prostate-specific antigen and a number of clinical measures in 97 men who were about to receive a radical prostatectomy. The `prost` data, as I'll name it in R, contains 97 rows and 11 columns.

```
prost
```

```
# A tibble: 97 x 10
```

	subject	lpsa	lcavol	lweight	age	bph	svi	lcp	gleason	pgg45
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<chr>	<dbl>
1	1	-0.431	-0.580	2.77	50	Low	0	-1.39	6	0
2	2	-0.163	-0.994	3.32	58	Low	0	-1.39	6	0
3	3	-0.163	-0.511	2.69	74	Low	0	-1.39	7	20

```

4      4 -0.163 -1.20      3.28    58 Low      0 -1.39 6      0
5      5  0.372  0.751      3.43    62 Low      0 -1.39 6      0
6      6  0.765 -1.05      3.23    50 Low      0 -1.39 6      0
7      7  0.765  0.737      3.47    64 Medium    0 -1.39 6      0
8      8  0.854  0.693      3.54    58 High      0 -1.39 6      0
9      9  1.05  -0.777      3.54    47 Low      0 -1.39 6      0
10     10  1.05   0.223      3.24    63 Low      0 -1.39 6      0
# i 87 more rows

```

Note that a related `prost` data frame is also available as part of several R packages, including the `faraway` package, but there is an error in the `lweight` data for subject 32 in those presentations. The value of `lweight` for subject 32 should not be 6.1, corresponding to a prostate that is 449 grams in size, but instead the `lweight` value should be 3.804438, corresponding to a 44.9 gram prostate¹.

I've also changed the `gleason` and `bph` variables from their presentation in other settings, to let me teach some additional details.

15.3 Code Book

Variable	Description
<code>subject</code>	subject number (1 to 97)
<code>lpsa</code>	log(prostate specific antigen in ng/ml), our outcome
<code>lcavol</code>	log(cancer volume in cm ³)
<code>lweight</code>	log(prostate weight, in g)
<code>age</code>	age
<code>bph</code>	benign prostatic hyperplasia amount (Low, Medium, or High)
<code>svi</code>	seminal vesicle invasion (1 = yes, 0 = no)
<code>lcp</code>	log(capsular penetration, in cm)
<code>gleason</code>	combined Gleason score (6, 7, or > 7 here)
<code>pgg45</code>	percentage Gleason scores 4 or 5

Notes:

- in general, higher levels of PSA are stronger indicators of prostate cancer. An old standard (established almost exclusively with testing in white males, and definitely flawed) suggested that values below 4 were normal, and above 4 needed further testing. A PSA of 4 corresponds to an `lpsa` of 1.39.
- all logarithms are natural (base e) logarithms, obtained in R with the function `log()`

¹<https://statweb.stanford.edu/~tibs/ElemStatLearn/> attributes the correction to Professor Stephen W. Link.

- all variables other than `subject` and `lpsa` are candidate predictors
- the `gleason` variable captures the highest combined Gleason score [^Scores range (in these data) from 6 (a well-differentiated, or low-grade cancer) to 9 (a high-grade cancer), although the maximum possible score is 10. 6 is the lowest score used for cancerous prostates. As this combination value increases, the rate at which the cancer grows and spreads should increase. This score refers to the combined Gleason grade, which is based on the sum of two areas (each scored 1-5) that make up most of the cancer.] in a biopsy, and higher scores indicate more aggressive cancer cells. It's stored here as 6, 7, or > 7.
- the `pgg45` variable captures the percentage of individual Gleason scores [^The 1-5 scale for individual biopsies are defined so that 1 indicates something that looks like normal prostate tissue, and 5 indicates that the cells and their growth patterns look very abnormal. In this study, the percentage of 4s and 5s shown in the data appears to be based on 5-20 individual scores in most subjects.] that are 4 or 5, on a 1-5 scale, where higher scores indicate more abnormal cells.

15.4 Additions for Later Use

The code below adds to the `prost` tibble:

- a factor version of the `svi` variable, called `svi_f`, with levels No and Yes,
- a factor version of `gleason` called `gleason_f`, with the levels ordered > 7, 7, and finally 6,
- a factor version of `bph` called `bph_f`, with levels ordered Low, Medium, High,
- a centered version of `lcavol` called `lcavol_c`,
- exponentiated `cavol` and `psa` results derived from the natural logarithms `lcavol` and `lpsa`.

```
prost <- prost |>
  mutate(svi_f = fct_recode(factor(svi), "No" = "0", "Yes" = "1"),
         gleason_f = fct_relevel(gleason, c("> 7", "7", "6")),
         bph_f = fct_relevel(bph, c("Low", "Medium", "High")),
         lcavol_c = lcavol - mean(lcavol),
         cavol = exp(lcavol),
         psa = exp(lpsa))

glimpse(prost)
```

Rows: 97

Columns: 16

```
$ subject <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1~
$ lpsa    <dbl> -0.4307829, -0.1625189, -0.1625189, -0.1625189, 0.3715636, 0~
```



```

$ lcavol      <dbl> -0.5798185, -0.9942523, -0.5108256, -1.2039728, 0.7514161, -
~
$ lweight     <dbl> 2.769459, 3.319626, 2.691243, 3.282789, 3.432373, 3.228826, ~
$ age         <dbl> 50, 58, 74, 58, 62, 50, 64, 58, 47, 63, 65, 63, 63, 67, 57, ~
$ bph         <chr> "Low", "Low", "Low", "Low", "Low", "Low", "Low", "Medium", "High", ~
$ svi         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ lcp         <dbl> -1.3862944, -1.3862944, -1.3862944, -1.3862944, -1.3862944, ~
$ gleason     <chr> "6", "6", "7", "6", "6", "6", "6", "6", "6", "6", "6", "6", ~
$ pgg45       <dbl> 0, 0, 20, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 30, 5, 5, 0, 30, 0, ~
$ svi_f       <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No, No, ~
$ gleason_f   <fct> 6, 6, 7, 6, 6, 6, 6, 6, 6, 6, 6, 6, 7, 7, 7, 6, 7, 6, 6, ~
$ bph_f       <fct> Low, Low, Low, Low, Low, Low, Low, Medium, High, Low, Low, Low, M~
$ lcavol_c    <dbl> -1.9298281, -2.3442619, -1.8608352, -2.5539824, -0.5985935, ~
$ cavol       <dbl> 0.56, 0.37, 0.60, 0.30, 2.12, 0.35, 2.09, 2.00, 0.46, 1.25, ~
$ psa         <dbl> 0.65, 0.85, 0.85, 0.85, 1.45, 2.15, 2.15, 2.35, 2.85, 2.85, ~

```

15.5 Fitting and Evaluating a Two-Predictor Model

To begin, let's use two predictors (`lcavol` and `svi`) and their interaction in a linear regression model that predicts `lpsa`. I'll call this model `prost_A`

Earlier, we centered the `lcavol` values to facilitate interpretation of the terms. I'll use that centered version (called `lcavol_c`) of the quantitative predictor, and the 1/0 version of the `svi` variable¹. We could certainly use the factor version of `svi` here, but it won't change the model in any meaningful way. There's no distinction in model *fitting* via `lm` between a 0/1 numeric variable and a No/Yes factor variable. The factor version of this information will be useful elsewhere, for instance in plotting the model.

```

prost_A <- lm(lpsa ~ lcavol_c * svi, data = prost)
summary(prost_A)

```

Call:

```
lm(formula = lpsa ~ lcavol_c * svi, data = prost)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.6305	-0.5007	0.1266	0.4886	1.6847

Coefficients:

```
Estimate Std. Error t value Pr(>|t|)
```

```

(Intercept)    2.33134    0.09128   25.540 < 2e-16 ***
lcavol_c       0.58640    0.08207    7.145 1.98e-10 ***
svi            0.60132    0.35833    1.678 0.0967 .
lcavol_c:svi   0.06479    0.26614    0.243 0.8082
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7595 on 93 degrees of freedom
Multiple R-squared:  0.5806,    Adjusted R-squared:  0.5671
F-statistic: 42.92 on 3 and 93 DF,  p-value: < 2.2e-16

```

15.5.1 Using tidy

It can be very useful to build a data frame of the model's results. We can use the `tidy` function in the `broom` package to do so.

```
tidy(prost_A)
```

```

# A tibble: 4 x 5
  term          estimate std.error statistic  p.value
  <chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    2.33      0.0913     25.5 8.25e-44
2 lcavol_c       0.586     0.0821      7.15 1.98e-10
3 svi           0.601     0.358      1.68 9.67e- 2
4 lcavol_c:svi   0.0648    0.266      0.243 8.08e- 1

```

This makes it much easier to pull out individual elements of the model fit.

For example, to specify the coefficient for `svi`, rounded to three decimal places, I could use

```
tidy(prost_A) |> filter(term == "svi") |> select(estimate) |> round(3)
```

- The result is 0.601.
- If you look at the Markdown file, you'll see that the number shown in the bullet point above this one was generated using inline R code, and the function specified above.

15.5.2 Interpretation

1. The intercept, 2.33, for the model is the predicted value of `lpsa` when `lcavol` is at its average and there is no seminal vesicle invasion (e.g. `svi` = 0).
2. The coefficient for `lcavol_c`, 0.59, is the predicted change in `lpsa` associated with a one unit increase in `lcavol` (or `lcavol_c`) when there is no seminal vesicle invasion.
3. The coefficient for `svi`, 0.6, is the predicted change in `lpsa` associated with having no `svi` to having an `svi` while the `lcavol` remains at its average.
4. The coefficient for `lcavol_c:svi`, the product term, which is 0.06, is the difference in the slope of `lcavol_c` for a subject with `svi` as compared to one with no `svi`.

15.6 Exploring Model `prost_A`

The `glance` function from the `broom` package builds a nice one-row summary for the model.

```
glance(prost_A)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
  <dbl>      <dbl> <dbl>    <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1    0.581      0.567 0.759     42.9 1.68e-17     3  -109.  228.  241.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

This summary includes, in order,

- the model R^2 , adjusted R^2 and $\hat{\sigma}$, the residual standard deviation,
- the ANOVA F statistic and associated p value,
- the number of degrees of freedom used by the model, and its log-likelihood ratio
- the model's AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion)
- the model's deviance statistic and residual degrees of freedom

15.6.1 summary for Model `prost_A`

If necessary, we can also run `summary` on this `prost_A` object to pick up some additional summaries. Since the `svi` variable is binary, the interaction term is, too, so the t test here and the F test in the ANOVA yield the same result.

```
summary(prost_A)
```

Call:

```
lm(formula = lpsa ~ lcavol_c * svi, data = prost)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.6305	-0.5007	0.1266	0.4886	1.6847

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.33134	0.09128	25.540	< 2e-16 ***
lcavol_c	0.58640	0.08207	7.145	1.98e-10 ***
svi	0.60132	0.35833	1.678	0.0967 .
lcavol_c:svi	0.06479	0.26614	0.243	0.8082

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7595 on 93 degrees of freedom

Multiple R-squared: 0.5806, Adjusted R-squared: 0.5671

F-statistic: 42.92 on 3 and 93 DF, p-value: < 2.2e-16

If you've forgotten the details of the pieces of this summary, review the Part C Notes from 431.

15.6.2 Adjusted R^2

R^2 is greedy.

- R^2 will always suggest that we make our models as big as possible, often including variables of dubious predictive value.
- As a result, there are various methods for penalizing R^2 so that we wind up with smaller models.
- The **adjusted** R^2 is often a useful way to compare multiple models for the same response.
 - $R_{adj}^2 = 1 - \frac{(1-R^2)(n-1)}{n-k}$, where n = the number of observations and k is the number of coefficients estimated by the regression (including the intercept and any slopes).
 - So, in this case, $R_{adj}^2 = 1 - \frac{(1-0.5806)(97-1)}{97-4} = 0.5671$
 - The adjusted R^2 value is not, technically, a proportion of anything, but it is comparable across models for the same outcome.
 - The adjusted R^2 will always be less than the (unadjusted) R^2 .

15.6.3 Coefficient Confidence Intervals

Here are the 90% confidence intervals for the coefficients in Model A. Adjust the `level` to get different intervals.

```
confint(prost_A, level = 0.90)
```

	5 %	95 %
(Intercept)	2.17968697	2.4830012
lcavol_c	0.45004577	0.7227462
svi	0.00599401	1.1966454
lcavol_c:svi	-0.37737623	0.5069622

What can we conclude from this about the utility of the interaction term?

15.6.4 ANOVA for Model `prost_A`

The interaction term appears unnecessary. We might wind up fitting the model without it. A complete ANOVA test is available, including a p value, if you want it.

```
anova(prost_A)
```

Analysis of Variance Table

Response: `lpsa`

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
lcavol_c	1	69.003	69.003	119.6289	< 2.2e-16 ***
svi	1	5.237	5.237	9.0801	0.003329 **
lcavol_c:svi	1	0.034	0.034	0.0593	0.808191
Residuals	93	53.643	0.577		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note that the `anova` approach for a `lm` object is sequential. The first row shows the impact of `lcavol_c` as compared to a model with no predictors (just an intercept). The second row shows the impact of adding `svi` to a model that already contains `lcavol_c`. The third row shows the impact of adding the interaction (product) term to the model with the two main effects. So the order in which the variables are added to the regression model matters for this ANOVA. The F tests here describe the incremental impact of each covariate in turn.

15.6.5 Residuals, Fitted Values and Standard Errors with augment

The `augment` function in the `broom` package builds a data frame including the data used in the model, along with predictions (fitted values), residuals and other useful information.

```
prost_A_aug <- augment(prost_A)
summary(prost_A_aug)
```

lpsa	lcavol_c	svi	.fitted
Min. : -0.4308	Min. : -2.69708	Min. : 0.0000	Min. : 0.7498
1st Qu.: 1.7317	1st Qu.: -0.83719	1st Qu.: 0.0000	1st Qu.: 1.8404
Median : 2.5915	Median : 0.09691	Median : 0.0000	Median : 2.3950
Mean : 2.4784	Mean : 0.00000	Mean : 0.2165	Mean : 2.4784
3rd Qu.: 3.0564	3rd Qu.: 0.77703	3rd Qu.: 0.0000	3rd Qu.: 3.0709
Max. : 5.5829	Max. : 2.47099	Max. : 1.0000	Max. : 4.5417
.resid	.hat	.sigma	.cooks
Min. : -1.6305	Min. : 0.01316	Min. : 0.7423	Min. : 6.899e-06
1st Qu.: -0.5007	1st Qu.: 0.01562	1st Qu.: 0.7569	1st Qu.: 7.837e-04
Median : 0.1266	Median : 0.02498	Median : 0.7617	Median : 3.470e-03
Mean : 0.0000	Mean : 0.04124	Mean : 0.7595	Mean : 1.113e-02
3rd Qu.: 0.4886	3rd Qu.: 0.04939	3rd Qu.: 0.7631	3rd Qu.: 1.035e-02
Max. : 1.6847	Max. : 0.24627	Max. : 0.7636	Max. : 1.341e-01
.std.resid			
Min. : -2.194508			
1st Qu.: -0.687945			
Median : 0.168917			
Mean : 0.001249			
3rd Qu.: 0.653611			
Max. : 2.261830			

Elements shown here include:

- `.fitted` Fitted values of model (or predicted values)
- `.se.fit` Standard errors of fitted values
- `.resid` Residuals (observed - fitted values)
- `.hat` Diagonal of the hat matrix (these indicate *leverage* - points with high leverage indicate unusual combinations of predictors - values more than 2-3 times the mean leverage are worth some study - leverage is always between 0 and 1, and measures the amount by which the predicted value would change if the observation's y value was increased by one unit - a point with leverage 1 would cause the line to follow that point perfectly)
- `.sigma` Estimate of residual standard deviation when corresponding observation is dropped from model

- `.cooks` Cook's distance, which helps identify influential points (values of Cook's $d > 0.5$ may be influential, values > 1.0 almost certainly are - an influential point changes the fit substantially when it is removed from the data)
- `.std.resid` Standardized residuals (values above 2 in absolute value are worth some study - treat these as normal deviates [Z scores], essentially)

See `?augment.lm` in R for more details.

15.6.6 Making Predictions with `prost_A`

Suppose we want to predict the `lpsa` for a patient with cancer volume equal to this group's mean, for both a patient with and without seminal vesicle invasion, and in each case, we want to use a 90% prediction interval?

```
newdata <- data.frame(lcavol_c = c(0,0), svi = c(0,1))
predict(prost_A, newdata, interval = "prediction", level = 0.90)
```

	fit	lwr	upr
1	2.331344	1.060462	3.602226
2	2.932664	1.545742	4.319586

Since the predicted value in `fit` refers to the natural logarithm of PSA, to make the predictions in terms of PSA, we would need to exponentiate. The code below will accomplish that task.

```
pred <- predict(prost_A, newdata, interval = "prediction", level = 0.90)
exp(pred)
```

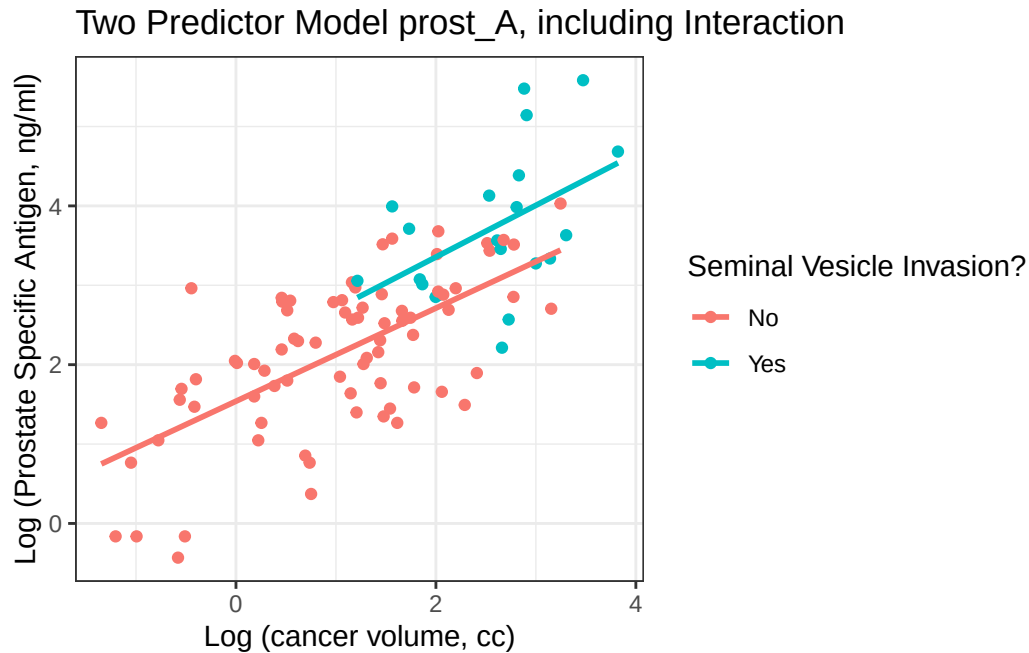
	fit	lwr	upr
1	10.29177	2.887706	36.67978
2	18.77758	4.691450	75.15750

15.7 Plotting Model `prost_A`

15.7.0.1 Plot logs conventionally

Here, we'll use `ggplot2` to plot the logarithms of the variables as they came to us, on a conventional coordinate scale. Note that the lines are nearly parallel. What does this suggest about our Model A?

```
ggplot(prost, aes(x = lcavol, y = lpsa, group = svi_f, color = svi_f)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  scale_color_discrete(name = "Seminal Vesicle Invasion?") +
  theme_bw() +
  labs(x = "Log (cancer volume, cc)",
       y = "Log (Prostate Specific Antigen, ng/ml)",
       title = "Two Predictor Model prost_A, including Interaction")
```



15.7.0.2 Plot on log-log scale

Another approach (which might be easier in some settings) would be to plot the raw values of Cancer Volume and PSA, but use logarithmic axes, again using the natural (base e) logarithm, as follows. If we use the default choice with `trans = "log"`, we'll find a need to select some useful break points for the grid, as I've done in what follows.

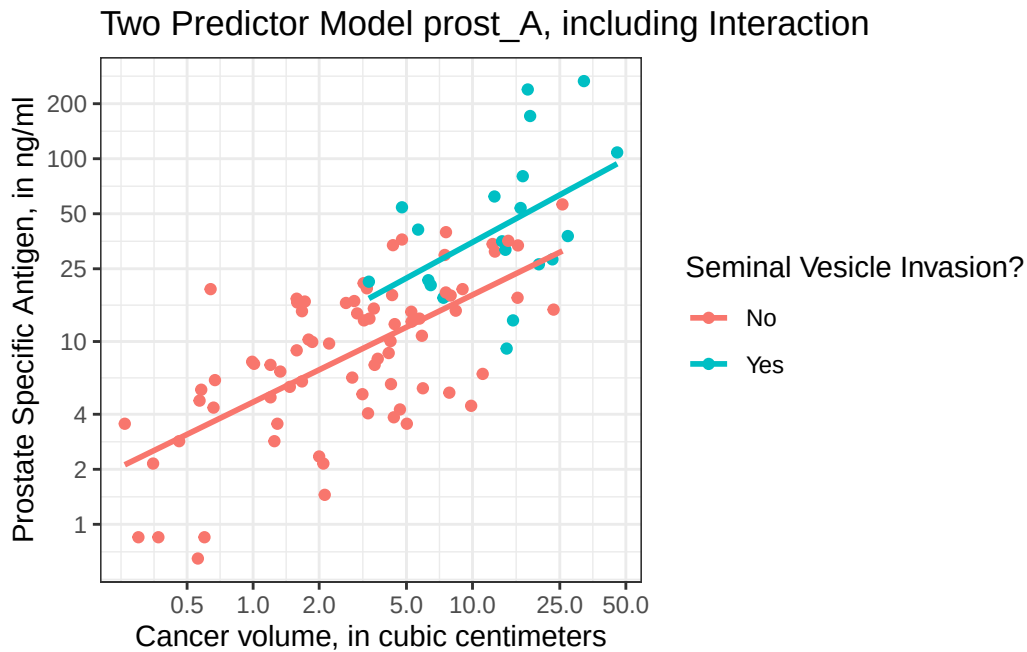
```
ggplot(prost, aes(x = cavol, y = psa, group = svi_f, color = svi_f)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  scale_color_discrete(name = "Seminal Vesicle Invasion?") +
  scale_x_continuous(trans = "log",
```



```

breaks = c(0.5, 1, 2, 5, 10, 25, 50)) +
scale_y_continuous(trans = "log",
breaks = c(1, 2, 4, 10, 25, 50, 100, 200)) +
theme_bw() +
labs(x = "Cancer volume, in cubic centimeters",
y = "Prostate Specific Antigen, in ng/ml",
title = "Two Predictor Model prost_A, including Interaction")

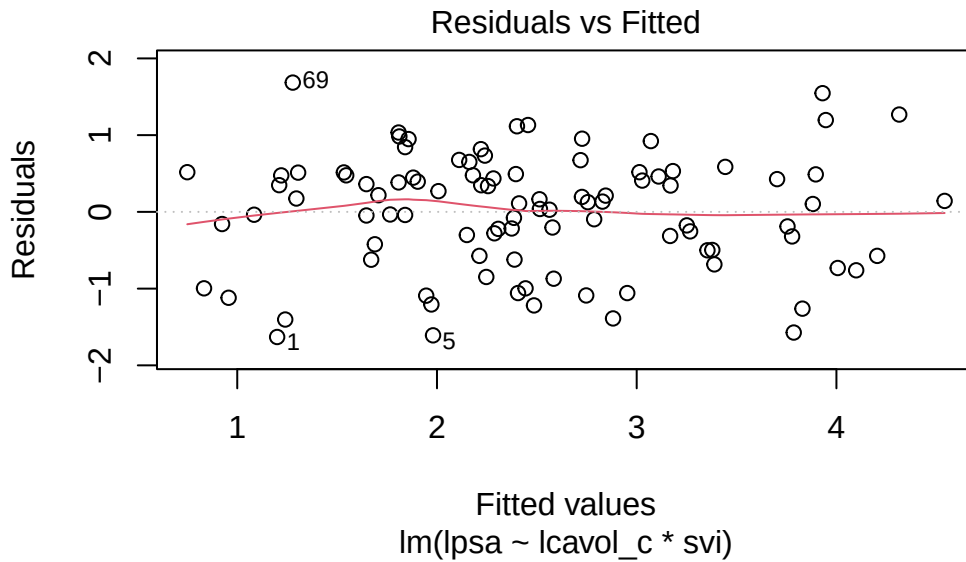
```



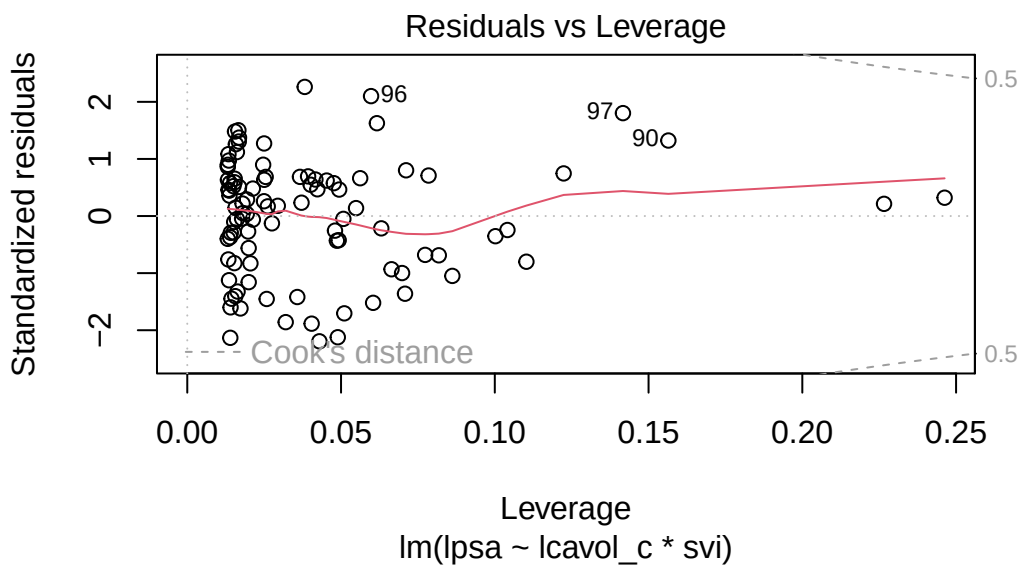
I've used the break point of 4 on the Y axis because of the old rule suggesting further testing for asymptomatic men with PSA of 4 or higher, but the other break points are arbitrary - they seemed to work for me, and used round numbers.

15.7.1 Residual Plots of prost_A

```
plot(prost_A, which = 1)
```



```
plot(prost_A, which = 5)
```



In our next Chapter, we'll see how well this model can be validated.

16 Validating our Prostate Cancer Model

16.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(rsample)
library(yardstick)
library(caret)
library(tidyverse)

theme_set(theme_bw())
```

16.1.1 Data Load

```
prost <- read_csv("data/prost.csv", show_col_types = FALSE)
```

We'll repeat the data cleaning and model-fitting from our previous chapter.

16.2 Data Cleaning

```
prost <- prost |>
  mutate(svi_f = fct_recode(factor(svi), "No" = "0", "Yes" = "1"),
         gleason_f = fct_relevel(gleason, c("> 7", "7", "6")),
         bph_f = fct_relevel(bph, c("Low", "Medium", "High")),
         lcavol_c = lcavol - mean(lcavol),
         cavol = exp(lcavol),
         psa = exp(lpsa))
```

16.3 Fitting the `prostA` model

```
prost_A <- lm(lpsa ~ lcavol_c * svi, data = prost)
```

16.4 Split Validation of Model `prost_A`

Suppose we want to evaluate whether our model `prost_A` predicts effectively in new data.

We'll first demonstrate a validation split approach (used, for instance, in 431) which splits our sample into a separate training (perhaps 70% of the data) and test (perhaps 30% of the data) samples, and then:

- fit the model in the training sample,
- use the resulting model to make predictions for `lpsa` in the test sample, and
- evaluate the quality of those predictions, perhaps by comparing the results to what we'd get using a different model.

Our goal will be to cross-validate model `prost_A`, which, you'll recall, uses `lcavol_c`, `svi` and their interaction, to predict `lpsa` in the `prost` data.

We'll start by identifying a random sample of 70% of our `prost` data in a training sample (which we'll call `prost_train`, and leave the rest as our test sample, called `prost_test`. To do this, we'll use functions from the `rsample` package.

```
set.seed(432432)

prost_split <- initial_split(prost, prop = 0.7)

prost_train <- training(prost_split)
prost_test <- testing(prost_split)
```

- Don't forget to pre-specify the random seed, for replicability, as I've done here.

Let's verify that we now have the samples we expect...

```
dim(prost_train)
```

```
[1] 67 16
```

```
dim(prost_test)
```

```
[1] 30 16
```

OK. Next, we'll run the `prost_A` model in the training sample.

```
prost_A_train <- lm(lpsa ~ lcavol_c * svi, data = prost_train)

prost_A_train
```

Call:

```
lm(formula = lpsa ~ lcavol_c * svi, data = prost_train)
```

Coefficients:

(Intercept)	lcavol_c	svi	lcavol_c:svi
2.2900	0.6922	1.1317	-0.4269

Then we'll use the coefficients from this model to obtain predicted `lpsa` values in the test sample.

```
prost_A_test_aug <- augment(prost_A, newdata = prost_test)
```

Now, we can use the functions from the `yardstick` package to obtain several key summaries of fit quality for our model. These summary statistics are:

- the RMSE or root mean squared error, which measures the average difference (i.e. prediction error) between the observed known outcome values and the values predicted by the model by first squaring all of the errors, averaging them, and then taking the square root of the result. The lower the RMSE, the better the model.
- the Rsquared or R^2 , which is just the square of the Pearson correlation coefficient relating the predicted and observed values, so we'd like this to be as large as possible, and
- the MAE or mean absolute error, which is a bit less sensitive to outliers than the RMSE, because it measures the average prediction error by taking the absolute value of each error, and then grabbing the average of those values. The lower the MAE, the better the model.

These statistics are more helpful, generally, for comparing multiple models to each other, than for making final decisions on their own. The `yardstick` package provides individual functions to summarize performance, as follows.

```
rmse(data = prost_A_test_aug, truth = lpsa, estimate = .fitted)
```

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>       <dbl>
1 rmse    standard      0.813
```

```
rsq(data = prost_A_test_aug, truth = lpsa, estimate = .fitted)
```

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>       <dbl>
1 rsq     standard      0.515
```

```
mae(data = prost_A_test_aug, truth = lpsa, estimate = .fitted)
```

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>       <dbl>
1 mae     standard      0.672
```

16.5 V-fold Cross-Validation Approach for model prostA

V-fold cross-validation (also known as k-fold cross-validation) randomly splits the data into V groups of roughly equal size (called “folds”). A resample of the analysis data consists of V-1 of the folds while the assessment set contains the final fold. In basic V-fold cross-validation (i.e. no repeats), the number of resamples is equal to V.

- https://rsample.tidymodels.org/reference/vfold_cv.html

The idea of, for instance, 5-fold cross-validation in this case is to create five different subgroups (or folds) of the data, and then select 4 of the folds to be used as a model training sample, leaving the remaining fold as the model testing sample. We then repeat this over each of the five possible selections of testing sample, and summarize the results. This is very straightforward using the **caret** package, so we’ll demonstrate that approach here.

First, we use the `trainControl()` function from **caret** to set up five-fold cross-validation.

```
set.seed(432432)
ctrl <- trainControl(method = "cv", number = 5)
```

Next, we train our model on these five folds:

```
pros_model <- train(lpsa ~ lcavol_c * svi, data = prost,
                    method = "lm", trControl = ctrl)
```

Now, we can view a summary of the k-fold cross-validation

```
pros_model
```

Linear Regression

97 samples
2 predictor

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 79, 78, 77, 77, 77
Resampling results:

RMSE	Rsquared	MAE
0.7777655	0.5946201	0.6411997

Tuning parameter 'intercept' was held constant at a value of TRUE

- No pre-processing means we didn't scale the data before fitting models.
- We used 5-fold cross-validation
- The sample size for these training sets was between 77 and 79 for each pass.
- The validated root mean squared error (averaged across the five resamplings) was 0.7778
- The cross-validated R-squared is 0.595
- The cross-validated mean absolute error is 0.641

To examine the final fitted model, we have:

```
pros_model$finalModel
```

Call:
lm(formula = .outcome ~ ., data = dat)

```

Coefficients:
  (Intercept)      lcavol_c          svi  `lcavol_c:svi`
      2.33134        0.58640        0.60132        0.06479

```

This model can be presented using all of our usual tools from the **broom** package.

```
tidy(pros_model$finalModel)
```

```

# A tibble: 4 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    2.33      0.0913    25.5 8.25e-44
2 lcavol_c       0.586     0.0821     7.15 1.98e-10
3 svi            0.601     0.358     1.68 9.67e- 2
4 `lcavol_c:svi` 0.0648    0.266     0.243 8.08e- 1

```

```
glance(pros_model$finalModel)
```

```

# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.581      0.567 0.759    42.9 1.68e-17     3 -109.  228.  241.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

```

We can also review the model predictions made within each fold:

```
pros_model$resample
```

```

      RMSE Rsquared      MAE Resample
1 0.7447413 0.7030717 0.6124787 Fold1
2 0.8208461 0.3071658 0.6699472 Fold2
3 0.6510721 0.7944721 0.5589196 Fold3
4 0.8514246 0.4731454 0.6885641 Fold4
5 0.8207437 0.6952452 0.6760887 Fold5

```


17 Multiple Imputation and Linear Regression

17.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(car)
library(knitr)
library(mosaic)
library(mice)
library(rms)
library(naniar)
library(tidyverse)

theme_set(theme_bw())
```

17.2 Data Load

In this chapter, we'll return to the `smart_ohio` file based on data from BRFSS 2017 that we built and cleaned back in Chapter 6.

```
smart_ohio <- readRDS("data/smart_ohio.Rds")
```

17.3 Developing a `smart_16` data set

We're going to look at a selection of variables from this tibble, among subjects who have been told they have diabetes, and who also provided a response to our `physhealth` (Number of Days Physical Health Not Good) variable, which asks "Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?" We'll build two models. In this chapter, we'll look at a linear

model for `physhealth` and in the next chapter, we'll look at a logistic regression describing whether or not the subject's `physhealth` response was at least 1.

```
smart_16 <- smart_ohio |>
  filter(dm_status == "Diabetes") |>
  filter(complete.cases(physhealth)) |>
  mutate(bad_phys = ifelse(physhealth > 0, 1, 0),
         comor = hx_mi + hx_chd + hx_stroke + hx_asthma +
                 hx_skinc + hx_otherc + hx_copd + hx_arthr) |>
  select(SEQNO, mmsa, physhealth, bad_phys, age_imp, smoke100,
         comor, hx_depress, bmi, activity)
```

The variables included in this `smart_16` tibble are:

Variable	Description
SEQNO	respondent identification number (all begin with 2016)
mmsa	
physhealth	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?
bad_phys	Is <code>physhealth</code> 1 or more?
age_imp	Age in years (imputed from age categories)
smoke100	Have you smoked at least 100 cigarettes in your life? (1 = yes, 0 = no)
hx_depress	Has a doctor, nurse, or other health professional ever told you that you have a depressive disorder, including depression, major depression, dysthymia, or minor depression?
bmi	Body mass index, in kg/m ²
activity	Physical activity (Highly Active, Active, Insufficiently Active, Inactive)
comor	Sum of 8 potential groups of comorbidities (see below)

The `comor` variable is the sum of the following 8 variables, each of which is measured on a 1 = Yes, 0 = No scale, and begin with “Has a doctor, nurse, or other health professional ever told you that you had ...”

- `hx_mi`: a heart attack, also called a myocardial infarction?
- `hx_chd`: angina or coronary heart disease?
- `hx_stroke`: a stroke?
- `hx_asthma`: asthma?
- `hx_skinc`: skin cancer?
- `hx_otherc`: any other types of cancer?
- `hx_copd`: Chronic Obstructive Pulmonary Disease or COPD, emphysema or chronic bronchitis?

- `hx_arthr`: some form of arthritis, rheumatoid arthritis, gout, lupus, or fibromyalgia?

```
smart_16 |> tabyl(comor)
```

comor	n	percent	valid_percent
0	224	0.211920530	0.221782178
1	315	0.298013245	0.311881188
2	228	0.215704825	0.225742574
3	130	0.122989593	0.128712871
4	72	0.068117313	0.071287129
5	29	0.027436140	0.028712871
6	9	0.008514664	0.008910891
7	3	0.002838221	0.002970297
NA	47	0.044465468	NA

17.3.1 Any missing values?

We have 1057 observations (rows) in the `smart_16` data set, of whom 860 have complete data on all variables.

```
dim(smart_16)
```

```
[1] 1057  10
```

```
n_case_complete(smart_16)
```

```
[1] 860
```

Which variables are missing?

```
miss_var_summary(smart_16)
```

```
# A tibble: 10 x 3
  variable  n_miss pct_miss
  <chr>      <int>   <num>
1 activity     85    8.04
2 bmi         84    7.95
3 comor        47    4.45
4 smoke100     24    2.27
```

5	age_imp	12	1.14
6	hx_depress	3	0.284
7	SEQNO	0	0
8	mmsa	0	0
9	physhealth	0	0
10	bad_phys	0	0

Note that our outcomes (`physhealth` and the derived `bad_phys`) have no missing values here, by design. We will be performing multiple imputation to account appropriately for missingness in the predictors with missing values.

17.4 Obtaining a Simple Imputation with `mice`

The `mice` package provides several approaches we can use for imputation in building models of all kinds. Here, we'll use it just to obtain a single set of imputed results that we can apply to “complete” our data for the purposes of thinking about (a) transforming our outcome and (b) considering the addition of non-linear predictor terms.

```
# requires library(mice)

set.seed(432)

# create small data set including only variables to
# be used in building the imputation model

sm16 <- smart_16 |>
  select(physhealth, activity, age_imp, bmi, comor,
         hx_depress, smoke100)

smart_16_mice1 <- mice(sm16, m = 1)
```

iter	imp	variable						
1	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
5	1	activity	age_imp	bmi	comor	hx_depress	smoke100	

```
smart_16_imp1 <- mice::complete(smart_16_mice1)

n_case_miss(smart_16_imp1)
```

```
[1] 0
```

And now we'll use this completed `smart_16_imp1` data set (the product of just a single imputation) to help us address the next two issues.

17.5 Linear Regression: Considering a Transformation of the Outcome

A plausible strategy here would be to try to identify an outcome transformation only after some accounting for missing predictor values, perhaps through a simple imputation approach. However, to keep things simple here, I'll just use the complete cases in this section.

Recall that our outcome here, `physhealth` can take the value 0, and is thus not strictly positive.

```
favstats(~ physhealth, data = smart_16_imp1)
```

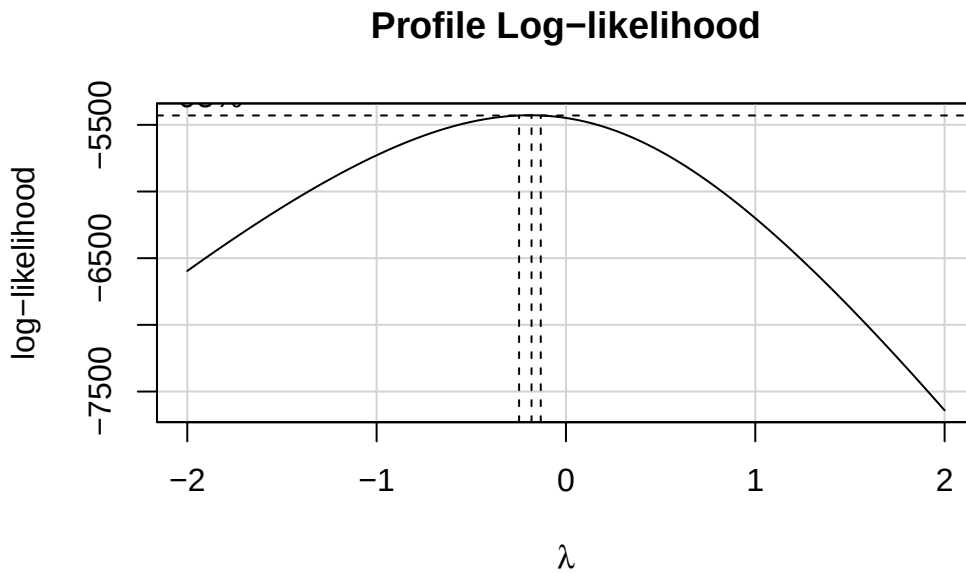
min	Q1	median	Q3	max	mean	sd	n	missing
0	0	2	20	30	9.227058	11.92676	1057	0

So, if we want to investigate a potential transformation with a Box-Cox plot, we'll have to add a small value to each `physhealth` value. We'll add 1, so that the range of potential values is now from 1-31.

```
smart_16_imp1 <- smart_16_imp1 |>
  mutate(phplus1 = physhealth + 1)

test_model <- lm(phplus1 ~ age_imp + comor + smoke100 +
  hx_depress + bmi + activity, data = smart_16_imp1)

boxCox(test_model)
```



It looks like the logarithm is a reasonable transformation in this setting. So we'll create a new outcome, that is the natural logarithm of $(\text{physhealth} + 1)$, which we'll call **phys_tr** to remind us that a transformation is involved that we'll eventually need to back out of to make predictions. We'll build this new variable in both our original **smart_16** data set and in the simply imputed data set we're using for just these early stages.

```
smart_16_imp1 <- smart_16_imp1 |>
  mutate(phys_tr = log(physhealth + 1))

smart_16 <- smart_16 |>
  mutate(phys_tr = log(physhealth + 1))
```

So we have $\text{phys_tr} = \log(\text{physhealth} + 1)$

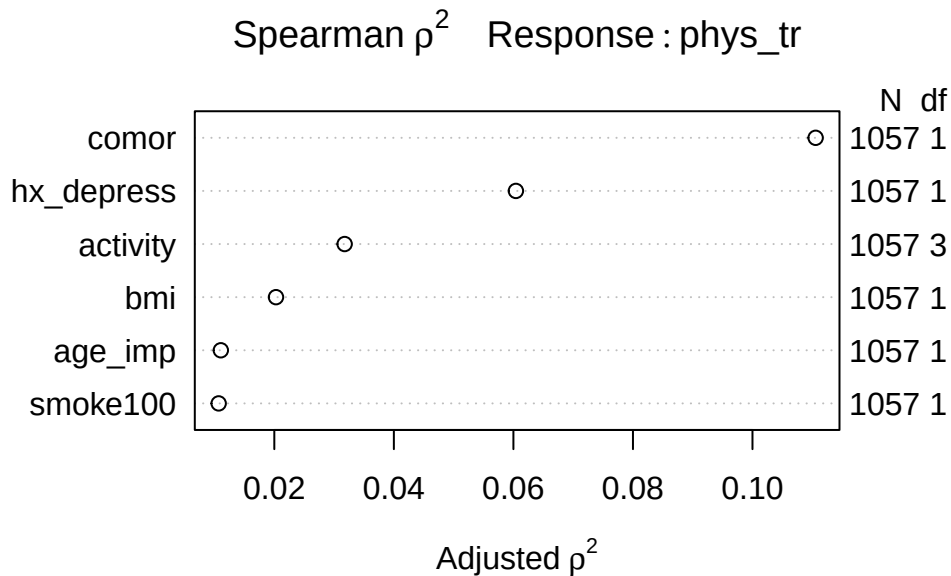
- where we are referring above to the natural (base e logarithm).

We can also specify our back-transformation to the original **physhealth** values from our new **phys_tr** as $\text{physhealth} = \exp(\text{phys_tr}) - 1$.

17.6 Linear Regression: Considering Non-Linearity in the Predictors

Consider the following Spearman ρ^2 plot.

```
plot(spearman2(phys_tr ~ age_imp + comor + smoke100 +
  hx_depress + bmi + activity, data = smart_16_imp1))
```



After our single imputation, we have the same N value in all rows of this plot, which is what we want to see. It appears that in considering potential non-linear terms, `comor` and `hx_depress` and perhaps `activity` are worthy of increased attention. I'll make a couple of arbitrary choices, to add a raw cubic polynomial to represent the `comor` information, and we'll add an interaction term between `hx_depress` and `activity`.

17.7 “Main Effects” Linear Regression with `lm` on the Complete Cases

Recall that we have 860 complete cases in our `smart_16` data, out of a total of 1057 observations in total. A model using only the complete cases should thus drop the remaining 197 subjects. Let's see if a main effects only model for our newly transformed `phys_tr` outcome does in fact do this.

```
m_1cc <-
  lm(phys_tr ~ age_imp + comor + smoke100 +
    hx_depress + bmi + activity, data = smart_16)
```

```
summary(m_1cc)
```

Call:

```
lm(formula = phys_tr ~ age_imp + comor + smoke100 + hx_depress +  
    bmi + activity, data = smart_16)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.0801	-1.0389	-0.2918	1.1029	2.8478

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.581959	0.370847	1.569	0.11696
age_imp	-0.007043	0.003813	-1.847	0.06511 .
comor	0.301773	0.033105	9.116	< 2e-16 ***
smoke100	0.099038	0.090280	1.097	0.27295
hx_depress	0.471949	0.104232	4.528	6.81e-06 ***
bmi	0.016375	0.006295	2.601	0.00945 **
activityActive	-0.229927	0.154912	-1.484	0.13812
activityInsufficiently_Active	-0.116998	0.139440	-0.839	0.40168
activityInactive	0.256118	0.115266	2.222	0.02655 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.303 on 851 degrees of freedom

(197 observations deleted due to missingness)

Multiple R-squared: 0.1806, Adjusted R-squared: 0.1729

F-statistic: 23.45 on 8 and 851 DF, p-value: < 2.2e-16

Note that the appropriate number of observations are listed as “deleted due to missingness.”

17.7.1 Quality of Fit Statistics

```
glance(m_1cc) |>  
  select(r.squared, adj.r.squared, sigma, AIC, BIC) |>  
  kable(digits = c(3, 3, 2, 1, 1))
```


r.squared	adj.r.squared	sigma	AIC	BIC
0.181	0.173	1.3	2906.3	2953.8

17.7.2 Interpreting Effect Sizes

```
tidy(m_1cc, conf.int = TRUE) |>
  select(term, estimate, std.error, conf.low, conf.high) |>
  kable(digits = 3)
```

term	estimate	std.error	conf.low	conf.high
(Intercept)	0.582	0.371	-0.146	1.310
age_imp	-0.007	0.004	-0.015	0.000
comor	0.302	0.033	0.237	0.367
smoke100	0.099	0.090	-0.078	0.276
hx_depress	0.472	0.104	0.267	0.677
bmi	0.016	0.006	0.004	0.029
activityActive	-0.230	0.155	-0.534	0.074
activityInsufficiently_Active	-0.117	0.139	-0.391	0.157
activityInactive	0.256	0.115	0.030	0.482

We'll interpret three of the predictors here to demonstrate ideas: `comor`, `hx_depress` and `activity`.

- If we have two subjects with the same values of `age_imp`, `smoke100`, `hx_depress`, `bmi`, and `activity`, but Harry has a `comor` score that is one point higher than Sally's, then the model predicts that Harry's transformed outcome (specifically the natural logarithm of (his `physhealth` days + 1)) will be 0.302 higher than Sally's, with a 95% confidence interval around that estimate ranging from (0.237, 0.367).
- If we have two subjects with the same values of `age_imp`, `comor`, `smoke100`, `bmi`, and `activity`, but Harry has a history of depression (`hx_depress` = 1) while Sally does not have such a history (so Sally's `hx_depress` = 0), then the model predicts that Harry's transformed outcome (specifically the natural logarithm of (his `physhealth` days + 1)) will be 0.472 higher than Sally's, with a 95% confidence interval around that estimate ranging from (0.267, 0.677).
- The `activity` variable has four categories as indicated in the table below. The model uses the "Highly_Active" category as the reference group.

```
smart_16_imp1 |> tabyl(activity)
```

	activity	n	percent
	Highly_Active	252	0.2384106
	Active	135	0.1277200
	Insufficiently_Active	193	0.1825922
	Inactive	477	0.4512772

- From the tidied set of coefficients, we can describe the `activity` effects as follows.
 - If Sally is “Highly Active” and Harry is “Active” but they otherwise have the same values of all predictors, then our prediction is that Harry’s transformed outcome (specifically the natural logarithm of (his `physhealth` days + 1)) will be 0.230 lower than Sally’s, with a 95% confidence interval around that estimate ranging from (0.534 lower than Sally’s to 0.074 higher than Sally’s).
 - If instead Harry is “Insufficiently Active” but nothing else changes, then our prediction is that Harry’s transformed outcome will be 0.117 lower than Sally’s, with a 95% confidence interval around that estimate ranging from (0.391 lower to 0.157 higher than Sally’s.)
 - If instead Harry is “Inactive” but nothing else changes, then our prediction is that Harry’s transformed outcome will be 0.256 higher than Sally’s, with a 95% confidence interval around that estimate ranging from (0.030 to 0.482 higher than Sally’s.)

17.7.3 Making Predictions with the Model

Let’s describe two subjects, and use this model (and the ones that follow) to predict their `physhealth` values.

- Sheena is age 50, has 2 comorbidities, has smoked 100 cigarettes in her life, has no history of depression, a BMI of 25, and is Highly Active.
- Jacob is age 65, has 4 comorbidities, has never smoked, has a history of depression, a BMI of 32 and is Inactive.

We’ll first build predictions for Sheena and Jacob (with 95% prediction intervals) for `phys_tr`.

```
new2 <- tibble(  
  name = c("Sheena", "Jacob"),  
  age_imp = c(50, 65),  
  comor = c(2, 4),  
  smoke100 = c(1, 0),
```

```

    hx_depress = c(0, 1),
    bmi = c(25, 32),
    activity = c("Highly_Active", "Inactive")
)

preds_m_1cc <- predict(m_1cc, newdata = new2,
                      interval = "prediction")

preds_m_1cc

```

```

      fit      lwr      upr
1 1.341778 -1.22937 3.912925
2 2.583336  0.01399 5.152681

```

The model makes predictions for our transformed outcome, `phys_tr`. Now, we need to back-transform the predictions and the confidence intervals to build predictions for `physhealth`.

```

preds_m_1cc <- preds_m_1cc |>
  tbl_df() |>
  mutate(names = c("Sheena", "Jacob"),
         pred_physhealth = exp(fit) - 1,
         conf_low = exp(lwr) - 1,
         conf_high = exp(upr) - 1) |>
  select(names, pred_physhealth, conf_low, conf_high,
         everything())

```

Warning: `tbl\_df()` was deprecated in dplyr 1.0.0.
 i Please use `tibble::as\_tibble()` instead.

```
preds_m_1cc |> kable(digits = 3)
```

names	pred_physhealth	conf_low	conf_high	fit	lwr	upr
Sheena	2.826	-0.708	49.045	1.342	-1.229	3.913
Jacob	12.241	0.014	171.894	2.583	0.014	5.153

17.8 “Augmented” Linear Regression with `lm` on the Complete Cases

Now, we’ll add the non-linear terms we discussed earlier. We’ll add a (raw) cubic polynomial to represent the `comor` information, and we’ll add an interaction term between `hx_depress` and `activity`.

```
m_2cc <-  
  lm(phys_tr ~ age_imp + pol(comor, 3) + smoke100 +  
      bmi + hx_depress*activity, data = smart_16)  
  
summary(m_2cc)
```

Call:

```
lm(formula = phys_tr ~ age_imp + pol(comor, 3) + smoke100 + bmi +  
    hx_depress * activity, data = smart_16)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.907	-1.063	-0.267	1.143	2.924

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.514823	0.376203	1.368	0.17153
age_imp	-0.008100	0.003865	-2.096	0.03640
pol(comor, 3)comor	0.634274	0.160630	3.949	8.51e-
05				
pol(comor, 3)comor^2	-0.130626	0.073525	-1.777	0.07599
pol(comor, 3)comor^3	0.012508	0.008977	1.393	0.16386
smoke100	0.089345	0.090336	0.989	0.32294
bmi	0.015203	0.006315	2.408	0.01627
hx_depress	0.647054	0.229696	2.817	0.00496
activityActive	-0.202196	0.172300	-1.174	0.24092
activityInsufficiently_Active	-0.005815	0.166221	-0.035	0.97210
activityInactive	0.290380	0.132198	2.197	0.02832
hx_depress:activityActive	-0.124836	0.395415	-0.316	0.75230
hx_depress:activityInsufficiently_Active	-0.376355	0.310160	-1.213	0.22531
hx_depress:activityInactive	-0.172952	0.267427	-0.647	0.51798

(Intercept)

```

age_imp *
pol(comor, 3)comor ***
pol(comor, 3)comor^2 .
pol(comor, 3)comor^3
smoke100
bmi *
hx_depress **
activityActive
activityInsufficiently_Active
activityInactive *
hx_depress:activityActive
hx_depress:activityInsufficiently_Active
hx_depress:activityInactive
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 1.301 on 846 degrees of freedom
 (197 observations deleted due to missingness)
 Multiple R-squared: 0.187, Adjusted R-squared: 0.1745
 F-statistic: 14.97 on 13 and 846 DF, p-value: < 2.2e-16

Note again that the appropriate number of observations are listed as “deleted due to missingness.”

17.8.1 Quality of Fit Statistics

```

glance(m_2cc) |>
  select(r.squared, adj.r.squared, sigma, AIC, BIC) |>
  kable(digits = c(3, 3, 2, 1, 1))

```

r.squared	adj.r.squared	sigma	AIC	BIC
0.187	0.175	1.3	2909.5	2980.9

17.8.2 ANOVA assessing the impact of the non-linear terms

```
anova(m_1cc, m_2cc)
```

Analysis of Variance Table

Model 1: `phys_tr ~ age_imp + comor + smoke100 + hx_depress + bmi + activity`

Model 2: `phys_tr ~ age_imp + pol(comor, 3) + smoke100 + bmi + hx_depress * activity`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	851	1444.0				
2	846	1432.8	5	11.265	1.3303	0.249

The difference between the models doesn't meet the standard for statistical detectability at our usual α levels.

17.8.3 Interpreting Effect Sizes

```
tidy(m_2cc, conf.int = TRUE) |>
  select(term, estimate, std.error, conf.low, conf.high) |>
  kable(digits = 3)
```

term	estimate	std.error	conf.low	conf.high
(Intercept)	0.515	0.376	-0.224	1.253
age_imp	-0.008	0.004	-0.016	-0.001
pol(comor, 3)comor	0.634	0.161	0.319	0.950
pol(comor, 3)comor^2	-0.131	0.074	-0.275	0.014
pol(comor, 3)comor^3	0.013	0.009	-0.005	0.030
smoke100	0.089	0.090	-0.088	0.267
bmi	0.015	0.006	0.003	0.028
hx_depress	0.647	0.230	0.196	1.098
activityActive	-0.202	0.172	-0.540	0.136
activityInsufficiently_Active	-0.006	0.166	-0.332	0.320
activityInactive	0.290	0.132	0.031	0.550
hx_depress:activityActive	-0.125	0.395	-0.901	0.651
hx_depress:activityInsufficiently_Active	-0.376	0.310	-0.985	0.232
hx_depress:activityInactive	-0.173	0.267	-0.698	0.352

Let's focus first on interpreting the interaction terms between `hx_depress` and `activity`.

Assume first that we have a set of subjects with the same values of `age_imp`, `smoke100`, `bmi`, and `comor`.

- Arnold has `hx_depress = 1` and is Inactive

- Betty has `hx_depress` = 1 and is Insufficiently Active
- Carlos has `hx_depress` = 1 and is Active
- Debbie has `hx_depress` = 1 and is Highly Active
- Eamon has `hx_depress` = 0 and is Inactive
- Florence has `hx_depress` = 0 and is Insufficiently Active
- Garry has `hx_depress` = 0 and is Active
- Harry has `hx_depress` = 0 and is Highly Active

So the model, essentially can be used to compare each of the first seven people on that list to Harry (who has the reference levels of both `hx_depress` and `activity`.) Let's compare Arnold to Harry.

For instance, as compared to Harry, Arnold is expected to have a transformed outcome (specifically the natural logarithm of (his `physhealth` days + 1)) that is:

- 0.647 higher because Arnold's `hx_depress` = 1, and
- 0.29 higher still because Arnold's `activity` is "Inactive", and
- 0.173 lower because of the combination (see the '`hx_depress:activityInactive`' row)

So, in total, we expect Arnold's transformed outcome to be $0.647 + 0.29 + (-0.173)$, or 0.764 higher than Harry's.

If we want to compare Arnold to, for instance, Betty, we first calculate Betty's difference from Harry, and then compare the two differences.

As compared to Harry, Betty is expected to have a transformed outcome (specifically the natural logarithm of (her `physhealth` days + 1)) that is:

- 0.647 higher because Betty's `hx_depress` = 1, and
- 0.006 lower still because Betty's `activity` is "Insufficiently Active", and
- 0.376 lower because of the combination (see the '`hx_depress:activityInsufficiently_Active`' row)

So, in total, we expect Betty's transformed outcome to be $0.647 + (-0.006) + (-0.376)$, or 0.265 higher than Harry's.

And thus we can compare Betty and Arnold directly.

- Arnold is predicted to have an outcome that is 0.764 higher than Harry's.
- Betty is predicted to have an outcome that is 0.265 higher than Harry's.
- And so Arnold's predicted outcome (`phys_tr`) is 0.499 larger than Betty's.

Now, suppose we want to look at our cubic polynomial in `comor`.

- Suppose Harry and Sally have the same values for all other predictors in the model, but Harry has 1 comorbidity where Sally has none. Then the three terms in the model related to `comor` will be 1 for Harry and 0 for Sally, and the interpretation becomes pretty straightforward.
- But suppose instead that nothing has changed except Harry has 2 comorbidities and Sally has just 1. The size of the impact of this Harry - Sally difference is far larger in this situation, because the `comor` variable enters the model in a non-linear way. This is an area where fitting the model using `ols` can be helpful because of the ability to generate plots (of effects, nomograms, etc.) that can show this non-linearity in a clear way.

Suppose for instance, that Harry and Sally share the following values for the other predictors: each is age 40, has never smoked, has no history of depression, a BMI of 30 and is Highly Active.

- Now, if Harry has 1 comorbidity and Sally has none, the predicted `phys_tr` values for Harry and Sally are as indicated below.

```
hands1 <- tibble(
  name = c("Harry", "Sally"),
  age_imp = c(40, 40),
  comor = c(1, 0),
  smoke100 = c(0, 0),
  hx_depress = c(0, 0),
  bmi = c(30, 30),
  activity = c("Highly_Active", "Highly_Active")
)

predict(m_2cc, newdata = hands1)
```

```
      1      2
1.1630840 0.6469282
```

But if Harry has 2 comorbidities and Sally 1, the predictions are:

```
hands2 <- tibble(
  name = c("Harry", "Sally"),
  age_imp = c(40, 40),
  comor = c(2, 1), # only thing that changes
  smoke100 = c(0, 0),
  hx_depress = c(0, 0),
  bmi = c(30, 30),
```



```

    activity = c("Highly_Active", "Highly_Active")
  )

predict(m_2cc, newdata = hands2)

```

```

      1      2
1.493035 1.163084

```

Note that the difference in predictions between Harry and Sally is much smaller now than it was previously.

17.8.4 Making Predictions with the Model

As before, we'll use the new model to predict `physhealth` values for Sheena and Jacob.

- Sheena is age 50, has 2 comorbidities, has smoked 100 cigarettes in her life, has no history of depression, a BMI of 25, and is Highly Active.
- Jacob is age 65, has 4 comorbidities, has never smoked, has a history of depression, a BMI of 32 and is Inactive.

We'll first build predictions for Sheena and Jacob (with 95% prediction intervals) for `phys_tr`.

```

new2 <- tibble(
  name = c("Sheena", "Jacob"),
  age_imp = c(50, 65),
  comor = c(2, 4),
  smoke100 = c(1, 0),
  hx_depress = c(0, 1),
  bmi = c(25, 32),
  activity = c("Highly_Active", "Inactive")
)

preds_m_2cc <- predict(m_2cc, newdata = new2,
                      interval = "prediction")

preds_m_2cc

```

```

      fit      lwr      upr
1 1.425362 -1.14707613 3.997801
2 2.486907 -0.08635658 5.060171

```

Now, we need to back-transform the predictions and the confidence intervals that describe `phys_tr` to build predictions for `physhealth`.

```
preds_m_2cc <- preds_m_2cc |>
  tbl_df() |>
  mutate(names = c("Sheena", "Jacob"),
         pred_physhealth = exp(fit) - 1,
         conf_low = exp(lwr) - 1,
         conf_high = exp(upr) - 1) |>
  select(names, pred_physhealth, conf_low, conf_high,
         everything())
```

Warning: `tbl\_df()` was deprecated in dplyr 1.0.0.
i Please use `tibble::as\_tibble()` instead.

```
preds_m_2cc |> kable(digits = 3)
```

names	pred_physhealth	conf_low	conf_high	fit	lwr	upr
Sheena	3.159	-0.682	53.478	1.425	-1.147	3.998
Jacob	11.024	-0.083	156.617	2.487	-0.086	5.060

17.9 Using mice to perform Multiple Imputation

Let's focus on the main effects model, and look at the impact of performing multiple imputation to account for the missing data. Recall that in our `smart_16` data, the most "missingness" is shown in the `activity` variable, which is still missing less than 10% of the time. So we'll try a set of 10 imputations, using the default settings in the `mice` package.

```
# requires library(mice)

set.seed(432)

# create small data set including only variables to
# be used in building the imputation model

sm16 <- smart_16 |>
  select(physhealth, phys_tr, activity, age_imp, bmi, comor,
         hx_depress, smoke100)

smart_16_mice10 <- mice(sm16, m = 10)
```

iter	imp	variable						
1	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	2	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	3	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	4	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	5	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	6	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	7	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	8	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	9	activity	age_imp	bmi	comor	hx_depress	smoke100	
1	10	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	2	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	3	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	4	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	5	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	6	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	7	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	8	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	9	activity	age_imp	bmi	comor	hx_depress	smoke100	
2	10	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	2	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	3	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	4	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	5	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	6	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	7	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	8	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	9	activity	age_imp	bmi	comor	hx_depress	smoke100	
3	10	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	1	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	2	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	3	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	4	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	5	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	6	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	7	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	8	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	9	activity	age_imp	bmi	comor	hx_depress	smoke100	
4	10	activity	age_imp	bmi	comor	hx_depress	smoke100	
5	1	activity	age_imp	bmi	comor	hx_depress	smoke100	

```

5 2 activity age_imp bmi comor hx_depress smoke100
5 3 activity age_imp bmi comor hx_depress smoke100
5 4 activity age_imp bmi comor hx_depress smoke100
5 5 activity age_imp bmi comor hx_depress smoke100
5 6 activity age_imp bmi comor hx_depress smoke100
5 7 activity age_imp bmi comor hx_depress smoke100
5 8 activity age_imp bmi comor hx_depress smoke100
5 9 activity age_imp bmi comor hx_depress smoke100
5 10 activity age_imp bmi comor hx_depress smoke100

```

```
summary(smart_16_mice10)
```

Class: mids

Number of multiple imputations: 10

Imputation methods:

```

physhealth    phys_tr    activity    age_imp    bmi    comor hx_depress
      ""          ""    "polyreg"    "pmm"    "pmm"    "pmm"    "pmm"
      smoke100
      "pmm"

```

PredictorMatrix:

	physhealth	phys_tr	activity	age_imp	bmi	comor	hx_depress	smoke100
physhealth	0	1	1	1	1	1	1	1
phys_tr	1	0	1	1	1	1	1	1
activity	1	1	0	1	1	1	1	1
age_imp	1	1	1	0	1	1	1	1
bmi	1	1	1	1	0	1	1	1
comor	1	1	1	1	1	0	1	1

17.10 Running the Linear Regression in lm with Multiple Imputation

Next, we'll run the linear model (main effects) on each of the 10 imputed data sets.

```

m10_mods <-
  with(smart_16_mice10, lm(phys_tr ~ age_imp + comor +
    smoke100 + hx_depress +
    bmi + activity))

summary(m10_mods)

```

A tibble: 90 x 7

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>	nobs <int>	df.residual <dbl>
1	(Intercept)	0.317	0.326	0.971	3.32e- 1	1057	1048
2	age_imp	-0.00489	0.00334	-1.47	1.43e- 1	1057	1048
3	comor	0.313	0.0295	10.6	4.72e-25	1057	1048
4	smoke100	0.135	0.0799	1.69	9.22e- 2	1057	1048
5	hx_depress	0.500	0.0929	5.38	9.14e- 8	1057	1048
6	bmi	0.0187	0.00564	3.31	9.64e- 4	1057	1048
7	activityActive	-0.202	0.138	-1.46	1.44e- 1	1057	1048
8	activityInsufficient~	-0.0695	0.124	-0.561	5.75e- 1	1057	1048
9	activityInactive	0.262	0.103	2.54	1.11e- 2	1057	1048
10	(Intercept)	0.363	0.332	1.10	2.74e- 1	1057	1048

i 80 more rows

Then, we'll pool results across the 10 imputations

```
m10_pool <- pool(m10_mods)
summary(m10_pool, conf.int = TRUE) |>
  select(-statistic, -df) |>
  kable(digits = 3)
```

term	esti- mate	std.er- ror	p.value	2.5 %	97.5 %	conf.low	conf.high
(Intercept)	0.444	0.342	0.194	-	1.114	-0.227	1.114
age_imp	-0.005	0.003	0.128	-	0.002	-0.012	0.002
comor	0.309	0.031	0.000	0.249	0.369	0.249	0.369
smoke100	0.114	0.083	0.171	-	0.278	-0.049	0.278
hx_depress	0.512	0.094	0.000	0.327	0.696	0.327	0.696
bmi	0.016	0.006	0.009	0.004	0.027	0.004	0.027
activityActive	-0.204	0.140	0.146	-	0.071	-0.479	0.071
activityInsuffi- ciently_Active	-0.044	0.129	0.735	-	0.210	-0.298	0.210
activityInactive	0.260	0.106	0.014	0.052	0.469	0.052	0.469

And we can compare these results to the complete case analysis we completed earlier.

```
tidy(m_1cc, conf.int = TRUE) |>
  select(term, estimate, std.error, p.value, conf.low, conf.high) |>
  kable(digits = 3)
```

term	estimate	std.error	p.value	conf.low	conf.high
(Intercept)	0.582	0.371	0.117	-0.146	1.310
age_imp	-0.007	0.004	0.065	-0.015	0.000
comor	0.302	0.033	0.000	0.237	0.367
smoke100	0.099	0.090	0.273	-0.078	0.276
hx_depress	0.472	0.104	0.000	0.267	0.677
bmi	0.016	0.006	0.009	0.004	0.029
activityActive	-0.230	0.155	0.138	-0.534	0.074
activityInsufficiently_Active	-0.117	0.139	0.402	-0.391	0.157
activityInactive	0.256	0.115	0.027	0.030	0.482

Note that there are some sizeable differences here, although nothing enormous.

If we want the pooled R^2 or pooled adjusted R^2 after imputation, R will provide it (and a 95% confidence interval around the estimate) with ...

```
pool.r.squared(m10_mods)
```

```
      est      lo 95      hi 95      fmi
R^2 0.1912561 0.1482819 0.2369623 0.08061427
```

```
pool.r.squared(m10_mods, adjusted = TRUE)
```

```
      est      lo 95      hi 95      fmi
adj R^2 0.1850807 0.1425132 0.2305277 0.08312639
```

We can see the fraction of missing information about each coefficient due to non-response (fmi) and other details with the following code...

```
m10_pool
```

```
Class: mipo      m = 10
      term m estimate      ubar      b
1      (Intercept) 10 0.44377168 1.078194e-01 8.002900e-03
```

```

2          age_imp 10 -0.00522810 1.123668e-05 4.824290e-07
3          comor 10  0.30871888 8.801039e-04 5.466082e-05
4          smoke100 10  0.11415718 6.474388e-03 4.130673e-04
5          hx_depress 10  0.51155722 8.669413e-03 1.582684e-04
6          bmi 10  0.01576150 3.182381e-05 3.425191e-06
7          activityActive 10 -0.20412627 1.914250e-02 4.851040e-04
8 activityInsufficiently_Active 10 -0.04383739 1.565925e-02 9.855183e-04
9          activityInactive 10  0.26046070 1.069627e-02 5.023887e-04
      t dfcom      df      riv      lambda      fmi
1 1.166226e-01 1048 599.8172 0.08164758 0.07548446 0.07855177
2 1.176735e-05 1048 814.9042 0.04722675 0.04509697 0.04743197
3 9.402308e-04 1048 677.6361 0.06831796 0.06394909 0.06669961
4 6.928762e-03 1048 666.2487 0.07018023 0.06557795 0.06837041
5 8.843508e-03 1048 982.0512 0.02008154 0.01968621 0.02167660
6 3.559152e-05 1048 432.0874 0.11839279 0.10585976 0.10996992
7 1.967611e-02 1048 939.5064 0.02787591 0.02711992 0.02918437
8 1.674332e-02 1048 672.0473 0.06922874 0.06474643 0.06751736
9 1.124890e-02 1048 785.1905 0.05166545 0.04912727 0.05154007

```

17.11 Fit the Multiple Imputation Model with `aregImpute`

Here, we'll use `aregImpute` to deal with missing values through multiple imputation, and use the `ols` function in the `rms` package to fit the model.

The first step is to fit the multiple imputation model. We'll use `n.impute = 10` imputations, with `B = 10` bootstrap samples for the predictive mean matching, and fit both linear models and models with restricted cubic splines with 3 knots (`nk = c(0, 3)`) allowing the target variable to have a non-linear transformation when `nk` is 3, via `tlinear = FALSE`.

```

set.seed(43201602)
dd <- datadist(smart_16)
options(datadist = "dd")

fit16_imp <-
  aregImpute(~ phys_tr + age_imp + comor + smoke100 +
             hx_depress + bmi + activity,
             nk = c(0, 3), tlinear = FALSE,
             data = smart_16, B = 10, n.impute = 10)

```

```

Iteration 1
Iteration 2

```

```

Iteration 3
Iteration 4
Iteration 5
Iteration 6
Iteration 7
Iteration 8
Iteration 9
Iteration 10
Iteration 11
Iteration 12
Iteration 13

```

Here are the results of that imputation model.

```
fit16_imp
```

Multiple Imputation using Bootstrap and PMM

```

aregImpute(formula = ~phys_tr + age_imp + comor + smoke100 +
  hx_depress + bmi + activity, data = smart_16, n.impute = 10,
  nk = c(0, 3), tlinear = FALSE, B = 10)

```

```
n: 1057      p: 7      Imputations: 10      nk: 0
```

Number of NAs:

phys_tr	age_imp	comor	smoke100	hx_depress	bmi	activity
0	12	47	24	3	84	85

	type	d.f.
phys_tr	s	1
age_imp	s	1
comor	s	1
smoke100	l	1
hx_depress	l	1
bmi	s	1
activity	c	3

R-squares for Predicting Non-Missing Values for Each Variable
Using Last Imputations of Predictors

age_imp	comor	smoke100	hx_depress	bmi	activity
0.224	0.206	0.059	0.167	0.169	0.057

Resampling results for determining the complexity of imputation models

Variable being imputed: age_imp

		nk=0	nk=3
Bootstrap bias-corrected	R ²	0.186	0.215
10-fold cross-validated	R ²	0.211	0.215
Bootstrap bias-corrected	mean error	9.108	10.894
10-fold cross-validated	mean error	65.169	10.919
Bootstrap bias-corrected	median error	7.290	8.784
10-fold cross-validated	median error	66.006	8.613

Variable being imputed: comor

		nk=0	nk=3
Bootstrap bias-corrected	R ²	0.183	0.182
10-fold cross-validated	R ²	0.184	0.193
Bootstrap bias-corrected	mean error	0.987	1.184
10-fold cross-validated	mean error	1.759	1.171
Bootstrap bias-corrected	median error	0.828	0.910
10-fold cross-validated	median error	1.574	0.892

Variable being imputed: smoke100

		nk=0	nk=3
Bootstrap bias-corrected	R ²	0.0224	0.0187
10-fold cross-validated	R ²	0.0358	0.0217
Bootstrap bias-corrected	mean error	0.4853	0.4866
10-fold cross-validated	mean error	0.9462	0.9561
Bootstrap bias-corrected	median error	0.4788	0.4772
10-fold cross-validated	median error	0.8479	0.8706

Variable being imputed: hx_depress

		nk=0	nk=3
Bootstrap bias-corrected	R ²	0.157	0.138
10-fold cross-validated	R ²	0.147	0.148
Bootstrap bias-corrected	mean error	0.355	0.360
10-fold cross-validated	mean error	0.801	0.783
Bootstrap bias-corrected	median error	0.333	0.337
10-fold cross-validated	median error	0.711	0.673

Variable being imputed: bmi

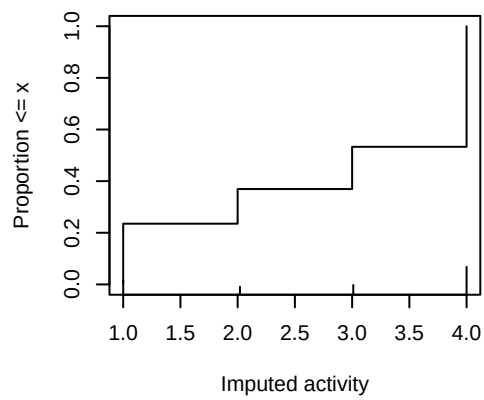
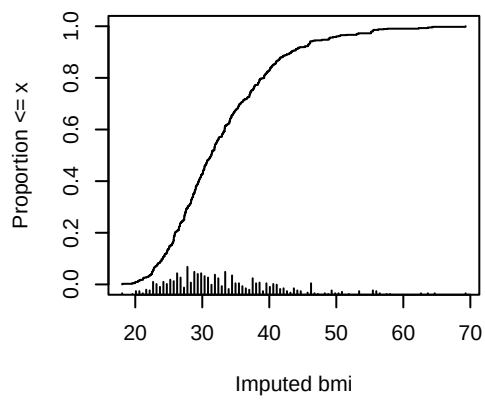
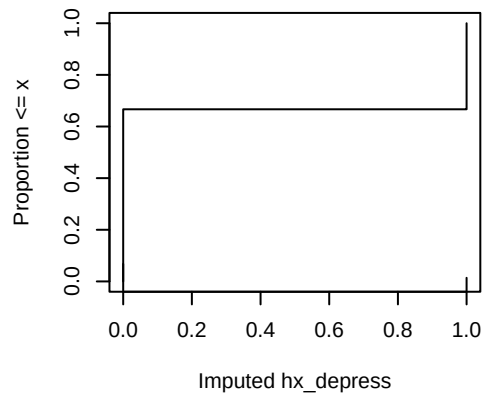
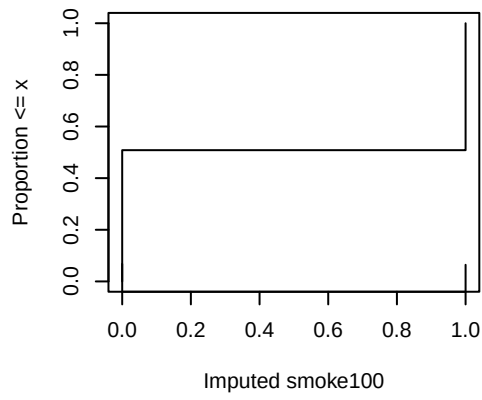
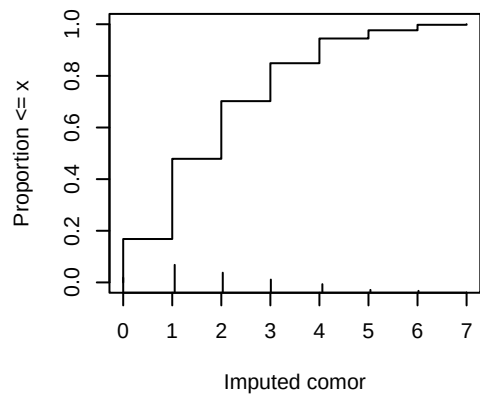
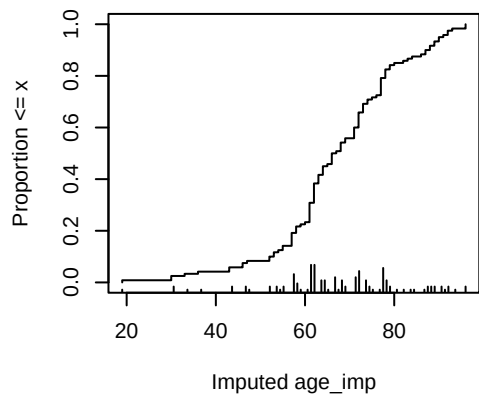
		nk=0	nk=3
Bootstrap bias-corrected	R ²	0.125	0.122
10-fold cross-validated	R ²	0.134	0.133

Bootstrap bias-corrected mean	error	5.221	6.822
10-fold cross-validated mean	error	32.458	6.884
Bootstrap bias-corrected median	error	4.178	5.782
10-fold cross-validated median	error	31.330	5.932

Variable being imputed: activity

		nk=0	nk=3
Bootstrap bias-corrected	R ²	0.0312	0.0275
10-fold cross-validated	R ²	0.0450	0.0402
Bootstrap bias-corrected mean	error	1.8774	1.8884
10-fold cross-validated mean	error	1.1121	1.1043
Bootstrap bias-corrected median	error	2.0000	2.0000
10-fold cross-validated median	error	1.0000	1.0000

```
par(mfrow = c(3,2))
plot(fit16_imp)
```



```
par(mfrow = c(1,1))
```

The plot helps us see where the imputations are happening.

17.12 Fit Linear Regression using `ols` and `fit.mult.impute`

```
m16_imp <-
  fit.mult.impute(phys_tr ~ age_imp + comor + smoke100 +
    hx_depress + bmi + activity,
    fitter = ols, xtrans = fit16_imp,
    data = smart_16, fitargs=list(x=TRUE,y=TRUE))
```

Wald Statistic Information

Variance Inflation Factors Due to Imputation:

Intercept	age_imp
1.03	1.01
comor	smoke100
1.03	1.06
hx_depress	bmi
1.02	1.06
activity=Active	activity=Insufficiently_Active
1.19	1.14
activity=Inactive	
1.23	

Fraction of Missing Information:

Intercept	age_imp
0.03	0.01
comor	smoke100
0.03	0.06
hx_depress	bmi
0.02	0.06
activity=Active	activity=Insufficiently_Active
0.16	0.13
activity=Inactive	

0.19

d.f. for t-distribution for Tests of Single Coefficients:

Intercept	age_imp
8176.67	45410.80
comor	smoke100
13030.27	2670.64
hx_depress	bmi
28199.30	2935.89
activity=Active	activity=Insufficiently_Active
354.62	571.56
activity=Inactive	
258.42	

The following fit components were averaged over the 10 model fits:

```
fitted.values stats linear.predictors
```

17.12.1 Summaries and Coefficients

Here are the results:

```
m16_imp
```

Linear Regression Model

```
fit.mult.impute(formula = phys_tr ~ age_imp + comor + smoke100 +  
  hx_depress + bmi + activity, fitter = ols, xtrans = fit16_imp,  
  data = smart_16, fitargs = list(x = TRUE, y = TRUE))
```

		Model Likelihood		Discrimination	
		Ratio Test		Indexes	
Obs	1057	LR chi2	219.94	R2	0.188
sigma1	2881	d.f.	8	R2 adj	0.182
d.f.	1048	Pr(> chi2)	0.0000	g	0.687

Residuals

Min	1Q	Median	3Q	Max
-3.0621	-1.0327	-0.2878	1.1104	2.8018

	Coef	S.E.	t	Pr(> t)
Intercept	0.4052	0.3352	1.21	0.2271
age_imp	-0.0049	0.0034	-1.46	0.1437
comor	0.3078	0.0302	10.20	<0.0001
smoke100	0.1259	0.0830	1.52	0.1296
hx_depress	0.5120	0.0940	5.45	<0.0001
bmi	0.0164	0.0058	2.83	0.0048
activity=Active	-0.1773	0.1513	-1.17	0.2416
activity=Insufficiently_Active	-0.0396	0.1342	-0.30	0.7680
activity=Inactive	0.2401	0.1144	2.10	0.0360

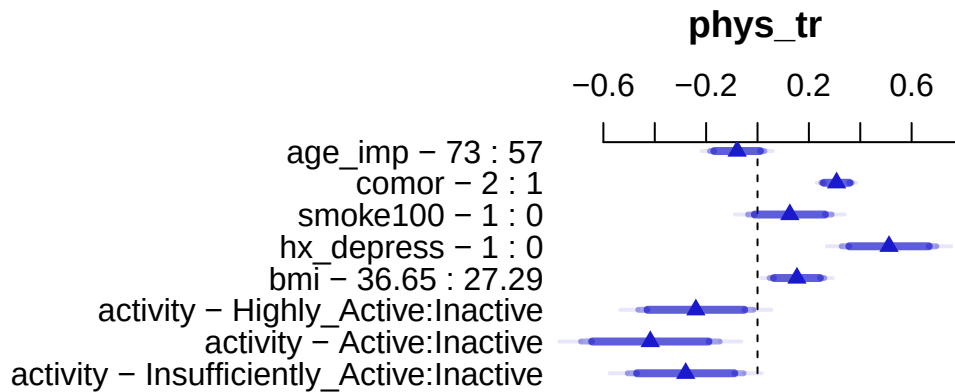
17.12.2 Effect Sizes

We can plot and summarize the effect sizes using the usual `ols` tools:

```
summary(m16_imp)
```

Effects	Response : phys_tr				
Factor	Low	High	Diff.	Effect	S.E.
age_imp	57.00	73.00	16.00	-0.079163	0.054107
comor	1.00	2.00	1.00	0.307790	0.030171
smoke100	0.00	1.00	1.00	0.125890	0.083000
hx_depress	0.00	1.00	1.00	0.511980	0.094007
bmi	27.29	36.65	9.36	0.153530	0.054322
activity - Highly_Active:Inactive	4.00	1.00	NA	-0.240070	0.114350
activity - Active:Inactive	4.00	2.00	NA	-0.417320	0.137640
activity - Insufficiently_Active:Inactive	4.00	3.00	NA	-0.279650	0.115000
Lower 0.95	0.95	Upper	0.95		
-0.185330	0.027008				
0.248590	0.366990				
-0.036973	0.288760				
0.327520	0.696450				
0.046932	0.260120				
-0.464450	-0.015686				
-0.687400	-0.147250				
-0.505310	-0.054002				

```
plot(summary(m16_imp))
```



17.12.3 Making Predictions with this Model

Once again, let's make predictions for our two subjects, and use this model (and the ones that follow) to predict their `physhealth` values.

- Sheena is age 50, has 2 comorbidities, has smoked 100 cigarettes in her life, has no history of depression, a BMI of 25, and is Highly Active.
- Jacob is age 65, has 4 comorbidities, has never smoked, has a history of depression, a BMI of 32 and is Inactive.

```
new2 <- tibble(
  name = c("Sheena", "Jacob"),
  age_imp = c(50, 65),
  comor = c(2, 4),
  smoke100 = c(1, 0),
  hx_depress = c(0, 1),
  bmi = c(25, 32),
  activity = c("Highly_Active", "Inactive")
)

preds_m_16imp <- predict(m16_imp,
  newdata = data.frame(new2))
```

```
preds_m_16imp
```

```
      1      2  
1.309306 2.591649
```

```
preds_m_16imp <- preds_m_16imp |>  
  tbl_df() |>  
  mutate(names = c("Sheena", "Jacob"),  
         pred_physhealth = exp(value) - 1) |>  
  select(names, pred_physhealth)
```

Warning: `tbl\_df()` was deprecated in dplyr 1.0.0.
i Please use `tibble::as\_tibble()` instead.

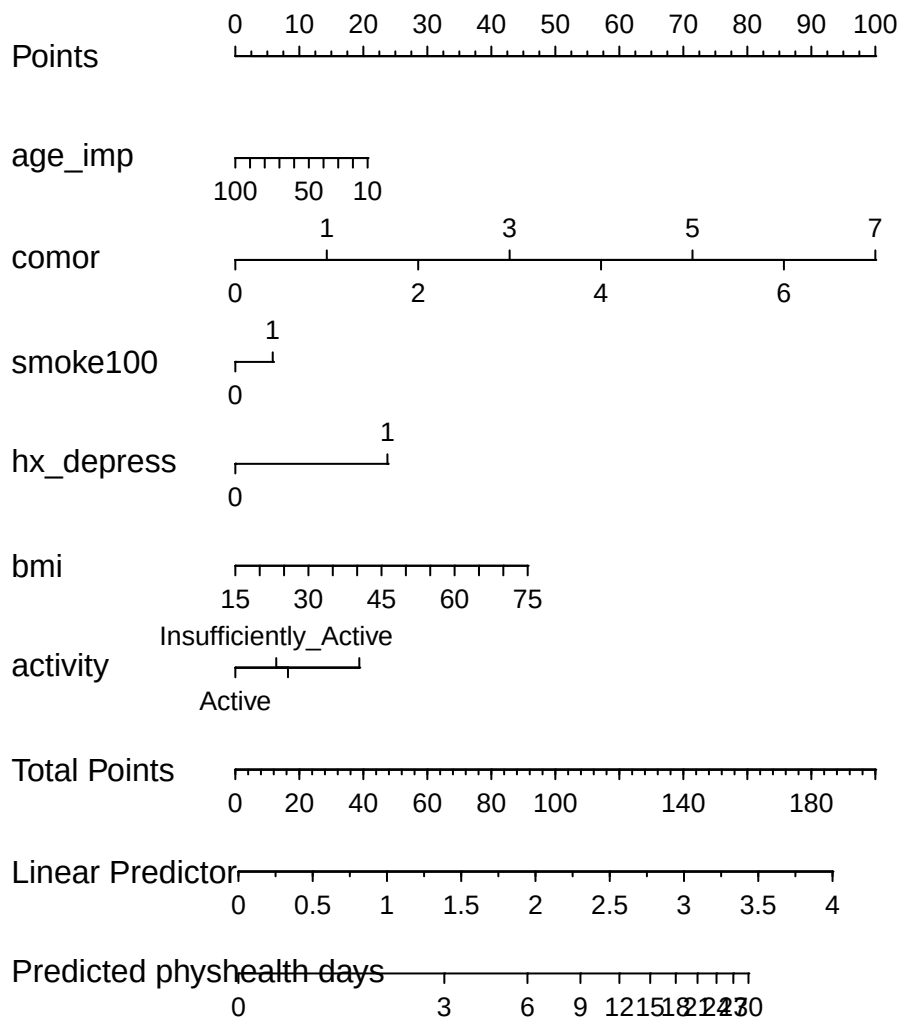
```
preds_m_16imp |> kable(digits = 3)
```

names	pred_physhealth
Sheena	2.704
Jacob	12.352

17.12.4 Nomogram

We can also develop a nomogram, if we like. As a special touch, we'll add a prediction at the bottom which back-transforms out of the predicted `phys_tr` back to the `physhealth` days.

```
plot(nomogram(m16_imp,  
  fun = list(function(x) exp(x) - 1),  
  funlabel = "Predicted physhealth days",  
  fun.at = seq(0, 30, 3)))
```

We can see the big role of `comor` and `hx_depress` in this model.

17.12.5 Validating Summary Statistics

We can cross-validate summary measures, like R^2 ...

```
validate(m16_imp)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper	n
R-square	0.1867	0.1984	0.1793	0.0192	0.1676	0.1189	0.2091	40
MSE	1.6472	1.6168	1.6623	-0.0455	1.6927	1.5993	1.7895	40
g	0.6876	0.7031	0.6749	0.0282	0.6594	0.5848	0.7506	40
Intercept	0.0000	0.0000	0.0677	-0.0677	0.0677	-0.1179	0.2291	40
Slope	1.0000	1.0000	0.9636	0.0364	0.9636	0.8474	1.0951	40

18 Building Table 1

18.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(tableone)
library(knitr)
library(tidyverse)

theme_set(theme_bw())
```

Many scientific articles involve direct comparison of results from various exposures, perhaps treatments. In 431, we studied numerous methods, including various sorts of hypothesis tests, confidence intervals, and descriptive summaries, which can help us to understand and compare outcomes in such a setting. One common approach is to present what's often called Table 1. Table 1 provides a summary of the characteristics of a sample, or of groups of samples, which is most commonly used to help understand the nature of the data being compared.

18.2 Data Load

```
fakestroke <- read_csv("data/fakestroke.csv", show_col_types = FALSE)
bloodbrain <- read_csv("data/bloodbrain.csv", show_col_types = FALSE)
```

18.3 Two examples from the *New England Journal of Medicine*

18.3.1 A simple Table 1

Table 1 is especially common in the context of clinical research. Consider the excerpt below, from a January 2015 article in the *New England Journal of Medicine* [Tolaney2015].

Table 1. Baseline Characteristics of the Patients.*	
Characteristic	Patients (N=406)
	<i>no. (%)</i>
Age group	
<50 yr	132 (32.5)
50–59 yr	137 (33.7)
60–69 yr	96 (23.6)
≥70 yr	41 (10.1)
Sex	
Female	405 (99.8)
Male	1 (0.2)
Race†	
White	351 (86.5)
Black	28 (6.9)
Asian	11 (2.7)
Other	16 (3.9)

This (partial) table reports baseline characteristics on age group, sex and race, describing 406 patients with HER2-positive¹ invasive breast cancer that began the protocol therapy. Age, sex and race (along with severity of illness) are the most commonly identified characteristics in a Table 1.

In addition to the measures shown in this excerpt, the full Table also includes detailed information on the primary tumor for each patient, including its size, nodal status and histologic grade. Footnotes tell us that the percentages shown are subject to rounding, and may not total 100, and that the race information was self-reported.

18.3.2 A group comparison

A more typical Table 1 involves a group comparison, for example in this excerpt from @Roy2008. This Table 1 describes a multi-center randomized clinical trial comparing two different approaches to caring for patients with heart failure and atrial fibrillation².

¹HER2 = human epidermal growth factor receptor type 2. Over-expression of this occurs in 15-20% of invasive breast cancers, and has been associated with poor outcomes.

²The complete Table 1 appears on pages 2668-2669 of @Roy2008, but I have only reproduced the first page and the footnote in this excerpt.

Table 1. Baseline Characteristics of the Patients.*		
Variable	Rhythm-Control Group (N=682)	Rate-Control Group (N=694)
Male sex (%)	78	85
Age (yr)	66±11	67±11
Body-mass index†	27.8±5.4	28.0±5.1
Nonwhite race (%)‡	16	13
NYHA class III or IV (%)		
At baseline	32	31
During previous 6 mo	76	76
Predominant cardiac diagnosis (%)§		
Coronary artery disease	48	48
Valvular heart disease	5	5
Nonischemic cardiomyopathy	36	39
Congenital heart disease	1	1
Hypertensive heart disease	10	7

The article provides percentages, means and standard deviations across groups, but note that it does not provide p values for the comparison of baseline characteristics. This is a common feature of NEJM reports on randomized clinical trials, where we anticipate that the two groups will be well matched at baseline. Note that the patients in this study were *randomly* assigned to either the rhythm-control group or to the rate-control group, using blocked randomization stratified by study center.

18.4 The MR CLEAN trial

@Berkhemer2015 reported on the MR CLEAN trial, involving 500 patients with acute ischemic stroke caused by a proximal intracranial arterial occlusion. The trial was conducted at 16 medical centers in the Netherlands, where 233 were randomly assigned to the intervention (intraarterial treatment plus usual care) and 267 to control (usual care alone.) The primary outcome was the modified Rankin scale score at 90 days; this categorical scale measures functional outcome, with scores ranging from 0 (no symptoms) to 6 (death). The fundamental conclusion of @Berkhemer2015 was that in patients with acute ischemic stroke caused by a proximal intracranial occlusion of the anterior circulation, intraarterial treatment administered within 6 hours after stroke onset was effective and safe.

Here's the Table 1 from @Berkhemer2015.

Table 1. Baseline Characteristics of the 500 Patients.*		
Characteristic	Intervention (N = 233)	Control (N = 267)
Age — yr		
Median	65.8	65.7
Interquartile range	54.5–76.0	55.5–76.4
Male sex — no. (%)	135 (57.9)	157 (58.8)
NIHSS score†		
Median (interquartile range)	17 (14–21)	18 (14–22)
Range	3–30	4–38
Location of stroke in left hemisphere — no. (%)	116 (49.8)	153 (57.3)
History of ischemic stroke — no. (%)	29 (12.4)	25 (9.4)
Atrial fibrillation — no. (%)	66 (28.3)	69 (25.8)
Diabetes mellitus — no. (%)	34 (14.6)	34 (12.7)
Prestroke modified Rankin scale score — no. (%)‡		
0	190 (81.5)	214 (80.1)
1	21 (9.0)	29 (10.9)
2	12 (5.2)	13 (4.9)
>2	10 (4.3)	11 (4.1)
Systolic blood pressure — mm Hg§	146±26.0	145±24.4
Treatment with IV alteplase — no. (%)	203 (87.1)	242 (90.6)
Time from stroke onset to start of IV alteplase — min		
Median	85	87
Interquartile range	67–110	65–116
ASPECTS — median (interquartile range)¶	9 (7–10)	9 (8–10)
Intracranial arterial occlusion — no./total no. (%)		
Intracranial ICA	1/233 (0.4)	3/266 (1.1)
ICA with involvement of the M1 middle cerebral artery segment	59/233 (25.3)	75/266 (28.2)
M1 middle cerebral artery segment	154/233 (66.1)	165/266 (62.0)
M2 middle cerebral artery segment	18/233 (7.7)	21/266 (7.9)
A1 or A2 anterior cerebral artery segment	1/233 (0.4)	2/266 (0.8)
Extracranial ICA occlusion — no./total no. (%) **	75/233 (32.2)	70/266 (26.3)
Time from stroke onset to randomization — min††		
Median	204	196
Interquartile range	152–251	149–266
Time from stroke onset to groin puncture — min		
Median	260	NA
Interquartile range	210–313	

The Table was accompanied by the following notes.

- * The intervention group was assigned to intraarterial treatment plus usual care, and the control group was assigned to usual care alone. Plus-minus values are means $\pm$ SD. ICA denotes internal carotid artery, IV intravenous, and NA not applicable.
- † Scores on the National Institutes of Health Stroke Scale (NIHSS) range from 0 to 42, with higher scores indicating more severe neurologic deficits. The NIHSS is a 15-item scale, and values for 30 of the 7500 items were missing (0.4%). The highest number of missing items for a single patient was 6.
- ‡ Scores on the modified Rankin scale of functional disability range from 0 (no symptoms) to 6 (death). A score of 2 or less indicates functional independence.
- § Data on systolic blood pressure at baseline were missing for one patient assigned to the control group.
- ¶ The Alberta Stroke Program Early Computed Tomography Score (ASPECTS) is a measure of the extent of stroke. Scores range from 0 to 10, with higher scores indicating fewer early ischemic changes. Scores were not available for four patients assigned to the control group: noncontrast computed tomography was not performed in one patient, and three patients had strokes in the territory of the anterior cerebral artery.
- || Vessel imaging was not performed in one patient in the control group, so the level of occlusion was not known.
- ** Extracranial ICA occlusions were reported by local investigators.
- †† Data were missing for two patients in the intervention group.

18.5 Simulated fakestroke data

Consider the simulated data, available on our [Data and Code website](#) in the `fakestroke.csv` file, which I built to let us mirror the Table 1 for MR CLEAN [Berkhemer2015]. The `fakestroke.csv` file contains the following 18 variables for 500 patients.

Variable	Description
<code>studyid</code>	Study ID # (z001 through z500)
<code>trt</code>	Treatment group (Intervention or Control)
<code>age</code>	Age in years
<code>sex</code>	Male or Female
<code>nihss</code>	NIH Stroke Scale Score (can range from 0-42; higher scores indicate more severe neurological deficits)
<code>location</code>	Stroke Location - Left or Right Hemisphere
<code>hx.isch</code>	History of Ischemic Stroke (Yes/No)
<code>afib</code>	Atrial Fibrillation (1 = Yes, 0 = No)
<code>dm</code>	Diabetes Mellitus (1 = Yes, 0 = No)
<code>mrankin</code>	Pre-stroke modified Rankin scale score (0, 1, 2 or > 2) indicating functional disability - complete range is 0 (no symptoms) to 6 (death)
<code>sbp</code>	Systolic blood pressure, in mm Hg
<code>iv.altep</code>	Treatment with IV alteplase (Yes/No)
<code>time.iv</code>	Time from stroke onset to start of IV alteplase (minutes) if iv.altep=Yes
<code>aspects</code>	Alberta Stroke Program Early Computed Tomography score, which measures extent of stroke from 0 - 10; higher scores indicate fewer early ischemic changes

Variable	Description
ia.occlus	Intracranial arterial occlusion, based on vessel imaging - five categories ³
extra.ica	Extracranial ICA occlusion (1 = Yes, 0 = No)
time.rand	Time from stroke onset to study randomization, in minutes
time.punc	Time from stroke onset to groin puncture, in minutes (only if Intervention)

Here's a quick look at the simulated data in `fakestroke`.

`fakestroke`

```
# A tibble: 500 x 18
  studyid trt      age sex  nihss location hx.isch afib  dm mrankin sbp
  <chr>   <chr>   <dbl> <chr> <dbl> <chr>   <chr> <dbl> <dbl> <chr>   <dbl>
1 z001   Control    53 Male    21 Right    No        0      0 2      127
2 z002   Interve~    51 Male    23 Left     No        1      0 0      137
3 z003   Control    68 Fema~    11 Right    No        0      0 0      138
4 z004   Control    28 Male    22 Left     No        0      0 0      122
5 z005   Control    91 Male    24 Right    No        0      0 0      162
6 z006   Control    34 Fema~    18 Left     No        0      0 2      166
7 z007   Interve~    75 Male    25 Right    No        0      0 0      140
8 z008   Control    89 Fema~    18 Right    No        0      0 0      157
9 z009   Control    75 Male    25 Left     No        1      0 2      129
10 z010  Interve~    26 Fema~    27 Right    No        0      0 0      143
# i 490 more rows
# i 7 more variables: iv.altep <chr>, time.iv <dbl>, aspects <dbl>,
#   ia.occlus <chr>, extra.ica <dbl>, time.rand <dbl>, time.punc <dbl>
```

18.6 Building Table 1 for fakestroke: Attempt 1

Our goal, then, is to take the data in `fakestroke.csv` and use it to generate a Table 1 for the study that compares the 233 patients in the Intervention group to the 267 patients in the Control group, on all of the other variables (except study ID #) available. I'll use the `tableone` package of functions available in R to help me complete this task. We'll make a first attempt, using the `CreateTableOne` function in the `tableone` package. To use the function, we'll need to specify:

³The five categories are Intracranial ICA, ICA with involvement of the M1 middle cerebral artery segment, M1 middle cerebral artery segment, M2 middle cerebral artery segment, A1 or A2 anterior cerebral artery segment

- the `vars` or variables we want to place in the rows of our Table 1 (which will include just about everything in the `fakestroke` data except the `studyid` code and the `trt` variable for which we have other plans, and the `time.punc` which applies only to subjects in the Intervention group.)
 - A useful trick here is to use the `dput` function, specifically something like `dput(names(fakestroke))` can be used to generate a list of all of the variables included in the `fakestroke` tibble, and then this can be copied and pasted into the `vars` specification, saving some typing.
- the `strata` which indicates the levels want to use in the columns of our Table 1 (for us, that's `trt`)

```
fs.vars <- c("age", "sex", "nihss", "location",
            "hx.isch", "afib", "dm", "mrankin", "sbp",
            "iv.altep", "time.iv", "aspects",
            "ia.occlus", "extra.ica", "time.rand")

fs.trt <- c("trt")

att1 <- CreateTableOne(data = fakestroke,
                      vars = fs.vars,
                      strata = fs.trt)

print(att1)
```

	Stratified by trt			
	Control	Intervention	p	test
n	267	233		
age (mean (SD))	65.38 (16.10)	63.93 (18.09)	0.343	
sex = Male (%)	157 (58.8)	135 (57.9)	0.917	
nihss (mean (SD))	18.08 (4.32)	17.97 (5.04)	0.787	
location = Right (%)	114 (42.7)	117 (50.2)	0.111	
hx.isch = Yes (%)	25 (9.4)	29 (12.4)	0.335	
afib (mean (SD))	0.26 (0.44)	0.28 (0.45)	0.534	
dm (mean (SD))	0.13 (0.33)	0.12 (0.33)	0.923	
mrankin (%)			0.922	
> 2	11 (4.1)	10 (4.3)		
0	214 (80.1)	190 (81.5)		
1	29 (10.9)	21 (9.0)		
2	13 (4.9)	12 (5.2)		
sbp (mean (SD))	145.00 (24.40)	146.03 (26.00)	0.647	
iv.altep = Yes (%)	242 (90.6)	203 (87.1)	0.267	
time.iv (mean (SD))	87.96 (26.01)	98.22 (45.48)	0.003	

aspects (mean (SD))	8.65 (1.47)	8.35 (1.64)	0.033
ia.occlus (%)			0.795
A1 or A2	2 (0.8)	1 (0.4)	
ICA with M1	75 (28.2)	59 (25.3)	
Intracranial ICA	3 (1.1)	1 (0.4)	
M1	165 (62.0)	154 (66.1)	
M2	21 (7.9)	18 (7.7)	
extra.ica (mean (SD))	0.26 (0.44)	0.32 (0.47)	0.150
time.rand (mean (SD))	213.88 (70.29)	202.51 (57.33)	0.051

18.6.1 Some of this is very useful, and other parts need to be fixed.

1. The 1/0 variables (`afib`, `dm`, `extra.ica`) might be better if they were treated as the factors they are, and reported as the Yes/No variables are reported, with counts and percentages rather than with means and standard deviations.
2. In some cases, we may prefer to re-order the levels of the categorical (factor) variables, particularly the `mrarkin` variable, but also the `ia.occlus` variable. It would also be more typical to put the Intervention group to the left and the Control group to the right, so we may need to adjust our `trt` variable's levels accordingly.
3. For each of the quantitative variables (`age`, `nihss`, `sbp`, `time.iv`, `aspects`, `extra.ica`, `time.rand` and `time.punc`) we should make a decision whether a summary with mean and standard deviation is appropriate, or whether we should instead summarize with, say, the median and quartiles. A mean and standard deviation really only yields an appropriate summary when the data are least approximately Normally distributed. This will make the *p* values a bit more reasonable, too. The `test` column in the first attempt will soon have something useful to tell us.
4. If we'd left in the `time.punc` variable, we'd get some warnings, having to do with the fact that `time.punc` is only relevant to patients in the Intervention group.

18.6.2 fakestroke Cleaning Up Categorical Variables

Let's specify each of the categorical variables as categorical explicitly. This helps the `CreateTableOne` function treat them appropriately, and display them with counts and percentages. This includes all of the 1/0, Yes/No and multi-categorical variables.

```
fs.factorvars <- c("sex", "location", "hx.isch", "afib", "dm",
                  "mrarkin", "iv.altep", "ia.occlus", "extra.ica")
```

Then we simply add a `factorVars = fs.factorvars` call to the `CreateTableOne` function.

We also want to re-order some of those categorical variables, so that the levels are more useful to us. Specifically, we want to:

- place Intervention before Control in the `trt` variable,
- reorder the `mraink` scale as 0, 1, 2, > 2, and
- rearrange the `ia.occlus` variable to the order⁴ presented in @Berkhemer2015.

To accomplish this, we'll use the `fct_relevel` function from the `forcats` package (loaded with the rest of the core `tidyverse` packages) to reorder our levels manually.

```
fakestroke <- fakestroke %>%
  mutate(trt = fct_relevel(trt, "Intervention", "Control"),
         mraink = fct_relevel(mraink, "0", "1", "2", "> 2"),
         ia.occlus = fct_relevel(ia.occlus, "Intracranial ICA",
                                "ICA with M1", "M1", "M2",
                                "A1 or A2")
  )
```

18.7 fakestroke Table 1: Attempt 2

```
att2 <- CreateTableOne(data = fakestroke,
                      vars = fs.vars,
                      factorVars = fs.factorvars,
                      strata = fs.trt)
print(att2)
```

	Stratified by trt			
	Intervention	Control	p	test
n	233	267		
age (mean (SD))	63.93 (18.09)	65.38 (16.10)	0.343	
sex = Male (%)	135 (57.9)	157 (58.8)	0.917	
nihss (mean (SD))	17.97 (5.04)	18.08 (4.32)	0.787	
location = Right (%)	117 (50.2)	114 (42.7)	0.111	
hx.isch = Yes (%)	29 (12.4)	25 (9.4)	0.335	
afib = 1 (%)	66 (28.3)	69 (25.8)	0.601	
dm = 1 (%)	29 (12.4)	34 (12.7)	1.000	
mraink (%)			0.922	
0	190 (81.5)	214 (80.1)		
1	21 (9.0)	29 (10.9)		
2	12 (5.2)	13 (4.9)		

⁴We might also have considered reordering the `ia.occlus` factor by its frequency, using the `fct_infreq` function

> 2	10 (4.3)	11 (4.1)	
sbp (mean (SD))	146.03 (26.00)	145.00 (24.40)	0.647
iv.altep = Yes (%)	203 (87.1)	242 (90.6)	0.267
time.iv (mean (SD))	98.22 (45.48)	87.96 (26.01)	0.003
aspects (mean (SD))	8.35 (1.64)	8.65 (1.47)	0.033
ia.occlus (%)			0.795
Intracranial ICA	1 (0.4)	3 (1.1)	
ICA with M1	59 (25.3)	75 (28.2)	
M1	154 (66.1)	165 (62.0)	
M2	18 (7.7)	21 (7.9)	
A1 or A2	1 (0.4)	2 (0.8)	
extra.ica = 1 (%)	75 (32.2)	70 (26.3)	0.179
time.rand (mean (SD))	202.51 (57.33)	213.88 (70.29)	0.051

The categorical data presentation looks much improved.

18.7.1 What summaries should we show?

Now, we'll move on to the issue of making a decision about what type of summary to show for the quantitative variables. Since the `fakestroke` data are just simulated and only match the summary statistics of the original results, not the details, we'll adopt the decisions made by @Berkhemer2015, which were to use medians and interquartile ranges to summarize the distributions of all of the continuous variables **except** systolic blood pressure.

- Specifying certain quantitative variables as *non-normal* causes R to show them with medians and the 25th and 75th percentiles, rather than means and standard deviations, and also causes those variables to be tested using non-parametric tests, like the Wilcoxon signed rank test, rather than the t test. The `test` column indicates this with the word `nonnorm`.
 - In real data situations, what should we do? The answer is to look at the data. I would not make the decision as to which approach to take without first plotting (perhaps in a histogram or a Normal Q-Q plot) the observed distributions in each of the two samples, so that I could make a sound decision about whether Normality was a reasonable assumption. If the means and medians are meaningfully different from each other, this is especially important.
 - To be honest, though, if the variable in question is a relatively unimportant covariate and the *p* values for the two approaches are nearly the same, I'd say that further investigation is rarely important,
- Specifying *exact* tests for certain categorical variables (we'll try this for the `location` and `mrankin` variables) can be done, and these changes will be noted in the `test` column, as well.

- In real data situations, I would rarely be concerned about this issue, and often choose Pearson (approximate) options across the board. This is reasonable so long as the number of subjects falling in each category is reasonably large, say above 10. If not, then an exact test may be a tiny improvement.
- Paraphrasing @Rosenbaum2017, having an exact rather than an approximate test result is about as valuable as having a nice crease in your trousers.

To finish our Table 1, then, we need to specify which variables should be treated as non-Normal in the `print` statement - notice that we don't need to redo the `CreateTableOne` for this change.

```
print(att2,
      nonnormal = c("age", "nihss", "time.iv", "aspects", "time.rand"),
      exact = c("location", "mrankin"))
```

	Stratified by trt	
	Intervention	Control
n	233	267
age (median [IQR])	65.80 [54.50, 76.00]	65.70 [55.75, 76.20]
sex = Male (%)	135 (57.9)	157 (58.8)
nihss (median [IQR])	17.00 [14.00, 21.00]	18.00 [14.00, 22.00]
location = Right (%)	117 (50.2)	114 (42.7)
hx.isch = Yes (%)	29 (12.4)	25 (9.4)
afib = 1 (%)	66 (28.3)	69 (25.8)
dm = 1 (%)	29 (12.4)	34 (12.7)
mrankin (%)		
0	190 (81.5)	214 (80.1)
1	21 (9.0)	29 (10.9)
2	12 (5.2)	13 (4.9)
> 2	10 (4.3)	11 (4.1)
sbp (mean (SD))	146.03 (26.00)	145.00 (24.40)
iv.altep = Yes (%)	203 (87.1)	242 (90.6)
time.iv (median [IQR])	85.00 [67.00, 110.00]	87.00 [65.00, 116.00]
aspects (median [IQR])	9.00 [7.00, 10.00]	9.00 [8.00, 10.00]
ia.occlus (%)		
Intracranial ICA	1 (0.4)	3 (1.1)
ICA with M1	59 (25.3)	75 (28.2)
M1	154 (66.1)	165 (62.0)
M2	18 (7.7)	21 (7.9)
A1 or A2	1 (0.4)	2 (0.8)
extra.ica = 1 (%)	75 (32.2)	70 (26.3)
time.rand (median [IQR])	204.00 [152.00, 249.50]	196.00 [149.00, 266.00]

	Stratified by trt	
	p	test
n		
age (median [IQR])	0.579	nonnorm
sex = Male (%)	0.917	
nihss (median [IQR])	0.453	nonnorm
location = Right (%)	0.106	exact
hx.isch = Yes (%)	0.335	
afib = 1 (%)	0.601	
dm = 1 (%)	1.000	
mrankin (%)	0.917	exact
0		
1		
2		
> 2		
sbp (mean (SD))	0.647	
iv.altep = Yes (%)	0.267	
time.iv (median [IQR])	0.596	nonnorm
aspects (median [IQR])	0.075	nonnorm
ia.occlus (%)	0.795	
Intracranial ICA		
ICA with M1		
M1		
M2		
A1 or A2		
extra.ica = 1 (%)	0.179	
time.rand (median [IQR])	0.251	nonnorm

18.8 Obtaining a more detailed Summary

If this was a real data set, we'd want to get a more detailed description of the data to make decisions about things like potentially collapsing categories of a variable, or whether or not a normal distribution was useful for a particular continuous variable, etc. You can do this with the `summary` command applied to a created Table 1, which shows, among other things, the effect of changing from normal to non-normal p values for continuous variables, and from approximate to "exact" p values for categorical factors.

Again, as noted above, in a real data situation, we'd want to plot the quantitative variables (within each group) to make a smart decision about whether a t test or Wilcoxon approach is more appropriate.

Note in the summary below that we have some missing values here. Often, we'll present this information within the Table 1, as well.

```
summary(att2)
```

```
### Summary of continuous variables ###
```

```
trt: Intervention
```

	n	miss	p.miss	mean	sd	median	p25	p75	min	max	skew	kurt
age	233	0	0.0	64	18	66	54	76	23	96	-0.34	-0.52
nihss	233	0	0.0	18	5	17	14	21	10	28	0.48	-0.74
sbp	233	0	0.0	146	26	146	129	164	78	214	-0.07	-0.22
time.iv	233	30	12.9	98	45	85	67	110	42	218	1.03	0.08
aspects	233	0	0.0	8	2	9	7	10	5	10	-0.56	-0.98
time.rand	233	2	0.9	203	57	204	152	250	100	300	0.01	-1.16

```
-----
```

```
trt: Control
```

	n	miss	p.miss	mean	sd	median	p25	p75	min	max	skew	kurt
age	267	0	0.0	65	16	66	56	76	24	94	-0.296	-0.28
nihss	267	0	0.0	18	4	18	14	22	11	25	0.017	-1.24
sbp	267	1	0.4	145	24	145	128	161	82	231	0.156	0.08
time.iv	267	25	9.4	88	26	87	65	116	44	130	0.001	-1.32
aspects	267	4	1.5	9	1	9	8	10	5	10	-1.071	0.36
time.rand	267	0	0.0	214	70	196	149	266	120	360	0.508	-0.93

```
p-values
```

	pNormal	pNonNormal
age	0.342813660	0.57856976
nihss	0.787487252	0.45311695
sbp	0.647157646	0.51346132
time.iv	0.003073372	0.59641104
aspects	0.032662901	0.07464683
time.rand	0.050803672	0.25134327

```
Standardize mean differences
```

	1 vs 2
age	0.08478764
nihss	0.02405390
sbp	0.04100833
time.iv	0.27691223
aspects	0.19210662

time.rand 0.17720957

=====

Summary of categorical variables

trt: Intervention

var	n	miss	p.miss	level	freq	percent	cum.percent
sex	233	0	0.0	Female	98	42.1	42.1
				Male	135	57.9	100.0
location	233	0	0.0	Left	116	49.8	49.8
				Right	117	50.2	100.0
hx.isch	233	0	0.0	No	204	87.6	87.6
				Yes	29	12.4	100.0
afib	233	0	0.0	0	167	71.7	71.7
				1	66	28.3	100.0
dm	233	0	0.0	0	204	87.6	87.6
				1	29	12.4	100.0
mrankin	233	0	0.0	0	190	81.5	81.5
				1	21	9.0	90.6
				2	12	5.2	95.7
				> 2	10	4.3	100.0
iv.altep	233	0	0.0	No	30	12.9	12.9
				Yes	203	87.1	100.0
ia.occlus	233	0	0.0	Intracranial ICA	1	0.4	0.4
				ICA with M1	59	25.3	25.8
				M1	154	66.1	91.8
				M2	18	7.7	99.6
				A1 or A2	1	0.4	100.0
extra.ica	233	0	0.0	0	158	67.8	67.8
				1	75	32.2	100.0

trt: Control

var	n	miss	p.miss	level	freq	percent	cum.percent
-----	---	------	--------	-------	------	---------	-------------

sex	267	0	0.0	Female	110	41.2	41.2
				Male	157	58.8	100.0
location	267	0	0.0	Left	153	57.3	57.3
				Right	114	42.7	100.0
hx.isch	267	0	0.0	No	242	90.6	90.6
				Yes	25	9.4	100.0
afib	267	0	0.0	0	198	74.2	74.2
				1	69	25.8	100.0
dm	267	0	0.0	0	233	87.3	87.3
				1	34	12.7	100.0
mrankin	267	0	0.0	0	214	80.1	80.1
				1	29	10.9	91.0
				2	13	4.9	95.9
				> 2	11	4.1	100.0
iv.altep	267	0	0.0	No	25	9.4	9.4
				Yes	242	90.6	100.0
ia.occlus	267	1	0.4	Intracranial ICA	3	1.1	1.1
				ICA with M1	75	28.2	29.3
				M1	165	62.0	91.4
				M2	21	7.9	99.2
				A1 or A2	2	0.8	100.0
extra.ica	267	1	0.4	0	196	73.7	73.7
				1	70	26.3	100.0

p-values

	pApprox	pExact
sex	0.9171387	0.8561188
location	0.1113553	0.1056020
hx.isch	0.3352617	0.3124683
afib	0.6009691	0.5460206
dm	1.0000000	1.0000000
mrankin	0.9224798	0.9173657
iv.altep	0.2674968	0.2518374
ia.occlus	0.7945580	0.8189090

```
extra.ica 0.1793385 0.1667574
```

```
Standardize mean differences
```

```
1 vs 2
```

```
sex      0.017479025
location 0.151168444
hx.isch  0.099032275
afib     0.055906317
dm       0.008673478
mrankin  0.062543164
iv.altep 0.111897009
ia.occlus 0.117394890
extra.ica 0.129370206
```

In this case, I have simulated the data to mirror the results in the published Table 1 for this study. In no way have I captured the full range of the real data, or any of the relationships in that data, so it's more important here to see what's available in the analysis, rather than to interpret it closely in the clinical context.

18.9 Exporting the Completed Table 1 from R to Excel or Word

Once you've built the table and are generally satisfied with it, you'll probably want to be able to drop it into Excel or Word for final cleanup.

18.9.1 Approach A: Save and open in Excel

One option is to **save the Table 1** to a `.csv` file within our `data` subfolder (note that the `data` folder must already exist), which you can then open directly in Excel. This is the approach I generally use. Note the addition of some `quote`, `noSpaces` and `printToggle` selections here.

```
fs.table1save <- print(att2,
  nonnormal = c("age", "nihss", "time.iv", "aspects", "time.rand"),
  exact = c("location", "mrankin"),
  quote = FALSE, noSpaces = TRUE, printToggle = FALSE)

write.csv(fs.table1save, file = "data/fs-table1.csv")
```

When I then open the `fs-table1.csv` file in Excel, it looks like this:

	A	B	C	D	E
1		Intervention	Control	p	test
2	n	233	267		
3	age (median [IQR])	65.80 [54.50, 76.00]	65.70 [55.75, 76.20]	0.579	nonnorm
4	sex = Male (%)	135 (57.9)	157 (58.8)	0.917	
5	nihss (median [IQR])	17.00 [14.00, 21.00]	18.00 [14.00, 22.00]	0.453	nonnorm
6	location = Right (%)	117 (50.2)	114 (42.7)	0.111	
7	hx.isch = Yes (%)	29 (12.4)	25 (9.4)	0.335	
8	afib = 1 (%)	66 (28.3)	69 (25.8)	0.601	
9	dm = 1 (%)	29 (12.4)	34 (12.7)	1	
10	mrakin (%)			0.922	
11	0	190 (81.5)	214 (80.1)		
12	1	21 (9.0)	29 (10.9)		
13	2	12 (5.2)	13 (4.9)		
14	> 2	10 (4.3)	11 (4.1)		
15	sbp (mean (sd))	146.03 (26.00)	145.00 (24.40)	0.647	
16	iv.altep = Yes (%)	203 (87.1)	242 (90.6)	0.267	
17	time.iv (median [IQR])	85.00 [67.00, 110.00]	87.00 [65.00, 116.00]	0.596	nonnorm
18	aspects (median [IQR])	9.00 [7.00, 10.00]	9.00 [8.00, 10.00]	0.075	nonnorm
19	ia.occlus (%)			0.795	
20	Intracranial ICA	1 (0.4)	3 (1.1)		
21	ICA with M1	59 (25.3)	75 (28.2)		
22	M1	154 (66.1)	165 (62.0)		
23	M2	18 (7.7)	21 (7.9)		
24	A1 or A2	1 (0.4)	2 (0.8)		
25	extra.ica = 1 (%)	75 (32.2)	70 (26.3)	0.179	
26	time.rand (median [IQR])	204.00 [152.00, 249.50]	196.00 [149.00, 266.00]	0.251	nonnorm
27	time.punc (median [IQR])	260.00 [212.00, 313.00]	NA [NA, NA]	NA	nonnorm

And from here, I can either drop it directly into Word, or present it as is, or start tweaking it to meet formatting needs.

18.9.2 Approach B: Produce the Table so you can cut and paste it

```
print(att2,
      nonnormal = c("age", "nihss", "time.iv", "aspects", "time.rand"),
      exact = c("location", "mrakin"),
      quote = TRUE, noSpaces = TRUE)
```

This will look like a mess by itself, but if you:

1. copy and paste that mess into Excel
2. select Text to Columns from the Data menu
3. select Delimited, then Space and select Treat consecutive delimiters as one

you should get something usable again.

Or, in Word,

1. insert the text
2. select the text with your mouse
3. select Insert ... Table ... Convert Text to Table
4. place a quotation mark in the “Other” area under Separate text at ...

After dropping blank columns, the result looks pretty good.

18.10 A Controlled Biological Experiment - The Blood-Brain Barrier

My source for the data and the following explanatory paragraph is page 307 from @RamseySchafer2002. The original data come from @Barnett1995.

The human brain (and that of rats, coincidentally) is protected from the bacteria and toxins that course through the bloodstream by something called the blood-brain barrier. After a method of disrupting the barrier was developed, researchers tested this new mechanism, as follows. A series of 34 rats were inoculated with human lung cancer cells to induce brain tumors. After 9-11 days they were infused with either the barrier disruption (BD) solution or, as a control, a normal saline (NS) solution. Fifteen minutes later, the rats received a standard dose of a particular therapeutic antibody (L6-F(ab')₂). The key measure of the effectiveness of transmission across the brain-blood barrier is the ratio of the antibody concentration in the brain tumor to the antibody concentration in normal tissue outside the brain. The rats were then sacrificed, and the amounts of antibody in the brain tumor and in normal tissue from the liver were measured. The study's primary objective is to determine whether the antibody concentration in the tumor increased when the blood-barrier disruption infusion was given, and if so, by how much?

18.11 The bloodbrain.csv file

Consider the data, available on [our Data and Code website](#) in the `bloodbrain.csv` file, which includes the following variables:

Variable	Description
<code>case</code>	identification number for the rat (1 - 34)
<code>brain</code>	an outcome: Brain tumor antibody count (per gram)
<code>liver</code>	an outcome: Liver antibody count (per gram)
<code>tlratio</code>	an outcome: tumor / liver concentration ratio
<code>solution</code>	the treatment: BD (barrier disruption) or NS (normal saline)
<code>sactime</code>	a design variable: Sacrifice time (hours; either 0.5, 3, 24 or 72)
<code>postin</code>	covariate: Days post-inoculation of lung cancer cells (9, 10 or 11)
<code>sex</code>	covariate: M or F
<code>wt.init</code>	covariate: Initial weight (grams)
<code>wt.loss</code>	covariate: Weight loss (grams)
<code>wt.tumor</code>	covariate: Tumor weight (10^{-4} grams)

And here's what the data look like in R.

```
bloodbrain
```

```
# A tibble: 34 x 11
  case brain  liver tlratio solution sactime postin sex  wt.init wt.loss
  <dbl> <dbl>   <dbl>   <dbl> <chr>      <dbl>   <dbl> <chr>   <dbl>   <dbl>
1     1     1 41081 1456164 0.0282 BD         0.5     10 F      239     5.9
2     2     2 44286 1602171 0.0276 BD         0.5     10 F      225      4
3     3     3 102926 1601936 0.0642 BD         0.5     10 F      224    -4.9
4     4     4 25927 1776411 0.0146 BD         0.5     10 F      184     9.8
5     5     5 42643 1351184 0.0316 BD         0.5     10 F      250      6
6     6     6 31342 1790863 0.0175 NS         0.5     10 F      196     7.7
7     7     7 22815 1633386 0.0140 NS         0.5     10 F      200     0.5
8     8     8 16629 1618757 0.0103 NS         0.5     10 F      273      4
9     9     9 22315 1567602 0.0142 NS         0.5     10 F      216     2.8
10    10    10 77961 1060057 0.0735 BD         3       10 F      267     2.6
# i 24 more rows
# i 1 more variable: wt.tumor <dbl>
```

18.12 A Table 1 for bloodbrain

@Barnett1995 did not provide a Table 1 for these data, so let's build one to compare the two `solutions` (BD vs. NS) on the covariates and outcomes, plus the natural logarithm of the tumor/liver concentration ratio (`tlratio`). We'll opt to treat the sacrifice time (`sactime`) and the days post-inoculation of lung cancer cells (`postin`) as categorical rather than quantitative variables.

```
bloodbrain <- bloodbrain %>%
  mutate(logTL = log(tlratio))

dput(names(bloodbrain))
```

```
c("case", "brain", "liver", "tlratio", "solution", "sactime",
  "postin", "sex", "wt.init", "wt.loss", "wt.tumor", "logTL")
```

OK - there's the list of variables we'll need. I'll put the outcomes at the bottom of the table.

```
bb.vars <- c("sactime", "postin", "sex", "wt.init", "wt.loss",
  "wt.tumor", "brain", "liver", "tlratio", "logTL")

bb.factors <- c("sactime", "sex", "postin")

bb.att1 <- CreateTableOne(data = bloodbrain,
  vars = bb.vars,
  factorVars = bb.factors,
  strata = c("solution"))

summary(bb.att1)
```

Summary of continuous variables

```
solution: BD
```

	n	miss	p.miss	mean	sd	median	p25	p75	min	max	skew
wt.init	17	0	0	243	3e+01	2e+02	2e+02	3e+02	2e+02	3e+02	-0.39
wt.loss	17	0	0	3	5e+00	4e+00	1e+00	6e+00	-5e+00	1e+01	-0.10
wt.tumor	17	0	0	157	8e+01	2e+02	1e+02	2e+02	2e+01	4e+02	0.53
brain	17	0	0	56043	3e+04	5e+04	4e+04	8e+04	6e+03	1e+05	0.29
liver	17	0	0	672577	7e+05	6e+05	2e+04	1e+06	2e+03	2e+06	0.35
tlratio	17	0	0	2	3e+00	1e-01	6e-02	3e+00	1e-02	9e+00	1.58
logTL	17	0	0	-1	2e+00	-2e+00	-3e+00	1e+00	-4e+00	2e+00	0.08

```

      kurt
wt.init  0.7
wt.loss  0.2
wt.tumor 1.0
brain   -0.6
liver   -1.7
tlratio  1.7
logTL   -1.7
```

solution: NS

	n	miss	p.miss	mean	sd	median	p25	p75	min	max	skew
wt.init	17	0	0	240	3e+01	2e+02	2e+02	3e+02	2e+02	3e+02	0.33
wt.loss	17	0	0	4	4e+00	3e+00	2e+00	7e+00	-4e+00	1e+01	-0.09
wt.tumor	17	0	0	209	1e+02	2e+02	2e+02	3e+02	3e+01	5e+02	0.63
brain	17	0	0	23887	1e+04	2e+04	1e+04	3e+04	1e+03	5e+04	0.30
liver	17	0	0	664975	7e+05	7e+05	2e+04	1e+06	9e+02	2e+06	0.40
tlratio	17	0	0	1	2e+00	5e-02	3e-02	9e-01	1e-02	7e+00	2.27
logTL	17	0	0	-2	2e+00	-3e+00	-3e+00	-7e-02	-5e+00	2e+00	0.27

kurt

wt.init	-0.48
wt.loss	0.08
wt.tumor	0.77
brain	-0.35
liver	-1.56
tlratio	4.84
logTL	-1.61

p-values

pNormal pNonNormal

wt.init	0.807308940	0.641940278
wt.loss	0.683756156	0.876749808
wt.tumor	0.151510151	0.190482094
brain	0.001027678	0.002579901
liver	0.974853609	0.904045603
tlratio	0.320501715	0.221425879
logTL	0.351633525	0.221425879

Standardize mean differences

1 vs 2

wt.init	0.08435244
wt.loss	0.14099823
wt.tumor	0.50397184
brain	1.23884159
liver	0.01089667
tlratio	0.34611465
logTL	0.32420504

=====
Summary of categorical variables

solution: BD

var	n	miss	p.miss	level	freq	percent	cum.percent
sactime	17	0	0.0	0.5	5	29.4	29.4
				3	4	23.5	52.9
				24	4	23.5	76.5
				72	4	23.5	100.0
postin	17	0	0.0	9	1	5.9	5.9
				10	14	82.4	88.2
				11	2	11.8	100.0
sex	17	0	0.0	F	13	76.5	76.5
				M	4	23.5	100.0

solution: NS

var	n	miss	p.miss	level	freq	percent	cum.percent
sactime	17	0	0.0	0.5	4	23.5	23.5
				3	5	29.4	52.9
				24	4	23.5	76.5
				72	4	23.5	100.0
postin	17	0	0.0	9	2	11.8	11.8
				10	13	76.5	88.2
				11	2	11.8	100.0
sex	17	0	0.0	F	13	76.5	76.5
				M	4	23.5	100.0

p-values

	pApprox	pExact
sactime	0.9739246	1
postin	0.8309504	1
sex	1.0000000	1

Standardize mean differences

	1 vs 2
sactime	0.1622214
postin	0.2098877
sex	0.0000000

Note that, in this particular case, the decisions we make about normality vs. non-normality

(for quantitative variables) and the decisions we make about approximate vs. exact testing (for categorical variables) won't actually change the implications of the p values. Each approach gives similar results for each variable. Of course, that's not always true.

18.12.1 Generate final Table 1 for bloodbrain

I'll choose to treat `tlratio` and its logarithm as non-Normal, but otherwise, use t tests, but admittedly, that's an arbitrary decision, really.

```
print(bb.att1, nonnormal = c("tlratio", "logTL"))
```

	Stratified by solution	
	BD	NS
n	17	17
sactime (%)		
0.5	5 (29.4)	4 (23.5)
3	4 (23.5)	5 (29.4)
24	4 (23.5)	4 (23.5)
72	4 (23.5)	4 (23.5)
postin (%)		
9	1 (5.9)	2 (11.8)
10	14 (82.4)	13 (76.5)
11	2 (11.8)	2 (11.8)
sex = M (%)	4 (23.5)	4 (23.5)
wt.init (mean (SD))	242.82 (27.23)	240.47 (28.54)
wt.loss (mean (SD))	3.34 (4.68)	3.94 (3.88)
wt.tumor (mean (SD))	157.29 (84.00)	208.53 (116.68)
brain (mean (SD))	56043.41 (33675.40)	23887.18 (14610.53)
liver (mean (SD))	672577.35 (694479.58)	664975.47 (700773.13)
tlratio (median [IQR])	0.12 [0.06, 2.84]	0.05 [0.03, 0.94]
logTL (median [IQR])	-2.10 [-2.74, 1.04]	-2.95 [-3.41, -0.07]
	Stratified by solution	
	p	test
n		
sactime (%)	0.974	
0.5		
3		
24		
72		
postin (%)	0.831	
9		

```

10
11
sex = M (%)          1.000
wt.init (mean (SD))  0.807
wt.loss (mean (SD))  0.684
wt.tumor (mean (SD)) 0.152
brain (mean (SD))    0.001
liver (mean (SD))    0.975
tlratio (median [IQR]) 0.221 nonnorm
logTL (median [IQR])  0.221 nonnorm

```

Or, we can get an Excel-readable version placed in a `data` subfolder, using

```

bb.t1 <- print(bb.att1, nonnormal = c("tlratio", "logTL"), quote = FALSE,
               noSpaces = TRUE, printToggle = FALSE)

write.csv(bb.t1, file = "data/bb-table1.csv")

```

which, when dropped into Excel, will look like this:

	A	B	C	D	E
1		BD	NS	p	test
2	n	17	17		
3	sex = M (%)	4 (23.5)	4 (23.5)	1	
4	sactime (%)			0.974	
5		0.5 5 (29.4)	4 (23.5)		
6		3 4 (23.5)	5 (29.4)		
7		24 4 (23.5)	4 (23.5)		
8		72 4 (23.5)	4 (23.5)		
9	postin (%)			0.831	
10		9 1 (5.9)	2 (11.8)		
11		10 14 (82.4)	13 (76.5)		
12		11 2 (11.8)	2 (11.8)		
13	wt.init (mean (sd))	242.82 (27.23)	240.47 (28.54)	0.807	
14	wt.loss (mean (sd))	3.34 (4.68)	3.94 (3.88)	0.684	
15	wt.tumor (mean (sd))	157.29 (84.00)	208.53 (116.68)	0.152	
16	brain (mean (sd))	56043.41 (33675.40)	23887.18 (14610.53)	0.001	
17	liver (mean (sd))	672577.35 (694479.58)	664975.47 (700773.13)	0.975	
18	tlratio (median [IQR])	0.12 [0.06, 2.84]	0.05 [0.03, 0.94]	0.221 nonnorm	
19	logTL (median [IQR])	-2.10 [-2.74, 1.04]	-2.95 [-3.41, -0.07]	0.221 nonnorm	
20					

One thing I would definitely clean up here, in practice, is to change the presentation of the p value for **sex** from 1 to > 0.99 , or just omit it altogether. I'd also drop the **computer-ese** where possible, add units for the measures, round **a lot**, identify the outcomes carefully, and use notes to indicate deviations from the main approach.

18.12.2 A More Finished Version (after Cleanup in Word)

Table 1. Comparing Rats Receiving BD to those Receiving NS on Available Covariates and Design Variables, and Key Outcomes

	Barrier Disruption (BD: treatment)	Normal Saline (NS: control)	p
# of Rats	17	17	
Sex = Male	4 (23.5)	4 (23.5)	-
Sacrifice Time (hours)			0.97
0.5	5 (29.4)	4 (23.5)	
3	4 (23.5)	5 (29.4)	
24	4 (23.5)	4 (23.5)	
72	4 (23.5)	4 (23.5)	
Days post-inoculation of lung cancer cells			0.83
9	1 (5.9)	2 (11.8)	
10	14 (82.4)	13 (76.5)	
11	2 (11.8)	2 (11.8)	
Initial Weight (g)	243 (27)	240 (29)	0.81
Weight Loss (g)	3.3 (4.7)	3.9 (3.9)	0.68
Tumor Weight (10 ⁻⁴ g)	157.3 (84.0)	208.5 (116.7)	0.15
Key Outcomes: mean (sd) unless otherwise indicated			
Brain Tumor Antibody Count (per g)	56,043 (33,675)	23,887 (14,611)	0.001
Liver Antibody Count (per g)	672,577 (694,480)	664,975 (700,773)	0.98
Tumor/Liver Ratio (median [Q25, Q75])	0.12 [0.06, 2.84]	0.05 [0.03, 0.94]	0.22
Natural Log of Tumor/Liver Ratio (median [Q25, Q75])	-2.10 [-2.74, 1.04]	-2.95 [-3.41, -0.07]	0.22

Table 1 Notes:

- Categorical variables are summarized with counts, percentages and p values based on approximate chi-square tests.
- Continuous variables, unless otherwise indicated, are summarized with means, standard deviations and p values based on t tests.
- The Tumor / Liver ratio and its natural logarithm are summarized with the median and quartiles and a p value from a non-parametric (Wilcoxon signed rank) test.

19 Logistic Regression: The Foundations

Sources for this material include @Harrell2001, @HarrellRMSnotes, @RamseySchafer2002 (chapters 20-21), @Vittinghoff2012 (chapter 5) and @Faraway2006 (chapter 2).

19.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(tidyverse)

theme_set(theme_bw())
```

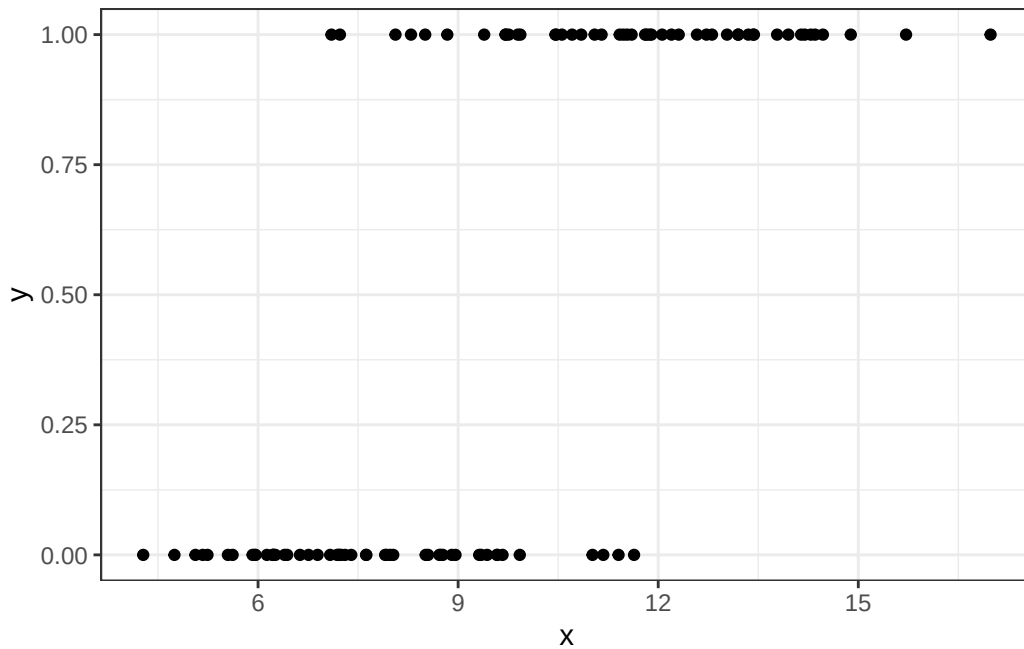
19.2 A First Attempt: A Linear Probability Model

Suppose we want to predict a binary outcome which takes on the value 1 or 0, based on a single quantitative predictor. Let y be a 1/0 outcome, and x be a quantitative predictor in the following simulation.

```
set.seed(432)
sim12 <- tibble(x = rnorm(100, 10, 3),
               err = rnorm(100, 0, 2),
               y = ifelse(x + err > 10, 1, 0))

sim12 <- select(sim12, x, y)

ggplot(sim12, aes(x = x, y = y)) + geom_point()
```



Now, we want to use our variable x here to predict our variable y (which takes on the values 0 and 1).

One approach to doing this would be a linear probability model, as follows:

```
mod12a <- lm(y ~ x, data = sim12)
summary(mod12a)
```

Call:

```
lm(formula = y ~ x, data = sim12)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.74104	-0.23411	-0.02894	0.23117	0.83153

Coefficients:

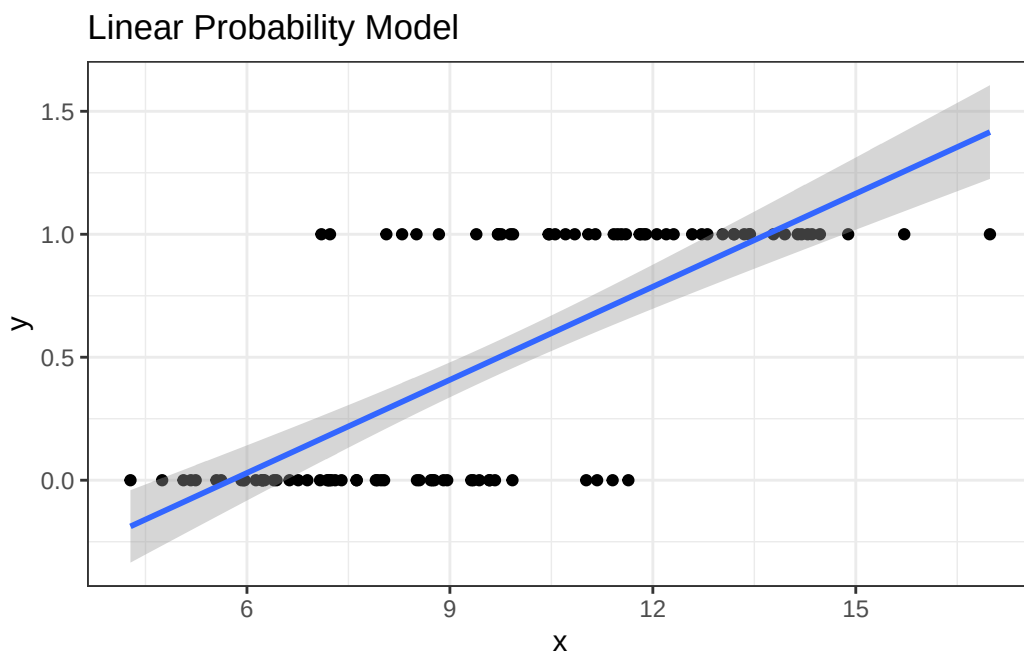
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.72761	0.12272	-5.929	4.57e-08 ***
x	0.12620	0.01219	10.349	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3491 on 98 degrees of freedom
Multiple R-squared: 0.5222, Adjusted R-squared: 0.5173
F-statistic: 107.1 on 1 and 98 DF, p-value: < 2.2e-16

Here's a picture of this model. What's wrong here?

```
ggplot(sim12, aes(x = x, y = y)) +  
  geom_point() +  
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE) +  
  labs(title = "Linear Probability Model")
```



If y can only take the values 0 and 1 (or, more precisely, if we're trying to predict the value $\pi = \Pr(y = 1)$) then what do we do with the predictions that are outside the range of $(0, 1)$?

19.3 Logistic Regression

Logistic regression is the most common model used when the outcome is binary. Our response variable is assumed to take on two values, zero or one, and we then describe the probability of a “one” response, given a linear function of explanatory predictors. We use logistic regression rather than linear regression for predicting binary outcomes. Linear regression approaches to

the problem of predicting probabilities are problematic for several reasons - not least of which being that they predict probabilities greater than one and less than zero. There are several available alternatives, including probit regression and binomial regression, for the problem of predicting a binary outcome.

Logistic regression is part of a class called **generalized linear models** which extend the linear regression model in a variety of ways. There are also several extensions to the logistic regression model, including multinomial logistic regression (which is used for nominal categorical outcomes with more than two levels) and ordered logistic regression (used for ordered multi-categorical outcomes.) The methods involved in binary logistic regression may also be extended to the case where the outcomes are proportions based on counts, often through grouped binary responses (the proportion of cells with chromosomal aberrations, or the proportion of subjects who develop a particular condition.)

Although the models are different in some crucial ways, the practical use of logistic regression tracks well with much of what we've learned about linear regression.

19.4 The Logistic Regression Model

A generalized linear model (or GLM) is a probability model in which the mean of an outcome is related to predictors through a regression equation. A link function g is used to relate the mean, μ , to a linear regression of the predictors $X_1, X_2, \dots, X_k$.

$$g(\mu) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

In the case of a logistic regression model,

- the mean μ of our 0/1 outcome is represented by π which describes the probability of a “1” outcome.
- the linking function we use in logistic regression makes use of the logit function, which is built on the natural logarithm.

19.5 The Link Function

Logistic regression is a non-linear regression approach, since the equation for the mean of the 0/1 Y values conditioned on the values of our predictors $X_1, X_2, \dots, X_k$ turns out to be non-linear in the β coefficients. Its nonlinearity, however, is solely found in its link function, hence the term *generalized* linear model.

The particular link function we use in logistic regression is called the **logit link**.

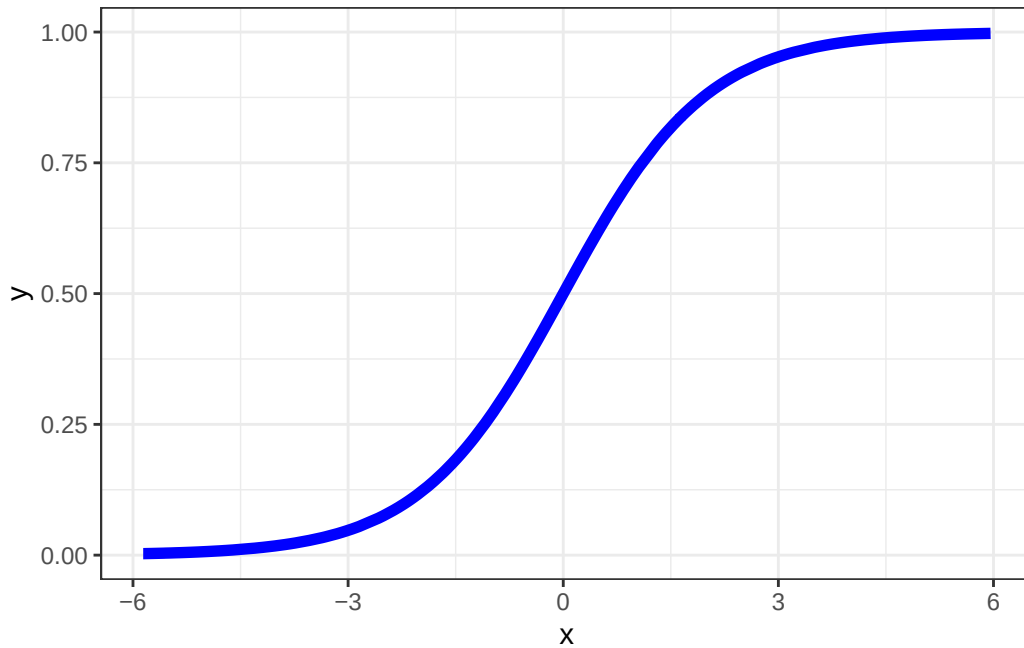
$$\text{logit}(\pi) = \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

The inverse of the logit function is called the **logistic function**. If $\text{logit}(\pi) = \eta$, then $\pi = \frac{\exp(\eta)}{1+\exp(\eta)}$

The plot below displays the logistic function $y = \frac{e^x}{1+e^x}$

```
set.seed(43201)
temp <- tibble(
  x = runif(200, min = -6, max = 6),
  y = exp(x) / (1 + exp(x)))

ggplot(temp, aes(x = x, y = y)) +
  geom_line(linewidth = 2, col = "blue")
```



As you can see in the figure above, the logistic function $\frac{e^x}{1+e^x}$ takes any value x in the real numbers and returns a value between 0 and 1.

19.6 The logit or log odds

We usually focus on the **logit** in statistical work, which is the inverse of the logistic function.

- If we have a probability $\pi < 0.5$, then $\text{logit}(\pi) < 0$.
- If our probability $\pi > 0.5$, then $\text{logit}(\pi) > 0$.
- Finally, if $\pi = 0.5$, then $\text{logit}(\pi) = 0$.

19.7 Interpreting the Coefficients of a Logistic Regression Model

The critical thing to remember in interpreting a logistic regression model is that the logit is the log odds function. Exponentiating the logit yields the odds.

So, suppose we have a yes/no outcome variable, where yes = 1, and no = 0, and $\pi = \Pr(y = 1)$. Our model holds that:

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

The odds of a yes response (the odds that $Y = 1$) at the level $X_1, X_2, \dots, X_k$ are:

$$\text{Odds}(Y = 1) = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)$$

The **probability** of a yes response ($\Pr(y = 1)$, or π) is just

$$\pi = \Pr(Y = 1) = \frac{\text{Odds}(Y = 1)}{1 + \text{Odds}(Y = 1)} = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}$$

19.8 The Logistic Regression has non-constant variance

In ordinary least squares regression, the variance $\text{Var}(Y|X_1, X_2, \dots, X_k) = \sigma^2$ is a constant that does not depend on the predictor values. This is not the case in logistic regression. The mean and variance specifications of the logistic regression model are quite different.

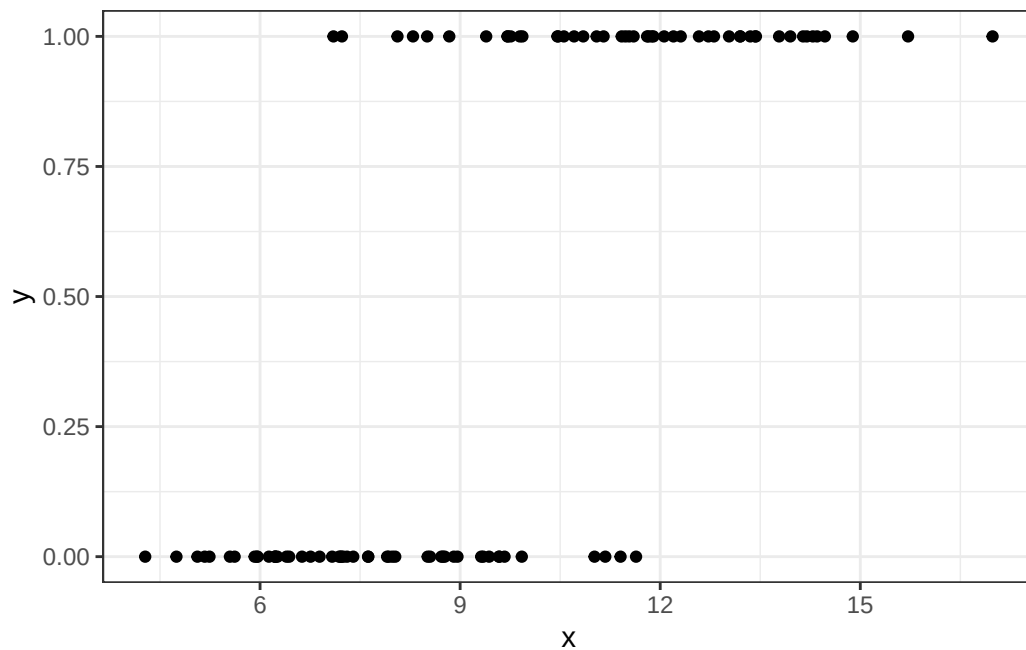
$$\text{logit}(\pi) = \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k \quad \mu[Y|X_1, \dots, X_k] = \pi, \quad \text{Var}[Y|X_1, \dots, X_k] = \pi(1-\pi)$$

The variance is now a function of the mean, and contains no additional parameter for us to estimate.

19.9 Fitting a Logistic Regression Model to our Simulated Data

Recall the `sim12` data we built earlier.

```
ggplot(sim12, aes(x = x, y = y)) + geom_point()
```



Here is the fitted logistic regression model.

```
model12b <- glm(y ~ x, data = sim12, family = binomial)
```

```
model12b
```

```
Call: glm(formula = y ~ x, family = binomial, data = sim12)
```

```
Coefficients:
```

```
(Intercept)          x  
    -9.1955      0.9566
```

```
Degrees of Freedom: 99 Total (i.e. Null); 98 Residual
```

```
Null Deviance:      138.6
```

```
Residual Deviance: 70.03    AIC: 74.03
```

The logistic regression equation is:

$$\text{logit}(\Pr(y = 1)) = \log\left(\frac{\Pr(y = 1)}{1 - \Pr(y = 1)}\right) = -9.1955 + 0.9566x$$

We can exponentiate the results of this model to get to an equation about odds, and eventually, a prediction about probabilities. Suppose, for instance, that we are interested in the prediction when $x = 12$.

$$\text{logit}(\Pr(y = 1)|X = 12) = \log\left(\frac{\Pr(y = 1)}{1 - \Pr(y = 1)}\right) = -9.1955 + 0.9566 * 12 = 2.2837$$

And we can also get this from the `predict` function applied to our model, although the `predict` approach retains a few more decimal places internally:

```
predict(model12b, newdata = data.frame(x = 12))
```

```
1  
2.284069
```

$$\text{Odds}(Y = 1|X = 12) = \exp(-9.20 + 0.96 * 12) = \exp(2.2837) = 9.812921$$

```
exp(predict(model12b, newdata = data.frame(x = 12)))
```

```
1  
9.81654
```

The estimated **probability** of a yes response ($\Pr(y = 1)$, or π) if $x = 12$ is just

$$\pi = \Pr(Y = 1|X = 12) = \frac{\text{Odds}(Y = 1|X = 12)}{1 + \text{Odds}(Y = 1|X = 12)} = \frac{\exp(-9.20 + 0.96x)}{1 + \exp(-9.20 + 0.96x)} = \frac{9.812921}{1 + 9.812921} = 0.908$$

Does this work out?

```
exp(predict(model12b, newdata = data.frame(x = 12))) /  
(1 + exp(predict(model12b, newdata = data.frame(x = 12))))
```

```
1
0.907549
```

which is also directly available by running `predict` with `type = "response"`.

```
predict(model12b, newdata = data.frame(x = 12), type = "response")
```

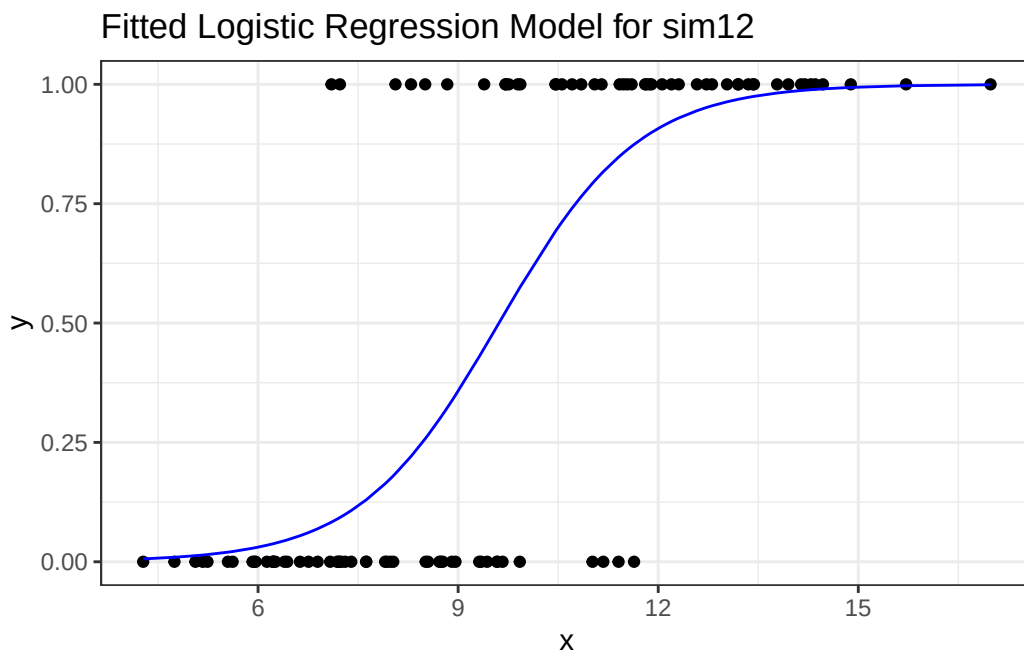
```
1
0.907549
```

19.10 Plotting the Logistic Regression Model

We can use the `augment` function from the `broom` package to get our fitted probabilities included in the data.

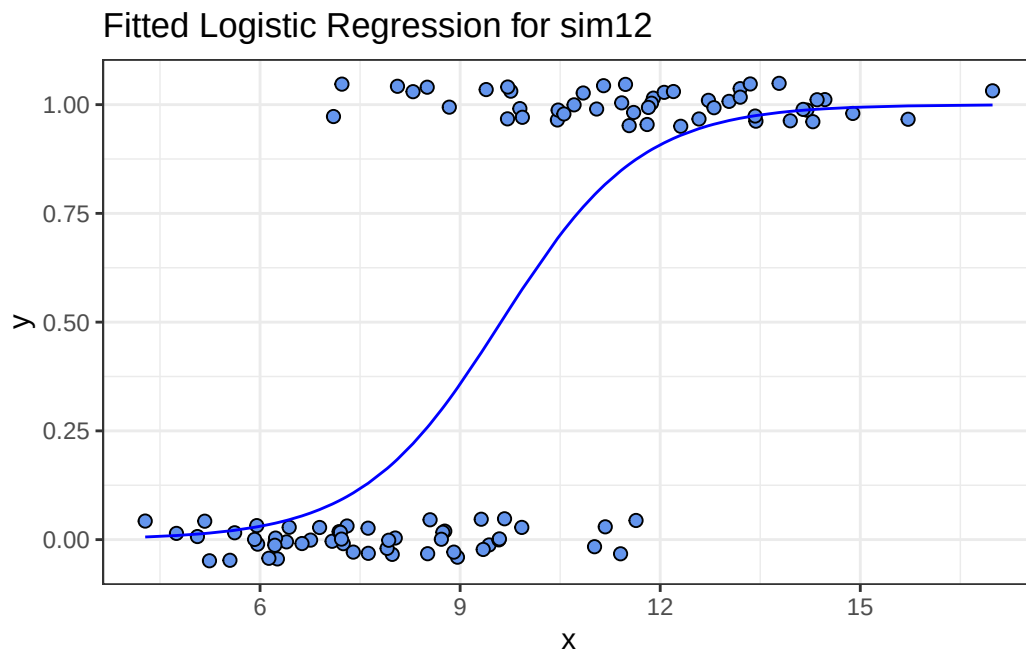
```
m12b_aug <- augment(model12b, sim12, type.predict = "response")

ggplot(m12b_aug, aes(x = x, y = y)) +
  geom_point() +
  geom_line(aes(x = x, y = .fitted), col = "blue") +
  labs(title = "Fitted Logistic Regression Model for sim12")
```



I'll add a little jitter on the vertical scale to the points, so we can avoid overlap, and also make the points a little bigger.

```
ggplot(m12b_aug, aes(x = x, y = y)) +  
  geom_jitter(height = 0.05, size = 2, pch = 21,  
             fill = "cornflowerblue") +  
  geom_line(aes(x = x, y = .fitted), col = "blue") +  
  labs(title = "Fitted Logistic Regression for sim12") +  
  theme_bw()
```



All right, it's time to move on to fitting models. We'll do that next.

20 Logistic Regression with glm

20.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(caret)
library(ROCR)
library(pROC)
library(broom)
library(mosaic)
library(naniar)
library(tidyverse)

theme_set(theme_bw())
```

20.1.1 Data Load

```
resect <- read_csv("data/resect.csv", show_col_types = FALSE)
```

20.2 The resect data

My source for these data was @Riffenburgh2006. The data describe 134 patients who had undergone resection of the tracheal carina (most often this is done to address tumors in the trachea), and the `resect.csv` data file contains the following variables:

- `id` = a patient ID #,
 - `age` = the patient's age at surgery,
 - `prior` = prior tracheal surgery (1 = yes, 0 = no),
 - `resection` = extent of the resection (in cm),
 - `intubated` = whether intubation was required at the end of surgery (1 = yes, 0 = no),
- and

- died = the patient's death status (1 = dead, 0 = alive).

```
miss_var_summary(resect)
```

```
# A tibble: 6 x 3
  variable n_miss pct_miss
  <chr>      <int>    <num>
1 subj_id      0        0
2 age          0        0
3 prior        0        0
4 resection    0        0
5 intubated    0        0
6 died         0        0
```

```
resect |> count(died, prior)
```

```
# A tibble: 4 x 3
  died prior    n
  <dbl> <dbl> <int>
1     0     0   89
2     0     1   28
3     1     0   11
4     1     1    6
```

```
resect |> inspect()
```

quantitative variables:

	name	class	min	Q1	median	Q3	max	mean	sd	n
1	subj_id	numeric	1	34.25	67.5	100.75	134	67.5000000	38.8265373	134
2	age	numeric	8	36.00	51.0	61.00	80	47.8432836	15.7775202	134
3	prior	numeric	0	0.00	0.0	0.75	1	0.2537313	0.4367785	134
4	resection	numeric	1	2.00	2.5	4.00	6	2.9634328	1.2402123	134
5	intubated	numeric	0	0.00	0.0	0.00	1	0.1417910	0.3501447	134
6	died	numeric	0	0.00	0.0	0.00	1	0.1268657	0.3340713	134

missing

1	0
2	0
3	0
4	0
5	0
6	0

We have no missing data, and 17 of the 134 patients died. Our goal will be to understand the characteristics of the patients, and how they relate to the binary outcome of interest, death.

20.3 Running A Simple Logistic Regression Model

In the most common scenario, a logistic regression model is used to predict a binary outcome (which can take on the values 0 or 1.) We will eventually fit a logistic regression model in two ways.

1. Through the `glm` function in the base package of R (similar to `lm` for linear regression)
2. Through the `lrm` function available in the `rms` package (similar to `ols` for linear regression)

We'll focus on the `glm` approach first, and save the `lrm` ideas for later in this Chapter.

20.3.1 Logistic Regression Can Be Harder than Linear Regression

- Logistic regression models are fitted using the method of maximum likelihood in `glm`, which requires multiple iterations until convergence is reached.
- Logistic regression models are harder to interpret (for most people) than linear regressions.
- Logistic regression models don't have the same set of assumptions as linear models.
- Logistic regression outcomes (yes/no) carry much less information than quantitative outcomes. As a result, fitting a reasonable logistic regression requires more data than a linear model of similar size.
 - The rule I learned in graduate school was that a logistic regression requires 100 observations to fit an intercept plus another 15 observations for each candidate predictor. That's not terrible, but it's a very large sample size.
 - Frank Harrell recommends that 96 observations + a function of the number of candidate predictors (which depends on the amount of variation in the predictors, but 15 x the number of such predictors isn't too bad if the signal to noise ratio is pretty good) are required just to get reasonable confidence intervals around your predictions.
 - * In a [twitter note](#), Frank suggests that $96 + 8 \times \text{number of candidate parameters}$ might be reasonable so long as the smallest cell of interest (combination of an outcome and a split of the covariates) is 96 or more observations.
 - @Peduzzi1996 suggest that if we let π be the smaller of the proportions of “yes” or “no” cases in the population of interest, and k be the number of inputs under consideration, then $N = 10k/\pi$ is the minimum number of cases to include, except that if $N < 100$ by this standard, you should increase it to 100, according to @Long1997.

- * That suggests that if you have an outcome that happens 10% of the time, and you are running a model with 3 predictors, then you could get away with $(10 \times 3)/(0.10) = 300$ observations, but if your outcome happened 40% of the time, you could get away with only $(10 \times 3)/(0.40) = 75$ observations, which you'd round up to 100.

20.4 Logistic Regression using glm

We'll begin by attempting to predict death based on the extent of the resection.

```
res_modA <- glm(died ~ resection, data=resect,  
               family="binomial"(link="logit"))  
  
res_modA
```

```
Call:  glm(formula = died ~ resection, family = binomial(link = "logit"),  
          data = resect)
```

Coefficients:

(Intercept)	resection
-4.4337	0.7417

Degrees of Freedom: 133 Total (i.e. Null); 132 Residual

Null Deviance: 101.9

Residual Deviance: 89.49 AIC: 93.49

Note that the logit link is the default approach with the binomial family, so we could also have used:

```
res_modA <- glm(died ~ resection, data = resect,  
               family = "binomial")
```

which yields the same model.

20.4.1 Interpreting the Coefficients of a Logistic Regression Model

Our model is:

$$\begin{aligned}\text{logit}(\text{died} = 1) &= \log \left(\frac{\text{Pr}(\text{died} = 1)}{1 - \text{Pr}(\text{died} = 1)} \right) \\ &= \beta_0 + \beta_1 x = -4.4337 + 0.7417 \times \text{resection}\end{aligned}$$

The predicted log odds of death for a subject with a resection of 4 cm is:

$$\log \left(\frac{\text{Pr}(\text{died} = 1)}{1 - \text{Pr}(\text{died} = 1)} \right) = -4.4337 + 0.7417 \times 4 = -1.467$$

The predicted odds of death for a subject with a resection of 4 cm is thus:

$$\frac{\text{Pr}(\text{died} = 1)}{1 - \text{Pr}(\text{died} = 1)} = e^{-4.4337 + 0.7417 \times 4} = e^{-1.467} = 0.2306$$

Since the odds are less than 1, we should find that the probability of death is less than 1/2. With a little algebra, we see that the predicted probability of death for a subject with a resection of 4 cm is:

$$\text{Pr}(\text{died} = 1) = \frac{e^{-4.4337 + 0.7417 \times 4}}{1 + e^{-4.4337 + 0.7417 \times 4}} = \frac{e^{-1.467}}{1 + e^{-1.467}} = \frac{0.2306}{1.2306} = 0.187$$

In general, we can frame the model in terms of a statement about probabilities, like this:

$$\text{Pr}(\text{died} = 1) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} = \frac{e^{-4.4337 + 0.7417 \times \text{resection}}}{1 + e^{-4.4337 + 0.7417 \times \text{resection}}}$$

and so by substituting in values for **resection**, we can estimate the model's fitted probabilities of death.

20.4.2 Using predict to describe the model's fits

To obtain these fitted odds and probabilities in R, we can use the `predict` function.

- The default predictions are on the scale of the log odds. These predictions are also available through the `type = "link"` command within the `predict` function for a generalized linear model like logistic regression.
- Here are the predicted log odds of death for a subject (Sally) with a 4 cm resection and a subject (Harry) who had a 5 cm resection.

```
predict(res_modA, newdata = tibble(resection = c(4,5)))
```

```
      1      2  
-1.4669912 -0.7253027
```

- We can also obtain predictions for each subject on the original response (here, probability) scale, backing out of the logit link.

```
predict(res_modA, newdata = tibble(resection = c(4, 5)),  
       type = "response")
```

```
      1      2  
0.1874004 0.3262264
```

So the predicted probability of death is 0.187 for Sally, the subject with a 4 cm resection, and 0.326 for Harry, the subject with a 5 cm resection.

20.4.3 Odds Ratio interpretation of Coefficients

Often, we will exponentiate the estimated slope coefficients of a logistic regression model to help us understand the impact of changing a predictor on the odds of our outcome.

```
exp(coef(res_modA))
```

```
(Intercept)  resection  
0.01186995  2.09947754
```

To interpret this finding, suppose we have two subjects, Harry and Sally. Harry had a resection that was 1 cm larger than Sally. This estimated coefficient suggests that the estimated odds for death associated with Harry is 2.099 times larger than the odds for death associated with Sally. In general, the odds ratio comparing two subjects who differ by 1 cm on the resection length is 2.099.

To illustrate, again let's assume that Harry's resection was 5 cm, and Sally's was 4 cm. Then we have:

$$\log \left(\frac{Pr(Harrydied)}{1 - Pr(Harrydied)} \right) = -4.4337 + 0.7417 \times 5 = -0.7253$$

$$\log \left(\frac{Pr(Sallydied)}{1 - Pr(Sallydied)} \right) = -4.4337 + 0.7417 \times 4 = -1.4667.$$

which implies that our estimated odds of death for Harry and Sally are:

$$Odds(Harrydied) = \frac{Pr(Harrydied)}{1 - Pr(Harrydied)} = e^{-4.4337+0.7417 \times 5} = e^{-0.7253} = 0.4842$$

$$Odds(Sallydied) = \frac{Pr(Sallydied)}{1 - Pr(Sallydied)} = e^{-4.4337+0.7417 \times 4} = e^{-1.4667} = 0.2307$$

and so the odds ratio is:

$$OR = \frac{Odds(Harrydied)}{Odds(Sallydied)} = \frac{0.4842}{0.2307} = 2.099$$

- If the odds ratio was 1, that would mean that Harry and Sally had the same estimated odds of death, and thus the same estimated probability of death, despite having different sizes of resections.
- Since the odds ratio is greater than 1, it means that Harry has a higher estimated odds of death than Sally, and thus that Harry has a higher estimated probability of death than Sally.
- If the odds ratio was less than 1, it would mean that Harry had a lower estimated odds of death than Sally, and thus that Harry had a lower estimated probability of death than Sally.

Remember that the odds ratio is a fraction describing two positive numbers (odds can only be non-negative) so that the smallest possible odds ratio is 0.

20.4.4 Interpreting the rest of the model output from glm

```
res_modA
```

```
Call: glm(formula = died ~ resection, family = "binomial", data = resect)
```

```
Coefficients:
```

```
(Intercept)    resection  
    -4.4337      0.7417
```

```
Degrees of Freedom: 133 Total (i.e. Null);  132 Residual
```

```
Null Deviance:      101.9
```

```
Residual Deviance: 89.49    AIC: 93.49
```

In addition to specifying the logistic regression coefficients, we are also presented with information on degrees of freedom, deviance (null and residual) and AIC.

- The degrees of freedom indicate the sample size.
 - Recall that we had $n = 134$ subjects in the data. The “Null” model includes only an intercept term (which uses 1 df) and we thus have $n - 1$ (here 133) degrees of freedom available for estimation.
 - In our `res_modA` model, a logistic regression is fit including a single slope (resection) and an intercept term. Each uses up one degree of freedom to build an estimate, so we have $n - 2 = 134 - 2 = 132$ residual df remaining.
- The AIC or Akaike Information Criterion (lower values are better) is also provided. This is helpful if we’re comparing multiple models for the same outcome.

20.4.5 Deviance and Comparing Our Model to the Null Model

- The deviance (a measure of the model’s *lack of fit*) is available for both the null model (the model with only an intercept) and for our model (`res_modA`) predicting our outcome, mortality.
- The deviance test, though available in R (see below) isn’t really a test of whether the model works well. Instead, it assumes the model is true, and then tests to see if the coefficients are different from zero. So it isn’t of much practical use.
 - To compare the **deviance** statistics, we can subtract the residual deviance from the null deviance to describe the impact of our model on fit.

- Null Deviance - Residual Deviance can be compared to a χ^2 distribution with Null DF - Residual DF degrees of freedom to obtain a global test of the in-sample predictive power of our model.
- We can see this comparison more directly by running `anova` on our model:

```
anova(res_modA, test = "LRT")
```

Analysis of Deviance Table

Model: binomial, link: logit

Response: died

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			133	101.943	
resection	1	12.45	132	89.493	0.0004179 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The `test = "LRT"` section completes a deviance test and provides a p value, which just estimates the probability that a chi-square distribution with a single degree of freedom would exhibit an improvement in deviance as large as 12.45.

The p value for the deviance test here is about 0.0004. But, again, this isn't a test of whether the model is any good - it assumes the model is true, and then tests some consequences.

- Specifically, it tests whether (if the model is TRUE) some of the model's coefficients are non-zero.
- That's not so practically useful, so I discourage you from performing global tests of a logistic regression model with a deviance test.

20.4.6 Using `glance` with a logistic regression model

We can use the `glance` function from the `broom` package to obtain the null and residual deviance and degrees of freedom. Note that the deviance for our model is related to the log likelihood by $-2 \times \log \text{Lik}$.

```
glance(res_modA)
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik   AIC   BIC deviance df.residual  nobs
      <dbl>     <int>  <dbl> <dbl> <dbl>   <dbl>       <int> <int>
1      102.       133  -44.7  93.5  99.3    89.5        132   134
```

The `glance` result also provides the AIC, and the BIC (Bayes Information Criterion), each of which is helpful in understanding comparisons between multiple models for the same outcome (with smaller values of either criterion indicating better models.) The AIC is based on the deviance, but penalizes you for making the model more complicated. The BIC does the same sort of thing but with a different penalty.

Again we see that we have a null deviance of 101.94 on 133 degrees of freedom. Including the `resection` information in the model decreased the deviance to 89.49 points on 132 degrees of freedom, so that's a decrease of 12.45 points while using one degree of freedom, which looks like a meaningful reduction in deviance.

20.5 Interpreting the Model Summary

Let's get a more detailed summary of our `res_modA` model, including 95% confidence intervals for the coefficients:

```
summary(res_modA)
```

Call:

```
glm(formula = died ~ resection, family = "binomial", data = resect)
```

Coefficients:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -4.4337      0.8799  -5.039 4.67e-07 ***
resection      0.7417      0.2230   3.327 0.000879 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 101.943  on 133  degrees of freedom
Residual deviance:  89.493  on 132  degrees of freedom
AIC: 93.493
```

```
Number of Fisher Scoring iterations: 5
```



```
confint(res_modA, level = 0.95)
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-6.344472	-2.855856
resection	0.322898	1.208311

Some elements of this summary are very familiar from our work with linear models.

- We still have a five-number summary of residuals, although these are called *deviance* residuals.
- We have a table of coefficients with standard errors, and hypothesis tests, although these are Wald z-tests, rather than the t tests we saw in linear modeling.
- We have a summary of global fit in the comparison of null deviance and residual deviance, but without a formal p value. And we have the AIC, as discussed above.
- We also have some new items related to a *dispersion* parameter and to the number of Fisher Scoring Iterations.

Let's walk through each of these elements.

20.5.1 Wald Z tests for Coefficients in a Logistic Regression

The coefficients output provides the estimated coefficients, and their standard errors, plus a Wald Z statistic, which is just the estimated coefficient divided by its standard error. This is compared to a standard Normal distribution to obtain the two-tailed p values summarized in the $\Pr(>|z|)$ column.

- The interesting result is **resection**, which has a Wald $Z = 3.327$, yielding a p value of 0.00088.
- The p value assesses whether the estimated coefficient of **resection**, 0.7417, is different from 0. If the coefficient (on the logit scale) for **resection** was truly 0, this would mean that:
 - the log odds of death did not change based on the **resection** size,
 - the odds of death were unchanged based on the **resection** size (the odds ratio would be 1), and
 - the probability of death was unchanged based on the **resection** size.

In our case, we have a change in the log odds of **died** associated with changes in **resection**, according to this p value. We conclude that **resection** size is associated with a positive impact on death rates (death rates are generally higher for people with larger resections.)

20.5.2 Confidence Intervals for the Coefficients

As in linear regression, we can obtain 95% confidence intervals (to get other levels, change the `level` parameter in `confint`) for the intercept and slope coefficients.

Here, we see, for example, that the coefficient of `resection` has a point estimate of 0.7417, and a confidence interval of (0.3229, 1.208). Since this is on the logit scale, it's not that interpretable, but we will often exponentiate the model and its confidence interval to obtain a more interpretable result on the odds ratio scale.

```
tidy(res_modA, exponentiate = TRUE, conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high)
```

```
# A tibble: 2 x 4
  term      estimate conf.low conf.high
  <chr>      <dbl>   <dbl>   <dbl>
1 (Intercept) 0.0119 0.00176 0.0575
2 resection   2.10   1.38    3.35
```

From this output, we can estimate the odds ratio for death associated with a 1 cm increase in resection size is 2.099, with a 95% CI of (1.38, 3.35). - If the odds ratio was 1, it would indicate that the odds of death did not change based on the change in resection size. - Here, it's clear that the estimated odds of death will be larger (odds > 1) for subjects with larger resection sizes. Larger odds of death also indicate larger probabilities of death. This confidence interval indicates that with 95% confidence, we conclude that increases in resection size are associated with increases in the odds of death. - If the odds ratio was less than 1 (remember that it cannot be less than 0) that would mean that subjects with larger resection sizes were associated with smaller estimated odds of death.

20.5.3 Deviance Residuals

In logistic regression, it's certainly a good idea to check to see how well the model fits the data. However, there are a few different types of residuals. The residuals presented here by default are called deviance residuals. Other types of residuals are available for generalized linear models, such as Pearson residuals, working residuals, and response residuals. Logistic regression model diagnostics often make use of multiple types of residuals.

The deviance residuals for each individual subject sum up to the deviance statistic for the model, and describe the contribution of each point to the model likelihood function.

The deviance residual, d_i , for the i^{th} observation in a model predicting y_i (a binary variable), with the estimate being $\hat{\pi}_i$ is:

$$d_i = s_i \sqrt{-2[y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i)]},$$

where s_i is 1 if $y_i = 1$ and $s_i = -1$ if $y_i = 0$.

Again, the sum of the deviance residuals is the deviance.

20.5.4 Dispersion Parameter

The dispersion parameter is taken to be 1 for `glm` fit using either the `binomial` or `Poisson` families. For other sorts of generalized linear models, the dispersion parameter will be of some importance in estimating standard errors sensibly.

20.5.5 Fisher Scoring iterations

The solution of a logistic regression model involves maximizing a likelihood function. Fisher's scoring algorithm in our `res_modA` needed five iterations to perform the logistic regression fit. All that this tells you is that the model converged, and didn't require a lot of time to do so.

20.6 Plotting a Simple Logistic Regression Model

Let's plot the logistic regression model `res_modA` for `died` using the extent of the resection in terms of probabilities. We can use either of two different approaches:

- we can plot the fitted values from our specific model against the original data, using the `augment` function from the `broom` package, or
- we can create a smooth function called `binomial_smooth` that plots a simple logistic model in an analogous way to `geom_smooth(method = "lm")` for a simple linear regression.

20.6.1 Using `augment` to capture the fitted probabilities

```
res_A_aug <- augment(res_modA, resect,
                     type.predict = "response")
head(res_A_aug)
```

```
# A tibble: 6 x 12
  subj_id age prior resection intubated died .fitted .resid .hat .sigma
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1    34     1     2.5         0     0  0.0705 -0.382 0.0100  0.826
2     2    57     0     5         0     0  0.326  -0.889 0.0337  0.823
3     3    60     1     4         1     1  0.187   1.83  0.0120  0.811
4     4    62     1     4.2       0     0  0.211  -0.689 0.0143  0.824
5     5    28     0     6         1     1  0.504   1.17  0.0818  0.820
6     6    52     0     3         0     0  0.0990 -0.457 0.00922  0.826
# i 2 more variables: .cooksd <dbl>, .std.resid <dbl>
```

This approach augments the `resect` data set with fitted, residual and other summaries of each observation's impact on the fit, using the “response” type of prediction, which yields the fitted probabilities in the `.fitted` column.

20.6.2 Plotting a Logistic Regression Model's Fitted Values

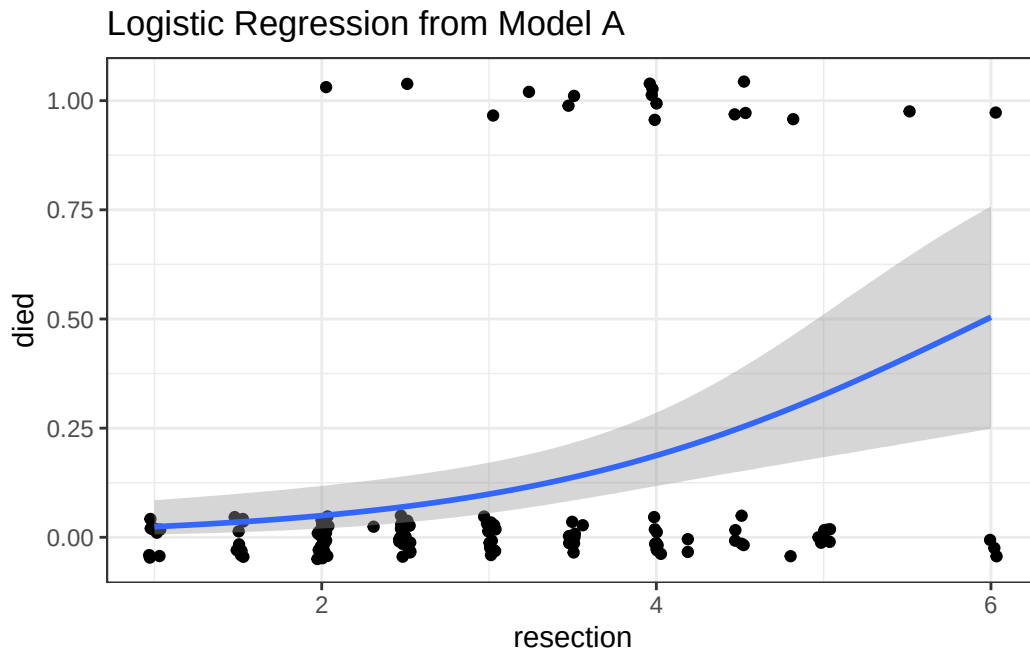
```
ggplot(res_A_aug, aes(x = resection, y = died)) +
  geom_jitter(height = 0.05) +
  geom_line(aes(x = resection, y = .fitted),
            col = "blue") +
  labs(title = "Logistic Regression from Model res_modA")
```



20.6.3 Plotting a Simple Logistic Model using `binomial_smooth`

```
binomial_smooth <- function(...) {
  geom_smooth(method = "glm", formula = y ~ x,
             method.args = list(family = "binomial"), ...)
}

ggplot(resect, aes(x = resection, y = died)) +
  geom_jitter(height = 0.05) +
  binomial_smooth() + ## ...smooth(se=FALSE) to leave out interval
  labs(title = "Logistic Regression from Model A")
```



As expected, we see an increase in the model probability of death as the extent of the resection grows larger.

20.7 How well does Model A classify subjects?

A natural question to ask is how well does our model classify patients in terms of likelihood of death.

We could specify a particular rule, for example: if the predicted probability of death is 0.5 or greater, then predict “Died”.

```
res_A_aug$rule.5 <- ifelse(res_A_aug$.fitted >= 0.5,
                           "Predict Died", "Predict Alive")

table(res_A_aug$rule.5, res_A_aug$died)
```

	0	1
Predict Alive	114	16
Predict Died	3	1

And perhaps build the linked table of row probabilities which tells us, for example, that 87.69% of the patients predicted by the model to be alive actually did survive.

```
round(100*prop.table(
  table(res_A_aug$rule.5, res_A_aug$died), 1), 2)
```

	0	1
Predict Alive	87.69	12.31
Predict Died	75.00	25.00

Or the table of column probabilities which tell us, for example, that 97.44% of those who actually survived were predicted by the model to be alive.

```
round(100*prop.table(
  table(res_A_aug$rule.5, res_A_aug$died), 2), 2)
```

	0	1
Predict Alive	97.44	94.12
Predict Died	2.56	5.88

We'll discuss various measures of concordance derived from this sort of classification later.

20.8 The Confusion Matrix

Let's build this misclassification table in standard epidemiological format.

```
res_A_aug <- res_A_aug |>
  mutate(death_predicted = factor(.fitted >= 0.5),
         death_actual = factor(died == "1"),
         death_predicted = fct_relevel(death_predicted, "TRUE"),
         death_actual = fct_relevel(death_actual, "TRUE"))

confuseA_small <- table(res_A_aug$death_predicted, res_A_aug$death_actual)

confuseA_small
```

	TRUE	FALSE
TRUE	1	3
FALSE	16	114

In total, we have 134 observations.

- 115 correct predictions, or 85.8% accuracy
- 17 subjects who died, or 12.6% prevalence of death
- 4 subjects who were predicted to die, or 3.0% detection prevalence.

The sensitivity (also called recall) here is $1 / (1 + 16) = 5.9\%$.

- 5.9% of the subjects who actually died were predicted to die by the model.

The specificity here is $114 / (114 + 3) = 97.4\%$.

- 97.4% of the subjects who actually survived were predicted to survive by the model.

The positive predictive value (PPV: also called precision) is $1 / (1 + 3) = 25\%$

- Our predictions of death were correct 25% of the time.

The negative predictive value (NPV) is $114 / (114 + 16) = 87.7\%$

- Our predictions of survival were correct 87.7% of the time.

20.9 Using the confusionMatrix tool from the caret package

This provides a more detailed summary of the classification results from our logistic regression model.

```
confusionMatrix(
  data = factor(res_A_aug$.fitted >= 0.5),
  reference = factor(res_A_aug$died == 1),
  positive = "TRUE"
)
```

Confusion Matrix and Statistics

	Reference	
Prediction	FALSE	TRUE
FALSE	114	16


```

TRUE      3      1

      Accuracy : 0.8582
      95% CI : (0.7875, 0.9124)
No Information Rate : 0.8731
P-Value [Acc > NIR] : 0.747802

      Kappa : 0.0493

McNemar's Test P-Value : 0.005905

      Sensitivity : 0.058824
      Specificity : 0.974359
Pos Pred Value : 0.250000
Neg Pred Value : 0.876923
Prevalence : 0.126866
Detection Rate : 0.007463
Detection Prevalence : 0.029851
Balanced Accuracy : 0.516591

'Positive' Class : TRUE

```

- The No Information Rate or NIR is just the percentage of correct predictions we'd get if we just predicted the more common classification (not dead) for every subject.
- Kappa is a correlation statistic ranging from -1 to +1. It measures the inter-rater reliability of our predictions and the true classifications, in this context. Complete agreement would be +1, and complete disagreement would be -1.

20.10 Receiver Operating Characteristic Curve Analysis

One way to assess the predictive accuracy within the model development sample in a logistic regression is to consider an analyses based on the receiver operating characteristic (ROC) curve. ROC curves are commonly used in assessing diagnoses in medical settings, and in signal detection applications.

The accuracy of a “test” can be evaluated by considering two types of errors: false positives and false negatives.

In our `res_modA` model, we use `resection` size to predict whether the patient died. Suppose we established a value `R`, so that if the resection size was larger than `R` cm, we would predict that the patient died, and otherwise we would predict that the patient did not die.

A good outcome of our model's "test", then, would be when the resection size is larger than R for a patient who actually died. Another good outcome would be when the resection size is smaller than R for a patient who survived.

But we can make errors, too.

- A false positive error in this setting would occur when the resection size is larger than R (so we predict the patient dies) but in fact the patient does not die.
- A false negative error in this case would occur when the resection size is smaller than R (so we predict the patient survives) but in fact the patient dies.

Formally, the true positive fraction (TPF) for a specific resection cutoff R , is the probability of a positive test (a prediction that the patient will die) among the people who have the outcome died = 1 (those who actually die).

$$TPF(R) = Pr(\text{resection} > R | \text{subjectdied})$$

Similarly, the false positive fraction (FPF) for a specific cutoff R is the probability of a positive test (prediction that the patient will die) among the people with died = 0 (those who don't actually die)

$$FPF(R) = Pr(\text{resection} > R | \text{subjectdidnotdie})$$

The True Positive Rate is referred to as the sensitivity of a diagnostic test, and the True Negative rate (1 - the False Positive rate) is referred to as the specificity of a diagnostic test.

Since the cutoff R is not fixed in advanced, we can plot the value of TPF (on the y axis) against FPF (on the x axis) for all possible values of R , and this is what the ROC curve is. Others refer to the Sensitivity on the Y axis, and 1-Specificity on the X axis, and this is the same idea.

Before we get too far into the weeds, let me show you some simple situations so you can understand what you might learn from the ROC curve. The web page <http://blog.yhat.com/posts/roc-curves.html> provides source materials.

20.10.1 Interpreting the Area under the ROC curve

The AUC or Area under the ROC curve is the amount of space underneath the ROC curve. Often referred to as the c statistic, the AUC represents the quality of your TPR and FPR overall in a single number. The C statistic ranges from 0 to 1, with $C = 0.5$ for a prediction that is no better than random guessing, and $C = 1$ for a perfect prediction model.

Next, I'll build a simulation to demonstrate several possible ROC curves in the sections that follow.

```

set.seed(432999)
sim.temp <- tibble(x = rnorm(n = 200),
                  prob = exp(x)/(1 + exp(x)),
                  y = as.numeric(1 * runif(200) < prob))

sim.temp <- sim.temp |>
  mutate(p_guess = 1,
         p_perfect = y,
         p_bad = exp(-2*x) / (1 + exp(-2*x)),
         p_ok = prob + (1-y)*runif(1, 0, 0.05),
         p_good = prob + y*runif(1, 0, 0.27))

```

20.10.1.1 What if we are guessing?

If we're guessing completely at random, then the model should correctly classify a subject (as died or not died) about 50% of the time, so the TPR and FPR will be equal. This yields a diagonal line in the ROC curve, and an area under the curve (C statistic) of 0.5.

There are several ways to do this on the web, but I'll show this one, which has some bizarre code, but that's a function of using a package called `ROCR` to do the work. It comes from [this link](#)

```

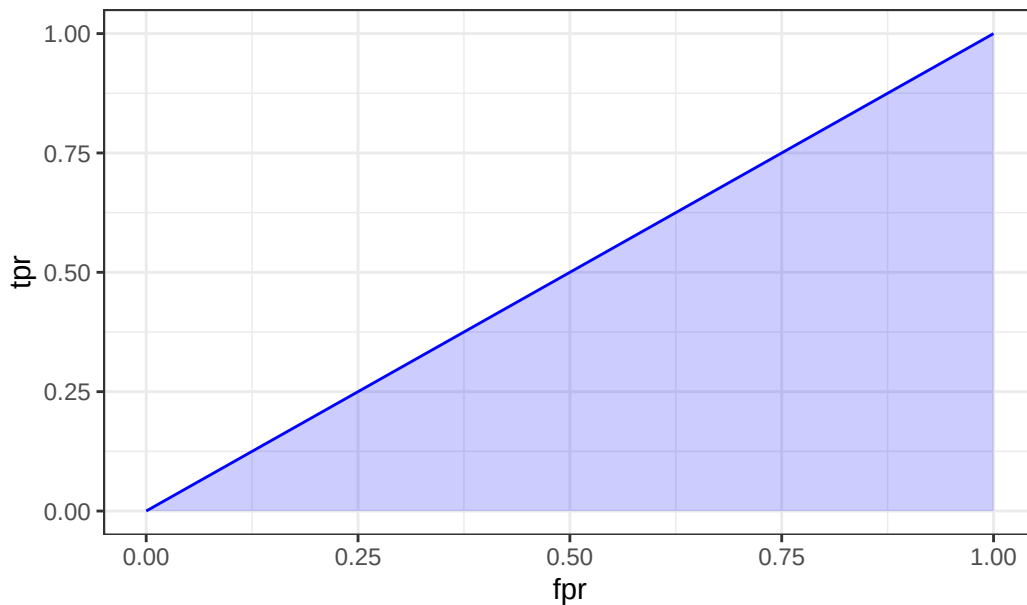
pred_guess <- prediction(sim.temp$p_guess, sim.temp$y)
perf_guess <- performance(pred_guess, measure = "tpr", x.measure = "fpr")
auc_guess <- performance(pred_guess, measure="auc")

auc_guess <- round(auc_guess@y.values[[1]],3)
roc_guess <- data.frame(fpr=unlist(perf_guess@x.values),
                      tpr=unlist(perf_guess@y.values),
                      model="GLM")

ggplot(roc_guess, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  labs(title = paste0("Guessing: ROC Curve w/ AUC=", auc_guess)) +
  theme_bw()

```

Guessing: ROC Curve w/ AUC=0.5



20.10.1.2 What if we classify things perfectly?

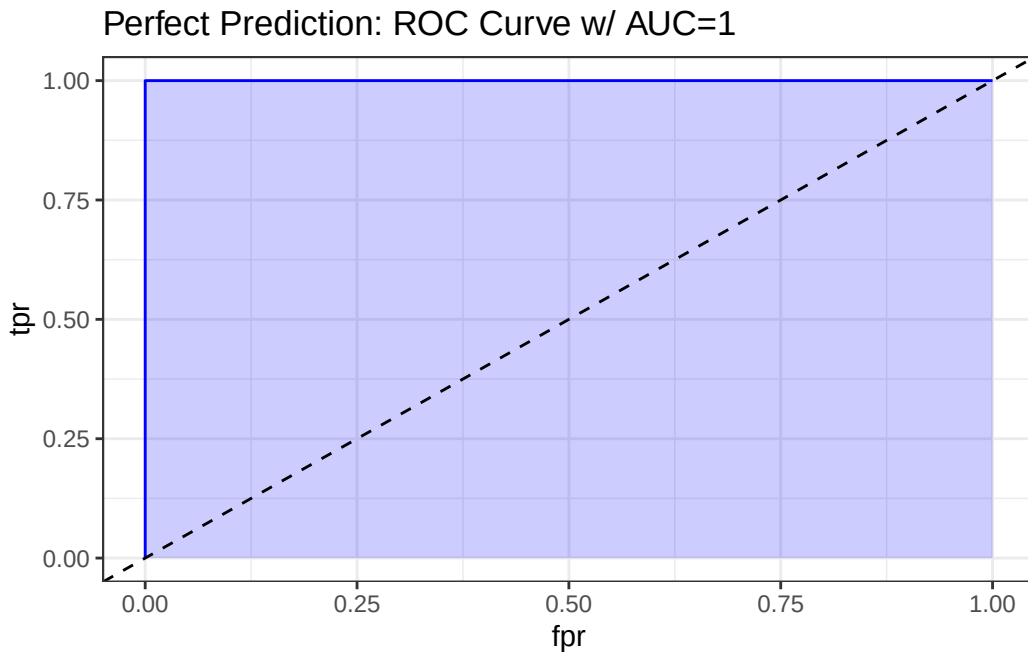
If we're classifying subjects perfectly, then we have a TPR of 1 and an FPR of 0. That yields an ROC curve that looks like the upper and left edges of a box. If our model correctly classifies a subject (as died or not died) 100% of the time, the area under the curve (c statistic) will be 1.0. We'll add in the diagonal line here (in a dashed black line) to show how this model compares to random guessing.

```
pred_perf <- prediction(sim.temp$p_perfect, sim.temp$y)
perf_perf <- performance(pred_perf, measure = "tpr", x.measure = "fpr")
auc_perf <- performance(pred_perf, measure="auc")

auc_perf <- round(auc_perf@y.values[[1]],3)
roc_perf <- data.frame(fpr=unlist(perf_perf@x.values),
                      tpr=unlist(perf_perf@y.values),
                      model="GLM")

ggplot(roc_perf, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
```

```
labs(title = paste0("Perfect Prediction: ROC Curve w/ AUC=", auc_perf)) +
theme_bw()
```



20.10.1.3 What does “worse than guessing” look like?

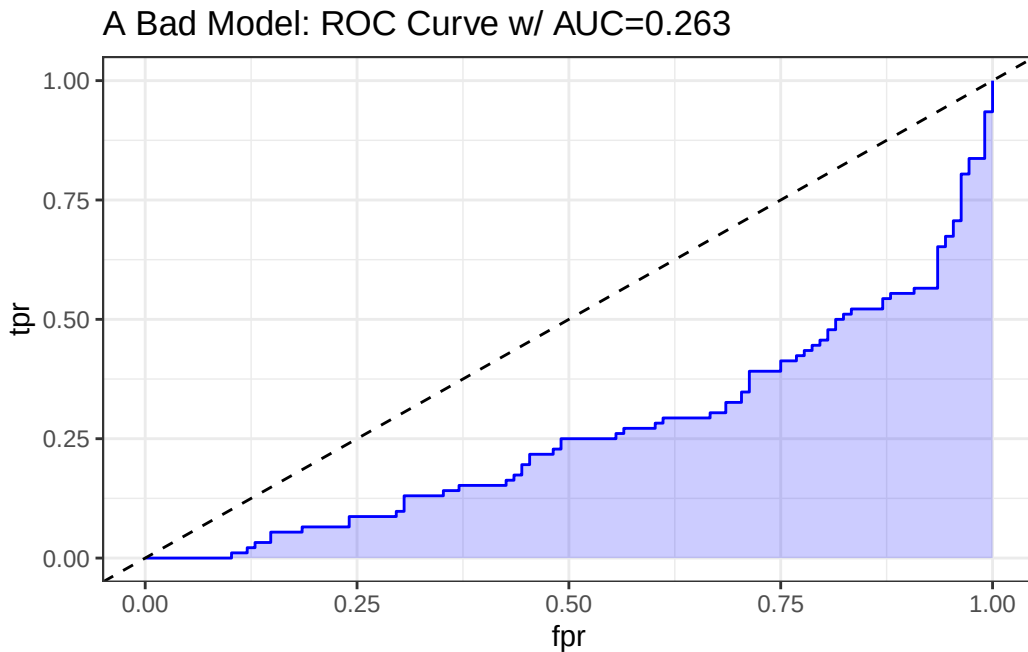
A bad classifier will appear below and to the right of the diagonal line we’d see if we were completely guessing. Such a model will have a c statistic below 0.5, and will be valueless.

```
pred_bad <- prediction(sim.temp$p_bad, sim.temp$y)
perf_bad <- performance(pred_bad, measure = "tpr", x.measure = "fpr")
auc_bad <- performance(pred_bad, measure="auc")

auc_bad <- round(auc_bad@y.values[[1]],3)
roc_bad <- data.frame(fpr=unlist(perf_bad@x.values),
                     tpr=unlist(perf_bad@y.values),
                     model="GLM")

ggplot(roc_bad, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
```

```
labs(title = paste0("A Bad Model: ROC Curve w/ AUC=", auc_bad)) +
theme_bw()
```



20.10.1.4 What does “better than guessing” look like?

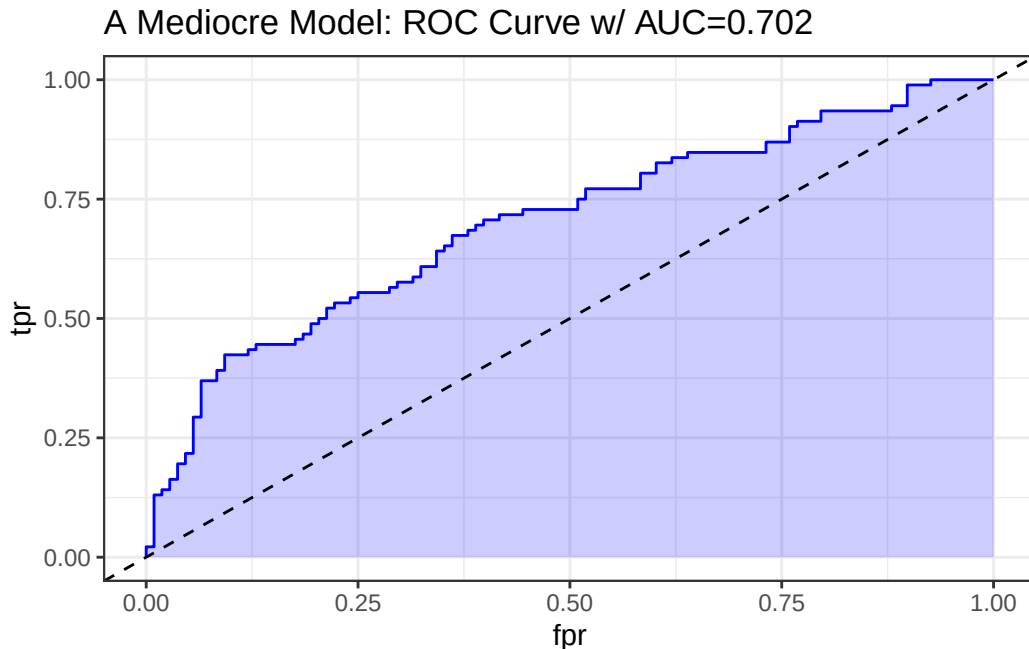
An “OK” classifier will appear above and to the left of the diagonal line we’d see if we were completely guessing. Such a model will have a c statistic above 0.5, and might have some value. The plot below shows a very fairly poor model, but at least it’s better than guessing.

```
pred_ok <- prediction(sim.temp$p_ok, sim.temp$y)
perf_ok <- performance(pred_ok, measure = "tpr", x.measure = "fpr")
auc_ok <- performance(pred_ok, measure="auc")

auc_ok <- round(auc_ok@y.values[[1]],3)
roc_ok <- data.frame(fpr=unlist(perf_ok@x.values),
                    tpr=unlist(perf_ok@y.values),
                    model="GLM")

ggplot(roc_ok, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
```

```
geom_abline(intercept = 0, slope = 1, lty = "dashed") +
labs(title = paste0("A Mediocre Model: ROC Curve w/ AUC=", auc_ok)) +
theme_bw()
```



Sometimes people grasp for a rough guide as to the accuracy of a model’s predictions based on the area under the ROC curve. A common thought is to assess the C statistic much like you would a class grade.

C statistic	Interpretation
0.90 to 1.00	model does an excellent job at discriminating “yes” from “no” (A)
0.80 to 0.90	model does a good job (B)
0.70 to 0.80	model does a fair job (C)
0.60 to 0.70	model does a poor job (D)
0.50 to 0.60	model fails (F)
below 0.50	model is worse than random guessing

20.10.1.5 What does “pretty good” look like?

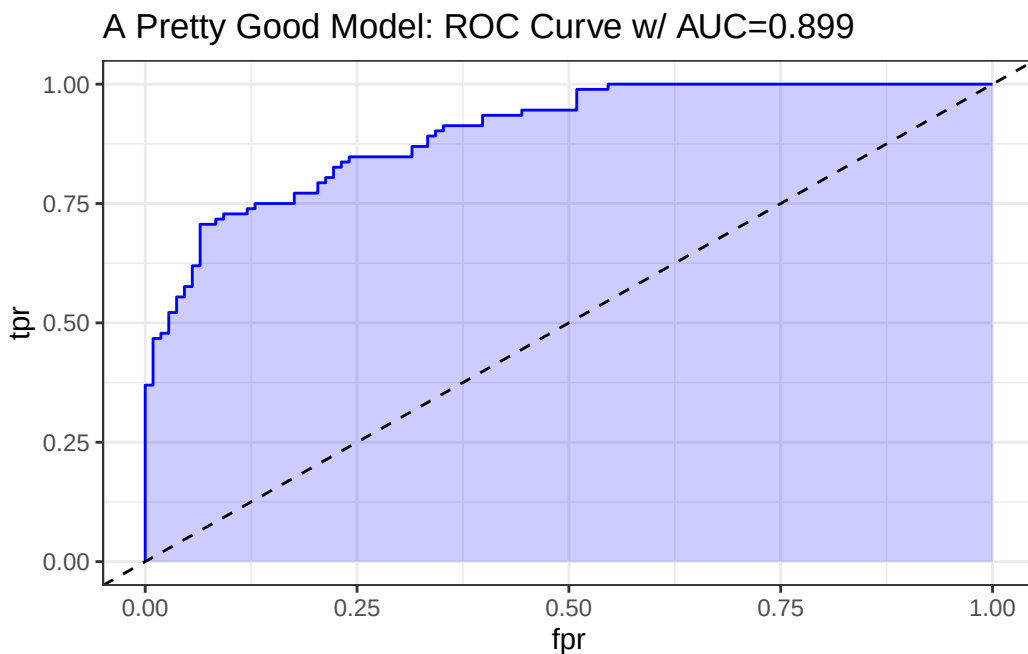
A strong and good classifier will appear above and to the left of the diagonal line we’d see if we were completely guessing, often with a nice curve that is continually increasing and appears to be pulled up towards the top left. Such a model will have a c statistic well above 0.5, but

not as large as 1. The plot below shows a stronger model, which appears substantially better than guessing.

```
pred_good <- prediction(sim.temp$p_good, sim.temp$y)
perf_good <- performance(pred_good, measure = "tpr", x.measure = "fpr")
auc_good <- performance(pred_good, measure="auc")

auc_good <- round(auc_good@y.values[[1]],3)
roc_good <- data.frame(fpr=unlist(perf_good@x.values),
                      tpr=unlist(perf_good@y.values),
                      model="GLM")

ggplot(roc_good, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
  labs(title = paste0("A Pretty Good Model: ROC Curve w/ AUC=", auc_good)) +
  theme_bw()
```



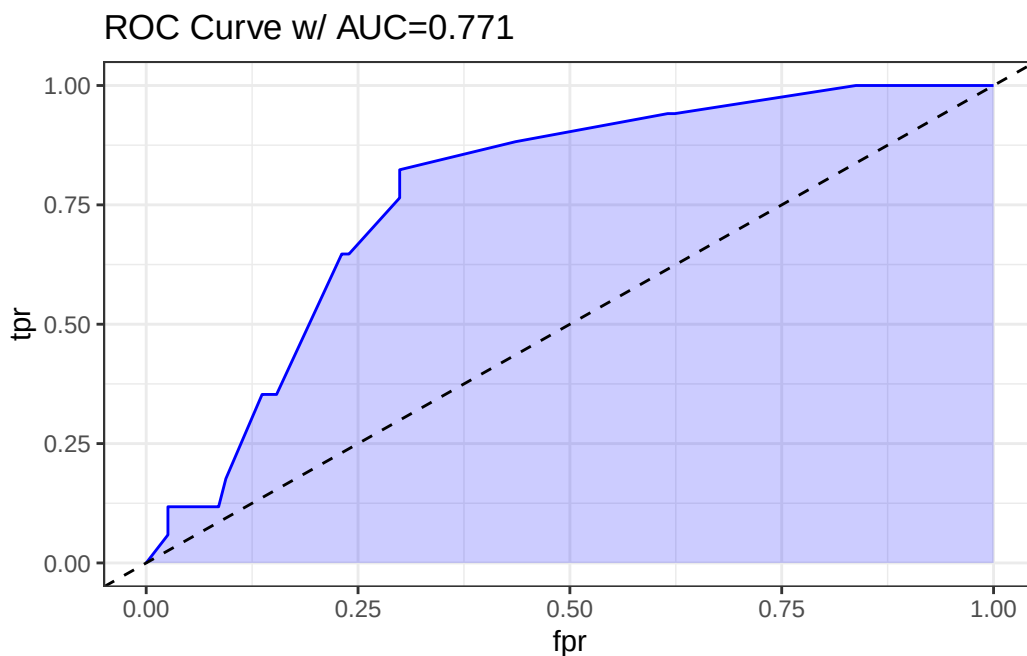
20.11 The ROC Plot for res_modA

Let me show you the ROC curve for our `res_modA` model.


```
## requires ROCR package
prob <- predict(res_modA, resect, type="response")
pred <- prediction(prob, resect$died)
perf <- performance(pred, measure = "tpr", x.measure = "fpr")
auc <- performance(pred, measure="auc")

auc <- round(auc@y.values[[1]],3)
roc.data <- data.frame(fpr=unlist(perf@x.values),
                      tpr=unlist(perf@y.values),
                      model="GLM")

ggplot(roc.data, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
  labs(title = paste0("ROC Curve w/ AUC=", auc)) +
  theme_bw()
```



Based on the C statistic (AUC = 0.771) this would rank somewhere near the high end of a “fair” predictive model by this standard, not quite to the level of a “good” model.

20.11.1 Another way to plot the ROC Curve

If we've loaded the `pROC` package, we can also use the following (admittedly simpler) approach to plot the ROC curve, without `ggplot2`, and to obtain the C statistic, and a 95% confidence interval around that C statistic.

```
## requires pROC package
roc.modA <-
  roc(resect$died ~ predict(res_modA, type="response"),
      ci = TRUE)
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

```
roc.modA
```

Call:

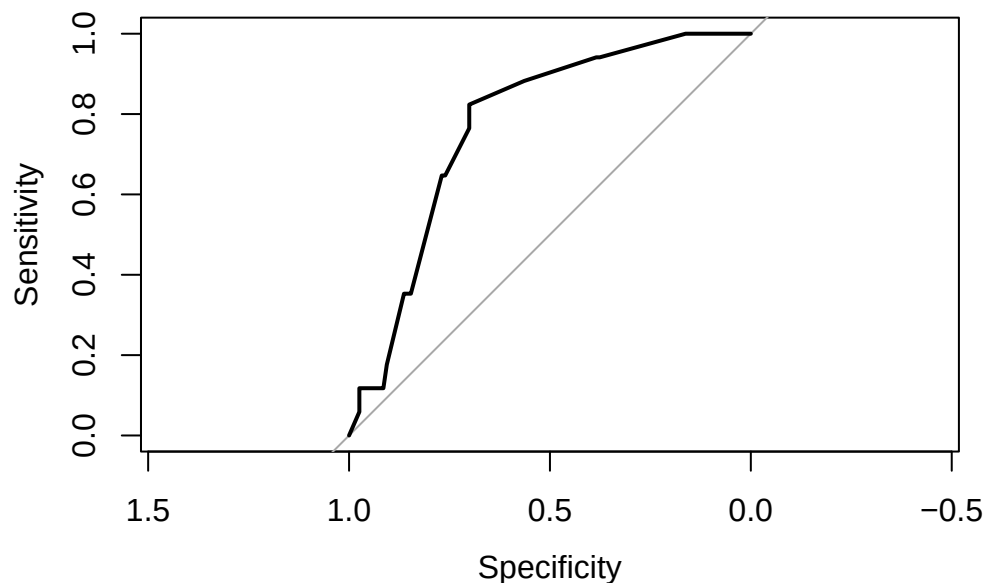
```
roc.formula(formula = resect$died ~ predict(res_modA, type = "response"),      ci = TRUE)
```

Data: predict(res_modA, type = "response") in 117 controls (resect\$died 0) < 17 cases (resect\$died 1)

Area under the curve: 0.7707

95% CI: 0.67-0.8715 (DeLong)

```
plot(roc.modA)
```



20.12 Assessing Residual Plots from Model A

Residuals are certainly less informative for logistic regression than they are for linear regression: not only do yes/no outcomes inherently contain less information than continuous ones, but the fact that the adjusted response depends on the fit hampers our ability to use residuals as external checks on the model.

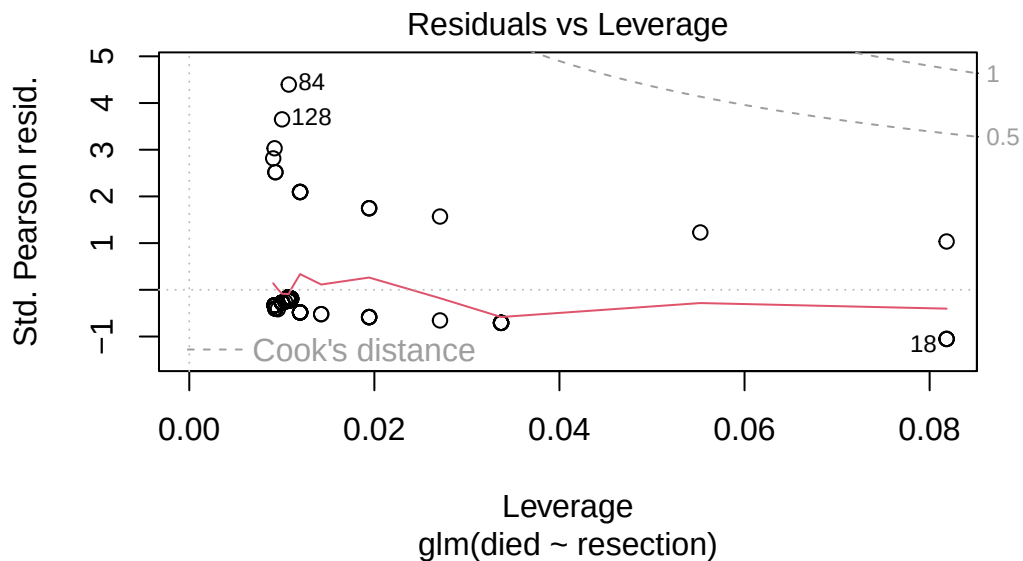
This is mitigated to some extent, however, by the fact that we are also making fewer distributional assumptions in logistic regression, so there is no need to inspect residuals for, say, skewness or heteroskedasticity.

- Patrick Breheny, University of Kentucky, Slides on GLM Residuals and Diagnostics (no longer online, alas.)

The usual residual plots are available in R for a logistic regression model, but most of them are irrelevant in the logistic regression setting. The residuals shouldn't follow a standard Normal distribution, and they will not show constant variance over the range of the predictor variables, so plots looking into those issues aren't helpful.

The only plot from the standard set that we'll look at in many settings is plot 5, which helps us assess influence (via Cook's distance contours), and a measure related to leverage (how unusual an observation is in terms of the predictors) and standardized Pearson residuals.

```
plot(res_modA, which = 5)
```



In this case, I don't see any highly influential points, as no points fall outside of the Cook's distance (0.5 or 1) contours.

20.13 Model B: A “Kitchen Sink” Logistic Regression Model

```
res_modB <- glm(died ~ resection + age + prior + intubated,
               data = resect, family = binomial)
```

```
res_modB
```

```
Call: glm(formula = died ~ resection + age + prior + intubated, family = binomial,
          data = resect)
```

Coefficients:

(Intercept)	resection	age	prior	intubated
-5.152886	0.612211	0.001173	0.814691	2.810797

```
Degrees of Freedom: 133 Total (i.e. Null); 129 Residual
Null Deviance:      101.9
Residual Deviance: 67.36    AIC: 77.36
```

20.13.1 Comparing Model A to Model B

```
anova(res_modA, res_modB)
```

Analysis of Deviance Table

Model 1: died ~ resection

Model 2: died ~ resection + age + prior + intubated

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	132	89.493			
2	129	67.359	3	22.134	6.116e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The addition of age, prior and intubated reduces the lack of fit by 22.134 points, at a cost of 3 degrees of freedom.

```
glance(res_modA)
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik    AIC    BIC deviance df.residual  nobs
      <dbl>    <int>  <dbl> <dbl> <dbl>   <dbl>      <int> <int>
1      102.      133  -44.7   93.5   99.3    89.5       132   134
```

```
glance(res_modB)
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik    AIC    BIC deviance df.residual  nobs
      <dbl>    <int>  <dbl> <dbl> <dbl>   <dbl>      <int> <int>
1      102.      133  -33.7   77.4   91.8    67.4       129   134
```

By either AIC or BIC, the larger model (res_modB) looks more effective.

20.13.2 Interpreting Model B

```
summary(res_modB)
```

Call:

```
glm(formula = died ~ resection + age + prior + intubated, family = binomial,  
     data = resect)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.152886	1.469453	-3.507	0.000454	***
resection	0.612211	0.282807	2.165	0.030406	*
age	0.001173	0.020646	0.057	0.954700	
prior	0.814691	0.704785	1.156	0.247705	
intubated	2.810797	0.658395	4.269	1.96e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 101.943 on 133 degrees of freedom
Residual deviance: 67.359 on 129 degrees of freedom
AIC: 77.359

Number of Fisher Scoring iterations: 6

It appears that the `intubated` predictor adds some value to the model, by the Wald test.

Let's focus on the impact of these variables through odds ratios.

```
tidy(res_modB, exponentiate = TRUE, conf.int = TRUE) |>  
  select(term, estimate, conf.low, conf.high)
```

A tibble: 5 x 4

	term	estimate	conf.low	conf.high
	<chr>	<dbl>	<dbl>	<dbl>
1	(Intercept)	0.00578	0.000241	0.0837
2	resection	1.84	1.08	3.35
3	age	1.00	0.962	1.04
4	prior	2.26	0.549	9.17
5	intubated	16.6	4.75	64.6

At a 5% significance level, we might conclude that:

- larger sized **resections** are associated with a meaningful rise (est OR: 1.84, 95% CI 1.08, 3.35) in the odds of death, holding all other predictors constant,
- the need for **intubation** at the end of surgery is associated with a substantial rise (est OR: 16.6, 95% CI 4.7, 64.7) in the odds of death, holding all other predictors constant, but that
- older **age** as well as having a **prior** tracheal surgery appears to be associated with an increase in death risk, but not with a small p value.

20.14 Plotting Model B

Let's think about plotting the fitted values from our model, in terms of probabilities.

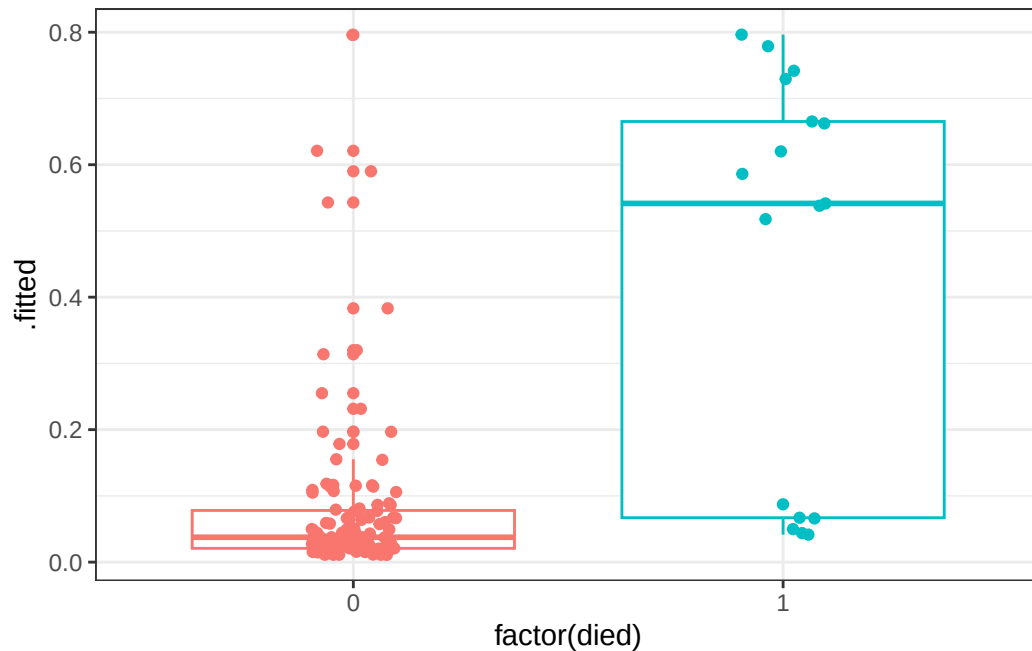
20.14.1 Using `augment` to capture the fitted probabilities

```
res_B_aug <- augment(res_modB, resect,  
                     type.predict = "response")  
head(res_B_aug)
```

```
# A tibble: 6 x 12  
  subj_id age prior resection intubated died .fitted .resid .hat .sigma  
    <dbl> <dbl> <dbl>    <dbl>    <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl>  
1       1   34     1      2.5        0     0  0.0591 -0.349 0.0267 0.725  
2       2   57     0       5        0     0  0.117  -0.498 0.0380 0.724  
3       3   60     1       4        1     1  0.729   0.794 0.114 0.722  
4       4   62     1      4.2        0     0  0.155  -0.581 0.0704 0.723  
5       5   28     0       6        1     1  0.796   0.675 0.131 0.723  
6       6   52     0       3        0     0  0.0371 -0.275 0.0105 0.725  
# i 2 more variables: .cooksd <dbl>, .std.resid <dbl>
```

20.14.2 Plotting Model B Fits by Observed Mortality

```
ggplot(res_B_aug, aes(x = factor(died), y = .fitted, col = factor(died))) +  
  geom_boxplot() +  
  geom_jitter(width = 0.1) +  
  guides(col = "none")
```



Certainly it appears as though most of our predicted probabilities (of death) for the subjects who actually survived are quite small, but not all of them. We also have at least 6 big “misses” among the 17 subjects who actually died.

20.14.3 Confusion Matrix for Model B

```
confusionMatrix(
  data = factor(res_B_aug$.fitted >= 0.5),
  reference = factor(res_B_aug$died == 1),
  positive = "TRUE"
)
```

Confusion Matrix and Statistics

	Reference	
Prediction	FALSE	TRUE
FALSE	113	6
TRUE	4	11

Accuracy : 0.9254
95% CI : (0.867, 0.9636)

No Information Rate : 0.8731
P-Value [Acc > NIR] : 0.03897

Kappa : 0.6453

McNemar's Test P-Value : 0.75183

Sensitivity : 0.64706
Specificity : 0.96581
Pos Pred Value : 0.73333
Neg Pred Value : 0.94958
Prevalence : 0.12687
Detection Rate : 0.08209
Detection Prevalence : 0.11194
Balanced Accuracy : 0.80644

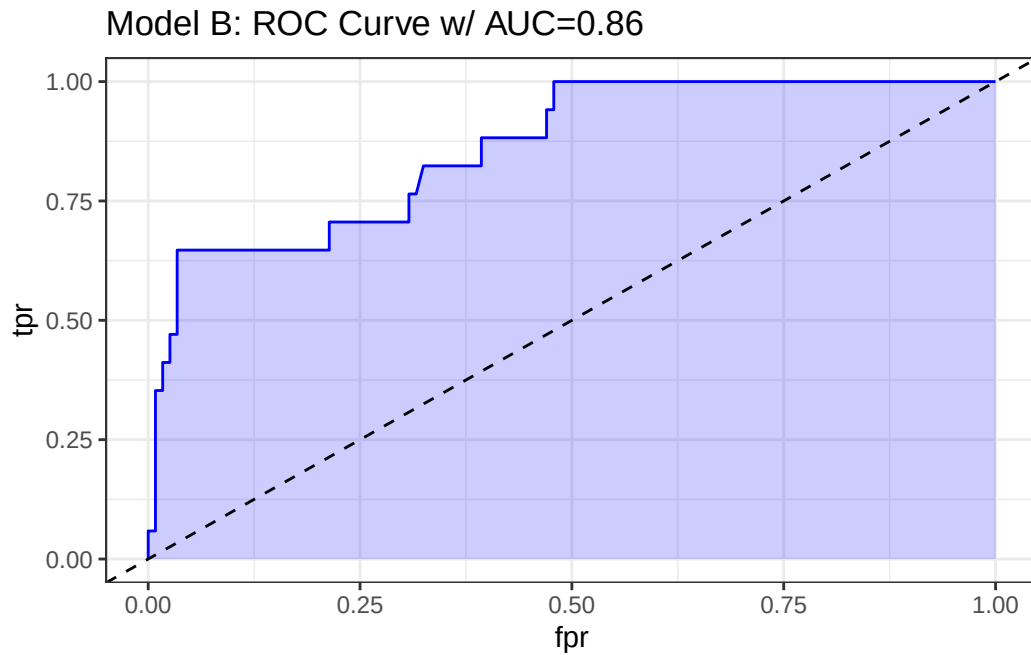
'Positive' Class : TRUE

20.14.4 The ROC curve for Model B

```
## requires ROCR package
prob <- predict(res_modB, resect, type="response")
pred <- prediction(prob, resect$died)
perf <- performance(pred, measure = "tpr", x.measure = "fpr")
auc <- performance(pred, measure="auc")

auc <- round(auc@y.values[[1]],3)
roc.data <- data.frame(fpr=unlist(perf@x.values),
                      tpr=unlist(perf@y.values),
                      model="GLM")

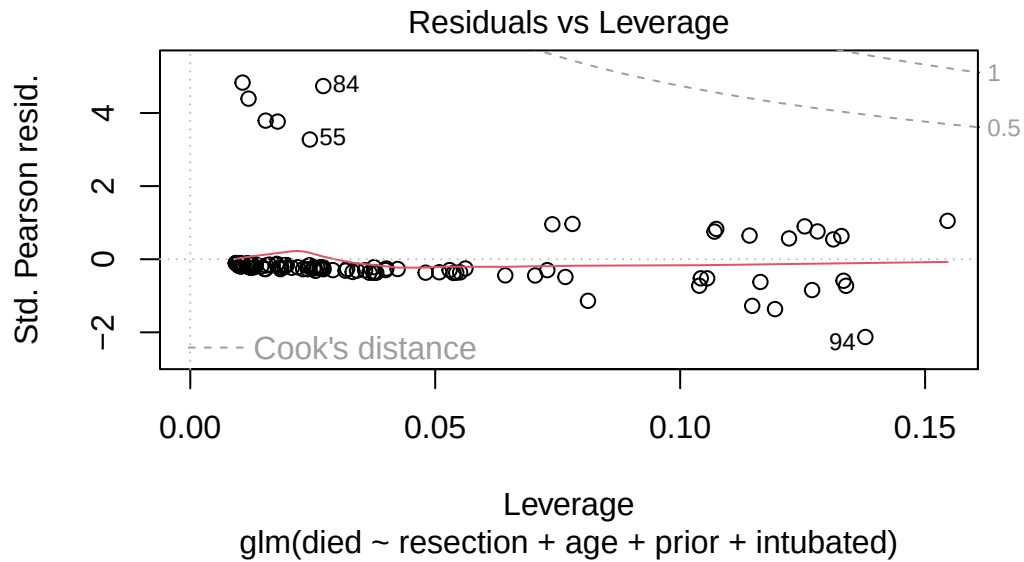
ggplot(roc.data, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
  labs(title = paste0("Model B: ROC Curve w/ AUC=", auc)) +
  theme_bw()
```



The area under the curve (C-statistic) is 0.86, which certainly looks like a more discriminating fit than model A with resection alone.

20.14.5 Residuals, Leverage and Influence

```
plot(res_modB, which = 5)
```



Again, we see no signs of deeply influential points in this model.

We'll continue working with these `resect` data as we fit logistic regression models with the help of the `rms` package in our next Chapter.

21 Logistic Regression with lrm

21.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(ROCR)
library(pROC)
library(naniar)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

21.1.1 Data Load

```
resect <- read_csv("data/resect.csv", show_col_types = FALSE)
```

21.2 Logistic Regression using lrm

To obtain the Nagelkerke R^2 and the C statistic, as well as some other summaries, I'll now demonstrate the use of `lrm` from the `rms` package to fit a logistic regression model.

We'll return to the original model, predicting death using resection size alone.

```
dd <- datadist(resect)
options(datadist="dd")

res_modC <- lrm(died ~ resection, data=resect, x=TRUE, y=TRUE)
res_modC
```

Logistic Regression Model

```
lrm(formula = died ~ resection, data = resect, x = TRUE, y = TRUE)
```

		Model Likelihood		Discrimination		Rank Discrim.	
		Ratio Test		Indexes		Indexes	
Obs	134	LR chi2	12.45	R2	0.167	C	0.771
0	117	d.f.	1	R2(1,134)	0.082	Dxy	0.541
1	17	Pr(> chi2)	0.0004	R2(1,44.5)	0.227	gamma	0.582
max deriv	2e-06			Brier	0.103	tau-a	0.121

	Coef	S.E.	Wald Z	Pr(> Z)
Intercept	-4.4337	0.8799	-5.04	<0.0001
resection	0.7417	0.2230	3.33	0.0009

This output specifies the following:

- **Obs** = The number of observations used to fit the model, with 0 = the number of zeros and 1 = the number of ones in our outcome, **died**. Also specified is the maximum absolute value of the derivative at the point where the maximum likelihood function was estimated. I wouldn't worry about that practically, as all you will care about is whether the iterative function-fitting process converged, and R will warn you in other ways if it doesn't.
- A likelihood ratio test (drop in deviance test) subtracting the residual deviance from the null deviance obtain the Likelihood Ratio χ^2 statistic, subtracting residual df from null df to obtain degrees of freedom, and comparing the resulting test statistic to a χ^2 distribution with the appropriate degrees of freedom to determine a p value.
- A series of discrimination indexes, including the Nagelkerke R^2 , symbolized R2, and several others we'll discuss shortly.
- A series of rank discrimination indexes, including the C statistic (area under the ROC curve) and Somers' D (Dxy), and several others.
- A table of coefficients, standard errors, Wald Z statistics and p values based on those Wald statistics.

The C statistic is estimated to be 0.771, with an associated (Nagelkerke) $R^2 = 0.167$, both indicating at best mediocre performance for this model, as it turns out.

21.2.1 Interpreting Nagelkerke R^2

There are many ways to calculate R^2 for logistic regression.

- At the [URL linked here](#) there is a nice summary of the key issue, which is that there are at least three different ways to think about R^2 in linear regression that are equivalent in that context, but when you move to a categorical outcome, which interpretation you use leads you down a different path for extension to the new type of outcome. In linear regression...
 - You might think of R^2 as a measure of the proportion of variability explained.
 - You might think of R^2 as measuring the improvement from a null model to a fitted model.
 - You might think of R^2 as the square of the correlation coefficient.
- Paul Allison, for instance, describes several at [this link](#) in a post entitled “What’s the Best R-Squared for Logistic Regression?”
- Jonathan Bartlett looks at McFadden’s pseudo R^2 in some detail (including some R code) at [this link](#), in a post entitled “R squared in logistic regression”

The Nagelkerke approach that is presented as `R2` in the `lrm` output is as good as most of the available approaches, and has the positive feature that it does reach 1 if the fitted model shows as much improvement as possible over the null model (which predicts the mean response for all subjects, and has $R^2 = 0$). The greater the improvement, the higher the Nagelkerke R^2 .

For model A, our Nagelkerke $R^2 = 0.167$, which is pretty poor. It doesn’t technically mean that 16.7% of any sort of variation has been explained, though.

21.2.2 Interpreting the C statistic and Plotting the ROC Curve

The C statistic is a measure of the area under the receiver operating characteristic curve. [This link](#) has some nice material that provides some insight into the C statistic and ROC curve.

- Recall that C ranges from 0 to 1. 0 = BAD, 1 = GOOD.
 - values of C less than 0.5 indicate that your prediction model is not even as good as simple random guessing of “yes” or “no” for your response.
 - $C = 0.5$ for random guessing
 - $C = 1$ indicates a perfect classification scheme - one that correctly guesses “yes” for all “yes” patients, and for none of the “no” patients.
- The closer C is to 1, the happier we’ll be, most of the time.
 - Often we’ll call models with $0.5 < C < 0.8$ poor or weak in terms of predictive ability by this measure
 - $0.8 \leq C < 0.9$ are moderately strong in terms of predictive power (indicate good discrimination)
 - $C \geq 0.9$ usually indicates a very strong model in this regard (indicate excellent discrimination)

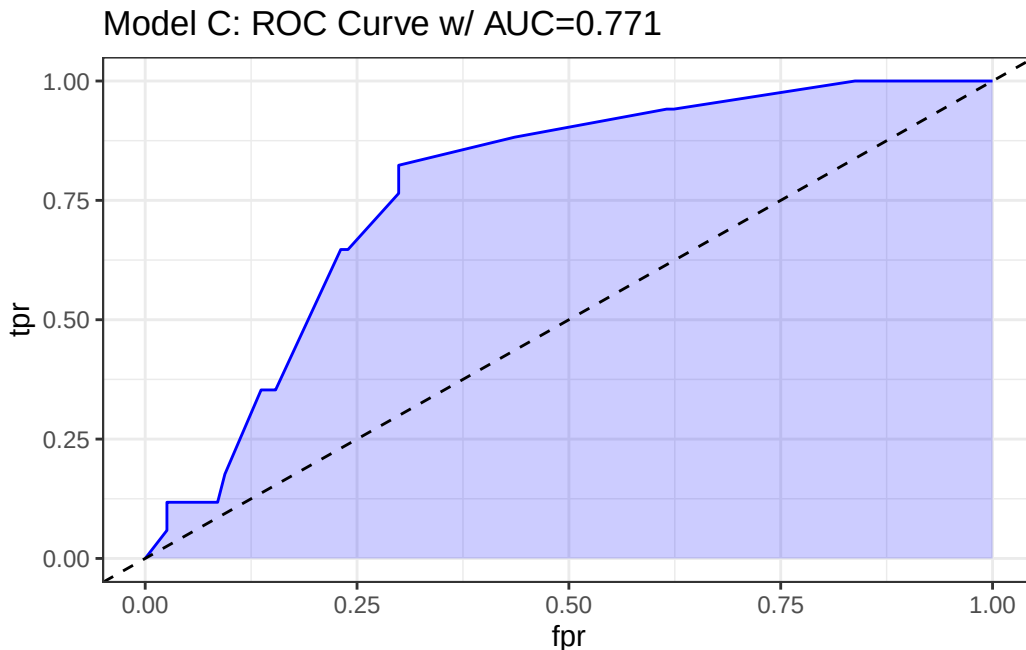
We've seen the ROC curve for this model before, when we looked at model `res_modA` fitted using `glm` in the previous chapter. But, just for completeness, I'll include it.

Note. I change the initial `predict` call from `type = "response"` for a `glm` fit to `type = "fitted"` in a `lrm` fit. Otherwise, this is the same approach.

```
## requires ROCR package
prob <- predict(res_modC, resect, type="fitted")
pred <- prediction(prob, resect$died)
perf <- performance(pred, measure = "tpr", x.measure = "fpr")
auc <- performance(pred, measure="auc")

auc <- round(auc@y.values[[1]],3)
roc.data <- data.frame(fpr=unlist(perf@x.values),
                      tpr=unlist(perf@y.values),
                      model="GLM")

ggplot(roc.data, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
  labs(title = paste0("Model C: ROC Curve w/ AUC=", auc)) +
  theme_bw()
```



21.2.3 The C statistic and Somers' D

- The C statistic is directly related to **Somers' D statistic**, abbreviated D_{xy} , by the equation $C = 0.5 + (D/2)$.
 - Somers' D and the ROC area only measure how well predicted values from the model can rank-order the responses. For example, predicted probabilities of 0.01 and 0.99 for a pair of subjects are no better than probabilities of 0.2 and 0.8 using rank measures, if the first subject had a lower response value than the second.
 - Thus, the C statistic (or D_{xy}) may not be very sensitive ways to choose between models, even though they provide reasonable summaries of the models individually.
 - This is especially true when the models are strong. The Nagelkerke R^2 may be more sensitive.
- But as it turns out, we sometimes have to look at the ROC shapes, as the summary statistic alone isn't enough.

In our case, Somers D (D_{xy}) = .541, so the C statistic is 0.771.

21.2.4 Validating the Logistic Regression Model Summary Statistics

Like other regression-fitting tools in **rms**, the **lrm** function has a special **validate** tool to help perform resampling validation of a model, with or without backwards step-wise variable selection. Here, we'll validate our model's summary statistics using 100 bootstrap replications.

```
set.seed(432001)
validate(res_modC, B = 100)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper
Dxy	0.5415	0.5422	0.5415	0.0007	0.5408	0.3466	0.7518
R2	0.1666	0.1748	0.1666	0.0083	0.1583	-0.0190	0.2973
Intercept	0.0000	0.0000	0.1631	-0.1631	0.1631	-1.2683	2.4073
Slope	1.0000	1.0000	1.0463	-0.0463	1.0463	0.4032	2.0222
Emax	0.0000	0.0000	0.1493	-0.1493	0.1493	-0.0388	0.4623
D	0.0854	0.0909	0.0854	0.0055	0.0800	-0.0480	0.1628
U	-0.0149	-0.0149	0.0017	-0.0167	0.0017	-0.0246	0.0841
Q	0.1004	0.1058	0.0837	0.0221	0.0783	-0.0891	0.1662
B	0.1025	0.0986	0.1051	-0.0065	0.1090	0.0742	0.1518
g	1.0369	1.0677	1.0369	0.0308	1.0061	0.3482	1.5654
gp	0.1101	0.1080	0.1101	-0.0021	0.1122	0.0348	0.1746
	n						
Dxy	100						
R2	100						

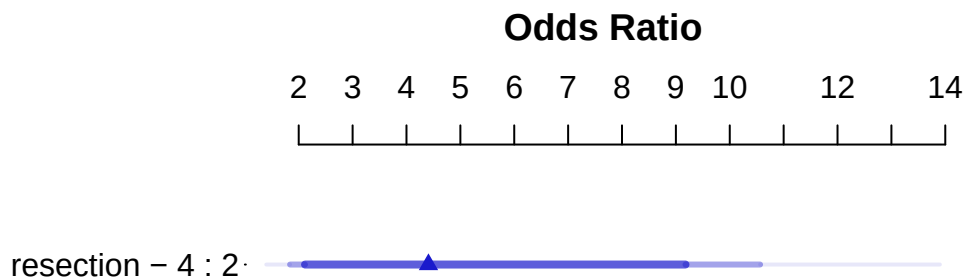
Intercept	100
Slope	100
E _{max}	100
D	100
U	100
Q	100
B	100
g	100
gp	100

Recall that our area under the curve (C statistic) = $0.5 + (D_{xy}/2)$, so that we can also use the first row of statistics to validate the C statistic. Accounting for optimism in this manner, our validation-corrected estimates are $D_{xy} = 0.5408$, so $C = 0.7704$, and, from the second row of statistics, we can read off the validated Nagelkerke R^2 , which is 0.1583.

21.2.5 Plotting the Summary of the lrm approach

The `summary` function applied to an `lrm` fit shows the effect size comparing the 25th percentile to the 75th percentile of resection.

```
plot(summary(res_modC))
```



```
summary(res_modC)
```

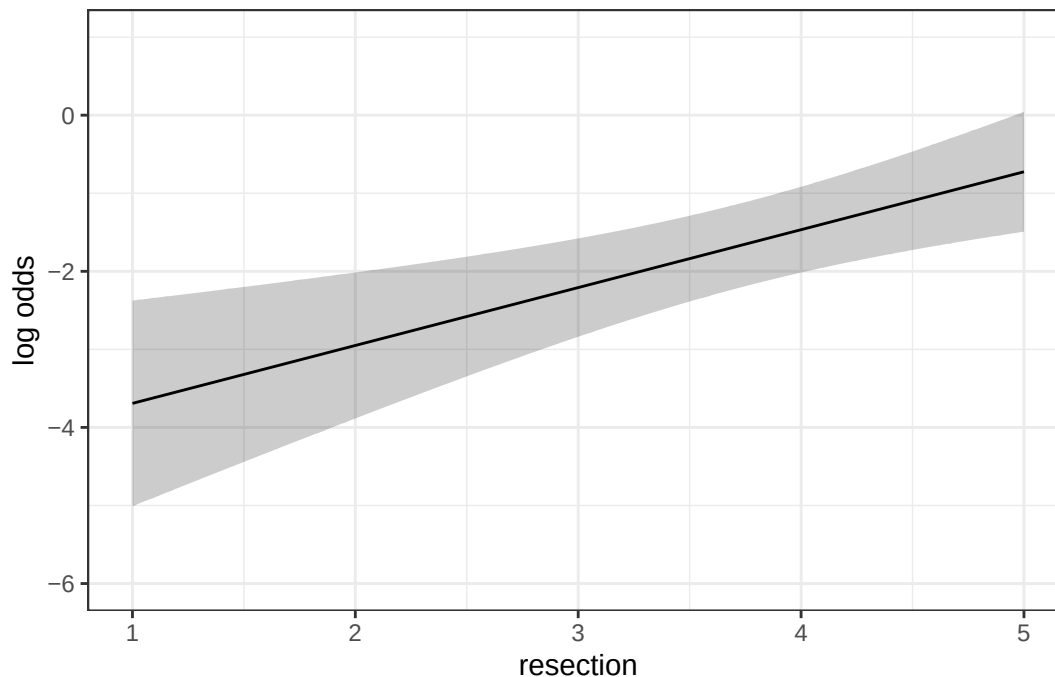
Effects		Response : died						
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	
resection	2	4	2	1.4834	0.44591	0.6094	2.3574	
Odds Ratio	2	4	2	4.4078	NA	1.8393	10.5630	

So, a move from a resection of 2 cm to a resection of 4 cm is associated with an estimated effect on the log odds of death of 1.48 (with standard error 0.45), or with an estimated effect on the odds ratio for death of 4.41, with 95% CI (1.84, 10.56).

21.2.6 Plot In-Sample Predictions for Model C

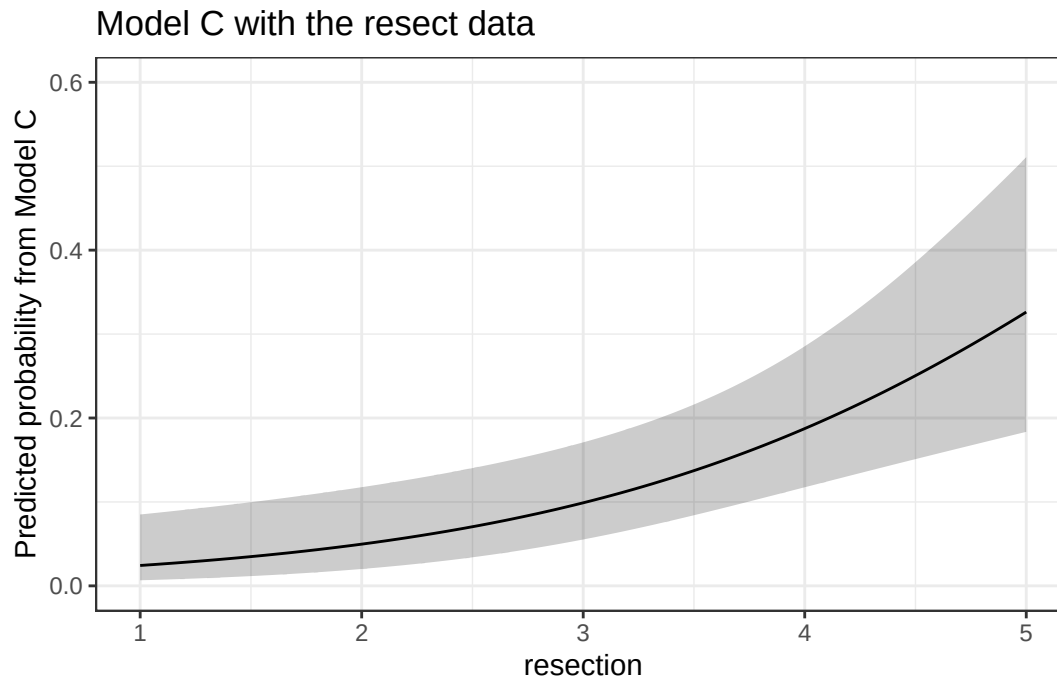
Here we plot the effect of **resection** (and 95% confidence intervals) across the range of observed values of **resection** on the log odds of death. Note the linear effect of **resection** size on the log odds scale.

```
ggplot(Predict(res_modC))
```



By applying the `plogis` function within the `Predict` command, we can plot the effect of `resection` on the estimated probability of death. Note the non-linear effect on this probability in this logistic regression model.

```
ggplot(Predict(res_modC, fun = plogis)) +
  labs(y = "Predicted probability from Model C",
       title = "Model C with the resect data")
```



The `Predict` function itself provides the raw material being captured in this plot.

```
head(Predict(res_modC, fun = plogis))
```

	resection	yhat	lower	upper	.predictor.
resection.1	1.000000	0.02431475	0.006636500	0.08505222	resection
resection.2	1.020101	0.02467095	0.006789311	0.08559055	resection
resection.3	1.040201	0.02503223	0.006945547	0.08613276	resection
resection.4	1.060302	0.02539867	0.007105281	0.08667888	resection
resection.5	1.080402	0.02577032	0.007268587	0.08722895	resection
resection.6	1.100503	0.02614727	0.007435540	0.08778303	resection

Response variable (y):

Limits are 0.95 confidence limits

21.2.7 ANOVA from the lrm approach

```
anova(res_modC)
```

	Wald Statistics			Response: died
Factor	Chi-Square	d.f.	P	
resection	11.07	1	9e-04	
TOTAL	11.07	1	9e-04	

The ANOVA approach applied to a `lrm` fit provides a Wald test for the model as a whole. Here, the use of `resection` is a significant improvement over a null (intercept-only) model. The p value is 9×10^{-4} .

21.2.8 Are any points particularly influential?

I'll use a cutoff for `dfbeta` here of 0.3, instead of the default 0.2, because I want to focus on truly influential points. Note that we have to use the data frame version of `resect` as `show.influence` isn't tibble-friendly.

```
inf.C <- which.influence(res_modC, cutoff=0.3)
inf.C
```

```
$Intercept
[1] 84 128
```

```
$resection
[1] 84
```

```
show.influence(object = inf.C, dframe = data.frame(resect))
```

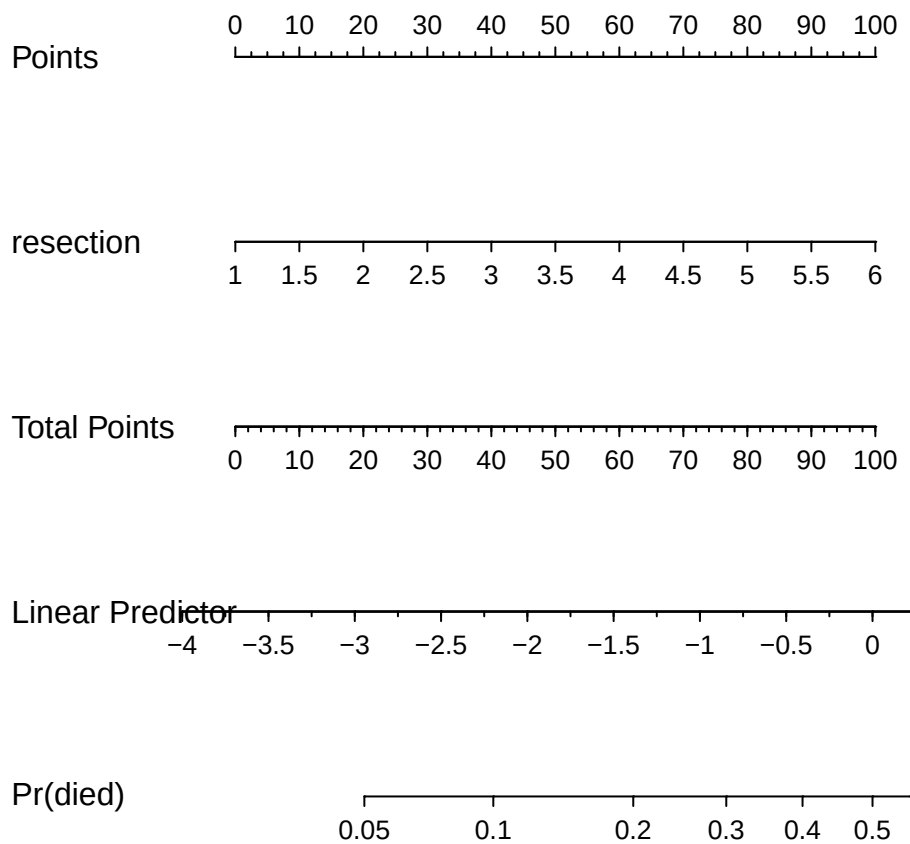
	Count	resection	
84	2		*2.0
128	1		2.5

It appears that observation 84 may have a meaningful effect on both the intercept and the coefficient for `resection`.

21.2.9 A Nomogram for Model C

We use the `plogis` function within a nomogram call to get R to produce fitted probabilities (of our outcome, `died`) in this case.

```
plot(nomogram(res_modC, fun=plogis,  
             fun.at=c(0.05, seq(0.1, 0.9, by = 0.1), 0.95),  
             funlabel="Pr(died)"))
```



Since there's no non-linearity in the right hand side of our simple logistic regression model, the nomogram is straightforward. We calculate the points based on the resection by traveling up, and then travel down in a straight vertical line from total points through the linear (log odds) predictor straight to a fitted probability. Note that fitted probabilities above 0.5 are not possible within the range of observed `resection` values in this case.

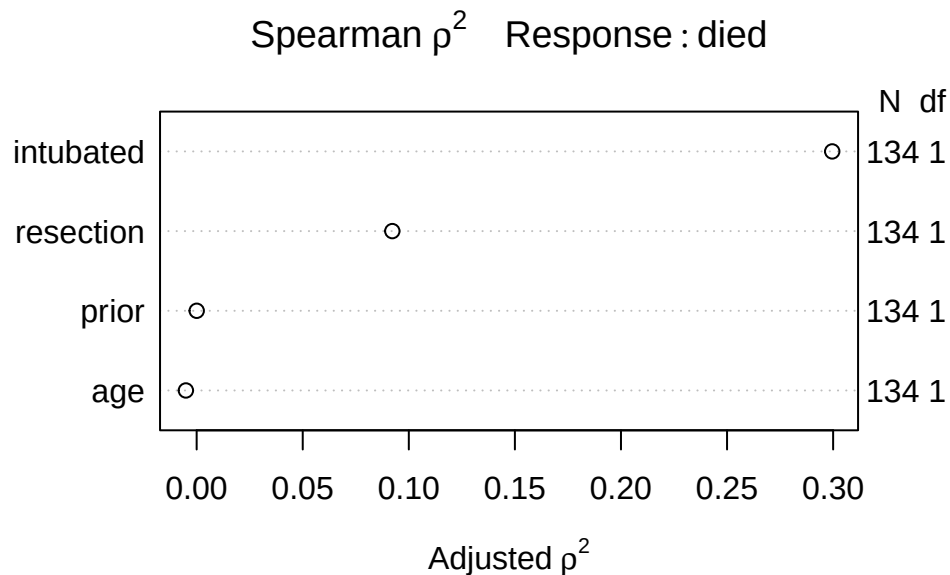
21.3 Model D: An Augmented Kitchen Sink Model

Can we predict survival from the patient's age, whether the patient had prior tracheal surgery or not, the extent of the resection, and whether intubation was required at the end of surgery?

21.3.1 Spearman ρ^2 Plot

Let's start by considering the limited use of non-linear terms for predictors that look important in a Spearman ρ^2 plot.

```
plot(spearman2(died ~ age + prior + resection + intubated, data=resect))
```



The most important variable appears to be whether intubation was required, so I'll include `intubated`'s interaction with the linear effect of the next most (apparently) important variable, `resection`, and also a cubic spline for `resection`, with three knots. Since `prior` and `age` look less important, I'll simply add them as linear terms.

21.3.2 Fitting Model D using lrm

Note the use of %ia% here. This insures that only the linear part of the `resection` term will be used in the interaction with `intubated`.

```
dd <- datadist(resect)
options(datadist="dd")

res_modD <- lrm(died ~ age + prior + rcs(resection, 3) +
               intubated + intubated %ia% resection,
               data=resect, x=TRUE, y=TRUE)
```

21.3.3 Assessing Model D using lrm's tools

```
res_modD
```

Logistic Regression Model

```
lrm(formula = died ~ age + prior + rcs(resection, 3) + intubated +
     intubated %ia% resection, data = resect, x = TRUE, y = TRUE)
```

		Model Likelihood	Discrimination	Rank Discrim.			
		Ratio Test	Indexes	Indexes			
Obs	134	LR chi2	38.08	R2	0.464	C	0.883
0	117	d.f.	6	R2(6,134)	0.213	Dxy	0.765
1	17	Pr(> chi2)	<0.0001	R2(6,44.5)	0.513	gamma	0.766
max deriv	9e-08			Brier	0.067	tau-a	0.171

	Coef	S.E.	Wald Z	Pr(> Z)
Intercept	-11.3636	4.9099	-2.31	0.0206
age	0.0000	0.0210	0.00	0.9988
prior	0.6269	0.7367	0.85	0.3947
resection	3.3799	1.9700	1.72	0.0862
resection'	-4.2104	2.7035	-1.56	0.1194
intubated	0.4576	2.7848	0.16	0.8695
intubated * resection	0.6188	0.7306	0.85	0.3970

- The model likelihood ratio test suggests that at least some of these predictors are helpful.
- The Nagelkerke R^2 of 0.46, and the C statistic of 0.88 indicate a meaningful improvement in discrimination over our model with `resection` alone.

- The Wald Z tests see some potential need to prune the model, as none of the elements reaches statistical significance without the others. The product term between **intubated** and **resection**, in particular, doesn't appear to have helped much, once we already had the main effects.

21.3.4 ANOVA and Wald Tests for Model D

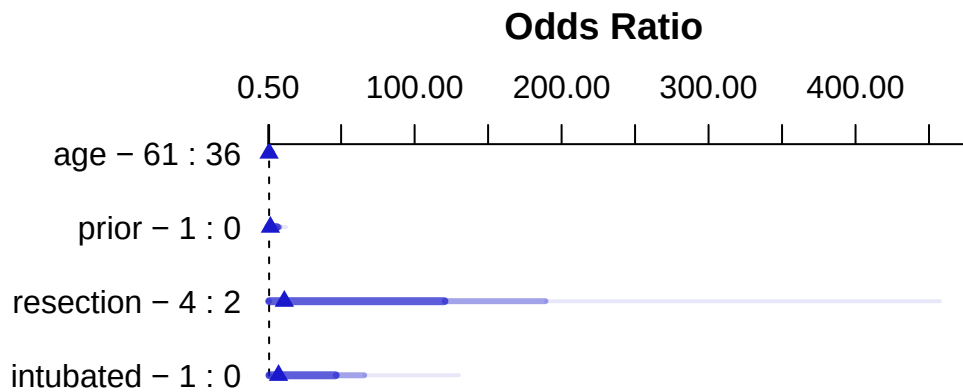
```
anova(res_modD)
```

Wald Statistics		Response: died		
Factor		Chi-Square	d.f.	P
age		0.00	1	0.9988
prior		0.72	1	0.3947
resection	(Factor+Higher Order Factors)	4.95	3	0.1753
All Interactions		0.72	1	0.3970
Nonlinear		2.43	1	0.1194
intubated	(Factor+Higher Order Factors)	16.45	2	0.0003
All Interactions		0.72	1	0.3970
intubated * resection	(Factor+Higher Order Factors)	0.72	1	0.3970
TOTAL NONLINEAR + INTERACTION		2.56	2	0.2783
TOTAL		23.90	6	0.0005

Neither the interaction term nor the non-linearity from the cubic spline appears to be statistically significant, based on the Wald tests via ANOVA. However it is clear that **intubated** has a significant impact as a main effect.

21.3.5 Effect Sizes in Model D

```
plot(summary(res_modD))
```

Adjusted to: resection=2.5 intubated=0

```
summary(res_modD)
```

Effects				Response : died			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
age	36	61	25	-0.00080933	0.52409	-1.02800	1.0264
Odds Ratio	36	61	25	0.99919000	NA	0.35772	2.7910
prior	0	1	1	0.62693000	0.73665	-0.81688	2.0707
Odds Ratio	0	1	1	1.87190000	NA	0.44181	7.9307
resection	2	4	2	2.42930000	1.43510	-0.38342	5.2419
Odds Ratio	2	4	2	11.35000000	NA	0.68153	189.0400
intubated	0	1	1	2.00470000	1.11220	-0.17513	4.1845
Odds Ratio	0	1	1	7.42380000	NA	0.83934	65.6610

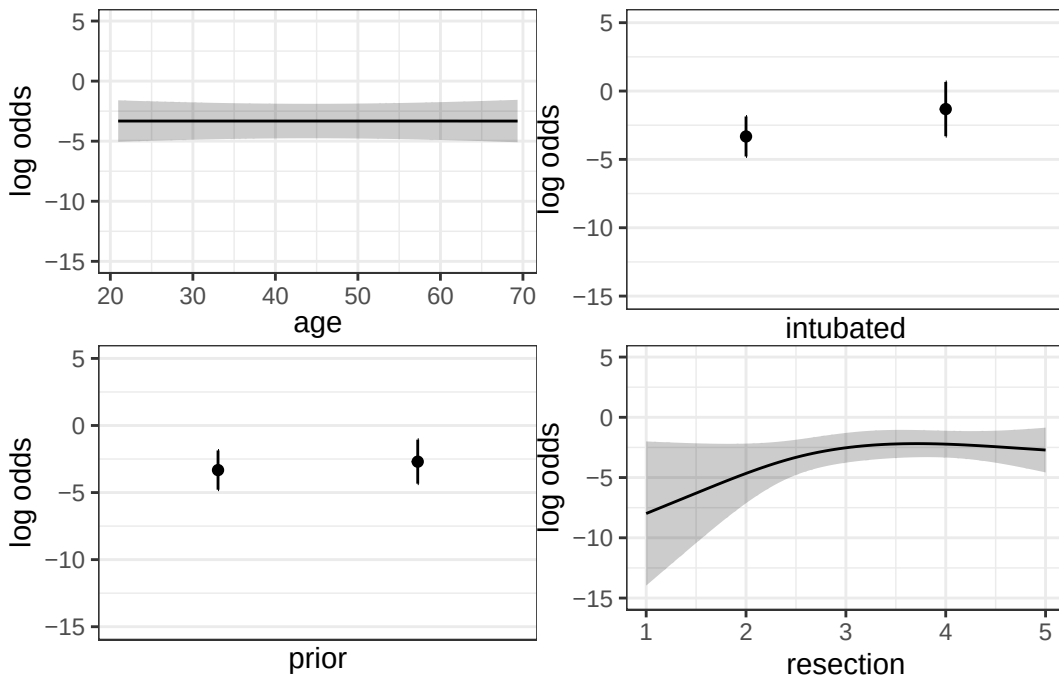
Adjusted to: resection=2.5 intubated=0

The effect sizes are perhaps best described in terms of odds ratios. The odds ratio for death isn't significantly different from 1 for any variable, but the impact of **resection** and **intubated**, though not strong enough to be significant, look more substantial (if poorly estimated) than the effects of **age** and **prior**.

21.3.6 Plot In-Sample Predictions for Model D

Here are plots of the effects across the range of each predictor (holding the others constant) on the log odds scale. Note the non-linear effect of resection implied by the use of a spline there.

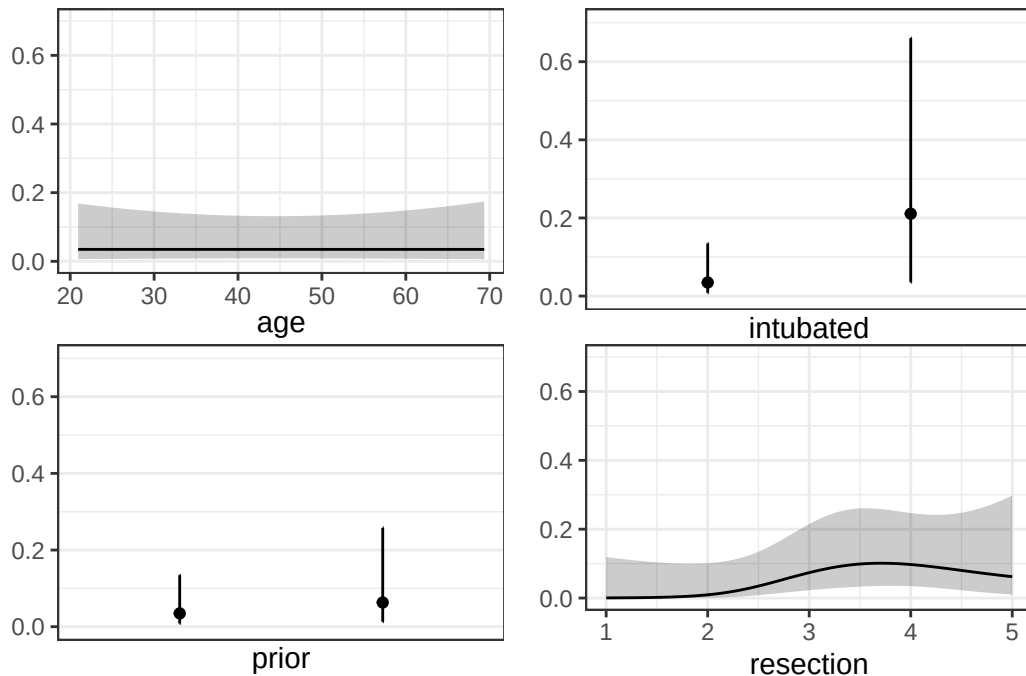
```
ggplot(Predict(res_modD))
```



We can also capture and plot these results on the probability scale, as follows¹.

```
ggplot(Predict(res_modD, fun = plogis))
```

¹Although I've yet to figure out how to get the y axis relabeled properly without simply dumping the Predict results into a new tibble and starting over with creating the plots.



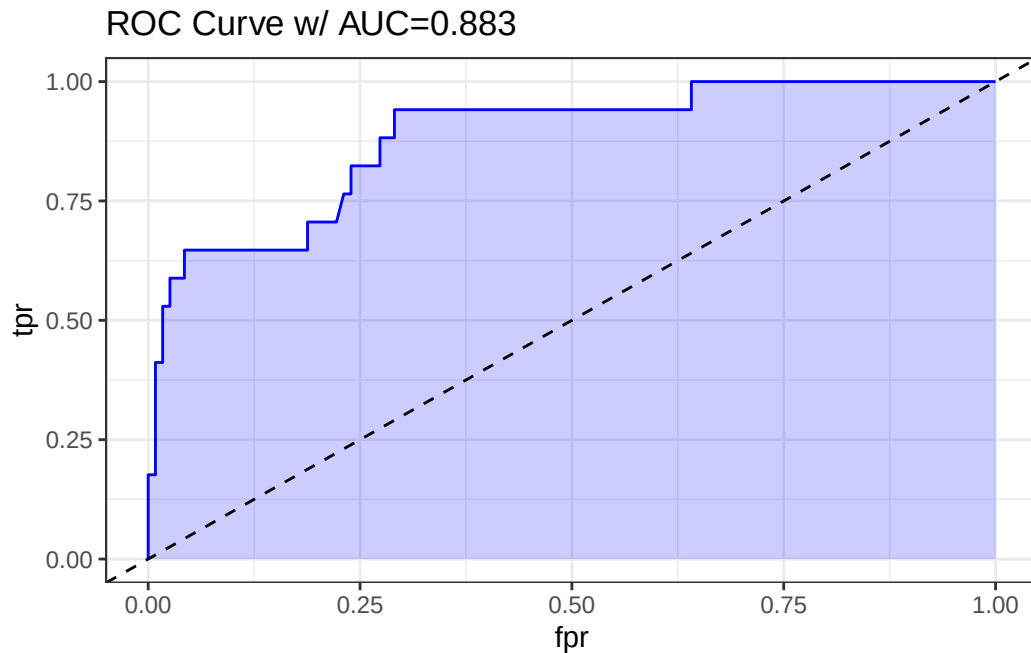
21.3.7 Plotting the ROC curve for Model D

Again, remember to use `type = "fitted"` with a `lm` fit.

```
## requires ROCR package
prob <- predict(res_modD, resect, type="fitted")
pred <- prediction(prob, resect$died)
perf <- performance(pred, measure = "tpr", x.measure = "fpr")
auc <- performance(pred, measure="auc")

auc <- round(auc@y.values[[1]],3)
roc.data <- data.frame(fpr=unlist(perf@x.values),
                      tpr=unlist(perf@y.values),
                      model="GLM")

ggplot(roc.data, aes(x=fpr, ymin=0, ymax=tpr)) +
  geom_ribbon(alpha=0.2, fill = "blue") +
  geom_line(aes(y=tpr), col = "blue") +
  geom_abline(intercept = 0, slope = 1, lty = "dashed") +
  labs(title = paste0("ROC Curve w/ AUC=", auc)) +
  theme_bw()
```



The AUC fitted with `ROCR` (0.883) is slightly different than what `lrm` has told us (0.880), and this also happens if we use the `pROC` approach, demonstrated below.

```
## requires pROC package
roc.modD <-
  roc(resect$died ~ predict(res_modD, type="fitted"),
      ci = TRUE)
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

```
roc.modD
```

Call:

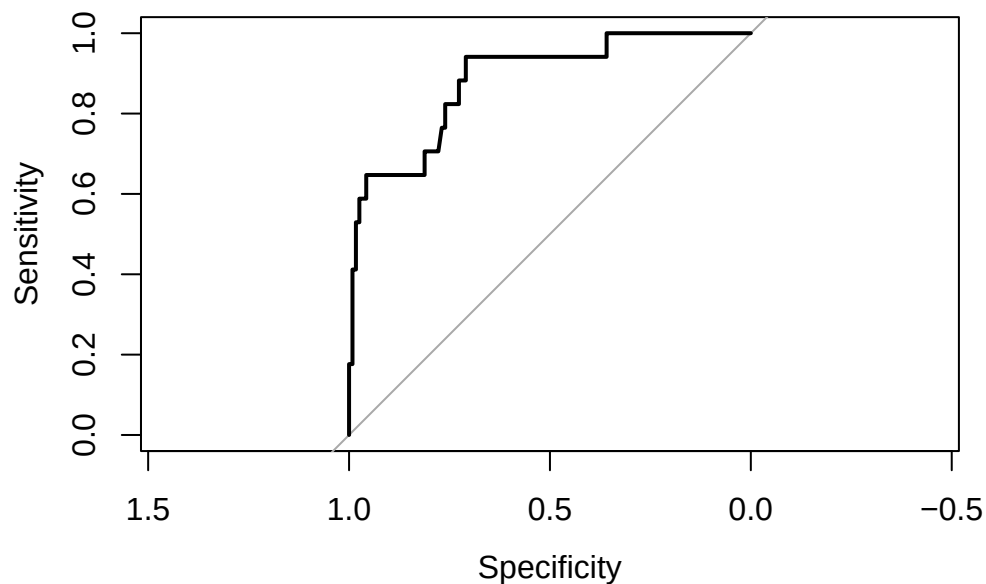
```
roc.formula(formula = resect$died ~ predict(res_modD, type = "fitted"),      ci = TRUE)
```

Data: predict(res_modD, type = "fitted") in 117 controls (resect\$died 0) < 17 cases (resect\$died 1)

Area under the curve: 0.8826

95% CI: 0.7952-0.97 (DeLong)

```
plot(roc.modD)
```



21.3.8 Validation of Model D summaries

```
set.seed(432002)
validate(res_modD, B = 100)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper
Dxy	0.7652	0.8145	0.7344	0.0801	0.6851	0.4995	0.8649
R2	0.4643	0.5440	0.4083	0.1357	0.3286	0.0097	0.5739
Intercept	0.0000	0.0000	-0.3648	0.3648	-0.3648	-1.7583	0.4712
Slope	1.0000	1.0000	0.7375	0.2625	0.7375	-0.0389	1.3076
Emax	0.0000	0.0000	0.1810	-0.1810	0.1810	-0.0723	0.8181
D	0.2767	0.3440	0.2383	0.1058	0.1709	-0.0800	0.3555
U	-0.0149	-0.0149	0.1288	-0.1437	0.1288	-0.1716	3.7024
Q	0.2916	0.3589	0.1094	0.2495	0.0421	-3.2578	0.4550
B	0.0673	0.0623	0.0738	-0.0115	0.0788	0.0460	0.1217
g	2.3819	7.8843	2.5223	5.3620	-2.9801	-185.6315	6.2611
gp	0.1720	0.1892	0.1627	0.0265	0.1455	0.0523	0.2094

n

Dxy	100
R2	100
Intercept	100
Slope	100
Emax	100
D	100
U	100
Q	100
B	100
g	100
gp	100

The C statistic indicates fairly strong discrimination, at $C = 0.88$, although after validation, this looks much weaker (based on $Dxy = 0.6851$, we would have $C = 0.5 + 0.6851/2 = 0.84$) and the Nagelkerke R^2 is also reasonably good, at 0.46, although this, too, is overly optimistic, and we bias-correct through our validation study to 0.33.

21.4 Model E: Fitting a Reduced Model in light of Model D

Can you suggest a reduced model (using a subset of the independent variables in model D) that adequately predicts survival?

Based on the anova for model D and the Spearman rho-squared plot, it appears that a two-predictor model using intubation and resection may be sufficient. Neither of the other potential predictors shows a statistically detectable effect in its Wald test.

```
res_modE <- lrm(died ~ intubated + resection, data=resect,
               x=TRUE, y=TRUE)
res_modE
```

Logistic Regression Model

```
lrm(formula = died ~ intubated + resection, data = resect, x = TRUE,
    y = TRUE)
```

		Model Likelihood		Discrimination		Rank Discrim.	
		Ratio Test		Indexes		Indexes	
Obs	134	LR chi2	33.27	R2	0.413	C	0.867
0	117	d.f.	2	R2(2,134)	0.208	Dxy	0.734
1	17	Pr(> chi2)	<0.0001	R2(2,44.5)	0.505	gamma	0.757
max deriv	5e-10			Brier	0.073	tau-a	0.164

	Coef	S.E.	Wald Z	Pr(> Z)
Intercept	-4.6370	1.0430	-4.45	<0.0001
intubated	2.8640	0.6479	4.42	<0.0001
resection	0.5475	0.2689	2.04	0.0418

The model equation is that the log odds of death is $-4.637 + 2.864 \text{ intubated} + 0.548 \text{ resection}$.

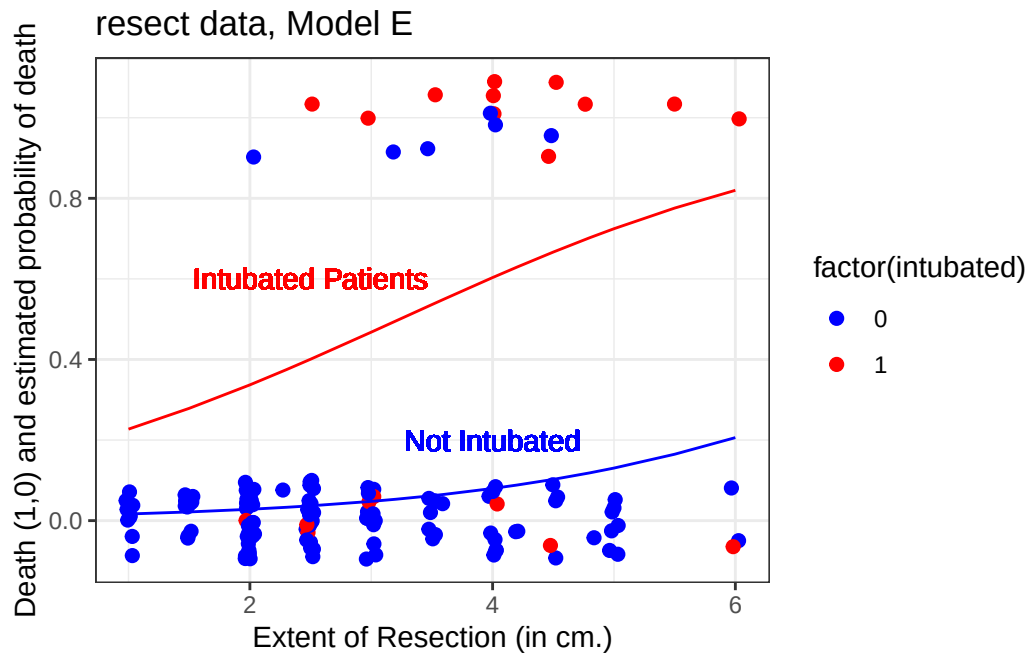
This implies that:

- for intubated patients, the equation is $-1.773 + 0.548 \text{ resection}$, while
- for non-intubated patients, the equation is $-4.637 + 0.548 \text{ resection}$.

We can use the `ilogit` function within the `faraway` package to help plot this.

21.4.1 A Plot comparing the two intubation groups

```
ggplot(resect, aes(x = resection, y = died,
                  col = factor(intubated))) +
  scale_color_manual(values = c("blue", "red")) +
  geom_jitter(size = 2, height = 0.1) +
  geom_line(aes(x = resection,
                y = faraway::ilogit(-4.637 + 0.548*resection)),
            col = "blue") +
  geom_line(aes(x = resection,
                y = faraway::ilogit(-1.773 + 0.548*resection)),
            col = "red") +
  geom_text(x = 4, y = 0.2, label = "Not Intubated",
            col="blue") +
  geom_text(x = 2.5, y = 0.6, label = "Intubated Patients",
            col="red") +
  labs(x = "Extent of Resection (in cm.)",
       y = "Death (1,0) and estimated probability of death",
       title = "resect data, Model E")
```

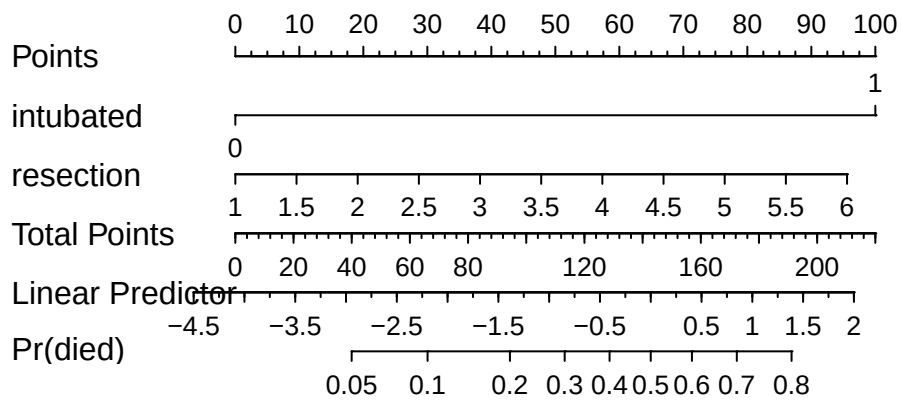


The effect of intubation appears to be very large, compared to the resection size effect.

21.4.2 Nomogram for Model E

A nomogram of the model would help, too.

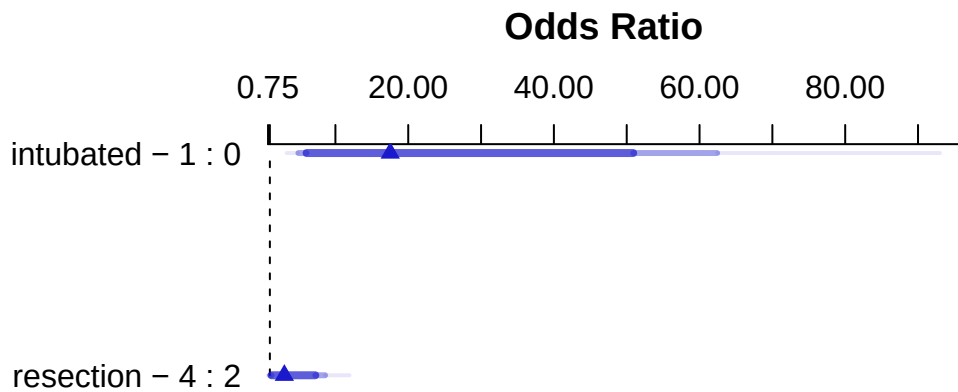
```
plot(nomogram(res_modE, fun=plogis,
  fun.at=c(0.05, seq(0.1, 0.9, by=0.1), 0.95),
  funlabel="Pr(died)"))
```

Again, we see that the effect of intubation is enormous, compared to the effect of resection. Another way to see this is to plot the effect sizes directly.

21.4.3 Effect Sizes from Model E

```
plot(summary(res_modE))
```



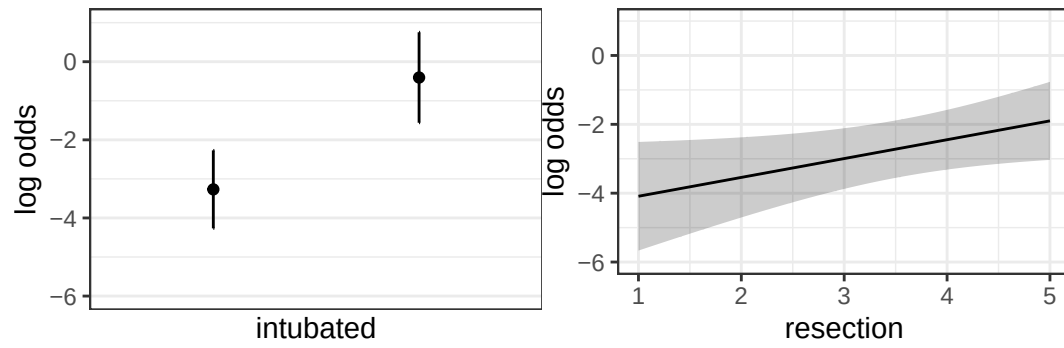
```
summary(res_modE)
```

Effects				Response : died			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
intubated	0	1	1	2.8640	0.64790	1.59410	4.1338
Odds Ratio	0	1	1	17.5310	NA	4.92390	62.4160
resection	2	4	2	1.0949	0.53783	0.04082	2.1491
Odds Ratio	2	4	2	2.9890	NA	1.04170	8.5769

21.4.4 Plot In-Sample Predictions for Model E

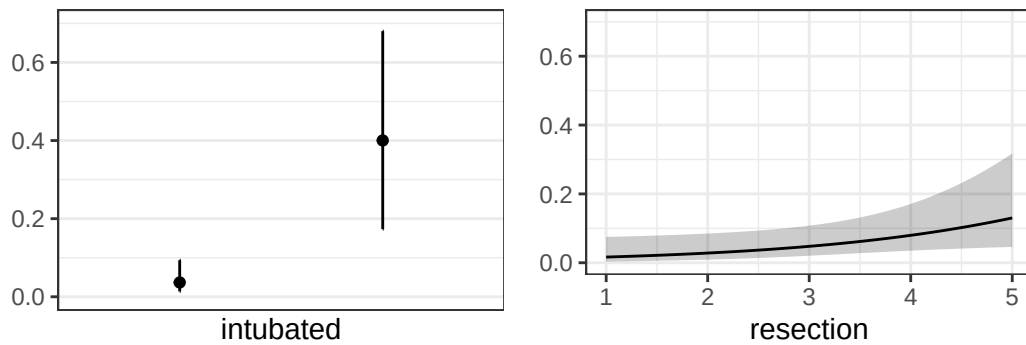
Here are plots of the effects across the range of each predictor (holding the other constant) on the log odds scale.

```
ggplot(Predict(res_modE))
```



We can also capture and plot these results on the probability scale, as follows.

```
ggplot(Predict(res_modE, fun = plogis))
```



21.4.5 ANOVA for Model E

```
anova(res_modE)
```

	Wald Statistics			Response: died
Factor	Chi-Square	d.f.	P	
intubated	19.54	1	<.0001	
resection	4.14	1	0.0418	
TOTAL	25.47	2	<.0001	

21.4.6 Validation of Model E

```
validate(res_modE, method="boot", B=40)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper
Dxy	0.7340	0.6896	0.7326	-0.0430	0.7771	0.5660	1.0568
R2	0.4128	0.3814	0.4025	-0.0211	0.4339	0.1986	0.6563
Intercept	0.0000	0.0000	0.1367	-0.1367	0.1367	-0.9788	1.4918
Slope	1.0000	1.0000	1.0472	-0.0472	1.0472	0.5648	1.8238
E _{max}	0.0000	0.0000	0.1089	-0.1089	0.1089	-0.0169	0.3573
D	0.2408	0.2183	0.2339	-0.0157	0.2565	0.0625	0.4013
U	-0.0149	-0.0149	-0.0001	-0.0148	-0.0001	-0.0229	0.0604
Q	0.2558	0.2332	0.2340	-0.0009	0.2566	0.0379	0.3997
B	0.0727	0.0727	0.0759	-0.0032	0.0759	0.0465	0.1029
g	1.3970	1.3391	1.3577	-0.0186	1.4156	0.7540	2.1219
gp	0.1597	0.1446	0.1563	-0.0117	0.1714	0.0970	0.2377
	n						
Dxy	40						
R2	40						
Intercept	40						
Slope	40						
E _{max}	40						
D	40						
U	40						
Q	40						
B	40						
g	40						
gp	40						

Our bootstrap validated assessments of discrimination and goodness of fit look somewhat more reasonable now.

21.4.7 Do any points seem particularly influential?

As a last step, I'll look at influence, and residuals, associated with model E.

```
inf.E <- which.influence(res_modE, cutoff=0.3)

inf.E
```

```
$Intercept
[1] 84 94
```

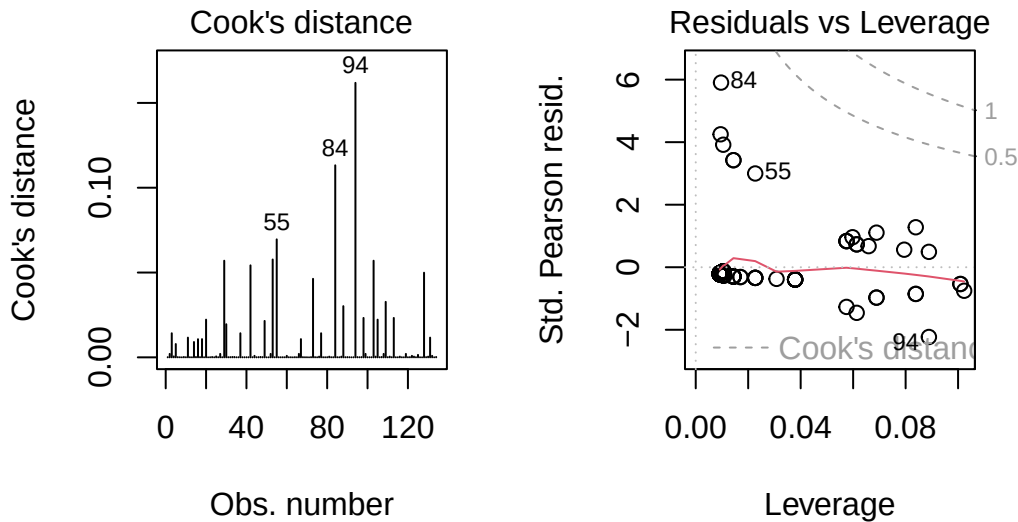
```
$resection
[1] 84 94
```

```
show.influence(inf.E, dframe = data.frame(resect))
```

```
      Count resection
84         2         *2
94         2         *6
```

21.4.8 Fitting Model E using glm to get plots about influence

```
res_modEglm <- glm(died ~ intubated + resection,
                  data=resect, family="binomial")
par(mfrow=c(1,2))
plot(res_modEglm, which=c(4:5))
```



Using this `glm` residuals approach, we again see that points 84 and 94 have the largest influence on our model E.

21.5 Concordance: Comparing Model C, D and E's predictions

To start, we'll gather the predictions made by each model (C, D and E) on the probability scale, in one place. Sadly, `augment` from `broom` doesn't work well with `lrm` fits, so we have to do this on our own.

```
resect_preds <- reselect |>
  mutate(C = predict(res_modC, type = "fitted"),
         D = predict(res_modD, type = "fitted"),
         E = predict(res_modE, type = "fitted"))
```

```
head(resect_preds)
```

```
# A tibble: 6 x 9
  subj_id age prior resection intubated died      C      D      E
  <dbl> <dbl> <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
1     1     34     1       2.5         0     0 0.0705 0.0632 0.0367
2     2     57     0        5         0     0 0.326  0.0620 0.130
```

3	3	60	1	4	1	1	0.187	0.791	0.603
4	4	62	1	4.2	0	0	0.211	0.158	0.0881
5	5	28	0	6	1	1	0.504	0.711	0.819
6	6	52	0	3	0	0	0.0990	0.0737	0.0477

And now, we'll use the `pivot_longer()` function from the **tidyr** package (part of the tidyverse) to arrange the models and predicted probabilities in a more useful manner for plotting.

```
res_p <- reselect_preds |>
  pivot_longer(cols = 7:9, names_to = "model", values_to = "prediction") |>
  select(subj_id, died, model, prediction)

head(res_p)
```

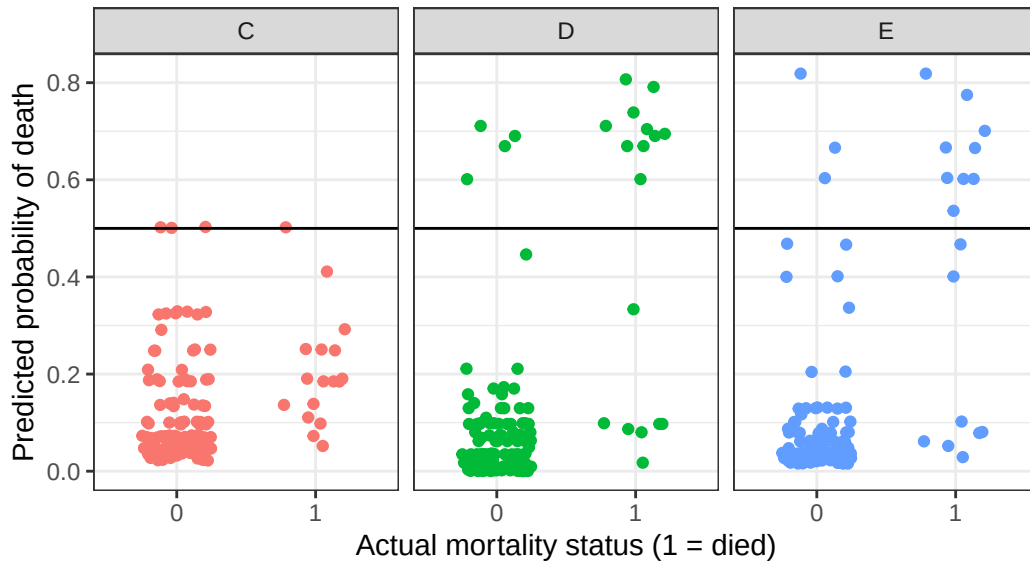
```
# A tibble: 6 x 4
  subj_id died model prediction
  <dbl> <dbl> <chr>      <dbl>
1       1     0 C         0.0705
2       1     0 D         0.0632
3       1     0 E         0.0367
4       2     0 C         0.326
5       2     0 D         0.0620
6       2     0 E         0.130
```

Here's the resulting plot.

```
ggplot(res_p, aes(x = factor(died), y = prediction,
                  group = model, col = model)) +
  geom_jitter(width = 0.25) +
  geom_hline(yintercept = 0.5) +
  facet_wrap(~ model) +
  guides(color = "none") +
  labs(title = "Comparing Predictions for our Three Models",
       subtitle = "A graphical view of concordance",
       x = "Actual mortality status (1 = died)",
       y = "Predicted probability of death")
```

Comparing Predictions for our Three Models

A graphical view of concordance



We could specify a particular rule, for example: if the predicted probability of death is 0.5 or greater, then predict “Died”.

```
res_p$rule.5 <- ifelse(res_p$prediction >= 0.5,
                        "Predict Died", "Predict Alive")

fable(table(res_p$model, res_p$rule.5, res_p$died))
```

	0	1
C Predict Alive	114	16
Predict Died	3	1
D Predict Alive	113	7
Predict Died	4	10
E Predict Alive	114	8
Predict Died	3	9

And perhaps build the linked table of row probabilities...

```
round(100*prop.table(
  fable(table(res_p$model, res_p$rule.5, res_p$died))
  ,1),2)
```


0 1

C Predict Alive	87.69	12.31
Predict Died	75.00	25.00
D Predict Alive	94.17	5.83
Predict Died	28.57	71.43
E Predict Alive	93.44	6.56
Predict Died	25.00	75.00

For example, in model E, 93.44% of those predicted to be alive actually survived, and 75% of those predicted to die actually died.

- Model D does a little better in one direction (94.17% of those predicted by Model D to be alive actually survived) but worse in the other (71.43% of those predicted by Model D to die actually died.)
- Model C does worse than each of the others in both predicting those who survive and those who die.

Note that the approaches discussed here would be useful if we had a new sample to predict on, as well. We could then compare the errors for that new data made by this sort of classification scheme either graphically or in a table.

21.6 Conclusions

It appears that **intubated** status and, to a lesser degree, the extent of the **resection** both play a meaningful role in predicting death associated with tracheal carina resection surgery. Patients who are intubated are associated with worse outcomes (greater risk of death) and more extensive resections are also associated with worse outcomes.

22 Estimating and Interpreting Effect Sizes

22.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(knitr)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

22.1.1 Data Load

Consider the `smalldat.csv` data available on our site, which is modeled on the public Framingham data set available [from BIOLINCC](#)¹. From the BIOLINCC documentation:

The Framingham Heart Study is a long term prospective study of the etiology of cardiovascular disease among a population of free living subjects in the community of Framingham, Massachusetts. The Framingham Heart Study was a landmark study in epidemiology in that it was the first prospective study of cardiovascular disease and identified the concept of risk factors and their joint effects.

```
smalldat <- read_csv("data/smalldat.csv", show_col_types = FALSE)
```

22.2 Available Variables

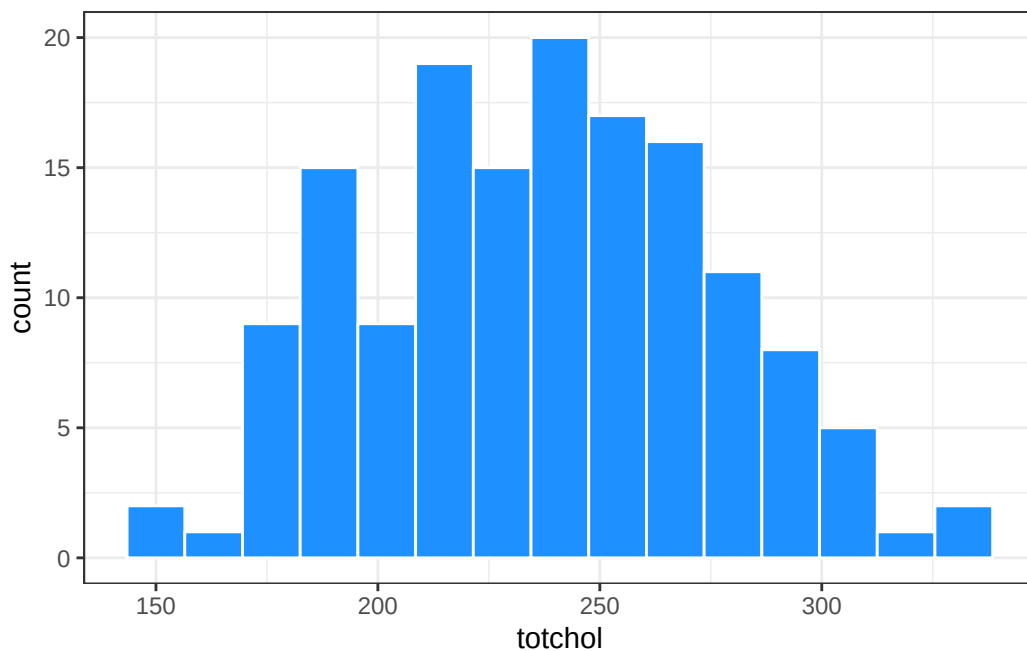
The `smalldat` data contains 150 observations on the following variables²:

¹The Framingham data from this source are not appropriate for publication or project use because they have been anonymized by permuting the results of individual subjects.

²The *educ* levels are: 1_Low, 2_Middle, 3_High and 4_VHigh, which stands for Very High

Variable	Description
<i>subject</i>	Subject identification code
<i>smoker</i>	1 = current smoker, 0 = not current smoker
<i>totchol</i>	total cholesterol, in mg/dl
<i>age</i>	age in years
<i>sex</i>	subject's sex (M or F)
<i>educ</i>	subject's educational attainment (4 levels)

```
ggplot(smallldat, aes(x = totchol)) +  
  geom_histogram(bins = 15, col = "white", fill = "dodgerblue")
```



22.3 Effect Interpretation in A Linear Regression Model

```
m1 <- lm(totchol ~ age + sex + factor(educ),  
         data = smallldat)  
  
kable(tidy(m1, conf.int = TRUE), digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	171.197	20.201	8.475	0.000	131.268	211.126
age	1.202	0.367	3.270	0.001	0.475	1.928
sexM	3.612	6.441	0.561	0.576	-9.119	16.343
factor(educ)2_Middle	11.044	7.702	1.434	0.154	-4.180	26.268
factor(educ)3_High	-2.459	9.390	-0.262	0.794	-21.019	16.101
factor(educ)4_VHigh	10.927	9.780	1.117	0.266	-8.405	30.258

1. What is the effect of *age* on *totchol* in Model `m1`?

```
temp.a <- tidy(m1, conf.int = TRUE) %>%
  filter(term == "age")

kable(temp.a, digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
age	1.202	0.367	3.27	0.001	0.475	1.928

The coefficient of the *age* effect on *totchol* is 1.202. Suppose we have two subjects, Doris and Emily, who are the same sex and have the same level of education, but Doris is one year older than Emily. Our model predicts that Doris' total cholesterol will be 1.202 mg/dl higher than Emily's.

The 95% confidence interval for this estimated *age* coefficient is (0.475, 1.928), so holding everything else constant, it seems that older *age* is associated with higher *totchol* in this model.

2. What is the effect of *sex* on *totchol* in Model `m1`?

```
temp.s <- tidy(m1, conf.int = TRUE) %>%
  filter(term == "sexM")

kable(temp.s, digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
sexM	3.612	6.441	0.561	0.576	-9.119	16.343

The model is parametrized to incorporate the *sex* information with an indicator (and factor) variable called *sexM* which is interpreted as taking the value 1 when *sex* = M, and 0 otherwise. The coefficient of the *sexM* effect on *totchol* is 3.612. Suppose we have two subjects, David and Emily, who are the same age, have the same level of education, but David is male and Emily is female. Our model predicts that David’s total cholesterol will be 3.612 mg/dl higher than Emily’s.

The 95% confidence interval for this estimated *sexM* coefficient is (-9.119, 16.343), which suggests that the effect of *sex* on *totchol* could be quite small, and that the data are consistent with a wide range of estimates for the *sexM* effect, some of which are negative.

3. What is the effect of *educ* on *totchol* in Model m1?

The *educ* variable splits the subjects into four categories. In this model the “1_Low” category is used as the baseline, and we have estimates for “2_Middle” (as compared to “1_Low”), for “3_High” (as compared to “1_Low”) and for “4_VHigh” (as compared to “1_Low”).

```
temp.ed <- tidy(m1, conf.int = TRUE) %>%
  filter(term %in% c("factor(educ)2_Middle",
                    "factor(educ)3_High",
                    "factor(educ)4_VHigh"))

kable(temp.ed,
      digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
factor(educ)2_Middle	11.044	7.702	1.434	0.154	-4.180	26.268
factor(educ)3_High	-2.459	9.390	-0.262	0.794	-21.019	16.101
factor(educ)4_VHigh	10.927	9.780	1.117	0.266	-8.405	30.258

The coefficient of the *educ* effect comparing the “2_Middle” group to the baseline “1_Low” group on *totchol* is 11.044.

Note that none of the *educ* levels show especially large differences from the baseline group, and each of their 95% confidence intervals contains zero.

- Suppose we have two subjects, Lola and Mina, who are the same age, and the same sex, but Lola is in the “1_Low” education group and Mina is in the “2_Middle” education group.
- Our model predicts that Mina’s total cholesterol will be 11.044 mg/dl higher than Lola’s.

The coefficient of the *educ* effect comparing the “3_High” group to the baseline “1_Low” group on *totchol* is -2.459.

- Suppose we have two subjects, Lola and Heidi, who are the same age, and the same sex, but Lola is in the “1_Low” education group and Heidi is in the “3_High” education group.
- Our model predicts that Heidi’s total cholesterol will be 2.459 mg/dl **lower** than Lola’s.

Finally, the coefficient of the *educ* effect comparing the “4_VHigh” group to the baseline “1_Low” group on *totchol* is 10.927.

- Suppose we have two subjects, Lola and Vera, who are the same age, and the same sex, but Lola is in the “1_Low” education group and Vera is in the “4_VHigh” education group.
- Our model predicts that Vera’s total cholesterol will be 10.927 mg/dl higher than Lola’s.

22.4 Making a prediction and building a prediction interval with an `lm` fit

Suppose we want to use `m1` to make a prediction for Lola and Vera, who we’ll now assume are each Female and 30 years of age, and we want to accompany this with a 90% prediction interval for each subject. Here’s one way to do that.

```
new1 <- tibble(  
  name = c("Lola", "Vera"),  
  age = c(30, 30),  
  sex = c("F", "F"),  
  educ = c("1_Low", "4_VHigh")  
)  
  
new1
```

```
# A tibble: 2 x 4  
  name    age sex  educ  
  <chr> <dbl> <chr> <chr>  
1 Lola    30 F    1_Low  
2 Vera    30 F    4_VHigh
```

```
res1 <- predict(m1, newdata = new1,  
  interval = "prediction", level = 0.9)  
  
res1
```

	fit	lwr	upr
1	207.2456	142.322	272.1691
2	218.1725	152.387	283.9580

```
new1_aug <- bind_cols(new1, fit = res1[, "fit"],
                      pi90.lo = res1[, "lwr"],
                      pi90.hi = res1[, "upr"])
```

```
new1_aug
```

```
# A tibble: 2 x 7
  name    age sex  educ    fit pi90.lo pi90.hi
  <chr> <dbl> <chr> <chr>   <dbl>   <dbl>   <dbl>
1 Lola    30 F    1_Low  207.    142.    272.
2 Vera    30 F    4_VHigh 218.    152.    284.
```

22.5 What if we include a Spline or an Interaction?

Suppose we fit a new model to predict *totchol* using a five-knot spline in *age* and the interaction of sex and educational attainment. How does that change our interpretation of the effect sizes?

None of these coefficients show particularly large effects, and zero is contained in each of the 95% confidence intervals provided in the table summarizing model m2.

```
d <- datadist(smalldat); options(datadist = "d")

m2 <- ols(totchol ~ rcs(age, 5) + sex*catg(educ),
          data = smalldat, x = TRUE, y = TRUE)

kable(summary(m2), digits = 2)
```

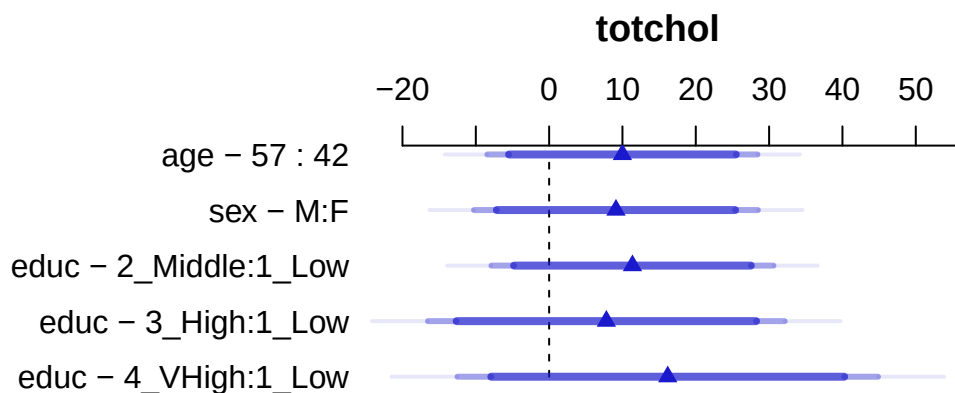
	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	Type
age	42	57	15	9.99	9.40	-8.59	28.57	1
sex - M:F	1	2	NA	9.11	9.87	-10.41	28.64	1
educ - 2_Middle:1_Low	1	2	NA	11.36	9.81	-8.04	30.76	1
educ - 3_High:1_Low	1	3	NA	7.80	12.41	-16.73	32.34	1

	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	Type
educ - 4_VHigh:1_Low	1	4	NA	16.17	14.63	-12.75	45.10	1

The `kable` approach I used in these notes hides the adjusted values specified at the bottom of the `summary` table for this `ols` model, but they are `Adjusted to: sex=F educ=1_Low`.

Now, how do we interpret these model `m2` results?

```
plot(summary(m2))
```



Adjusted to:sex=F educ=1_Low

- The **age** interpretation is that if we have two subjects, Al and Bob, who are the same sex and have the same education level, but Al is age 42 and Bob is age 57, then model `m2` projects that Bob's *totchol* will be 9.993 mg/dl higher than will Al's.
- Because of the interaction between *sex* and *educ* in our model `m2`, we must select an *educ* level in order to cleanly interpret the effect of *sex* on *totchol*. The **sex - M:F** interpretation compares M(ale) to F(emale) *sex* while requiring³ that *educ* = `1_Low`. The result is that if we have two subjects, Carl and Diane, who are the same age and

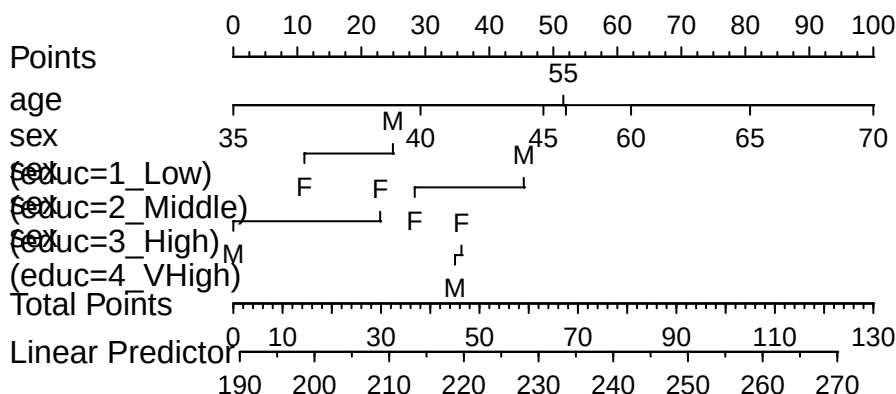
³We know this because of the `Adjusted to: sex = F, educ=1_Low` note at the bottom of the `summary` output for the `ols` model.

each is in the low education group, but Carl is Male and Diane is Female, then model `m2` predicts that Carl's *totchol* will be 9.115 mg/dl higher than will Diane's.

- Because of the interaction between *sex* and *educ* in our model `m2`, we must select a *sex* in order to cleanly interpret the effect of *educ* on *totchol*. The **`educ - 2_Middle:1_Low`** term, for instance, compares “2_Middle” education to “1_Low” education while requiring that *sex* is Female⁴. The result is that if we have two subjects, Lola and Mina, who are the same age and *each is Female*, but Lola is in the “1_Low” education group and Mina is in the “2_Middle” education group, then model `m2` predicts that Mina's *totchol* will be 11.363 mg/dl higher than will Lola's.

Here is a nomogram of model `m2`.

```
plot(nomogram(m2))
```



22.6 Making a prediction and building a prediction interval with an `ols` fit

Suppose we want to use `m2` to make a prediction for Lola and Vera, who we'll again assume are each Female and 30 years of age, and we want to accompany this with a 90% prediction interval for each subject. Here's one way to do that.

⁴Again, we know this because of the **Adjusted to:** `sex = F, educ=1_Low` note.

```
res2_lola <- Predict(m2,
                     age = 30, sex = "F", educ = "1_Low",
                     conf.int = 0.90,
                     conf.type = "individual")
res2_lola
```

	age	sex	educ	yhat	lower	upper
1	30	F	1_Low	176.7746	96.77241	256.7768

Response variable (y): totchol

Limits are 0.9 confidence limits

```
res2_vera <- Predict(m2,
                     age = 30, sex = "F", educ = "4_VHigh",
                     conf.int = 0.90,
                     conf.type = "individual")
res2_vera
```

	age	sex	educ	yhat	lower	upper
1	30	F	4_VHigh	192.9483	112.3727	273.5238

Response variable (y): totchol

Limits are 0.9 confidence limits

22.7 Effect Estimates in A Logistic Regression fit with glm

In a binary **logistic** model, where we predict the log odds of smoking ($smoker = 1$), we will exponentiate so as to interpret the odds ratio estimates associated with each coefficient.

```
m3 <- glm(smoker ~ age + sex + factor(educ),
          data = smalldat, family = binomial)

kable(tidy(m3, exponentiate = TRUE, conf.int = TRUE),
      digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	19.054	1.152	2.557	0.011	2.082	195.209
age	0.943	0.021	-2.782	0.005	0.903	0.982
sexM	1.795	0.356	1.643	0.100	0.897	3.637
factor(educ)2_Middle	0.690	0.428	-0.866	0.386	0.295	1.589
factor(educ)3_High	0.725	0.519	-0.619	0.536	0.258	2.005
factor(educ)4_VHigh	0.339	0.571	-1.895	0.058	0.105	1.008

1. What is the effect of *age* on the odds of being a *smoker* in Model *m3*?

```
temp.3a <- tidy(m3, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term == "age")

kable(temp.3a, digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
age	0.943	0.021	-2.782	0.005	0.903	0.982

The estimated odds ratio for the *age* effect on *smoker* is 0.943. Suppose we have two subjects, Doris and Emily, who are the same sex and have the same level of education, but Doris is one year older than Emily. Our model predicts that Doris' odds of smoking will be 0.943 times as high as Emily's. Another way to write this would be that Doris' odds of smoking are estimated to be 94.3% of Emily's. Yet another way would be to state that Doris' odds of smoking are estimated to be 5.7% smaller than Emily's odds.

The 95% confidence interval for this estimated odds ratio for the *age* effect on being a *smoker* is (0.903, 0.982). This confidence interval for the odds ratio does not include one, and again we see that holding everything else constant, older *age* is associated with detectably lower odds of being a *smoker* in this model.

2. What is the effect of *sex* on the odds of being a *smoker* in Model *m3*?

```
temp.3s <- tidy(m3, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term == "sexM")

kable(temp.3s, digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
sexM	1.795	0.356	1.643	0.1	0.897	3.637

The model incorporates the *sex* information with an indicator (and factor) variable called *sexM* which is interpreted as taking the value 1 when *sex* = M, and 0 otherwise. The estimated odds ratio describing the *sexM* effect on being a *smoker* is 1.795. Suppose we have two subjects, David and Emily, who are the same age, have the same level of education, but David is male and Emily is female. Our model predicts that David's odds of being a *smoker* are 1.795 times the odds that Emily is a *smoker*, or equivalently, that David's odds are 179.5% of Emily's odds. Another equivalent statement would be that David's odds are 79.5% larger than Emily's odds.

The 95% confidence interval for the odds ratio estimate of the effect of *sexM* on being a *smoker* is (0.897, 3.637). The effect of *sex* on the odds of being a *smoker* appears modest, and 1 is included in the confidence interval.

3. What is the effect of *educ* on the odds of being a *smoker* in Model m3?

Again, the *educ* variable splits the subjects into four categories. In this model the “1_Low” category is used as the baseline, and we have estimates for “2_Middle” (as compared to “1_Low”), for “3_High” (as compared to “1_Low”) and for “4_VHigh” (as compared to “1_Low”).

```
temp.3ed <- tidy(m3, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term %in% c("factor(educ)2_Middle",
                    "factor(educ)3_High",
                    "factor(educ)4_VHigh"))

kable(temp.3ed,
      digits = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
factor(educ)2_Middle	0.690	0.428	-0.866	0.386	0.295	1.589
factor(educ)3_High	0.725	0.519	-0.619	0.536	0.258	2.005
factor(educ)4_VHigh	0.339	0.571	-1.895	0.058	0.105	1.008

The estimated odds ratio describing the effect of *educ* being “2_Middle” instead of the baseline “1_Low” on the odds of being a *smoker* is 0.69, for people of the same age and sex.

No *educ* levels show meaningful differences from the baseline group, and their 95% confidence intervals all include 1, although the comparison of 4_VHigh to 1_Low only barely includes 1.

- Suppose we have two subjects, Lola and Mina, who are the same age, and the same sex, but Lola is in the “1_Low” education group and Mina is in the “2_Middle” education group.
- Our model predicts that Mina’s odds of being a smoker will be 0.69 times the odds of Lola being a smoker.

The estimated odds ratio comparing the *educ* = “3_High” group to the baseline *educ* = “1_Low” group on *smoker* is 0.725.

- Suppose we have two subjects, Lola and Heidi, who are the same age, and the same sex, but Lola is in the “1_Low” education group and Heidi is in the “3_High” education group.
- Our model predicts that Heidi’s odds of being a *smoker* will be 0.725 times the odds of Lola being a smoker.

Finally, The estimated odds ratio comparing the *educ* = “4_VHigh” group to the baseline *educ* = “1_Low” group on *smoker* is 0.339.

- Suppose we have two subjects, Lola and Vera, who are the same age, and the same sex, but Lola is in the “1_Low” education group and Vera is in the “4_VHigh” education group.
- Our model predicts that Vera’s odds of being a *smoker* will be 0.339 times the odds of Lola being a smoker.

22.8 Estimates in The Same Logistic Regression fit with `lrm`

When we fit the same model as `m3` using `lrm`, we get identical results as we get from the `glm` fit for the categorical predictors, but there’s a change in how the odds ratio for the quantitative predictor (*age*) is presented.

```
d <- datadist(smalldat); options(datadist = "d")

m3.lrm <- lrm(smoker ~ age + sex + educ,
             data = smalldat, x = TRUE, y = TRUE)

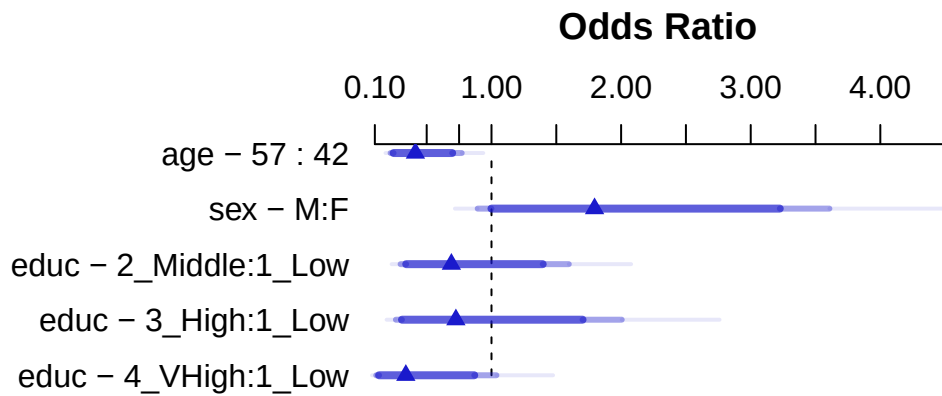
kable(summary(m3.lrm), digits = 3)
```

	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	Type
age	42	57	15	-0.885	0.318	-1.508	-0.261	1
Odds Ratio	42	57	15	0.413	NA	0.221	0.770	2
sex - M:F	1	2	NA	0.585	0.356	-0.113	1.283	1
Odds Ratio	1	2	NA	1.795	NA	0.893	3.607	2
educ - 2_Middle:1_Low	1	2	NA	-0.370	0.428	-1.209	0.468	1
Odds Ratio	1	2	NA	0.690	NA	0.299	1.596	2
educ - 3_High:1_Low	1	3	NA	-0.321	0.519	-1.338	0.696	1
Odds Ratio	1	3	NA	0.725	NA	0.262	2.005	2
educ - 4_VHigh:1_Low	1	4	NA	-1.082	0.571	-2.201	0.037	1
Odds Ratio	1	4	NA	0.339	NA	0.111	1.038	2

Note that the odds ratio effect sizes and confidence intervals are identical to what we saw in the `glm` fit for the *sex* and *educ* variables here, but the *age* result is presented differently.

- The **age** interpretation is that if we have two subjects, Al and Bob, who are the same sex and have the same education level, but Al is age 42 and Bob is age 57, then model `m3` projects that Bob's odds of being a *smoker* will be 0.413 times Al's odds of being a smoker. Bob's odds are 41.3% as large as Al's, equivalently.
- After adjustment for *sex* and *educ*, increasing age appears to be associated with decreasing odds of smoking. Note, too, that the effect of *age* on the odds of being a *smoker* has a confidence interval for the odds ratio entirely below 1.

```
plot(summary(m3.lrm))
```



22.9 Estimates in A New Logistic Regression fit with lrm

Now, suppose we fit a new model to predict the log odds of being a *smoker* using a five-knot spline in *age* and the interaction of sex and educational attainment. How does that change our interpretation of the effect sizes?

```
d <- datadist(smalldat); options(datadist = "d")

m4 <- lrm(smoker ~ rcs(age,5) + sex * catg(educ),
          data = smalldat, x = TRUE, y = TRUE)

kable(summary(m4), digits = 3)
```

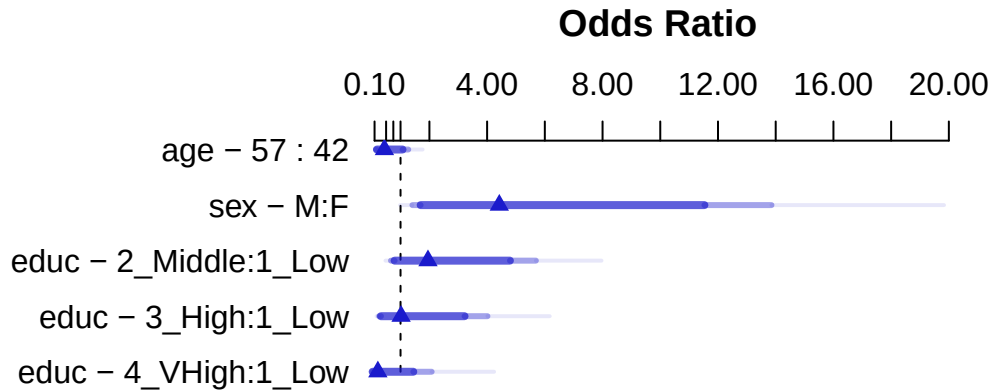
	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	Type
age	42	57	15	-0.817	0.538	-1.872	0.237	1
Odds Ratio	42	57	15	0.442	NA	0.154	1.268	2
sex - M:F	1	2	NA	1.487	0.583	0.345	2.629	1
Odds Ratio	1	2	NA	4.422	NA	1.412	13.853	2

	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	Type
educ - 2_Middle:1_Low	1	2	NA	0.668	0.546	-0.402	1.739	1
Odds Ratio	1	2	NA	1.951	NA	0.669	5.690	2
educ - 3_High:1_Low	1	3	NA	0.019	0.699	-1.351	1.389	1
Odds Ratio	1	3	NA	1.019	NA	0.259	4.011	2
educ - 4_VHigh:1_Low	1	4	NA	-1.541	1.159	-3.813	0.731	1
Odds Ratio	1	4	NA	0.214	NA	0.022	2.078	2

Again, the `kable` approach I used in these notes hides the adjusted values specified at the bottom of the `summary` table for this `lrm` model (model `m4`), but they are **Adjusted to: sex=F educ=1_Low**.

Now, how do we interpret these model `m4` results?

```
plot(summary(m4))
```



Adjusted to:sex=F educ=1_Low

- The **age** interpretation is that if we have two subjects, Al and Bob, who are the same sex and have the same education level, but Al is age 42 and Bob is age 57, then model

m4 projects that Bob's odds of being a smoker will be 0.442 times Al's odds of being a smoker. Equivalently, Bob's odds are 44.2% as large as Al's.

- Because of the interaction between *sex* and *educ* in our model m4, we must select an *educ* level in order to cleanly interpret the effect of *sex* on *smoker*. The **sex - M:F** interpretation compares M(ale) to F(emale) *sex* while requiring⁵ that *educ* = 1_Low. The result is that if we have two subjects, Carl and Diane, who are the same age and *each is in the low education group*, but Carl is Male and Diane is Female, then model m4 predicts that Carl's odds of being a *smoker* will be 4.422 times the odds for Diane.
- Because of the interaction between *sex* and *educ* in our model m4, we must select a *sex* in order to cleanly interpret the effect of *educ* on *totchol*. The **educ - 2_Middle:1_Low** term, for instance, compares "2_Middle" education to "1_Low" education while requiring that *sex* is Female⁶. The result is that if we have two subjects, Lola and Mina, who are the same age and *each is Female*, but Lola is in the "1_Low" education group and Mina is in the "2_Middle" education group, then model m4 predicts that Mina's odds of being a *smoker* will be 1.951 times Lola's odds, or, equivalently, that Mina will have 95.1% higher odds than Lola.

It should be easy to see that one is contained in each of the 95% confidence intervals summarizing model m4 except for the one for the main effect of *sex*, but we might also consider the impact of the interaction term, as described by the `anova` result for model m4.

```
anova(m4)
```

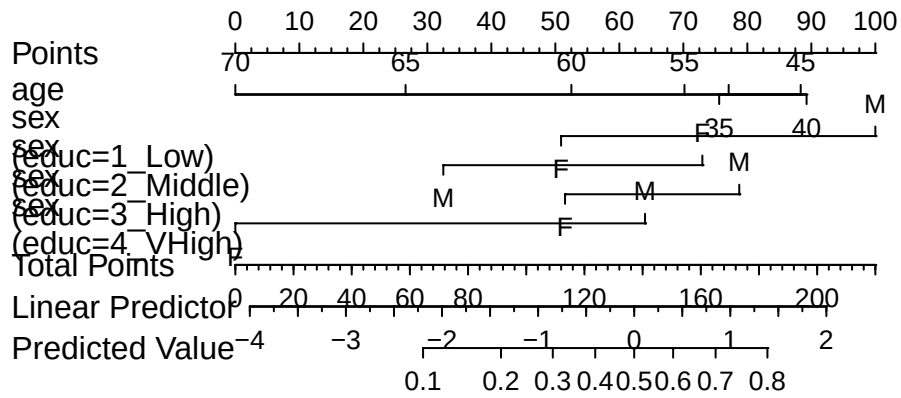
Wald Statistics		Response: smoker		
Factor		Chi-Square	d.f.	P
age		8.34	4	0.0799
Nonlinear		1.78	3	0.6196
sex (Factor+Higher Order Factors)		12.64	4	0.0132
All Interactions		10.48	3	0.0149
educ (Factor+Higher Order Factors)		12.43	6	0.0531
All Interactions		10.48	3	0.0149
sex * educ (Factor+Higher Order Factors)		10.48	3	0.0149
TOTAL NONLINEAR + INTERACTION		12.60	6	0.0498
TOTAL		21.69	11	0.0269

Finally, here is a nomogram of model m4.

⁵We know this because of the Adjusted to: *sex* = F, *educ*=1_Low note at the bottom of the `summary` output.

⁶Adjusted to: *sex* = F, *educ*=1_Low tells us this.

```
plot(nomogram(m4, fun = plogis))
```



23 Colorectal Cancer Screening and Some Special Cases

In this Chapter, we discuss two issues not yet raised regarding regression on a binary outcome.

1. What do we do if our binary outcome is not available for each subject individually, but instead aggregated?
2. What is probit regression, and how can we use it as an alternative to logistic regression on a binary outcome?

23.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

23.2 Data Load

```
colscr <- read_csv("data/screening.csv", show_col_types = FALSE)
colscr2 <- read_csv("data/screening2.csv", show_col_types = FALSE)
```

23.3 Logistic Regression for Aggregated Data

23.3.1 Colorectal Cancer Screening Data

The `screening.csv` data (imported into the R tibble `colscr` are simulated. They mirror a subset of the actual results from the [Better Health Partnership](#)'s pilot study of colorectal cancer screening in primary care clinics in Northeast Ohio, but the numbers have been fuzzed slightly, and the clinics have been de-identified and moved around from system to system.

Available to us are the following variables:

Variable	Description
<code>location</code>	clinic code
<code>subjects</code>	number of subjects reported by clinic
<code>screen_rate</code>	proportion of <code>subjects</code> who were screened
<code>screened</code>	number of <code>subjects</code> who were screened
<code>notscreened</code>	number of <code>subjects</code> not screened
<code>meanage</code>	mean age of clinic's subjects, years
<code>female</code>	% of clinic's subjects who are female
<code>pct_lowins</code>	% of clinic's subjects who have Medicaid or are uninsured
<code>system</code>	system code

```
describe(colscr)
```

```
colscr
```

```
9 Variables      26 Observations
```

```
-----
```

```
location
```

```
  n missing distinct
26      0         26
```

```
lowest : A B C D E, highest: V W X Y Z
```

```
-----
```

```
subjects
```

```
  n missing distinct    Info    Mean  pMedian    Gmd    .05
26      0         26      1    3247    2931    2134    1249
.10    .25    .50    .75    .90    .95
1475   1915   2766   3608   6523   7068
```

```
lowest : 803 1179 1459 1491 1512, highest: 5451 6061 6985 7095 7677
```

```

-----
screen_rate
      n  missing distinct      Info      Mean  pMedian      Gmd      .05
      26      0      26        1    0.7659    0.7658    0.08339    0.6682
      .10     .25     .50     .75      .90      .95
      0.6732    0.7179    0.7579    0.8088    0.8654    0.8899

```

```

lowest : 0.63549 0.666667 0.672891 0.673452 0.686992
highest: 0.823713 0.846911 0.883875 0.891911 0.904911
-----

```

```

screened
      n  missing distinct      Info      Mean  pMedian      Gmd      .05
      26      0      26        1    2584      2197      1895      843.5
      .10     .25     .50     .75      .90      .95
      993.0    1395.2    2169.5    2716.0    5293.5    6107.2

```

```

lowest : 572 794 992 994 1088, highest: 4818 4848 5739 6230 6947
-----

```

```

notscreened
      n  missing distinct      Info      Mean  pMedian      Gmd      .05
      26      0      26        1    663.2      641      303.5      336.0
      .10     .25     .50     .75      .90      .95
      352.5    508.8    611.0    791.0    989.0    1172.5

```

```

lowest : 231 335 339 366 371, highest: 881 927 1051 1213 1356
-----

```

```

meanage
      n  missing distinct      Info      Mean  pMedian      Gmd      .05
      26      0      23    0.999    60.58    60.45      2.186      58.23
      .10     .25     .50     .75      .90      .95
      58.35    58.82    60.50    61.98    62.50    62.90

```

```

lowest : 58 58.2 58.3 58.4 58.5, highest: 62.2 62.4 62.6 63 65.9
-----

```

```

female
      n  missing distinct      Info      Mean  pMedian      Gmd      .05
      26      0      23    0.999    58.72    59.1      7.118      46.93
      .10     .25     .50     .75      .90      .95
      48.45    55.42    60.05    62.62    64.90    67.50

```

```

lowest : 46.2 46.6 47.9 49 54.3, highest: 63.6 64.1 65.7 68.1 70.3
-----

```

```

pct_lowins

```

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
26	0	24	0.999	24.47	24.7	22.12	0.675
.10	.25	.50	.75	.90	.95		
1.800	4.800	23.950	44.025	49.500	49.950		

lowest : 0.3 0.4 1.5 2.1 3 , highest: 45.4 47.1 49.5 50.1 51.3

system

n	missing	distinct
26	0	4

Value	Sys_1	Sys_2	Sys_3	Sys_4
Frequency	7	7	6	6
Proportion	0.269	0.269	0.231	0.231

23.3.2 Fitting a Logistic Regression Model to Proportion Data

Here, we have a binary outcome (was the subject screened or not?) but we have aggregated results. We can use the counts of the numbers of subjects at each clinic (in `subjects`) and the proportion who were screened (in `screen_rate`) to fit a logistic regression model, as follows:

```
m_screen1 <- glm(screen_rate ~ meanage + female +
                  pct_lowins + system, family = binomial,
                  weights = subjects, data = colscr)

summary(m_screen1)
```

Call:

```
glm(formula = screen_rate ~ meanage + female + pct_lowins + system,
     family = binomial, data = colscr, weights = subjects)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.3270393	0.5530782	-2.399	0.0164 *
meanage	0.0679866	0.0089754	7.575	3.60e-14 ***
female	-0.0193142	0.0015831	-12.200	< 2e-16 ***
pct_lowins	-0.0134547	0.0008585	-15.672	< 2e-16 ***
systemSys_2	-0.1382189	0.0246591	-5.605	2.08e-08 ***
systemSys_3	-0.0400170	0.0254505	-1.572	0.1159

```

systemSys_4  0.0229273  0.0294207  0.779  0.4358
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2825.28  on 25  degrees of freedom
Residual deviance:  816.39  on 19  degrees of freedom
AIC: 1037.9

Number of Fisher Scoring iterations: 4

```

23.3.3 Fitting a Logistic Regression Model to Counts of Successes and Failures

```

m_screen2 <- glm(cbind(screened, notscreened) ~
                 meanage + female + pct_lowins + system,
                 family = binomial, data = colscr)
summary(m_screen2)

```

```

Call:
glm(formula = cbind(screened, notscreened) ~ meanage + female +
    pct_lowins + system, family = binomial, data = colscr)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.3270392	0.5530782	-2.399	0.0164	*
meanage	0.0679866	0.0089754	7.575	3.60e-14	***
female	-0.0193142	0.0015831	-12.200	< 2e-16	***
pct_lowins	-0.0134547	0.0008585	-15.672	< 2e-16	***
systemSys_2	-0.1382189	0.0246591	-5.605	2.08e-08	***
systemSys_3	-0.0400170	0.0254505	-1.572	0.1159	
systemSys_4	0.0229273	0.0294207	0.779	0.4358	

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

    Null deviance: 2825.28  on 25  degrees of freedom
Residual deviance:  816.39  on 19  degrees of freedom

```

AIC: 1037.9

Number of Fisher Scoring iterations: 4

23.3.4 How does one address this problem in rms?

We can use Glm. As an example to mirror `m_screen1`, we have the following..

```
d <- datadist(colscr)
options(datadist = "d")

mod_screen_1 <- Glm(screen_rate ~ meanage + female +
                    pct_lowins + system,
                    family = binomial, weights = subjects,
                    data = colscr, x = T, y = T)

mod_screen_1
```

General Linear Model

```
Glm(formula = screen_rate ~ meanage + female + pct_lowins + system,
    family = binomial, data = colscr, weights = subjects, x = T,
    y = T)
```

		Model Likelihood		
		Ratio Test		
Obs	26	LR chi2	2008.90	
Residual d.f.	19	d.f.	6	
g	0.461	Pr(> chi2)	<0.0001	

	Coef	S.E.	Wald Z	Pr(> Z)
Intercept	-1.3270	0.5531	-2.40	0.0164
meanage	0.0680	0.0090	7.57	<0.0001
female	-0.0193	0.0016	-12.20	<0.0001
pct_lowins	-0.0135	0.0009	-15.67	<0.0001
system=Sys_2	-0.1382	0.0247	-5.61	<0.0001
system=Sys_3	-0.0400	0.0255	-1.57	0.1159
system=Sys_4	0.0229	0.0294	0.78	0.4358

23.4 Probit Regression

23.4.1 Colorectal Cancer Screening Data on Individuals

The data in the `colscr2` data frame describe (disguised) data on the status of 172 adults who were eligible for colon cancer screening, with the following information included:

Variable	Description
<code>subject</code>	subject ID code
<code>age</code>	subject's age (years)
<code>race</code>	subject's race (White/Black/Other)
<code>hispanic</code>	subject of Hispanic ethnicity (1 = yes / 0 = no)
<code>insurance</code>	Commercial, Medicaid, Medicare, Uninsured
<code>bmi</code>	body mass index at most recent visit
<code>sbp</code>	systolic blood pressure at most recent visit
<code>up_to_date</code>	meets colon cancer screening standards

The goal is to use the other variables (besides subject ID) to predict whether or not a subject is up to date.

```
colscr2 %>% describe()
```

.

```
8 Variables      172 Observations
-----
subject
  n missing distinct    Info    Mean  pMedian    Gmd    .05
  172      0      172      1    186.5    186.5    57.67   109.6
  .10    .25    .50    .75     .90     .95
 118.1   143.8   186.5   229.2   254.9   263.4

lowest : 101 102 103 104 105, highest: 268 269 270 271 272
-----
age
  n missing distinct    Info    Mean  pMedian    Gmd    .05
  172      0      19    0.995    57.8    57.5    5.536   51.00
  .10    .25    .50    .75     .90     .95
 52.00   54.00   57.00   61.25   65.00   67.00

Value      51    52    53    54    55    56    57    58    59    60    61
```

Frequency	10	17	14	9	15	13	18	13	4	10	6
Proportion	0.058	0.099	0.081	0.052	0.087	0.076	0.105	0.076	0.023	0.058	0.035

Value	62	63	64	65	66	67	68	69
Frequency	11	4	5	7	6	3	4	3
Proportion	0.064	0.023	0.029	0.041	0.035	0.017	0.023	0.017

race

n	missing	distinct
172	0	3

Value	Black	Other	White
Frequency	118	9	45
Proportion	0.686	0.052	0.262

hispanic

n	missing	distinct	Info	Sum	Mean
172	0	2	0.18	11	0.06395

insurance

n	missing	distinct
172	0	4

Value	Commercial	Medicaid	Medicare	Uninsured
Frequency	32	81	46	13
Proportion	0.186	0.471	0.267	0.076

bmi

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
172	0	165	1	31.24	30.63	8.982	20.32
.10	.25	.50	.75	.90	.95		
21.88	25.48	30.05	36.03	43.06	45.68		

lowest : 17.2 17.59 17.85 18.09 18.44, highest: 49.41 50.83 53.28 54.66 55.41

sbp

n	missing	distinct	Info	Mean	pMedian	Gmd	.05
172	0	68	0.999	128.9	128.5	19.46	101.6
.10	.25	.50	.75	.90	.95		
109.1	118.0	127.0	138.0	150.9	162.0		

lowest : 89 90 94 95 96, highest: 166 168 169 170 198

```

-----
up_to_date
      n missing distinct      Info      Sum      Mean
    172         0         2    0.717     104    0.6047
-----

```

23.4.2 A logistic regression model

Here is a logistic regression model.

```

m_scr2_logistic <- glm(up_to_date ~ age + race + hispanic +
                        insurance + bmi + sbp,
                        family = binomial, data = colscr2)

summary(m_scr2_logistic)

```

Call:

```

glm(formula = up_to_date ~ age + race + hispanic + insurance +
     bmi + sbp, family = binomial, data = colscr2)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.7040470	2.7418625	0.986	0.3240
age	0.0204901	0.0396920	0.516	0.6057
raceOther	-1.9722351	1.0023237	-1.968	0.0491 *
raceWhite	-0.3210458	0.4001744	-0.802	0.4224
hispanic	0.0005855	0.7953482	0.001	0.9994
insuranceMedicaid	-1.0151860	0.4945169	-2.053	0.0401 *
insuranceMedicare	-0.5216006	0.5629935	-0.926	0.3542
insuranceUninsured	0.1099966	0.7906196	0.139	0.8893
bmi	0.0155894	0.0213547	0.730	0.4654
sbp	-0.0241777	0.0099138	-2.439	0.0147 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 230.85 on 171 degrees of freedom
Residual deviance: 210.55 on 162 degrees of freedom

```

AIC: 230.55

Number of Fisher Scoring iterations: 4

```
confint(m_scr2_logistic)
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-2.64274441	8.157738367
age	-0.05703934	0.099298904
raceOther	-4.24604744	-0.175953229
raceWhite	-1.10800168	0.469396096
hispanic	-1.53691431	1.688738936
insuranceMedicaid	-2.03640881	-0.079142467
insuranceMedicare	-1.66246932	0.564088260
insuranceUninsured	-1.40110091	1.759391281
bmi	-0.02568426	0.058563761
sbp	-0.04436143	-0.005282686

In this model, there appears to be some link between `sbp` and screening, as well as, perhaps, some statistically significant differences between some race groups and some insurance groups. We won't look at this much further for now, though. Instead, we'll simply describe predictions for two subjects, Harry and Sally.

23.4.3 Predicting status for Harry and Sally

- Harry is age 65, White, non-Hispanic, with Medicare insurance, a BMI of 28 and SBP of 135.
- Sally is age 60, Black, Hispanic, with Medicaid insurance, a BMI of 22 and SBP of 148.

```
newdat_s2 <- tibble(subject = c("Harry", "Sally"),
  age = c(65, 60),
  race = c("White", "Black"),
  hispanic = c(0, 1),
  insurance = c("Medicare", "Medicaid"),
  bmi = c(28, 22),
  sbp = c(135, 148))

predict(m_scr2_logistic, newdata = newdat_s2,
  type = "response")
```

```
      1      2
0.5904364 0.4215335
```

The prediction for Harry is 0.59, and for Sally, 0.42, by this logistic regression model.

23.4.4 A probit regression model

Now, consider a probit regression, fit by changing the default link for the `binomial` family as follows:

```
m_scr2_probit <- glm(up_to_date ~ age + race + hispanic +
                    insurance + bmi + sbp,
                    family = binomial(link = "probit"),
                    data = colscr2)

summary(m_scr2_probit)
```

Call:

```
glm(formula = up_to_date ~ age + race + hispanic + insurance +
     bmi + sbp, family = binomial(link = "probit"), data = colscr2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.584604	1.658489	0.955	0.3394
age	0.013461	0.024107	0.558	0.5766
raceOther	-1.238445	0.587093	-2.109	0.0349 *
raceWhite	-0.199260	0.243505	-0.818	0.4132
hispanic	0.029483	0.484819	0.061	0.9515
insuranceMedicaid	-0.619277	0.293205	-2.112	0.0347 *
insuranceMedicare	-0.322881	0.333549	-0.968	0.3330
insuranceUninsured	0.052776	0.463798	0.114	0.9094
bmi	0.009652	0.012887	0.749	0.4539
sbp	-0.014696	0.005944	-2.472	0.0134 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 230.85  on 171  degrees of freedom
Residual deviance: 210.49  on 162  degrees of freedom
```

AIC: 230.49

Number of Fisher Scoring iterations: 4

```
confint(m_scr2_probit)
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-1.70739455	4.894713408
age	-0.03400934	0.061437784
raceOther	-2.53819772	-0.119812915
raceWhite	-0.67910311	0.282566349
hispanic	-0.92476109	1.011089937
insuranceMedicaid	-1.21212866	-0.049749873
insuranceMedicare	-0.99315267	0.329851215
insuranceUninsured	-0.83335121	0.966504724
bmi	-0.01577917	0.035632196
sbp	-0.02659318	-0.003148488

23.4.5 Interpreting the Probit Model's Coefficients

It is possible to use any number of link functions to ensure that predicted values in a generalized linear model fall between 0 and 1. The probit regression model, for instance, uses the inverse of the cumulative distribution function of the Normal model as its link function. Let's look more closely at the coefficients of the probit model we just fit.

```
m_scr2_probit$coef
```

(Intercept)	age	raceOther	raceWhite
1.584603569	0.013461338	-1.238445198	-0.199260184
hispanic	insuranceMedicaid	insuranceMedicare	insuranceUninsured
0.029483051	-0.619276718	-0.322880519	0.052775722
bmi	sbp		
0.009652339	-0.014695526		

The probit regression coefficients give the change in the z-score of the outcome of interest (here, `up_to_date`) for a one-unit change in the target predictor, holding all other predictors constant.

- So, for a one-year increase in age, holding all other predictors constant, the z-score for `up_to_date` increases by 0.013
- And for a Medicaid subject as compared to a Commercial subject of the same age, race, ethnicity, bmi and sbp, the z-score for the Medicaid subject is predicted to be -0.619 lower, according to this model.

23.4.6 What about Harry and Sally?

Do the predictions for Harry and Sally change much with this probit model, as compared to the logistic regression?

```
predict(m_scr2_probit, newdata = newdat_s2, type = "response")
```

	1	2
	0.5885511	0.4364027

24 Modeling a Count Outcome

In this chapter, and the next two chapters, I use a count outcome (# of poor physical health days out of the last 30) in OHIO SMART data created in Chapter 6 to demonstrate regression models for count outcomes.

Methods discussed in the chapter include:

- Ordinary Least Squares
- Poisson Regression
- Overdispersed Quasi-Poisson Regression

24.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(boot)
library(countreg)
library(topmodels)
library(GGally)
library(lmtest)
library(rms)
library(sandwich)
library(tidyverse)

theme_set(theme_bw())
```

24.2 Data Load

```
smart_oh <- readRDS("data/smart_ohio.Rds")
```


24.3 Creating A Useful Analytic Subset, ohioA

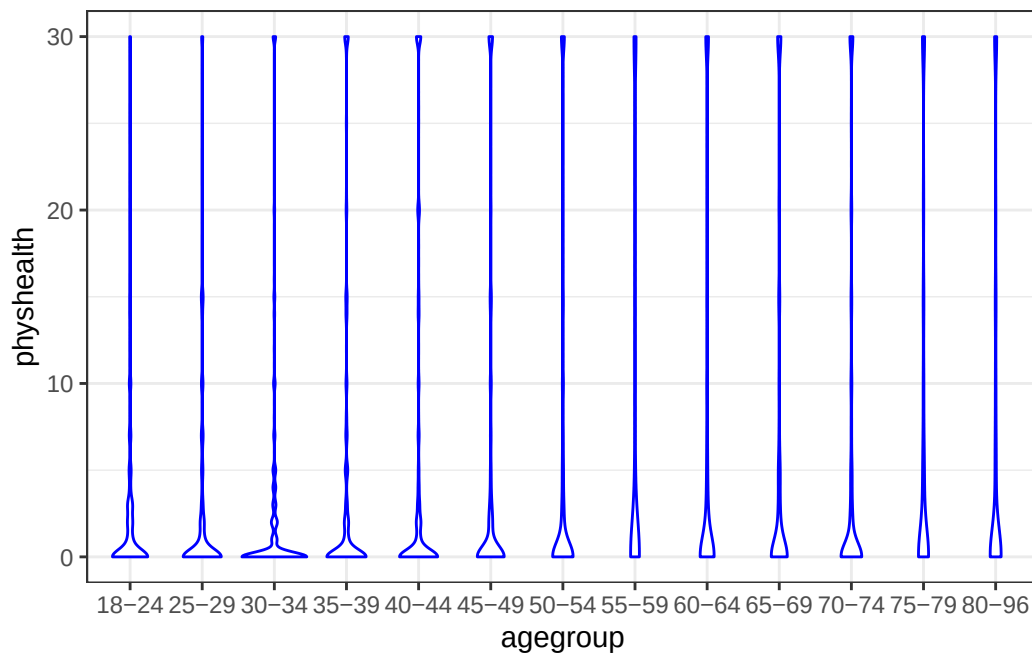
For this work, I'll include the subset of all observations in `smart_oh` with complete data on these 14 variables.

Variable	Description
SEQNO	Subject identification code
mmsa_name	Name of statistical area
genhealth	Five categories (E, VG, G, F, P) on general health
physhealth	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?
menthlth	Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?
healthplan	1 if the subject has any kind of health care coverage, 0 otherwise
costprob	1 indicates Yes to "Was there a time in the past 12 months when you needed to see a doctor but could not because of cost?"
agegroup	13 age groups from 18 through 80+
female	1 if subject is female
incomegroup	8 income groups from < 10,000 to 75,000 or more
bmi	body-mass index
smoke100	1 if Yes to "Have you smoked at least 100 cigarettes in your entire life?"
alcdays	# of days out of the past 30 on which the subject had at least one alcoholic drink

```
ohioA <- smart_oh |>
  select(SEQNO, mmsa_name, genhealth, physhealth,
         menthealth, healthplan, costprob,
         agegroup, female, incomegroup, bmi, smoke100,
         alcdays) |>
  drop_na()
```

24.3.1 Is age group associated with physhealth?

```
ggplot(ohioA, aes(x = agegroup, y = physhealth)) +
  geom_violin(col = "blue")
```



It's hard to see much of anything here. The main conclusion seems to be that 0 is by far the most common response.

Here's a table by age group of:

- the number of respondents in that age group,
- the group's mean **physhealth** response (remember that these are the number of poor physical health days in the last 30),
- their median **physhealth** response (which turns out to be 0 in each group), and
- the percentage of group members who responded 0.

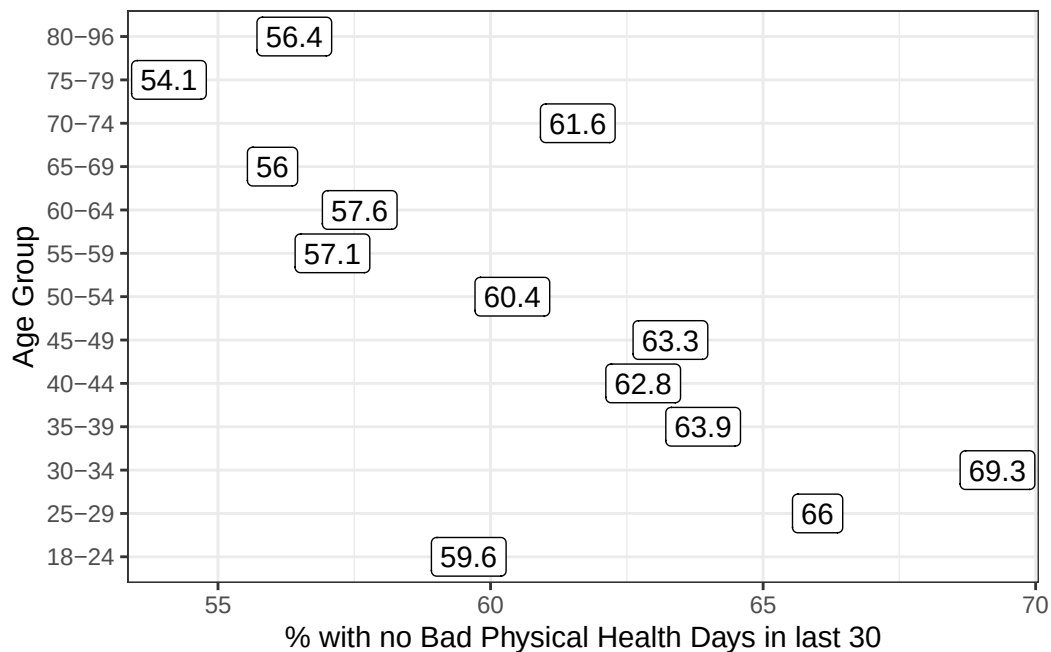
```
ohioA |> group_by(agegroup) |>
  summarize(n = n(), mean = round(mean(physhealth),2),
            median = median(physhealth),
            percent_0s = round(100*sum(physhealth == 0)/n,1))
```

```
# A tibble: 13 x 5
  agegroup      n mean median percent_0s
  <fct>    <int> <dbl>  <dbl>      <dbl>
1 18-24     297  2.31     0        59.6
2 25-29     259  2.53     0         66
3 30-34     296  2.14     0        69.3
4 35-39     366  3.67     0        63.9
```

5	40-44	347	4.18	0	62.8
6	45-49	409	4.46	0	63.3
7	50-54	472	4.76	0	60.4
8	55-59	608	6.71	0	57.1
9	60-64	648	5.9	0	57.6
10	65-69	604	6.09	0	56
11	70-74	490	4.89	0	61.6
12	75-79	338	6.38	0	54.1
13	80-96	374	6.1	0	56.4

We can see a real change between the 45-49 age group and the 50-54 age group. The mean difference is clear from the table above, and the plot below (of the percentage with a zero response) in each age group identifies the same story.

```
ohioA |> group_by(agegroup) |>
  summarize(n = n(),
            percent_0s = round(100*sum(physhealth == 0)/n,1)) |>
  ggplot(aes(y = agegroup, x = percent_0s)) +
  geom_label(aes(label = percent_0s)) +
  labs(x = "% with no Bad Physical Health Days in last 30",
       y = "Age Group")
```



It looks like we have a fairly consistent result in the younger age range (18-49) or the older range (50+). On the theory that most of the people reading this document are in that younger range, we'll focus on those respondents in what follows.

24.4 Exploratory Data Analysis (in the 18-49 group)

We want to predict the 0-30 `physhealth` count variable for the 18-49 year old respondents.

To start, we'll use two predictors:

- the respondent's body mass index, and
- whether the respondent has smoked 100 cigarettes in their lifetime.

We anticipate that each of these variables will have positive associations with the `physhealth` score. That is, heavier people, and those who have used tobacco will be less healthy, and thus have higher numbers of poor physical health days.

24.4.1 Build a subset of those ages 18-49

First, we'll identify the subset of respondents who are between 18 and 49 years of age.

```
ohioA_young.raw <- ohioA |>
  filter(agegroup %in% c("18-24", "25-29", "30-34",
                        "35-39", "40-44", "45-49")) |>
  droplevels()

ohioA_young.raw |>
  select(physhealth, bmi, smoke100, agegroup) |>
  summary()
```

physhealth	bmi	smoke100	agegroup
Min. : 0.000	Min. :14.00	Min. :0.0000	18-24:297
1st Qu.: 0.000	1st Qu.:23.74	1st Qu.:0.0000	25-29:259
Median : 0.000	Median :27.32	Median :0.0000	30-34:296
Mean : 3.337	Mean :28.79	Mean :0.4189	35-39:366
3rd Qu.: 2.000	3rd Qu.:32.43	3rd Qu.:1.0000	40-44:347
Max. :30.000	Max. :75.52	Max. :1.0000	45-49:409

24.4.2 Centering bmi

I'm going to center the `bmi` variable to help me interpret the final models later.

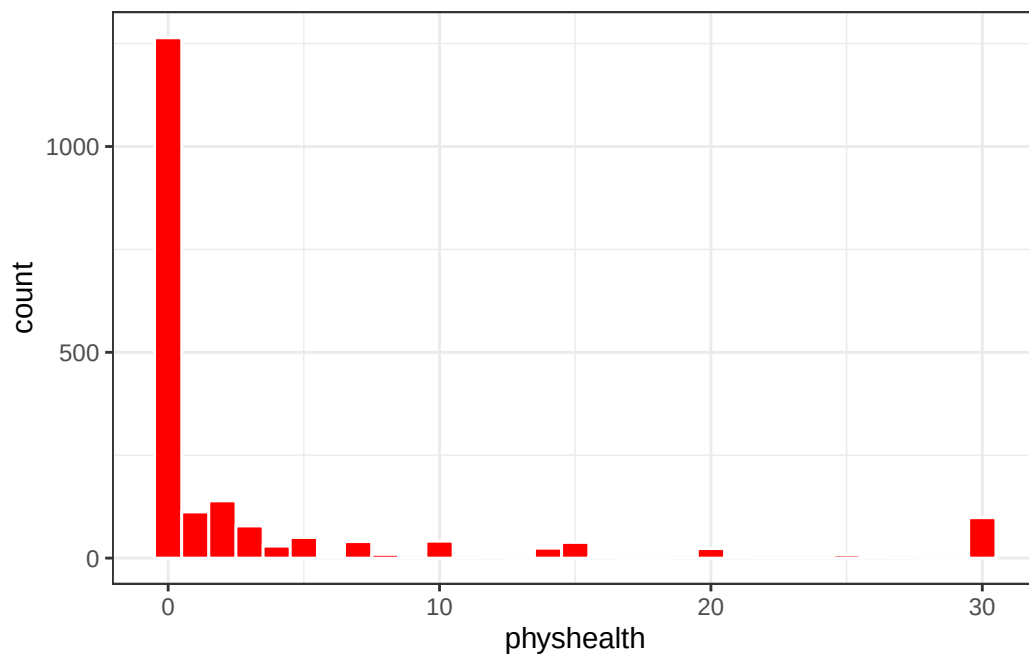
```
ohioA_young <- ohioA_young.raw |>
  mutate(bmi_c = bmi - mean(bmi))
```

Now, let's look more closely at the distribution of these variables, starting with our outcome.

24.4.3 Distribution of the Outcome

What's the distribution of `physhealth`?

```
ggplot(ohioA_young.raw, aes(x = physhealth)) +
  geom_histogram(binwidth = 1, fill = "red", col = "white")
```



```
ohioA_young.raw |>
  count(physhealth == 0, physhealth == 30)
```

```
# A tibble: 3 x 3
  `physhealth == 0` `physhealth == 30`      n
  <lgl>             <lgl>             <int>
1 FALSE            FALSE             612
2 FALSE            TRUE              98
3 TRUE             FALSE            1264
```

Most of our respondents said zero, the minimum allowable value, although there is also a much smaller bump at 30, the maximum value we will allow.

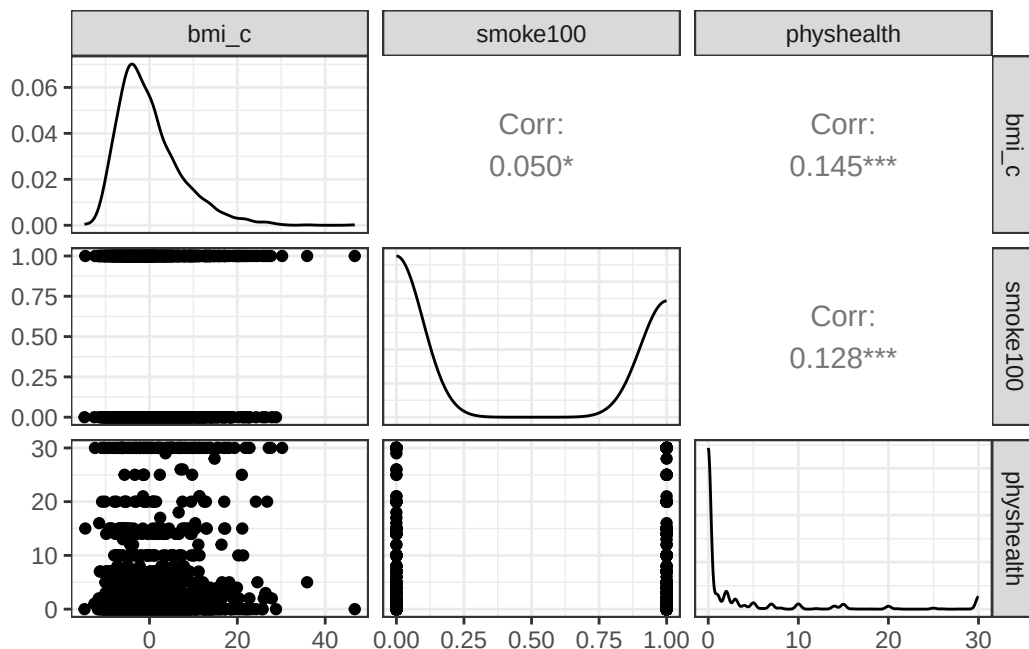
Dealing with this distribution is going to be a bit of a challenge. We will develop a series of potential modeling approaches for this sort of data, but before we do that, let's look at the distribution of our other two variables, and the pairwise associations, in a scatterplot matrix.

24.4.4 Scatterplot Matrix

Now, here's the scatterplot matrix for those 1974 subjects, using the centered `bmi` data captured in the `bmi_c` variable.

```
temp <- ohioA_young |> select(bmi_c, smoke100, physhealth)

ggpairs(temp)
```



So `bmi_c` and `smoke100` each have modest positive correlations with `physhealth` and only a very small correlation with each other. Here are some summary statistics for this final data.

24.4.5 Summary of the final subset of data

Remember that since the mean of `bmi` is 28.8, the `bmi_c` values are just `bmi - 28.8` for each subject.

```
ohioA_young |>
  select(bmi, bmi_c, smoke100, physhealth) |>
  summary()
```

	bmi	bmi_c	smoke100	physhealth
Min.	:14.00	Min. : -14.791	Min. :0.0000	Min. : 0.000
1st Qu.	:23.74	1st Qu.: -5.051	1st Qu.:0.0000	1st Qu.: 0.000
Median	:27.32	Median : -1.471	Median :0.0000	Median : 0.000
Mean	:28.79	Mean : 0.000	Mean :0.4189	Mean : 3.337
3rd Qu.	:32.43	3rd Qu.: 3.636	3rd Qu.:1.0000	3rd Qu.: 2.000
Max.	:75.52	Max. : 46.729	Max. :1.0000	Max. :30.000

24.5 Modeling Strategies Explored Here

We are going to predict `physhealth` using `bmi_c` and `smoke100`.

- Remember that `physhealth` is a count of the number of poor physical health days in the past 30.
- As a result, `physhealth` is restricted to taking values between 0 and 30.

We will demonstrate the use of each of the following regression models, some of which are better choices than others.

1. Ordinary Least Squares (OLS) predicting `physhealth`
2. OLS predicting the logarithm of (`physhealth + 1`)
3. Poisson regression, which is appropriate for predicting counts
4. Poisson regression, adjusted to account for overdispersion

and, in Chapter 25:

5. Negative binomial regression, also used for counts and which adjusts for overdispersion

and, in Chapter 26:

6. Zero-inflated models, in both the Poisson and Negative Binomial varieties, which allow us to fit counts that have lots of zero values
7. A “hurdle” model, which allows us to separately fit a model to predict the incidence of “0” and then a separate model to predict the value of `physhealth` when we know it is not zero
8. Tobit regression, where a lower (and upper) bound may be set, but the underlying model describes a latent variable which can extend beyond these boundaries

24.5.1 What Will We Demonstrate?

With each approach, we will fit the model and specify procedures for doing so in R. Then we will:

1. Specify the fitted model equation
2. Interpret the model’s coefficient estimates and 95% confidence intervals around those estimates.
3. Perform a test of whether each variable adds value to the model, when the other one is already included.
4. Store the fitted values and appropriate residuals for each model.
5. Summarize the model’s apparent R^2 value, the proportion of variation explained, and the model log likelihood.
6. Perform checks of model assumptions as appropriate.
7. Describe how predictions would be made for two new subjects.
 - Harry has a BMI that is 10 kg/m² higher than the average across all respondents and has smoked more than 100 cigarettes in his life.
 - Sally has a BMI that is 5 kg/m² less than the average across all respondents and has not smoked more than 100 cigarettes in her life.

In addition, for some of the new models, we provide a little of the mathematical background, and point to other resources you can use to learn more about the model.

24.5.2 Extra Data File for Harry and Sally

To make our lives a little easier, I’ll create a little tibble containing the necessary data for Harry and Sally.

```
hs_data <- tibble(subj = c("Harry", "Sally"),
                  bmi_c = c(10, -5), smoke100 = c(1, 0))
hs_data
```



```
# A tibble: 2 x 3
  subj bmi_c smoke100
  <chr> <dbl>   <dbl>
1 Harry    10         1
2 Sally    -5         0
```

24.6 The OLS Approach

```
mod_ols1 <- lm(physhealth ~ bmi_c + smoke100,
               data = ohioA_young)

summary(mod_ols1)
```

Call:

```
lm(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.1472	-3.6639	-2.2426	-0.7807	28.8777

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.57046	0.21648	11.874	< 2e-16 ***
bmi_c	0.14437	0.02305	6.263	4.61e-10 ***
smoke100	1.83061	0.33469	5.470	5.09e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.328 on 1971 degrees of freedom

Multiple R-squared: 0.0356, Adjusted R-squared: 0.03462

F-statistic: 36.37 on 2 and 1971 DF, p-value: 3.072e-16

```
confint(mod_ols1)
```

	2.5 %	97.5 %
(Intercept)	2.1459143	2.995005
bmi_c	0.0991636	0.189573
smoke100	1.1742213	2.486995

24.6.1 Interpreting the Coefficients

- The intercept, 2.57, is the predicted `physhealth` (in days) for a subject with average BMI who has not smoked 100 cigarettes or more.
- The `bmi_c` coefficient, 0.144, indicates that for each additional kg/m^2 of BMI, while holding `smoke100` constant, the predicted `physhealth` value increases by 0.144 day.
- The `smoke100` coefficient, 1.83, indicates that a subject who has smoked 100 cigarettes or more has a predicted `physhealth` value 1.83 days larger than another subject with the same `bmi` but who has not smoked 100 cigarettes.

24.6.2 Store fitted values and residuals

We can use `broom` to do this. Here, for instance, is a table of the first six predictions and residuals.

```
sm_ols_1 <- augment(mod_ols1, ohioA_young)

sm_ols_1 |> select(physhealth, .fitted, .resid) |> head()
```

```
# A tibble: 6 x 3
  physhealth .fitted .resid
    <dbl>    <dbl> <dbl>
1         0     2.13  -2.13
2         0     2.25  -2.25
3         0     3.14  -3.14
4        30     5.72  24.3
5         0     3.20  -3.20
6         0     3.83  -3.83
```

It turns out that 0 of the 1974 predictions that we make are below 0, and the largest prediction made by this model is 11.15 days.

24.6.3 Specify the R^2 and $\log(\text{likelihood})$ values

The `glance` function in the `broom` package gives us the raw and adjusted R^2 values, and the model $\log(\text{likelihood})$, among other summaries.

```
glance(mod_ols1) |> round(3)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>    <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>
1  0.036      0.035  7.33     36.4    0     2 -6731. 13470. 13492.
# i 3 more variables: deviance <dbl>, df.residual <dbl>, nobs <dbl>
```

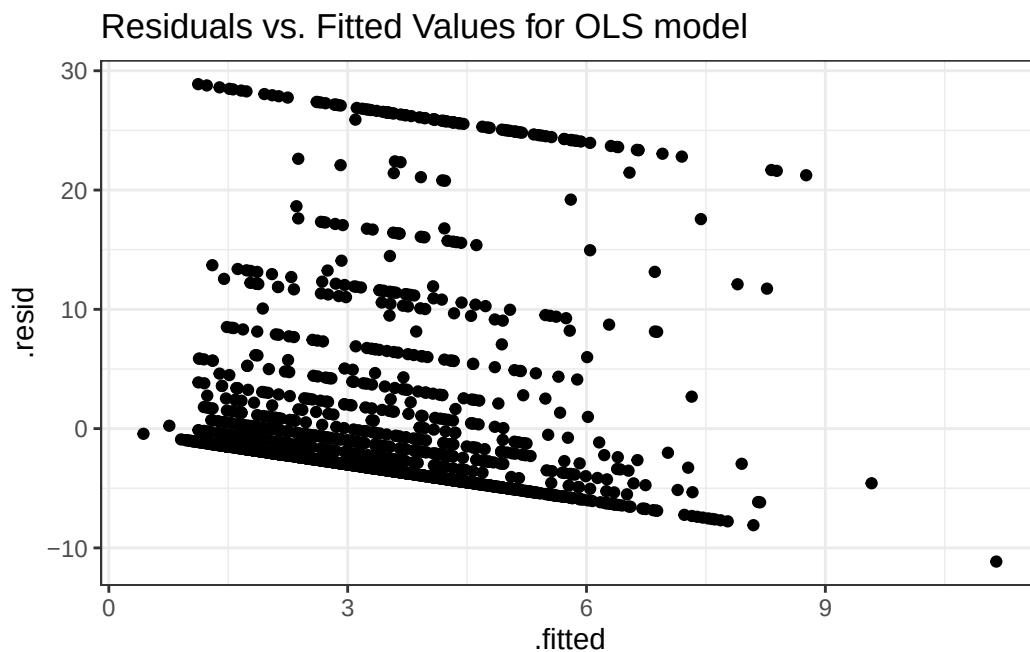
Here, we have

Model	R^2	log(likelihood)
OLS	0.036	-6730.98

24.6.4 Check model assumptions

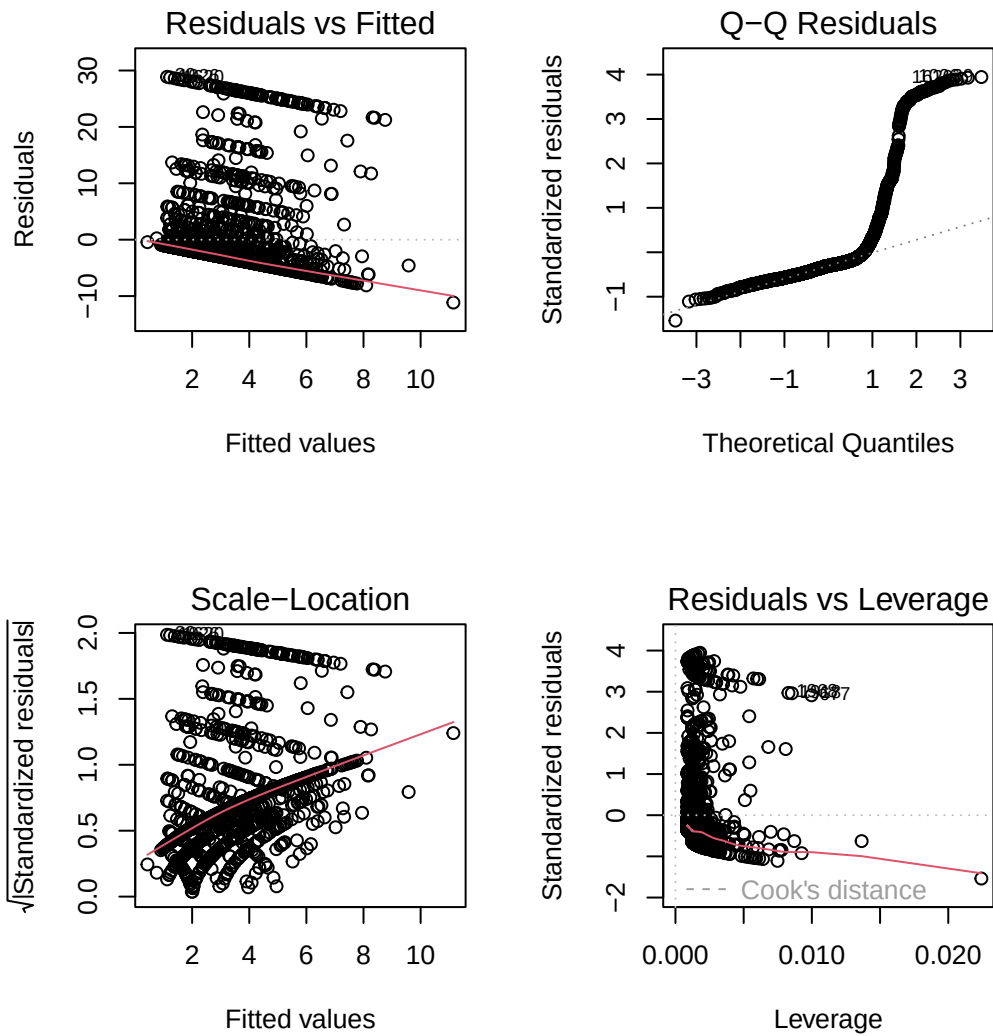
Here is a plot of the residuals vs. the fitted values for this OLS model.

```
ggplot(sm_ols_1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted Values for OLS model")
```



As usual, we can check OLS assumptions (linearity, homoscedasticity and normality) with R's complete set of residual plots.

```
par(mfrow = c(2,2))
plot(mod_ols1)
```



```
par(mfrow = c(1,1))
```

We see the problem with our residuals. They don't follow a Normal distribution.

24.6.5 Predictions for Harry and Sally

```
predict(mod_ols1, newdata = hs_data,  
        interval = "prediction")
```

	fit	lwr	upr
1	5.844750	-8.541164	20.23067
2	1.848618	-12.529923	16.22716

The prediction for Harry is 5.8 days, and for Sally is 1.8 days. The prediction intervals for each include some values below 0, even though 0 is the smallest possible value.

24.6.6 Notes

- This model could have been estimated using the `ols` function in the `rms` package, as well.

```
dd <- datadist(ohioA_young)  
options(datadist = "dd")  
  
(mod_ols1a <- ols(physhealth ~ bmi_c + smoke100,  
                  data = ohioA_young, x = TRUE, y = TRUE))
```

Linear Regression Model

```
ols(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young,  
     x = TRUE, y = TRUE)
```

		Model Likelihood	Discrimination
		Ratio Test	Indexes
Obs	1974	LR chi2	71.55
sigma	7.3276	d.f.	2
d.f.	1971	Pr(> chi2)	0.0000
			g
			1.570

Residuals

	Min	1Q	Median	3Q	Max
	-11.1472	-3.6639	-2.2426	-0.7807	28.8777

	Coef	S.E.	t	Pr(> t)
Intercept	2.5705	0.2165	11.87	<0.0001
bmi_c	0.1444	0.0230	6.26	<0.0001
smoke100	1.8306	0.3347	5.47	<0.0001

24.7 OLS model on log(physhealth + 1) days

We could try to solve the problem of fitting some predictions below 0 by log-transforming the data, so as to force values to be at least 0. Since we have undefined values when we take the log of 0, we'll add one to each of the `physhealth` values before taking logs, and then transform back when we want to make predictions.

```
mod_ols_log1 <- lm(log(physhealth + 1) ~ bmi_c + smoke100,
                   data = ohioA_young)

summary(mod_ols_log1)
```

Call:

```
lm(formula = log(physhealth + 1) ~ bmi_c + smoke100, data = ohioA_young)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.7079	-0.7058	-0.5051	0.5053	3.0484

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.57746	0.03099	18.634	< 2e-16 ***
bmi_c	0.01912	0.00330	5.796	7.91e-09 ***
smoke100	0.23679	0.04791	4.942	8.38e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.049 on 1971 degrees of freedom

Multiple R-squared: 0.03003, Adjusted R-squared: 0.02905

F-statistic: 30.51 on 2 and 1971 DF, p-value: 8.897e-14

```
confint(mod_ols_log1)
```

2.5 %	97.5 %

```
(Intercept) 0.51668682 0.63823590
bmi_c       0.01265192 0.02559421
smoke100    0.14282518 0.33075147
```

24.7.1 Interpreting the Coefficients

- The intercept, 0.58, is the predicted logarithm of `(physhealth + 1)` (in days) for a subject with average BMI who has not smoked 100 cigarettes or more.
 - We can exponentiate to see that the prediction for `(physhealth + 1)` here is $\exp(0.58) = 1.79$ so the predicted `physhealth` for a subject with average BMI who has not smoked 100 cigarettes is 0.79 days.
- The `bmi_c` coefficient, 0.019, indicates that for each additional kg/m^2 of BMI, while holding `smoke100` constant, the predicted logarithm of `(physhealth + 1)` increases by 0.019
- The `smoke100` coefficient, 0.24, indicates that a subject who has smoked 100 cigarettes or more has a predicted log of `(physhealth + 1)` value that is 0.24 larger than another subject with the same `bmi` but who has not smoked 100 cigarettes.

24.7.2 Store fitted values and residuals

We can use `broom` to help us with this. Here, for instance, is a table of the first six predictions and residuals, on the scale of our transformed response, `log(physhealth + 1)`.

```
sm_ols_log1 <- augment(mod_ols_log1, ohioA_young)

sm_ols_log1 <- sm_ols_log1 |>
  mutate(outcome = log(physhealth + 1))

sm_ols_log1 |>
  select(physhealth, outcome, .fitted, .resid) |>
  head()
```

```
# A tibble: 6 x 4
  physhealth outcome .fitted .resid
    <dbl>     <dbl>   <dbl> <dbl>
1         0         0     0.520 -0.520
2         0         0     0.535 -0.535
3         0         0     0.647 -0.647
4        30       3.43     0.995  2.44
5         0         0     0.656 -0.656
```

6	0	0	0.739	-0.739
---	---	---	-------	--------

Note that the `outcome` used in this model is `log(physhealth + 1)`, so the `.fitted` and `.resid` values react to that outcome, and not to our original `physhealth`.

Another option would be to calculate the model-predicted `physhealth`, which I'll call *ph* for a moment, with the formula:

$$ph = e^{\text{fitted}} - 1$$

```
sm_ols_log1 <- sm_ols_log1 |>
  mutate(pred.physhealth = exp(.fitted) - 1,
         res.physhealth = physhealth - pred.physhealth)

sm_ols_log1 |>
  select(physhealth, pred.physhealth, res.physhealth) |>
  head()
```

```
# A tibble: 6 x 3
  physhealth pred.physhealth res.physhealth
      <dbl>         <dbl>         <dbl>
1         0         0.681         -0.681
2         0         0.708         -0.708
3         0         0.910         -0.910
4        30         1.70          28.3
5         0         0.926         -0.926
6         0         1.09         -1.09
```

It turns out that 0 of the 1974 predictions that we make are below 0, and the largest prediction made by this model is 4.52 days.

24.7.3 Specify the R^2 and `log(likelihood)` values

The `glance` function in the `broom` package gives us the raw and adjusted R^2 values, and the model `log(likelihood)`, among other summaries.

```
glance(mod_ols_log1) |> round(3)
```



```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC
    <dbl>      <dbl> <dbl>      <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
1     0.03      0.029  1.05      30.5     0     2 -2894. 5796. 5818.
# i 3 more variables: deviance <dbl>, df.residual <dbl>, nobs <dbl>
```

Here, we have

Model	Scale	R^2	log(likelihood)
OLS on log log(physhealth + 1)	0.03		-2893.83

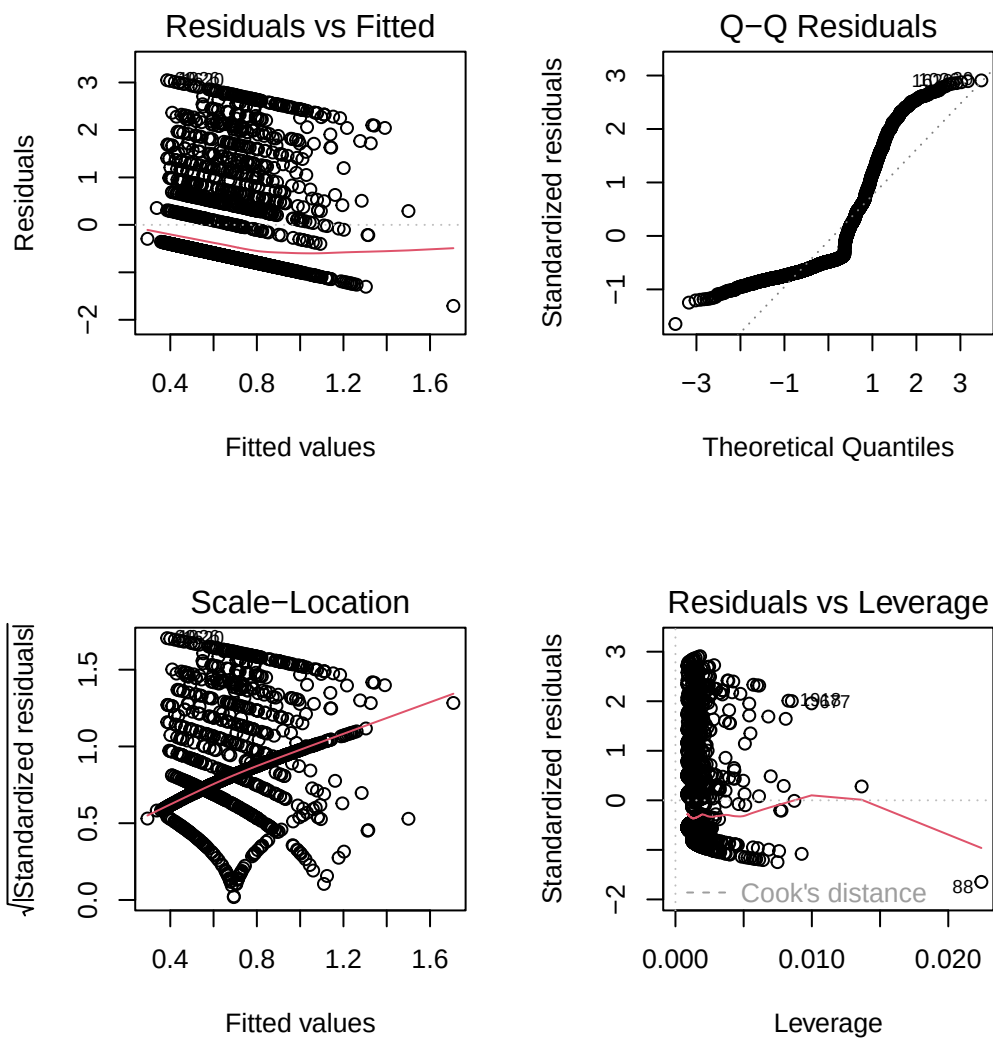
24.7.4 Getting R^2 on the scale of physhealth

We could find the correlation of our model-predicted `physhealth` values, after back-transformation, and our observed `physhealth` values, if we wanted to, and then square that to get a sort of R^2 value. But this model is not linear in `physhealth`, of course, so it's not completely comparable to our prior OLS model.

24.7.5 Check model assumptions

As usual, we can check OLS assumptions (linearity, homoscedasticity and normality) with R's complete set of residual plots. Of course, these residuals and fitted values are now on the `log(physhealth + 1)` scale.

```
par(mfrow = c(2,2))
plot(mod_ols_log1)
```



```
par(mfrow = c(1,1))
```

24.7.6 Predictions for Harry and Sally

```
predict(mod_ols_log1, newdata = hs_data,
        interval = "prediction", type = "response")
```

```
fit      lwr      upr
```

```
1 1.005480 -1.053893 3.064854
2 0.481846 -1.576472 2.540164
```

Again, these predictions are on the $\log(\text{physhealth} + 1)$ scale, and so we have to exponentiate them, and then subtract 1, to see them on the original `physhealth` scale.

```
exp(predict(mod_ols_log1, newdata = hs_data,
            interval = "prediction", type = "response")) - 1
```

```
      fit      lwr      upr
1 1.7332198 -0.6514221 20.43133
2 0.6190605 -0.7932970 11.68175
```

The prediction for Harry is now 1.73 days, and for Sally is 0.62 days. The prediction intervals for each again include some values below 0, which is the smallest possible value.

24.8 A Poisson Regression Model

The `physhealth` data describe a count. Specifically a count of the number of days where the subject felt poorly in the last 30. Why wouldn't we model this count with linear regression?

- A count can only be positive. Linear regression would estimate some subjects as having negative counts.
- A count is unlikely to follow a Normal distribution. In fact, it's far more likely that the log of the counts will follow a Poisson distribution.

So, we'll try that. The Poisson distribution is used to model a *count* outcome - that is, an outcome with possible values (0, 1, 2, ...). The model takes a somewhat familiar form to the models we've used for linear and logistic regression¹. If our outcome is y and our linear predictors X , then the model is:

$$y_i \sim \text{Poisson}(\lambda_i)$$

The parameter λ must be positive, so it makes sense to fit a linear regression on the logarithm of this...

$$\lambda_i = \exp(\beta_0 + \beta_1 X_1 + \dots \beta_k X_k)$$

¹This discussion is motivated by Section 6.2 of Gelman and Hill.

The coefficients β can be exponentiated and treated as multiplicative effects.

We'll run a generalized linear model with a log link function, ensuring that all of the predicted values will be positive, and using a Poisson error distribution. This is called **Poisson regression**.

Poisson regression may be appropriate when the dependent variable is a count of events. The events must be independent - the occurrence of one event must not make any other more or less likely. That's hard to justify in our case, but we can take a look.

```
mod_poiss1 <- glm(physhealth ~ bmi_c + smoke100,
                  family = poisson(),
                  data = ohioA_young)

summary(mod_poiss1)
```

Call:

```
glm(formula = physhealth ~ bmi_c + smoke100, family = poisson(),
     data = ohioA_young)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.906991	0.018699	48.51	<2e-16 ***
bmi_c	0.035051	0.001421	24.66	<2e-16 ***
smoke100	0.532505	0.024903	21.38	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 20222 on 1973 degrees of freedom
Residual deviance: 19151 on 1971 degrees of freedom
AIC: 21645

Number of Fisher Scoring iterations: 6

```
confint(mod_poiss1)
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	0.87011622	0.94342323
bmi_c	0.03225125	0.03782255
smoke100	0.48373882	0.58136497

24.8.1 The Fitted Equation

The model equation is

$$\log(\text{physhealth}) = 0.91 + 0.035 \text{ bmi_c} + 0.53 \text{ smoke100}$$

It looks like both `bmi` and `smoke_100` have confidence intervals excluding 0.

24.8.2 Interpreting the Coefficients

Our new model for y_i = counts of poor `physhealth` days in the last 30, follows the regression equation:

$$y_i \sim \text{Poisson}(\exp(0.91 + 0.035\text{bmi}_c + 0.53\text{smoke100}))$$

where `smoke100` is 1 if the subject has smoked 100 cigarettes (lifetime) and 0 otherwise, and `bmi_c` is just the centered body-mass index value in kg/m^2 . We interpret the coefficients as follows:

- The constant term, 0.91, gives us the intercept of the regression - the prediction if `smoke100` = 0 and `bmi_c` = 0. In this case, because we've centered BMI, it implies that $\exp(0.91) = 2.48$ is the predicted days of poor `physhealth` for a non-smoker with average BMI.
- The coefficient of `bmi_c`, 0.035, is the expected difference in count of poor `physhealth` days (on the log scale) for each additional kg/m^2 of body mass index. The expected multiplicative *increase* is $e^{0.035} = 1.036$, corresponding to a 3.6% difference in the count.
- The coefficient of `smoke100`, 0.53, tells us that the predictive difference between those who have and who have not smoked 100 cigarettes can be found by multiplying the `physhealth` count by $\exp(0.53) = 1.7$, yielding a 70% increase of the `physhealth` count.

As with linear or logistic regression, each coefficient is interpreted as a comparison where one predictor changes by one unit, while the others remain constant.

24.8.3 Testing the Predictors

We can use the Wald tests (z tests) provided with the Poisson regression output, or we can fit the model and then run an ANOVA to obtain a test based on the deviance (a simple transformation of the log likelihood ratio.)

- By the small p values in the Wald tests above, it seems that each predictor adds some predictive value to the model given the other predictor.
- The ANOVA approach for this model lets us check the impact of adding `smoke100` to a model already containing `bmi_c`.

```
anova(mod_poiss1, test = "Chisq")
```

Analysis of Deviance Table

Model: poisson, link: log

Response: physhealth

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			1973	20222	
bmi_c	1	609.46	1972	19612	< 2.2e-16 ***
smoke100	1	461.46	1971	19151	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

To obtain a p value for `smoke100`'s impact after `bmi_c` is accounted for, we compare the difference in deviance to a chi-square distribution with 1 degree of freedom. To check the effect of `bmi_c`, we could refit the model with `bmi_c` entering last, and again run an ANOVA.

We could also run a likelihood-ratio test for each predictor, by fitting the model with and without that predictor.

```
mod_poiss1_without_bmi <- glm(physhealth ~ smoke100,
                             family = poisson(),
                             data = ohioA_young)

anova(mod_poiss1, mod_poiss1_without_bmi, test = "Chisq")
```

Analysis of Deviance Table

```
Model 1: physhealth ~ bmi_c + smoke100
Model 2: physhealth ~ smoke100
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      1971      19151
2      1972      19692 -1  -540.98 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

24.8.4 Correcting for Overdispersion with `coeftest`/`coefci`

The main assumption we'll think about in a Poisson model is about **overdispersion**. We might deal with the overdispersion we see in this model by changing the nature of the tests we run within this model, using the `coeftest` or `coefci` approaches from the `lmtest` package, as I'll demonstrate next, or we might refit the model using a quasi-likelihood approach, as I'll show in the material to come.

Here, we'll use the `coeftest` and `coefci` approach from `lmtest` combined with robust sandwich estimation (via the `sandwich` package) to re-compute the Wald tests.

```
coeftest(mod_pois1, vcov. = sandwich)
```

z test of coefficients:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) 0.9069908  0.0717221 12.6459 < 2.2e-16 ***
bmi_c        0.0350508  0.0061178  5.7293 1.008e-08 ***
smoke100     0.5325053  0.1004300  5.3023 1.144e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coefci(mod_pois1, vcov. = sandwich)
```

```
              2.5 %      97.5 %
(Intercept) 0.76641801 1.04756366
bmi_c        0.02306016 0.04704145
smoke100     0.33566606 0.72934460
```

Both predictors still display small p values, but the standard errors are more appropriate. Later, we'll fit this approach by changing the estimation method to a quasi-likelihood approach.

24.8.5 Store fitted values and residuals

What happens if we try using the `broom` package in this case? We can, if we like, get our residuals and predicted values right on the scale of our `physhealth` response.

```
sm_poiss1 <- augment(mod_poiss1, ohioA_young,
                      type.predict = "response")

sm_poiss1 |>
  select(physhealth, .fitted) |>
  head()
```

```
# A tibble: 6 x 2
  physhealth .fitted
      <dbl>   <dbl>
1         0     2.23
2         0     2.29
3         0     3.10
4        30     5.32
5         0     3.15
6         0     3.68
```

24.8.6 Rootogram: see the fit of a count regression model

A **rootogram** is a very useful way to visualize the fit of a count regression model². The `rootogram` function in the `topmodels` package makes this pretty straightforward. By default, this fits a hanging rootogram on the square root of the frequencies.

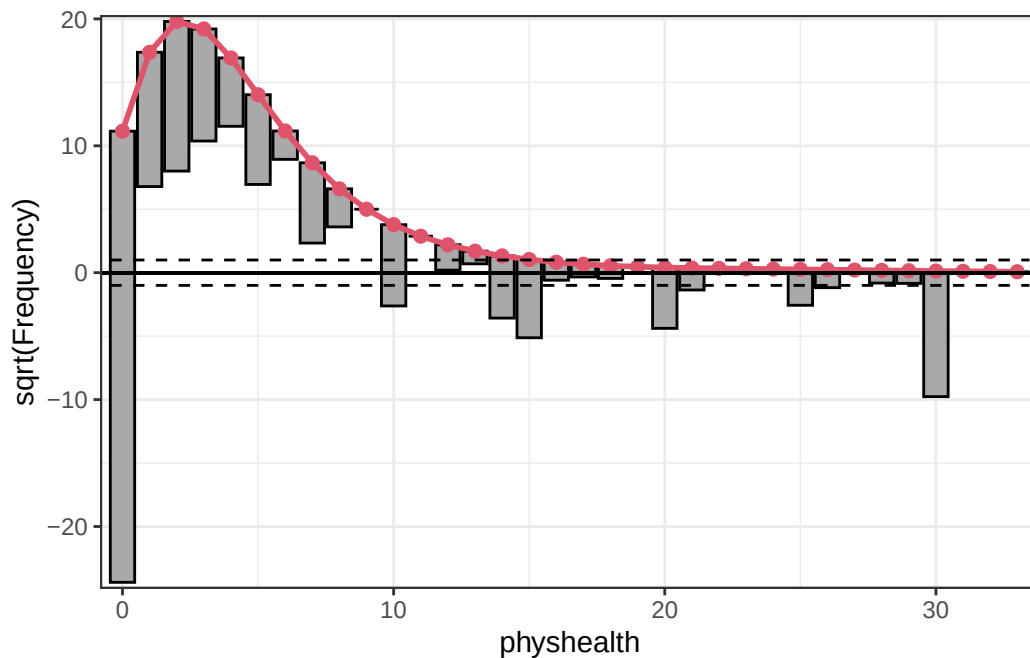
```
rootogram(mod_poiss1, max = 30)
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the topmodels package.
  Please report the issue to the authors.
```

²See <http://data.library.virginia.edu/getting-started-with-negative-binomial-regression-modeling/>

Warning: Using the `size` aesthetic in this geom was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` in the `default\_aes` field and elsewhere instead.
 i The deprecated feature was likely used in the topmodels package.
 Please report the issue to the authors.

Warning: Using the `size` aesthetic with geom_path was deprecated in ggplot2 3.4.0.
 i Please use the `linewidth` aesthetic instead.
 i The deprecated feature was likely used in the topmodels package.
 Please report the issue to the authors.



The red curved line is the theoretical Poisson fit. “Hanging” from each point on the red line is a bar, the height of which represents the difference between expected and observed counts. A bar hanging below 0 indicates underfitting. A bar hanging above 0 indicates overfitting. The counts have been transformed with a square root transformation to prevent smaller counts from getting obscured and overwhelmed by larger counts. We see a great deal of underfitting for counts of 0, and overfitting for most other counts, especially 1-6, with some underfitting again by `physhealth` above 14 days.

24.8.7 Specify the R^2 and $\log(\text{likelihood})$ values

We can calculate the R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(poiss_r <- with(sm_poiss1, cor(physhealth, .fitted)))
```

```
[1] 0.1846814
```

```
# R-square
poiss_r^2
```

```
[1] 0.03410723
```

The `glance` function in the `broom` package gives us model log(likelihood), among other summaries.

```
glance(mod_poiss1) |> round(3)
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik    AIC    BIC deviance df.residual nobs
      <dbl>    <dbl>   <dbl>  <dbl>  <dbl>   <dbl>      <dbl>  <dbl>
1      20222.    1973 -10820. 21645. 21662.   19151.      1971  1974
```

Here, we have

Model	Scale	R^2	log(likelihood)
Poisson	log(physhealth)	0.034	-10189.33

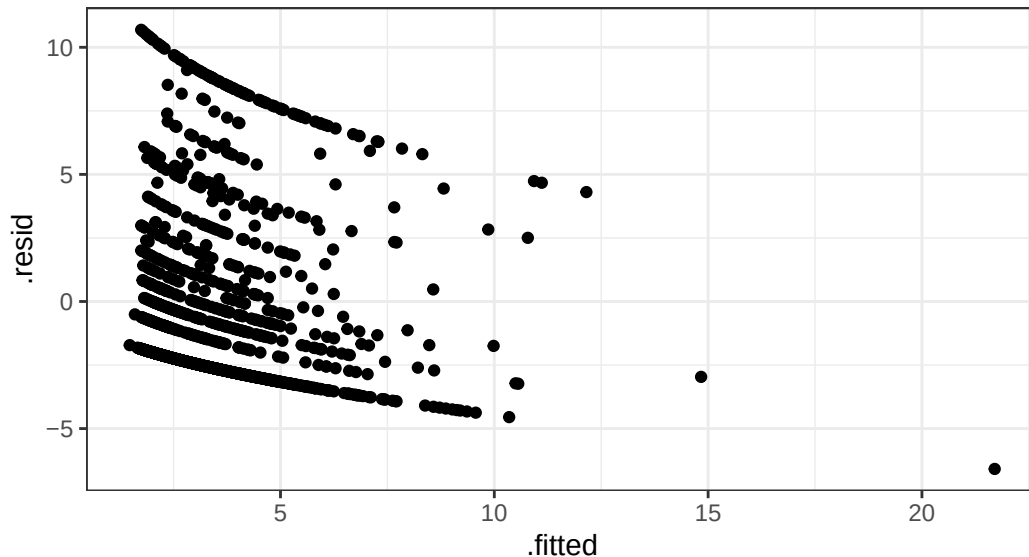
24.8.8 Check model assumptions

The Poisson model is a classical generalized linear model, estimated using the method of maximum likelihood. While the default `plot` option for a `glm` still shows the plots we would use to assess the assumptions of an OLS model, we don't actually get much from that, since our Poisson model has different assumptions. It can be useful to look at a plot of residuals vs. fitted values on the original `physhealth` scale.

```
ggplot(sm_poiss1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted `physhealth`",
       subtitle = "Original Poisson Regression model")
```

Residuals vs. Fitted `physhealth`

Original Poisson Regression model

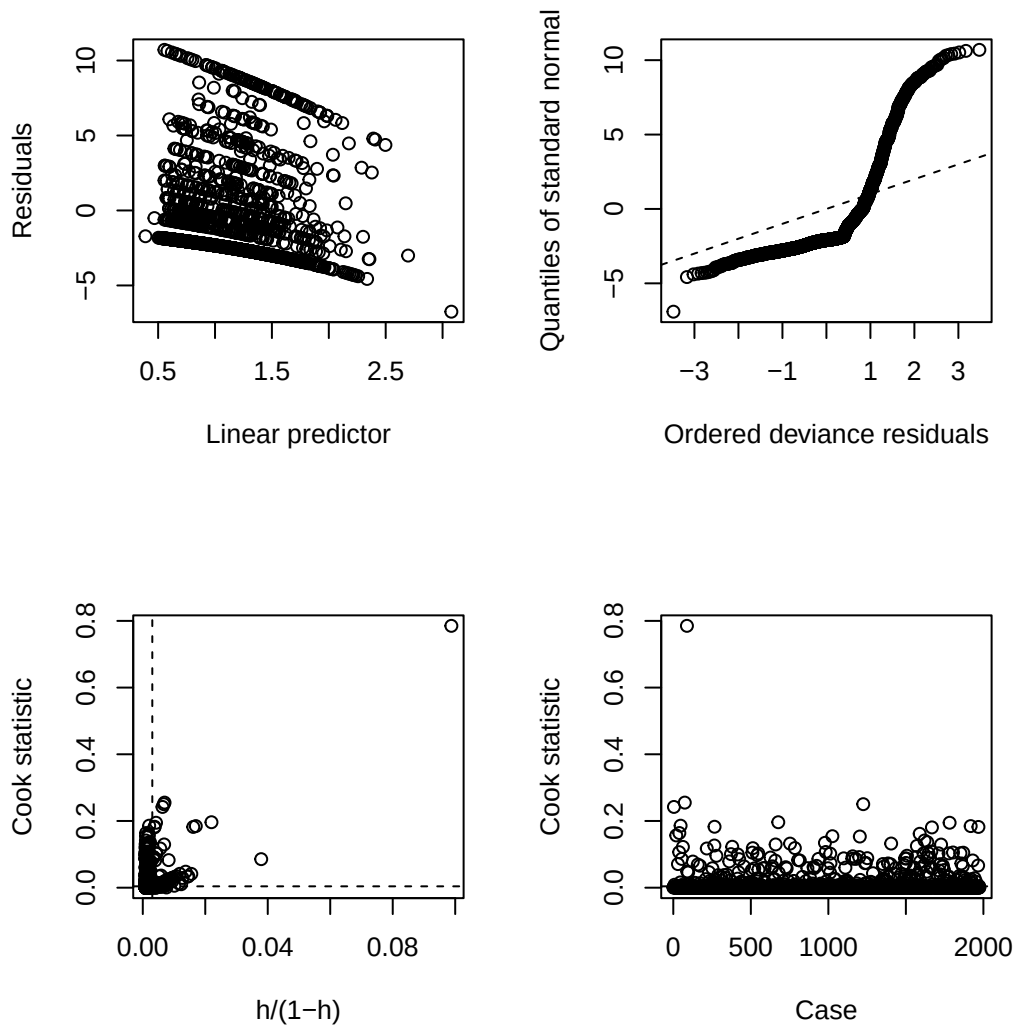


24.8.9 Using `glm.diag.plots` from the `boot` package

The `glm.diag.plots` function from the `boot` package makes a series of diagnostic plots for generalized linear models.

- (Top, Left) Jackknife deviance residuals against fitted values. This is essentially identical to what you obtain with `plot(mod_poiss1, which = 1)`. A *jackknife deviance* residual is also called a likelihood residual. It is the change in deviance when this observation is omitted from the data.
- (Top, Right) Normal Q-Q plot of standardized deviance residuals. (Dotted line shows expectation if those standardized residuals followed a Normal distribution, and these residuals generally should.) The result is similar to what you obtain with `plot(mod_poiss1, which = 2)`.
- (Bottom, Left) Cook statistic vs. standardized leverage
 - $n = \#$ of observations, $p = \#$ of parameters estimated
 - Horizontal dotted line is at $\frac{8}{n-2p}$. Points above the line have high influence on the model.
 - Vertical line is at $\frac{2p}{n-2p}$. Points to the right of the line have high leverage.
- (Bottom, Right) Index plot of Cook's statistic to help identify the observations with high influence. This is essentially the same plot as `plot(mod_poiss1, which = 4)`

```
glm.diag.plots(mod_pois1)
```



When working with these plots, it is possible to use the `iden` command to perform some interactive identification of points in your R terminal. But that doesn't play out effectively in an HTML summary document like this, so we'll leave that out.

24.8.10 Predictions for Harry and Sally

The predictions from a `glm` fit like this don't include prediction intervals. But we can get predictions on the scale of our original response variable, `physhealth`, like this.

```
predict(mod_pois1, newdata = hs_data, se.fit = TRUE,  
        type = "response")
```

```
$fit  
      1      2  
5.989478 2.078688  
  
$se.fit  
      1      2  
0.11544326 0.04314273  
  
$residual.scale  
[1] 1
```

By using `response` as the type, these predictions fall on the original `physhealth` scale. The prediction for Harry is now 5.99 days, and for Sally is 2.08 days.

24.9 Overdispersion in a Poisson Model

Poisson regressions do not supply an independent variance parameter σ , and as a result can be overdispersed, and usually are. Under the Poisson distribution, the variance equals the mean - so the standard deviation equals the square root of the mean. The notion of **overdispersion** arises here. When fitting generalized linear models with Poisson error distributions, the residual deviance and its degrees of freedom should be approximately equal if the model fits well.

If the residual deviance is far greater than the degrees of freedom, then overdispersion may well be a problem. In this case, the residual deviance is about 8.5 times the size of the residual degrees of freedom, so that's a clear indication of overdispersion. We saw earlier that the Poisson regression model requires that the outcome (here the `physhealth` counts) be independent. A possible reason for the overdispersion we see here is that `physhealth` on different days likely do not occur independently of one another but likely “cluster” together.

24.9.1 Testing for Overdispersion?

Gelman and Hill provide an overdispersion test in R for a Poisson model as follows...

```
yhat <- predict(mod_poiss1, type = "response")
n <- arm::display(mod_poiss1)$n
```

```
glm(formula = physhealth ~ bmi_c + smoke100, family = poisson(),
     data = ohioA_young)
               coef.est coef.se
(Intercept)  0.91      0.02
bmi_c         0.04      0.00
smoke100      0.53      0.02
---
n = 1974, k = 3
residual deviance = 19150.9, null deviance = 20221.8 (difference = 1070.9)
```

```
k <- arm::display(mod_poiss1)$k
```

```
glm(formula = physhealth ~ bmi_c + smoke100, family = poisson(),
     data = ohioA_young)
               coef.est coef.se
(Intercept)  0.91      0.02
bmi_c         0.04      0.00
smoke100      0.53      0.02
---
n = 1974, k = 3
residual deviance = 19150.9, null deviance = 20221.8 (difference = 1070.9)
```

```
z <- (ohioA_young$physhealth - yhat) / sqrt(yhat)
cat("overdispersion ratio is ", sum(z^2)/ (n - k), "\n")
```

```
overdispersion ratio is  15.58261
```

```
cat("p value of overdispersion test is ",
    pchisq(sum(z^2), df = n-k, lower.tail = FALSE), "\n")
```

```
p value of overdispersion test is  0
```

The p value is very small, indicating that the probability is essentially zero that a random variable from a χ^2 distribution with $(n - k) = 1971$ degrees of freedom would be as large as what we observed in this case. So there is almost surely some overdispersion.

In summary, the `physhealth` counts are overdispersed by a factor of 15.581, which is enormous (even a factor of 2 would be considered large.) The basic correction for overdispersion is to multiply all regression standard errors by $\sqrt{15.581} = 3.95$.

The `quasipoisson` model and the negative binomial model that we'll fit below are very similar. We write the overdispersed “quasiPoisson” model as:

$$y_i \sim \text{overdispersed Poisson}(\mu_i \exp(X_i \beta), \omega)$$

where ω is the overdispersion parameter, 15.581, in our case. The Poisson model we saw previously is then just the overdispersed Poisson model with $\omega = 1$.

24.10 Fitting the Quasi-Poisson Model

To deal with overdispersion, one useful approach is to apply a **quasi-likelihood estimation procedure**, as follows:

```
mod_poiss_od1 <- glm(physhealth ~ bmi_c + smoke100,
                     family = quasipoisson(),
                     data = ohioA_young)

summary(mod_poiss_od1)
```

Call:

```
glm(formula = physhealth ~ bmi_c + smoke100, family = quasipoisson(),
    data = ohioA_young)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.906991	0.073818	12.287	< 2e-16 ***
bmi_c	0.035051	0.005611	6.247	5.11e-10 ***
smoke100	0.532505	0.098308	5.417	6.81e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 15.58436)

```
Null deviance: 20222  on 1973  degrees of freedom
Residual deviance: 19151  on 1971  degrees of freedom
AIC: NA
```

```
Number of Fisher Scoring iterations: 6
```

```
confint(mod_poisson1)
```

```
Waiting for profiling to be done...
```

	2.5 %	97.5 %
(Intercept)	0.75877174	1.04831740
bmi_c	0.02383574	0.04583252
smoke100	0.34039368	0.72606948

This “quasi-Poisson regression” model uses the same mean function as Poisson regression, but now estimated by quasi-maximum likelihood estimation or, equivalently, through the method of generalized estimating equations, where the inference is adjusted by an estimated dispersion parameter. Sometimes, though I won’t demonstrate this here, people fit an “adjusted” Poisson regression model, where this estimation by quasi-ML is augmented to adjust the inference via sandwich estimates of the covariances³.

24.10.1 The Fitted Equation

The model equation is still $\log(\text{physhealth}) = 0.91 + 0.035 \text{ bmi_c} + 0.53 \text{ smoke100}$. The estimated coefficients still have small p values, but the standard errors for each coefficient are considerably larger when we account for overdispersion.

The dispersion parameter for the quasi-Poisson family is now taken to be a bit less than the square root of the ratio of the residual deviance and its degrees of freedom. This is a much more believable model, as a result.

24.10.2 Interpreting the Coefficients

No meaningful change from the Poisson model we saw previously.

³See Zeileis A Kleiber C Jackman S *Regression Models for Count Data in R* Vignette at <https://cran.r-project.org/web/packages/pscl/vignettes/countreg.pdf>

24.10.3 Testing the Predictors

Again, we can use the Wald tests (z tests) provided with the Poisson regression output, or we can fit the model and then run an ANOVA to obtain a test based on the deviance (a simple transformation of the log likelihood ratio.)

- By the Wald tests shown above, each predictor appears to add some predictive value to the model given the other predictor.
- The ANOVA approach for this model lets us check the impact of adding `smoke100` to a model already containing `bmi_c`.

```
anova(mod_pois_od1, test = "Chisq")
```

Analysis of Deviance Table

Model: quasipoisson, link: log

Response: physhealth

Terms added sequentially (first to last)

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			1973	20222	
bmi_c	1	609.46	1972	19612	4.011e-10 ***
smoke100	1	461.46	1971	19151	5.282e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The result is unchanged. To obtain a p value for `smoke100`'s impact after `bmi_c` is accounted for, we compare the difference in deviance to a chi-square distribution with 1 degree of freedom. The resulting p value is very small.

To check the effect of `bmi_c`, we could refit the model with and without `bmi_c`, and again run an ANOVA. I'll skip that here.

24.10.4 Store fitted values and residuals

What happens if we try using the `broom` package in this case? We can, if we like, get our predicted values right on the scale of our `physhealth` response.

```
sm_poiss_od1 <- augment(mod_poiss_od1, ohioA_young,
                        type.predict = "response")

sm_poiss_od1 |>
  select(physhealth, .fitted) |>
  head()
```

```
# A tibble: 6 x 2
  physhealth .fitted
    <dbl>    <dbl>
1         0     2.23
2         0     2.29
3         0     3.10
4        30     5.32
5         0     3.15
6         0     3.68
```

It turns out that 0 of the 1974 predictions that we make are below 0, and the largest prediction made by this model is 21.7 days.

The `rootogram` function we've shown doesn't support overdispersed Poisson models at the moment.

24.10.5 Specify the R^2 and $\log(\text{likelihood})$ values

We can calculate the R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(poiss_od_r <- with(sm_poiss_od1, cor(physhealth, .fitted)))
```

```
[1] 0.1846814
```

```
# R-square
poiss_od_r^2
```

```
[1] 0.03410723
```

The `glance` function in the `broom` package gives us model log(likelihood), among other summaries.

```
glance(mod_poiss_od1) |> round(3)
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik   AIC   BIC deviance df.residual  nobs
    <dbl>      <dbl>  <dbl> <dbl> <dbl>   <dbl>      <dbl> <dbl>
1    20222.    1973    NA    NA    NA   19151.    1971  1974
```

Here, we have

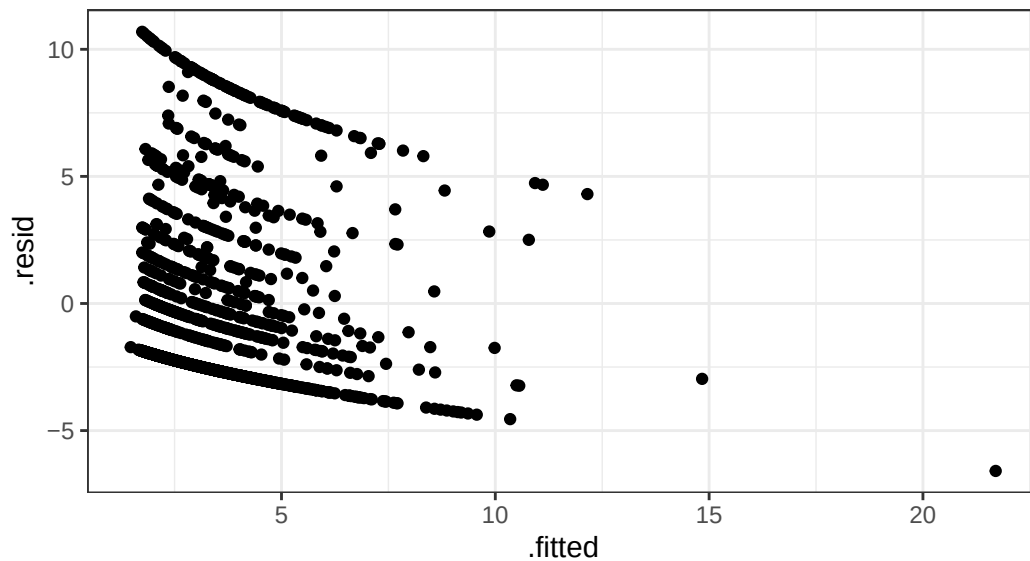
Model	Scale	R^2	log(likelihood)
Poisson	log(physhealth)	0.034	NA

24.10.6 Check model assumptions

Having dealt with the overdispersion, this should be a cleaner model in some ways, but the diagnostics (other than the dispersion) will be the same. Here is a plot of residuals vs. fitted values on the original `physhealth` scale.

```
ggplot(sm_poiss_od1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted `physhealth`",
       subtitle = "Overdispersed Poisson Regression model")
```

Residuals vs. Fitted `physhealth` Overdispersed Poisson Regression model



I'll skip the `glm.diag.plots` results, since you've already seen them.

24.10.7 Predictions for Harry and Sally

The predictions from this overdispersed Poisson regression will match those in the original Poisson regression, but the standard error will be larger.

```
predict(mod_pois_od1, newdata = hs_data, se.fit = TRUE,  
        type = "response")
```

```
$fit
```

```
      1      2  
5.989478 2.078688
```

```
$se.fit
```

```
      1      2  
0.4557357 0.1703147
```

```
$residual.scale
```

```
[1] 3.947703
```

By using `response` as the type, these predictions fall on the original `physhealth` scale. Again, the prediction for Harry is 5.99 days, and for Sally is 2.08 days.

24.11 Poisson and Quasi-Poisson models using Glm from the rms package

The `Glm` function in the `rms` package can be used to fit both the original Poisson regression and the quasi-Poisson model accounting for overdispersion.

24.11.1 Refitting the original Poisson regression with Glm

```
d <- datadist(ohioA_young)
options(datadist = "d")

mod_poi_Glm_1 <- Glm(physhealth ~ bmi_c + smoke100,
                     family = poisson(),
                     data = ohioA_young,
                     x = T, y = T)

mod_poi_Glm_1
```

General Linear Model

```
Glm(formula = physhealth ~ bmi_c + smoke100, family = poisson(),
     data = ohioA_young, x = T, y = T)
```

		Model Likelihood	
		Ratio Test	
	Obs1974	LR chi2	1070.93
Residual	d.f.1971	d.f.	2
	g 0.418	Pr(> chi2)	<0.0001
	Coef	S.E.	Wald Z Pr(> Z)
Intercept	0.9070	0.0187	48.50 <0.0001
bmi_c	0.0351	0.0014	24.66 <0.0001
smoke100	0.5325	0.0249	21.38 <0.0001

24.11.2 Refitting the overdispersed Poisson regression with Glm

```
d <- datadist(ohioA_young)
options(datadist = "d")

mod_poi_od_Glm_1 <- Glm(physhealth ~ bmi_c + smoke100,
                        family = quasipoisson(),
                        data = ohioA_young,
                        x = T, y = T)

mod_poi_od_Glm_1
```

General Linear Model

```
Glm(formula = physhealth ~ bmi_c + smoke100, family = quasipoisson(),
    data = ohioA_young, x = T, y = T)
```

		Model Likelihood	Ratio Test
	Obs1974	LR chi2	1070.93
Residual	d.f.1971	d.f.	2
	g 0.418	Pr(> chi2)	<0.0001

	Coef	S.E.	Wald Z	Pr(> Z)
Intercept	0.9070	0.0738	12.29	<0.0001
bmi_c	0.0351	0.0056	6.25	<0.0001
smoke100	0.5325	0.0983	5.42	<0.0001

The big advantage here is that we have access to the usual ANOVA, `summary`, and `nomogram` features that `rms` brings to fitting models.

24.11.3 ANOVA on a Glm fit

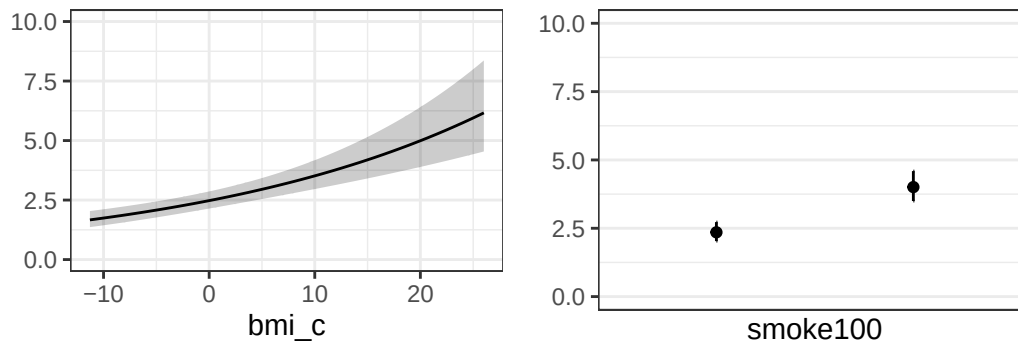
```
anova(mod_poi_od_Glm_1)
```

	Wald Statistics			Response: physhealth
Factor	Chi-Square	d.f.	P	
bmi_c	39.03	1	<.0001	
smoke100	29.34	1	<.0001	
TOTAL	74.39	2	<.0001	

This shows the individual Wald χ^2 tests without having to refit the model.

24.11.4 ggplots from Glm fit

```
ggplot(Predict(mod_poi_od_Glm_1, fun = exp))
```



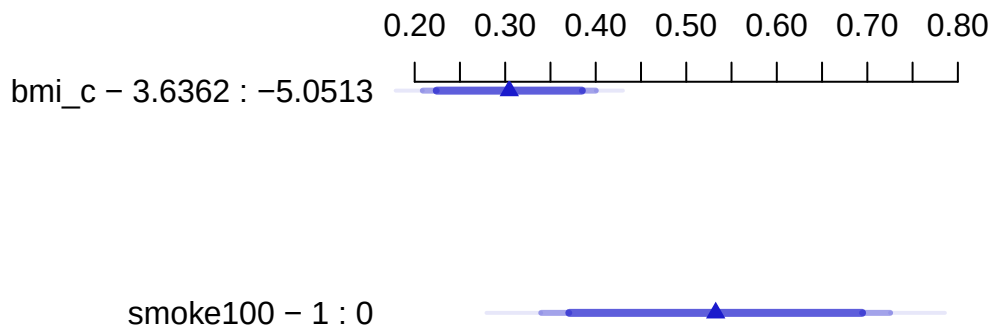
24.11.5 Summary of a Glm fit

```
summary(mod_poi_od_Glm_1)
```

Effects				Response : physhealth			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
bmi_c	-5.0513	3.6362	8.6875	0.30450	0.048742	0.20891	0.4001
smoke100	0.0000	1.0000	1.0000	0.53251	0.098308	0.33971	0.7253

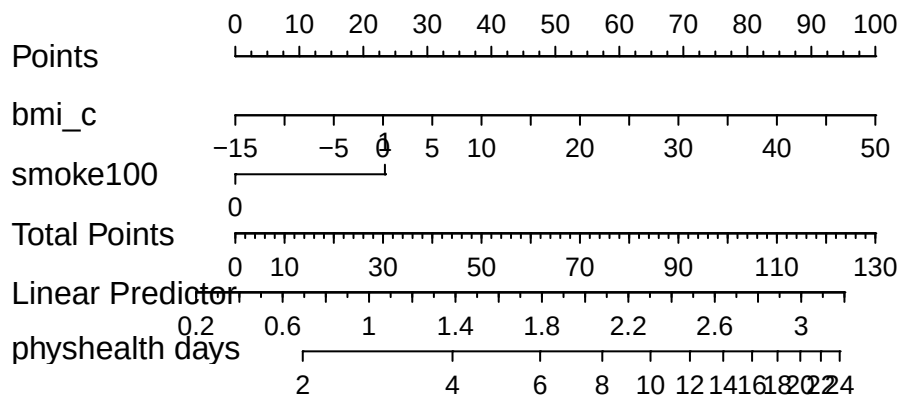
24.11.6 Plot of the Summary

```
plot(summary(mod_poi_od_Glm_1))
```



24.11.7 Nomogram of a Glm fit

```
plot(nomogram(mod_poi_od_Glm_1, fun = exp,  
             funlabel = "physhealth days"))
```

Note the use of `fun=exp` in both the `ggplot` of `Predict` and the nomogram. What's that doing?

In the next chapter, we'll expand beyond Poisson regression to consider a Negative Binomial model.

25 Negative Binomial Models for Count Data

We will continue to use a count outcome (# of poor physical health days out of the last 30) in OHIO SMART data created in Chapter 6 to demonstrate regression models for count outcomes, as we did in Chapter 24.

Methods discussed in the chapter include:

- Negative Binomial Regression

25.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(boot)
library(countreg)
library(topmodels)
library(MASS)
library(tidyverse)

theme_set(theme_bw())
```

25.2 Data Load and Subset Creation

```
smart_oh <- readRDS("data/smart_ohio.Rds")
```

As in Chapter 24, we'll create a subset of these data for analysis.

```
ohioA <- smart_oh |>
  select(SEQN0, mmsa_name, genhealth, physhealth,
         menthealth, healthplan, costprob,
         agegroup, female, incomegroup, bmi, smoke100,
         alcdays) |>
  drop_na()

ohioA_young <- ohioA |>
  filter(agegroup %in% c("18-24", "25-29", "30-34",
                        "35-39", "40-44", "45-49")) |>
  droplevels() |>
  mutate(bmi_c = bmi - mean(bmi))
```

25.3 Setup for this Chapter

Again, we're going to predict `physhealth` using `bmi_c` and `smoke100`.

- Remember that `physhealth` is a count of the number of poor physical health days in the past 30.
- As a result, `physhealth` is restricted to taking values between 0 and 30.

In this chapter, we demonstrate Negative binomial regression, which (like Poisson regression discussed in Chapter 24) is also used for counts and adjusts for overdispersion.

25.3.1 What Will We Demonstrate?

With this new approach, we will again fit the model and specify procedures for doing so in R. Then we will:

1. Specify the fitted model equation
2. Interpret the model's coefficient estimates and 95% confidence intervals around those estimates.
3. Perform a test of whether each variable adds value to the model, when the other one is already included.
4. Store the fitted values and appropriate residuals for each model.
5. Summarize the model's apparent R^2 value, the proportion of variation explained, and the model log likelihood.
6. Perform checks of model assumptions as appropriate.
7. Describe how predictions would be made for two new subjects.

- Harry has a BMI that is 10 kg/m² higher than the average across all respondents and has smoked more than 100 cigarettes in his life.
- Sally has a BMI that is 5 kg/m² less than the average across all respondents and has not smoked more than 100 cigarettes in her life.

In addition, for some of the new models, we provide a little of the mathematical background, and point to other resources you can use to learn more about the model.

25.3.2 Extra Data File for Harry and Sally

To make our lives a little easier, I'll create a little tibble containing the necessary data for Harry and Sally.

```
hs_data <- tibble(subj = c("Harry", "Sally"),
                  bmi_c = c(10, -5),
                  smoke100 = c(1, 0))
hs_data
```

```
# A tibble: 2 x 3
  subj   bmi_c smoke100
  <chr> <dbl>    <dbl>
1 Harry     10         1
2 Sally     -5         0
```

25.3.3 Our Poisson Model (for comparison)

```
mod_poiss1 <- glm(physhealth ~ bmi_c + smoke100,
                  family = poisson(),
                  data = ohioA_young)
```

25.4 Negative Binomial Model

Another approach to dealing with overdispersion is to fit a negative binomial model¹ to predict the log(`physhealth`) counts. This involves the fitting of an additional parameter, θ . That's our dispersion parameter²

¹See <https://cran.r-project.org/web/packages/pscl/vignettes/countreg.pdf> for more details.

²This θ is the inverse of the dispersion parameter estimated for these models by most other software packages, like SAS, Stata and SPSS. See <https://stats.idre.ucla.edu/r/dae/negative-binomial-regression/> for more details.

Sometimes, people will fit a model where θ is known, for instance a geometric model (where $\theta = 1$), and then this can be directly plugged into a `glm()` fit, but the more common scenario is that we are going to iteratively estimate the β coefficients and θ . To do this, I'll use the `glm.nb` function from the **MASS** package.

```
mod_nb1 <- glm.nb(physhealth ~ bmi_c + smoke100, link = log,
                  data = ohioA_young)

summary(mod_nb1)
```

Call:

```
glm.nb(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young,
       link = log, init.theta = 0.1487673114)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.874530	0.078994	11.071	< 2e-16 ***
bmi_c	0.035712	0.008317	4.294	1.76e-05 ***
smoke100	0.596396	0.121166	4.922	8.56e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.1488) family taken to be 1)

Null deviance: 1468.1 on 1973 degrees of freedom
Residual deviance: 1422.6 on 1971 degrees of freedom
AIC: 6976.6

Number of Fisher Scoring iterations: 1

Theta: 0.14877
Std. Err.: 0.00705

2 x log-likelihood: -6968.55300

```
confint(mod_nb1)
```

Waiting for profiling to be done...

2.5 % 97.5 %

```
(Intercept) 0.72304817 1.03304660
bmi_c        0.02072601 0.05124925
smoke100     0.35977590 0.83584673
```

25.4.1 The Fitted Equation

The form of the model equation for a negative binomial regression is the same as that for Poisson regression.

```
log(physhealth) = 0.87 + 0.036 bmi_c + 0.60 smoke100
```

25.4.2 Comparison with the (raw) Poisson model

To compare the negative binomial model to the Poisson model (without the overdispersion) we can use the `logLik` function to make a comparison. Note that the Poisson model is a subset of the negative binomial.

```
logLik(mod_nb1)
```

```
'log Lik.' -3484.277 (df=4)
```

```
logLik(mod_poiss1)
```

```
'log Lik.' -10819.6 (df=3)
```

```
2 * (logLik(mod_nb1) - logLik(mod_poiss1))
```

```
'log Lik.' 14670.65 (df=4)
```

```
pchisq(2 * (logLik(mod_nb1) - logLik(mod_poiss1)), df = 1, lower.tail = FALSE)
```

```
'log Lik.' 0 (df=4)
```

Here, the difference in the log likelihoods is large enough that the resulting p value is very small. This strongly suggests that the negative binomial model, which adds the dispersion parameter, is more appropriate than the raw Poisson model.

However, both the regression coefficients and the standard errors are rather similar to the quasi-Poisson and the sandwich-adjusted Poisson results above. Thus, in terms of predicted means, all three models give very similar results; the associated Wald tests also lead to the same conclusions.

25.4.3 Interpreting the Coefficients

There's only a small change here from the Poisson models we saw previously.

- The constant term, 0.87, gives us the intercept of the regression - the prediction if `smoke100 = 0` and `bmi_c = 0`. In this case, because we've centered BMI, it implies that $\exp(0.87) = 2.39$ is the predicted days of poor `physhealth` for a non-smoker with average BMI.
- The coefficient of `bmi_c`, 0.036, is the expected difference in count of poor `physhealth` days (on the log scale) for each additional kg/m^2 of body mass index. The expected multiplicative *increase* is $e^{0.036} = 1.037$, corresponding to a 3.7% difference in the count.
- The coefficient of `smoke100`, 0.60, tells us that the predictive difference between those who have and who have not smoked 100 cigarettes can be found by multiplying the `physhealth` count by $\exp(0.6) = 1.82$, yielding essentially an 82% increase of the `physhealth` count.

25.4.4 Interpretation of Coefficients in terms of IRRs

We might be interested in looking at incident rate ratios rather than coefficients. The coefficients have an additive effect in the $\log(y)$ scale, and the IRR have a multiplicative effect in the y scale. To do this, we can exponentiate our model coefficients. This also applies to the confidence intervals.

```
exp(coef(mod_nb1))
```

(Intercept)	bmi_c	smoke100
2.397748	1.036357	1.815563

```
exp(confint(mod_nb1))
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	2.060705	2.809613
bmi_c	1.020942	1.052585
smoke100	1.433008	2.306766

As an example, then, the incident rate for `smoke100 = 1` is 1.82 times the incident rate of `physhealth` days for the reference group (`smoke100 = 0`). The percent change in the incident rate of `physhealth` is a 3.6% increase for every kg/m^2 increase in centered `bmi`.

25.4.5 Testing the Predictors

Again, we can use the Wald tests (z tests) provided with the negative binomial regression output.

As an alternative, we probably should not use the standard `anova` process, because the models there don't re-estimate θ for each new model, as the warning message below indicates.

```
anova(mod_nb1)
```

```
Warning in anova.negbin(mod_nb1): tests made without re-estimating 'theta'
```

```
Analysis of Deviance Table
```

```
Model: Negative Binomial(0.1488), link: log
```

```
Response: physhealth
```

```
Terms added sequentially (first to last)
```

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			1973	1468.0	
bmi_c	1	20.837	1972	1447.2	5.001e-06 ***
smoke100	1	24.584	1971	1422.6	7.115e-07 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

So, instead, if we want, for instance to assess the significance of `bmi_c`, after `smoke100` is already included in the model, we fit both models (with and without `bmi_c`) and then compare those models with a likelihood ratio test.

```
mod_nb1_without_bmi <- glm.nb(physhealth ~ smoke100,  
                              link = log,  
                              data = ohioA_young)  
  
anova(mod_nb1, mod_nb1_without_bmi)
```

```
Likelihood ratio tests of Negative Binomial Models
```

```
Response: physhealth
```

Model	theta	Resid. df	2 x log-lik.	Test	df	LR stat.
-------	-------	-----------	--------------	------	----	----------


```

1          smoke100 0.1452219      1972      -6991.195
2 bmi_c + smoke100 0.1487673      1971      -6968.553 1 vs 2      1  22.6421
      Pr(Chi)
1
2 1.951615e-06

```

And we could compare the negative binomial models with and without `smoke100` in a similar way.

```

mod_nb1_without_smoke <- glm.nb(physhealth ~ bmi_c,
                                link = log,
                                data = ohioA_young)

anova(mod_nb1, mod_nb1_without_smoke)

```

Likelihood ratio tests of Negative Binomial Models

```

Response: physhealth
      Model      theta Resid. df    2 x log-lik.  Test      df LR stat.
1          bmi_c 0.1449966     1972     -6992.839
2 bmi_c + smoke100 0.1487673     1971     -6968.553 1 vs 2      1 24.28569
      Pr(Chi)
1
2 8.305388e-07

```

25.4.6 Store fitted values and residuals

The `broom` package works in this case, too. We'll look here at predicted (fitted) values on the scale of our `physhealth` response.

```

sm_nb1 <- augment(mod_nb1, ohioA_young,
                  type.predict = "response")

```

Warning: The ``augment()`` method for objects of class ``negbin`` is not maintained by the broom

This warning is displayed once per session.

```

sm_nb1 |>
  select(physhealth, .fitted) |>
  head()

```

```
# A tibble: 6 x 2
  physhealth .fitted
    <dbl>    <dbl>
1         0     2.15
2         0     2.22
3         0     3.18
4        30     5.23
5         0     3.24
6         0     3.78
```

25.4.7 Rootogram for Negative Binomial model

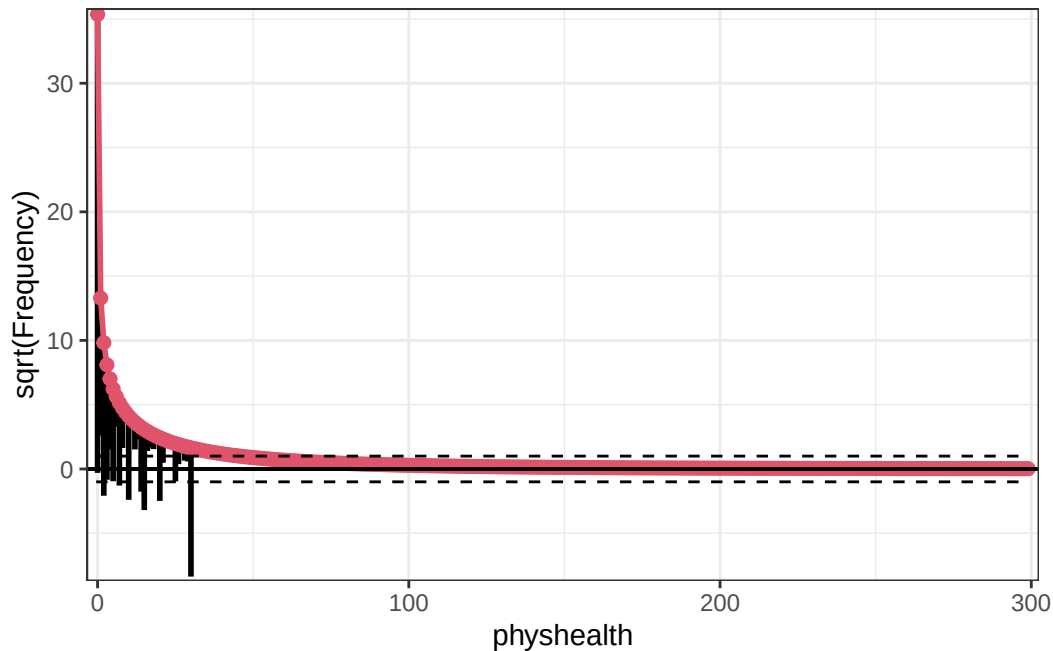
Here's the rootogram for the negative binomial model.

```
rootogram(mod_nb1, max = 30)
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the topmodels package.
  Please report the issue to the authors.
```

```
Warning: Using the `size` aesthetic in this geom was deprecated in ggplot2 3.4.0.
i Please use `linewidth` in the `default_aes` field and elsewhere instead.
i The deprecated feature was likely used in the topmodels package.
  Please report the issue to the authors.
```

```
Warning: Using the `size` aesthetic with geom_path was deprecated in ggplot2 3.4.0.
i Please use the `linewidth` aesthetic instead.
i The deprecated feature was likely used in the topmodels package.
  Please report the issue to the authors.
```



Again, the red curved line is the theoretical (negative binomial) fit. “Hanging” from each point on the red line is a bar, the height of which represents the difference between expected and observed counts. A bar hanging below 0 indicates underfitting. A bar hanging above 0 indicates overfitting. The counts have been transformed with a square root transformation to prevent smaller counts from getting obscured and overwhelmed by larger counts.

The match looks much better than the Poisson model, which is a sign that accounting for overdispersion is very important. Even this model badly underfits the number of 30 values, however.

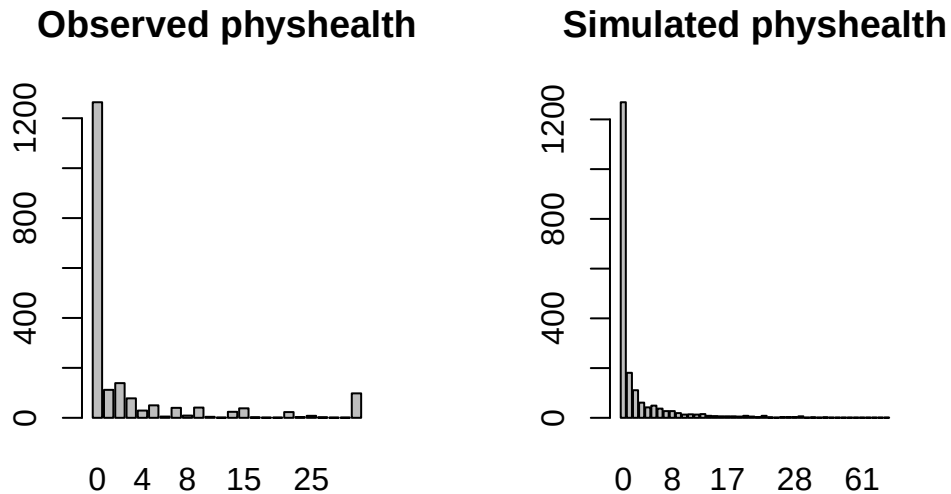
25.4.8 Simulating what the Negative Binomial model predicts

We can use the parameters of the negative binomial model to simulate data³ and compare the simulated results to our observed `physhealth` data.

```
par(mfrow=c(1,2))
ohioA_young$physhealth |>
  table() |> barplot(main = "Observed physhealth")
set.seed(432122)
rnbino(n = nrow(ohioA_young),
       size = mod_nb1$theta,
```

³See <http://data.library.virginia.edu/getting-started-with-negative-binomial-regression-modeling/>

```
mu = exp(coef(mod_nb1)[1])) |>
table() |> barplot(main = "Simulated physhealth")
```



Again we see that the simulated data badly underfits the 30 values, and includes some predictions larger than 30.

25.4.9 Specify the R^2 and log(likelihood) values

We can calculate the R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(nb_r <- with(sm_nb1, cor(physhealth, .fitted)))
```

```
[1] 0.183675
```

```
# R-square
nb_r^2
```

```
[1] 0.03373649
```

The `glance` function in the `broom` package gives us model log(likelihood), among other summaries.

```
glance(mod_nb1) |> round(3)
```

```
# A tibble: 1 x 8
  null.deviance df.null logLik      AIC      BIC deviance df.residual nobs
      <dbl>    <dbl> <logLik>   <dbl>   <dbl>   <dbl>      <dbl>  <dbl>
1      1468.    1973 -3484.277 6977.  6999.   1423.      1971   1974
```

Here, we have

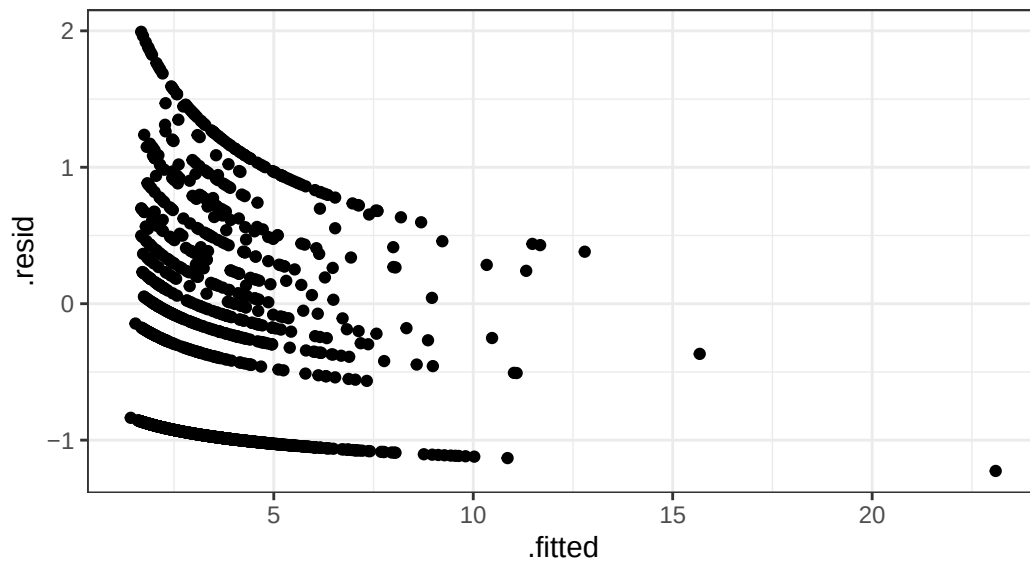
Model	Scale	R^2	log(likelihood)
Negative Binomial	log(physhealth)	.034	-3484.27

25.4.10 Check model assumptions

Here is a plot of residuals vs. fitted values on the original `physhealth` scale.

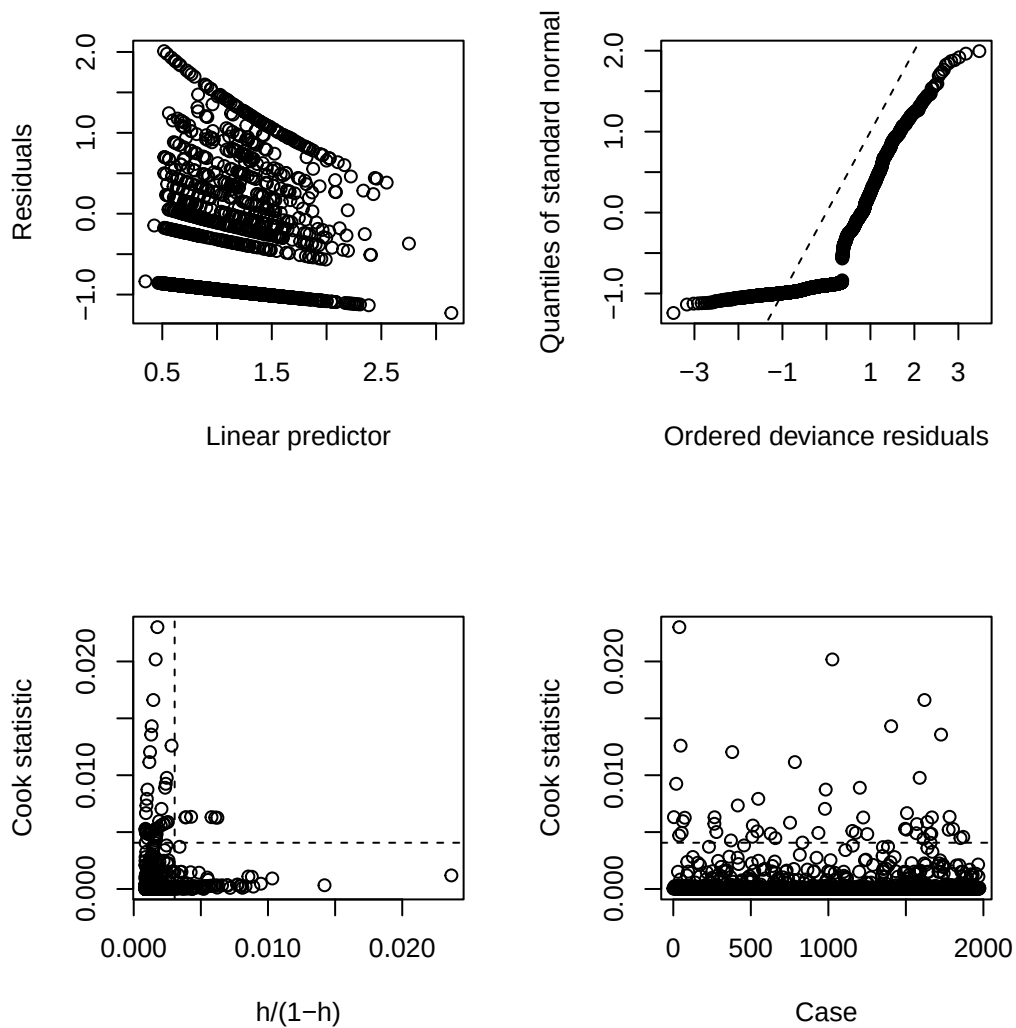
```
ggplot(sm_nb1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted `physhealth`",
       subtitle = "Negative Binomial Regression model")
```

Residuals vs. Fitted `physhealth`
Negative Binomial Regression model



Here are the glm diagnostic plots from the `boot` package.

```
glm.diag.plots(mod_nb1)
```



From the lower left plot, we see fewer points with large values of both Cook's distance and leverage, so that's a step in the right direction. The upper right plot still has some issues, but we're closer to a desirable result there, too.

25.4.11 Predictions for Harry and Sally

The predictions from this negative binomial regression model will be only a little different than those from the Poisson models.

```
predict(mod_nb1, newdata = hs_data, se.fit = TRUE,
        type = "response")
```

```
$fit
      1      2
6.221696 2.005657

$se.fit
      1      2
0.7537535 0.1773595

$residual.scale
[1] 1
```

As we've seen in the past, when we use `response` as the type, the predictions fall on the original `physhealth` scale. The prediction for Harry is 6.2 days, and for Sally is 2.0 days.

25.5 The Problem: Too Few Zeros

Remember that we observe more than 1000 zeros in our `physhealth` data.

```
ohioA_young |> count(physhealth == 0)
```

```
# A tibble: 2 x 2
  `physhealth == 0`      n
  <lgl>             <int>
1 FALSE              710
2 TRUE             1264
```

Let's go back to our Poisson model (without overdispersion) for a moment, and concentrate on the zero values.

```
# predict expected mean physhealth for each subject
mu <- predict(mod_pois1, type = "response")

# sum the probabilities of a zero count for each mean
exp <- sum(dpois(x = 0, lambda = mu))

# predicted number of zeros from Poisson model
round(exp)
```


[1] 124

As we've seen previously, we're severely underfitting zero counts. We can compare the observed number of zero `physhealth` results to the expected number of zero values from the likelihood-based models.

```
round(c("Obs" = sum(ohioA_young$physhealth == 0),  
      "Poisson" = sum(dpois(0, fitted(mod_poiss1))),  
      "NB" = sum(dnbinom(0, mu = fitted(mod_nb1), size = mod_nb1$theta))),0)
```

Obs	Poisson	NB
1264	124	1250

There are at least two ways to tackle this problem.

- Fitting a model which deliberately inflates the number of zeros that are fitted
- Fitting a hurdle model

We'll look at those options, next.

26 Zero-Inflated Models for Count Data

We will continue to use a count outcome (# of poor physical health days out of the last 30) in OHIO SMART data created in Chapter 6 to demonstrate regression models for count outcomes, as we did in Chapter 24 and Chapter 25.

Methods discussed in the chapter include:

- Zero-inflated Poisson Regression
- Zero-inflated Negative Binomial Regression
- Two different types of “hurdle” model
- A Tobit (Censored) Regression Model

26.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(lmtest)
library(MASS)
library(pscl)
library(VGAM)
library(tidyverse)

theme_set(theme_bw())
```

26.2 Data Load and Subset Creation

```
smart_oh <- readRDS("data/smart_ohio.Rds")
```

As in Chapter 24 and Chapter 25, we’ll create a subset of these data for analysis.

```
ohioA <- smart_oh |>
  select(SEQN0, mmsa_name, genhealth, physhealth,
         menthealth, healthplan, costprob,
         agegroup, female, incomegroup, bmi, smoke100,
         alcdays) |>
  drop_na()

ohioA_young <- ohioA |>
  filter(agegroup %in% c("18-24", "25-29", "30-34",
                       "35-39", "40-44", "45-49")) |>
  droplevels() |>
  mutate(bmi_c = bmi - mean(bmi))
```

26.3 Setup for this Chapter

Again, we're going to predict `physhealth` using `bmi_c` and `smoke100`.

- Remember that `physhealth` is a count of the number of poor physical health days in the past 30.
- As a result, `physhealth` is restricted to taking values between 0 and 30.

In this chapter, we demonstrate:

- Zero-inflated Poisson models
- Zero-inflated Negative Binomial models
- Hurdle models
- Tobit models

26.3.1 What Will We Demonstrate?

With each new approach, we again will fit the model and specify procedures for doing so in R. Then we will:

1. Specify the fitted model equation
2. Interpret the model's coefficient estimates and 95% confidence intervals around those estimates.
3. Perform a test of whether each variable adds value to the model, when the other one is already included.
4. Store the fitted values and appropriate residuals for each model.
5. Summarize the model's apparent R^2 value, the proportion of variation explained, and the model log likelihood.

6. Perform checks of model assumptions as appropriate.
7. Describe how predictions would be made for two new subjects.
 - Harry has a BMI that is 10 kg/m² higher than the average across all respondents and has smoked more than 100 cigarettes in his life.
 - Sally has a BMI that is 5 kg/m² less than the average across all respondents and has not smoked more than 100 cigarettes in her life.

In addition, for some of the new models, we provide a little of the mathematical background, and point to other resources you can use to learn more about the model.

26.3.2 Extra Data File for Harry and Sally

To make our lives a little easier, I'll create a little tibble containing the necessary data for Harry and Sally.

```
hs_data <- tibble(subj = c("Harry", "Sally"),
                  bmi_c = c(10, -5),
                  smoke100 = c(1, 0))
hs_data
```

```
# A tibble: 2 x 3
  subj  bmi_c smoke100
<chr> <dbl>   <dbl>
1 Harry    10         1
2 Sally    -5         0
```

26.3.3 Previous Models (for comparison)

```
mod_poiss1 <- glm(physhealth ~ bmi_c + smoke100,
                 family = poisson(),
                 data = ohioA_young)

mod_nb1 <- glm.nb(physhealth ~ bmi_c + smoke100,
                 link = log,
                 data = ohioA_young)
```

26.4 The Zero-Inflated Poisson Regression Model

There are at least two ways to tackle the problem of not predicting enough 0 values.

- Fitting a model which deliberately inflates the number of zeros that are fitted
- Fitting a hurdle model

The zero-inflated Poisson or (ZIP) model is used to describe count data with an excess of zero counts¹. The model posits that there are two processes involved:

- a logit model is used to predict excess zeros
- while a Poisson model is used to predict the counts, generally

The `pscl` package is used here, which can conflict with the `topmodels` package we used to fit rootograms.

To run the zero-inflated Poisson model, we use the following:

```
mod_zip1 <- zeroinfl(physhealth ~ bmi_c + smoke100,  
                     data = ohioA_young)  
  
summary(mod_zip1)
```

Call:

```
zeroinfl(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young)
```

Pearson residuals:

	Min	1Q	Median	3Q	Max
	-1.4387	-0.6987	-0.6138	-0.1947	9.3029

Count model coefficients (poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.997619	0.018797	106.3	<2e-16 ***
bmi_c	0.018177	0.001398	13.0	<2e-16 ***
smoke100	0.393355	0.024889	15.8	<2e-16 ***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.682306	0.062852	10.856	< 2e-16 ***
bmi_c	-0.027725	0.006509	-4.260	2.05e-05 ***
smoke100	-0.237181	0.095316	-2.488	0.0128 *

¹See <https://stats.idre.ucla.edu/r/dae/zip/> for more on the zero-inflated poisson model.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 14

Log-likelihood: -5680 on 6 Df

```
confint(mod_zip1)
```

	2.5 %	97.5 %
count_(Intercept)	1.96077775	2.03445952
count_bmi_c	0.01543587	0.02091727
count_smoke100	0.34457348	0.44213651
zero_(Intercept)	0.55911859	0.80549344
zero_bmi_c	-0.04048184	-0.01496748
zero_smoke100	-0.42399711	-0.05036477

The output describes two separate regression models. Below the model call, we see information on a Poisson regression model. Then we see another block describing the inflation model.

Each predictor (bmi_c and smoke100) yields a small p value in each part of the model.

26.4.1 Comparison to a null model

To show that this model fits better than the null model (the model with intercept only), we can compare them directly with a chi-squared test. Since we have two predictors in the full model, the degrees of freedom for this test is 2.

```
mod_zipnull <- pscl::zeroinfl(physhealth ~ 1,
                             data = ohioA_young)

summary(mod_zipnull)
```

Call:

```
pscl::zeroinfl(formula = physhealth ~ 1, data = ohioA_young)
```

Pearson residuals:

Min	1Q	Median	3Q	Max
-0.6934	-0.6934	-0.6934	-0.2779	5.5399

Count model coefficients (poisson with log link):

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.22765     0.01233   180.7   <2e-16 ***

Zero-inflation model coefficients (binomial with logit link):
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.5766     0.0469   12.29   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 9
Log-likelihood: -5908 on 2 Df

pchisq(2 * (logLik(mod_zip1) - logLik(mod_zipnull)), df = 2, lower.tail = FALSE)

'log Lik.' 8.596735e-100 (df=6)

```

26.4.2 Comparison to a Poisson Model with the Vuong test

```
vuong(mod_zip1, mod_poiss1)
```

Vuong Non-Nested Hypothesis Test-Statistic:
(test-statistic is asymptotically distributed $N(0,1)$ under the null that the models are indistinguishable)

```

-----
              Vuong z-statistic              H_A      p-value
Raw              19.90824 model1 > model2 < 2.22e-16
AIC-corrected    19.89662 model1 > model2 < 2.22e-16
BIC-corrected    19.86415 model1 > model2 < 2.22e-16

```

Certainly, the ZIP model seems to improve the standard Poisson model, according to the Vuong test.

26.4.3 The Fitted Equation

The form of the model equation for a zero-inflated Poisson regression requires us to take two separate models into account. First we have a logistic regression model to predict the log odds of zero `physhealth` days. That takes care of the *extra* zeros. Then, to predict the number of `physhealth` days, we have a Poisson model, which may produce some additional zero count estimates.

26.4.4 Interpreting the Coefficients

We can exponentiate the logistic regression coefficients to obtain results in terms of odds ratios for that model, and that can be of some help in understanding the process behind excess zeros.

Also, exponentiating the coefficients of the count model help us describe those counts on the original scale of `physhealth`.

```
exp(coef(mod_zip1))
```

count_(Intercept)	count_bmi_c	count_smoke100	zero_(Intercept)
7.3714810	1.0183428	1.4819444	1.9784348
zero_bmi_c	zero_smoke100		
0.9726561	0.7888485		

For example,

- in the model for `physhealth = 0`, the odds of `physhealth = 0` are 79% as high for subjects with `smoke100 = 1` as for non-smokers with the same BMI.
- in the Poisson model for `physhealth`, the `physhealth` count is estimated to increase by 1.48 for smokers as compared to non-smokers with the same BMI.

26.4.5 Testing the Predictors

We can test the model with and without `bmi_c`, for example, by fitting the model both ways, and comparing the results with either a Wald or Likelihood Ratio test, each of which is available in the `lmtest` package.

```
mod_zip1_nobmi <- zeroinfl(physhealth ~ smoke100,  
                           data = ohioA_young)  
  
lmtest::waldtest(mod_zip1, mod_zip1_nobmi)
```

Wald test

```
Model 1: physhealth ~ bmi_c + smoke100  
Model 2: physhealth ~ smoke100  
  Res.Df Df  Chisq Pr(>Chisq)  
1    1968  
2    1970 -2 187.2  < 2.2e-16 ***
```



```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmtest::lrtest(mod_zip1, mod_zip1_nobmi)
```

Likelihood ratio test

Model 1: physhealth ~ bmi_c + smoke100

Model 2: physhealth ~ smoke100

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	6	-5679.8			
2	4	-5769.5	-2	179.35	< 2.2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

26.4.6 Store fitted values and residuals

The `broom` package does not work with the `zeroinfl` tool. So we need to build up the fitted values and residuals ourselves.

```
sm_zip1 <- ohioA_young |>
  mutate(fitted = fitted(mod_zip1, type = "response"),
         resid = resid(mod_zip1, type = "response"))

sm_zip1 |>
  dplyr::select(physhealth, fitted, resid) |>
  head()
```

```
# A tibble: 6 x 3
  physhealth fitted resid
    <dbl>    <dbl> <dbl>
1         0     2.21 -2.21
2         0     2.28 -2.28
3         0     3.12 -3.12
4        30     5.27 24.7
5         0     3.17 -3.17
6         0     3.71 -3.71
```

26.4.7 Modeled Number of Zero Counts

The zero-inflated model is designed to perfectly match the number of observed zeros. We can compare the observed number of zero `physhealth` results to the expected number of zero values from the likelihood-based models.

```
round(c("Obs" = sum(ohioA_young$physhealth == 0),  
      "Poisson" = sum(dpois(0, fitted(mod_pois1))),  
      "NB" = sum(dnbinom(0, mu = fitted(mod_nb1), size = mod_nb1$theta)),  
      "ZIP" = sum(predict(mod_zip1, type = "prob")[,1])), 0)
```

Obs	Poisson	NB	ZIP
1264	124	1250	1264

26.4.8 Rootogram for ZIP model

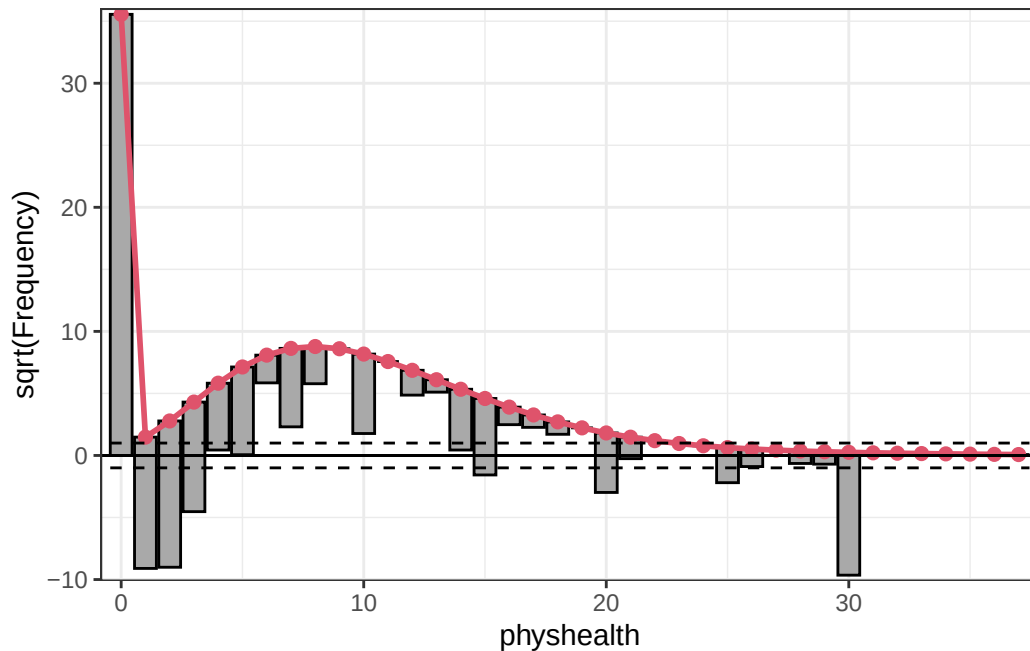
Here's the rootogram for the zero-inflated Poisson model.

```
topmodels::rootogram(mod_zip1, max = 30)
```

Warning: `aes\_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the topmodels package.
Please report the issue to the authors.

Warning: Using the `size` aesthetic in this geom was deprecated in ggplot2 3.4.0.
i Please use `linewidth` in the `default\_aes` field and elsewhere instead.
i The deprecated feature was likely used in the topmodels package.
Please report the issue to the authors.

Warning: Using the `size` aesthetic with geom_path was deprecated in ggplot2 3.4.0.
i Please use the `linewidth` aesthetic instead.
i The deprecated feature was likely used in the topmodels package.
Please report the issue to the authors.



The zero frequencies are perfectly matched here, but we can see that counts of 1 and 2 are now substantially underfit, and values between 6 and 13 are overfit.

26.4.9 Specify the R^2 and log (likelihood) values

We can calculate a proxy for R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(zip_r <- with(sm_zip1, cor(physhealth, fitted)))
```

```
[1] 0.1873284
```

```
# R-square
zip_r^2
```

```
[1] 0.03509194
```

```
logLik(mod_zip1)
```

```
'log Lik.' -5679.794 (df=6)
```

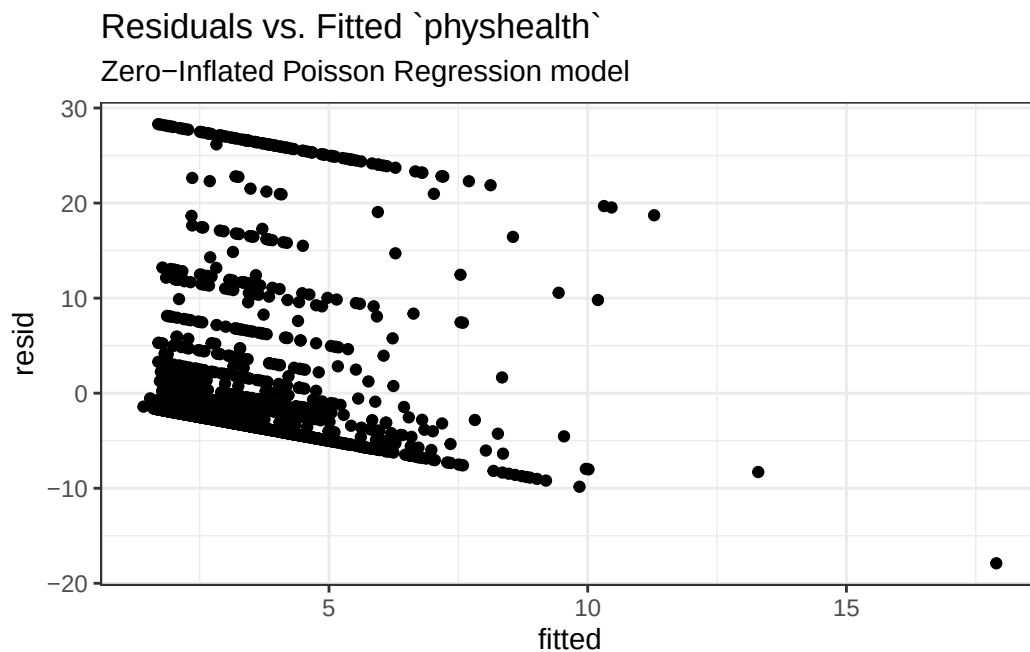
Here, we have

Model	Scale	R^2	log(likelihood)
Zero-Inflated Poisson Complex: log(physhealth)		.035	-5679.83

26.4.10 Check model assumptions

Here is a plot of residuals vs. fitted values on the original `physhealth` scale.

```
ggplot(sm_zip1, aes(x = fitted, y = resid)) +  
  geom_point() +  
  labs(title = "Residuals vs. Fitted `physhealth`",  
        subtitle = "Zero-Inflated Poisson Regression model")
```



26.4.11 Predictions for Harry and Sally

The predictions from this ZIP regression model are obtained as follows...

```
predict(mod_zip1, newdata = hs_data, type = "response")
```

```
      1      2
6.002249 2.056792
```

As we've seen in the past, when we use `response` as the type, the predictions fall on the original `physhealth` scale. The prediction for Harry is 6.0 days, and for Sally is 2.1 days.

26.5 The Zero-Inflated Negative Binomial Regression Model

As an alternative to the ZIP model, we might consider a zero-inflated negative binomial regression². This will involve a logistic regression to predict the probability of a 0, and then a negative binomial model to describe the counts of `physhealth`.

To run the zero-inflated negative binomial model, we use the following code:

```
mod_zinb1 <- pscl::zeroinfl(physhealth ~ bmi_c + smoke100,
                           dist = "negbin", data = ohioA_young)

summary(mod_zinb1)
```

Call:

```
pscl::zeroinfl(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young,
               dist = "negbin")
```

Pearson residuals:

	Min	1Q	Median	3Q	Max
	-0.5579	-0.4192	-0.3957	-0.1166	6.4355

Count model coefficients (negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.545639	0.099787	15.489	< 2e-16 ***
bmi_c	0.024617	0.006641	3.707	0.00021 ***
smoke100	0.517302	0.110770	4.670	3.01e-06 ***
Log(theta)	-0.874127	0.143900	-6.075	1.24e-09 ***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.071616	0.164055	-0.437	0.66245
bmi_c	-0.027609	0.009314	-2.964	0.00303 **

²See <https://stats.idre.ucla.edu/r/dae/zinb/>

```
smoke100    -0.127260    0.137180   -0.928    0.35357
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta = 0.4172
Number of iterations in BFGS optimization: 16
Log-likelihood: -3469 on 7 Df
```

```
confint(mod_zinb1)
```

	2.5 %	97.5 %
count_(Intercept)	1.35005972	1.741217558
count_bmi_c	0.01160172	0.037633119
count_smoke100	0.30019616	0.734408526
zero_(Intercept)	-0.39315782	0.249925968
zero_bmi_c	-0.04586350	-0.009355106
zero_smoke100	-0.39612706	0.141607082

26.5.1 Comparison to a null model

To show that this model fits better than the null model (the model with intercept only), we can compare them directly with a chi-squared test. Since we have two predictors in the full model, the degrees of freedom for this test is 2.

```
mod_zinbnull <- pscl::zeroinfl(physhealth ~ 1, dist = "negbin",
                               data = ohioA_young)

summary(mod_zinbnull)
```

Call:

```
pscl::zeroinfl(formula = physhealth ~ 1, data = ohioA_young, dist = "negbin")
```

Pearson residuals:

Min	1Q	Median	3Q	Max
-0.4048	-0.4048	-0.4048	-0.1622	3.2340

Count model coefficients (negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.7766	0.0964	18.429	< 2e-16 ***

```

Log(theta)    -1.0445      0.1612   -6.479 9.25e-11 ***

Zero-inflation model coefficients (binomial with logit link):
      Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.2605      0.1920  -1.357   0.175
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta = 0.3519
Number of iterations in BFGS optimization: 13
Log-likelihood: -3498 on 3 Df

pchisq(2 * (logLik(mod_nb1) - logLik(mod_zinbnull)), df = 2, lower.tail = FALSE)

'log Lik.' 8.538917e-07 (df=4)

```

26.5.2 Comparison to a Negative Binomial Model: Vuong test

```
vuong(mod_zinb1, mod_nb1)
```

Vuong Non-Nested Hypothesis Test-Statistic:
(test-statistic is asymptotically distributed $N(0,1)$ under the null that the models are indistinguishable)

```

-----
              Vuong z-statistic              H_A    p-value
Raw              3.0756577 model1 > model2 0.0010502
AIC-corrected    2.4599576 model1 > model2 0.0069477
BIC-corrected    0.7397477 model1 > model2 0.2297265

```

The zero-inflated negative binomial model appears to be an improvement over the standard negative binomial model according to the the raw or AIC-corrected Vuong tests, but not according to the BIC-corrected test.

26.5.3 The Fitted Equation

Like the ZIP, the zero-inflated negative binomial regression also requires us to take two separate models into account. First we have a logistic regression model to predict the log odds of zero `physhealth` days. That takes care of the *extra* zeros. Then, to predict the number of `physhealth` days, we have a negative binomial regression, with a θ term, and this negative binomial regression model may also produce some additional zero count estimates.

26.5.4 Interpreting the Coefficients

As with the zip, we can exponentiate the logistic regression coefficients to obtain results in terms of odds ratios for that model, and that can be of some help in understanding the process behind excess zeros.

```
exp(coef(mod_zinb1))
```

count_(Intercept)	count_bmi_c	count_smoke100	zero_(Intercept)
4.6909665	1.0249229	1.6774962	0.9308884
zero_bmi_c	zero_smoke100		
0.9727684	0.8805047		

For example,

- in the model for `physhealth = 0`, the odds of `physhealth = 0` are 88.1% as high for subjects with `smoke100 = 1` as for non-smokers with the same BMI.

Interpreting the negative binomial piece works the same way as it did in the negative binomial regression.

26.5.5 Testing the Predictors

We can test the model with and without `bmi_c`, for example, by fitting the model both ways, and comparing the results with either a Wald or Likelihood Ratio test, each of which is available in the `lmtest` package.

```
mod_zinb1_nobmi <- pscl::zeroinfl(physhealth ~ smoke100,  
                                dist = "negbin",  
                                data = ohioA_young)  
  
lmtest::waldtest(mod_zinb1, mod_zinb1_nobmi)
```

Wald test

```
Model 1: physhealth ~ bmi_c + smoke100  
Model 2: physhealth ~ smoke100  
  Res.Df Df    Chisq Pr(>Chisq)  
1    1967  
2    1969 -2  29.593   3.75e-07 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
lmtest::lrtest(mod_zinb1, mod_zinb1_nobmi)
```

Likelihood ratio test

Model 1: physhealth ~ bmi_c + smoke100

Model 2: physhealth ~ smoke100

```
#Df  LogLik Df  Chisq Pr(>Chisq)
```

```
1    7 -3469.3
```

```
2    5 -3485.0 -2 31.418  1.506e-07 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

26.5.6 Store fitted values and residuals

Again, we need to build up the fitted values and residuals without the **broom** package.

```
sm_zinb1 <- ohioA_young |>
  mutate(fitted = fitted(mod_zinb1, type = "response"),
         resid = resid(mod_zinb1, type = "response"))

sm_zip1 |>
  dplyr::select(physhealth, fitted, resid) |>
  head()
```

```
# A tibble: 6 x 3
  physhealth fitted resid
    <dbl>    <dbl> <dbl>
1         0    2.21 -2.21
2         0    2.28 -2.28
3         0    3.12 -3.12
4        30    5.27 24.7
5         0    3.17 -3.17
6         0    3.71 -3.71
```

26.5.7 Modeled Number of Zero Counts

Once again, we can compare the observed number of zero **physhealth** results to the expected number of zero values from the likelihood-based models.

```
round(c("Obs" = sum(ohioA_young$physhealth == 0),
      "Poisson" = sum(dpois(0, fitted(mod_pois1))),
      "NB" = sum(dnbinom(0, mu = fitted(mod_nb1), size = mod_nb1$theta)),
      "ZIP" = sum(predict(mod_zip1, type = "prob")[,1]),
      "ZINB" = sum(predict(mod_zinb1, type = "prob")[,1]),0)
```

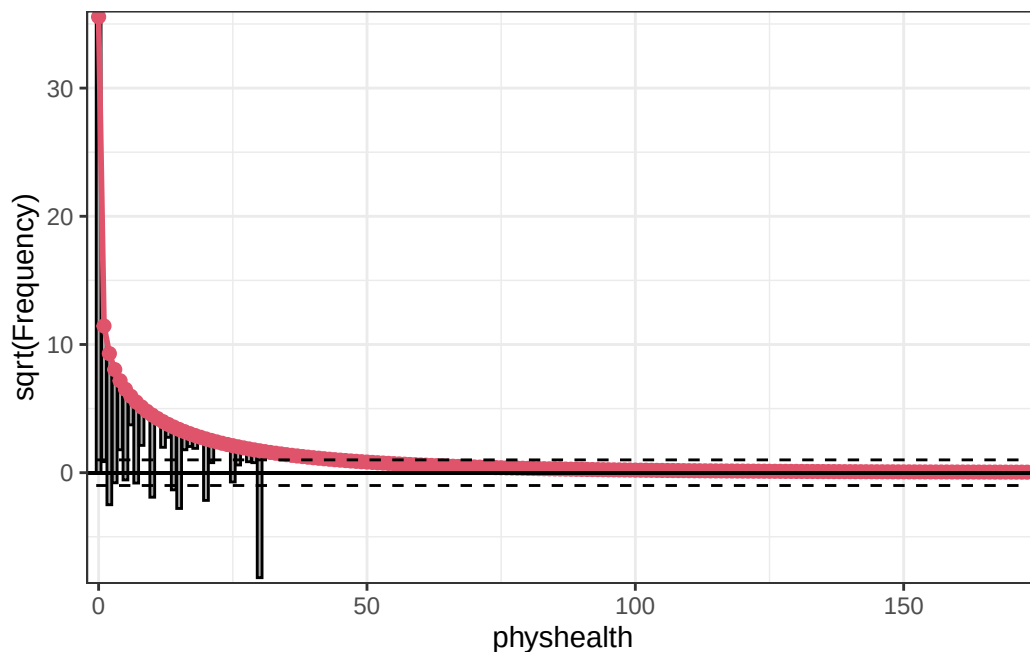
Obs	Poisson	NB	ZIP	ZINB
1264	124	1250	1264	1264

So, the Poisson model is clearly inappropriate, but the zero-inflated (Poisson and NB) and the negative binomial model all give reasonable fits in this regard.

26.5.8 Rootogram for Zero-Inflated Negative Binomial model

Here's the rootogram for the zero-inflated negative binomial model.

```
topmodels::rootogram(mod_zinb1, max = 30)
```



As in the ZIP model, the zero frequencies are perfectly matched here, but we can see that counts of 1 and 2 are now closer to the data we observe than in the ZIP model. We are still substantially underfitting values of 30.

26.5.9 Specify the R^2 and log (likelihood) values

We can calculate a proxy for R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(zinb_r <- with(sm_zinb1, cor(physhealth, fitted)))
```

```
[1] 0.1858996
```

```
# R-square
zinb_r^2
```

```
[1] 0.03455865
```

```
logLik(mod_zinb1)
```

```
'log Lik.' -3469.29 (df=7)
```

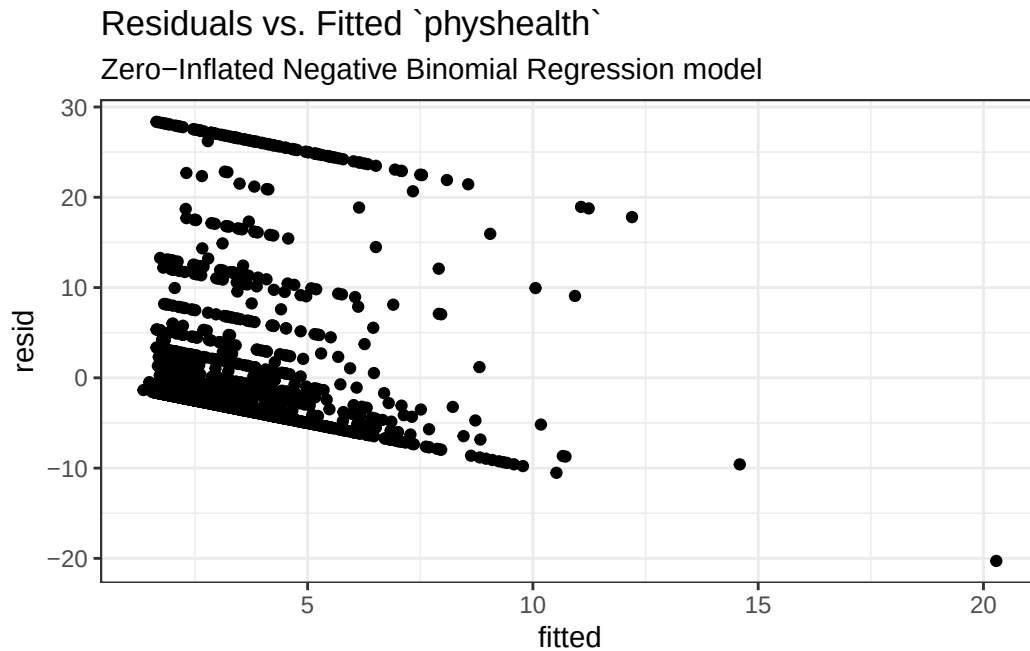
Here, we have

Model	Scale	R^2	log(likelihood)
Zero-Inflated Negative Binomial	Complex: log(physhealth)	.035	-3469.27

26.5.10 Check model assumptions

Here is a plot of residuals vs. fitted values on the original `physhealth` scale.

```
ggplot(sm_zinb1, aes(x = fitted, y = resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted `physhealth`",
        subtitle = "Zero-Inflated Negative Binomial Regression model")
```



26.5.11 Predictions for Harry and Sally

The predictions from this zero-inflated negative binomial regression model are obtained as follows...

```
predict(mod_zinb1, newdata = hs_data, type = "response")
```

```
      1      2
6.205985 2.004987
```

As we've seen in the past, when we use `response` as the type, the predictions fall on the original `physhealth` scale. The prediction for Harry is 6.2 days, and for Sally is 2.0 days.

26.6 A “hurdle” model (with Poisson)

Much of the discussion here of hurdle models comes from Clay Ford at the University of Virginia³. Ford describes a hurdle model as follows:

³<http://data.library.virginia.edu/getting-started-with-hurdle-models/> is an excellent introduction, by Clay Ford, a Statistical Research Consultant at the University of Virginia Library. I can also recommend

The hurdle model is a two-part model that specifies one process for zero counts and another process for positive counts. The idea is that positive counts occur once a threshold is crossed, or put another way, a hurdle is cleared. If the hurdle is not cleared, then we have a count of 0.

The first part of the model is typically a binary logit model. This models whether an observation takes a positive count or not. The second part of the model is usually a truncated Poisson or Negative Binomial model. Truncated means we're only fitting positive counts. If we were to fit a hurdle model to our [medicare] data, the interpretation would be that one process governs whether a patient visits a doctor or not, and another process governs how many visits are made.

To fit a hurdle model, we'll use the `hurdle` function in the `pscl` package.

```
mod_hur1 <- pscl::hurdle(physhealth ~ bmi_c + smoke100,
                        dist = "poisson", zero.dist = "binomial",
                        data = ohioA_young)

summary(mod_hur1)
```

Call:

```
pscl::hurdle(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young,
             dist = "poisson", zero.dist = "binomial")
```

Pearson residuals:

	Min	1Q	Median	3Q	Max
	-1.4403	-0.6987	-0.6139	-0.1946	9.2997

Count model coefficients (truncated poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.997612	0.018797	106.3	<2e-16 ***
bmi_c	0.018178	0.001398	13.0	<2e-16 ***
smoke100	0.393348	0.024889	15.8	<2e-16 ***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.683509	0.062827	-10.879	< 2e-16 ***
bmi_c	0.027777	0.006508	4.268	1.97e-05 ***

https://rpubs.com/kaz_yos/pscl-2 as a place to learn more about the `pscl` package, and the fitting and interpretation of both hurdle and zero-inflated regression models. That `rpubs` site has a link to [this article by Hu, Pavlicova and Nunes from the Am J Drug Alcohol Abuse](#) which provides a real set of examples from a trial of a behavioral health intervention meant to reduce the risk of unprotected sexual occasions as part of a strategy to reduce HIV risk.

```
smoke100      0.238300   0.095301   2.500   0.0124 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 14
Log-likelihood: -5680 on 6 Df
```

```
confint(mod_hur1)
```

	2.5 %	97.5 %
count_(Intercept)	1.96077105	2.03445261
count_bmi_c	0.01543722	0.02091901
count_smoke100	0.34456599	0.44212932
zero_(Intercept)	-0.80664805	-0.56037031
zero_bmi_c	0.01502124	0.04053230
zero_smoke100	0.05151314	0.42508607

We are using the default settings here, using the same predictors for both models:

- a **Binomial** model to predict the probability of `physhealth = 0` given our predictors, as specified by the `zero.dist` argument in the `hurdle` function, and
- a (truncated) **Poisson** model to predict the positive-count of `physhealth` given those same predictors, as specified by the `dist` argument in the `hurdle` function.

26.6.1 Comparison to a null model

To show that this model fits better than the null model (the model with intercept only), we can compare them directly with a chi-squared test. Since we have two predictors in the full model, the degrees of freedom for this test is 2.

```
mod_hurnull <- pscl::hurdle(physhealth ~ 1, dist = "poisson",
                           zero.dist = "binomial",
                           data = ohioA_young)

summary(mod_hurnull)
```

Call:

```
pscl::hurdle(formula = physhealth ~ 1, data = ohioA_young, dist = "poisson",
             zero.dist = "binomial")
```

Pearson residuals:

Min	1Q	Median	3Q	Max
-0.6934	-0.6934	-0.6934	-0.2779	5.5399

Count model coefficients (truncated poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.22765	0.01233	180.7	<2e-16 ***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.5768	0.0469	-12.3	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 8

Log-likelihood: -5908 on 2 Df

```
pchisq(2 * (logLik(mod_hur1) - logLik(mod_hurnull)), df = 2, lower.tail = FALSE)
```

'log Lik.' 8.577393e-100 (df=6)

26.6.2 Comparison to a Poisson Model: Vuong test

```
vuong(mod_hur1, mod_poiss1)
```

Vuong Non-Nested Hypothesis Test-Statistic:

(test-statistic is asymptotically distributed $N(0,1)$ under the null that the models are indistinguishable)

```
-----
```

	Vuong z-statistic	H_A	p-value
Raw	19.90847	model1 > model2	< 2.22e-16
AIC-corrected	19.89685	model1 > model2	< 2.22e-16
BIC-corrected	19.86438	model1 > model2	< 2.22e-16

The hurdle model shows a detectable improvement over the standard Poisson model according to this test.

26.6.3 Comparison to a Zero-Inflated Poisson Model: Vuong test

Is the hurdle model comparable to the zero-inflated Poisson?

```
vuong(mod_hur1, mod_zip1)
```

Vuong Non-Nested Hypothesis Test-Statistic:
(test-statistic is asymptotically distributed $N(0,1)$ under the null that the models are indistinguishable)

```
-----  
              Vuong z-statistic          H_A p-value  
Raw              0.2269041 model1 > model2 0.41025  
AIC-corrected    0.2269041 model1 > model2 0.41025  
BIC-corrected    0.2269041 model1 > model2 0.41025
```

The hurdle model doesn't show a substantial improvement over the zero-inflated Poisson model according to this test.

26.6.4 The Fitted Equation

The form of the model equation for this hurdle also requires us to take two separate models into account. First we have a logistic regression model to predict the log odds of zero `physhealth` days. That takes care of the zeros. Then, to predict the number of `physhealth` days, we use a truncated Poisson model, which is truncated to produce only estimates greater than zero.

26.6.5 Interpreting the Coefficients

We can exponentiate the logistic regression coefficients to obtain results in terms of odds ratios for that model, and that can be of some help in understanding the process behind excess zeros.

Also, exponentiating the coefficients of the count model help us describe those counts on the original scale of `physhealth`.

```
exp(coef(mod_hur1))
```

```
count_(Intercept)    count_bmi_c    count_smoke100    zero_(Intercept)  
      7.3714308         1.0183443         1.4819335         0.5048423  
zero_bmi_c          zero_smoke100  
      1.0281661         1.2690894
```


For example,

- in the model for `physhealth = 0`, the odds of `physhealth = 0` are 127% as high for subjects with `smoke100 = 1` as for non-smokers with the same BMI.
- in the Poisson model for `physhealth`, the `physhealth` count is estimated to increase by 1.48 for smokers as compared to non-smokers with the same BMI.

26.6.6 Testing the Predictors

We can test the model with and without `bmi_c`, for example, by fitting the model both ways, and comparing the results with either a Wald or Likelihood Ratio test, each of which is available in the `lmtest` package.

```
mod_hur1_nobmi <- pscl::hurdle(physhealth ~ smoke100,
                               dist = "poisson",
                               zero.dist = "binomial",
                               data = ohioA_young)

lmtest::waldtest(mod_hur1, mod_hur1_nobmi)
```

Wald test

```
Model 1: physhealth ~ bmi_c + smoke100
Model 2: physhealth ~ smoke100
   Res.Df Df    Chisq Pr(>Chisq)
1     1968
2     1970 -2  187.19  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmtest::lrtest(mod_hur1, mod_hur1_nobmi)
```

Likelihood ratio test

```
Model 1: physhealth ~ bmi_c + smoke100
Model 2: physhealth ~ smoke100
   #Df LogLik Df    Chisq Pr(>Chisq)
1     6 -5679.8
2     4 -5769.5 -2  179.35  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

26.6.7 Store fitted values and residuals

The `broom` package does not work with the `hurdle` class of models. Again we need to build up the fitted values and residuals ourselves.

```
sm_hur1 <- ohioA_young |>
  mutate(fitted = fitted(mod_hur1, type = "response"),
         resid = resid(mod_hur1, type = "response"))

sm_hur1 |>
  dplyr::select(physhealth, fitted, resid) |>
  head()
```

```
# A tibble: 6 x 3
  physhealth fitted resid
    <dbl>    <dbl> <dbl>
1         0     2.21 -2.21
2         0     2.28 -2.28
3         0     3.12 -3.12
4        30     5.27 24.7
5         0     3.17 -3.17
6         0     3.71 -3.71
```

26.6.8 Modeled Number of Zero Counts

Once again, we can compare the observed number of zero `physhealth` results to the expected number of zero values from the likelihood-based models.

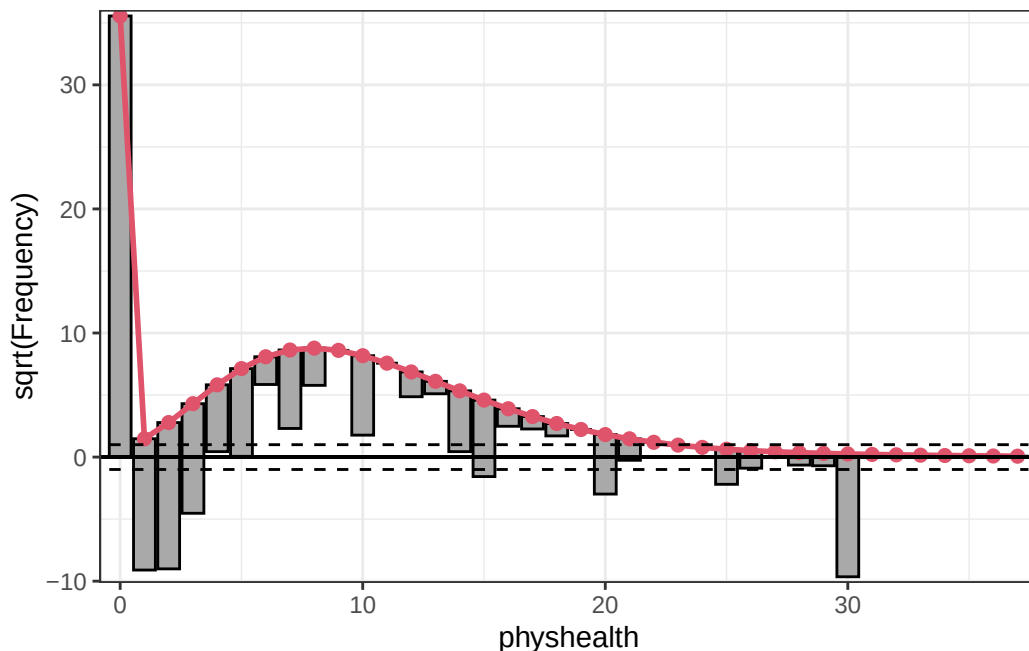
```
round(c("Obs" = sum(ohioA_young$physhealth == 0),
      "Poisson" = sum(dpois(0, fitted(mod_poiss1))),
      "NB" = sum(dnbinom(0, mu = fitted(mod_nb1), size = mod_nb1$theta)),
      "ZIP" = sum(predict(mod_zip1, type = "prob")[,1]),
      "ZINB" = sum(predict(mod_zinb1, type = "prob")[,1]),
      "Hurdle" = sum(predict(mod_hur1, type = "prob")[,1])), 0)
```

Obs	Poisson	NB	ZIP	ZINB	Hurdle
1264	124	1250	1264	1264	1264

The hurdle model does about as well as the negative binomial and zero-inflated models. All but the Poisson give reasonable fits in this regard.

26.6.9 Rootogram for Hurdle Model

```
topmodels::rootogram(mod_hur1, max = 30)
```



The results are still not perfect, of course. In fitting the zeros exactly, we're underfitting counts of 1, 2, and 30, and overfitting many of the counts between 6 and 20. We still have a problem here with overdispersion. That's why we'll consider a hurdle model with a negative binomial regression for the counts in a moment.

26.6.10 Understanding the Modeled Counts in Detail

The expected mean count uses both parts of the hurdle model. Mathematically, we want...

$$E[y|x] = \frac{1 - f_1(0|x)}{1 - f_2(0|x)} \mu_2(x)$$

where

- our count of `physhealth` is y
- our predictors are represented by $\mathbf{x}$
- and the expected count is the product of a ratio and a mean.

The ratio is the probability of a non-zero in the first process divided the probability of a non-zero in the second untruncated process. The f symbols represent distributions. Recall these are logistic and Poisson, respectively, by default but can be others. The mean is for the untruncated version of the positive-count process.

If we want to see the expected hurdle counts, we can get them using some clever applications of the `predict` function.

The first six expected mean counts ($E[y|x]$ from the equation above) are:

```
head(predict(mod_hur1, type = "response"))
```

```
      1      2      3      4      5      6
2.214179 2.281482 3.116124 5.267881 3.167890 3.712120
```

The ratio of non-zero probabilities, $\frac{1-f_1(0|x)}{1-f_2(0|x)}$, from the mathematical expression above can be extracted by:

```
head(predict(mod_hur1, type = "zero"))
```

```
      1      2      3      4      5      6
0.3173312 0.3221978 0.3344433 0.4806542 0.3372292 0.3649208
```

The mean for the untruncated process, $\mu_2(x)$, can also be obtained by:

```
head(predict(mod_hur1, type = "count"))
```

```
      1      2      3      4      5      6
6.977501 7.081000 9.317347 10.959814 9.393877 10.172401
```

and we can multiply these last two pieces together to verify that they match our expected hurdle counts.

```
head(predict(mod_hur1, type = "zero") * predict(mod_hur1, type = "count"), 5)
```

```
      1      2      3      4      5
2.214179 2.281482 3.116124 5.267881 3.167890
```

26.6.11 Specify the R^2 and log (likelihood) values

We can calculate a proxy for R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(hur1_r <- with(sm_hur1, cor(physhealth, fitted)))
```

```
[1] 0.1873104
```

```
# R-square
hur1_r^2
```

```
[1] 0.03508517
```

```
logLik(mod_hur1)
```

```
'log Lik.' -5679.792 (df=6)
```

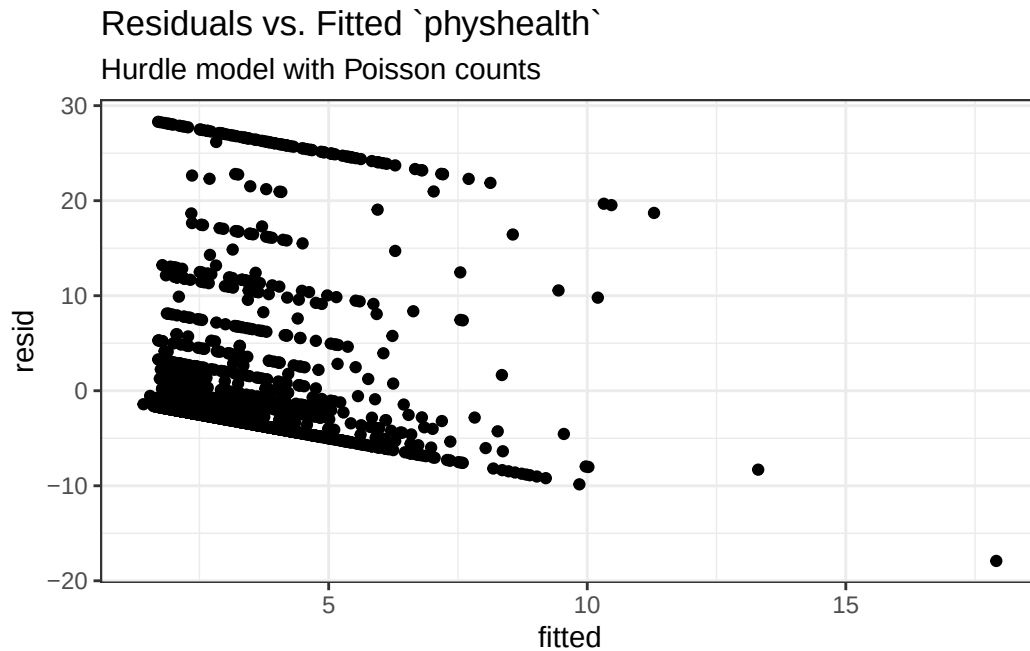
Here, we have

Model	Scale	R^2	log(likelihood)
Hurdle Model (Poisson)	Complex: log(physhealth)	.035	-5679.83

26.6.12 Check model assumptions

Here is a plot of residuals vs. fitted values on the original `physhealth` scale.

```
ggplot(sm_hur1, aes(x = fitted, y = resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted `physhealth`",
        subtitle = "Hurdle model with Poisson counts")
```



26.6.13 Predictions for Harry and Sally

The predictions from this zero-inflated negative binomial regression model are obtained as follows...

```
predict(mod_hur1, newdata = hs_data, type = "response")
```

```
      1      2
6.003689 2.057127
```

As we've seen in the past, when we use `response` as the type, the predictions fall on the original `physhealth` scale. The prediction for Harry is 6.0 days, and for Sally is 2.1 days.

26.7 A “hurdle” model (with negative binomial for overdispersion)

Let's account for overdispersion better with a negative binomial model for the counts in our hurdle model. We specify that the positive-count process be fit with this NB model using `dist = negbin`.

```
mod_hur_nb1 <- pscl::hurdle(physhealth ~ bmi_c + smoke100,
  dist = "negbin", zero.dist = "binomial",
  data = ohioA_young)

summary(mod_hur_nb1)
```

Call:

```
pscl::hurdle(formula = physhealth ~ bmi_c + smoke100, data = ohioA_young,
  dist = "negbin", zero.dist = "binomial")
```

Pearson residuals:

	Min	1Q	Median	3Q	Max
	-0.5749	-0.4178	-0.3948	-0.1165	6.4023

Count model coefficients (truncated negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.54170	0.10101	15.263	< 2e-16 ***
bmi_c	0.02434	0.00677	3.595	0.000324 ***
smoke100	0.51792	0.11101	4.666	3.08e-06 ***
Log(theta)	-0.88245	0.14653	-6.023	1.72e-09 ***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.683509	0.062827	-10.879	< 2e-16 ***
bmi_c	0.027777	0.006508	4.268	1.97e-05 ***
smoke100	0.238300	0.095301	2.500	0.0124 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta: count = 0.4138

Number of iterations in BFGS optimization: 17

Log-likelihood: -3469 on 7 Df

```
confint(mod_hur_nb1)
```

	2.5 %	97.5 %
count_(Intercept)	1.34373051	1.73966938
count_bmi_c	0.01107002	0.03760872
count_smoke100	0.30035006	0.73549531
zero_(Intercept)	-0.80664805	-0.56037031
zero_bmi_c	0.01502124	0.04053230
zero_smoke100	0.05151314	0.42508607

26.7.1 Comparison to a null model

To show that this model fits better than the null model (the model with intercept only), we can compare them directly with a chi-squared test. Since we have two predictors in the full model, the degrees of freedom for this test is 2.

```
mod_hur_nb_null <- pscl::hurdle(physhealth ~ 1, dist = "negbin",
                                zero.dist = "binomial",
                                data = ohioA_young)

summary(mod_hur_nb_null)
```

Call:

```
pscl::hurdle(formula = physhealth ~ 1, data = ohioA_young, dist = "negbin",
              zero.dist = "binomial")
```

Pearson residuals:

Min	1Q	Median	3Q	Max
-0.4048	-0.4048	-0.4048	-0.1622	3.2340

Count model coefficients (truncated negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.77653	0.09641	18.427	< 2e-16 ***
Log(theta)	-1.04455	0.16123	-6.479	9.25e-11 ***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.5768	0.0469	-12.3	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta: count = 0.3518

Number of iterations in BFGS optimization: 12

Log-likelihood: -3498 on 3 Df

```
pchisq(2 * (logLik(mod_hur_nb1) - logLik(mod_hur_nb_null)), df = 2, lower.tail = FALSE)
```

'log Lik.' 2.17419e-13 (df=7)

26.7.2 Comparison to a Negative Binomial Model: Vuong test

```
vuong(mod_hur_nb1, mod_nb1)
```

Vuong Non-Nested Hypothesis Test-Statistic:

(test-statistic is asymptotically distributed $N(0,1)$ under the null that the models are indistinguishable)

```
-----  
              Vuong z-statistic              H_A    p-value  
Raw              3.0953118 model1 > model2 0.00098303  
AIC-corrected    2.4837306 model1 > model2 0.00650071  
BIC-corrected    0.7750288 model1 > model2 0.21916133
```

The hurdle model improves the standard negative binomial model according to the raw and AIC-corrected versions of this test, but not the BIC-corrected version.

26.7.3 Comparison to a Zero-Inflated NB Model: Vuong test

Is the hurdle model comparable to the zero-inflated Poisson?

```
vuong(mod_hur_nb1, mod_zinb1)
```

Vuong Non-Nested Hypothesis Test-Statistic:

(test-statistic is asymptotically distributed $N(0,1)$ under the null that the models are indistinguishable)

```
-----  
              Vuong z-statistic              H_A p-value  
Raw              0.9126037 model1 > model2 0.18073  
AIC-corrected    0.9126037 model1 > model2 0.18073  
BIC-corrected    0.9126037 model1 > model2 0.18073
```

The hurdle model appears just barely improved over the zero-inflated Negative Binomial model.

26.7.4 Comparing the Hurdle Models with AIC and BIC

```
AIC(mod_hur1); BIC(mod_hur1)
```

```
[1] 11371.58
```

```
[1] 11405.11
```

```
AIC(mod_hur_nb1); BIC(mod_hur_nb1)
```

```
[1] 6952.186
```

```
[1] 6991.301
```

The negative binomial approach certainly looks better than the Poisson here.

26.7.5 The Fitted Equation

The form of the model equation for this hurdle also requires us to take two separate models into account. First we have a logistic regression model to predict the log odds of zero `physhealth` days. That takes care of the zeros. Then, to predict the number of `physhealth` days, we use a truncated negative binomial model, which is truncated to produce only estimates greater than zero, with θ estimated as $\exp(-1.123)$ or 0.325.

26.7.6 Interpreting the Coefficients

We can exponentiate the logistic regression coefficients to obtain results in terms of odds ratios for that model, and that can be of some help in understanding the process behind excess zeros.

```
exp(coef(mod_hur_nb1))
```

count_(Intercept)	count_bmi_c	count_smoke100	zero_(Intercept)
4.6725266	1.0246380	1.6785372	0.5048423
zero_bmi_c	zero_smoke100		
1.0281661	1.2690894		

For example,

- in the model for `physhealth` = 0, the odds of `physhealth` = 0 are 127% as high for subjects with `smoke100` = 1 as for non-smokers with the same BMI.

26.7.7 Testing the Predictors

We can test the model with and without `bmi_c`, for example, by fitting the model both ways, and comparing the results with either a Wald or Likelihood Ratio test, each of which is available in the `lmtest` package.

```
mod_hurnb1_nobmi <- pscl::hurdle(physhealth ~ smoke100,
                                dist = "negbin",
                                zero.dist = "binomial",
                                data = ohioA_young)

lmtest::waldtest(mod_hur_nb1, mod_hurnb1_nobmi)
```

Wald test

```
Model 1: physhealth ~ bmi_c + smoke100
Model 2: physhealth ~ smoke100
   Res.Df Df    Chisq Pr(>Chisq)
1     1967
2     1969 -2  31.141  1.729e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmtest::lrtest(mod_hur_nb1, mod_hurnb1_nobmi)
```

Likelihood ratio test

```
Model 1: physhealth ~ bmi_c + smoke100
Model 2: physhealth ~ smoke100
   #Df  LogLik Df  Chisq Pr(>Chisq)
1     7 -3469.1
2     5 -3485.0 -2  31.812  1.236e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

26.7.8 Store fitted values and residuals

Again we need to build up the fitted values and residuals, without `broom` to help.

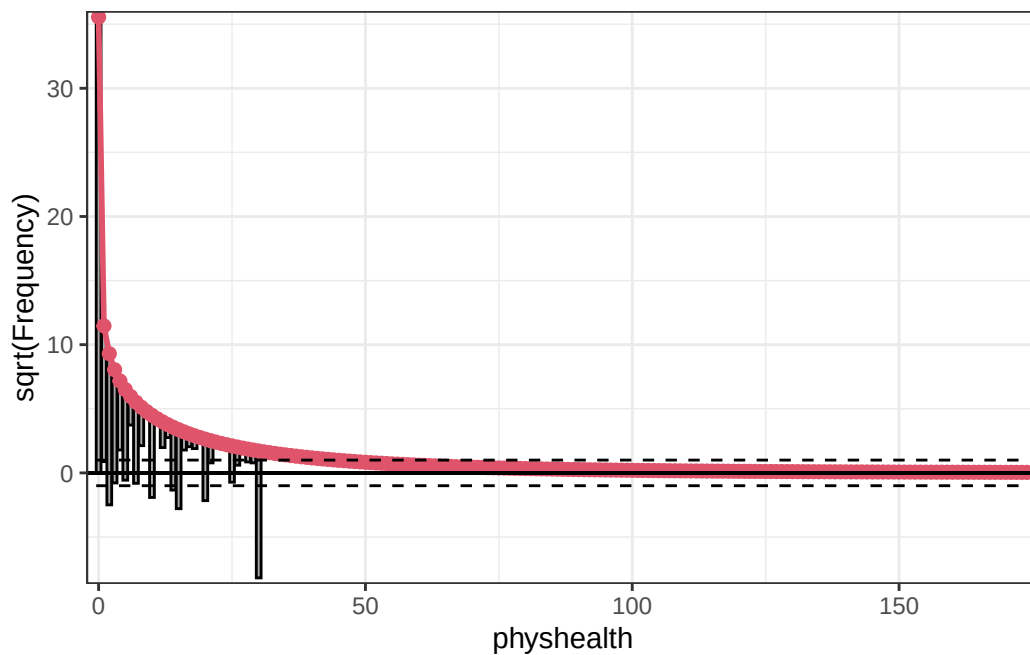
```
sm_hur_nb1 <- ohioA_young |>
  mutate(fitted = fitted(mod_hur_nb1, type = "response"),
         resid = resid(mod_hur_nb1, type = "response"))

sm_hur_nb1 |>
  dplyr::select(physhealth, fitted, resid) |>
  head()
```

```
# A tibble: 6 x 3
  physhealth fitted resid
    <dbl>    <dbl> <dbl>
1         0     2.16 -2.16
2         0     2.23 -2.23
3         0     3.09 -3.09
4        30     5.37 24.6
5         0     3.15 -3.15
6         0     3.72 -3.72
```

26.7.9 Rootogram for NB Hurdle Model

```
topmodels::rootogram(mod_hur_nb1, max = 30)
```



This improves the situation, but we're still underfitting the 30s.

26.7.10 Specify the R^2 and log (likelihood) values

We can calculate a proxy for R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(hurnb1_r <- with(sm_hur_nb1, cor(physhealth, fitted)))
```

```
[1] 0.1856393
```

```
# R-square
hurnb1_r^2
```

```
[1] 0.03446194
```

```
logLik(mod_hur_nb1)
```

```
'log Lik.' -3469.093 (df=7)
```

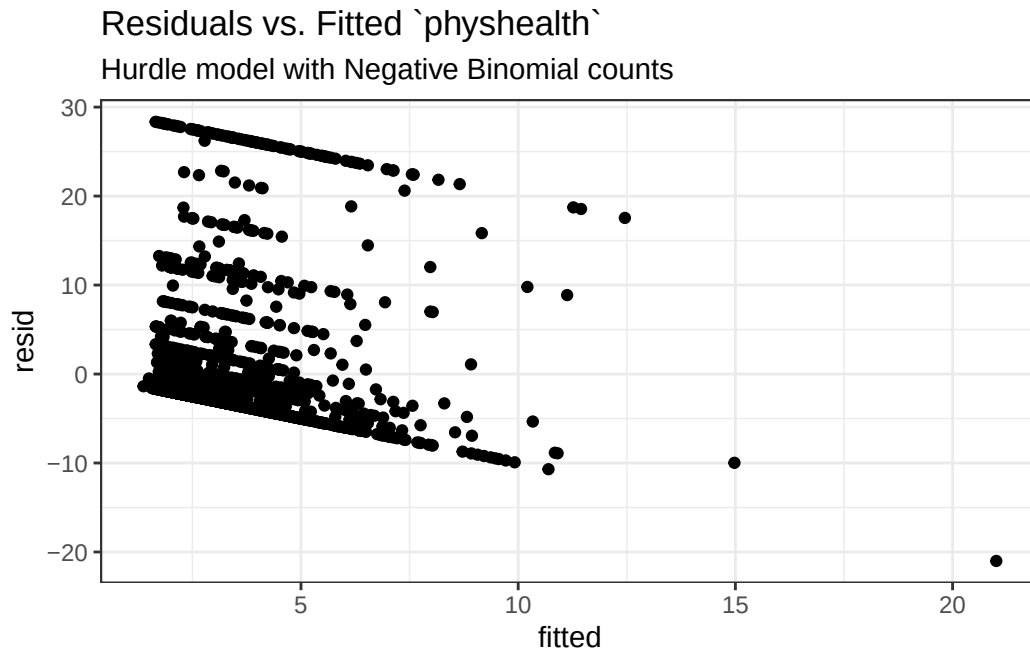
Here, we have

Model	Scale	R^2	log(likelihood)
Hurdle Model (Neg. Bin.)	Complex: log(physhealth)	.035	-3469.07

26.7.11 Check model assumptions

Here is a plot of residuals vs. fitted values on the original `physhealth` scale.

```
ggplot(sm_hur_nb1, aes(x = fitted, y = resid)) +
  geom_point() +
  labs(title = "Residuals vs. Fitted `physhealth`",
        subtitle = "Hurdle model with Negative Binomial counts")
```



26.7.12 Predictions for Harry and Sally

The predictions from this zero-inflated negative binomial regression model are obtained as follows...

```
predict(mod_hur_nb1, newdata = hs_data, type = "response")
```

```
      1      2
6.222041 2.007094
```

The prediction for Harry is 6.22 days, and for Sally is 2.01 days.

26.7.13 Note: Fitting a Different Hurdle Model for Counts and $\Pr(\text{zero})$

Suppose we wanted to use only `bmi_c` to predict the probability of a zero count, but use both predictors in the model for the positive counts. We use the `|` command.

```
mod_hur_new1 <-
  pscl::hurdle(physhealth ~ bmi_c + smoke100 | bmi_c,
    dist = "negbin", zero.dist = "binomial",
    data = ohioA_young)

summary(mod_hur_new1)
```

Call:

```
pscl::hurdle(formula = physhealth ~ bmi_c + smoke100 | bmi_c, data = ohioA_young,
  dist = "negbin", zero.dist = "binomial")
```

Pearson residuals:

Min	1Q	Median	3Q	Max
-0.5630	-0.4179	-0.3971	-0.1150	6.1958

Count model coefficients (truncated negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.54170	0.10101	15.263	< 2e-16 ***
bmi_c	0.02434	0.00677	3.595	0.000324 ***
smoke100	0.51792	0.11101	4.666	3.08e-06 ***
Log(theta)	-0.88245	0.14653	-6.023	1.72e-09 ***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.58186	0.04718	-12.332	< 2e-16 ***
bmi_c	0.02853	0.00649	4.396	1.1e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

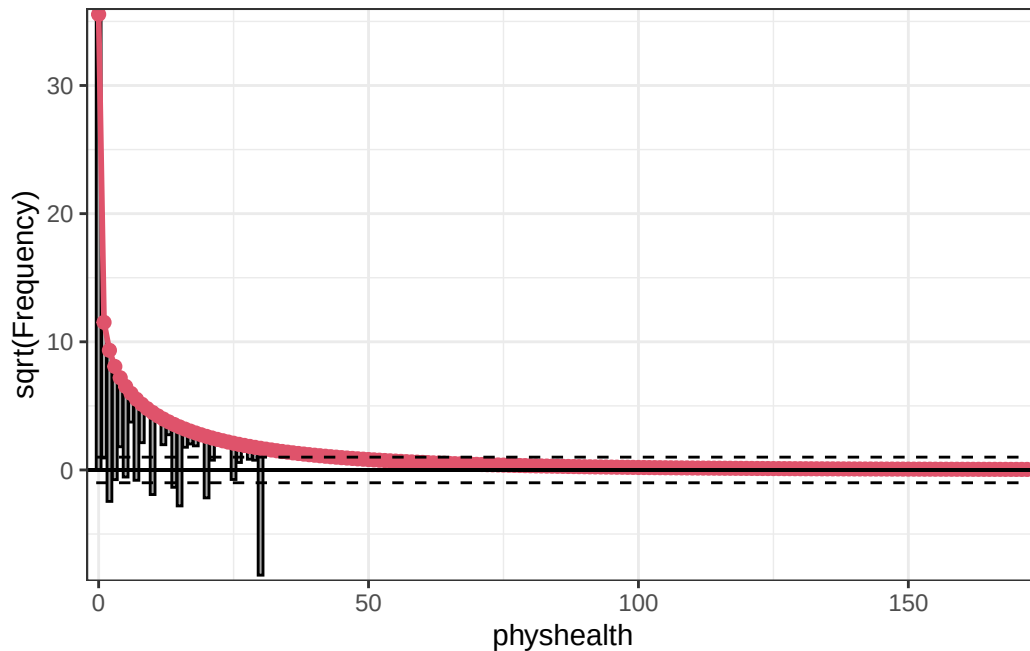
Theta: count = 0.4138

Number of iterations in BFGS optimization: 17

Log-likelihood: -3472 on 6 Df

26.7.14 Hanging Rootogram for this new Hurdle Model

```
topmodels::rootogram(mod_hur_new1, max = 30)
```



Not a meaningful improvement, certainly.

26.8 A Tobit (Censored) Regression Model

The idea of the **tobit** model (sometimes called a **censored regression** model) is to estimate associations for outcomes where we can see either left-censoring (censoring from below) or right-censoring (censoring from above.)

Consider the variable **physhealth**, which is restricted to fall between 0 and 30 by the way the measure was constructed. But supposed we think about a broader and latent (unobserved) variable describing physical health. Among the people with **physhealth** = 0, some would be incredible athletes and others would be in much poorer physical health but still healthy enough to truthfully answer 0. On the other end, some of the people responding 30 are in substantially worse physical health than others with that same response.

- Censoring from below takes place when values at or below a threshold (in this case 0) take that value.
- Censoring from above takes place when values at or above a threshold (here, 30) take that value.

Several examples of tobit analysis are available at <https://stats.idre.ucla.edu/r/dae/tobit-models/>, which is my primary source for the material here on those models.

The tobit model postulates that the value 0 in our model is just the lower limit of the underlying measure of poor physical health that we would actually observe in the population if we had a stronger measure. Similarly, we'll postulate that 30 is just the upper limit of "poor health" that we can see. The approach I'll take to run the tobit model comes from the `vglm` function in the VGAM package.

Here's the model, and its summary. Note that the default Lower value for a tobit model is 0, so we didn't technically have to list that here.

```
mod_tob1 <- vglm(physhealth ~ bmi_c + smoke100,
  tobit(Lower = 0, Upper = 30),
  type.fitted = "censored",
  data = ohioA_young)

summary(mod_tob1)
```

Call:

```
vglm(formula = physhealth ~ bmi_c + smoke100, family = tobit(Lower = 0,
  Upper = 30), data = ohioA_young, type.fitted = "censored")
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept):1	-9.29104	0.75509	-12.305	< 2e-16 ***
(Intercept):2	2.87081	0.03028	94.818	< 2e-16 ***
bmi_c	0.37434	0.06537	5.727	1.02e-08 ***
smoke100	4.35390	0.97223	4.478	7.52e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: mu, loglink(sd)

Log-likelihood: -3420.605 on 3944 degrees of freedom

Number of Fisher scoring iterations: 6

```
confint(mod_tob1)
```

	2.5 %	97.5 %
(Intercept):1	-10.7709932	-7.8110877
(Intercept):2	2.8114721	2.9301560
bmi_c	0.2462244	0.5024551
smoke100	2.4483714	6.2594277

26.8.1 The Fitted Equation

Because we've used the censoring approach, our model will limit its predictions to the range of $[0, 30]$, where any predictions outside that range are censored. Values below 0 are fitted as 0, and values above 30 are fitted as 30.

The model equation is

$$\text{physhealth} = -9.29 + 0.37 \text{ bmi_c} + 4.35 \text{ smoke100}$$

26.8.2 Interpreting the Coefficients

Tobit model regression coefficients are interpreted as we would a set of OLS coefficients, *except* that the linear effect is on the uncensored *latent variable*, rather than on the observed outcome.

In our case,

- a one-unit increase in `bmi_c` is associated with a 0.37 day increase in the predicted value of `physhealth`, holding `smoke100` constant
- a move from `smoke100 = 0` to 1 is associated with a 4.35 day increase in the predicted value of `physhealth`, holding `bmi_c` constant
- the coefficient labeled `(Intercept):1` is the intercept for the model and is the predicted value of `physhealth` when `smoke100 = 0` and `bmi_c = 0`. Note that this value is -9.29, which is outside the range of `physhealth` values we observed.
- the coefficient labeled `(Intercept):2` is a statistic we can use after we exponentiate it, as follows:
 - here `(Intercept):2 = 2.87`, and $\exp(2.87) = 17.6370182$, which is analogous to the square root of the residual variance in OLS regression, which is summarized for our OLS model as `Residual standard error: 17.64`.

26.8.3 Testing the Predictors

We can test the model with and without `bmi_c`, for example, by fitting the model both ways, and comparing the results with either a Wald or Likelihood Ratio test, each of which is available in the `lmtest` package.

```
mod_tob_nobmi <- vglm(physhealth ~ smoke100,
                      tobit(Lower = 0, Upper = 30),
                      type.fitted = "censored",
                      data = ohioA_young)
```

```
lmtest::waldtest(mod_tob1, mod_tob_nobmi)
```

Wald test

```
Model 1: physhealth ~ bmi_c + smoke100
Model 2: physhealth ~ smoke100
   Res.Df Df    Chisq Pr(>Chisq)
1     3944
2     3945 -1 32.796  1.023e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The likelihood ratio test we have used in some other settings isn't available here.

26.8.4 Store fitted values and residuals

The residuals and fitted values from the tobit model can be stored and then summarized in several ways:

```
sm_tob1 <- ohioA_young |>
  mutate(fitted = fitted(mod_tob1,
                        type.fitted = "censored"),
         resid = physhealth - fitted)

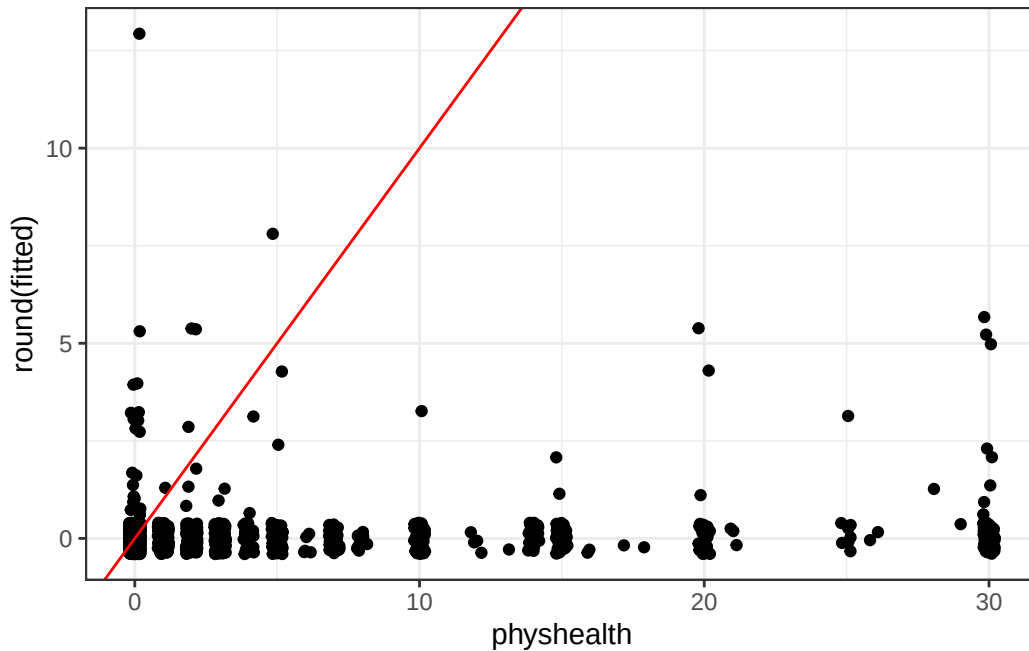
sm_tob1 |>
  dplyr::select(physhealth, fitted, resid) |>
  head()
```

```
# A tibble: 6 x 3
  physhealth fitted[,1] resid[,1]
    <dbl>      <dbl>      <dbl>
1         0         0         0
2         0         0         0
3         0         0         0
4        30         0        30
5         0         0         0
6         0         0         0
```

26.8.5 Building Something Like a Rootogram

Building a rootogram is tricky for a tobit model, to say the least, but we can approximate a piece of the issue by plotting the rounded fitted values against the observed `physhealth` data.

```
ggplot(sm_tob1, aes(x = physhealth, y = round(fitted))) +
  geom_jitter(width = 0.2) +
  geom_abline(intercept = 0, slope = 1, col = "red")
```



Note that the model never predicts a subject to have an underlying `physhealth` worse than about 13 (remember that larger numbers indicate worse health here.)

26.8.6 Tables of the Observed and Fitted `physhealth` from Tobit

```
addmargins(table(round(sm_tob1$physhealth)))
```

0	1	2	3	4	5	6	7	8	10	12	13	14	15	16	17
1264	112	139	78	29	50	5	40	9	41	4	1	24	38	2	1
18	20	21	25	26	28	29	30	Sum							
1	23	3	8	2	1	1	98	1974							

```
addmargins(table(round(sm_tob1$fitted)))
```

0	1	2	3	4	5	6	8	13	Sum
1924	20	7	10	4	6	1	1	1	1974

26.8.7 Specify the R^2 and log (likelihood) values

We can calculate a proxy for R^2 as the squared correlation of the fitted values and the observed values.

```
# The correlation of observed and fitted values
(tob1_r <- with(sm_tob1, cor(physhealth, fitted)))
```

```
      [,1]
[1,] 0.09329619
```

```
# R-square
tob1_r^2
```

```
      [,1]
[1,] 0.008704178
```

```
logLik(mod_tob1)
```

```
[1] -3420.605
```

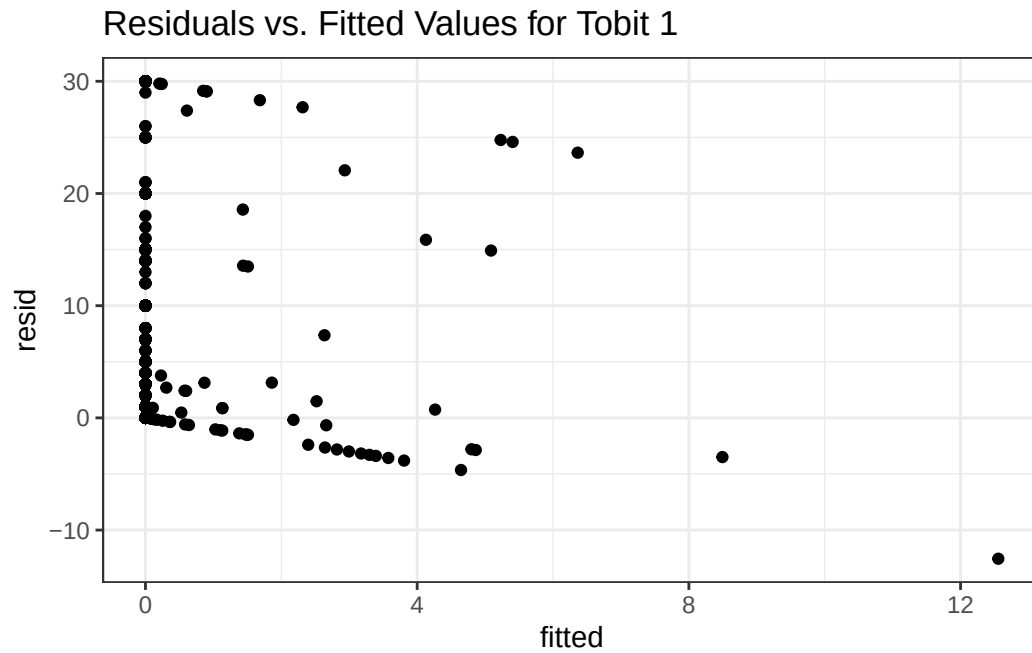
Here, we have

Model	Scale	R^2	log(likelihood)
Tobit	physhealth	.008	-3420.58

26.8.8 Check model assumptions

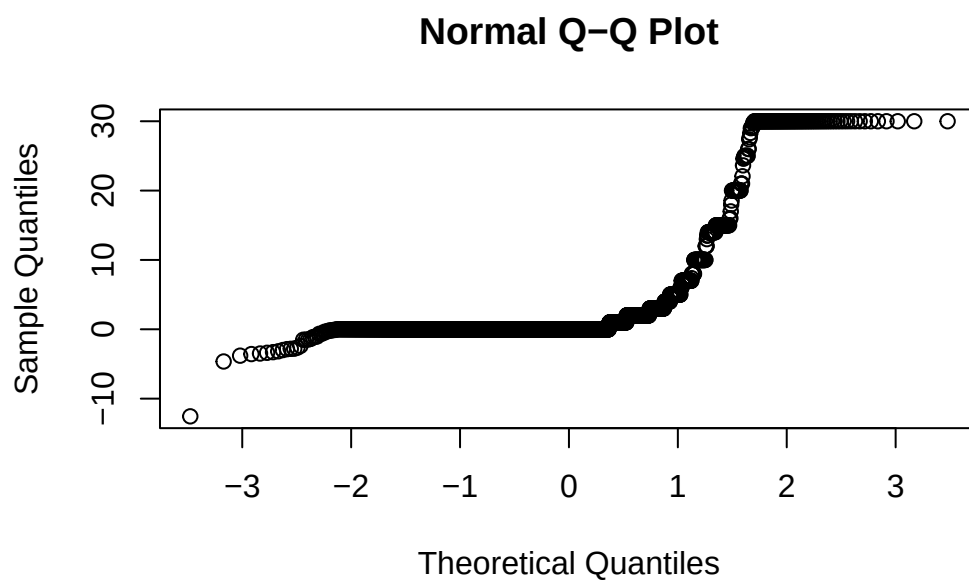
Here is a plot of residuals vs. fitted values.

```
ggplot(sm_tob1, aes(x = fitted, y = resid)) +  
  geom_point() +  
  labs(title = "Residuals vs. Fitted Values for Tobit 1")
```



Here is a normal Q-Q plot of the Tobit Model 1 residuals.

```
qqnorm(sm_tob1$resid)
```



26.8.9 Predictions for Harry and Sally

The predictions from this tobit model are obtained as follows...

```
predict(mod_tob1, newdata = hs_data, type = "response")
```

```
      [,1]
1  -1.193743
2 -11.162739
```

The prediction for both Harry and Sally under the tobit model would be truncated to 0 days.

27 Modeling an Ordinal Categorical Outcome

27.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(gmodels)
library(MASS)
library(nnet)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

27.1.1 Data Load

```
smart_oh <- readRDS("data/smart_ohio.Rds")
```

27.2 A subset of the Ohio SMART data

Let's consider the following data, which uses part of the `smart_oh` data we built in Chapter 6. The outcome we'll study now is `genhealth`, which has five ordered categories. I'll include the subset of all observations in `smart_oh` with complete data on these 7 variables.

Variable	Description
SEQNO	Subject identification code
genhealth	Five categories (1 = Excellent, 2 = Very Good, 3 = Good, 4 = Fair, 5 = Poor) on general health

Variable	Description
<code>physhealth</code>	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?
<code>veg_day</code>	mean number of vegetable servings consumed per day
<code>costprob</code>	1 indicates Yes to “Was there a time in the past 12 months when you needed to see a doctor but could not because of cost?”, and 0 otherwise.
<code>incomegroup</code>	8 income groups from < 10,000 to 75,000 or more
<code>bmi</code>	body-mass index

To make my life easier later, I’m going to drop any subjects with missing data on these variables. I’m also going to drop the subjects who have no missing data, but have a listed `bmi` above 60.

```
sm1 <- smart_oh |>
  select(SEQN0, genhealth, physhealth, costprob, veg_day,
         incomegroup, bmi) |>
  filter(bmi <= 60) |>
  drop_na()
```

In total, we have 5394 subjects in the `sm1` sample.

27.2.1 Several Ways of Storing Multi-Categorical data

We will store the information in our outcome, `genhealth` in both a numeric form (`gen_n`) and an ordered factor (`gen_h`) with some abbreviated labels) because we’ll have some use for each approach in this material.

```
sm1 <- sm1 |>
  mutate(genh = fct_recode(genhealth,
                           "1-E" = "1_Excellent",
                           "2_VG" = "2_VeryGood",
                           "3_G" = "3_Good",
                           "4_F" = "4_Fair",
                           "5_P" = "5_Poor"),
         genh = factor(genh, ordered = TRUE),
         gen_n = as.numeric(genhealth))

sm1 |> count(genh, gen_n, genhealth)
```

```
# A tibble: 5 x 4
  genh   gen_n genhealth     n
  <ord> <dbl> <fct>    <int>
1 1-E         1 1_Excellent  822
2 2_VG        2 2_VeryGood 1805
3 3_G         3 3_Good    1667
4 4_F         4 4_Fair     801
5 5_P         5 5_Poor     299
```

27.3 Building Cross-Tabulations

Is income group associated with general health?

27.3.1 Using base table functions

```
addmargins(table(sm1$incomegroup, sm1$genh))
```

	1-E	2_VG	3_G	4_F	5_P	Sum
0-9K	14	45	60	74	40	233
10-14K	12	41	66	93	46	258
15-19K	40	76	119	96	61	392
20-24K	51	129	175	100	50	505
25-34K	51	172	215	123	36	597
35-49K	97	270	303	118	24	812
50-74K	128	337	265	94	16	840
75K+	429	735	464	103	26	1757
Sum	822	1805	1667	801	299	5394

More people answer Very Good and Good than choose the other categories. It might be easier to look at percentages here.

27.3.1.1 Adding percentages within each row

Here are the percentages giving each `genhealth` response within each income group.

```
addmargins(
  round(100*prop.table(
    table(sm1$incomegroup, sm1$genh)
    ,1)
    ,1)
  )
)
```

	1-E	2_VG	3_G	4_F	5_P	Sum
0-9K	6.0	19.3	25.8	31.8	17.2	100.1
10-14K	4.7	15.9	25.6	36.0	17.8	100.0
15-19K	10.2	19.4	30.4	24.5	15.6	100.1
20-24K	10.1	25.5	34.7	19.8	9.9	100.0
25-34K	8.5	28.8	36.0	20.6	6.0	99.9
35-49K	11.9	33.3	37.3	14.5	3.0	100.0
50-74K	15.2	40.1	31.5	11.2	1.9	99.9
75K+	24.4	41.8	26.4	5.9	1.5	100.0
Sum	91.0	224.1	247.7	164.3	72.9	800.0

So, for example, 11.3% of the `genhealth` responses in subjects with incomes between 25 and 34 thousand dollars were Excellent.

27.3.1.2 Adding percentages within each column

Here are the percentages in each `incomegroup` within each `genhealth` response.

```
addmargins(
  round(100*prop.table(
    table(sm1$incomegroup, sm1$genh)
    ,2)
    ,1)
  )
)
```

	1-E	2_VG	3_G	4_F	5_P	Sum
0-9K	1.7	2.5	3.6	9.2	13.4	30.4
10-14K	1.5	2.3	4.0	11.6	15.4	34.8
15-19K	4.9	4.2	7.1	12.0	20.4	48.6
20-24K	6.2	7.1	10.5	12.5	16.7	53.0
25-34K	6.2	9.5	12.9	15.4	12.0	56.0

35-49K	11.8	15.0	18.2	14.7	8.0	67.7
50-74K	15.6	18.7	15.9	11.7	5.4	67.3
75K+	52.2	40.7	27.8	12.9	8.7	142.3
Sum	100.1	100.0	100.0	100.0	100.0	500.1

From this table, we see that 7.4% of the Excellent `genhealth` responses were given by people with incomes between 25 and 34 thousand dollars.

27.3.2 Using `xtabs`

The `xtabs` function provides a formula method for obtaining cross-tabulations.

```
xtabs(~ incomegroup + genh, data = sm1)
```

	genh				
incomegroup	1-E	2_VG	3_G	4_F	5_P
0-9K	14	45	60	74	40
10-14K	12	41	66	93	46
15-19K	40	76	119	96	61
20-24K	51	129	175	100	50
25-34K	51	172	215	123	36
35-49K	97	270	303	118	24
50-74K	128	337	265	94	16
75K+	429	735	464	103	26

27.3.3 Storing a table in a tibble

We can store the elements of a cross-tabulation in a tibble, like this:

```
(sm1.tableA <- sm1 |> count(incomegroup, genh))
```

```
# A tibble: 40 x 3
  incomegroup genh      n
  <fct>       <ord> <int>
1 0-9K       1-E      14
2 0-9K       2_VG     45
3 0-9K       3_G      60
4 0-9K       4_F      74
5 0-9K       5_P      40
6 10-14K     1-E      12
```

```

7 10-14K      2_VG      41
8 10-14K      3_G       66
9 10-14K      4_F       93
10 10-14K     5_P       46
# i 30 more rows

```

From such a tibble, we can visualize the data in many ways, but we can also return to `xtabs` and include the frequencies (`n`) in that setup.

```
xtabs(n ~ incomegroup + genh, data = sm1.tableA)
```

	genh				
incomegroup	1-E	2_VG	3_G	4_F	5_P
0-9K	14	45	60	74	40
10-14K	12	41	66	93	46
15-19K	40	76	119	96	61
20-24K	51	129	175	100	50
25-34K	51	172	215	123	36
35-49K	97	270	303	118	24
50-74K	128	337	265	94	16
75K+	429	735	464	103	26

And, we can get the χ^2 test of independence, with:

```
summary(xtabs(n ~ incomegroup + genh, data = sm1.tableA))
```

```

Call: xtabs(formula = n ~ incomegroup + genh, data = sm1.tableA)
Number of cases in table: 5394
Number of factors: 2
Test for independence of all factors:
    Chisq = 894.2, df = 28, p-value = 3.216e-170

```

27.3.4 Using CrossTable from the gmodels package

The `CrossTable` function from the `gmodels` package produces a cross-tabulation with various counts and proportions like people often generate with SPSS and SAS.

```
CrossTable(sm1$incomegroup, sm1$genh, chisq = T)
```

```

Cell Contents
|-----|
|              N |
| Chi-square contribution |
|      N / Row Total |
|      N / Col Total |
|      N / Table Total |
|-----|

```

Total Observations in Table: 5394

sm1\$incomegroup	sm1\$genh					Row Total
	1-E	2_VG	3_G	4_F	5_P	

0-9K	14	45	60	74	40	233
	13.027	13.941	2.002	44.865	56.796	
	0.060	0.193	0.258	0.318	0.172	0.043
	0.017	0.025	0.036	0.092	0.134	
	0.003	0.008	0.011	0.014	0.007	

10-14K	12	41	66	93	46	258
	18.980	23.806	2.366	78.061	70.259	
	0.047	0.159	0.256	0.360	0.178	0.048
	0.015	0.023	0.040	0.116	0.154	
	0.002	0.008	0.012	0.017	0.009	

15-19K	40	76	119	96	61	392
	6.521	23.208	0.038	24.531	70.973	
	0.102	0.194	0.304	0.245	0.156	0.073
	0.049	0.042	0.071	0.120	0.204	
	0.007	0.014	0.022	0.018	0.011	

20-24K	51	129	175	100	50	505
	8.756	9.463	2.296	8.340	17.301	
	0.101	0.255	0.347	0.198	0.099	0.094

		0.062	0.071	0.105	0.125	0.167	
		0.009	0.024	0.032	0.019	0.009	

25-34K	51	172	215	123	36	597	
	17.567	3.862	5.042	13.307	0.255		
	0.085	0.288	0.360	0.206	0.060	0.111	
	0.062	0.095	0.129	0.154	0.120		
	0.009	0.032	0.040	0.023	0.007		

35-49K	97	270	303	118	24	812	
	5.779	0.011	10.798	0.055	9.808		
	0.119	0.333	0.373	0.145	0.030	0.151	
	0.118	0.150	0.182	0.147	0.080		
	0.018	0.050	0.056	0.022	0.004		

50-74K	128	337	265	94	16	840	
	0.000	11.121	0.112	7.575	20.061		
	0.152	0.401	0.315	0.112	0.019	0.156	
	0.156	0.187	0.159	0.117	0.054		
	0.024	0.062	0.049	0.017	0.003		

75K+	429	735	464	103	26	1757	
	97.108	36.780	11.492	95.573	52.335		
	0.244	0.418	0.264	0.059	0.015	0.326	
	0.522	0.407	0.278	0.129	0.087		
	0.080	0.136	0.086	0.019	0.005		

Column Total	822	1805	1667	801	299	5394	
	0.152	0.335	0.309	0.148	0.055		

Statistics for All Table Factors

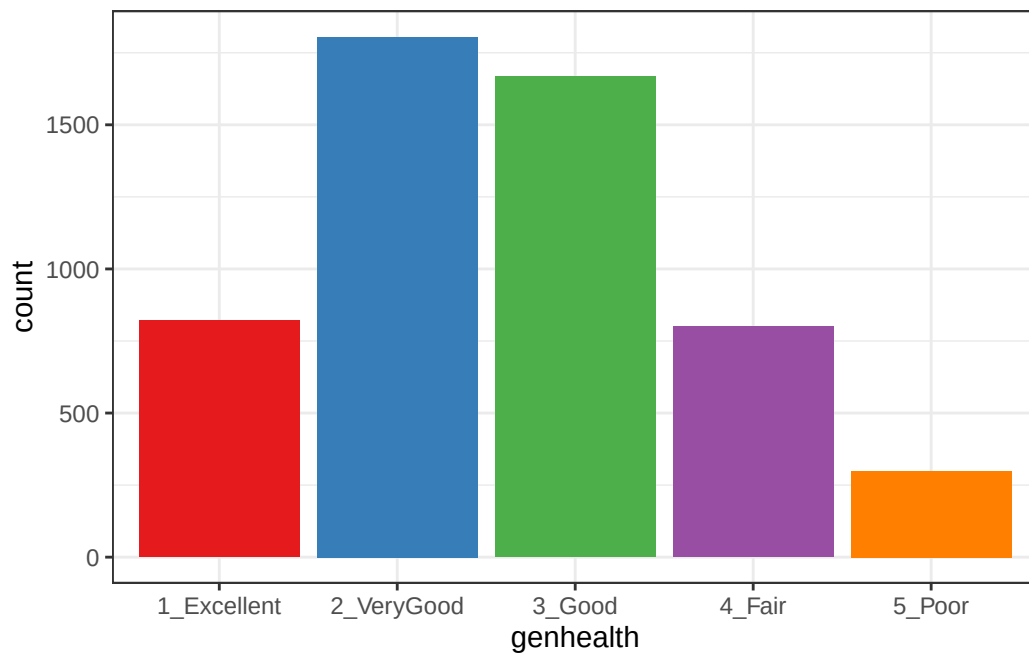
Pearson's Chi-squared test

Chi² = 894.1685 d.f. = 28 p = 3.216132e-170

27.4 Graphing Categorical Data

27.4.1 A Bar Chart for a Single Variable

```
ggplot(sm1, aes(x = genhealth, fill = genhealth)) +  
  geom_bar() +  
  scale_fill_brewer(palette = "Set1") +  
  guides(fill = "none")
```



or, you might prefer to plot percentages, perhaps like this:

```
ggplot(sm1, aes(x = genhealth, fill = genhealth)) +  
  geom_bar(aes(y = (..count..)/sum(..count..))) +  
  geom_text(aes(y = (..count..)/sum(..count..),  
                label = scales::percent(..count..) /
```

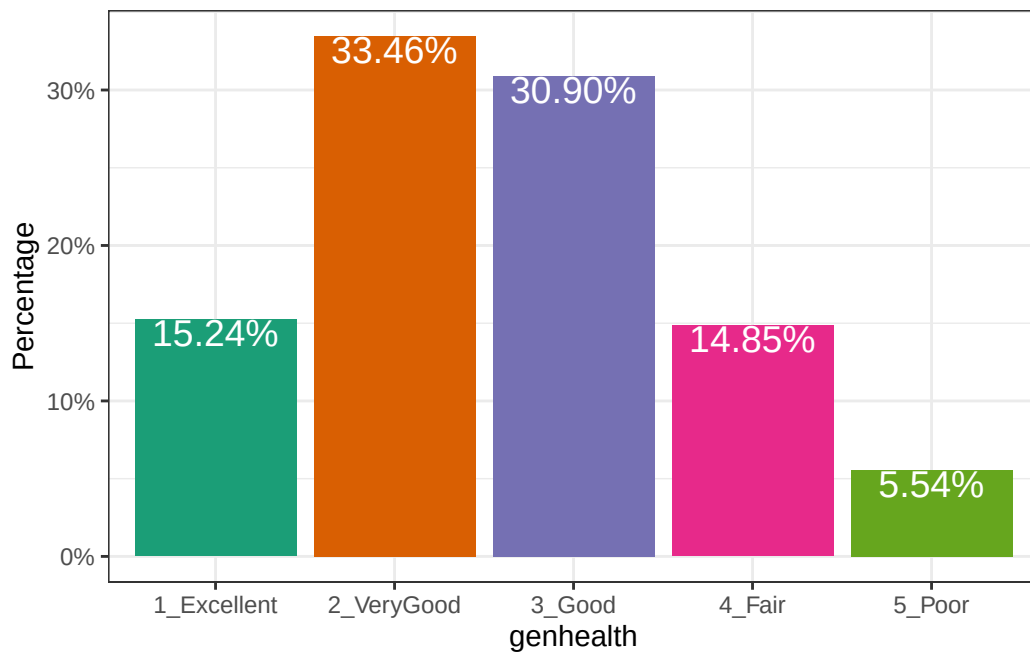


```

                                sum(..count..))),
    stat = "count", vjust = 1,
    color = "white", size = 5) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_brewer(palette = "Dark2") +
  guides(fill = "none") +
  labs(y = "Percentage")

```

Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
 i Please use `after\_stat(count)` instead.



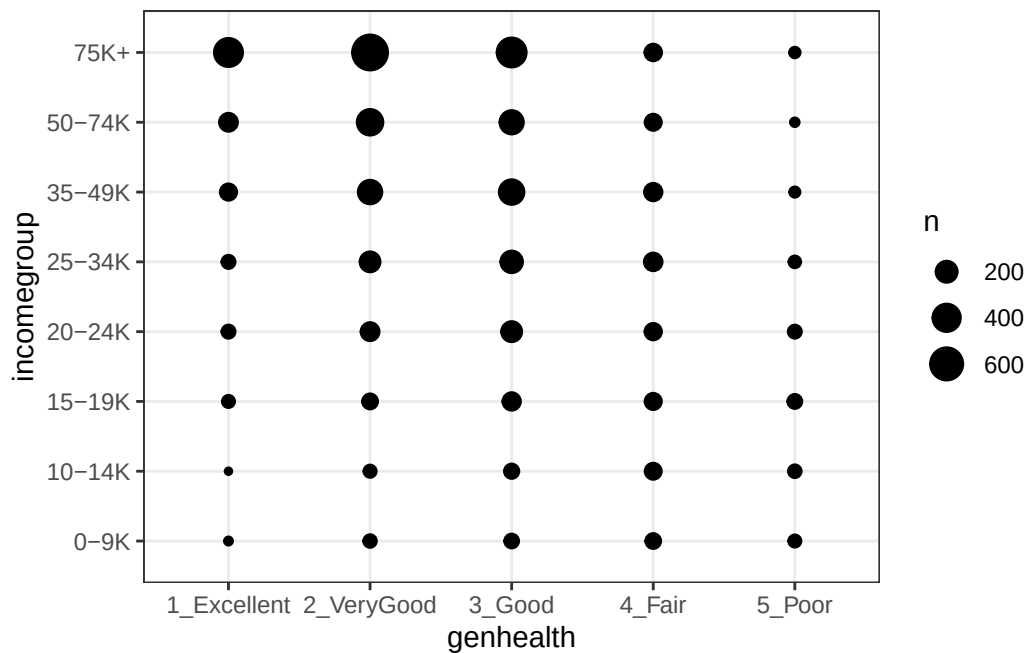
Use bar charts, rather than pie charts.

27.4.2 A Counts Chart for a 2-Way Cross-Tabulation

```

ggplot(sm1, aes(x = genhealth, y = incomegroup)) +
  geom_count()

```



27.5 Building a Model for genh using veg_day

To begin, we'll predict each subject's `genh` response using just one predictor, `veg_day`.

27.5.1 A little EDA

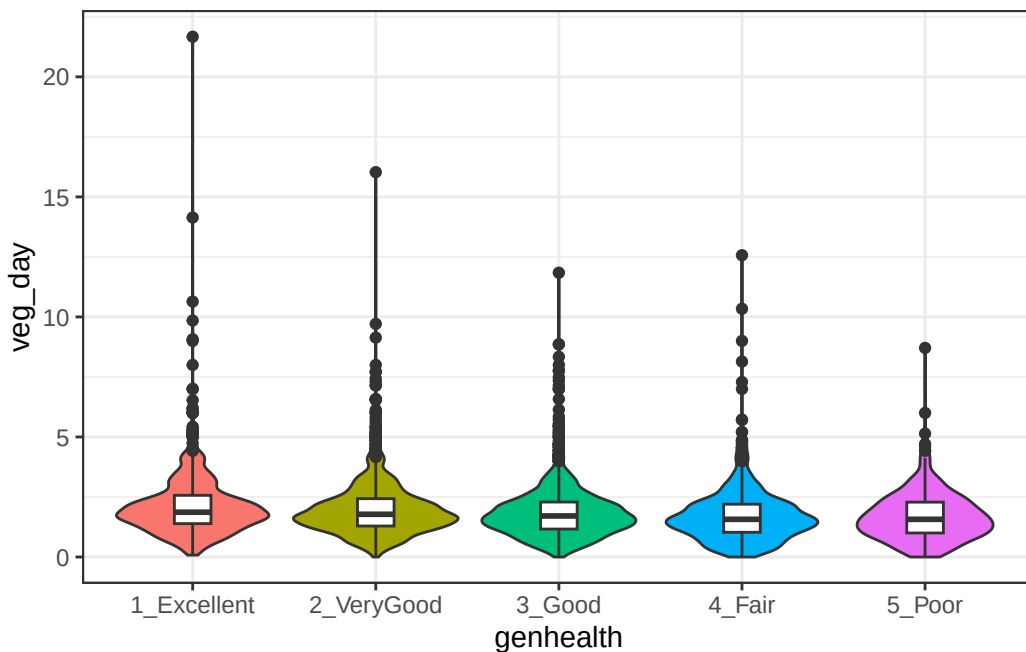
Let's start with a quick table of summary statistics.

```
sm1 |> group_by(genh) |>
  summarize(n(), mean(veg_day), sd(veg_day), median(veg_day))
```

```
# A tibble: 5 x 5
  genh   `n()` `mean(veg_day)` `sd(veg_day)` `median(veg_day)`
  <ord> <int>    <dbl>         <dbl>         <dbl>
1 1-E     822    2.16          1.46          1.87
2 2_VG   1805    1.99          1.13          1.78
3 3_G    1667    1.86          1.11          1.71
4 4_F     801    1.74          1.18          1.57
5 5_P     299    1.71          1.06          1.57
```

To actually see what's going on, we might build a comparison boxplot, or violin plot. The plot below shows both, together, with the violin plot helping to indicate the skewed nature of the `veg_day` data and the boxplot indicating quartiles and outlying values within each `genhealth` category.

```
ggplot(sm1, aes(x = genhealth, y = veg_day)) +  
  geom_violin(aes(fill = genhealth), trim = TRUE) +  
  geom_boxplot(width = 0.2) +  
  guides(fill = "none", color = "none") +  
  theme_bw()
```



27.5.2 Describing the Proportional-Odds Cumulative Logit Model

To fit the ordinal logistic regression model (specifically, a proportional-odds cumulative-logit model) in this situation, we'll use the `polr` function in the `MASS` package.

- Our outcome is `genh`, which has five ordered levels, with 1-E best and 5-P worst.
- Our model will include one quantitative predictor, `veg_day`.

The model will have four logit equations:

- one estimating the log odds that `genh` will be less than or equal to 1 (i.e. `genhealth = 1_Excellent`),

- one estimating the log odds that $\text{genh} \leq 2$ (i.e. $\text{genhealth} = 1_Excellent$ or $2_VeryGood$),
- another estimating the log odds that $\text{genh} \leq 3$ (i.e. $\text{genhealth} = 1_Excellent$, $2_VeryGood$ or 3_Good), and, finally,
- one estimating the log odds that $\text{genh} \leq 4$ (i.e. $\text{genhealth} = 1_Excellent$, $2_VeryGood$, 3_Good or 4_Fair)

That's all we need to estimate the five categories, since $\Pr(\text{genh} \leq 5) = 1$, because (5_Poor) is the maximum category for genhealth .

We'll have a total of five free parameters when we add in the slope for veg_day , and I'll label these parameters as $\zeta_1, \zeta_2, \zeta_3, \zeta_4$ and β_1 . The ζ s are read as “zeta” values, and the people who built the `polr` function use that term.

The four logistic equations that will be fit differ only by their intercepts. They are:

$$\text{logit}[\Pr(\text{genh} \leq 1)] = \log \frac{\Pr(\text{genh} \leq 1)}{\Pr(\text{genh} > 1)} = \zeta_1 - \beta_1 \text{veg_day}$$

which describes the log odds of a genh value of 1 (Excellent) as compared to a genh value greater than 1 (which includes Very Good, Good, Fair and Poor).

The second logit model is:

$$\text{logit}[\Pr(\text{genh} \leq 2)] = \log \frac{\Pr(\text{genh} \leq 2)}{\Pr(\text{genh} > 2)} = \zeta_2 - \beta_1 \text{veg_day}$$

which describes the log odds of a genh value of 1 (Excellent) or 2 (Very Good) as compared to a genh value greater than 2 (which includes Good, Fair and Poor).

Next we have:

$$\text{logit}[\Pr(\text{genh} \leq 3)] = \log \frac{\Pr(\text{genh} \leq 3)}{\Pr(\text{genh} > 3)} = \zeta_3 - \beta_1 \text{veg_day}$$

which describes the log odds of a genh value of 1 (Excellent) or 2 (Very Good) or 3 (Good) as compared to a genh value greater than 3 (which includes Fair and Poor).

Finally, we have

$$\text{logit}[\Pr(\text{genh} \leq 4)] = \log \frac{\Pr(\text{genh} \leq 4)}{\Pr(\text{genh} > 4)} = \zeta_4 - \beta_1 \text{veg_day}$$

which describes the log odds of a genh value of 4 or less, which includes Excellent, Very Good, Good and Fair as compared to a genh value greater than 4 (which is Poor).

Again, the intercept term is the only piece that varies across the four equations.

In this case, a positive coefficient β_1 for `veg_day` means that increasing the value of `veg_day` would increase the `genh` category (describing a worse level of general health, since higher values of `genh` are associated with worse health.)

27.5.3 Fitting a Proportional Odds Logistic Regression with `polr`

Our model `m1` will use proportional odds logistic regression (sometimes called an *ordered logit* model) to predict `genh` on the basis of `veg_day`. The `polr` function from the MASS package will be our main tool. Note that we include `Hess = TRUE` to retain what is called the *Hessian* matrix, which lets R calculate standard errors more effectively in `summary` and other follow-up descriptions of the model.

```
m1 <- polr(genh ~ veg_day,
           data = sm1, Hess = TRUE)

summary(m1)
```

Call:

```
polr(formula = genh ~ veg_day, data = sm1, Hess = TRUE)
```

Coefficients:

	Value	Std. Error	t value
veg_day	-0.1847	0.02178	-8.48

Intercepts:

	Value	Std. Error	t value
1-E 2_VG	-2.0866	0.0584	-35.7590
2_VG 3_G	-0.4065	0.0498	-8.1621
3_G 4_F	1.0202	0.0521	19.5771
4_F 5_P	2.5002	0.0710	35.2163

Residual Deviance: 15669.85

AIC: 15679.85

```
confint(m1)
```

Waiting for profiling to be done...

	2.5 %	97.5 %
	-0.2277073	-0.1423088

27.6 Interpreting Model m1

27.6.1 Looking at Predictions

Consider two individuals:

- Harry, who eats an average of 2.0 servings of vegetables per day, so Harry's `veg_day` = 2, and
- Sally, who eats an average of 1.0 serving of vegetables per day, so Sally's `veg_day` = 1.

We're going to start by using our model `m1` to predict the `genh` for Harry and Sally, so we can see the effect (on the predicted `genh` probabilities) of a change of one unit in `veg_day`.

For example, what are the log odds that Harry, with `veg_day` = 2, will describe his `genh` as Excellent (`genh` ≤ 1)?

$$\text{logit}[Pr(\text{genh} \leq 1)] = \zeta_1 - \beta_1 \text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 1)] = -2.0866 - (-0.1847)\text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 1)] = -2.0866 - (-0.1847)(2) = -1.7172$$

That's not much help. So we'll convert it to a probability by taking the inverse logit. The formula is

$$Pr(\text{genh} \leq 1) = \frac{\exp(\zeta_1 + \beta_1 \text{veg_day})}{1 + \exp(\zeta_1 + \beta_1 \text{veg_day})} = \frac{\exp(-1.7172)}{1 + \exp(-1.7172)} = \frac{0.180}{1.180} = 0.15$$

So the model estimates a 15% probability that Harry will describe his `genh` as Excellent.

OK. Now, what are the log odds that Harry, who eats 2 servings per day, will describe his `genh` as either Excellent or Very Good (`genh` ≤ 2)?

$$\text{logit}[Pr(\text{genh} \leq 2)] = \zeta_2 - \beta_1 \text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 2)] = -0.4065 - (-0.1847)\text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 2)] = -0.4065 - (-0.1847)(2) = -0.0371$$

Again, we'll convert this to a probability by taking the inverse logit.

$$Pr(genh \leq 2) = \frac{\exp(\zeta_2 + \beta_1 veg_{day})}{1 + \exp(\zeta_2 + \beta_1 veg_{day})} = \frac{\exp(-0.0371)}{1 + \exp(-0.0371)} = \frac{0.964}{1.964} = 0.49$$

So, the model estimates a probability of .49 that Harry will describe his **genh** as either Excellent or Very Good, so by subtraction, that's a probability of .34 that Harry describes his **genh** as Very Good.

Happily, that's the last time we'll calculate this by hand.

27.6.2 Making Predictions for Harry (and Sally) with predict

Suppose Harry eats 2 servings of vegetables per day on average, and Sally eats 1.

```
temp.dat <- data.frame(name = c("Harry", "Sally"),
                        veg_day = c(2,1))

predict(m1, temp.dat, type = "p")
```

	1-E	2_VG	3_G	4_F	5_P
1	0.1522351	0.3385119	0.3097906	0.1457864	0.05367596
2	0.1298931	0.3148971	0.3246105	0.1667285	0.06387071

The predicted probabilities of falling into each category of **genh** are:

Subject	veg_day	Pr(1_E)	Pr(2_VG)	Pr(3_G)	Pr(4_F)	Pr(5_P)
Harry	2	15.2	33.9	31.0	14.6	5.4
Sally	1	13.0	31.4	32.5	16.7	6.4

- Harry has a higher predicted probability of lower (healthier) values of **genh**. Specifically, Harry has a higher predicted probability than Sally of falling into the Excellent and Very Good categories, and a lower probability than Sally of falling into the Good, Fair and Poor categories.
- This means that Harry, with a higher **veg_day** is predicted to have, on average, a lower (that is to say, healthier) value of **genh**.
- As we'll see, this association will be indicated by a negative coefficient of **veg_day** in the proportional odds logistic regression model.

27.6.3 Predicting the actual classification of genh

The default prediction approach actually returns the predicted `genh` classification for Harry and Sally, which is just the classification with the largest predicted probability. Here, for Harry that is Very Good, and for Sally, that's Good.

```
predict(m1, temp.dat)
```

```
[1] 2_VG 3_G  
Levels: 1-E 2_VG 3_G 4_F 5_P
```

27.6.4 A Cross-Tabulation of Predictions?

```
addmargins(table(predict(m1), sm1$genh))
```

	1-E	2_VG	3_G	4_F	5_P	Sum
1-E	6	3	3	3	0	15
2_VG	647	1398	1198	525	192	3960
3_G	169	404	466	273	107	1419
4_F	0	0	0	0	0	0
5_P	0	0	0	0	0	0
Sum	822	1805	1667	801	299	5394

The `m1` model classifies all subjects in the `sm1` sample as either Excellent, Very Good or Good, and most subjects as Very Good or Good.

27.6.5 The Fitted Model Equations

```
summary(m1)
```

Call:

```
polr(formula = genh ~ veg_day, data = sm1, Hess = TRUE)
```

Coefficients:

	Value	Std. Error	t value
veg_day	-0.1847	0.02178	-8.48

Intercepts:

	Value	Std. Error	t value
1-E 2_VG	-2.0866	0.0584	-35.7590
2_VG 3_G	-0.4065	0.0498	-8.1621
3_G 4_F	1.0202	0.0521	19.5771
4_F 5_P	2.5002	0.0710	35.2163

Residual Deviance: 15669.85

AIC: 15679.85

The first part of the output provides coefficient estimates for the `veg_day` predictor, and these are followed by the estimates for the various model intercepts. Plugging in the estimates, we have:

$$\text{logit}[Pr(\text{genh} \leq 1)] = -2.0866 - (-0.1847)\text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 2)] = -0.4065 - (-0.1847)\text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 3)] = 1.0202 - (-0.1847)\text{veg_day}$$

$$\text{logit}[Pr(\text{genh} \leq 4)] = 2.5002 - (-0.1847)\text{veg_day}$$

Note that we can obtain these pieces separately as follows:

```
m1$zeta
```

```
      1-E|2_VG      2_VG|3_G      3_G|4_F      4_F|5_P
-2.0866313 -0.4064704  1.0202035  2.5001655
```

shows the boundary intercepts, and

```
m1$coefficients
```

```
      veg_day
-0.1847272
```

shows the regression coefficient for `veg_day`.

27.6.6 Interpreting the `veg_day` coefficient

The first part of the output provides coefficient estimates for the `veg_day` predictor.

- The estimated slope for `veg_day` is -0.1847
 - Remember Harry and Sally, who have the same values of `bmi` and `costprob`, but Harry eats one more serving than Sally does. We noted that Harry is predicted by the model to have a smaller (i.e. healthier) `genh` response than Sally.
 - So a negative coefficient here means that higher values of `veg_day` are associated with more of the probability distribution falling in lower values of `genh`.
 - We usually don't interpret this slope (on the log odds scale) directly, but rather exponentiate it.

27.6.7 Exponentiating the Slope Coefficient to facilitate Interpretation

We can compute the odds ratio associated with `veg_day` and its confidence interval as follows...

```
exp(coef(m1))
```

```
veg_day  
0.8313311
```

```
exp(confint(m1))
```

Waiting for profiling to be done...

```
2.5 %    97.5 %  
0.7963573 0.8673534
```

- So, if Harry eats one more serving of vegetables than Sally, our model predicts that Harry will have 83.1% of the odds of Sally of having a larger `genh` score. That means that Harry is likelier to have a smaller `genh` score.
 - Since `genh` gets larger as a person's general health gets worse (moves from Excellent towards Poor), this means that since Harry is predicted to have smaller odds of a larger `genh` score, he is also predicted to have smaller odds of worse general health.
 - Our 95% confidence interval around that estimated odds ratio of 0.831 is (0.796, 0.867). Since that interval is entirely below 1, the odds of having the larger (worse) `genh` for Harry are detectably lower than the odds for Sally.
 - So, an increase in `veg_day` is associated with smaller (better) `genh` scores.

27.6.8 Comparison to a Null Model

We can fit a model with intercepts only to assess the predictive value of `veg_day` in our model `m1`, using the `anova` function.

```
m0 <- polr(genh ~ 1, data = sm1)
anova(m1, m0)
```

Likelihood ratio tests of ordinal regression models

```
Response: genh
      Model Resid. df Resid. Dev   Test      Df LR stat. Pr(Chi)
1         1      5390   15744.89              1 75.04297      0
2 veg_day      5389   15669.85 1 vs 2      1 75.04297      0
```

We could also compare model `m1` to the null model `m0` with AIC or BIC.

```
AIC(m1, m0)
```

```
      df      AIC
m1    5 15679.85
m0    4 15752.89
```

```
BIC(m1,m0)
```

```
      df      BIC
m1    5 15712.81
m0    4 15779.26
```

Model `m1` looks like the better choice so far.

27.7 The Assumption of Proportional Odds

Let us calculate the odds for all levels of `genh` if a person eats two servings of vegetables. First, we'll get the probabilities, in another way, to demonstrate how to do so...

```
(prob.2 <- exp(m1$zeta - 2*m1$coefficients)/(1 + exp(m1$zeta - 2*m1$coefficients)))
```

```
1-E|2_VG  2_VG|3_G  3_G|4_F  4_F|5_P
0.1522351 0.4907471 0.8005376 0.9463240
```

```
(prob.1 <- exp(m1$zeta - 1*m1$coefficients)/(1 + exp(m1$zeta - 1*m1$coefficients)))
```

```
1-E|2_VG  2_VG|3_G  3_G|4_F  4_F|5_P
0.1298931 0.4447902 0.7694008 0.9361293
```

Now, we'll calculate the odds, first for a subject eating two servings:

```
(odds.2 = prob.2/(1-prob.2))
```

```
1-E|2_VG  2_VG|3_G  3_G|4_F  4_F|5_P
0.1795724 0.9636607 4.0134766 17.6303153
```

And here are the odds, for a subject eating one serving per day:

```
(odds.1 = prob.1/(1-prob.1))
```

```
1-E|2_VG  2_VG|3_G  3_G|4_F  4_F|5_P
0.1492841 0.8011211 3.3365277 14.6566285
```

Now, let's take the ratio of the odds for someone who eats two servings over the odds for someone who eats one.

```
odds.2/odds.1
```

```
1-E|2_VG 2_VG|3_G 3_G|4_F 4_F|5_P
1.20289  1.20289  1.20289  1.20289
```

They are all the same. The odds ratios are equal, which means they are proportional. For any level of **genh**, the estimated odds that a person who eats 2 servings has better (lower) **genh** is about 1.2 times the odds for someone who eats one serving. Those who eat more vegetables have higher odds of better (lower) **genh**. Less than 1 means lower odds, and more than 1 means greater odds.

Now, let's take the log of the odds ratios:

```
log(odds.2/odds.1)
```

```
1-E|2_VG  2_VG|3_G  3_G|4_F  4_F|5_P  
0.1847272 0.1847272 0.1847272 0.1847272
```

That should be familiar. It is the slope coefficient in the model summary, without the minus sign. R tacks on a minus sign so that higher levels of predictors correspond to the ordinal outcome falling in the higher end of its scale.

If we exponentiate the slope estimated by R (-0.1847), we get 0.83. If we have two people, and A eats one more serving of vegetables on average than B, then the estimated odds of A having a higher ‘genh’ (i.e. worse general health) are 83% as high as B’s.

27.7.1 Testing the Proportional Odds Assumption

One way to test the proportional odds assumption is to compare the fit of the proportional odds logistic regression to a model that does not make that assumption. A natural candidate is a **multinomial logit** model, which is typically used to model unordered multi-categorical outcomes, and fits a slope to each level of the **genh** outcome in this case, as opposed to the proportional odds logit, which fits only one slope across all levels.

Since the proportional odds logistic regression model is nested in the multinomial logit, we can perform a likelihood ratio test. To do this, we first fit the multinomial logit model, with the **multinom** function from the **nnet** package.

```
(m1_multi <- multinom(genh ~ veg_day, data = sm1))
```

```
# weights:  15 (8 variable)  
initial  value 8681.308100  
iter   10 value 7890.985276  
final   value 7835.248471  
converged
```

Call:

```
multinom(formula = genh ~ veg_day, data = sm1)
```

Coefficients:

```
      (Intercept)      veg_day  
2_VG    0.9791063 -0.09296694  
3_G     1.0911990 -0.19260067  
4_F     0.5708594 -0.31080687
```

```
5_P      -0.3583310 -0.34340619
```

```
Residual Deviance: 15670.5
```

```
AIC: 15686.5
```

The multinomial logit fits four intercepts and four slopes, for a total of 8 estimated parameters. The proportional odds logit, as we've seen, fits four intercepts and one slope, for a total of 5. The difference is 3, and we use that number in the sequence below to build our test of the proportional odds assumption.

```
LL_1 <- logLik(m1)
LL_1m <- logLik(m1_multi)
(G <- -2 * (LL_1[1] - LL_1m[1]))
```

```
[1] -0.6488392
```

```
pchisq(G, 3, lower.tail = FALSE)
```

```
[1] 1
```

The p value is very large, so it indicates that the proportional odds model fits about as well as the more complex multinomial logit. A large p value here isn't always the best way to assess the proportional odds assumption, but it does provide some evidence of model adequacy.

27.8 Can model m1 be fit using rms tools?

Yes.

```
d <- datadist(sm1)
options(datadist = "d")
m1_lrm <- lrm(genh ~ veg_day, data = sm1, x = T, y = T)

m1_lrm
```

Logistic Regression Model

```
lrm(formula = genh ~ veg_day, data = sm1, x = T, y = T)
```

Frequencies of Responses

```
1-E 2_VG 3_G 4_F 5_P
822 1805 1667 801 299
```

		Model Likelihood	Discrimination	Rank Discrim.
		Ratio Test	Indexes	Indexes
Obs	5394	LR chi2	R2	C
max deriv	4e-07	d.f.	R2(1,5394)	Dxy
		Pr(> chi2) <0.0001	R2(1,4995)	gamma
			Brier	tau-a
	Coef	S.E.	Wald Z	Pr(> Z)
y>=2_VG	2.0866	0.0584	35.76	<0.0001
y>=3_G	0.4065	0.0498	8.16	<0.0001
y>=4_F	-1.0202	0.0521	-19.58	<0.0001
y>=5_P	-2.5002	0.0710	-35.22	<0.0001
veg_day	-0.1847	0.0218	-8.48	<0.0001

The model has a small p value (remember the large sample size) but nonetheless appears very weak, with a Nagelkerke R^2 of 0.015, and a C statistic of 0.555.

```
summary(m1_lrm)
```

Effects				Response : genh			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
veg_day	1.21	2.36	1.15	-0.21243	0.025051	-0.26153	-0.16334
Odds Ratio	1.21	2.36	1.15	0.80861	NA	0.76987	0.84931

A change from 1.21 to 2.36 servings in `veg_day` is associated with an odds ratio of 0.81, with 95% confidence interval (0.77, 0.85). Since these values are all below 1, we have a clear indication of a statistically detectable effect of `veg_day` with higher `veg_day` associated with lower `genh`, which means, in this case, better health.

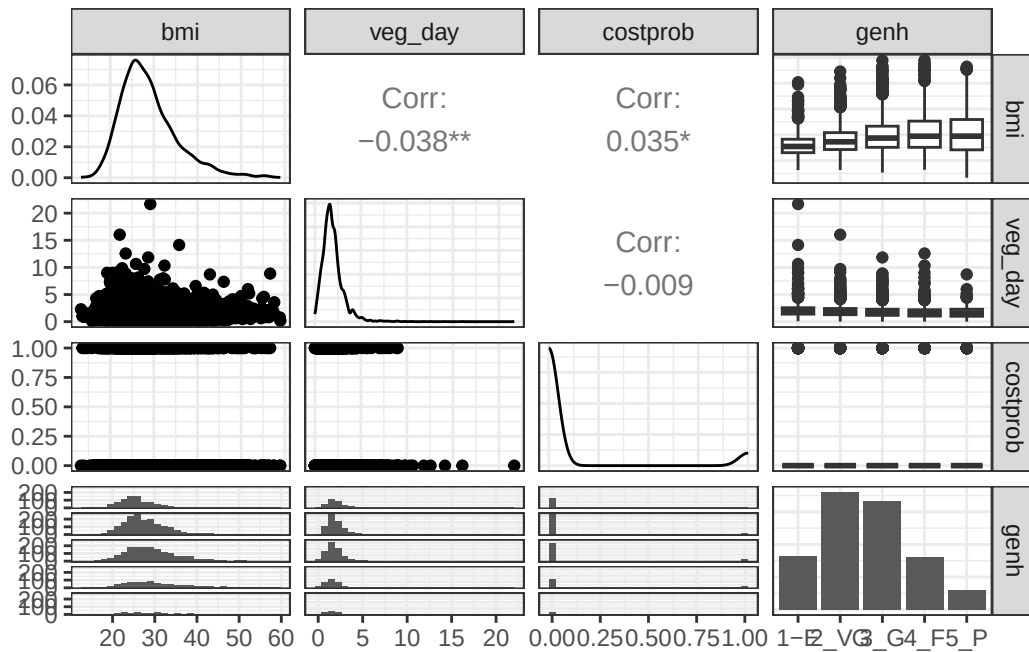
There is also a tool in `rms` called `orm` which may be used to fit a wide array of ordinal regression models. I suggest you read Frank Harrell's book on *Regression Modeling Strategies* if you want to learn more.

27.9 Building a Three-Predictor Model

Now, we'll model `genh` using `veg_day`, `bmi` and `costprob`.

27.9.1 Scatterplot Matrix

```
GGally::ggpairs(sm1 |>
  select(bmi, veg_day, costprob, genh))
```



We might choose to plot the `costprob` data as a binary factor, rather than the raw 0-1 numbers included above, but not at this time.

27.9.2 Our Three-Predictor Model, m2

```
m2 <- polr(genh ~ veg_day + bmi + costprob, data = sm1)

summary(m2)
```

Re-fitting to get Hessian

Call:

```
polr(formula = genh ~ veg_day + bmi + costprob, data = sm1)
```


Coefficients:

	Value	Std. Error	t value
veg_day	-0.17130	0.021783	-7.864
bmi	0.06673	0.003855	17.311
costprob	0.96825	0.084871	11.409

Intercepts:

	Value	Std. Error	t value
1-E 2_VG	-0.1252	0.1229	-1.0183
2_VG 3_G	1.6358	0.1234	13.2572
3_G 4_F	3.1534	0.1294	24.3755
4_F 5_P	4.6881	0.1412	33.1928

Residual Deviance: 15229.24

AIC: 15243.24

This model contains four intercepts (to cover the five `genh` categories) and three slopes (one each for `veg_day`, `bmi` and `costprob`.)

27.9.3 Does the three-predictor model outperform `m1`?

```
anova(m1, m2)
```

Likelihood ratio tests of ordinal regression models

Response: `genh`

	Model	Resid. df	Resid. Dev	Test	Df	LR stat.	Pr(Chi)
1	<code>veg_day</code>	5389	15669.85				
2	<code>veg_day + bmi + costprob</code>	5387	15229.24	1 vs 2	2	440.6041	0

It looks like the fit improves as we move from model 1 to model 2. The AIC and BIC are also better for the three-predictor model than they were for the model with `veg_day` alone.

```
AIC(m1, m2)
```

	df	AIC
<code>m1</code>	5	15679.85
<code>m2</code>	7	15243.24

```
BIC(m1, m2)
```

```
      df      BIC
m1  5 15712.81
m2  7 15289.40
```

27.9.4 Wald tests for individual predictors

To obtain the appropriate Wald tests, we can use `lrm` to fit the model instead.

```
d <- datadist(sm1)
options(datadist = "d")
m2_lrm <- lrm(genh ~ veg_day + bmi + costprob,
              data = sm1, x = T, y = T)
m2_lrm
```

Logistic Regression Model

```
lrm(formula = genh ~ veg_day + bmi + costprob, data = sm1, x = T,
     y = T)
```

Frequencies of Responses

```
1-E 2_VG  3_G  4_F  5_P
822 1805 1667  801  299
```

		Model Likelihood Ratio Test	Discrimination Indexes	Rank Discrim. Indexes
Obs	5394	LR chi2 515.65	R2 0.096	C 0.629
max deriv	4e-09	d.f. 3	R2(3,5394)0.091	Dxy 0.258
		Pr(> chi2) <0.0001	R2(3,4995)0.098	gamma 0.258
			Brier 0.231	tau-a 0.192

	Coef	S.E.	Wald Z	Pr(> Z)
y>=2_VG	0.1252	0.1229	1.02	0.3085
y>=3_G	-1.6358	0.1234	-13.26	<0.0001
y>=4_F	-3.1534	0.1294	-24.38	<0.0001
y>=5_P	-4.6881	0.1412	-33.19	<0.0001
veg_day	-0.1713	0.0218	-7.86	<0.0001

```
bmi          0.0667 0.0039 17.31 <0.0001
costprob     0.9683 0.0849 11.41 <0.0001
```

It appears that each of the added predictors (`bmi` and `costprob`) adds statistically detectable value to the model.

27.9.5 A Cross-Tabulation of Predictions?

```
addmargins(table(predict(m2), sm1$genh))
```

	1-E	2_VG	3_G	4_F	5_P	Sum
1-E	6	5	4	1	0	16
2_VG	686	1295	950	388	141	3460
3_G	128	495	672	374	135	1804
4_F	1	9	38	36	20	104
5_P	1	1	3	2	3	10
Sum	822	1805	1667	801	299	5394

At least the `m2` model predicted that a few of the cases will fall in the Fair and Poor categories, but still, this isn't impressive.

27.9.6 Interpreting the Effect Sizes

We can do this in two ways:

- By exponentiating the `polr` output, which shows the effect of increasing each predictor by a single unit
 - Increasing `veg_day` by 1 serving while holding the other predictors constant is associated with reducing the odds (by a factor of 0.84 with 95% CI 0.81, 0.88) of higher values of `genh`: hence increasing `veg_day` is associated with increasing the odds of a response indicating better health.
 - Increasing `bmi` by 1 kg/m² while holding the other predictors constant is associated with increasing the odds (by a factor of 1.07 with 95% CI 1.06, 1.08) of higher values of `genh`: hence increasing `bmi` is associated with reducing the odds of a response indicating better health.

- Increasing `costprob` from 0 to 1 while holding the other predictors constant is associated with an increase (by a factor of 2.63 with 95% CI 2.23, 3.11) of a higher `genh` value. Since higher `genh` values indicate worse health, those with `costprob` = 1 are modeled to have generally worse health.

```
exp(coef(m2))
```

```
      veg_day      bmi costprob
0.8425722 1.0690045 2.633356
```

```
exp(confint(m2))
```

Waiting for profiling to be done...

Re-fitting to get Hessian

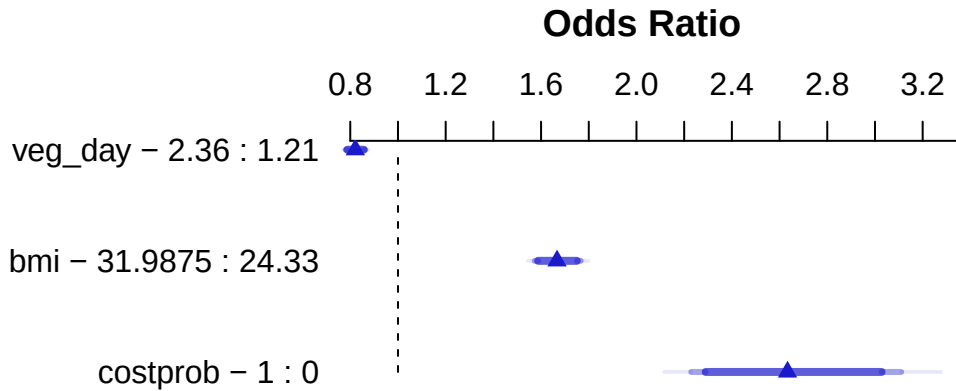
```
      2.5 %   97.5 %
veg_day 0.8071346 0.879096
bmi      1.0609722 1.077126
costprob 2.2301783 3.110633
```

- Or by looking at the summary provided by `lrm`, which like all such summaries produced by `rms` shows the impact of moving from the 25th to the 75th percentile on all continuous predictors.

```
summary(m2_lrm)
```

	Effects			Response : genh				
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95	
veg_day	1.21	2.360	1.1500	-0.19699	0.025051	-0.24609	-0.14789	
Odds Ratio	1.21	2.360	1.1500	0.82120	NA	0.78185	0.86252	
bmi	24.33	31.988	7.6575	0.51097	0.029515	0.45312	0.56882	
Odds Ratio	24.33	31.988	7.6575	1.66690	NA	1.57320	1.76620	
costprob	0.00	1.000	1.0000	0.96825	0.084871	0.80191	1.13460	
Odds Ratio	0.00	1.000	1.0000	2.63330	NA	2.22980	3.10990	

```
plot(summary(m2_lrm))
```



27.9.7 Quality of the Model Fit

Model `m2`, as we can see from the `m2_lrm` output, is still weak, with a Nagelkerke R^2 of 0.10, and a C statistic of 0.63.

27.9.8 Validating the Summary Statistics in `m2_lrm`

```
set.seed(43203); validate(m2_lrm)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper
Dxy	0.2583	0.2625	0.2579	0.0046	0.2537	0.2329	0.2730
R2	0.0964	0.0989	0.0960	0.0029	0.0935	0.0795	0.1074
Intercept	0.0000	0.0000	-0.0023	0.0023	-0.0023	-0.0736	0.0666
Slope	1.0000	1.0000	0.9876	0.0124	0.9876	0.9056	1.0987
Emax	0.0000	0.0000	0.0118	-0.0118	0.0118	-0.0008	0.0271
D	0.0954	0.0980	0.0950	0.0030	0.0924	0.0778	0.1069

U	-0.0004	-0.0004	-1.5149	1.5146	-1.5149	-1.5159	-
1.5138							
Q	0.0958	0.0984	1.6099	-1.5115	1.6073	1.5932	1.6213
B	0.2314	0.2308	0.2315	-0.0007	0.2321	0.2283	0.2351
g	0.6199	0.6278	0.6182	0.0096	0.6103	0.5593	0.6562
gp	0.1434	0.1429	0.1430	-0.0001	0.1436	0.1322	0.1768
	n						
Dxy	40						
R2	40						
Intercept	40						
Slope	40						
Emax	40						
D	40						
U	40						
Q	40						
B	40						
g	40						
gp	40						

As in our work with binary logistic regression, we can convert the index-corrected Dxy to an index-corrected C with $C = 0.5 + (Dxy/2)$. Both the R^2 and C statistics are pretty consistent with what we saw above.

27.9.9 Testing the Proportional Odds Assumption

Again, we'll fit the analogous multinomial logit model, with the `multinom` function from the `nnet` package.

```
(m2_multi <- multinom(genh ~ veg_day + bmi + costprob,
                      data = sm1))
```

```
# weights:  25 (16 variable)
initial  value 8681.308100
iter   10 value 8025.745934
iter   20 value 7605.878993
final   value 7595.767250
converged
```

Call:

```
multinom(formula = genh ~ veg_day + bmi + costprob, data = sm1)
```

Coefficients:

	(Intercept)	veg_day	bmi	costprob
2_VG	-0.9126285	-0.0905958	0.06947231	0.3258568
3_G	-2.1886806	-0.1893454	0.11552563	1.0488262
4_F	-3.4095145	-0.3056028	0.13679908	1.4422074
5_P	-4.2629564	-0.3384199	0.13178846	1.8612088

Residual Deviance: 15191.53

AIC: 15223.53

The multinomial logit fits four intercepts and 12 slopes, for a total of 16 estimated parameters. The proportional odds logit in model `m2`, as we've seen, fits four intercepts and three slopes, for a total of 7. The difference is 9, and we use that number in the sequence below to build our test of the proportional odds assumption.

```
LL_2 <- logLik(m2)
LL_2m <- logLik(m2_multi)
(G <- -2 * (LL_2[1] - LL_2m[1]))
```

```
[1] 37.70952
```

```
pchisq(G, 9, lower.tail = FALSE)
```

```
[1] 1.965186e-05
```

The resulting small p value suggests a problem with proportional odds assumption. When this happens, I suggest you build the following plot of score residuals:

```
par(mfrow = c(2,2))
resid(m2_lrm, 'score.binary', pl=TRUE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 2
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 1
```

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 0

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: There are other near singularities as well. 1

Warning in predLoess(object\$y, object\$x, newx = if (is.null(newdata)) object\$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : pseudoinverse used at 2

Warning in predLoess(object\$y, object\$x, newx = if (is.null(newdata)) object\$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : neighborhood radius 1

Warning in predLoess(object\$y, object\$x, newx = if (is.null(newdata)) object\$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : reciprocal condition
number 0

Warning in predLoess(object\$y, object\$x, newx = if (is.null(newdata)) object\$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : There are other near
singularities as well. 1

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 2

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 1

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 0

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: There are other near singularities as well. 1

Warning in predLoess(object\$y, object\$x, newx = if (is.null(newdata)) object\$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : pseudoinverse used at 2


```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : neighborhood radius 1
```

```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : reciprocal condition
number 0
```

```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : There are other near
singularities as well. 1
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 2
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 1
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 0
```

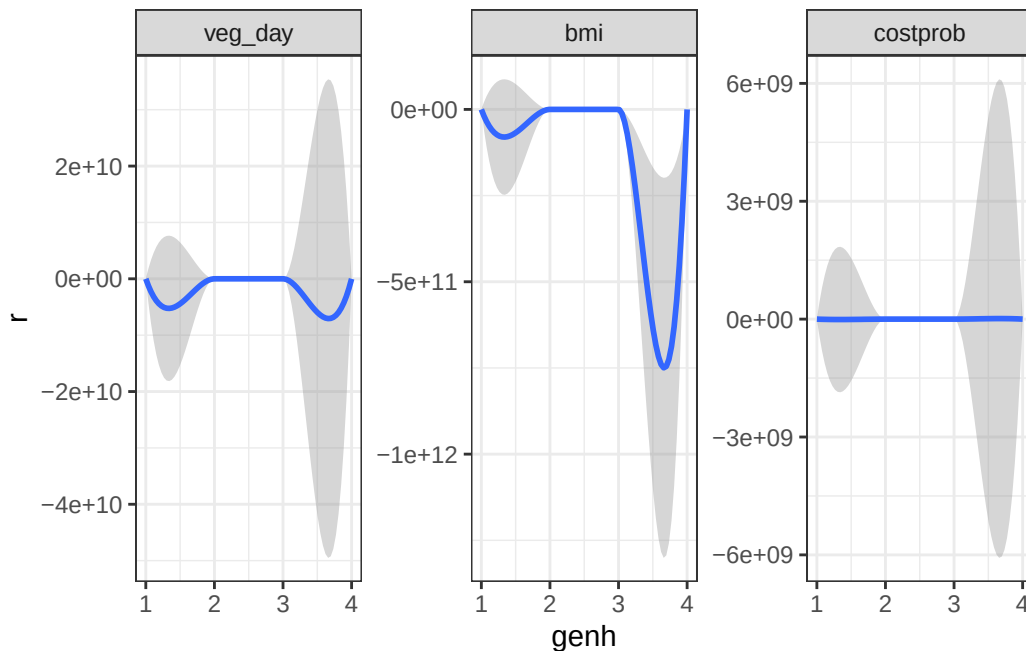
```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: There are other near singularities as well. 1
```

```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : pseudoinverse used at 2
```

```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : neighborhood radius 1
```

```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : reciprocal condition
number 0
```

```
Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object))), : There are other near
singularities as well. 1
```



```
par(mfrow= c(1,1))
```

From this plot, `bmi` (especially) and `costprob` vary as we move from the Very Good toward the Poor cutpoints, relative to `veg_day`, which is more stable.

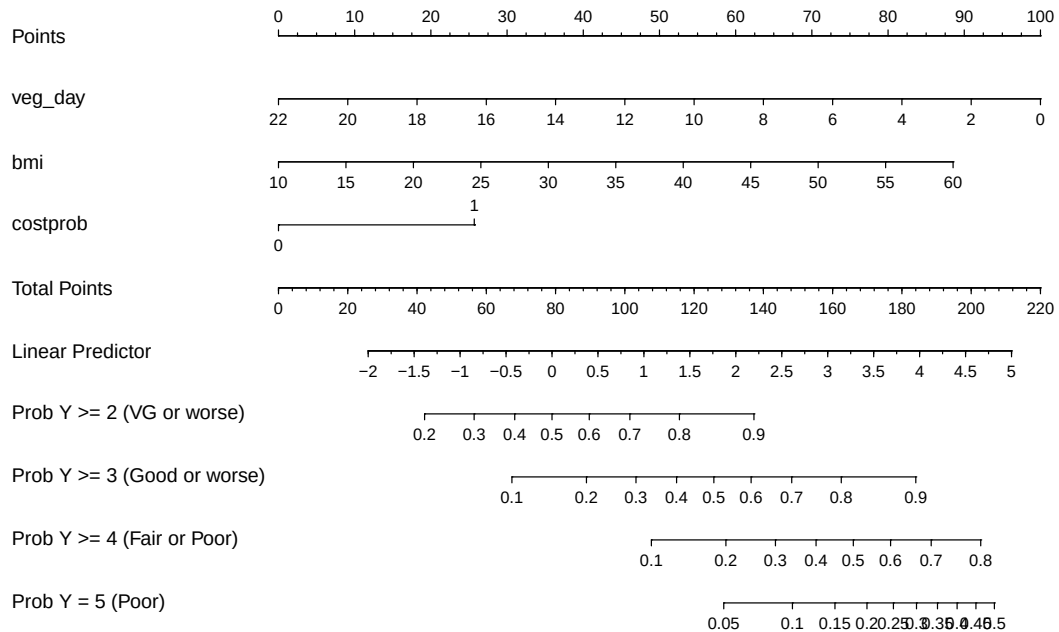
27.9.10 Plotting the Fitted Model

27.9.10.1 Nomogram

```
fun.ge3 <- function(x) plogis(x - m2_lrm$coef[1] + m2_lrm$coef[2])
fun.ge4 <- function(x) plogis(x - m2_lrm$coef[1] + m2_lrm$coef[3])
fun.ge5 <- function(x) plogis(x - m2_lrm$coef[1] + m2_lrm$coef[4])

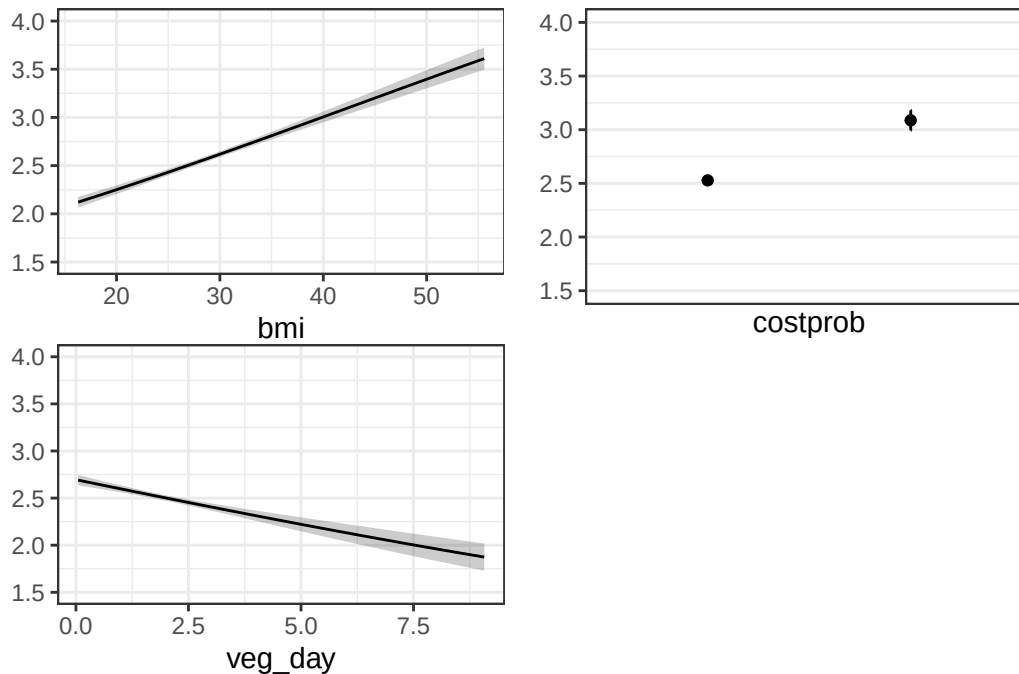
plot(nomogram(m2_lrm, fun=list('Prob Y >= 2 (VG or worse)' = plogis,
                              'Prob Y >= 3 (Good or worse)' = fun.ge3,
```

```
'Prob Y >= 4 (Fair or Poor)' = fun.ge4,  
'Prob Y = 5 (Poor)' = fun.ge5)))
```



27.9.10.2 Using Predict and showing mean prediction on 1-5 scale

```
ggplot(Predict(m2_lrm, fun = Mean(m2_lrm, code = TRUE)))
```



The nomogram and Predict results would be more interesting, of course, if we included a spline or interaction term. Let's do that in model `m3_lrm`, and also add the `incomegroup` information.

27.10 A Larger Model, including income group

```
m3_lrm <- lrm(gen_n ~ rcs(veg_day,3) + rcs(bmi, 4) +
              incomegroup + catg(costprob) +
              bmi %ia% costprob,
              data = sm1, x = T, y = T)

m3_lrm
```

Logistic Regression Model

```
lrm(formula = gen_n ~ rcs(veg_day, 3) + rcs(bmi, 4) + incomegroup +
     catg(costprob) + bmi %ia% costprob, data = sm1, x = T, y = T)
```

Frequencies of Responses

1	2	3	4	5
822	1805	1667	801	299

		Model Likelihood	Discrimination	Rank Discrim.
		Ratio Test	Indexes	Indexes
Obs	5394	LR chi2	R2	C
max deriv	2e-06	d.f.	R2(14,5394)	Dxy
		Pr(> chi2)	R2(14,4995)	gamma
			Brier	tau-a
			0.214	0.291

	Coef	S.E.	Wald Z	Pr(> Z)
y>=2	3.7535	0.4852	7.74	<0.0001
y>=3	1.8717	0.4838	3.87	0.0001
y>=4	0.2035	0.4831	0.42	0.6737
y>=5	-1.4386	0.4846	-2.97	0.0030
veg_day	-0.2602	0.0633	-4.11	<0.0001
veg_day'	0.1756	0.0693	2.53	0.0113
bmi	-0.0325	0.0203	-1.60	0.1086
bmi'	0.5422	0.0989	5.48	<0.0001
bmi''	-1.4579	0.2663	-5.47	<0.0001
incomegroup=10-14K	0.2445	0.1705	1.43	0.1516
incomegroup=15-19K	-0.2626	0.1582	-1.66	0.0969
incomegroup=20-24K	-0.6434	0.1501	-4.29	<0.0001
incomegroup=25-34K	-0.7427	0.1459	-5.09	<0.0001
incomegroup=35-49K	-1.1621	0.1415	-8.21	<0.0001
incomegroup=50-74K	-1.4579	0.1418	-10.28	<0.0001
incomegroup=75K+	-1.8592	0.1361	-13.66	<0.0001
costprob=1	1.4576	0.3528	4.13	<0.0001
bmi * costprob	-0.0259	0.0116	-2.24	0.0250

Another option here would have been to consider building `incomegroup` as a scored variable, with an order on its own, but I won't force that here. Here's the `polr` version...

```
m3 <- polr(genh ~ rcs(veg_day,3) + rcs(bmi, 4) +
  incomegroup + costprob +
  bmi %ia% costprob, data = sm1)
```

27.10.1 Cross-Tabulation of Predicted/Observed Classifications

```
addmargins(table(predict(m3), sm1$genh))
```

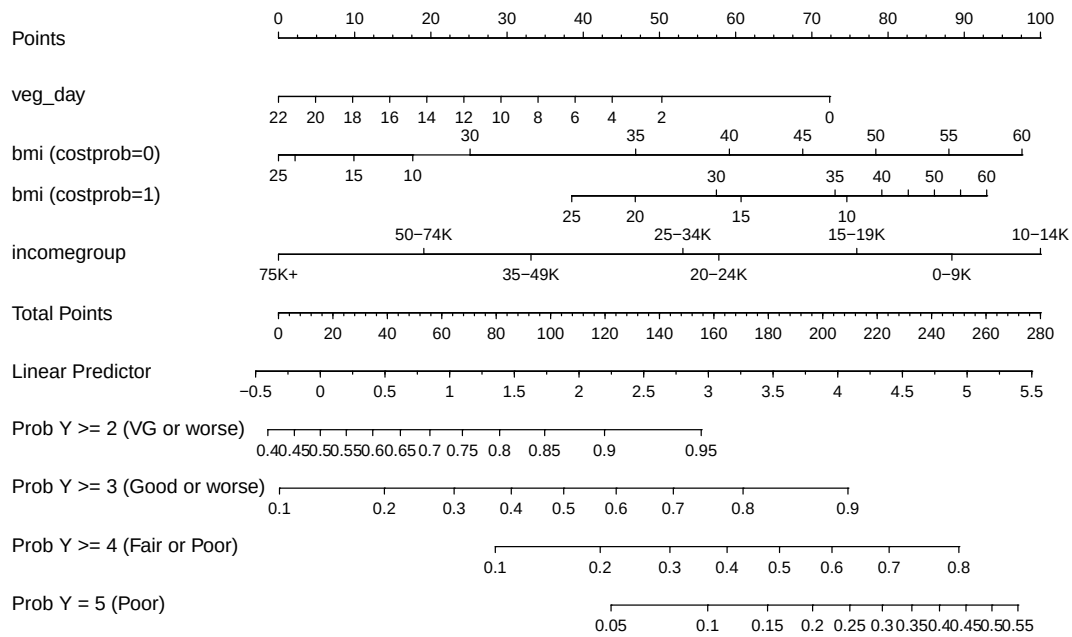
	1-E	2_VG	3_G	4_F	5_P	Sum
1-E	3	2	0	0	0	5
2_VG	642	1200	815	221	49	2927
3_G	170	565	754	468	182	2139
4_F	7	37	96	108	65	313
5_P	0	1	2	4	3	10
Sum	822	1805	1667	801	299	5394

This model predicts more Fair results, but still far too many Very Good with no Excellent at all.

27.10.2 Nomogram

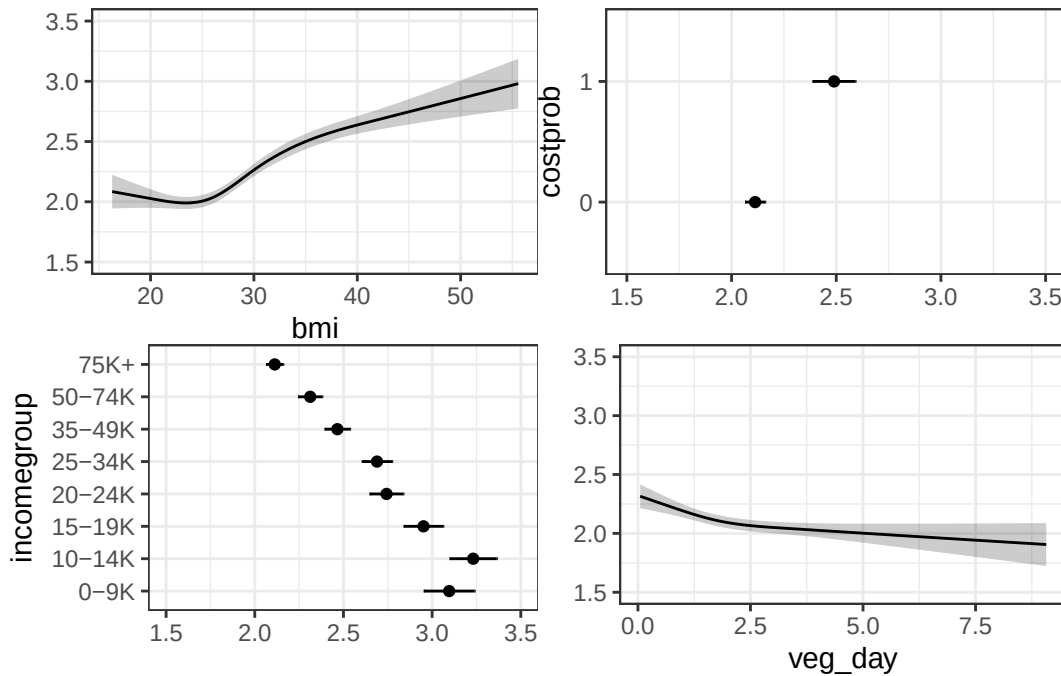
```
fun.ge3 <- function(x) plogis(x - m3_lrm$coef[1] + m3_lrm$coef[2])
fun.ge4 <- function(x) plogis(x - m3_lrm$coef[1] + m3_lrm$coef[3])
fun.ge5 <- function(x) plogis(x - m3_lrm$coef[1] + m3_lrm$coef[4])

plot(nomogram(m3_lrm, fun=list('Prob Y >= 2 (VG or worse)' = plogis,
                              'Prob Y >= 3 (Good or worse)' = fun.ge3,
                              'Prob Y >= 4 (Fair or Poor)' = fun.ge4,
                              'Prob Y = 5 (Poor)' = fun.ge5)))
```



27.10.3 Using Predict and showing mean prediction on 1-5 scale

```
ggplot(Predict(m3_lrm, fun = Mean(m3_lrm, code = TRUE)))
```



Here, we're plotting the mean score on the 1-5 `gen_n` scale.

27.10.4 Validating the Summary Statistics in `m3_lrm`

```
set.seed(43221); validate(m3_lrm)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper
Dxy	0.3915	0.3928	0.3899	0.0029	0.3886	0.3671	0.4081
R2	0.2093	0.2112	0.2071	0.0041	0.2053	0.1850	0.2258
Intercept	0.0000	0.0000	0.0019	-0.0019	0.0019	-0.0954	0.0768
Slope	1.0000	1.0000	0.9871	0.0129	0.9871	0.9234	1.0657
Emax	0.0000	0.0000	0.0107	-0.0107	0.0107	-0.0021	0.0304
D	0.2205	0.2227	0.2179	0.0049	0.2156	0.1916	0.2399
U	-0.0004	-0.0004	-1.4667	1.4663	-1.4667	-1.4695	-
1.4630							
Q	0.2209	0.2231	1.6845	-1.4614	1.6823	1.6571	1.7086
B	0.2137	0.2134	0.2142	-0.0007	0.2145	0.2086	0.2190
g	1.0363	1.0410	1.0276	0.0133	1.0229	0.9582	1.0867
gp	0.2262	0.2237	0.2244	-0.0007	0.2268	0.2134	0.2887
n							
Dxy	40						

R2	40
Intercept	40
Slope	40
Emax	40
D	40
U	40
Q	40
B	40
g	40
gp	40

Still not very impressive, but much better than where we started. It's not crazy to suggest that in new data, we might expect a Nagelkerke R^2 of 0.205 and a C statistic of $0.5 + (0.3886/2) = 0.6943$.

27.11 References for this Chapter

1. Some of the material here is adapted from <http://stats.idre.ucla.edu/r/dae/ordinal-logistic-regression/>.
2. I also found great guidance at <http://data.library.virginia.edu/fitting-and-interpreting-a-proportional-odds-model/>
3. Other parts are based on the work of Jeffrey S. Simonoff (2003) *Analyzing Categorical Data* in Chapter 10. Related data and R code are available at <http://people.stern.nyu.edu/jsimonof/AnalCatData/Splus/>.
4. Another good source for a simple example is <https://onlinecourses.science.psu.edu/stat504/node/177>.
5. Also helpful is <https://onlinecourses.science.psu.edu/stat504/node/178> which shows a more complex example nicely.

28 Multinomial Logistic Regression

28.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(knitr)
library(MASS)
library(nnet)
library(tidyverse)

theme_set(theme_bw())
```

28.1.1 Data Load

```
authorship <- read_csv("data/authorship.csv", show_col_types = FALSE)
```

28.2 The Authorship Example

This example is based on the work of Jeffrey S. Simonoff (2003) *Analyzing Categorical Data* in Chapter 10. Related data and R code are available at [this link](#). Meanwhile, the data set and analysis are based on the work of Peng RD and Hengartner NW (2002) Quantitative analysis of literary styles, *The American Statistician*, 56, 175-185.

The `authorship.csv` data file contains 841 rows. Each row describes a block of text that contains 1700 total words from one of several books by four authors: Jane Austen (samples from 7 books), Jack London (6 books), John Milton (2 books), or William Shakespeare (12 books). The data include counts within the blocks of text of 69 function words, such as “a”, “by”, “no”, “that” and “with”. The goal of our analysis, mirroring that of Simonoff, will be to use the incidence of these function words to build a model that distinguishes the authors.

```
authorship$Author <- factor(authorship$Author,
  levels = c("Shakespeare", "Austen", "London", "Milton"))

authorship
```

```
# A tibble: 841 x 71
  BookID Author      a  all  also  an  and  any  are  as  at  be
  <dbl> <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1      1  Austen    46   12    0    3   66    9    4   16   13   13
2      1  Austen    35   10    0    7   44    4    3   18   16    9
3      1  Austen    46    2    0    3   40    1   13   11    9   23
4      1  Austen    40    7    0    4   64    3    3   20   13   20
5      1  Austen    29    5    0    6   52    5   14   17    6   16
6      1  Austen    27    8    0    3   42    2   15   11   14   12
7      1  Austen    34    8    0   15   44    2    6   16   14   11
8      1  Austen    38    6    1    2   67    3    6   17    4   21
9      1  Austen    34   12    0    5   50    2    8    7   13    7
10     1  Austen    54    8    0    6   44    4    8   13   17   14

# i 831 more rows
# i 59 more variables: been <dbl>, but <dbl>, by <dbl>, can <dbl>, do <dbl>,
# down <dbl>, even <dbl>, every <dbl>, `for` <dbl>, from <dbl>, had <dbl>,
# has <dbl>, have <dbl>, her <dbl>, his <dbl>, `if` <dbl>, `in` <dbl>,
# into <dbl>, is <dbl>, it <dbl>, its <dbl>, may <dbl>, more <dbl>,
# must <dbl>, my <dbl>, no <dbl>, not <dbl>, now <dbl>, of <dbl>, on <dbl>,
# one <dbl>, only <dbl>, or <dbl>, our <dbl>, should <dbl>, so <dbl>, ...
```

To-morrow, and to-morrow, and to-morrow, Creeps in this petty pace from day to day, To the last syllable of recorded time; And all our yesterdays have lighted fools The way to dusty death. Out, out, brief candle! Life's but a walking shadow, a poor player, That struts and frets his hour upon the stage, And then is heard no more. It is a tale Told by an idiot, full of sound and fury, Signifying nothing.

28.3 Focus on 11 key words

Again, following Simonoff, we will focus on 11 words from the set of 69 potential predictors in the data, specifically...

- “be”, “been”, “had”, “it”, “may”, “not”, “on”, “the”, “upon”, “was” and “which”

```
auth2 <- authorship |>
  select(Author, BookID, be, been, had, it, may, not,
         on, the, upon, was, which)

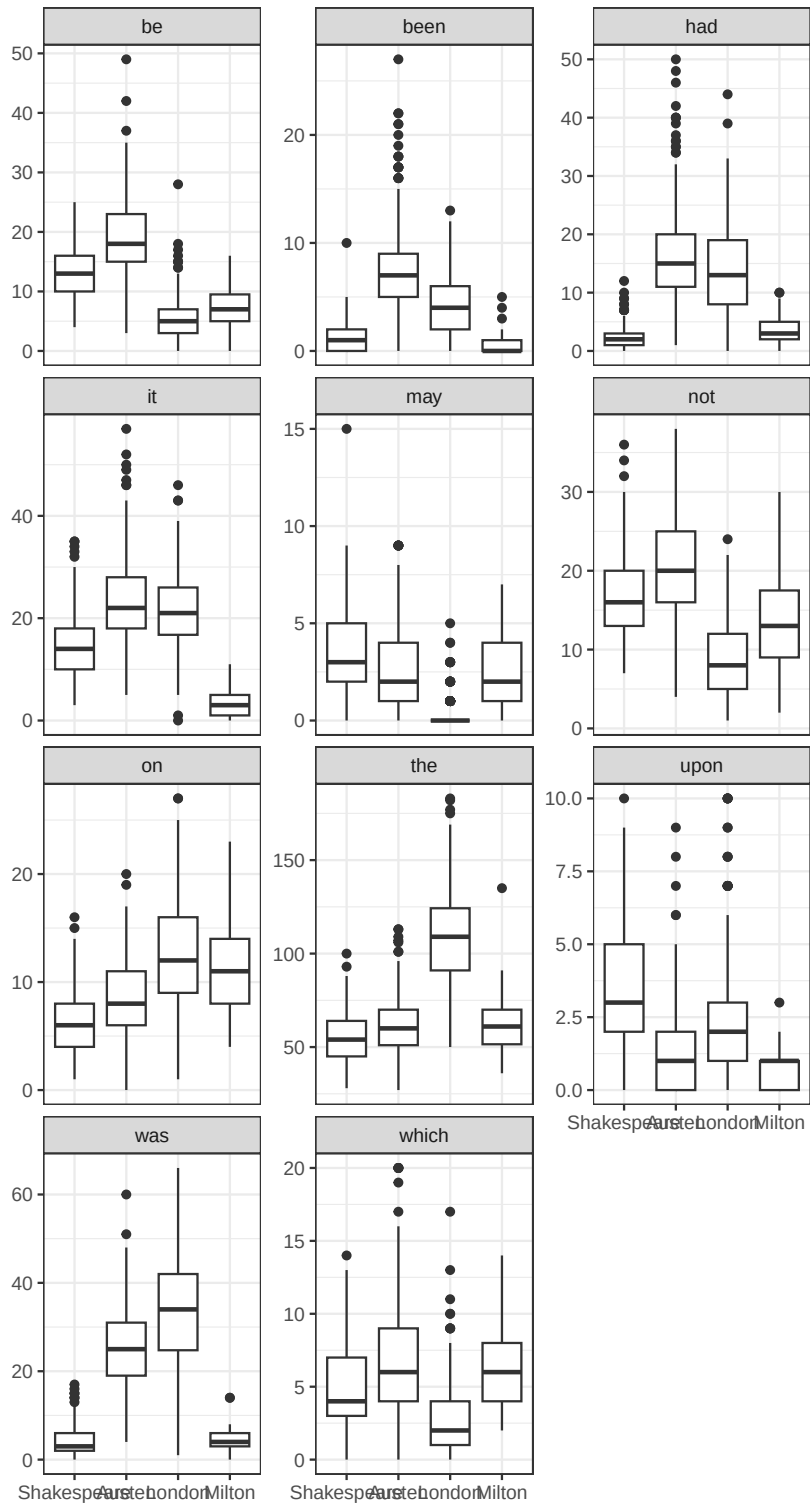
auth2.long <- auth2 |>
  gather("word", "n", 3:13)

auth2.long
```

```
# A tibble: 9,251 x 4
  Author BookID word      n
  <fct>   <dbl> <chr> <dbl>
1 Austen     1 be      13
2 Austen     1 be       9
3 Austen     1 be      23
4 Austen     1 be      20
5 Austen     1 be      16
6 Austen     1 be      12
7 Austen     1 be      11
8 Austen     1 be      21
9 Austen     1 be       7
10 Austen    1 be      14
# i 9,241 more rows
```

28.3.1 Side by Side Boxplots

```
ggplot(auth2.long, aes(x = Author, y = n)) +
  geom_boxplot() +
  facet_wrap(~ word, ncol = 3, scales = "free_y") +
  labs(x = "", y = "")
```



Oh! do not attack me with your watch. A watch is always too fast or too slow. I cannot be dictated to by a watch.

28.4 A Multinomial Logistic Regression Model

Let's start with a multinomial model to predict Author on the basis of these 11 key predictors, using the `multinom` function from the `nnet` package.

```
authnom1 <- multinom(Author ~ be + been + had + it + may + not + on +  
                      the + upon + was + which, data=authorship,  
                      maxit=200)
```

```
# weights: 52 (36 variable)  
initial value 1165.873558  
iter 10 value 293.806160  
iter 20 value 273.554538  
iter 30 value 192.309644  
iter 40 value 71.091334  
iter 50 value 48.419335  
iter 60 value 46.808141  
iter 70 value 46.184752  
iter 80 value 46.026162  
iter 90 value 45.932823  
iter 100 value 45.897793  
iter 110 value 45.868017  
iter 120 value 45.863256  
final value 45.863228  
converged
```

```
summary(authnom1)
```

Call:

```
multinom(formula = Author ~ be + been + had + it + may + not +  
          on + the + upon + was + which, data = authorship, maxit = 200)
```

Coefficients:

	(Intercept)	be	been	had	it	may
Austen	-15.504834	0.48974946	0.5380318	0.4620513	0.00388835	-0.15025084
London	-14.671720	-0.07497073	0.1733116	0.4842272	0.08674782	-0.01590702
Milton	-1.776866	-0.10891178	-0.9127155	0.5319573	-0.82046587	-0.06760436

	not	on	the	upon	was	which
Austen	-0.08861462	0.5967404	-0.02361614	-2.119001	0.7021371	0.10370827
London	-0.32567063	0.5749969	0.12039782	-1.914428	0.6767581	-0.59121054
Milton	0.05575887	0.5198173	0.08739368	-2.042475	0.3048202	-0.05939104

Std. Errors:

	(Intercept)	be	been	had	it	may	not
Austen	4.892258	0.1643694	0.3117357	0.2695081	0.09554376	0.3008164	0.1078329
London	5.372898	0.1916618	0.3308759	0.2812555	0.11355697	0.4946804	0.1440760
Milton	5.417300	0.1613282	0.5910561	0.3187304	0.23015421	0.3500753	0.1309306

	on	the	upon	was	which
Austen	0.2213827	0.03457288	0.6426484	0.1808681	0.1646472
London	0.2251642	0.03088881	0.7072129	0.1768901	0.2542886
Milton	0.2223575	0.04783646	0.6436399	0.1820885	0.2111105

Residual Deviance: 91.72646

AIC: 163.7265

28.4.1 Testing Model 1

```
z1 <- summary(authnom1)$coefficients/summary(authnom1)$standard.errors
round(z1,2)
```

	(Intercept)	be	been	had	it	may	not	on	the	upon	was
Austen	-3.17	2.98	1.73	1.71	0.04	-0.50	-0.82	2.70	-0.68	-3.30	3.88
London	-2.73	-0.39	0.52	1.72	0.76	-0.03	-2.26	2.55	3.90	-2.71	3.83
Milton	-0.33	-0.68	-1.54	1.67	-3.56	-0.19	0.43	2.34	1.83	-3.17	1.67

	which
Austen	0.63
London	-2.32
Milton	-0.28

```
p1 <- (1 - pnorm(abs(z1), 0, 1)) * 2
kable(round(p1,3))
```

	(Inter- cept)	be	been	had	it	may	not	on	the	upon	was	which
Austen	0.002	0.003	0.084	0.086	0.968	0.617	0.411	0.007	0.495	0.001	0.000	0.529

	(Inter- cept)	be	been	had	it	may	not	on	the	upon	was	which
Lon- don	0.006	0.696	0.600	0.085	0.445	0.974	0.024	0.011	0.000	0.007	0.000	0.020
Mil- ton	0.743	0.500	0.123	0.095	0.000	0.847	0.670	0.019	0.068	0.002	0.094	0.778

Simonoff suggests that “been” and “may” can be dropped. What do we think?

The proper function of man is to live, not to exist. I shall not waste my days in trying to prolong them. I shall use my time.

28.5 Model 2

```
authnom2 <- multinom(Author ~ be + had + it + not + on +
                      the + upon + was + which, data=authorship,
                      maxit=200)
```

```
# weights:  44 (30 variable)
initial value 1165.873558
iter  10 value 304.985478
iter  20 value 285.428679
iter  30 value 143.301103
iter  40 value  54.589791
iter  50 value  52.140470
iter  60 value  51.421454
iter  70 value  51.012790
iter  80 value  50.888718
iter  90 value  50.834262
iter 100 value  50.743136
final value 50.743111
converged
```

```
summary(authnom2)
```

Call:

```
multinom(formula = Author ~ be + had + it + not + on + the +
          upon + was + which, data = authorship, maxit = 200)
```


Coefficients:

	(Intercept)	be	had	it	not	on
Austen	-16.55647	0.45995950	0.6698612	0.02621612	-0.03684654	0.4676716
London	-16.06419	-0.13378141	0.6052164	0.10517792	-0.27934022	0.4958923
Milton	-2.22344	-0.07031256	0.1737526	-0.81984885	0.05444678	0.5363108

	the	upon	was	which
Austen	-0.001852454	-1.950761	0.6543956	0.06363998
London	0.128565811	-1.643829	0.6418607	-0.54690144
Milton	0.074236636	-1.762533	0.2932065	-0.08748272

Std. Errors:

	(Intercept)	be	had	it	not	on
Austen	4.723001	0.1293729	0.2201823	0.08657746	0.08771157	0.1949021
London	5.202732	0.1587639	0.2306803	0.10117217	0.11608348	0.2072383
Milton	4.593806	0.1499103	0.2057258	0.21551377	0.12103678	0.1895226

	the	upon	was	which
Austen	0.02945139	0.5620273	0.1524982	0.1466250
London	0.02739965	0.6219927	0.1512911	0.2087120
Milton	0.04463721	0.6246766	0.1601393	0.1928361

Residual Deviance: 101.4862

AIC: 161.4862

28.5.1 Comparing Model 2 to Model 1

```
anova(authnom1, authnom2)
```

Likelihood ratio tests of Multinomial Models

Response: Author

						Model Resid. df
1		be + had + it + not + on + the + upon + was + which				2493
2	be + been + had + it + may + not + on + the + upon + was + which					2487
	Resid. Dev	Test	Df	LR stat.	Pr(Chi)	
1	101.48622					
2	91.72646	1 vs 2	6	9.759767	0.1351402	

28.5.2 Testing Model 2

```
z2 <- summary(authnom2)$coefficients/summary(authnom2)$standard.errors  
round(z2,2)
```

	(Intercept)	be	had	it	not	on	the	upon	was	which
Austen	-3.51	3.56	3.04	0.30	-0.42	2.40	-0.06	-3.47	4.29	0.43
London	-3.09	-0.84	2.62	1.04	-2.41	2.39	4.69	-2.64	4.24	-2.62
Milton	-0.48	-0.47	0.84	-3.80	0.45	2.83	1.66	-2.82	1.83	-0.45

```
p2 <- (1 - pnorm(abs(z2), 0, 1)) * 2  
round(p2,3)
```

	(Intercept)	be	had	it	not	on	the	upon	was	which
Austen	0.000	0.000	0.002	0.762	0.674	0.016	0.950	0.001	0.000	0.664
London	0.002	0.399	0.009	0.299	0.016	0.017	0.000	0.008	0.000	0.009
Milton	0.628	0.639	0.398	0.000	0.653	0.005	0.096	0.005	0.067	0.650

28.5.3 A little history

Simonoff has an interesting note: Consider the lifetimes of these four authors:

- William Shakespeare was born in 1564 and died in 1616
- John Milton was born in 1608 (44 years after Shakespeare) and died in 1674
- Jane Austen was born in 1775 (211 years after Shakespeare) and died in 1817
- Jack London was born in 1876 (312 years after Shakespeare) and died in 1916

How many large coefficients does each author display relative to Shakespeare?

28.6 Classification Table

How well does this model (model 2) distinguish these authors based on blocks of 1700 words of text?

```
table(authorship$Author, predict(authnom2))
```

	Shakespeare	Austen	London	Milton
Shakespeare	168	3	1	1
Austen	4	308	5	0
London	0	1	294	1
Milton	2	0	1	52

Based on this classification table, I'd say it does a nice job. Almost 98% of the blocks of text are correctly classified.

Fly, envious Time, till thou run out thy race; Call on the lazy leaden-stepping hours, Whose speed is but the heavy plummet's pace; And glut thyself with what thy womb devours, Which is no more then what is false and vain, And merely mortal dross; So little is our loss, So little is thy gain. For when, as each thing bad thou hast entomb'd And last of all thy greedy self consumed, Then long Eternity shall greet our bliss, With an individual kiss; And Joy shall overtake us, as a flood, When every thing that is sincerely good, And perfectly divine, With truth, and peace, and love, shall ever shine, About the supreme throne Of Him, to whose happy-making sight, alone, When once our heavenly-guided soul shall climb, Then all this earthly grossness quit, Attired with stars, we shall for ever sit, Triumphant over Death, and Chance, and thee, O Time!

28.7 Probability Curves based on a Single Predictor

In situations where only one predictor is used, we can develop nice plots of estimated probabilities for each group as a function of the predictor. Suppose we look at the single word “been” (note that this was left out of Model 2.)

Note that the possible values for counts of “been” in the data range from 0 to 27...

```
summary(authorship$been)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	2.000	4.000	4.614	7.000	27.000

Now, we'll build a model to predict the author based solely on the counts of the word “been”.

```
authnom3 <- multinom(Author ~ been,
                      data=authorship, maxit=200)
```

```
# weights: 12 (6 variable)
initial value 1165.873558
iter 10 value 757.915093
iter 20 value 755.454631
final value 755.454551
converged
```

Next, we'll build a grid of the predicted log odds for each author (as compared to Shakespeare) using the fitted coefficients. The grid will cover every possible value from 0 to 27, increasing by 0.1, using the following trick in R.

```
beengrid <- cbind(1,c(0:270)/10)
austenlogit <- beengrid %*% coef(authnom3)[1,]
londonlogit <- beengrid %*% coef(authnom3)[2,]
miltonlogit <- beengrid %*% coef(authnom3)[3,]
```

Next, we'll use that grid of logit values to estimate the fitted probabilities for each value of "been" between 0 and 27.

```
austenprob <- exp(austenlogit)/
  (exp(austenlogit) + exp(londonlogit) +
   exp(miltonlogit) + 1)
londonprob <- exp(londonlogit)/
  (exp(austenlogit) + exp(londonlogit) +
   exp(miltonlogit) + 1)
miltonprob <- exp(miltonlogit)/
  (exp(austenlogit) + exp(londonlogit) +
   exp(miltonlogit) + 1)
shakesprob <- 1 - austenprob - londonprob - miltonprob

been_dat <- data_frame(been_count = beengrid[,2],
                       austen = austenprob[,1],
                       london = londonprob[,1],
                       milton = miltonprob[,1],
                       shakespeare = shakesprob[,1])
```

Warning: `data\_frame()` was deprecated in tibble 1.1.0.
i Please use `tibble()` instead.

```
been_dat
```

```
# A tibble: 271 x 5
  been_count austen london milton shakespeare
    <dbl>    <dbl>   <dbl>   <dbl>       <dbl>
1         0  0.0258  0.136  0.285       0.553
2        0.1  0.0288  0.147  0.272       0.553
3        0.2  0.0321  0.158  0.258       0.551
4        0.3  0.0357  0.171  0.245       0.548
5        0.4  0.0396  0.184  0.232       0.545
6        0.5  0.0438  0.197  0.219       0.540
7        0.6  0.0484  0.211  0.207       0.534
8        0.7  0.0534  0.225  0.195       0.527
9        0.8  0.0587  0.240  0.183       0.518
10       0.9  0.0644  0.256  0.171       0.509
# i 261 more rows
```

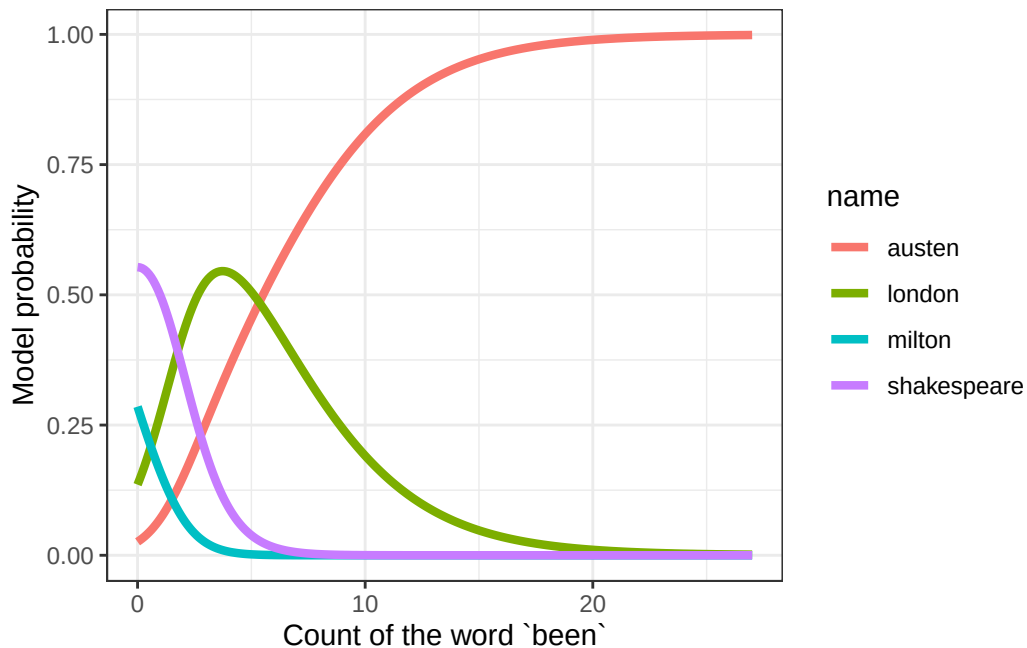
Now, we gather the data by author name and probability

```
been_dat_long <- been_dat |>
  gather("name", "prob", 2:5)
been_dat_long
```

```
# A tibble: 1,084 x 3
  been_count name      prob
    <dbl>   <chr>    <dbl>
1         0  austen 0.0258
2        0.1  austen 0.0288
3        0.2  austen 0.0321
4        0.3  austen 0.0357
5        0.4  austen 0.0396
6        0.5  austen 0.0438
7        0.6  austen 0.0484
8        0.7  austen 0.0534
9        0.8  austen 0.0587
10       0.9  austen 0.0644
# i 1,074 more rows
```

28.7.1 Produce the Plot of Estimated Probabilities based on “been” counts

```
ggplot(been_dat_long, aes(x = been_count, y = prob,
                          col = name)) +
  geom_line(linewidth = 1.5) +
  labs(x = "Count of the word `been`",
       y = "Model probability")
```

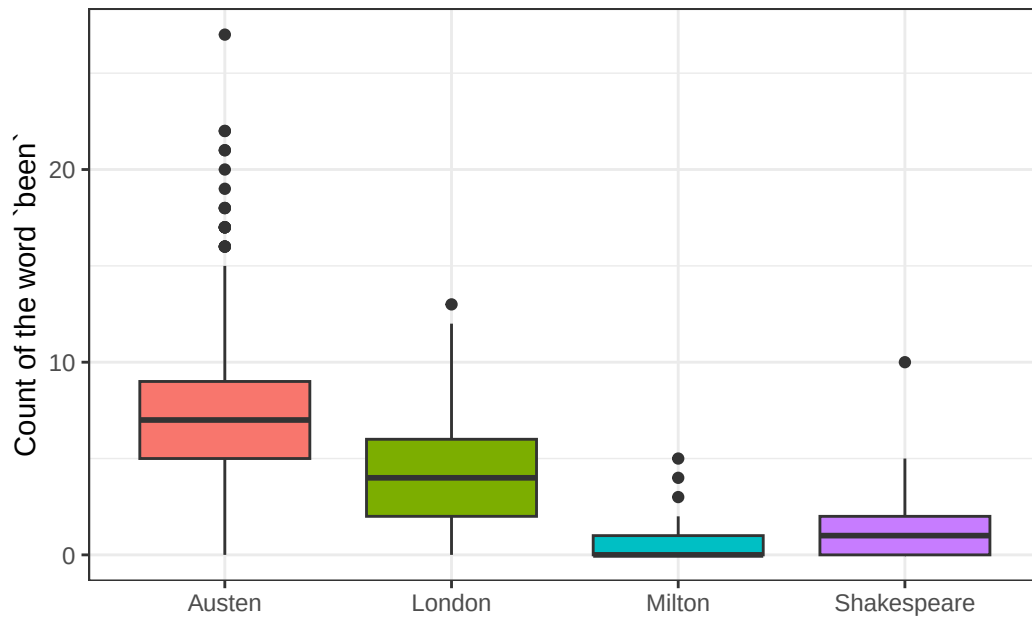


28.7.2 Boxplot of “been” counts

Compare this to what we see in the raw counts of the word “been”.

```
been.long <- filter(auth2.long, word == "been")
been.long$Auth <- fct_relevel(been.long$Author,
                             "Austen", "London", "Milton", "Shakespeare")
# releveled to make the colors match the model plot

ggplot(been.long, aes(x = Auth, y = n, fill = Auth)) +
  geom_boxplot() +
  guides(fill = "none") +
  labs(x = "", y = "Count of the word `been`")
```



28.7.3 Quote Sources

1. To-morrow, and to-morrow, and to-morrow ... Shakespeare *Macbeth* Act 5.
2. Oh! do not attack me with your watch. ... Jane Austen *Mansfield Park*
3. The proper function of man is to live, not to exist. ... Jack London *The Bulletin* San Francisco 1916-12-02.
4. Fly, envious Time, till thou run out thy race ... John Milton *On Time*

29 Time To Event / Survival Data

In many medical studies, the main outcome variable is the time to the occurrence of a particular event.

- In a randomized controlled trial of cancer, for instance, surgery, radiation, and chemotherapy might be compared with respect to time from randomization and the start of therapy until death.
 - In this case, the event of interest is the death of a patient, but in other situations it might be remission from a disease, relief from symptoms or the recurrence of a particular condition.
 - Such observations are generally referred to by the generic term **survival data** even when the endpoint or event being considered is not death but something else.

These notes on survival analysis are just an introduction to the key ideas of the field. The PQHS department offers an entire course on survival analysis (PQHS 435) and I recommend that to those of you interested in deeper learning about the approaches we'll discuss, or in learning more about other approaches to survival analysis.

The OpenIntro Statistics extra material on [Survival Analysis in R](#), written by David Diez is a very useful guide to survival analysis in R, using the **survival** package and supplemented by the **KMsurv** and **OIsurv** packages. A PDF version of that material is available, along with a full set of the code used in that guide, along with instructions to obtain the packages.

29.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(mosaic)
library(survival)
library(survminer)
library(tidyverse)

theme_set(theme_bw())
```


29.1.1 Data Load

```
hem <- read_csv("data/hem.csv", show_col_types = FALSE)
```

29.2 An Outline of Key Topics Discussed in these Notes

In this chapter, we tackle the building blocks of survival analysis, and use R to work with survival objects.

- The Survival Function, $S(t)$
 - The Kaplan-Meier Estimate/Plot
 - Comparing Survival Functions with log rank test
- The Hazard Function, $H(t) = -\log(S(t))$
- Using `survival` and related packages in R

In the next chapter, we introduce the the Cox Proportional Hazards Regression Model, one of several available models for fitting regressions to time-to-event (survival) outcomes.

29.3 Foundations of Survival Analysis

Survival analysis is concerned with *prospective* studies, where we start with a cohort of subjects and follow them forwards in time to determine some clinical outcome. Follow-up continues until either some event of interest occurs, the study ends, or further observation becomes impossible.

The outcomes in a survival analysis consist of the subject's **fate** and **length of follow-up** at the end of the study.

- For some patients, the outcome of interest may not occur during follow-up.
- For such patients, whose follow-up time is *censored*, we know only that this event did not occur while the patient was being followed. We do not know whether or not it will occur at some later time.

The primary problems with survival data are *non-normality* and *censoring*...

1. Survival data are quantitative, but not symmetrically distributed. They will often appear positively skewed, with a few people surviving a very long time compared with the majority; so assuming a normal distribution will not be reasonable.

2. At the completion of the study, some patients may not have reached the endpoint of interest (death, relapse, etc.). Consequently, the exact survival times are not known.
 - All that is known is that the survival times are greater than the amount of time the individual has been in the study.
 - The survival times of these individuals are said to be **censored** (precisely, they are right-censored).

29.3.1 The Survival Function, $S(t)$

The **survival function**, $S(t)$ (sometimes called the survivor function) is the probability that the survival time, T , is greater than or equal to a particular time, t .

- $S(t)$ = proportion of people surviving to time t or beyond

If there's no censoring, the survival function is easy to estimate.

$$\hat{S}(t) = \frac{\# \text{ of subjects with survival times } \geq t}{n}$$

but this won't work if there is censoring.

29.3.2 Kaplan-Meier Estimator of the Survival Function

The survival function $S(t)$ is the probability of surviving until at least time t . It is essentially estimated by the number of patients alive at time t divided by the total number of study subjects remaining at that time.

The Kaplan-Meier estimator first orders the (unique) survival times from smallest to largest, then estimates the survival function at each unique survival time.

- The survival function at the second death time, $t_{(2)}$ is equal to the estimated probability of not dying at time $t_{(2)}$ conditional on the individual being still at risk at time $t_{(2)}$.

In the presence of censoring, the survival function is estimated as follows.

1. Order the survival times from smallest to largest, where $t_{(j)}$ is the j th largest unique survival time, so we have...

$$t_{(1)} \leq t_{(2)} \leq t_{(3)} \leq \dots t_{(n)}$$

2. The Kaplan-Meier estimate of the survival function is

$$\hat{S}(t) = \prod_{j:t_{(j)} \leq t} (1 - \frac{d_j}{r_j})$$

where r_j is the number of people at risk just before $t_{(j)}$, including those censored at time $t_{(j)}$, and d_j is the number of people who experience the event at time $t_{(j)}$.

When we want to compare survival functions (or their Kaplan-Meier estimates, at least) we'll use a **log rank test** or one of several extensions of that test.

29.3.3 Creating a Survival Object in R

To do survival analysis in R, we're going to start with three main functions, all in the **survival** package:

- **Surv** creates a survival object
- **survfit** builds a Kaplan-Meier test, and the results may be plotted, as we've seen.
- **survdif** builds a log rank test, that will let us compare two survival functions, as well as running several alternatives.

Plus, we'll build out some estimates of the hazard function.

The **Surv** function, part of the **survival** package in R, will create a **survival object** from two arguments:

1. **time** = follow-up time
2. **event** = a status indicator, where
 - **event** = 1 or TRUE means the event was observed (for instance, the patient died)
 - **event** = 0 or FALSE means the follow-up time was censored

29.4 A First Example: Recurrent Lobar Intracerebral Hemorrhage

O'Donnell et al. (2000) studied the effect of the apolipoprotein E gene on the risk of recurrent lobar intracerebral hemorrhage in 70 patients who survived such a hemorrhage¹. Patients in the study are classified by:

- **time** = follow-up time, in months
- **recur** = indicator of whether or not they had a recurrent event (1 = subject had a recurrence, 0 subject did not have a recurrence), and

¹These data come from Dupont WD (2002) Statistical Modeling for Biomedical Researchers. New York: Cambridge U. Press, section 6.4.

- **genotype** = the subject's apolipoprotein E genotype (0 = Homozygous ϵ_3/ϵ_3 and 1 = At least one ϵ_2 or ϵ_4 allele)

```
hem |> head(4)
```

```
# A tibble: 4 x 4
  id genotype time recur
<dbl>   <dbl> <dbl> <dbl>
1     1       0 0.230     1
2     2       0 1.05     0
3     3       1 1.38     0
4     4       1 1.41     1
```

```
table(hem$recur)
```

```
0 1
52 18
```

We have 70 patients at the start, and observe 18 events (rest are censored.)

```
favstats(time ~ recur, data = hem)
```

	recur	min	Q1	median	Q3	max	mean	sd	n
1	0	1.0513350	14.414783	23.01437	38.64477	53.88091	25.51129	14.68242	52
2	1	0.2299795	3.367556	10.92403	24.84600	42.87474	14.98517	13.88893	18
	missing								
1	0								
2	0								

The median survival time looks like 23 weeks in the patients who do not exhibit a recurrence, but only 11 weeks in those who do.

29.5 Building a Survival Object

```
hemsurv <- Surv(time = hem$time, event = hem$recur)
head(hemsurv, 4)
```

```
[1] 0.2299795 1.0513350+ 1.3798770+ 1.4127311
```

This object both displays the survival time for each subject, and indicates whether or not the subject's follow-up was *censored* before a recurrent event occurred. Survival times with a + sign indicate censoring.

- Subject 1 lived for 0.23 months hemorrhage-free and then had a recurrence.
- Subject 2 lived for 1.05 months hemorrhage-free, at which point they were censored (perhaps because the study ended, or perhaps because the subject was no longer available for follow-up)

Remember that 18 of the subjects experienced a recurrent hemorrhage, and the other 52 are therefore censored.

29.6 Kaplan-Meier Estimate of the Survival Function

To build a Kaplan-Meier estimate of the survival function (to account properly for censoring), we take the survival object we have created, and use the `survfit` function from the `survival` package.

```
hemfit1 <- survfit(hemsurv ~ 1)
```

We can look at the `hemfit1` object directly, although the K-M estimate is usually plotted.

```
print(hemfit1, print.rmean=TRUE)
```

```
Call: survfit(formula = hemsurv ~ 1)
```

```
      n events rmean* se(rmean) median 0.95LCL 0.95UCL
[1,] 70      18  40.4      2.63    NA    42.9      NA
* restricted mean with upper limit = 53.9
```

We see that 18 events occurred out of a total of 70 subjects. The median survival time is listed as NA (missing) which implies it cannot be estimated by this simple model.

- This is because only 18 of our 70 subjects have a known recurrence-free survival time (the rest are censored), so we don't actually know what the median survival time will be across our 70 subjects. Apparently, R can produce a lower bound on a 95% confidence interval for the median survival time, but not the actual point estimate.

We also observe a **restricted mean survival time** estimate. The restricted mean uses as its upper limit the largest observed or censored survival time, which here is a censored value: 53.9 months. So it is the mean survival time, assuming all censored subjects lived hemorrhage-free for 53.9 months.

```
summary(hemfit1)
```

```
Call: survfit(formula = hemsurv ~ 1)
```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
0.23	70	1	0.986	0.0142	0.958	1.000
1.41	67	1	0.971	0.0202	0.932	1.000
1.58	65	1	0.956	0.0248	0.909	1.000
3.06	63	1	0.941	0.0287	0.886	0.999
3.32	62	1	0.926	0.0320	0.865	0.991
3.52	61	1	0.911	0.0349	0.845	0.982
3.55	60	1	0.895	0.0375	0.825	0.972
4.76	57	1	0.880	0.0400	0.805	0.962
9.53	54	1	0.863	0.0424	0.784	0.951
12.32	50	1	0.846	0.0449	0.762	0.939
15.57	46	1	0.828	0.0476	0.740	0.926
19.15	38	1	0.806	0.0511	0.712	0.912
24.77	32	1	0.781	0.0553	0.679	0.897
24.87	31	1	0.756	0.0590	0.648	0.881
28.09	26	1	0.726	0.0635	0.612	0.862
33.61	22	1	0.693	0.0687	0.571	0.842
37.52	17	1	0.653	0.0758	0.520	0.819
42.87	8	1	0.571	0.1011	0.404	0.808

This written summary provides us with lots of detail on the Kaplan-Meier estimate. In particular, the first two lines of this summary can be read to indicate the following.

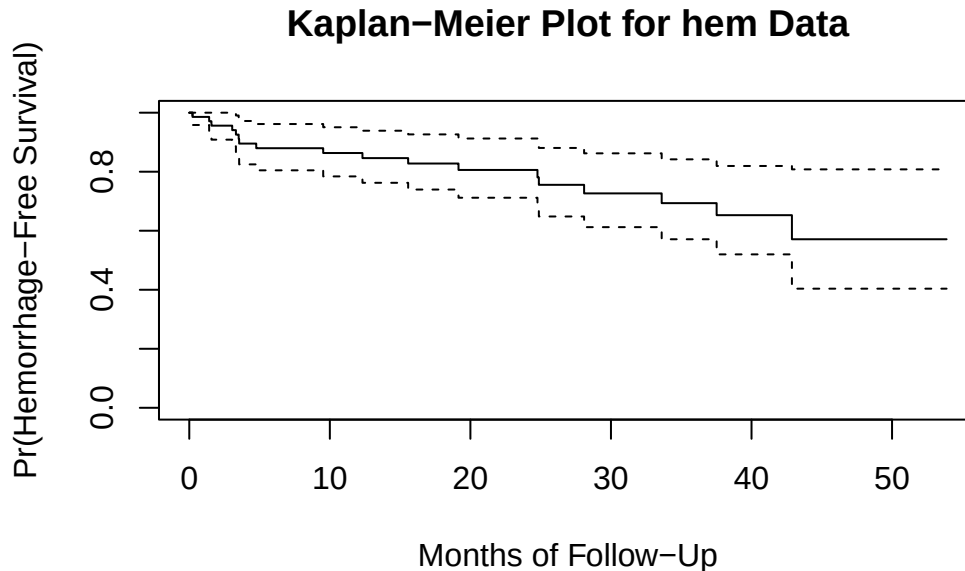
- Up to time 0.23 months, no patients had a recurrence. Then, an event occurred, and the estimated survival (i.e. non-recurrence) probability is reduced from 1 to 0.986.
- By time 1.41 months, when the next event occurred, only 67 patients remained at risk. This is because one of them had a recurrent hemorrhage already (at 0.23 months) and two others had been right-censored. The estimated hemorrhage-free survival probability estimate starting at time 1.41 months is now 0.971.

A Kaplan-Meier plot graphically represents this summary.

29.6.1 The Kaplan-Meier Plot, using Base R

Now, let's plot the Kaplan-Meier estimate, so we can see what is going on.

```
plot(hemfit1, ylab="Pr(Hemorrhage-Free Survival)",  
     xlab="Months of Follow-Up",  
     main="Kaplan-Meier Plot for hem Data")
```



The solid line indicates estimated hemorrhage-free survival probability. The dotted lines identify pointwise confidence intervals (default 95%).

- For example, we see that the estimated probability of hemorrhage-free survival to 20 months is estimated to be about 0.8
- The estimated probability of hemorrhage-free survival to 50 months is estimated to be about 0.6

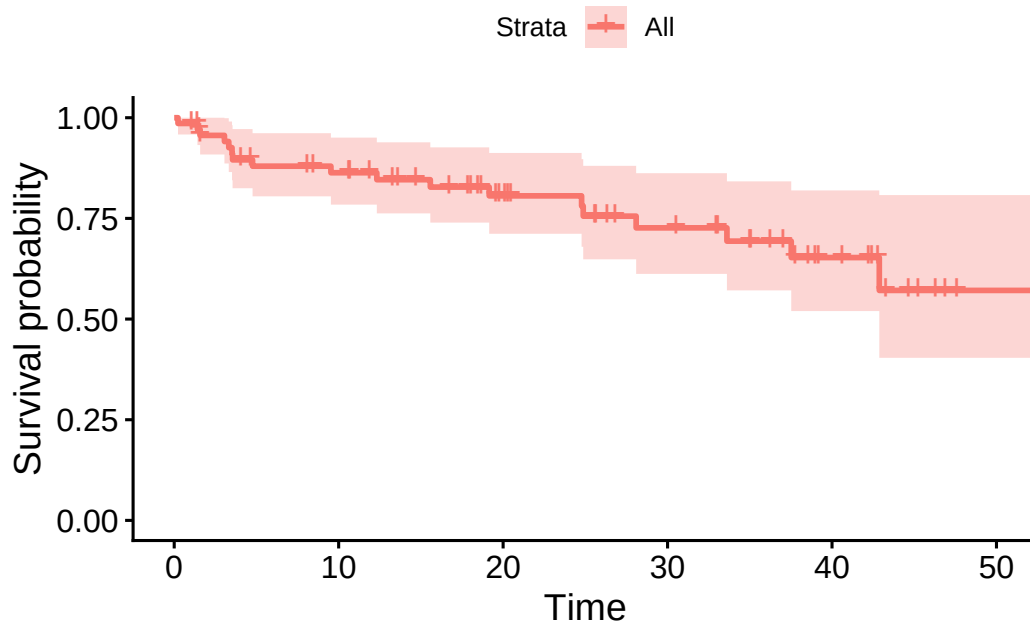
The steps down indicate events (recurrences.) The estimated probability of survival to 0 months starts at 1, and drops down at each time point where an event (or more than one event) is observed.

29.6.2 Using `survminer` to draw survival curves

Another approach to plotting the Kaplan-Meier estimate comes from `ggsurvplot`, from the `survminer` package.

```
ggsurvplot(hemfit1, data = hem)
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the ggpubr package.
Please report the issue at <<https://github.com/kassambara/ggpubr/issues>>.



Again, the solid line indicates estimated hemorrhage-free survival probability. The crosses indicate censoring. The steps down indicate events (recurrences,) and the shading indicates (default 95%) pointwise confidence intervals. By **pointwise** confidence intervals, I mean that these bounds apply only to individual points in the time scale.

For more on an alternative approach, using *simultaneous confidence bands*, visit the OpenIntro Statistics [Survival Analysis in R materials](#), written by David Diez, which are also posted on [our web site](#).

29.6.3 A “Fancy” K-M Plot with a number at risk table

We can do a lot more with these plots. Following the suggestions at <https://github.com/kassambara/survminer/> we can create the following...


```

ggsurvplot(hemfit1, data = hem,
  conf.int = TRUE, # Add confidence interval
  risk.table = TRUE, # Add risk table
  xlab = "Time in months", # adjust X axis label
  break.time.by = 12 # add tick every 12 months
)

```

Ignoring unknown labels:

* fill : "Strata"

Ignoring unknown labels:

* fill : "Strata"

Ignoring unknown labels:

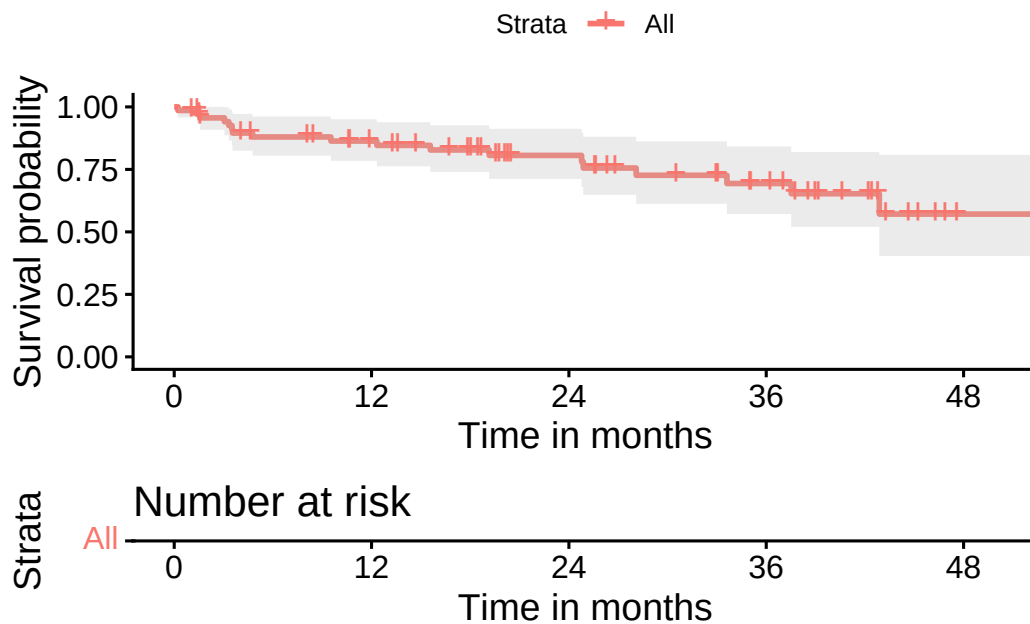
* fill : "Strata"

Ignoring unknown labels:

* fill : "Strata"

Ignoring unknown labels:

* colour : "Strata"



This sort of plot is really designed to work best when we compare multiple groups in terms of their survival. So let's do that.

29.7 Comparing Survival Across the Two Genotypes

Now, suppose we want to compare the hemorrhage-free survival functions for subjects classified by their apolipoprotein E genotype. Working with the same survival object `hemsurv` we now run the `survfit` function to compare across the two `genotype` groups.

```
hemfit2 <- survfit(hemsurv ~ hem$genotype)
print(hemfit2, print.rmean=TRUE)
```

Call: `survfit(formula = hemsurv ~ hem$genotype)`

	n	events	rmean*	se(rmean)	median	0.95LCL	0.95UCL
hem\$genotype=0	32	4	47.8	2.87	NA	NA	NA
hem\$genotype=1	38	14	33.9	3.77	37.5	24.9	NA

* restricted mean with upper limit = 53.9

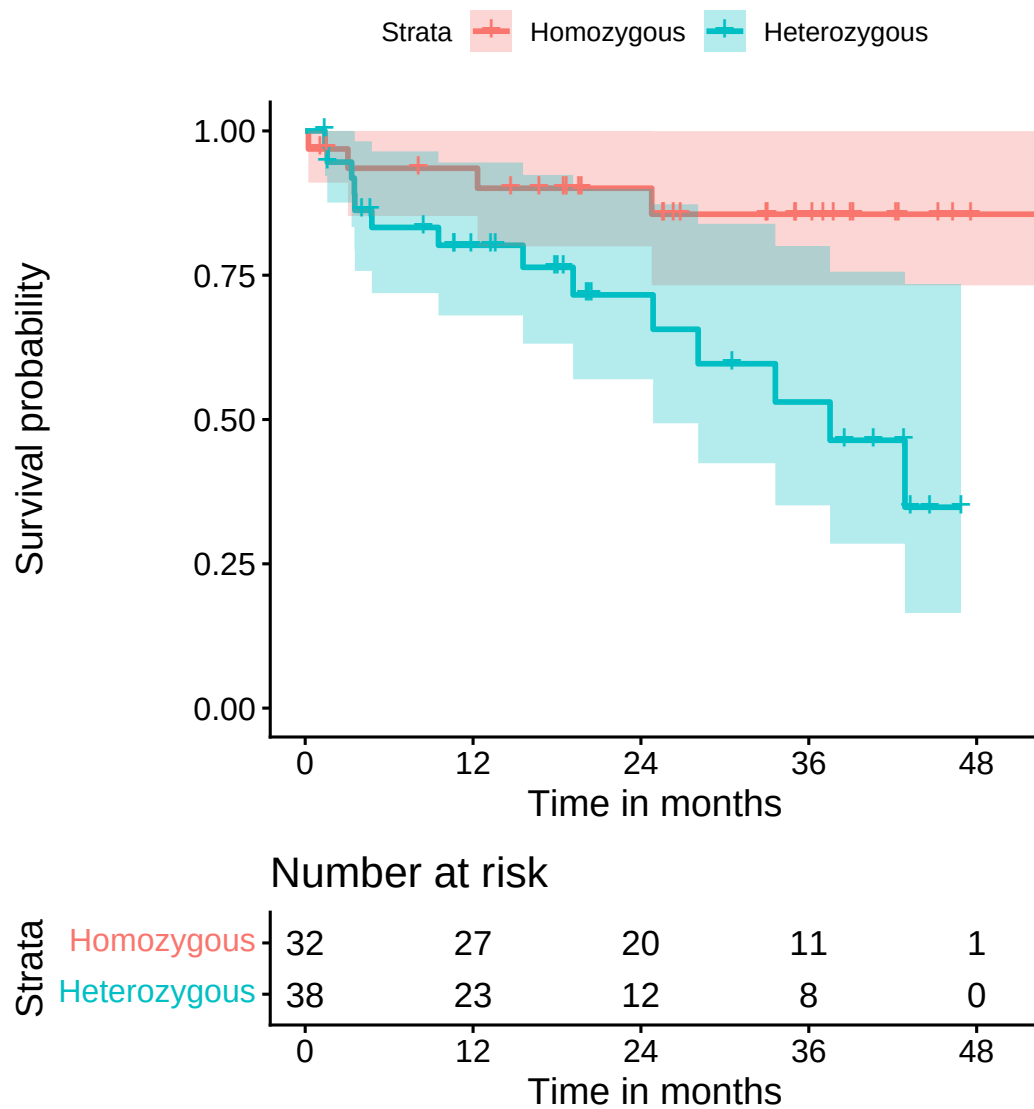
- In `genotype = 0` (the subjects who are Homozygous $\epsilon 3/\epsilon 3$), we had 32 subjects, and observed 4 recurrent hemorrhages. Our estimated restricted mean survival time in those subjects is 47.8 months and we cannot estimate a median survival time because only a small fraction of our subjects were not censored.
- In `genotype = 1` (subjects who have at least one $\epsilon 2$ or $\epsilon 4$ allele,) we had 38 subjects and observed 14 recurrences. The estimated restricted mean survival time is 33.9 months in these subjects, and we can (it seems) estimate a median survival time in this group of 37.5 months. Note that we don't actually need to observe the event in half of the subjects to estimate a median survival time.

29.7.1 Kaplan-Meier Survival Function Estimates, by Genotype

I find I have to crank the figure height in Quarto up to at least 6 to get the risk table to show up nicely in this setting.

```
ggsurvplot(hemfit2, data = hem,
  conf.int = TRUE,
  xlab = "Time in months",
  break.time.by = 12,
  legend.labs = c("Homozygous", "Heterozygous"),
  risk.table = TRUE,
  risk.table.height = 0.25
)
```

Ignoring unknown labels:
 * colour : "Strata"



It appears that patients who were homozygous for the $\epsilon 3$ allele of this gene (i.e. `genotype = 0` in the hemorrhage data) had a much better prognosis than others (`genotype = 1`.)

29.8 Testing the difference between two survival curves

To obtain a significance test comparing these two survival curves, we turn to a *log rank* test, which tests the null hypothesis $H_0 : S_1(t) = S_2(t)$ for all t where the two exposures have survival functions $S_1(t)$ and $S_2(t)$. We use the `survdif` function to explore this test, which uses a χ^2 statistic to do the testing.

```
survdif(hemsurv ~ hem$genotype)
```

Call:

```
survdif(formula = hemsurv ~ hem$genotype)
```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
hem\$genotype=0	32	4	9.28	3.00	6.28
hem\$genotype=1	38	14	8.72	3.19	6.28

Chisq= 6.3 on 1 degrees of freedom, p= 0.01

Based on the log rank test, we see a pretty small p value ($p = .0122$) for the comparison of the hemorrhage-free survival curves for the two genotypes, as shown in the Kaplan-Meier plot.

- The log rank test generalizes to permit survival comparisons across more than two groups, with the test statistic having an asymptotic chi-squared distribution with one degree of freedom less than the number of patient groups being compared.

29.8.1 Alternative log rank tests

An alternative approach to testing is the *Peto and Peto modification of the Gehan-Wilcoxon test*, which results from adding `rho=1` to the `survdif` function (`rho=0`, the default, yields the log rank test.)

```
survdif(hemsurv ~ hem$genotype, rho=1)
```

Call:

```
survdif(formula = hemsurv ~ hem$genotype, rho = 1)
```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
hem\$genotype=0	32	3.63	7.87	2.29	5.46
hem\$genotype=1	38	11.79	7.54	2.39	5.46

Chisq= 5.5 on 1 degrees of freedom, p= 0.02

As compared to the log rank test, this Peto-Peto modification (and others using `rho > 0`) give greater weight to the left hand (earlier) side of the survival curves.

- To obtain chi-square tests that give greater weight to the right hand (later) side of the survival curves than the log rank test, use a `rho` value which is less than 0.

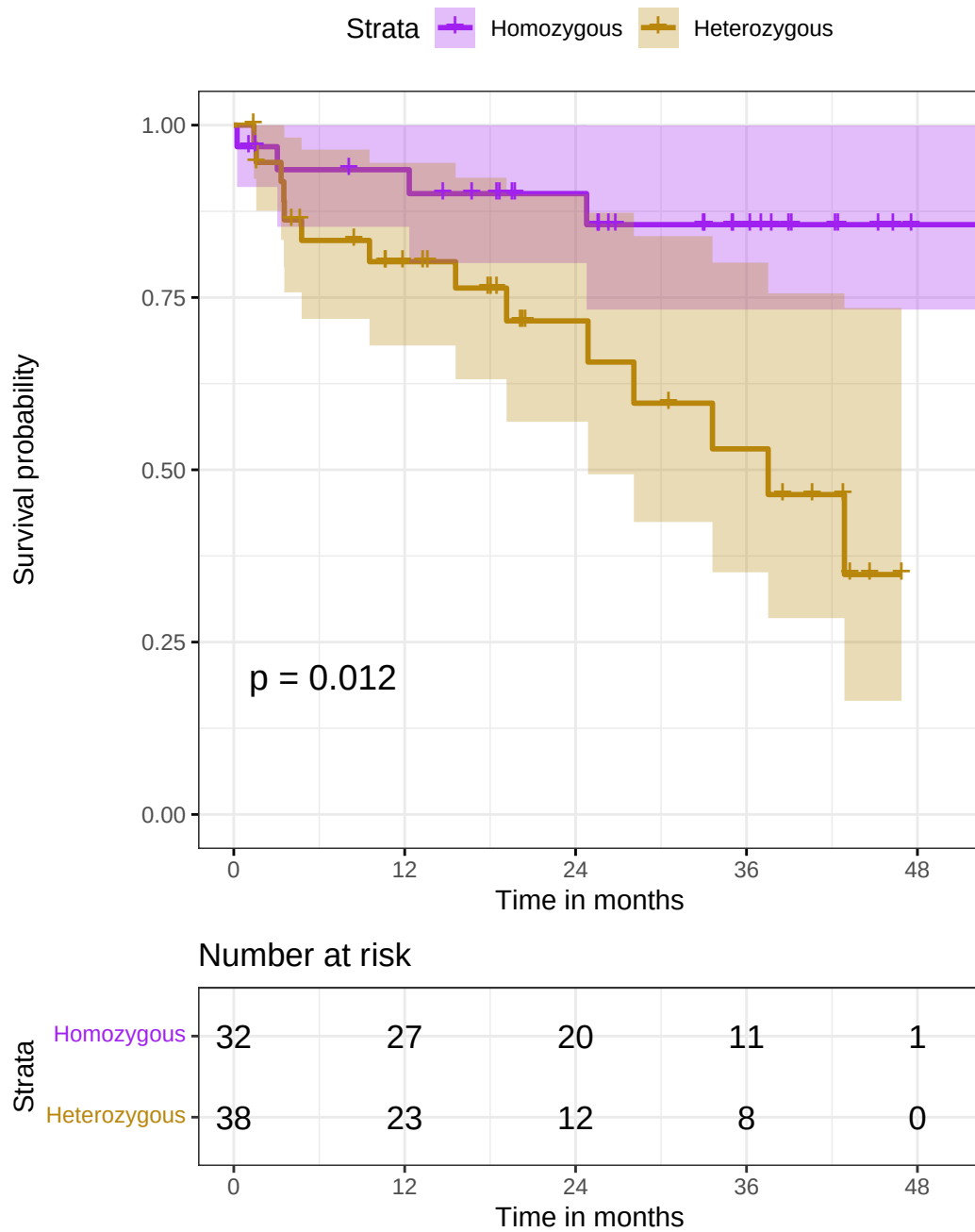
29.9 A “Fancy” K-M Plot with a number at risk table

We can add the log rank test result to our “fancy” K-M plot. Visit <https://github.com/kas-sambara/survminer/> for more options.

```
ggsurvplot(hemfit2, data = hem, size = 1,
  palette = c("purple", "darkgoldenrod"), # custom colors
  conf.int = TRUE, # Add confidence interval
  pval = TRUE, # Add p-value
  risk.table = TRUE, # Add risk table
  risk.table.height = 0.25, # change if you have >2 groups
  risk.table.y.text.col = T, # show colors in table listing
  xlab = "Time in months", # adjust X axis label
  break.time.by = 12, # break X axis in time intervals
  legend.labs = c("Homozygous", "Heterozygous"), # labels
  ggtheme = theme_bw() # Change ggplot2 theme
)
```

Ignoring unknown labels:

```
* colour : "Strata"
```



29.9.1 Customizing the Kaplan-Meier Plot Presentation Further

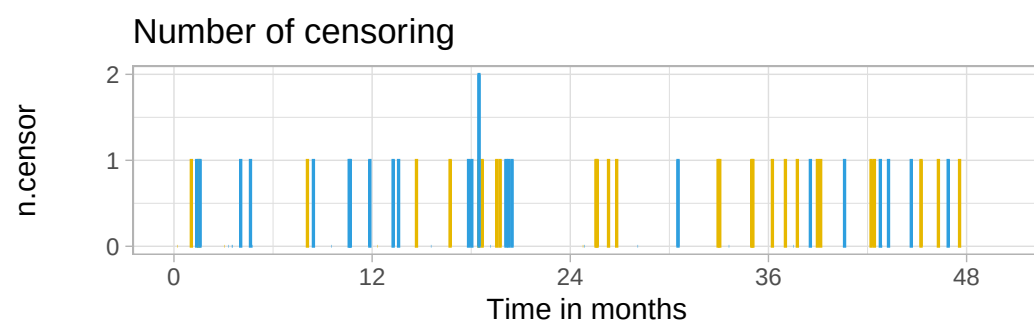
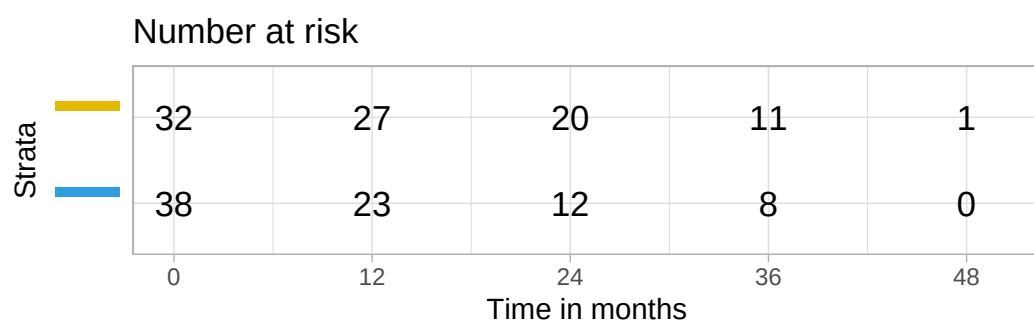
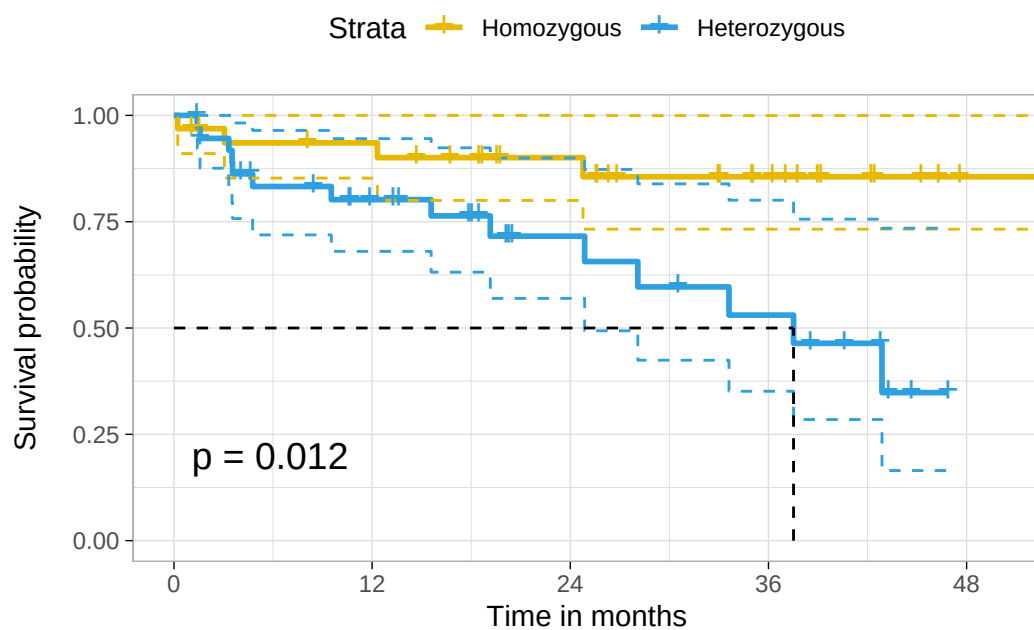
We can even add a plot of the number of censored subjects at each time point, as well as a median survival pointer (which, of course, we've seen that we can't estimate in one of the groups),

and customize the style of the confidence intervals. Again, see <https://github.com/kassambara/survminer/> for even more customized results.

```
ggsurvplot(hemfit2,
  data = hem,
  palette = c("#E7B800", "#2E9FDF"),
  risk.table = TRUE,
  pval = TRUE,
  conf.int = TRUE,
  xlab = "Time in months",
  break.time.by = 12,
  ggtheme = theme_light(),
  risk.table.y.text.col = T,
  risk.table.height = 0.25,
  risk.table.y.text = FALSE,
  ncensor.plot = TRUE,
  ncensor.plot.height = 0.25,
  conf.int.style = "step",
  surv.median.line = "hv",
  legend.labs = c("Homozygous", "Heterozygous")
)
```

Ignoring unknown labels:

```
* colour : "Strata"
```



29.10 The Hazard Function

To build regression models for time-to-event data, we will need to introduce the **hazard function**. Consider a subject in the hemorrhage study who has a hemorrhage-free survival time of 9 months.

- For this subject to have had a recurrent hemorrhage at 9 months, they had to be hemorrhage-free for the first 8 months.
- The subject's hazard at 9 months is the failure rate “per month” conditional on the subject being hemorrhage-free for the first 8 months.

If $S(t)$ is the survival function, and time t is taken to be continuous, then $S(t) = e^{-H(t)}$ defines the hazard function $H(t)$.

- Note that $H(t) = -\ln(S(t))$.
- The function $H(t)$ is an important analytic tool.
 - It's used to describe the concept of the risk of “failure” in an interval after time t , conditioned on the subject having survived to time t .
 - It's often called the *cumulative hazard function*, to emphasize the fact that its value is the “sum” of the hazard up to time t .

There are several different methods to estimate $H(t)$, but we'll focus on two...

1. The inverse Kaplan-Meier estimator
2. The Nelson-Aalen estimator

29.10.1 The Inverse Kaplan-Meier Estimator of $H(t)$

Our first estimator of the hazard function, $H(t)$ will be the inverse Kaplan-Meier estimate, which I'll place in an R object called `H_est1`.

- To start, we will take the negative of the log of the Kaplan-Meier survival estimate. That takes care of the first $t-1$ levels of the eventual estimate.
- To complete the process, we will repeat the final one of those time-specific estimates at the end.

```
H_est1 <- -log(hemfit1$surv)
H_est1 <- c(H_est1, tail(H_est1, 1))
```

Here are the first five, and last five values of the hazard function estimate.

```
head(H_est1,5) # first 5 values
```

```
[1] 0.01438874 0.01438874 0.01438874 0.02942661 0.02942661
```

```
tail(H_est1, 5) # last 5
```

```
[1] 0.5602049 0.5602049 0.5602049 0.5602049 0.5602049
```

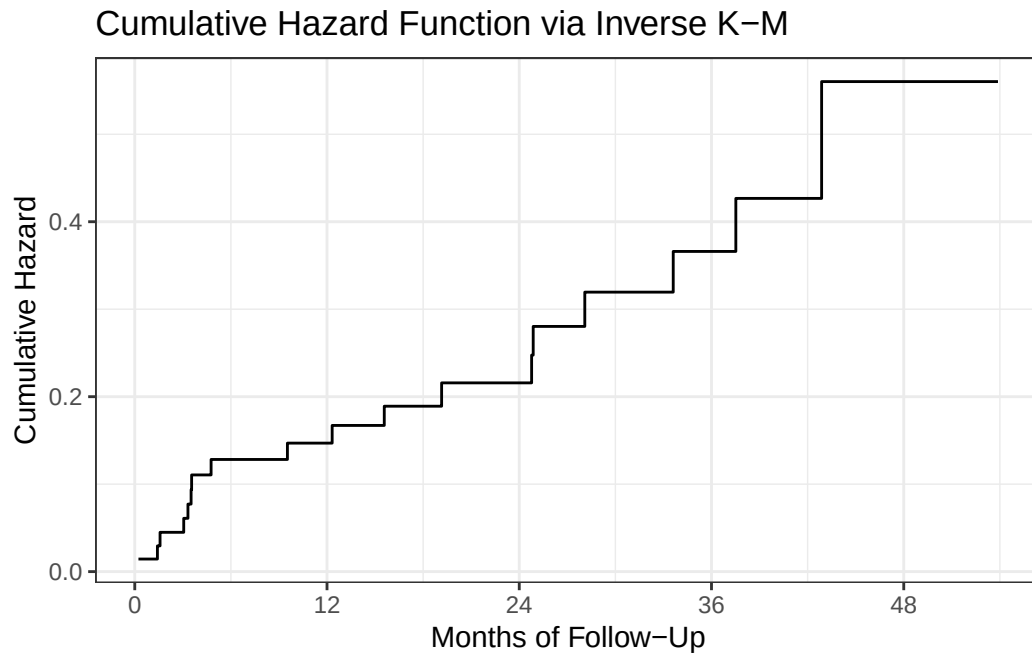
We can create a little tibble containing the times and hazard estimates, like this:

```
haz_hem <- tibble(  
  time = c(hemfit1$time, tail(hemfit1$time, 1)),  
  inverse_KM = H_est1  
)
```

29.10.2 Cumulative Hazard Function from Inverse K-M

Since we've built the data set of times and hazard values, we can use the `geom_step` function in `ggplot2`.

```
ggplot(haz_hem, aes(x = time, y = inverse_KM)) +  
  geom_step() +  
  scale_x_continuous(breaks = c(0, 12, 24, 36, 48)) +  
  labs(x = "Months of Follow-Up",  
       y = "Cumulative Hazard",  
       title = "Cumulative Hazard Function via Inverse K-M")
```



29.10.3 The Nelson-Aalen Estimator of $H(t)$

An alternative estimate of the cumulative hazard is called the Nelson-Aalen estimate, captured here in `H_est2`.

```
h_st <- hemfit1$n.event / hemfit1$n.risk
H_est2 <- cumsum(h_st)
H_est2 <- c(H_est2, tail(H_est2, 1))
haz_hem$Nelson_Aalen <- H_est2

head(haz_hem)
```

```
# A tibble: 6 x 3
  time inverse_KM Nelson_Aalen
  <dbl>      <dbl>      <dbl>
1 0.230    0.0144    0.0143
2 1.05     0.0144    0.0143
3 1.38     0.0144    0.0143
4 1.41     0.0294    0.0292
5 1.51     0.0294    0.0292
6 1.58     0.0449    0.0446
```

29.10.4 Convert Wide Data to Long

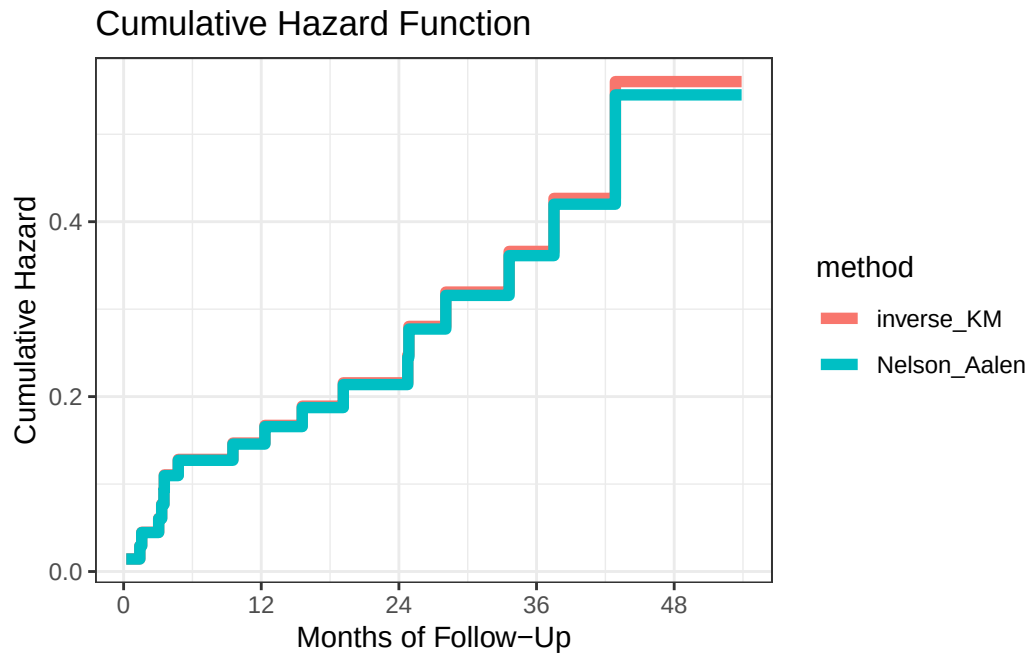
In order to easily plot the two hazard function estimates in the same graph, we'll want to convert these data from wide format to long format, with the `pivot_longer` function.

```
haz_hem_comp <- pivot_longer(data = haz_hem, cols = 2:3,  
                             names_to = "method", values_to = "hazardest")  
  
head(haz_hem_comp)
```

```
# A tibble: 6 x 3  
  time method      hazardest  
  <dbl> <chr>      <dbl>  
1 0.230 inverse_KM    0.0144  
2 0.230 Nelson_Aalen 0.0143  
3 1.05  inverse_KM    0.0144  
4 1.05  Nelson_Aalen 0.0143  
5 1.38  inverse_KM    0.0144  
6 1.38  Nelson_Aalen 0.0143
```

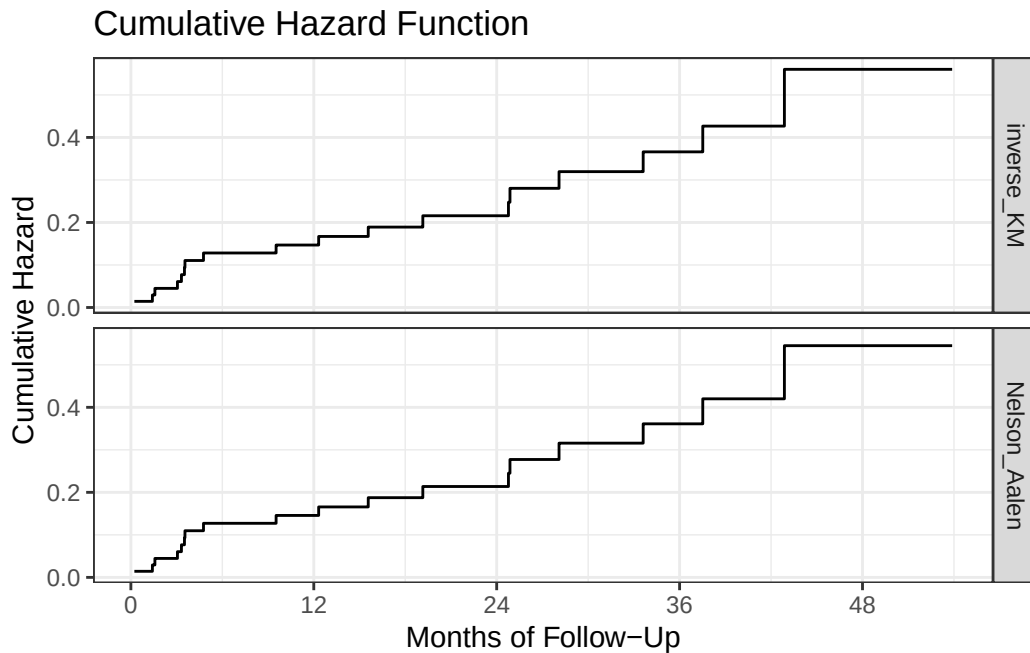
29.10.5 Plot Comparison of Hazard Estimates

```
ggplot(haz_hem_comp, aes(x = time, y = hazardest,  
                         col = method)) +  
  geom_step(linewidth = 2) +  
  scale_x_continuous(breaks = c(0, 12, 24, 36, 48)) +  
  labs(x = "Months of Follow-Up",  
       y = "Cumulative Hazard",  
       title = "Cumulative Hazard Function") +  
  theme_bw()
```



We can see that the two cumulative hazard function estimates are nearly identical in this case. We could instead compare the two functions in faceted plots, if that would be helpful.

```
ggplot(haz_hem_comp, aes(x = time, y = hazardest)) +  
  geom_step() +  
  scale_x_continuous(breaks = c(0, 12, 24, 36, 48)) +  
  labs(x = "Months of Follow-Up",  
       y = "Cumulative Hazard",  
       title = "Cumulative Hazard Function") +  
  facet_grid(method ~ .) + theme_bw()
```

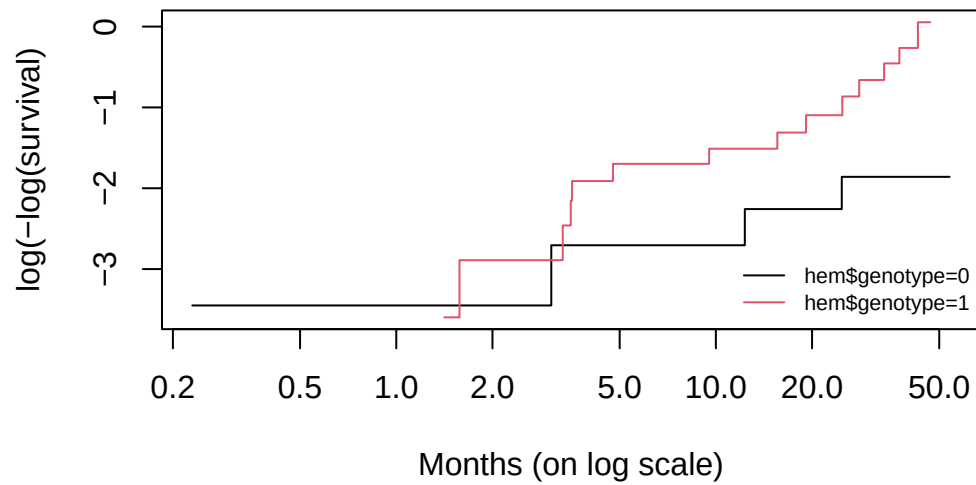


29.11 NEW!! Checking Assumptions Using a log (-log) plot

When fitting a log rank test, we may wish to consider whether the proportional hazards assumption required by such a test is satisfied. One common approach to doing this is to fit a log minus log plot, as follows.

```
plot(hemfit2, fun = "cloglog",
     main = "log minus log plot for hemfit2",
     xlab = "Months (on log scale)",
     ylab = "log(-log(survival)", col = c(1,2))
l1lab <- gsub("x = ","", names(hemfit2$strata))
legend("bottomright", legend=l1lab, col=1:2,
      lty = 1, horiz = FALSE, bty='n', cex = 0.7)
```

log minus log plot for hemfit2



Since the two curves shown in the plot cross during the observation period, this is an indication that the proportional hazard assumption made by the log rank test may be unsatisfied.

Next, we will consider the issue of modeling a survival outcome using Cox proportional hazards regression.

30 Cox Regression Models, Part 1

30.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(survival)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

30.1.1 Data Load

```
hem <- read_csv("data/hem.csv", show_col_types = FALSE)
```

The Cox proportional hazards (Cox regression) model fits survival data with a constant (i.e. not varying over time) covariate x to a hazard function of the form:

$$h(t|x) = h_0(t) \exp[\beta_1 x]$$

where we will estimate the unknown value of β_1 and where $h_0(t)$ is the baseline hazard, which is a non-parametric and unspecified value which depends on t but not on x .

- For particular x values, we will be able to estimate the survival function if we have an estimate of the baseline survival function, $\hat{S}_0(t)$.

The estimated survival function for an individual with covariate value x_k turns out to be

$$\hat{S}(t|x_k) = [\hat{S}_0(t)]^{\exp(\beta_1 x_k)}$$

From Wikipedia (yes, really) ...

Survival models can be viewed as consisting of two parts: the underlying hazard function, describing how the risk of event per time unit changes over time at baseline levels of covariates; and the effect parameters, describing how the hazard varies in response to explanatory covariates.

The key assumption in a Cox model is that the hazards are **proportional** - other types of survival models need not have this restriction. Quoting the always reliable (well, it's better than you think) Wikipedia ...

In a proportional hazards model, the unique effect of a unit increase in a covariate is multiplicative with respect to the hazard rate. For example, taking a drug may halve one's hazard rate for a stroke occurring, or, changing the material from which a manufactured component is constructed may double its hazard rate for failure.

There are two main approaches to fitting Cox models in R.

- the `coxph` function in the `survival` package, and
- the `cph` function in the `rms` package.

30.2 Sources used in building this material

- David Diez's excellent supplement for the OpenIntro Statistics project, on [Survival Analysis in R](#).
- Some tools in R to do some fancier work can be viewed at <https://cran.r-project.org/web/views/Survival.html>
- You might also look at these two blog posts, originally from the Just Another Data blog.
 - <https://www.r-bloggers.com/survival-analysis-1/>
 - <https://www.r-bloggers.com/survival-analysis-2/>
- <https://rpubs.com/daspringate/survival> has some great slides, and I've stolen from them quite a bit here.

30.3 Fitting a Cox Model in R with `coxph`

As a first example, I'll fit a model to predict time to recurrence in the `hem` data, on the basis of a single predictor: `genotype`.

```
cfit <- with(hem, coxph(Surv(time, recur) ~ genotype))
cfit
```

Call:

```
coxph(formula = Surv(time, recur) ~ genotype)
```

	coef	exp(coef)	se(coef)	z	p
genotype	1.3317	3.7874	0.5699	2.337	0.0195

Likelihood ratio test=6.61 on 1 df, p=0.01015

n= 70, number of events= 18

This summary provides an overall comparison of the two genotypes, using a proportional hazards model.

- The default approach in R is to use the “efron” method of breaking ties: other options include “breslow” and “exact”.

30.3.1 Summarizing the Fit

```
summary(cfit)
```

Call:

```
coxph(formula = Surv(time, recur) ~ genotype)
```

n= 70, number of events= 18

	coef	exp(coef)	se(coef)	z	Pr(> z)
genotype	1.3317	3.7874	0.5699	2.337	0.0195 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
genotype	3.787	0.264	1.239	11.57

Concordance= 0.622 (se = 0.061)

Likelihood ratio test= 6.61 on 1 df, p=0.01

Wald test = 5.46 on 1 df, p=0.02

Score (logrank) test = 6.28 on 1 df, p=0.01

This provides estimates of the β value for **genotype**, including standard errors and p values for a Wald test. Also included is an estimate of the hazard ratio and its confidence interval.

- Here we have a hazard ratio estimate of $\exp(\text{coef}) = 3.787$, with 95% CI (1.24, 11.57).
- The hazard ratio is the multiplicative effect of the covariate (here, having at least one of the $\epsilon 2$ or $\epsilon 4$ allele) on the hazard function for recurrent hemorrhage
 - A hazard ratio of 1 indicates no effect
 - A hazard ratio > 1 indicates an increase in the hazard as the covariate rises
 - A hazard ratio < 1 indicates a decrease in the hazard as the covariate rises

We can also tidy the hazard ratio estimate with the **broom** package.

```
tidy(cfit, exponentiate = TRUE)
```

```
# A tibble: 1 x 5
  term      estimate std.error statistic p.value
<chr>      <dbl>      <dbl>      <dbl>   <dbl>
1 genotype    3.79      0.570      2.34  0.0195
```

In addition, we have several other summaries:

- The *concordance* measure is only appropriate when we have at least one continuous predictor in our Cox model.
- The Cox & Snell pseudo- R^2 reflects the improvement of the model we've fit over the model with an intercept alone, but isn't a proportion of anything (hence the listing of the maximum possible value).
- The Likelihood ratio, Wald and Score (logrank) tests provide insight into the overall predictive value for the model.

We can obtain a more detailed description of the likelihood-ratio test of the model with **anova**.

```
anova(cfit)
```

Analysis of Deviance Table

Cox model: response is Surv(time, recur)
Terms added sequentially (first to last)

```
          loglik  Chisq Df Pr(>|Chi|)
NULL          -66.675
genotype -63.371 6.6078  1    0.01015 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

30.3.2 Glancing at the model?

```
glance(cfit)
```

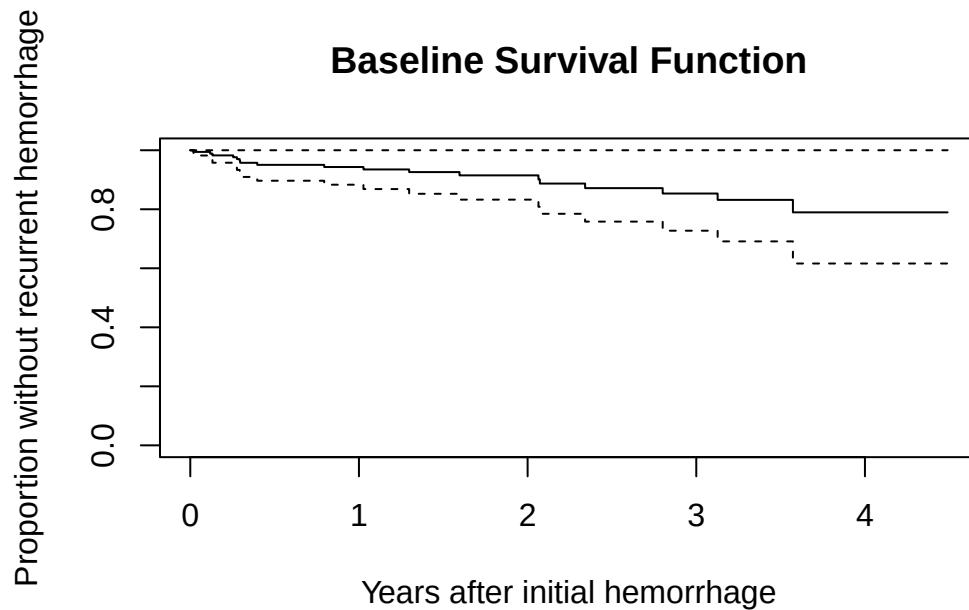
```
# A tibble: 1 x 18
      n nevent statistic.log p.value.log statistic.sc p.value.sc statistic.wald
  <int> <dbl>         <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
1    70     18          6.61          0.0102          6.28          0.0122          5.46
# i 11 more variables: p.value.wald <dbl>, statistic.robust <dbl>,
#   p.value.robust <dbl>, r.squared <dbl>, r.squared.max <dbl>,
#   concordance <dbl>, std.error.concordance <dbl>, logLik <dbl>, AIC <dbl>,
#   BIC <dbl>, nobs <dbl>
```

Here, we obtain several additional summaries of the model, including most of the important information from a summary of `cfit`.

30.3.3 Plot the baseline survival function

Here, we'll plot the time in terms of months, but scaled to 12 month (one year) groups.

```
plot(survfit(cfit), xscale = 12,
     xlab = "Years after initial hemorrhage",
     ylab = "Proportion without recurrent hemorrhage",
     main = "Baseline Survival Function")
```



30.3.4 Plot the genotype effect

There are several ways to build these plots. One approach follows. Another uses a `cph` fit and the `survplot` function from the `rms` package.

```
newdat <- with(hem,
  data.frame(
    genotype = c(1, 0)
  )
)

newdat
```

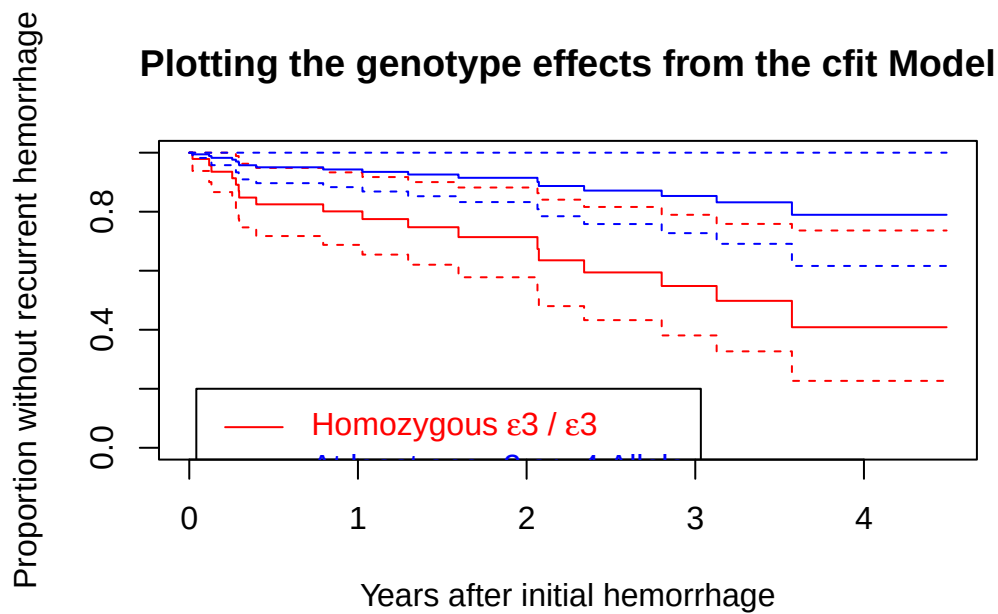
```
  genotype
1         1
2         0
```

```
plot(survfit(cfit, newdata = newdat), xscale = 12,
  conf.int = TRUE,
  col = c("red", "blue"),
  xlab = "Years after initial hemorrhage",
```

```

ylab = "Proportion without recurrent hemorrhage",
main = "Plotting the genotype effects from the cfit Model")
legend(0.5, 0.2,
  legend=c(
    expression(paste("Homozygous ", epsilon, "3 / ",
      epsilon, "3")),
    expression(paste("At least one ", epsilon, "2 or ",
      epsilon, "4 Allele"))
  ),
  lty = 1,
  col = c("red", "blue"),
  text.col = c("red", "blue"))

```



30.3.5 Testing the Key Assumption: Proportional Hazards

The `cox.zph` function in the `survival` package will test the proportionality of all of the predictors included in your model by creating interactions with time.

- A small p value would indicate a violation of the proportionality assumption.

```
cox.zph(cfit, transform="km", global=TRUE)
```

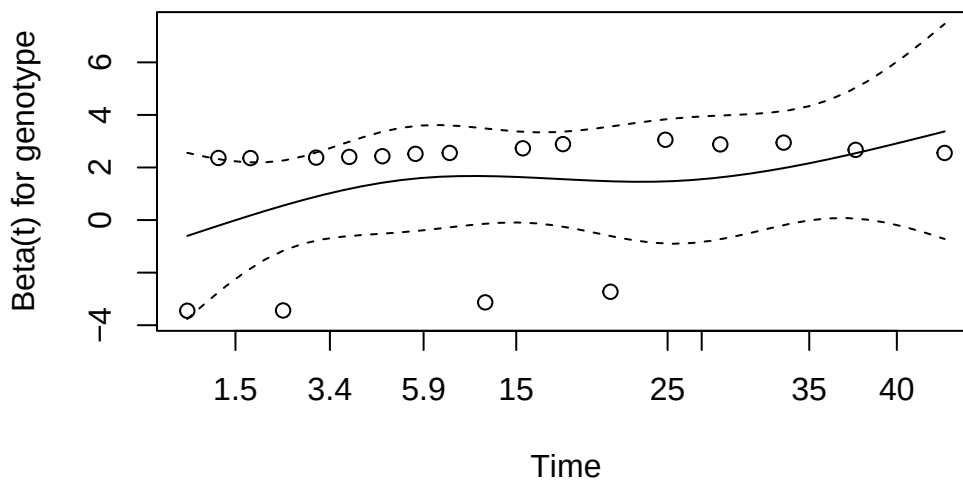
	chisq	df	p
genotype	2.09	1	0.15
GLOBAL	2.09	1	0.15

Since the p value here is quite large, we may be all right. But it's sensible to focus further on plots derived from the model fit, rather than relying solely on this test.

30.3.6 Plotting the `cox.zph` results for the `cfit` model

`cox.zph` function can be used to generate a plot for each of the individual predictors in the model. Of course, in this case, we have just one predictor: `genotype`. If the proportional hazards assumption is appropriate, then we should see a slope of essentially zero in each such plot. A slope that is seriously different from zero suggests a violation of the proportional hazards assumption.

```
plot(cox.zph(cfit, transform="km", global=TRUE))
```



The plot suggests only a slight rise in the plotted values over time, suggesting no serious problem with the proportional hazards assumption. This combined testing and plotting approach

is a reasonable starting place for assessing the proportional hazards assumption, but it's likely insufficient for good practical work.

Should the proportional hazards assumption fit less well, we have two main options: (1) fit a non-linear term in the covariate in question, and (2) fit a different type of regression model that doesn't require the proportional hazards assumption.

30.4 Fitting a Cox Model using `cph` from the `rms` package

To set up a `cph` fit for our comparison of genotypes in the `hem` data, we'll follow these steps.

```
units(hem$time) <- "month"
d <- datadist(hem)
options(datadist = "d")

hemsurv <- Surv(time = hem$time, event = hem$recur)

model_hem <- cph(hemsurv ~ genotype, data = hem,
                 x = TRUE, y = TRUE, surv = TRUE)
```

Note that the `surv = TRUE` piece is required to get some of the follow-up analyses to work smoothly.

30.4.1 The Main `cph` results

```
model_hem
```

Cox Proportional Hazards Model

```
cph(formula = hemsurv ~ genotype, data = hem, x = TRUE, y = TRUE,
     surv = TRUE)
```

		Model Tests		Discrimination Indexes	
Obs	70	LR chi2	6.61	R2	0.106
Events	18	d.f.	1	R2(1,70)	0.077
Center	0.7229	Pr(> chi2)	0.0102	R2(1,18)	0.268
		Score chi2	6.28	Dxy	0.244
		Pr(> chi2)	0.0122		

	Coef	S.E.	Wald Z	Pr(> Z)
genotype	1.3317	0.5699	2.34	0.0195

Included here are likelihood ratio and score tests for the model as a whole (as compared to the intercept-only model), as well as the usual discrimination indexes.

- These include both an R^2 analog due to Nagelkerke (which can go all the way up to 1), and
- Somers' D_{xy} , which can also produce an estimate of the C statistic (area under the curve) via the formula $C = 0.5 + D_{xy} / 2$, so here $C = 0.5 + (.244/2) = 0.622$
- For lots more on survival analysis C statistics, look at the **survAUC** package in R.

These results are followed by a table of Wald tests for each of the coefficients in the model.

30.4.2 Using anova with cph

As in other **rms** fits, we can use **anova** to obtain more detailed (in terms of combining nonlinear terms and, if available, interactions) tests.

```
anova(model_hem)
```

	Wald Statistics			Response: hemsurv
Factor	Chi-Square	d.f.	P	
genotype	5.46	1	0.0195	
TOTAL	5.46	1	0.0195	

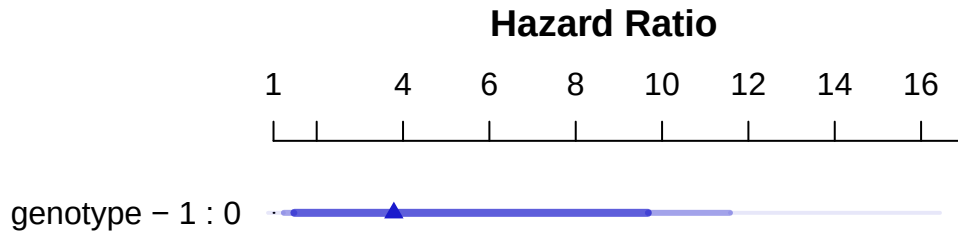
30.4.3 Effect Sizes after cph fit

We can use **summary** on a **cph** object to get and plot effect size estimates (here, these are hazard ratios.)

```
summary(model_hem)
```

	Effects			Response : hemsurv			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
genotype	0	1	1	1.3317	0.5699	0.21468	2.4486
Hazard Ratio	0	1	1	3.7873	NA	1.23950	11.5730

```
plot(summary(model_hem))
```



30.4.4 Validating cph summaries

For details on these last few indices (D, U, Q, etc.), visit `?validate.cph` in R.

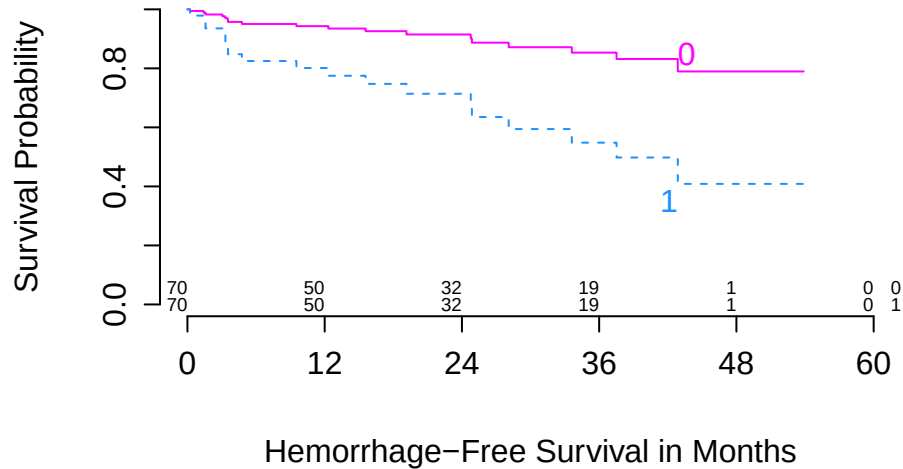
```
set.seed(43201); validate(model_hem)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper	n
Dxy	0.2441	0.2508	0.2197	0.0311	0.2130	-0.1341	0.4641	40
R2	0.1058	0.1203	0.1058	0.0144	0.0914	-0.0658	0.2363	40
Slope	1.0000	1.0000	-6.4747	7.4747	-6.4747	-Inf	12.2756	40
D	0.0421	0.0532	0.0421	0.0111	0.0310	-0.0657	0.1036	40
U	-0.0150	-0.0168	0.0198	-0.0366	0.0216	-0.0213	0.3859	40
Q	0.0571	0.0699	0.0222	0.0477	0.0094	-0.3547	0.0916	40
g	0.6705	0.7908	0.6705	0.1203	0.5502	-1.4896	1.2555	40

30.4.5 Plotting Survival Functions for each Genotype

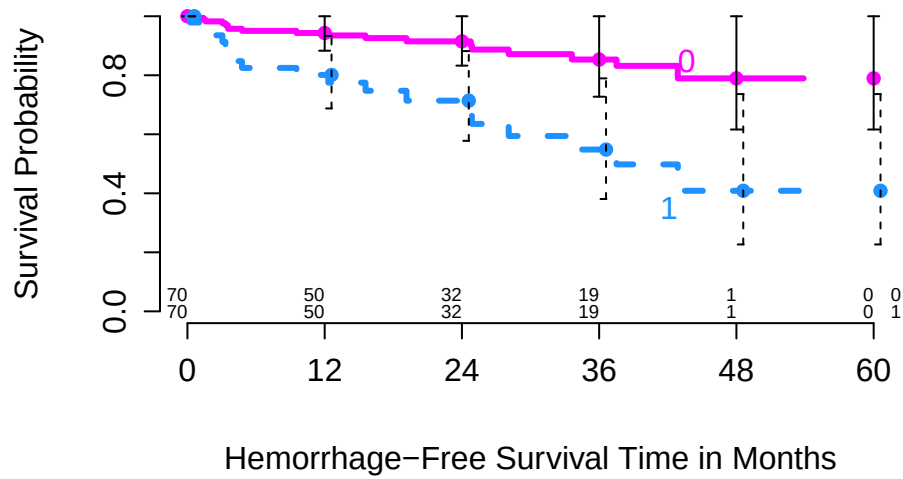
Here is the `survplot` approach I mentioned earlier.

```
survplot(model_hem, genotype,
         lty = c(1,2), n.risk=TRUE, time.inc=12,
         col=c("magenta", "dodgerblue"),
         xlab="Hemorrhage-Free Survival in Months")
```



We can add, for instance, confidence interval bars, with:

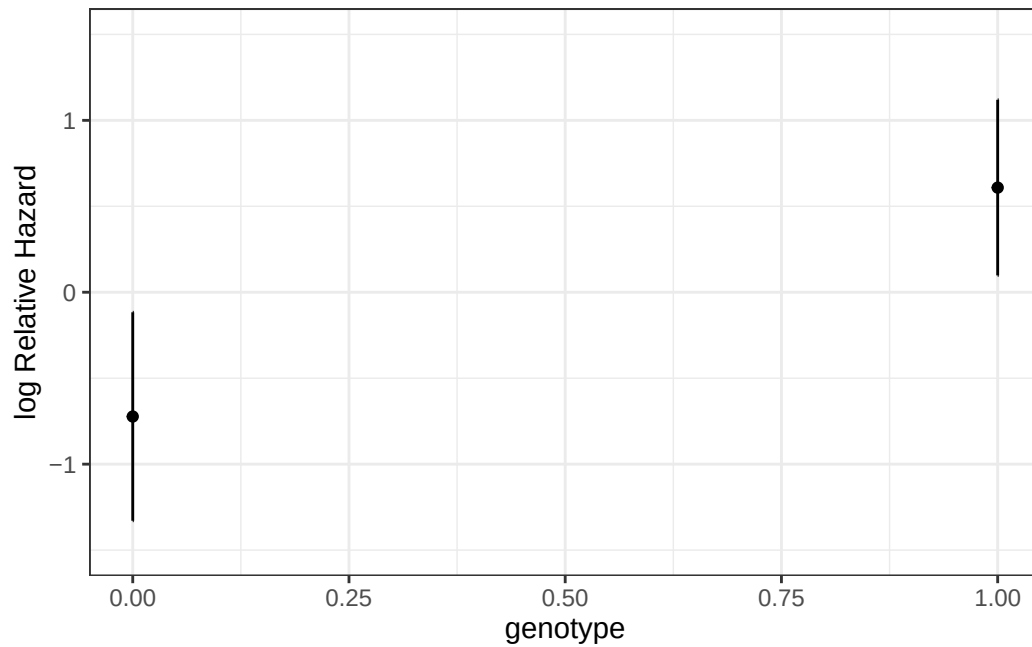
```
survplot(model_hem, genotype,
         lwd=3, lty = c(1,2), conf.int=.95,
         n.risk=TRUE, time.inc = 12, conf='bars',
         col=c("magenta", "dodgerblue"),
         xlab="Hemorrhage-Free Survival Time in Months")
```



For more details, check out R's help file on `survplot`.

30.4.6 Genotype's effect on log relative hazard

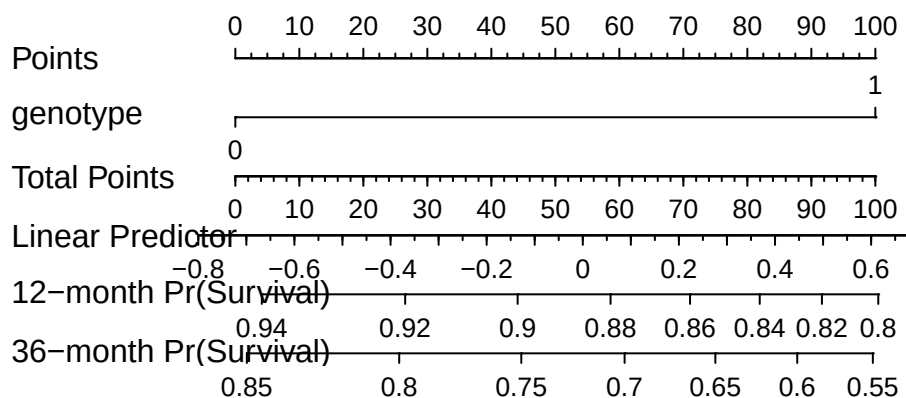
```
ggplot(Predict(model_hem, genotype))
```



30.4.7 Nomogram of our simple hem model

We can estimate 1-year and 3-year hemorrhage-free survival probabilities, for example, with this model, and incorporate these results into our nomogram.

```
survx <- Survival(model_hem)
plot(nomogram(model_hem, fun=list(function(x) survx(12, x),
                                function(x) survx(36, x)),
      funlabel=c("12-month Pr(Survival)",
                 "36-month Pr(Survival)")))
```



Again, this is just a very simple model, with one binary predictor.

30.4.8 Assessing the Proportional Hazards Assumption

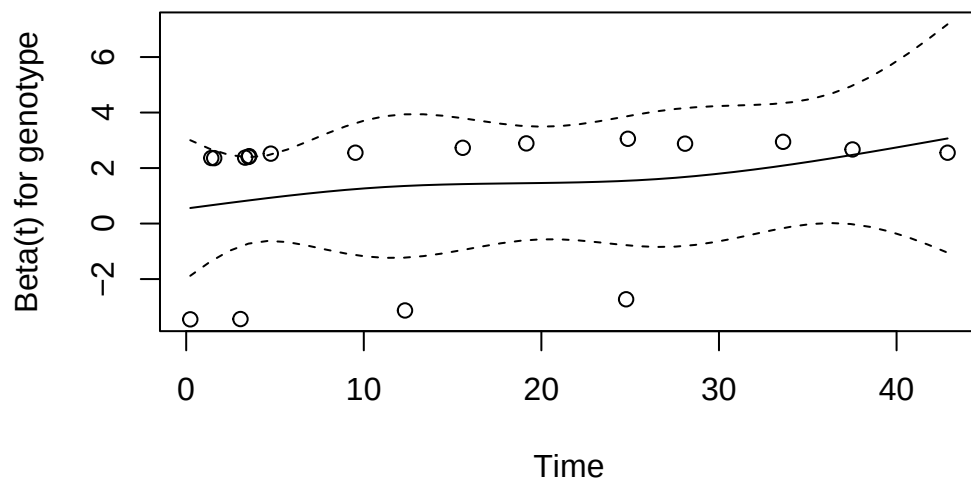
```
cox.zph(model_hem, transform="km")
```

	chisq	df	p
genotype	2.09	1	0.15
GLOBAL	2.09	1	0.15

- Consider using `transform="rank"` to transform the survival times by their ranks prior to performing the test.
- Or use `transform="identity"` as we'll do in the plot below.

30.4.9 Plot to Check PH Assumption

```
plot(cox.zph(model_hem, "identity"))
```



31 Cox Regression Models, Part 2

31.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(survival)
library(survminer)
library(rms)
library(tidyverse)

theme_set(theme_bw())
```

31.1.1 Data Load

```
leukem <- read_csv("data/leukem.csv", show_col_types = FALSE)
```

31.2 A Second Example: The leukem data

```
leukem
```

```
# A tibble: 51 x 8
   id   age pblasts  pinf  plab maxtemp months alive
  <dbl> <dbl>   <dbl> <dbl> <dbl>   <dbl>   <dbl> <dbl>
1     1    20     78    39     7     99     18     0
2     2    25     64    61    16    103     31     1
3     3    26     61    55    12    98.2     31     0
4     4    26     64    64    16    100     31     0
5     5    27     95    95     6     98     36     0
```



```

6      6      27      80      64      8    101      1      0
7      7      28      88      88     20    98.6      9      0
8      8      28      70      70     14    101      39      1
9      9      31      72      72      5    98.8      20      1
10     10     33      58      58      7    98.6      4      0
# i 41 more rows

```

The data describe 51 leukemia patients. The variables are:

- `id`, a patient identification code
- `age`, age at diagnosis
- `pblasts`, the Smear differential percentage of blasts
- `pinf`, the Percentage of absolute marrow leukemia infiltrate
- `plab`, the Percentage labeling index of the bone marrow leukemia cells
- `maxtemp`, Highest temperature prior to treatment (in °F)
- `months`, which is Survival time from diagnosis (in months)
- `alive`, which indicates Status as of the end of the study (1 = alive and thus censored, 0 = dead)

```
glimpse(leukem)
```

```

Rows: 51
Columns: 8
$ id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18,~
$ age     <dbl> 20, 25, 26, 26, 27, 27, 28, 28, 31, 33, 33, 33, 33, 34, 36, 37, 40~
$ pblasts <dbl> 78, 64, 61, 64, 95, 80, 88, 70, 72, 58, 92, 42, 26, 55, 71, 91~
$ pinf    <dbl> 39, 61, 55, 64, 95, 64, 88, 70, 72, 58, 92, 38, 26, 55, 71, 91~
$ plab    <dbl> 7, 16, 12, 16, 6, 8, 20, 14, 5, 7, 5, 12, 7, 14, 15, 9, 12, 4,~
$ maxtemp <dbl> 99.0, 103.0, 98.2, 100.0, 98.0, 101.0, 98.6, 101.0, 98.8, 98.6~
$ months  <dbl> 18, 31, 31, 31, 36, 1, 9, 39, 20, 4, 45, 36, 12, 8, 1, 15, 24,~
$ alive   <dbl> 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0,~

```

31.2.1 Creating our response: A survival time object

Regardless of how we're going to fit a survival model, we start by creating a *survival time* object that combines the information in `months` (the survival times, possibly censored) and `alive` (the censoring indicator) into a single variable we'll call `stime` in this example.

The function below correctly registers the survival time, and censors subjects who are alive at the end of the study (we need to indicate those whose times are known, and they are identified by `alive == 0`). All other subjects are alive for at least as long as we observe them, but their exact survival times are *right-censored*.

```
stime <- Surv(leukem$months, leukem$alive == 0)
stime
```

```
[1] 18 31+ 31 31 36 1 9 39+ 20+ 4 45+ 36 12 8 1 15 24 2 33
[20] 29+ 7 0 1 2 12 9 1 1 9 5 27+ 1 13 1 5 1 3 4
[39] 1 18 1 2 1 8 3 4 14 3 13 13 1
```

31.2.2 Models We'll Fit

We'll fit several models here, including:

- Model A: A model for survival time using **age** at diagnosis alone.
- Model B: A model for survival time using the main effects of 5 predictors, specifically, **age**, **pblasts**, **pinf**, **plab**, and **maxtemp**.
- Model B2: The model we get after applying stepwise variable selection to Model B, which will include **age**, **pinf** and **plab**.
- Model C: A model using **age** (with a restricted cubic spline), **plab** and **maxtemp**

31.3 Model A: coxph Model for Survival Time using age at diagnosis

We'll start by using **age** at diagnosis to predict our survival object (survival time, accounting for censoring).

```
modA <- coxph(Surv(months, alive==0) ~ age,
              data=leukem, model=TRUE)

summary(modA)
```

Call:

```
coxph(formula = Surv(months, alive == 0) ~ age, data = leukem,
      model = TRUE)
```

```
n= 51, number of events= 45
```

```
      coef exp(coef) se(coef)      z Pr(>|z|)
age 0.032397  1.032927 0.009521  3.403 0.000667 ***
```

```
----
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

      exp(coef) exp(-coef) lower .95 upper .95
age      1.033      0.9681      1.014      1.052

Concordance= 0.65  (se = 0.047 )
Likelihood ratio test= 11.85  on 1 df,   p=6e-04
Wald test              = 11.58  on 1 df,   p=7e-04
Score (logrank) test = 12.29  on 1 df,   p=5e-04

```

```

glance(modA) %>%
  select(r.squared, r.squared.max,
         concordance, std.error.concordance)

```

```

# A tibble: 1 x 4
  r.squared r.squared.max concordance std.error.concordance
    <dbl>         <dbl>         <dbl>             <dbl>
1   0.207         0.996         0.650             0.0465

```

Across these 51 subjects, we observe 45 events (deaths) and 6 subjects are censored. The hazard ratio (shown under `exp(coef)`) is 1.0329272, and this means each additional year of age at diagnosis is associated with a 1.03-fold increase in the hazard of death.

For this simple Cox regression model, we will focus on interpreting

1. the **hazard ratio** (specified by the `exp(coef)` result and associated confidence interval) as a measure of effect size,
 - Here, the hazard ratio associated with a 1-year increase in **age** is 1.033, and its 95% confidence interval is: (1.014, 1.052).
2. the **concordance** and **Rsquare** as measures of fit quality, and
 - **concordance** is only appropriate when we have at least one continuous predictor in our Cox model, in which case it assesses the probability of agreement between the survival time and the risk score generated by the predictor (or set of predictors.) A value of 1 indicates perfect agreement, but values of 0.6 to 0.7 are more common in survival data. 0.5 is an agreement that is no better than chance. Here, our concordance is 0.65, which is a fairly typical value.
 - **Rsquare** in this setting is Cox and Snell's pseudo- R^2 , which reflects the improvement of the model we have fit over the model with the intercept alone - a comparison that is tested by the likelihood ratio test. The maximum value of this statistic is often less than one, in which case R will tell you that. Here, our observed pseudo- R^2 is 0.207 and that is out of a possible maximum of 0.996.

3. the significance tests, particularly the **Wald** test (shown next to the coefficient estimates in the position of a t test in linear regression), and the **Likelihood ratio** test at the bottom of the output, which compares this model to a null model which predicts the mean survival time for all subjects.

- The Wald test for an individual predictor compares the coefficient to its standard error, just like a t test in linear regression.
- The likelihood ratio test compares the entire model to the null model (intercept-only). Again, run an ANOVA (technically an analysis of deviance) to get more details on the likelihood-ratio test.

```
anova(modA)
```

Analysis of Deviance Table

Cox model: response is Surv(months, alive == 0)

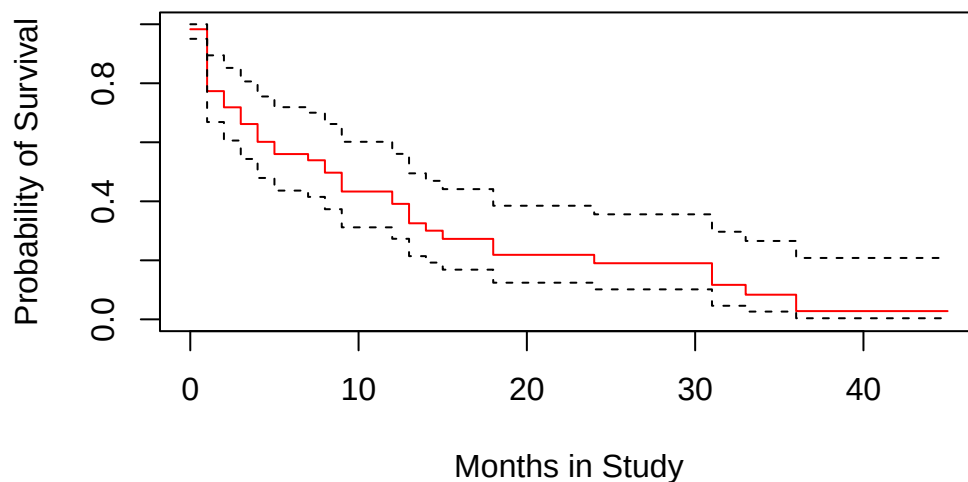
Terms added sequentially (first to last)

```
      loglik  Chisq Df Pr(>|Chi|)
NULL -142.94
age  -137.02 11.849  1   0.000577 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

31.3.1 Plotting the Survival Curve implied by Model A

```
plot(survfit(modA), ylab="Probability of Survival",
     xlab="Months in Study", col=c("red", "black", "black"))
```



31.3.2 Testing the Proportional Hazards Assumption

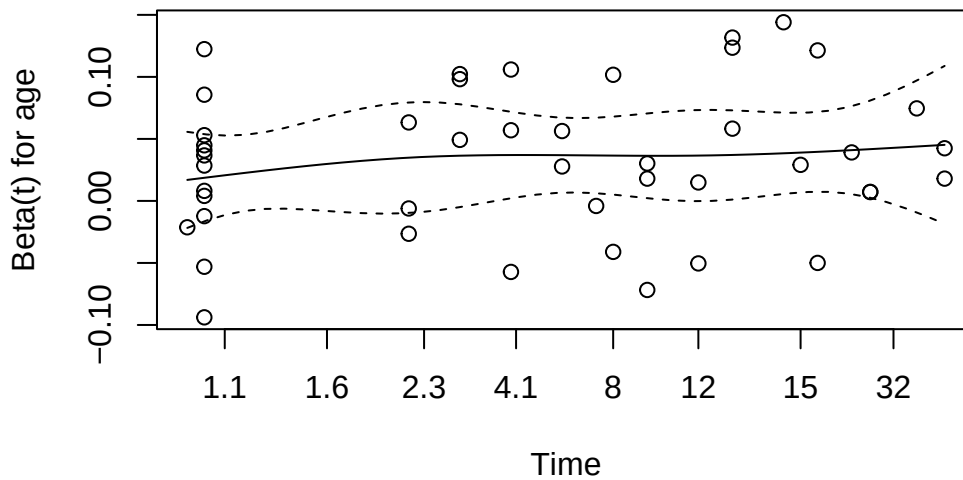
As we've noted, the key assumption in a Cox model is that the hazards are **proportional**.

```
cox.zph(modA)
```

	chisq	df	p
age	1.05	1	0.31
GLOBAL	1.05	1	0.31

A small p value would indicate a problem with the proportional hazards assumption - again, not the case here. We can also plot the results:

```
plot(cox.zph(modA))
```



We're looking for the smooth curve to be fairly level across the time horizon here, as opposed to substantially increasing or decreasing in level as time passes.

31.4 Building Model A with `cph` for the `leukem` data

```
units(leukem$months) <- "month"
d <- datadist(leukem)
options(datadist="d")
modA_cph <- cph(Surv(months, alive==0) ~ age, data=leukem,
                x=TRUE, y=TRUE, surv=TRUE, time.inc=12)
```

modA_cph

Cox Proportional Hazards Model

```
cph(formula = Surv(months, alive == 0) ~ age, data = leukem,
     x = TRUE, y = TRUE, surv = TRUE, time.inc = 12)
```

Model Tests Discrimination
 Indexes

Obs	51	LR chi2	11.85	R2	0.208
Events	45	d.f.	1	R2(1,51)	0.192
Center	1.6152	Pr(> chi2)	0.0006	R2(1,45)	0.214
		Score chi2	12.29	Dxy	0.301
		Pr(> chi2)	0.0005		

	Coef	S.E.	Wald Z	Pr(> Z)
age	0.0324	0.0095	3.40	0.0007

```
exp(coef(modA_cph)) # hazard ratio estimate
```

```
age
1.032923
```

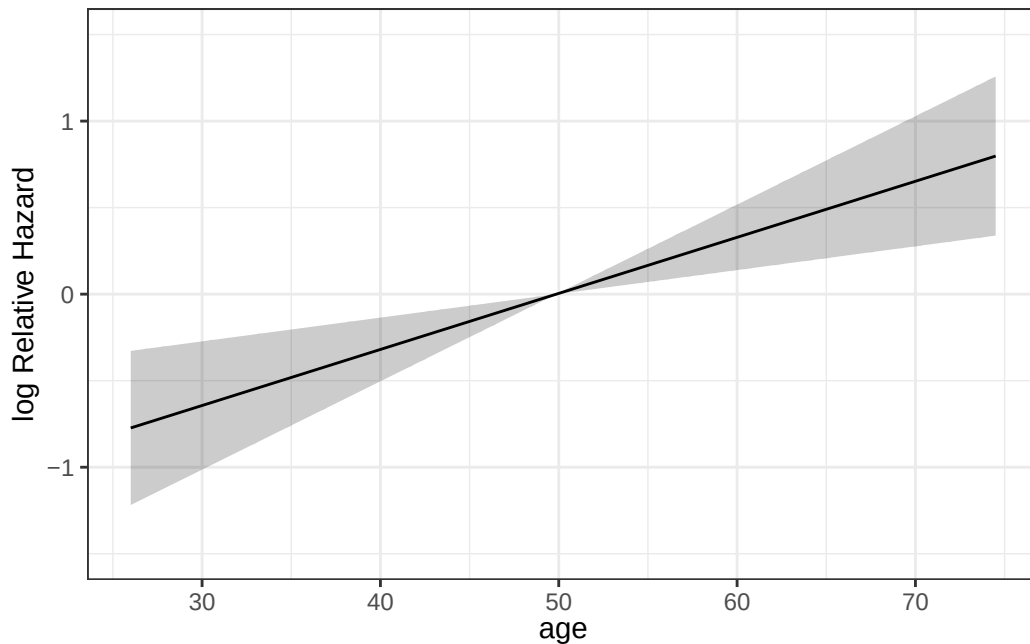
```
exp(confint(modA_cph)) # hazard ratio 95% CI
```

	2.5 %	97.5 %
age	1.013826	1.052379

31.4.1 Plotting the age effect implied by our model.

We can plot the age effect implied by the model, using `ggplot2`, as follows...

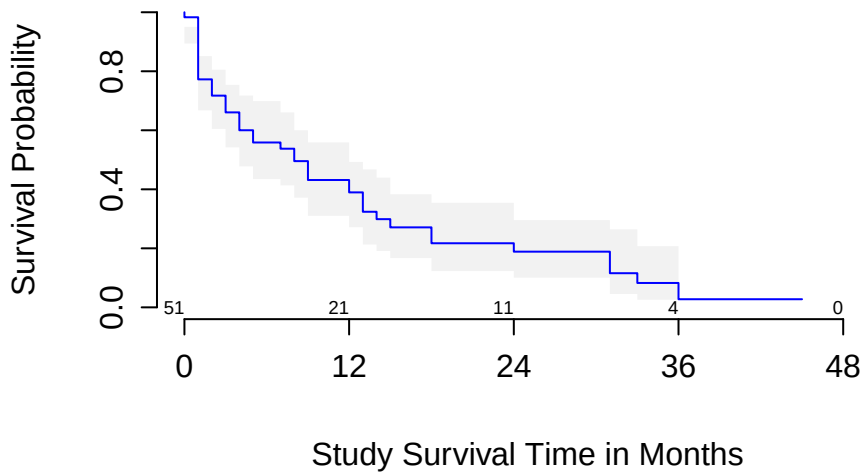
```
ggplot(Predict(modA_cph, age))
```



31.4.2 Survival Plots (Kaplan-Meier) of the age effect

The first survival plot I'll show displays 95% confidence intervals for the probability of survival at the median age at diagnosis in the sample, which turns out to be 50 years, with numbers of patients still at risk indicated every 12 months of time in the study. We can substitute in `conf = bars` to get a different flavor for this sort of plot.

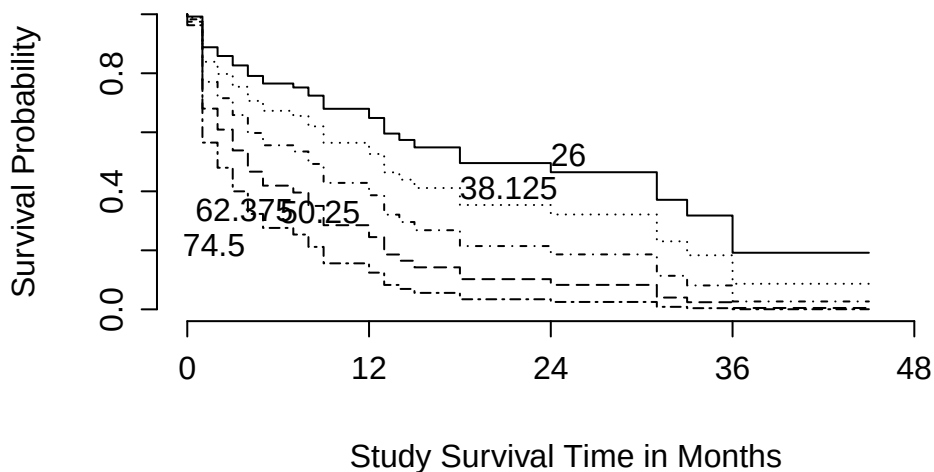
```
survplot(modA_cph, age=median(leukem$age), conf.int=.95,
         col='blue', time.inc=12, n.risk=TRUE,
         conf='bands', type="kaplan-meier",
         xlab="Study Survival Time in Months")
```

Or we can generate a survival plot that shows survival probabilities over time across a range of values for `age` at diagnosis, as follows...

```
survplot(modA_cph, levels.only=TRUE, time.inc=12,
         type="kaplan-meier",
         xlab="Study Survival Time in Months")
```

```
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
collapsing to unique 'x' values
```



This plot shows a series of modeled survival probabilities, for five different diagnosis **age** levels, as identified by the labels. Generally we see that the younger the subject is at diagnosis, the longer their survival time in the study.

31.4.3 ANOVA test for the `cph`-built model for `leukem`

We can run a likelihood-ratio (drop in deviance) test to consider the **age** effect...

```
anova(modA_cph)
```

	Wald Statistics			Response: Surv(months, alive == 0)
Factor	Chi-Square	d.f.	P	
age	11.57	1	7e-04	
TOTAL	11.57	1	7e-04	

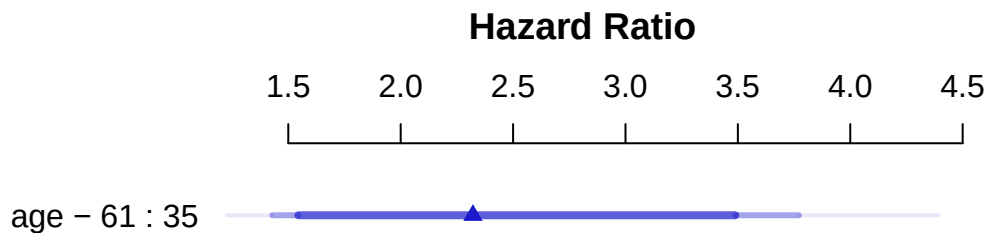
31.4.4 Summarizing the Effect Sizes from `modA_cph`

We can generate the usual summaries of effect size in this context, too.

```
summary(modA_cph)
```

Effects				Response : Surv(months, alive == 0)			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
age	35	61	26	0.8422	0.24755	0.35701	1.3274
Hazard Ratio	35	61	26	2.3215	NA	1.42910	3.7712

```
plot(summary(modA_cph))
```



As with all `rms` package effect estimates, this quantitative predictor (`age`) yields an effect comparing `age` at the 25th percentile of the sample (`age = 35`) to `age` at the 75th percentile (`age = 61`). So the hazard ratio is 2.32, with 95% CI (1.43, 3.77) for the effect of moving 26 years. Our `coxph` version of this same model showed a hazard ratio for the effect of moving just a single year.

31.4.5 Validating the Cox Model Summary Statistics

```
set.seed(432410); validate(modA_cph)
```

	index.orig	training	test	optimism	index.corrected	Lower	Upper	n
Dxy	0.3007	0.2950	0.3007	-0.0057	0.3064	0.1081	0.5890	40

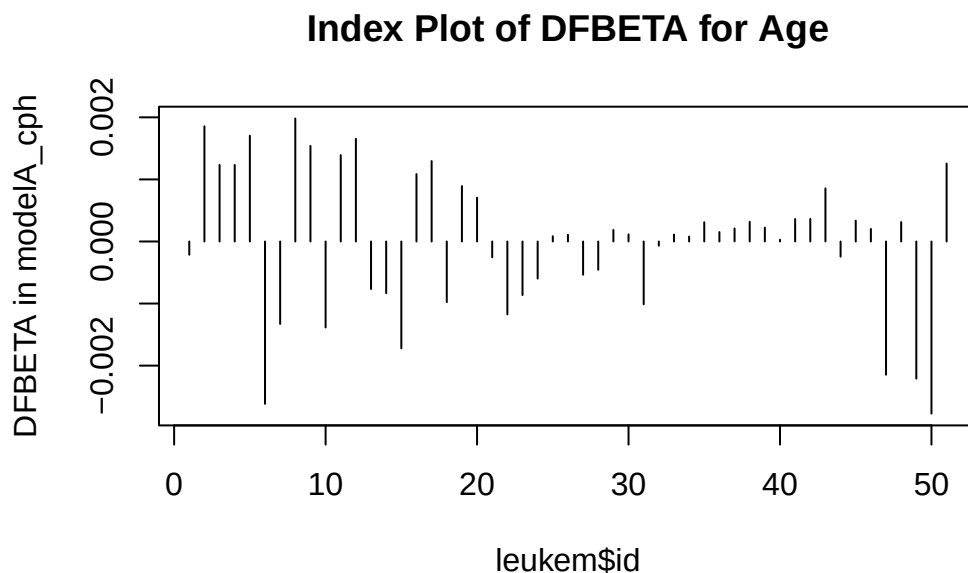
R2	0.2081	0.2226	0.2081	0.0145	0.1936	0.0310	0.3550	40
Slope	1.0000	1.0000	1.0253	-0.0253	1.0253	0.4670	2.3985	40
D	0.0379	0.0425	0.0379	0.0045	0.0334	-0.0078	0.0700	40
U	-0.0070	-0.0070	0.0037	-0.0107	0.0037	-0.0035	0.0300	40
Q	0.0449	0.0495	0.0343	0.0152	0.0297	-0.0282	0.0634	40
g	0.6210	0.6492	0.6210	0.0282	0.5928	0.2734	0.9779	40

The R^2 statistic barely moves, and neither does the Somers' d estimate, so at least in this simple model, the nominal summary statistics are likely to hold up pretty well in new data.

31.4.6 Looking for Influential Points

This plot shows the influence of each point, in terms of DFBETA - the impact on the coefficient of **age** in the model were that specific point to be removed from the data set. We can also identify the row numbers of the largest (positive and negative) DFBETAs.

```
plot(residuals(modA_cph, type="dfbeta",
              collapse = leukem$id) ~
      leukem$id, main="Index Plot of DFBETA for Age",
      type="h", ylab="DFBETA in modelA_cph")
```



```
which.max(residuals(modA_cph, type="dfbeta"))
```

```
8  
8
```

```
which.min(residuals(modA_cph, type="dfbeta"))
```

```
50  
50
```

The DFBETAs look very small here. Changes in the β estimates as large as 0.002 don't have a meaningful impact in this case, so I don't see anything particularly influential.

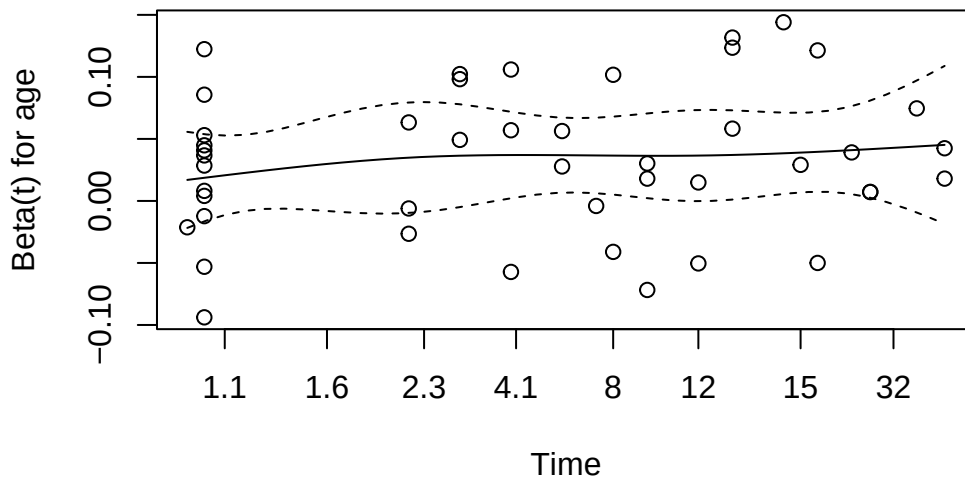
31.4.7 Checking the Proportional Hazards Assumption

As before, we can check the proportional hazards assumption with a test, or plot.

```
cox.zph(modA_cph)
```

	chisq	df	p
age	1.05	1	0.31
GLOBAL	1.05	1	0.31

```
plot(cox.zph(modA_cph))
```



Still no serious signs of trouble, of course. We'll see what happens when we fit a bigger model.

31.5 Model B: Fitting a 5-Predictor Model with coxph

Next, we use the `coxph` function from the `survival` package to apply a Cox regression model to predict the survival time using the main effects of the five predictors: `age`, `pblasts`, `pinf`, `plab` and `maxtemp`.

```
modB <- coxph(Surv(months, alive==0) ~
              age + pblasts + pinf + plab + maxtemp, data=leukem)
modB
```

Call:

```
coxph(formula = Surv(months, alive == 0) ~ age + pblasts + pinf +
      plab + maxtemp, data = leukem)
```

	coef	exp(coef)	se(coef)	z	p
age	0.033080	1.033633	0.010163	3.255	0.00113
pblasts	0.009452	1.009497	0.013959	0.677	0.49831
pinf	-0.017102	0.983043	0.012244	-1.397	0.16248
plab	-0.066000	0.936131	0.038651	-1.708	0.08771

```
maxtemp 0.155448 1.168182 0.111978 1.388 0.16507
```

```
Likelihood ratio test=18.48 on 5 df, p=0.002405
```

```
n= 51, number of events= 45
```

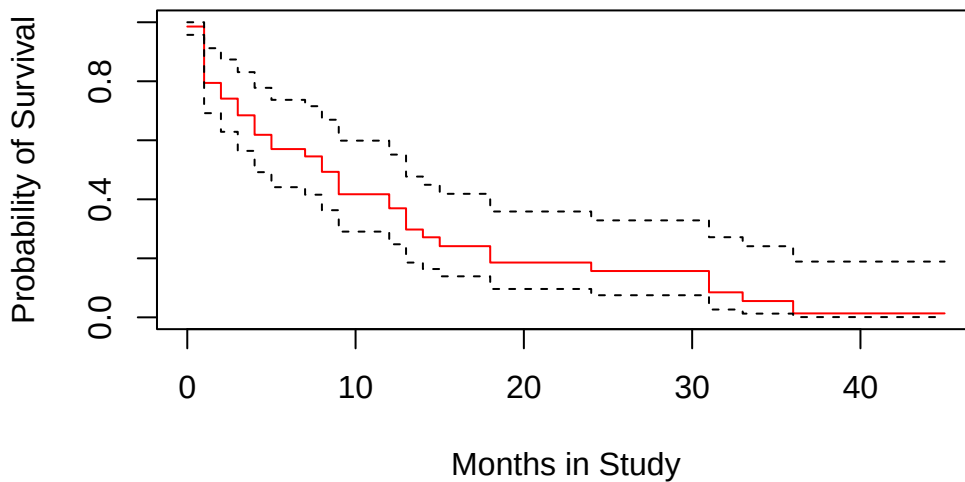
```
tidy(modB, exponentiate = TRUE, conf.int = TRUE, conf.level = 0.95) |>
  select(term, estimate, conf.low, conf.high)
```

```
# A tibble: 5 x 4
  term      estimate conf.low conf.high
  <chr>      <dbl>    <dbl>    <dbl>
1 age        1.03      1.01      1.05
2 pblasts    1.01      0.982     1.04
3 pinf       0.983     0.960     1.01
4 plab       0.936     0.868     1.01
5 maxtemp    1.17      0.938     1.45
```

The confidence intervals suggest that only the hazard ratio for **age** after accounting for all other predictors in this model seems to have a clear direction of its effect (higher hazard is associated with older age.)

31.5.1 Plotting the Survival Curve implied by Model B

```
plot(survfit(modB), ylab="Probability of Survival",
     xlab="Months in Study", col=c("red", "black", "black"))
```



The crosses in the plot indicate censoring points, while the drops indicate people who have died, and are thus no longer at risk.

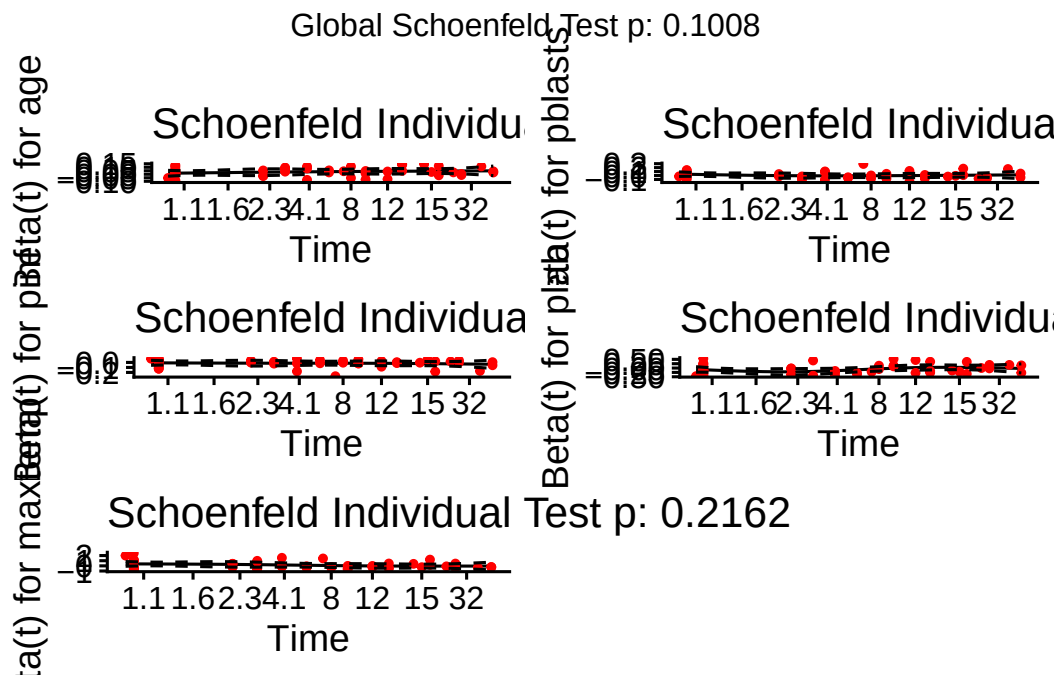
31.5.2 Testing the Proportional Hazards Assumption

```
cox.zph(modB, transform="km", global=TRUE)
```

	chisq	df	p
age	1.87	1	0.171
pblasts	4.37	1	0.037
pinf	3.51	1	0.061
plab	1.19	1	0.275
maxtemp	1.53	1	0.216
GLOBAL	9.22	5	0.101

Note that we get a global test, and a separate test for each predictor. None show substantial problems. We can plot the scaled Schoenfeld residuals directly with `ggcoxzph` from the `survminer` package.


```
ggcoxzph(cox.zph(modB))
```



31.5.3 Assessing Collinearity

Perhaps we have some collinearity here, which might imply that we could sensibly fit a smaller model, which would be appealing anyway, with only 45 actual events - we should probably be sticking to a model with no more than 2 or perhaps as many as 3 coefficients to be estimated.

```
rms::vif(modB)
```

```

      age  pblasts      pinf      plab  maxtemp
1.081775 3.029862 3.000944 1.035400 1.045249

```

The variance inflation factors don't look enormous - it may be that removing one of these variables will help us focus in on the most useful predictors. Let's consider a stepwise variable selection algorithm to see what results...

- Note that the `leaps` library, which generates best subsets output, is designed for linear regression, as is the `lars` library, which generates the lasso. Either could be used here for some guidance, but not with the survival object `stime = Surv(months, age)` as the response, but instead only with `months` as the outcome, which ignores the censoring. The `step` procedure can be used on the survival object, though.

31.6 Model B2: A Stepwise Reduction of Model B

```
stats::step(modB)
```

Start: AIC=277.4

Surv(months, alive == 0) ~ age + pblasts + pinf + plab + maxtemp

	Df	AIC
- pblasts	1	275.85
- pinf	1	277.17
- maxtemp	1	277.21
<none>		277.40
- plab	1	278.42
- age	1	286.47

Step: AIC=275.85

Surv(months, alive == 0) ~ age + pinf + plab + maxtemp

	Df	AIC
- maxtemp	1	275.63
<none>		275.85
- pinf	1	275.89
- plab	1	276.86
- age	1	284.47

Step: AIC=275.63

Surv(months, alive == 0) ~ age + pinf + plab

	Df	AIC
<none>		275.63
- pinf	1	275.69
- plab	1	275.95
- age	1	285.52

Call:

```
coxph(formula = Surv(months, alive == 0) ~ age + pinf + plab,  
      data = leukem)
```

	coef	exp(coef)	se(coef)	z	p
age	0.033171	1.033727	0.009733	3.408	0.000655

```
pinf -0.010147  0.989905  0.007088 -1.432 0.152246
plab -0.057558  0.944067  0.038476 -1.496 0.134662
```

```
Likelihood ratio test=16.25  on 3 df, p=0.001008
n= 51, number of events= 45
```

The stepwise procedure lands on a model with three predictors. How does this result look, practically?

```
modB2 <- coxph(Surv(months, alive==0) ~ age + pinf + plab, data=leukem)
summary(modB2)
```

Call:

```
coxph(formula = Surv(months, alive == 0) ~ age + pinf + plab,
      data = leukem)
```

```
n= 51, number of events= 45
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	0.033171	1.033727	0.009733	3.408	0.000655 ***
pinf	-0.010147	0.989905	0.007088	-1.432	0.152246
plab	-0.057558	0.944067	0.038476	-1.496	0.134662

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0337	0.9674	1.0142	1.054
pinf	0.9899	1.0102	0.9762	1.004
plab	0.9441	1.0592	0.8755	1.018

```
Concordance= 0.676 (se = 0.046 )
```

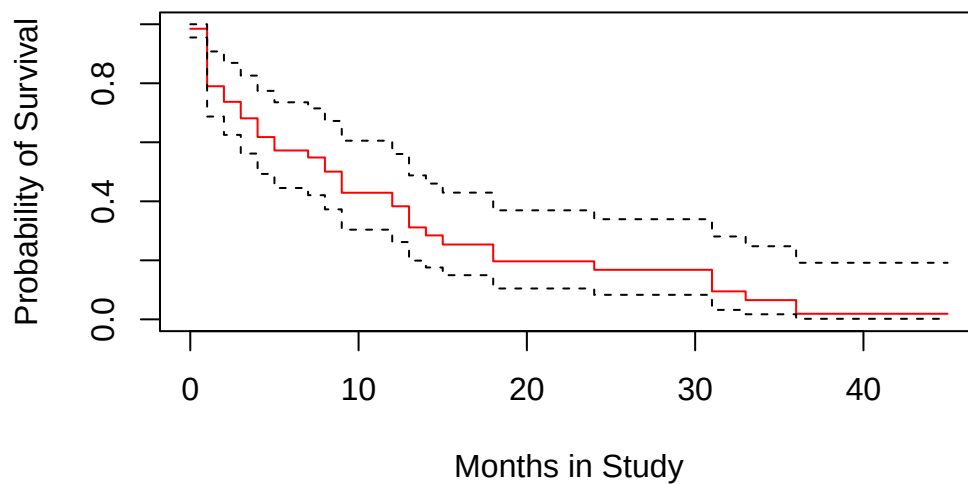
```
Likelihood ratio test= 16.25  on 3 df,  p=0.001
```

```
Wald test              = 15.28  on 3 df,  p=0.002
```

```
Score (logrank) test = 16.21  on 3 df,  p=0.001
```

31.6.1 The Survival Curve implied by Model B2

```
plot(survfit(modB2), ylab="Probability of Survival", xlab="Months in Study",
     col=c("red", "black", "black"))
```

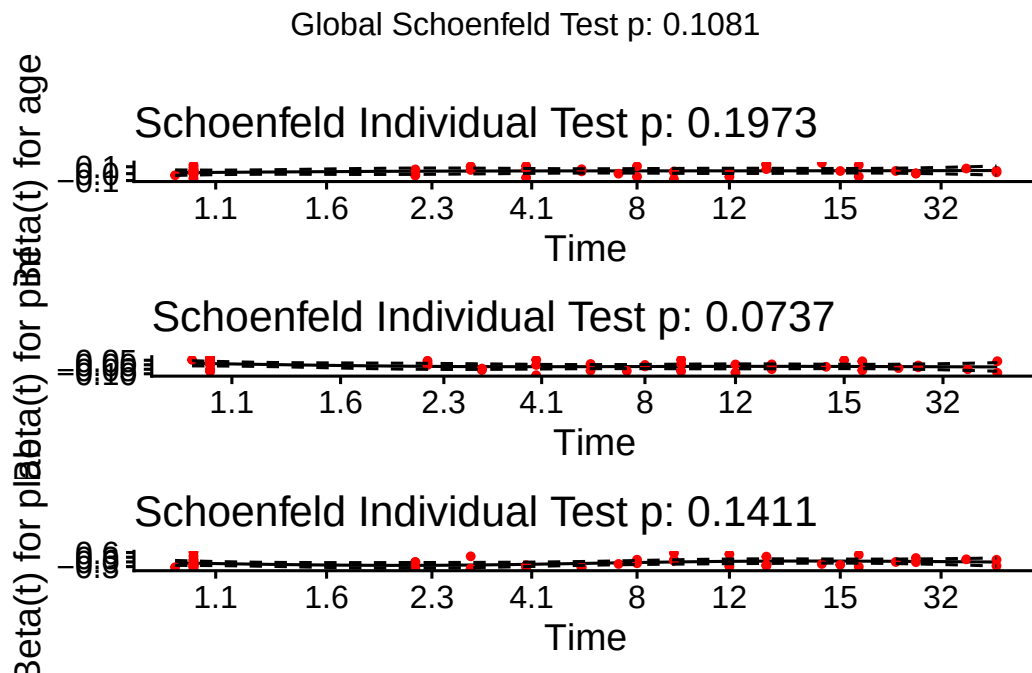


31.6.2 Checking Proportional Hazards for Model B2

```
cox.zph(modB2, transform="km", global=TRUE)
```

	chisq	df	p
age	1.66	1	0.197
pinf	3.20	1	0.074
plab	2.17	1	0.141
GLOBAL	6.07	3	0.108

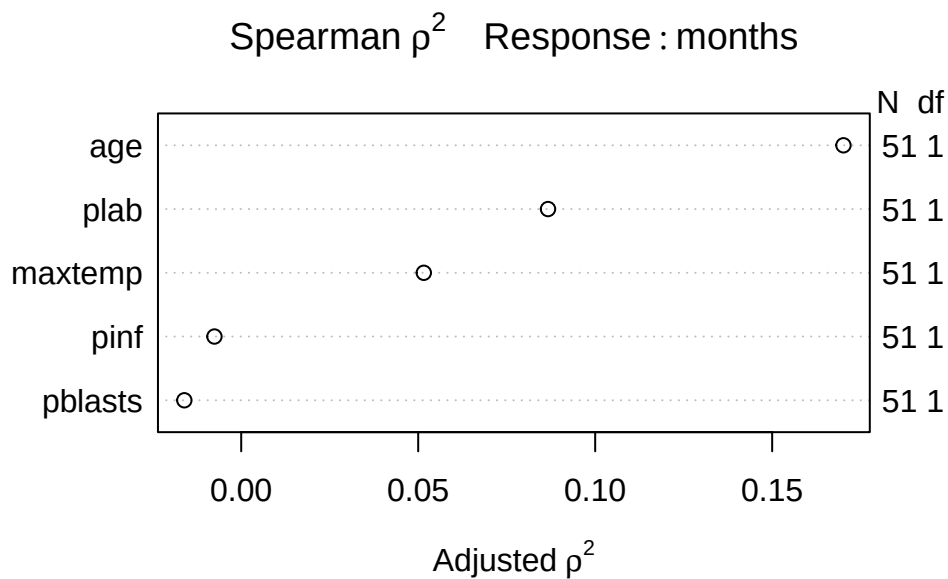
```
ggcoxzph(cox.zph(modB2))
```



31.7 Model C: Using a Spearman Plot to pick a model

If we want to use the **Spearman** ρ^2 plot to consider how we might perhaps incorporate non-linear terms describing any or all of the five potential predictors (`age`, `pblasts`, `pinf`, `plab` and `maxtemp`) for survival time, we need to do so on the raw `months` variable, rather than the survival object (`stime = Surv(months, alive==0)`) which accounts for censoring...

```
plot(spearman2(months ~ age + pblasts + pinf + plab + maxtemp, data=leukem))
```



Recognizing that we can probably only fit a small model safely (since we observe only 45 actual [uncensored] survival times) I will consider a non-linear term in **age** (specifically a restricted cubic spline with 3 knots), along with linear terms for **plab** and **maxtemp**. I'm mostly just looking for a new model to study for this example.

31.7.1 Fitting Model C

```
# still have datadist set up for leukemia
modC <- cph(Surv(months, alive==0) ~ rcs(age, 3) + plab + maxtemp,
            data=leukem, x=TRUE, y=TRUE, surv=TRUE, time.inc=12)
modC
```

Cox Proportional Hazards Model

```
cph(formula = Surv(months, alive == 0) ~ rcs(age, 3) + plab +
    maxtemp, data = leukem, x = TRUE, y = TRUE, surv = TRUE,
    time.inc = 12)
```

		Model Tests		Discrimination	
					Indexes
Obs	51	LR chi2	19.75	R2	0.322

Events	45	d.f.	4	R2(4,51)	0.266
Center	20.3856	Pr(> chi2)	0.0006	R2(4,45)	0.295
		Score chi2	18.66	Dxy	0.438
		Pr(> chi2)	0.0009		

	Coef	S.E.	Wald Z	Pr(> Z)
age	0.0804	0.0283	2.84	0.0045
age'	-0.0629	0.0332	-1.90	0.0580
plab	-0.0736	0.0381	-1.93	0.0536
maxtemp	0.1788	0.1145	1.56	0.1184

31.7.2 ANOVA for Model C

```
anova(modC)
```

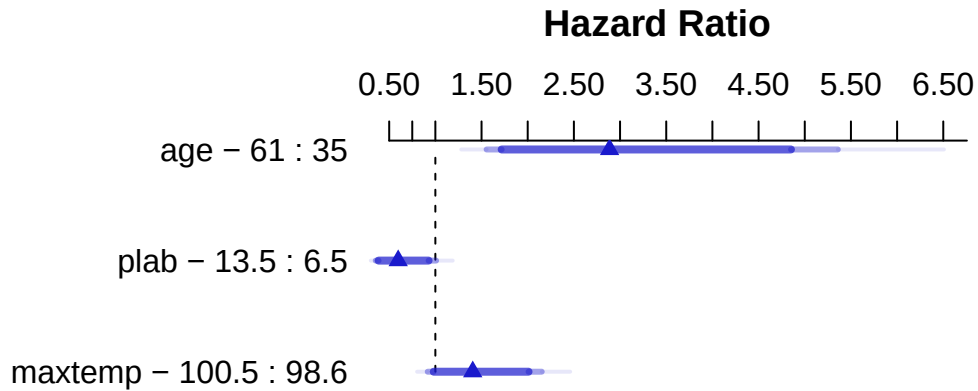
	Wald Statistics			Response: Surv(months, alive == 0)
Factor	Chi-Square	d.f.	P	
age	11.55	2	0.0031	
Nonlinear	3.59	1	0.0580	
plab	3.73	1	0.0536	
maxtemp	2.44	1	0.1184	
TOTAL	17.13	4	0.0018	

31.7.3 Summarizing Model C Effect Sizes

```
summary(modC)
```

	Effects			Response : Surv(months, alive == 0)			
Factor	Low	High	Diff.	Effect	S.E.	Lower 0.95	Upper 0.95
age	35.0	61.0	26.0	1.06010	0.31559	0.44152	1.6786000
Hazard Ratio	35.0	61.0	26.0	2.88650	NA	1.55510	5.3580000
plab	6.5	13.5	7.0	-0.51511	0.26686	-1.03820	0.0079317
Hazard Ratio	6.5	13.5	7.0	0.59744	NA	0.35411	1.0080000
maxtemp	98.6	100.5	1.9	0.33979	0.21758	-0.08666	0.7662400
Hazard Ratio	98.6	100.5	1.9	1.40470	NA	0.91699	2.1517000

```
plot(summary(modC))
```



31.7.4 Plotting the diagnosis age effect in Model C

Of course, we're no longer assuming that the log relative hazard is linear in `age`, once we include a restricted cubic spline for `age` in our Model C. So our hazard ratio and confidence intervals for `age` are a bit trickier to understand.

```
exp(coef(modC))
```

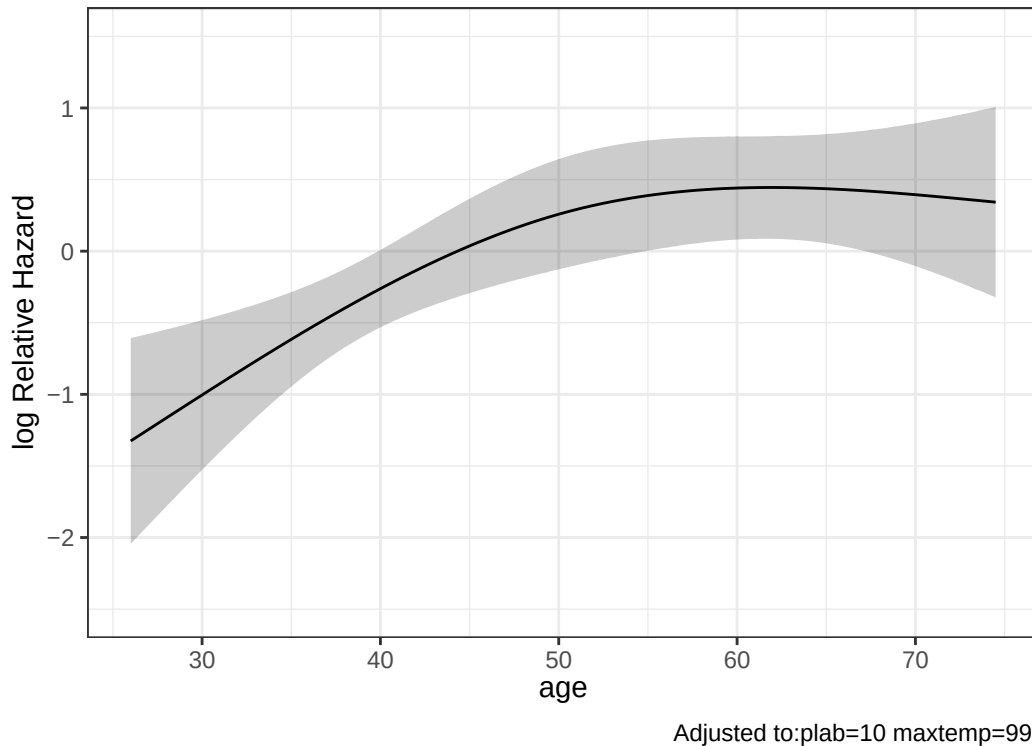
```
      age      age'      plab      maxtemp
1.0837479 0.9390008 0.9290553 1.1958265
```

```
exp(confint(modC))
```

```
      2.5 %   97.5 %
age      1.0252687 1.145563
age'     0.8798328 1.002148
plab     0.8621662 1.001134
maxtemp  0.9554141 1.496734
```


We can use `ggplot` and the `Predict` function to produce plots of the log Relative Hazard associated with any of our predictors, while holding the others constant at their medians. The effects of `maxtemp` and `plab` in our Model C are linear in the log Relative Hazard, but `age`, thanks to our use of a restricted cubic spline with 3 knots, shows a single bend.

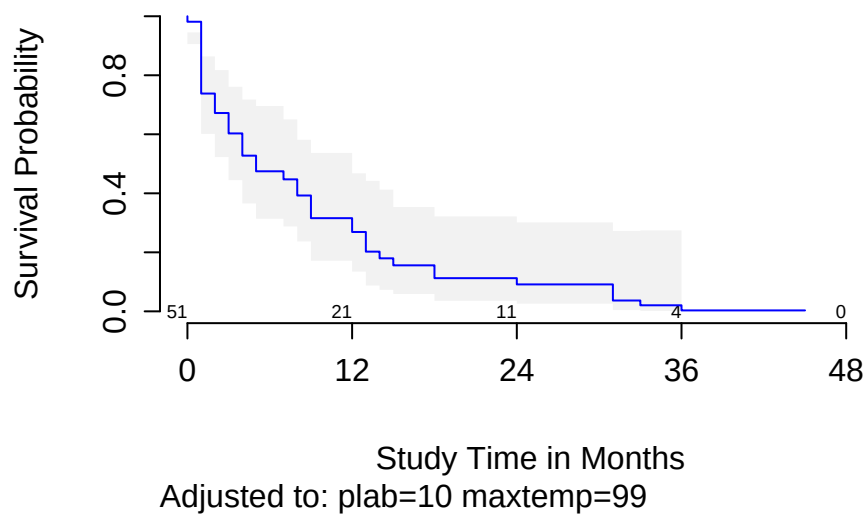
```
ggplot(Predict(modC, age))
```



31.7.5 Survival Plot associated with Model C

Let's look at a survival plot associated with Model C for a subject with median values of our three predictors.

```
survplot(modC, age=median(leukem$age), conf.int=0.95, col="blue",  
          time.inc=12, n.risk=TRUE, conf='bands',  
          xlab="Study Time in Months")
```



As before, we could fit such a plot to compare results across multiple `age` values, if desired.

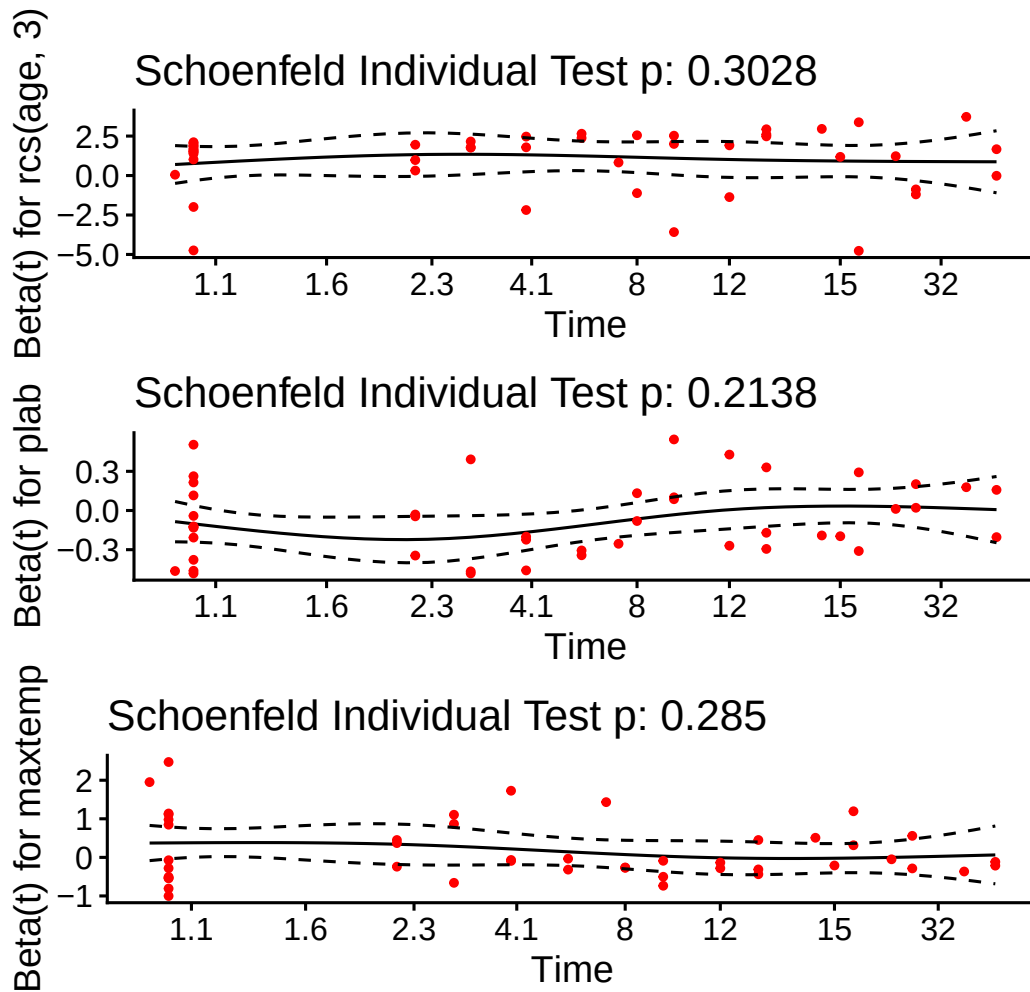
31.7.6 Checking the Proportional Hazards Assumption

```
cox.zph(modC, transform="km", global=TRUE)
```

	chisq	df	p
rcs(age, 3)	2.39	2	0.303
plab	1.55	1	0.214
maxtemp	1.14	1	0.285
GLOBAL	8.27	4	0.082

```
ggcoxzph(cox.zph(modC))
```

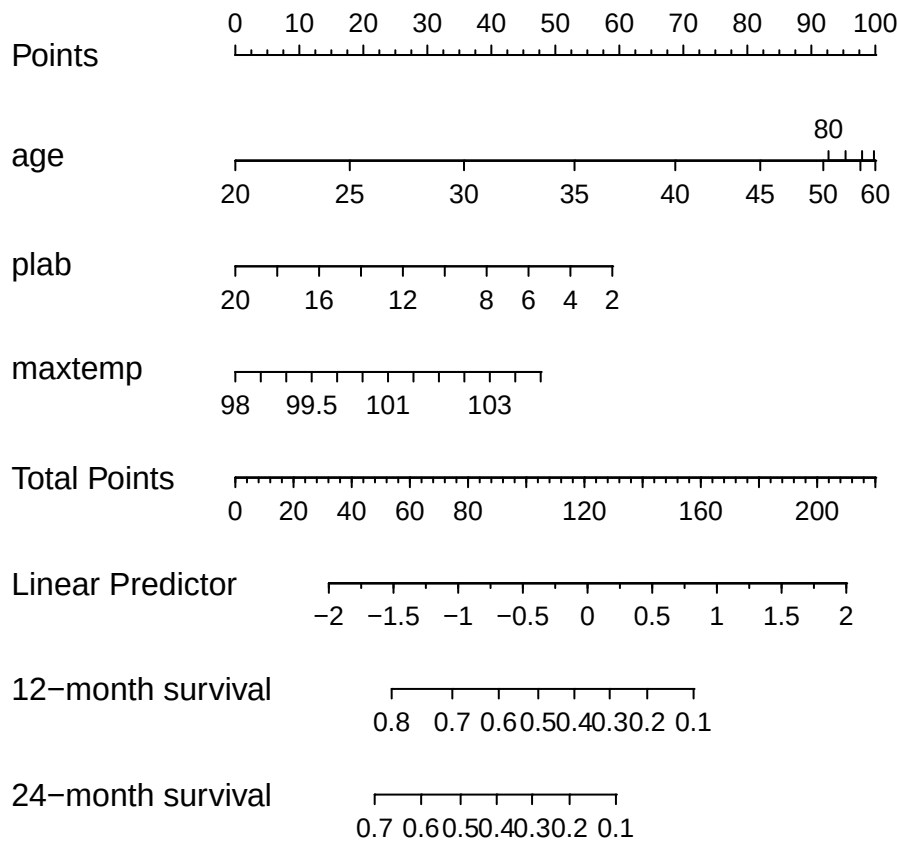
Global Schoenfeld Test p: 0.08221



31.7.7 Model C Nomogram

```
sv <- Survival(modC)
surv12 <- function(x) sv(12, lp=x)
surv24 <- function(x) sv(24, lp=x)

plot(nomogram(modC, fun=list(surv12, surv24),
  funlabel=c('12-month survival', '24-month survival')))
```



31.7.8 Validating Model C's Summary Statistics

We can validate the model for Somers' D_{xy} , which is the rank correlation between the predicted log hazard and observed survival time, and for slope shrinkage.

```
set.seed(43234); validate(modC, B=100)
```

index.orig	training	test	optimism	index.corrected	Lower	Upper	n
------------	----------	------	----------	-----------------	-------	-------	---

Dxy	0.4385	0.4749	0.4144	0.0605	0.3780	0.2034	0.5585	100
R2	0.3222	0.3716	0.2785	0.0931	0.2292	-0.0055	0.4436	100
Slope	1.0000	1.0000	0.8101	0.1899	0.8101	0.3388	1.3979	100
D	0.0656	0.0814	0.0547	0.0266	0.0390	-0.0303	0.0962	100
U	-0.0070	-0.0070	0.0101	-0.0171	0.0101	-0.0093	0.0836	100
Q	0.0726	0.0884	0.0446	0.0437	0.0288	-0.1076	0.0977	100
g	0.9466	1.0751	0.8364	0.2387	0.7078	0.0293	1.2119	100

31.7.9 Calibration of Model C (12-month survival estimates)

Finally, we validate the model for calibration accuracy in predicting the probability of surviving one year.

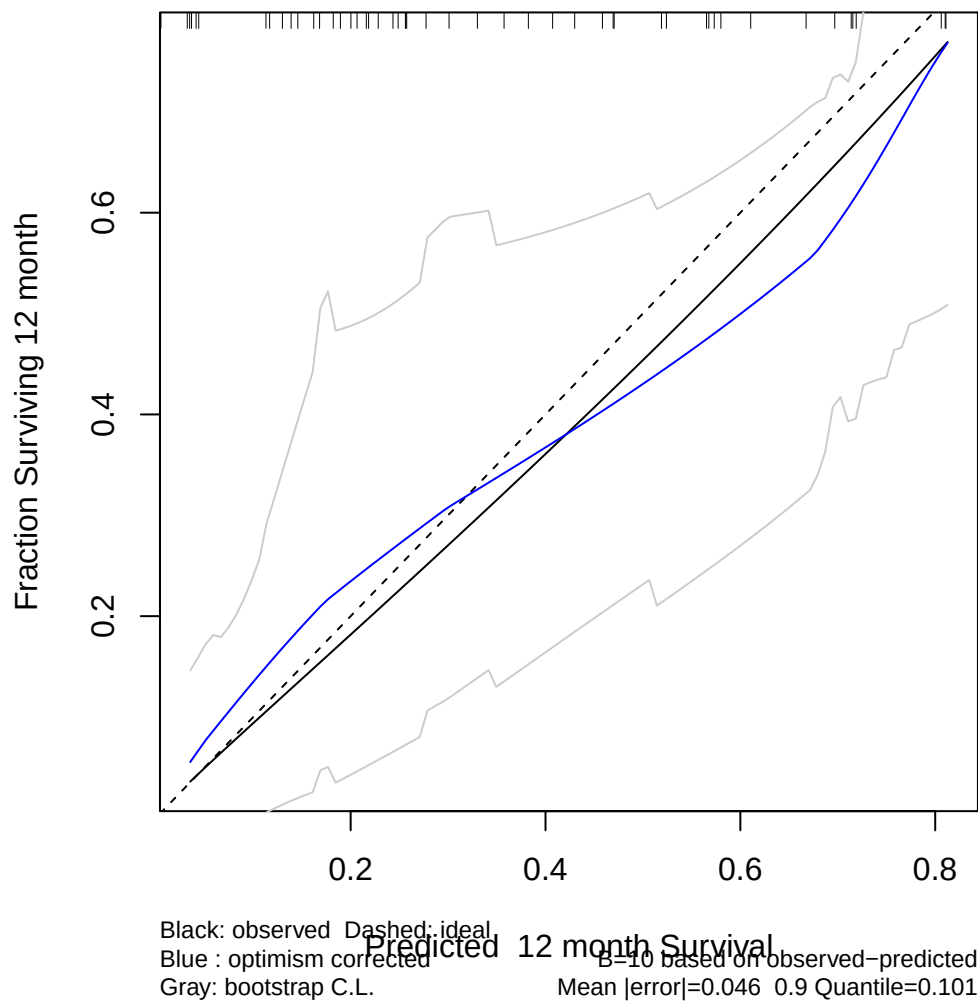
Quoting Harrell (page 529, RMS second edition):

The bootstrap is used to estimate the optimism in how well predicted [one]-year survival from the final Cox model tracks flexible smooth estimates, without any binning of predicted survival probabilities or assuming proportional hazards.

The `u` variable specifies the length of time at which we look at the calibration. I've specified the `units` earlier to be months.

```
set.seed(43233); plot(rms::calibrate(modC, B = 10, u = 12))
```

Using Cox survival estimates at 12 months



The model seems neither especially well calibrated nor especially poorly so - looking at the comparison of the blue curve to the gray, our predictions basically aren't aggressive enough - more people are surviving to a year in our low predicted probability of 12 month survival group, and fewer people are surviving on the high end of the x-axis than should be the case.

32 NEW!! A Few LASSO Ideas

32.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(janitor)
library(broom)
library(car)
library(gt)
library(Hmisc)
library(MASS)
library(mosaic)
library(glmnet)
library(conflicted)
library(patchwork)
library(tidyverse)

conflicts_prefer(dplyr::select, base::max)

theme_set(theme_bw())
```

32.1.1 Data Load

```
pollution <- read_csv("data/pollution.csv", show_col_types = FALSE)
```

32.2 The pollution data

Consider again the `pollution` data set we developed in Chapter 13 which contains 15 independent variables and a measure of mortality, describing 60 US metropolitan areas in 1959-1961.

pollution

```
# A tibble: 60 x 16
      x1    x2    x3    x4    x5    x6    x7    x8    x9    x10    x11    x12    x13
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1    13    49    68    7    3.36 12.2  90.7 2702    3    51.9    9.7   105    32
2    28    32    81    7    3.27 12.1   81   3665    7.5   51.6   13.2     4     2
3    10    55    70   7.3   3.11 12.1  88.9 3033    5.9    51    14   144    66
4    43    32    74  10.1   3.38  9.5  79.2 3214    2.9   43.7    12     11     7
5    25    12    73   9.2   3.28 12.1  83.1 2095     2   51.9    9.8    20    11
6    35    46    85   7.1   3.22 11.8  79.9 1441   14.8   51.2   16.1     1     1
7    60    67    82   10    2.98 11.5  88.6 4657   13.5   47.3   22.4     3     1
8    11    53    68   9.2   2.99 12.1  90.6 4700    7.8   48.9   12.3   648   319
9    31    24    72    9    3.37 10.9  82.8 3226    5.1   45.2   12.3     5     3
10   15    30    73   8.2   3.15 12.2  84.2 4824    4.7   53.1   12.7    17     8
# i 50 more rows
# i 3 more variables: x14 <dbl>, x15 <dbl>, y <dbl>
```

Here's a codebook:

Vari- able	Description
y	Total Age Adjusted Mortality Rate
x1	Mean annual precipitation in inches
x2	Mean January temperature in degrees Fahrenheit
x3	Mean July temperature in degrees Fahrenheit
x4	Percent of 1960 SMSA population that is 65 years of age or over
x5	Population per household, 1960 SMSA
x6	Median school years completed for those over 25 in 1960 SMSA
x7	Percent of housing units that are found with facilities
x8	Population per square mile in urbanized area in 1960
x9	Percent of 1960 urbanized area population that is non-white
x10	Percent employment in white-collar occupations in 1960 urbanized area
x11	Percent of families with income under \$30,000 in 1960 urbanized area
x12	Relative population potential of hydrocarbons, HC
x13	Relative pollution potential of oxides of nitrogen, NOx
x14	Relative pollution potential of sulfur dioxide, SO2
x15	Percent relative humidity, annual average at 1 p.m.

32.3 Should We Rescale any Predictors?

Let's get some basic summary statistics for our candidate predictors. When we do, we see that variable `x8` is roughly 100 times larger than all of our other predictors.

```
df_stats(~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 +  
          x10 + x11 + x12 + x13 + x14 + x15, data = pollution)
```

	response	min	Q1	median	Q3	max	mean	sd
1	x1	10.00	32.750	38.000	43.250	60.00	37.366667	9.9846775
2	x2	12.00	27.000	31.500	40.000	67.00	33.983333	10.1688985
3	x3	63.00	72.000	74.000	77.250	85.00	74.583333	4.7631768
4	x4	5.60	7.675	9.000	9.700	11.80	8.798333	1.4645520
5	x5	2.92	3.210	3.265	3.360	3.53	3.263167	0.1352523
6	x6	9.00	10.400	11.050	11.500	12.30	10.973333	0.8452994
7	x7	66.80	78.375	81.150	83.600	90.70	80.913333	5.1413731
8	x8	1441.00	3104.250	3567.000	4519.750	9699.00	3876.050000	1454.1023607
9	x9	0.80	4.950	10.400	15.650	38.50	11.870000	8.9211480
10	x10	33.80	43.250	45.500	49.525	59.70	46.081667	4.6130431
11	x11	9.40	12.000	13.200	15.150	26.40	14.373333	4.1600956
12	x12	1.00	7.000	14.500	30.250	648.00	37.850000	91.9776732
13	x13	1.00	4.000	9.000	23.750	319.00	22.650000	46.3332896
14	x14	1.00	11.000	30.000	69.000	278.00	53.766667	63.3904678
15	x15	38.00	55.000	57.000	60.000	73.00	57.666667	5.3699309
n missing								
1	60	0						
2	60	0						
3	60	0						
4	60	0						
5	60	0						
6	60	0						
7	60	0						
8	60	0						
9	60	0						
10	60	0						
11	60	0						
12	60	0						
13	60	0						
14	60	0						
15	60	0						

Let's dampen the size of that `x8` variable down a little to make our coefficient comparisons easier later, by dividing all of the `x8` values by 100.

```
pollution <- pollution |>
  mutate(x8 = x8/100)
```

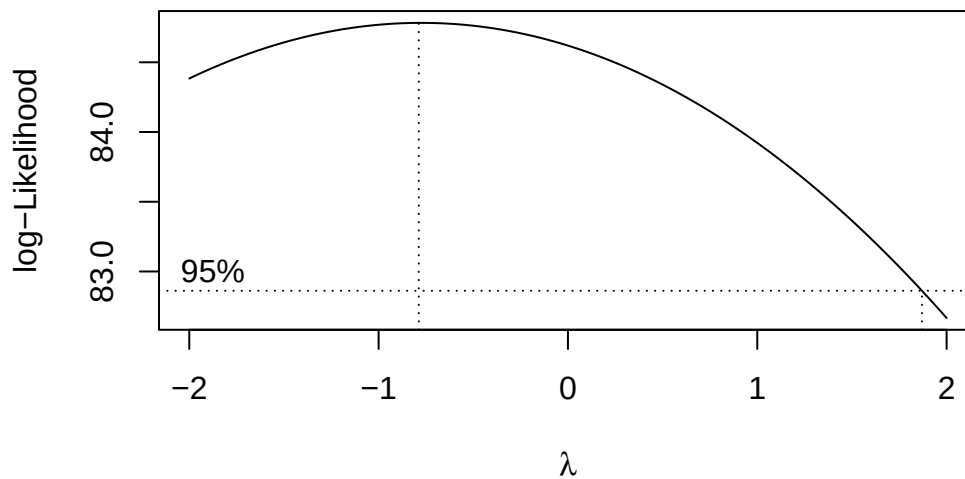
32.4 A Kitchen Sink Model

We'll begin by fitting the obviously underpowered model with all 15 main effects used to predict our outcome.

```
mod_sink <- lm(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 +
  x10 + x11 + x12 + x13 + x14 + x15,
  data = pollution)
```

32.4.1 Considering an Outcome Transformation

```
boxcox(mod_sink)
```

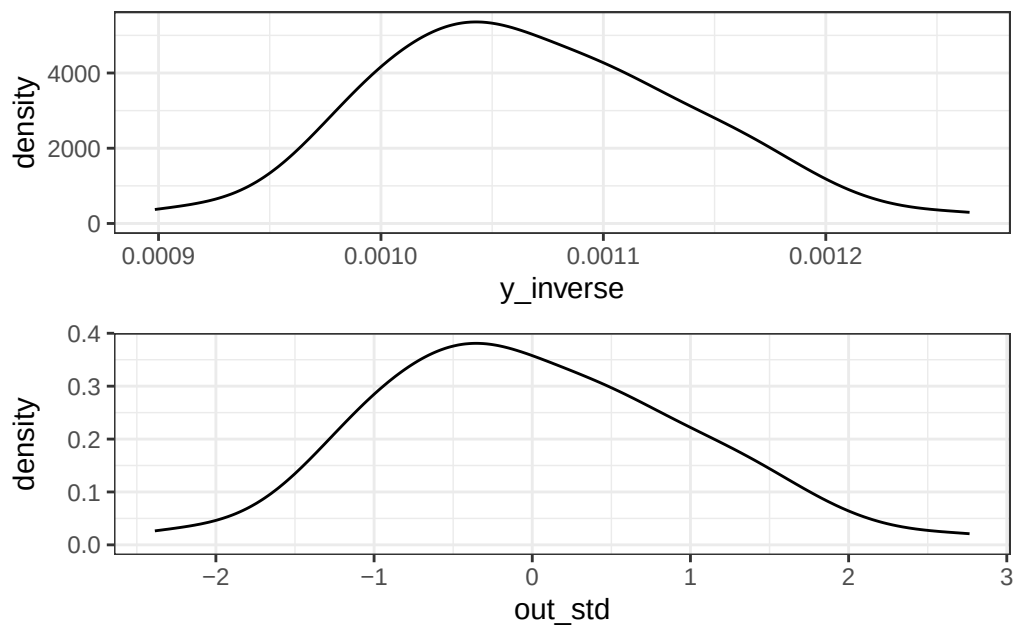


OK. In light of this Box-Cox plot, let's consider taking the inverse of our outcome here. I'll take that inverse and then standardize the result using the `scale()` function to both subtract the mean of the transformed outcome and divide by its standard deviation, so that our new outcome, which I'll call `out_std` has mean 0, standard deviation 1, and a shape similar to that of the inverse of our y .

```
pollution <- pollution |>
  mutate(y_inverse = 1/y,
         out_std = scale(1/y, center = TRUE, scale = TRUE))

p1 <- ggplot(pollution, aes(x = y_inverse)) + geom_density()
p2 <- ggplot(pollution, aes(x = out_std)) + geom_density()

p1 / p2
```



OK. So now, I'll build a revised kitchen sink model to use this new outcome.

```
mod_sink2 <- lm(out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 +
               x10 + x11 + x12 + x13 + x14 + x15, data = pollution)

tidy(mod_sink2, conf.int = TRUE, conf.level = 0.9) |>
  select(term, estimate, std.error, conf.low, conf.high, p.value) |>
  gt() |> fmt_number(decimals = 3)
```

term	estimate	std.error	conf.low	conf.high	p.value
(Intercept)	-15.899	6.886	-27.469	-4.328	0.026
x1	-0.035	0.015	-0.059	-0.011	0.020
x2	0.036	0.017	0.007	0.066	0.044
x3	0.055	0.030	0.005	0.105	0.074
x4	0.176	0.134	-0.049	0.400	0.196
x5	2.184	1.099	0.338	4.031	0.053
x6	0.225	0.187	-0.089	0.539	0.234
x7	0.016	0.028	-0.031	0.063	0.573
x8	-0.006	0.006	-0.017	0.005	0.351
x9	-0.070	0.021	-0.105	-0.035	0.002
x10	0.012	0.026	-0.031	0.056	0.636
x11	0.002	0.051	-0.084	0.087	0.974
x12	0.011	0.008	-0.002	0.024	0.160
x13	-0.022	0.016	-0.049	0.004	0.169
x14	-0.001	0.002	-0.005	0.003	0.549
x15	0.001	0.018	-0.029	0.032	0.936

r.squared	adj.r.squared	sigma	statistic	p.value	df	df.residual	nobs
0.774	0.698	0.550	10.070	0.000	15	44	60

Does this new model show a strong fit to the data?

```
glance(mod_sink2) |> select(r.squared:p.value, df, df.residual, nobs) |>
  gt() |> fmt_number(r.squared:p.value, decimals = 3)
```

```
glance(mod_sink2) |> select(logLik, AIC, BIC, deviance) |> gt() |>
  fmt_number(decimals = 2)
```

32.4.2 How much collinearity are we dealing with?

logLik	AIC	BIC	deviance
-39.96	113.92	149.52	13.31

```
vif(mod_sink2)
```

x1	x2	x3	x4	x5	x6	x7
4.113888	6.143551	3.967774	7.470045	4.307618	4.860538	3.994781
x8	x9	x10	x11	x12	x13	x14
1.658281	6.779599	2.841582	8.717068	98.639935	104.982405	4.228929
x15						
1.907092						

Clearly, we have some enormous collinearity to deal with, given that we have many VIFs over 5, some even over 100.

So a reduction in the size of the model seems appealing for multiple reasons.

32.5 Using the LASSO to suggest a smaller model

To begin, we will create a data matrix for our predictors, as follows:

```
pred_x <- model.matrix(mod_sink2)
```

Next, we create a matrix of our outcome.

```
out_y <- pollution |> select(out_std) |> as.matrix()
```

The LASSO involves both a cross-validation step, and a fitting step. Here's the code we'll use in this case:

```
set.seed(123456)

cv_poll1 <- cv.glmnet(pred_x, out_y, type.measure = "mse", nfolds = 10)

mod_las1 <- glmnet(pred_x, out_y, alpha = 1, lambda = cv_poll1$lambda.min)
```

Now, let's look at what the LASSO does. As we can see from the tidied output below, some predictors are dropped from the model, while others have their coefficients shrunk towards zero as compared to what we saw in the “kitchen sink” model.

term	step	estimate	lambda	dev.ratio
(Intercept)	1	-2.345226117	0.03783146	0.7335261
x1	1	-0.026289494	0.03783146	0.7335261
x2	1	0.020872173	0.03783146	0.7335261
x3	1	0.017666053	0.03783146	0.7335261
x6	1	0.101973313	0.03783146	0.7335261
x7	1	0.012515424	0.03783146	0.7335261
x8	1	-0.006425945	0.03783146	0.7335261
x9	1	-0.060226425	0.03783146	0.7335261
x10	1	0.007342706	0.03783146	0.7335261
x14	1	-0.003819796	0.03783146	0.7335261

```
tidy(mod_las1) |> gt()
```

This new LASSO model includes only 9 of the original 15 predictors.

32.6 Would the 9-predictor model be a big improvement?

Suppose we fit a new model inspired by this LASSO. It's still just a linear model, with no shrinkage, here.

```
mod_3 <- lm(out_std ~ x1 + x2 + x3 + x6 + x7 + x8 + x9 + x10 + x14,
             data = pollution)

vif(mod_3)
```

```
      x1      x2      x3      x6      x7      x8      x9      x10
1.850485 1.548039 1.867603 3.899194 2.318288 1.479122 2.281879 2.333836
      x14
1.609159
```

Well, the collinearity is certainly much improved.

```
tidy(mod_3, conf.int = TRUE, conf.level = 0.9) |>
  select(term, estimate, std.error, conf.low, conf.high, p.value) |>
  gt() |> fmt_number(decimals = 3)
```

Does this new model show a strong fit to the data?

term	estimate	std.error	conf.low	conf.high	p.value
(Intercept)	-4.122	2.254	-7.900	-0.344	0.073
x1	-0.031	0.010	-0.048	-0.015	0.002
x2	0.024	0.009	0.010	0.039	0.007
x3	0.040	0.020	0.005	0.074	0.058
x6	0.057	0.166	-0.222	0.335	0.734
x7	0.021	0.021	-0.014	0.056	0.328
x8	-0.009	0.006	-0.019	0.001	0.129
x9	-0.069	0.012	-0.090	-0.049	0.000
x10	0.013	0.024	-0.026	0.053	0.581
x14	-0.004	0.001	-0.006	-0.002	0.007

r.squared	adj.r.squared	sigma	statistic	p.value	df	df.residual	nobs
0.747	0.702	0.546	16.413	0.000	9	50	60

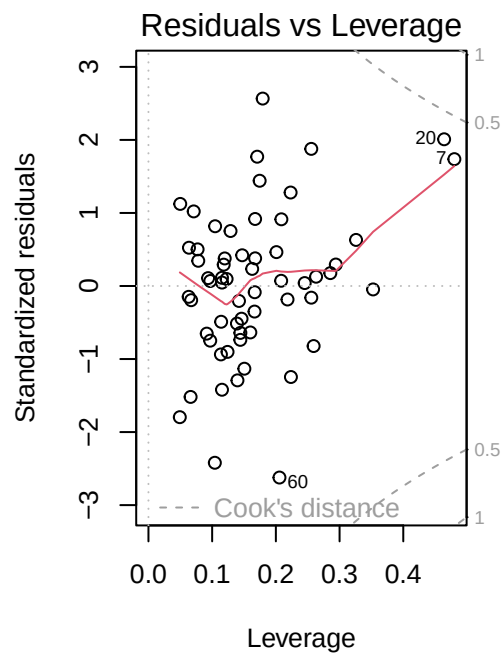
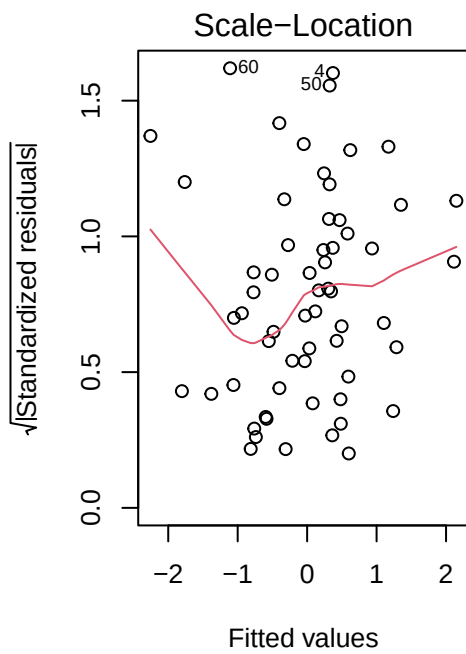
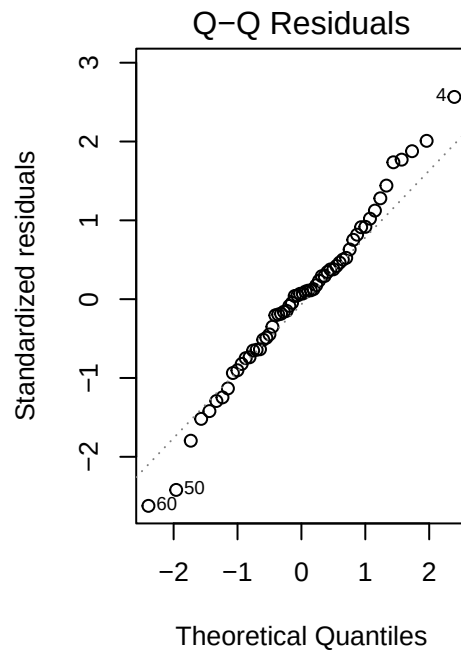
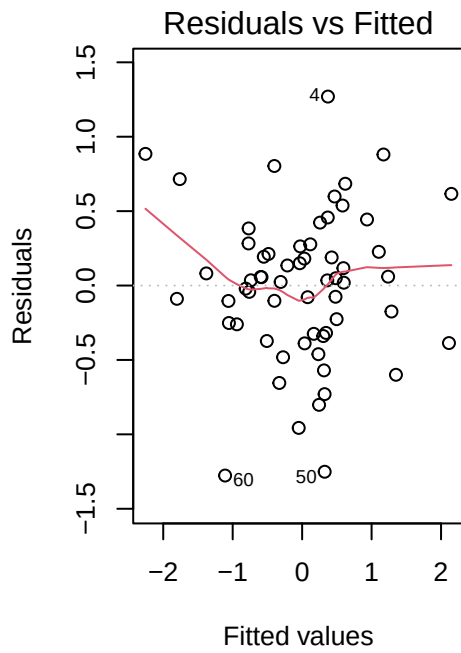
```
glance(mod_3) |> select(r.squared:p.value, df, df.residual, nobs) |>
  gt() |> fmt_number(r.squared:p.value, decimals = 3)
```

```
glance(mod_3) |> select(logLik, AIC, BIC, deviance) |> gt() |>
  fmt_number(decimals = 2)
```

Finally, here's a set of plots for regression diagnostics. How do things look?

```
par(mfrow = c(2,2))
plot(mod_3)
```

logLik	AIC	BIC	deviance
-43.39	108.78	131.81	14.92



32.7 Using Stepwise Regression to suggest a smaller model

```
mod_4 <- step(mod_sink2)
```

Start: AIC=-58.35

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x10 +
x11 + x12 + x13 + x14 + x15

	Df	Sum of Sq	RSS	AIC
- x11	1	0.0003	13.310	-60.351
- x15	1	0.0020	13.311	-60.344
- x10	1	0.0689	13.378	-60.043
- x7	1	0.0976	13.407	-59.914
- x14	1	0.1102	13.420	-59.858
- x8	1	0.2693	13.579	-59.151
- x6	1	0.4401	13.749	-58.401
<none>			13.309	-58.353
- x4	1	0.5226	13.832	-58.042
- x13	1	0.5906	13.900	-57.747
- x12	1	0.6186	13.928	-57.627
- x3	1	1.0145	14.324	-55.945
- x5	1	1.1956	14.505	-55.191
- x2	1	1.3063	14.616	-54.735
- x1	1	1.7574	15.067	-52.911
- x9	1	3.4141	16.723	-46.652

Step: AIC=-60.35

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x10 +
x12 + x13 + x14 + x15

	Df	Sum of Sq	RSS	AIC
- x15	1	0.0017	13.311	-62.343
- x10	1	0.0696	13.379	-62.038
- x14	1	0.1119	13.422	-61.849
- x7	1	0.1411	13.451	-61.719
- x8	1	0.2752	13.585	-61.123
- x6	1	0.4398	13.749	-60.401
<none>			13.310	-60.351
- x13	1	0.5905	13.900	-59.747
- x4	1	0.6048	13.915	-59.685
- x12	1	0.6202	13.930	-59.618

- x3	1	1.0419	14.352	-57.829
- x5	1	1.2132	14.523	-57.117
- x1	1	1.8672	15.177	-54.474
- x2	1	2.3995	15.709	-52.406
- x9	1	4.5934	17.903	-44.562

Step: AIC=-62.34

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x10 +
x12 + x13 + x14

	Df	Sum of Sq	RSS	AIC
- x10	1	0.0680	13.380	-64.037
- x14	1	0.1102	13.422	-63.849
- x7	1	0.1398	13.451	-63.717
- x8	1	0.2800	13.591	-63.094
<none>			13.311	-62.343
- x6	1	0.4538	13.765	-62.332
- x4	1	0.6066	13.918	-61.669
- x13	1	0.6080	13.919	-61.664
- x12	1	0.6251	13.937	-61.590
- x5	1	1.2142	14.526	-59.106
- x3	1	1.5567	14.868	-57.707
- x1	1	1.8661	15.178	-56.472
- x2	1	2.7077	16.019	-53.234
- x9	1	4.6990	18.010	-46.204

Step: AIC=-64.04

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x12 +
x13 + x14

	Df	Sum of Sq	RSS	AIC
- x14	1	0.0750	13.454	-65.702
- x7	1	0.1158	13.495	-65.520
- x8	1	0.2459	13.625	-64.945
<none>			13.380	-64.037
- x4	1	0.6708	14.050	-63.102
- x12	1	0.7497	14.129	-62.766
- x13	1	0.7575	14.137	-62.733
- x6	1	1.1206	14.500	-61.212
- x5	1	1.1650	14.544	-61.028
- x3	1	1.6471	15.027	-59.071
- x1	1	1.9708	15.350	-57.793
- x2	1	2.8845	16.264	-54.323

- x9 1 4.6459 18.025 -48.154

Step: AIC=-65.7

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x12 +
x13

	Df	Sum of Sq	RSS	AIC
- x7	1	0.0736	13.528	-67.375
- x8	1	0.2931	13.748	-66.409
<none>			13.454	-65.702
- x4	1	0.7633	14.218	-64.391
- x5	1	1.3276	14.782	-62.056
- x6	1	1.5095	14.964	-61.322
- x3	1	1.6936	15.148	-60.588
- x1	1	1.9047	15.359	-59.758
- x12	1	2.2433	15.698	-58.450
- x13	1	2.5554	16.010	-57.269
- x2	1	3.2927	16.747	-54.567
- x9	1	4.9296	18.384	-48.972

Step: AIC=-67.37

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x8 + x9 + x12 + x13

	Df	Sum of Sq	RSS	AIC
- x8	1	0.2268	13.755	-68.377
<none>			13.528	-67.375
- x4	1	0.6973	14.225	-66.359
- x5	1	1.2978	14.826	-63.878
- x3	1	1.6230	15.151	-62.576
- x1	1	1.8418	15.370	-61.716
- x6	1	2.1350	15.663	-60.582
- x12	1	2.2355	15.764	-60.199
- x13	1	2.5168	16.045	-59.137
- x2	1	3.3522	16.880	-56.092
- x9	1	6.1854	19.713	-46.783

Step: AIC=-68.38

out_std ~ x1 + x2 + x3 + x4 + x5 + x6 + x9 + x12 + x13

	Df	Sum of Sq	RSS	AIC
<none>			13.755	-68.377
- x4	1	0.8670	14.622	-66.710
- x3	1	1.6985	15.453	-63.391

term	estimate	std.error	conf.low	conf.high	p.value
(Intercept)	-17.332	5.257	-26.142	-8.522	0.002
x1	-0.034	0.013	-0.057	-0.012	0.012
x2	0.042	0.011	0.024	0.060	0.000
x3	0.055	0.022	0.018	0.092	0.016
x4	0.200	0.113	0.011	0.388	0.082
x5	2.530	0.942	0.951	4.109	0.010
x6	0.373	0.111	0.186	0.559	0.002
x9	-0.073	0.015	-0.099	-0.048	0.000
x12	0.015	0.005	0.007	0.023	0.004
x13	-0.031	0.010	-0.047	-0.015	0.002

r.squared	adj.r.squared	sigma	statistic	p.value	df	df.residual	nobs
0.767	0.725	0.524	18.274	0.000	9	50	60

```

- x1      1      1.8777 15.633 -62.700
- x5      1      1.9839 15.739 -62.293
- x12     1      2.4622 16.217 -60.497
- x13     1      2.8317 16.587 -59.145
- x6      1      3.0868 16.842 -58.229
- x2      1      4.0486 17.803 -54.897
- x9      1      6.3106 20.065 -47.721

```

Here's a summary of the fitted model after stepwise regression, which suggests a different set of 9 predictors.

```

tidy(mod_4, conf.int = TRUE, conf.level = 0.9) |>
  select(term, estimate, std.error, conf.low, conf.high, p.value) |>
  gt() |> fmt_number(decimals = 3)

```

```

glance(mod_4) |> select(r.squared:p.value, df, df.residual, nobs) |>
  gt() |> fmt_number(r.squared:p.value, decimals = 3)

```

```

glance(mod_4) |> select(logLik, AIC, BIC, deviance) |> gt() |>
  fmt_number(decimals = 2)

```

How is the collinearity in this model?

logLik	AIC	BIC	deviance
-40.95	103.90	126.93	13.75

```
vif(mod_4)
```

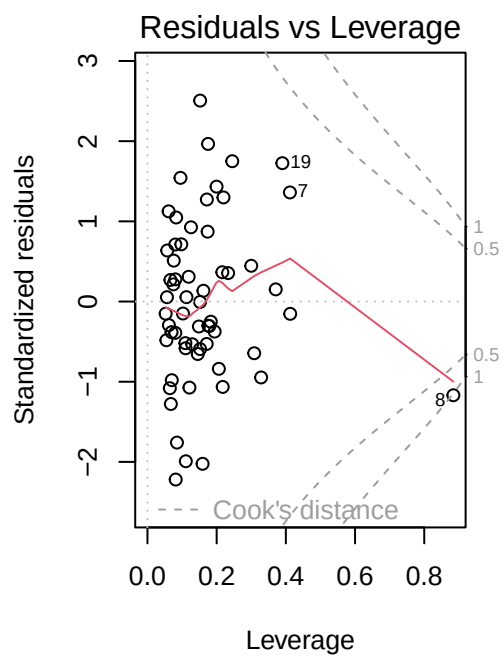
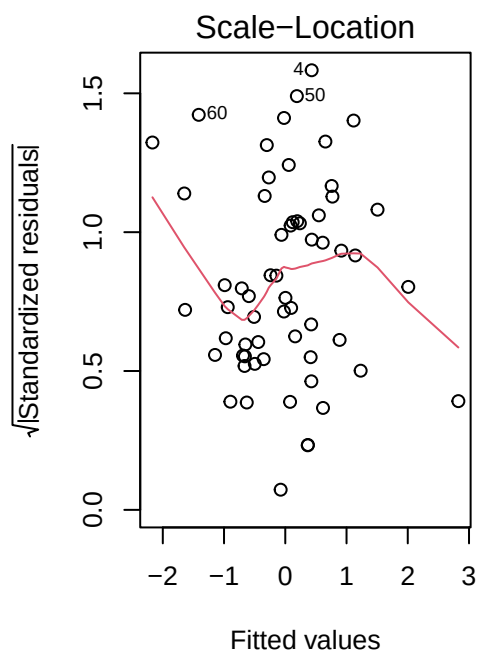
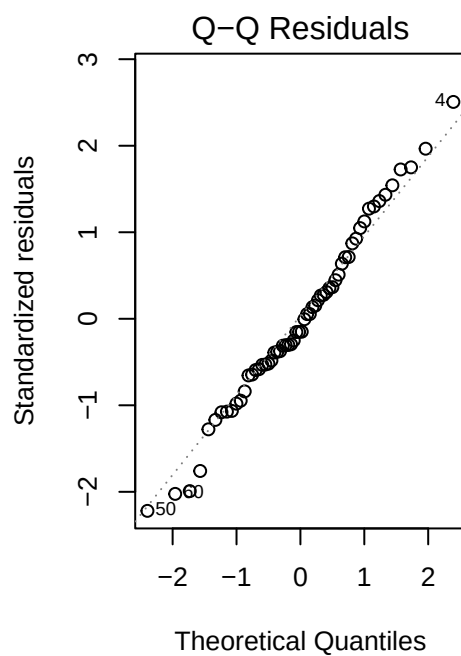
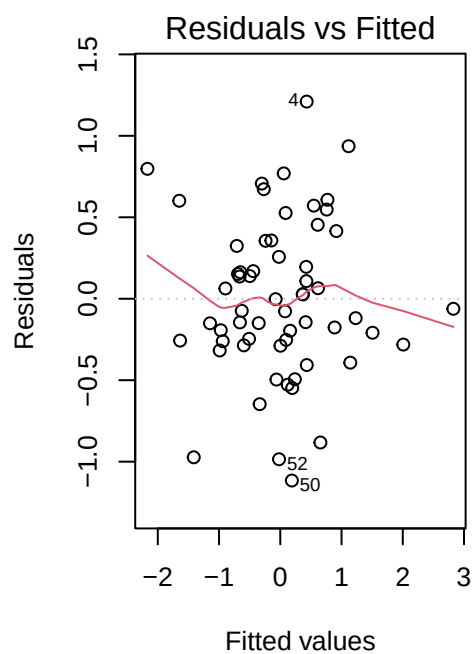
```

      x1      x2      x3      x4      x5      x6      x9      x12
3.724589 2.666789 2.359891 5.826631 3.483085 1.898922 3.981753 45.943057
      x13
42.500579

```

That looks more troubling to me, at least as compared to `mod_3`. How about the residual plots? Do those for `mod_4` below look meaningfully different from the ones we built for our LASSO-inspired model `mod_3`?

```
par(mfrow = c(2,2))
plot(mod_4)
```



We now seem to have a point with a pretty substantial Cook's distance, specifically the point from row 8 of the data.

```
pollution |> slice(8) |> select(x1:x12)
```

```
# A tibble: 1 x 12
   x1    x2    x3    x4    x5    x6    x7    x8    x9    x10   x11   x12
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1    11    53    68   9.2   2.99  12.1  90.6    47    7.8   48.9   12.3   648
```

```
pollution |> slice(8) |> select(x13:x15, y, y_inverse, out_std)
```

```
# A tibble: 1 x 6
   x13  x14  x15    y y_inverse out_std[,1]
<dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>
1   319   130    47 862.    0.00116    1.30
```

So what is unusual about Row 8? Well, it has an especially large value of x12 compared to the rest of the data.

```
describe(pollution$x12)
```

```
pollution$x12
  n missing distinct    Info    Mean  pMedian    Gmd    .05
  60      0      34   0.998   37.85     18    52.03   3.95
.10   .25   .50   .75    .90    .95
5.00   7.00  14.50  30.25  56.90  106.95

lowest :    1    3    4    5    6, highest:  88 105 144 311 648
```

That might be part of the problem, especially since stepwise regression maintains variable x12 whereas our LASSO-inspired model (`mod_3`) does not.

33 NEW!! Bayes and a Linear Model

Almost all of this material is based on

- <https://mc-stan.org/rstanarm/articles/continuous.html> and
- <https://easystats.github.io/bayestestR/articles/bayestestR.html> and
- <https://easystats.github.io/bayestestR/articles/example1.html>

There's not a lot that is truly original here. That's a summer project.

33.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(broom.mixed)
library(gt)
library(janitor)
library(mosaic)
library(bayestestR)
library(insight)
library(rstanarm)
library(conflicted)
library(tidyverse)

conflicts_prefer(dplyr::select, dplyr::filter,
                 base::mean, base::range, base::max)

theme_set(theme_bw())
```

33.2 Return to the `smalldat` Example

Consider the `smalldat.csv` data we discussed in Chapter 22. The data includes 150 observations on 6 variables, and one of our goals was to build a model to predict total cholesterol.

r.squared	adj.r.squared	sigma	AIC	BIC	df	df.residual	nobs
0.078	0.039	37.998	1525.772	1549.857	6	143	150

```
smalldat <- read_csv("data/smalldat.csv", show_col_types = FALSE)
```

Variable	Description
<i>subject</i>	Subject identification code
<i>smoker</i>	1 = current smoker, 0 = not current smoker
<i>totchol</i>	total cholesterol, in mg/dl
<i>age</i>	age in years
<i>sex</i>	subject's sex (M or F)
<i>educ</i>	subject's educational attainment (4 levels)

The *educ* levels are: 1_Low, 2_Middle, 3_High and 4_VHigh, which stands for Very High

33.3 The Distribution of Total Cholesterol

Across the 150 observations in the `smalldat` data, we have the following summaries of the distribution of total cholesterol, our outcome.

```
favstats(~ totchol, data = smalldat)
```

```
min      Q1 median      Q3 max    mean      sd    n missing
150 209.25   237 265.75 332 236.78 38.76809 150      0
```

33.4 Fitting a Linear Model with `lm()` for Total Cholesterol

```
m1 <- lm(totchol ~ age + sex + smoker + factor(educ), data = smalldat)

glance(m1) |> select(r.squared, adj.r.squared, sigma,
                    AIC, BIC, df, df.residual, nobs) |>
  gt() |> fmt_number(r.squared:sigma, decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	167.153	21.639	7.725	0.000	124.380	209.926
age	1.248	0.379	3.296	0.001	0.500	1.996
sexM	3.153	6.515	0.484	0.629	-9.725	16.030
smoker	3.462	6.526	0.531	0.597	-9.438	16.363
factor(educ)2_Middle	11.330	7.740	1.464	0.145	-3.970	26.631
factor(educ)3_High	-2.202	9.426	-0.234	0.816	-20.834	16.431
factor(educ)4_VHigh	11.743	9.925	1.183	0.239	-7.875	31.361

```
tidy(m1, conf.int = TRUE, conf.level = 0.95) |>
  gt() |> fmt_number(decimals = 3)
```

33.5 Fitting a Bayesian Linear Model

Can we fit a model for the same data using a Bayesian approach?

Yes, we can, for instance using the `stan_glm()` function from the `rstanarm` package.

```
set.seed(43211234) # best to set a random seed first

m2 <- stan_glm(totchol ~ age + sex + smoker + factor(educ),
  data = smalldat, refresh = 0)
```

Here the `refresh = 0` parameter stops the machine from printing out each of the updates it does while sampling, which is not generally something I need to look at. Here's what's placed in the `m2` object:

```
m2
```

```
stan_glm
family:      gaussian [identity]
formula:     totchol ~ age + sex + smoker + factor(educ)
observations: 150
predictors:  7
-----
              Median MAD_SD
(Intercept)   166.3    21.5
age            1.3     0.4
```

```
sexM          3.1    6.5
smoker        3.8    6.3
factor(educ)2_Middle 11.4  7.7
factor(educ)3_High  -2.4  9.5
factor(educ)4_VHigh 11.7  9.9
```

```
Auxiliary parameter(s):
      Median MAD_SD
sigma 38.1    2.4
```

```
* For help interpreting the printed output see ?print.stanreg
* For info on the priors used see ?prior_summary.stanreg
```

- The first few lines specify the fitting process.
- Next, for each coefficient, we find the median value from the posterior distribution, and the MAD_SD value, which is an indicator of variation derived from the estimated posterior distribution of the parameters, and is used as a standard error in what follows.
- Finally, we see the estimated root mean squared error (residual standard deviation) `sigma`, again estimated with the median of the `sigma` values in the posterior distribution.

Using the `summary()` function provides some additional information about the parameter estimates, but mostly some convergence diagnostics for the Markov Chain Monte Carlo procedure that the `rstanarm` package used to build the estimates.

```
summary(m2)
```

Model Info:

```
function:      stan_glm
family:        gaussian [identity]
formula:       totchol ~ age + sex + smoker + factor(educ)
algorithm:     sampling
sample:        4000 (posterior sample size)
priors:         see help('prior_summary')
observations:  150
predictors:    7
```

Estimates:

	mean	sd	10%	50%	90%
(Intercept)	167.0	21.8	139.6	166.3	195.3
age	1.2	0.4	0.8	1.3	1.7
sexM	3.1	6.7	-5.2	3.1	11.5

smoker	3.7	6.4	-4.5	3.8	12.0
factor(educ)2_Middle	11.3	7.9	1.2	11.4	21.3
factor(educ)3_High	-2.2	9.6	-14.3	-2.4	10.2
factor(educ)4_VHigh	11.8	10.0	-1.1	11.7	24.8
sigma	38.2	2.3	35.4	38.1	41.3

Fit Diagnostics:

	mean	sd	10%	50%	90%
mean_PPD	236.8	4.4	231.0	236.7	242.4

The mean_ppd is the sample average posterior predictive distribution of the outcome variable

MCMC diagnostics

	mcse	Rhat	n_eff
(Intercept)	0.4	1.0	3208
age	0.0	1.0	3858
sexM	0.1	1.0	4725
smoker	0.1	1.0	3941
factor(educ)2_Middle	0.1	1.0	3535
factor(educ)3_High	0.2	1.0	3947
factor(educ)4_VHigh	0.2	1.0	3980
sigma	0.0	1.0	4159
mean_PPD	0.1	1.0	4324
log-posterior	0.0	1.0	1773

For each parameter, mcse is Monte Carlo standard error, n_eff is a crude measure of effective

33.5.1 Extracting the Posterior

Let's extract the coefficients of our model, using the `get_parameters()` function from the `insight` package:

```
posteriors <- get_parameters(m2)
```

```
head(posteriors)
```

	(Intercept)	age	sexM	smoker	factor(educ)2_Middle
1	147.0370	1.4540875	2.104415	15.0169219	19.5402757
2	190.0645	0.9457021	6.863091	-12.4867763	8.4980822
3	136.3315	1.6844863	7.994193	9.1766741	24.8636150
4	169.9511	1.1722918	3.785779	0.5804774	6.5798777

5	141.0931	1.5370710	7.006277	12.6397405	17.0076561
6	153.6542	1.3556864	22.009667	11.4591142	-0.9136777
	factor(educ)3_High		factor(educ)4_VHigh		
1	5.941676		4.100910		
2	-9.449067		20.836287		
3	9.972711		12.456202		
4	-7.890619		16.454040		
5	7.221881		-9.123619		
6	-4.333041		7.401950		

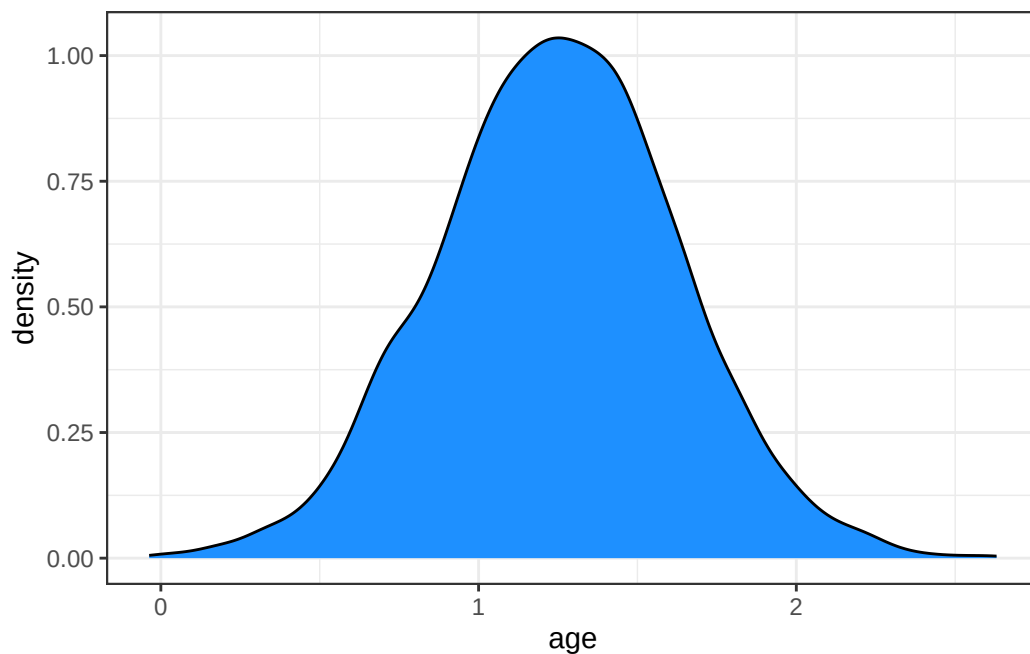
In all, we have 4000 observations of this posterior distribution:

```
nrow(posterior)
```

```
[1] 4000
```

Let's visualize the posterior distribution of our parameter for `age`.

```
ggplot(posterior, aes(x = age)) + geom_density(fill = "dodgerblue")
```



This distribution describes the probability (on the vertical axis) of various `age` effects (shown on the horizontal axis). Most of the distribution is between 0.5 and 2, with the peak being around 1.25.

Remember that our `m1` fit with `lm()` had an estimated β for `age` of 1.248, so, as is often the case, there is not a lot of difference between the two models in terms of the estimates they make.

Here's the mean and median of the `age` effect, across our 4000 simulations from the posterior distribution.

```
mean(posterior$age)
```

```
[1] 1.249122
```

```
median(posterior$age)
```

```
[1] 1.251402
```

Again, these are very close to what we obtained from least squares estimation.

Another option is to take the mode (peak) of the posterior distribution, and this is called the maximum a posteriori (MAP) estimate:

```
map_estimate(posterior$age)
```

MAP Estimate

Parameter	MAP_Estimate
x	1.26

Adding these estimates to our plot, we can see that they are all on top of each other:

```
ggplot(posterior, aes(x = age)) +  
  geom_density(fill = "dodgerblue") +  
  # The mean in yellow  
  geom_vline(xintercept = mean(posterior$age), color = "yellow", linewidth = 1) +  
  # The median in red  
  geom_vline(xintercept = median(posterior$age), color = "red", linewidth = 1) +  
  # The MAP in purple  
  geom_vline(xintercept = as.numeric(map_estimate(posterior$age)), color = "purple", linewidth = 1)
```



33.5.2 Describing Uncertainty

We might describe the range of estimates for the `age` effect.

```
range(posterior$age)
```

```
[1] -0.03663079  2.63023158
```

Instead of showing the whole range, we usually compute the highest density interval at some percentage level, for instance a 95% credible interval which shows the range containing the 95% most probable effect values.

```
hdi(posterior$age, ci = 0.95)
```

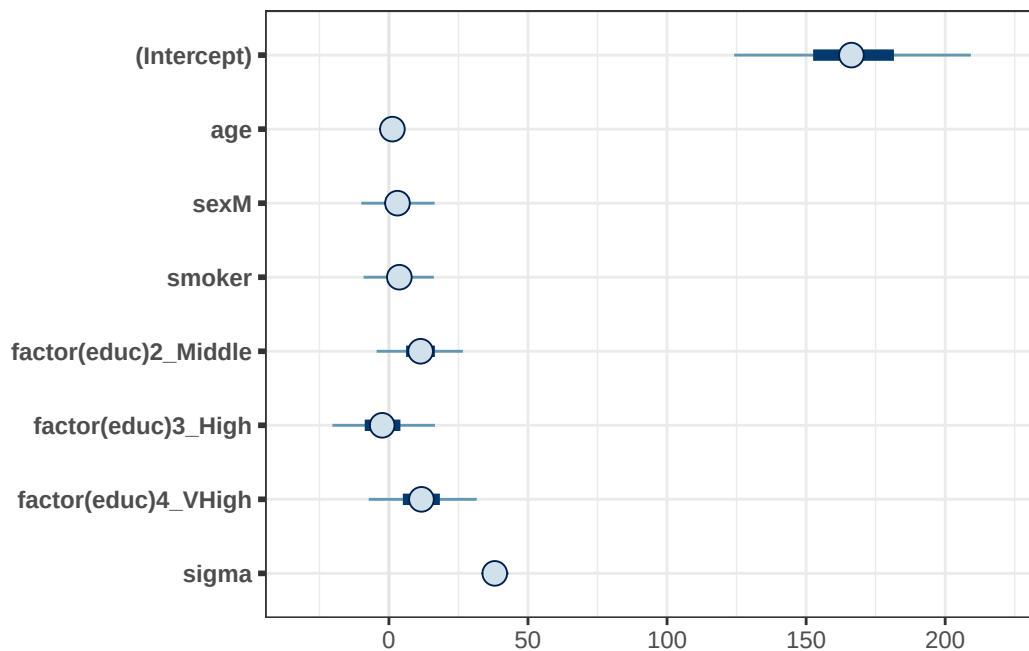
```
95% HDI: [0.50, 1.98]
```

So we conclude that the `age` effect has a 95% chance of falling within the `[0.50, 1.98]` range.

33.5.3 Visualizing the Coefficients and Credible Intervals

Here is a plot of the coefficients and parameters estimated in `m2`, along with credible intervals for their values. The inner interval (shaded region) here uses the default choice of 50%, and the outer interval (lines) uses a non-default choice of 95% (90% is the default choice here, as it turns out.) The point estimate shown here is the median of the posterior distribution, which is the default.

```
plot(m2, prob = 0.5, prob_outer = 0.95)
```



33.6 Summarizing the Posterior Distribution

A more detailed set of summaries for the posterior distribution can be obtained from the `describe_posterior()` function from the `bayestestR` package.

A brief tutorial on what is shown here is available at <https://easystats.github.io/bayestestR/articles/bayestestR.html> and <https://easystats.github.io/bayestestR/articles/example1.html> and this is the source for much of what I've built in this little chapter.

```
describe_posterior(m2) |> print_md(decimals = 3)
```


Table 33.2: Summary of Posterior Distribution

Parameter	Median	95% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	166.28	[124.14, 209.25]	100%	[-3.88, 3.88]	0%	1.000	3208
age	1.25	[0.51, 1.99]	99.95%	[-3.88, 3.88]	100%	1.000	3858
sexM	3.07	[-9.93, 16.46]	68.20%	[-3.88, 3.88]	42.32%	1.000	4725
smoker	3.75	[-9.12, 16.17]	71.67%	[-3.88, 3.88]	41.61%	1.000	3941
factor(educ)2_Middle	11.39	[-4.42, 26.57]	92.38%	[-3.88, 3.88]	14.45%	1.000	3535
factor(educ)3_High	-2.42	[-20.30, 16.54]	59.70%	[-3.88, 3.88]	31.45%	1.000	3947
factor(educ)4_VHigh	11.70	[-7.24, 31.59]	88.00%	[-3.88, 3.88]	16.58%	0.999	3980

Let's walk through all of this output.

33.6.1 Summarizing the Parameter values

For each parameter, we have:

- its estimated median across the posterior distribution
- its 95% credible interval (highest density interval of values within the posterior distribution)

as we've previously discussed.

33.6.2 Probability of Direction (pd) estimates.

The `pd` estimate helps us understand whether each effect is positive or negative. For instance, regarding `age`, we see the proportion of the posterior that is in the direction of the effect's point estimate, no matter what the "size" of the effect is, will be as follows.

```
n_positive <- posteriors |> filter(age > 0) |> nrow()
100 * n_positive / nrow(posteriors)
```

```
[1] 99.95
```

So we see that the effect of **age** is positive with a probability of 99.95%, and this is called the **probability of direction** and abbreviated **pd**.

We can also calculate this with

```
p_direction(posterior$age)
```

Probability of Direction

Parameter	pd

Posterior	99.95%

As it turns out, this **pd** index is usually highly correlated with the p-value from our **lm()** fit for **age**. We could almost roughly infer the corresponding p-value with a simple transformation:

```
pd <- 99.95
onesided_p <- 1 - pd / 100
twosided_p <- onesided_p * 2
twosided_p
```

```
[1] 0.001
```

This implies that the *p*-value for **age** in our **lm()** fit would be 0.001, and that is, in fact, what we got when we fit model **m1**.

The probability that each effect is in the direction shown by the point estimate is summarized in the **pd** column of our **describe_posterior()** results.

33.6.3 The ROPE estimates

Testing whether this distribution is different from 0 doesn't make sense, as 0 is a single value (and the probability that any distribution is different from a single value is infinite). However, one way to assess significance could be to define an area around 0, which will consider as practically equivalent to zero (i.e., absence of, or a negligible, effect). This is called the Region of Practical Equivalence (ROPE).

The default (semi-objective) way of defining the ROPE is to use the tenth (1/10) of the standard deviation (SD) of the outcome variable **totchol**. This is sometimes considered a “negligible” effect size.

In our case, **totchol** has a standard deviation of 38.8, so the ROPE range is approximately (-3.88, 3.88), or somewhat more precisely:

term	estimate	std.error	conf.low	conf.high
(Intercept)	166.284	21.546	124.137	209.245
age	1.251	0.374	0.511	1.992
sexM	3.074	6.538	-9.929	16.461
smoker	3.753	6.328	-9.119	16.175
factor(educ)2_Middle	11.388	7.705	-4.423	26.572
factor(educ)3_High	-2.421	9.495	-20.296	16.541
factor(educ)4_VHigh	11.702	9.889	-7.241	31.592

```
rope_range(m2)
```

```
[1] -3.876809  3.876809
```

So we then compute the **percentage in ROPE** as the percentage of the posterior distribution that falls within this ROPE range. When most of the posterior distribution does not overlap with ROPE, we might conclude that the effect is important enough to be noted.

In our case, 100% of the **age** effects are in the ROPE, so that's not really evidence of an important effect.

For **smoker**, though, only 42% of the effects are in the ROPE, so that's indicative of a somewhat more substantial effect, but we'd only get really excited if a much smaller fraction, say 1%, 5% or maybe 10% were in the ROPE.

33.6.4 Convergence Diagnostics

The value of Rhat should, ideally, be 1 for each element of the model, and indicate how well the MCMC procedure converged. Generally, values of Rhat below 1.05 are good, values between 1.05 and 1.1 are OK, but values above 1.1 are too high. Here, as you can see, we don't see any serious problems.

As for ESS, we'd like to see values of 1000 or more to ensure sufficiently stable estimates. Here, again, we're OK.

```
broom.mixed::tidy(m2, conf.int = TRUE, conf.level = 0.95) |>
  gt() |> fmt_number(decimals = 3)
```

33.7 Summarizing the Priors Used

From the `bayestestR` package, we also have the `describe_prior()` and `print_md()` functions to get the following summary of the priors we have assumed. Since we didn't specify anything about the priors in fitting model `m2`, we are looking at the default choices, which are weakly informative priors following Normal distributions. Details on the default prior choices can be found at <https://mc-stan.org/rstanarm/articles/priors.html>.

```
describe_prior(m2) |> print_md(decimals = 3)
```

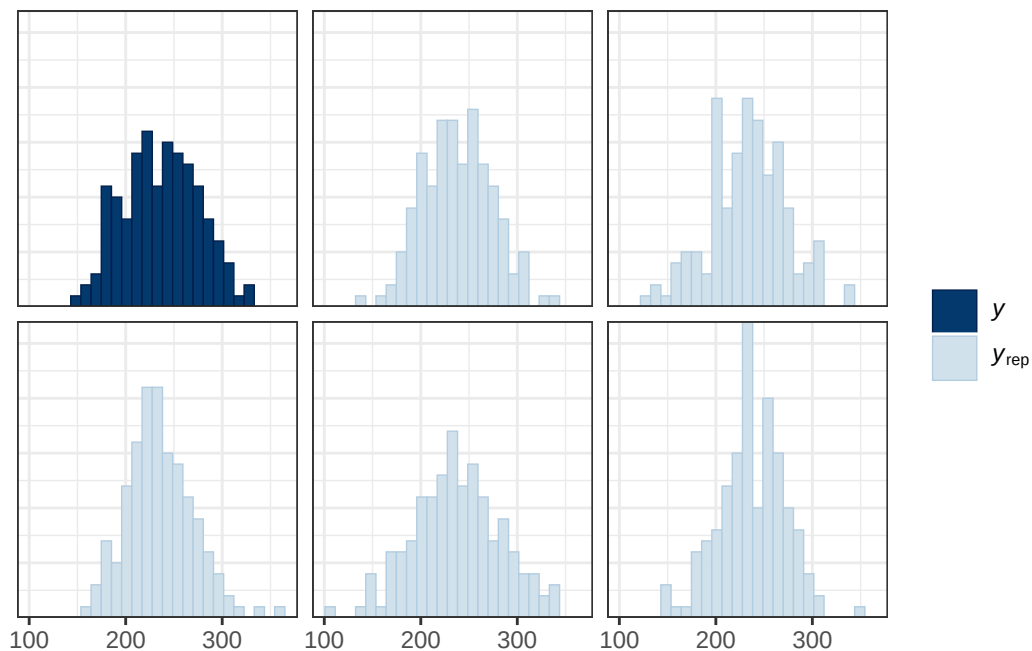
Parameter	Prior_Distribution	Prior_Location	Prior_Scale
(Intercept)	normal	236.78	96.92
age	normal	0.00	10.87
sexM	normal	0.00	195.71
smoker	normal	0.00	193.21
factor(educ)2_Middle	normal	0.00	210.79
factor(educ)3_High	normal	0.00	273.05
factor(educ)4_VHigh	normal	0.00	284.16

33.8 Graphical Posterior Predictive Checks

For more on these checks, visit <https://mc-stan.org/rstanarm/articles/continuous.html#the-posterior-predictive-distribution-1>, for example.

Here's the first plot which compares the distribution of the observed outcome y (`totchol`) to several of the simulated data sets y_{rep} from the posterior predictive distribution using the same predictor values as were used to fit the model.

```
pp_check(m2, plotfun = "hist", nreps = 5, bins = 25)
```

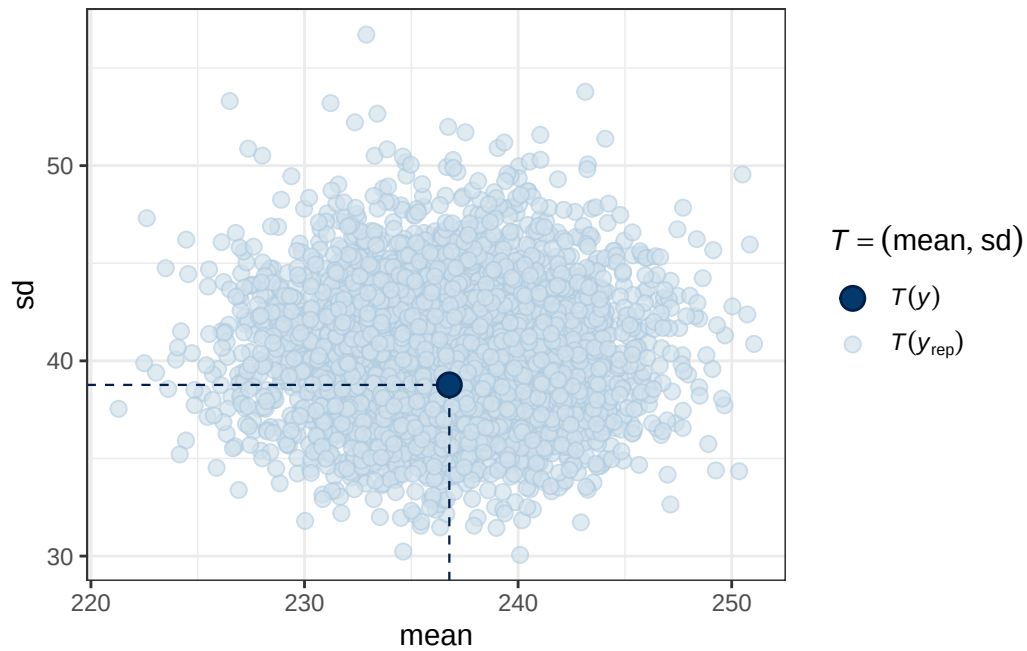


The idea is that if the model is a good fit to the data we should be able to generate data y_{rep} from the posterior predictive distribution that looks a lot like the observed data y . That is, given y , the y_{rep} we generate should look plausible. We'd worry a bit if this plot showed histograms that were wildly different from one another.

Another useful plot (shown below) made using `pp_check` shows the distribution of a test quantity $T(y_{rep})$ compared to $T(y)$, the value of the quantity in the observed data. I like this scatterplot version which allows us to look at where the simulations' mean and standard deviation fall compared to what the observed `totchol` values show us.

```
pp_check(m2, plotfun = "stat_2d", stat = c("mean", "sd"))
```

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.



We can see that the cloud of simulated means and standard deviations has the observed statistics near its center, although perhaps the standard deviations are a bit higher than we might like to see, typically. Ideally, the dot would be right in the center of this cloud of simulated results.

34 NEW!! Bayes and a Logistic Model

Almost all of this material is based on

- <https://mc-stan.org/rstanarm/articles/binomial.html> and
- <https://easystats.github.io/bayestestR/articles/bayestestR.html> and
- <https://easystats.github.io/bayestestR/articles/example1.html> and
- <https://easystats.github.io/bayestestR/articles/example2.html#logistic-model>

There's not a lot that is truly original here. That's a summer project.

34.1 R Setup Used Here

```
knitr::opts_chunk$set(comment = NA)

library(broom)
library(broom.mixed)
library(gt)
library(janitor)
library(mosaic)
library(bayestestR)
library(insight)
library(rstanarm)
library(conflicted)
library(tidyverse)

conflicts_prefer(dplyr::select, dplyr::filter,
                 base::mean, base::range, base::max)

theme_set(theme_bw())
```

34.2 Return to the `smalldat` Example

Consider the `smalldat.csv` data we discussed initially Chapter 22. The data includes 150 observations on 6 variables, and our goal here is to predict `smoker` given four predictors (`totchol`, `age`, `sex` and `educ`.)

```
smalldat <- read_csv("data/smalldat.csv", show_col_types = FALSE)
```

Variable	Description
<i>subject</i>	Subject identification code
<i>smoker</i>	1 = current smoker, 0 = not current smoker
<i>totchol</i>	total cholesterol, in mg/dl
<i>age</i>	age in years
<i>sex</i>	subject's sex (M or F)
<i>educ</i>	subject's educational attainment (4 levels)

The `educ` levels are: 1_Low, 2_Middle, 3_High and 4_VHigh, which stands for Very High

34.3 The Distribution of Smoking Status

Across the 150 observations in the `smalldat` data, we have 74 smokers.

```
smalldat |> tabyl(smoker) |> adorn_pct_formatting() |> adorn_totals()
```

smoker	n	percent
0	76	50.7%
1	74	49.3%
Total	150	-

34.4 Fitting a Logistic Regression Model with `glm()`

```
m1 <- glm((smoker == 1) ~ age + sex + totchol + factor(educ),
          data = smalldat)

glance(m1) |> select(nobs, df.residual, AIC, BIC, logLik, deviance, null.deviance) |>
  gt() |> fmt_number(AIC:null.deviance, decimals = 2)
```


nobs	df.residual	AIC	BIC	logLik	deviance	null.deviance
150	143	218.30	242.39	-101.15	33.83	37.49

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.071	0.317	3.374	0.001	0.449	1.693
age	-0.014	0.005	-2.872	0.005	-0.024	-0.004
sexM	0.131	0.083	1.578	0.117	-0.032	0.293
totchol	0.001	0.001	0.531	0.597	-0.002	0.003
factor(educ)2_Middle	-0.089	0.100	-0.894	0.373	-0.284	0.106
factor(educ)3_High	-0.073	0.121	-0.604	0.547	-0.309	0.163
factor(educ)4_VHigh	-0.242	0.126	-1.919	0.057	-0.489	0.005

```
## raw coefficients
```

```
tidy(m1, conf.int = TRUE, conf.level = 0.95) |>
  gt() |> fmt_number(decimals = 3)
```

```
## with exponentiated coefficients
```

```
tidy(m1, exponentiate = TRUE, conf.int = TRUE, conf.level = 0.95) |>
  gt() |> fmt_number(decimals = 3)
```

34.5 Fitting a Bayesian Logistic Regression

Can we fit a model for the same data using a Bayesian approach?

Yes, we can, for instance using the `stan_glm()` function from the `rstanarm` package.

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	2.918	0.317	3.374	0.001	1.566	5.435
age	0.986	0.005	-2.872	0.005	0.977	0.996
sexM	1.140	0.083	1.578	0.117	0.969	1.340
totchol	1.001	0.001	0.531	0.597	0.998	1.003
factor(educ)2_Middle	0.915	0.100	-0.894	0.373	0.753	1.112
factor(educ)3_High	0.930	0.121	-0.604	0.547	0.734	1.177
factor(educ)4_VHigh	0.785	0.126	-1.919	0.057	0.613	1.005

```
set.seed(43234231) # best to set a random seed first

m2 <- stan_glm((smoker == 1) ~ age + sex + totchol + factor(educ),
              data = smalldat, refresh = 0)
```

Here the `refresh = 0` parameter stops the machine from printing out each of the updates it does while sampling, which is not generally something I need to look at. Here's what's placed in the `m2` object:

```
m2

stan_glm
family:      gaussian [identity]
formula:     (smoker == 1) ~ age + sex + totchol + factor(educ)
observations: 150
predictors:  7
-----

              Median MAD_SD
(Intercept)    1.1    0.3
age             0.0    0.0
sexM            0.1    0.1
totchol         0.0    0.0
factor(educ)2_Middle -0.1  0.1
factor(educ)3_High  -0.1  0.1
factor(educ)4_VHigh -0.2  0.1

Auxiliary parameter(s):
      Median MAD_SD
sigma 0.5    0.0
-----

* For help interpreting the printed output see ?print.stanreg
* For info on the priors used see ?prior_summary.stanreg
```

- The first few lines specify the fitting process.
- Next, for each coefficient, we find the median value from the posterior distribution, and the `MAD_SD` value, which is an indicator of variation derived from the estimated posterior distribution of the parameters, and is used as a standard error in what follows.
- Finally, we see the estimated root mean squared error (residual standard deviation) `sigma`, again estimated with the median of the `sigma` values in the posterior distribution.

Using the `summary()` function provides some additional information about the parameter estimates, but mostly some convergence diagnostics for the Markov Chain Monte Carlo procedure that the `rstanarm` package used to build the estimates.

```
summary(m2)
```

Model Info:

```
function:      stan_glm
family:        gaussian [identity]
formula:       (smoker == 1) ~ age + sex + totchol + factor(educ)
algorithm:     sampling
sample:        4000 (posterior sample size)
priors:        see help('prior_summary')
observations:  150
predictors:    7
```

Estimates:

	mean	sd	10%	50%	90%
(Intercept)	1.1	0.3	0.7	1.1	1.5
age	0.0	0.0	0.0	0.0	0.0
sexM	0.1	0.1	0.0	0.1	0.2
totchol	0.0	0.0	0.0	0.0	0.0
factor(educ)2_Middle	-0.1	0.1	-0.2	-0.1	0.0
factor(educ)3_High	-0.1	0.1	-0.2	-0.1	0.1
factor(educ)4_VHigh	-0.2	0.1	-0.4	-0.2	-0.1
sigma	0.5	0.0	0.5	0.5	0.5

Fit Diagnostics:

	mean	sd	10%	50%	90%
mean_PPD	0.5	0.1	0.4	0.5	0.6

The `mean_ppd` is the sample average posterior predictive distribution of the outcome variable

MCMC diagnostics

	mcse	Rhat	n_eff
(Intercept)	0.0	1.0	4790
age	0.0	1.0	2700
sexM	0.0	1.0	3922
totchol	0.0	1.0	3983
factor(educ)2_Middle	0.0	1.0	2847
factor(educ)3_High	0.0	1.0	3724

```
factor(educ)4_VHigh 0.0 1.0 3550
sigma               0.0 1.0 4324
mean_PPD            0.0 1.0 4262
log-posterior       0.0 1.0 1996
```

For each parameter, `mcse` is Monte Carlo standard error, `n_eff` is a crude measure of effective

34.5.1 Extracting the Posterior

Let's extract the coefficients of our model, using the `get_parameters()` function from the `insight` package:

```
posteriors <- get_parameters(m2)

head(posteriors)
```

```
      (Intercept)      age      sexM      totchol factor(educ)2_Middle
1  1.0395289 -0.004619993 0.25120820 -0.0012609246      -0.07421785
2  1.3897359 -0.015504179 0.05308297 -0.0005806271      -0.05711548
3  0.6960912 -0.012991489 0.12013110  0.0015857650       0.08652352
4  0.3501038 -0.011626531 0.23179922  0.0024659073       0.12610455
5  1.5843855 -0.009405856 0.05310819 -0.0025427990      -0.05049668
6  1.0403138 -0.006376498 0.04919309 -0.0013779428       0.08575331
 factor(educ)3_High factor(educ)4_VHigh
1      -0.247278983      -0.34931693
2      -0.027945532      -0.04715445
3       0.098936630      -0.09453129
4       0.055186994      -0.08377399
5       0.003482159      -0.11669724
6       0.123287117      -0.06056735
```

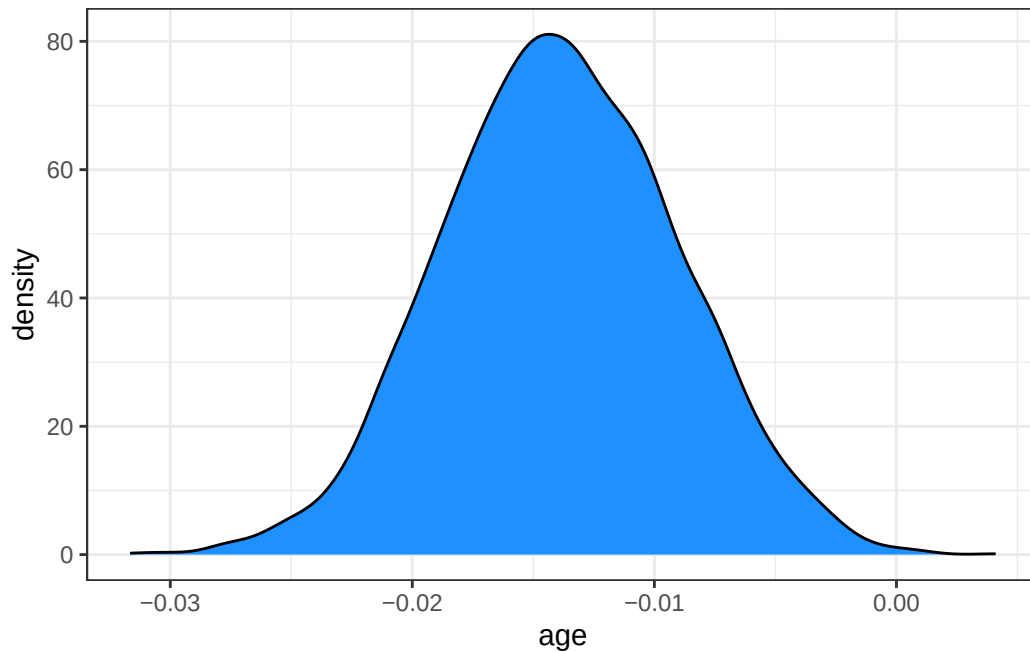
In all, we have 4000 observations of this posterior distribution:

```
nrow(posteriors)
```

```
[1] 4000
```

Let's visualize the posterior distribution of our parameter for `age`.

```
ggplot(posterioriors, aes(x = age)) + geom_density(fill = "dodgerblue")
```



This distribution describes the probability (on the vertical axis) of various **age** effects (shown on the horizontal axis). Most of the distribution is between -0.025 and -0.005, with the peak being around -0.015.

Remember that our `m1` fit with `glm()` had an estimated β for **age** of -0.014, so, as is often the case, there is not a lot of difference between the two models in terms of the estimates they make.

Here's the mean and median of the **age** effect, across our 4000 simulations from the posterior distribution.

```
mean(posterioriors$age)
```

```
[1] -0.01389409
```

```
median(posterioriors$age)
```

```
[1] -0.01391701
```

And here are the results after exponentiation, so that the estimates describe odds ratios:

```
mean(exp(posterior$age))
```

```
[1] 0.9862135
```

```
median(exp(posterior$age))
```

```
[1] 0.9861794
```

Again, these are very close to what we obtained from least squares estimation.

Another option is to take the mode (peak) of the posterior distribution, and this is called the maximum a posteriori (MAP) estimate:

```
map_estimate(posterior$age)
```

MAP Estimate

Parameter	MAP_Estimate
x	-0.01

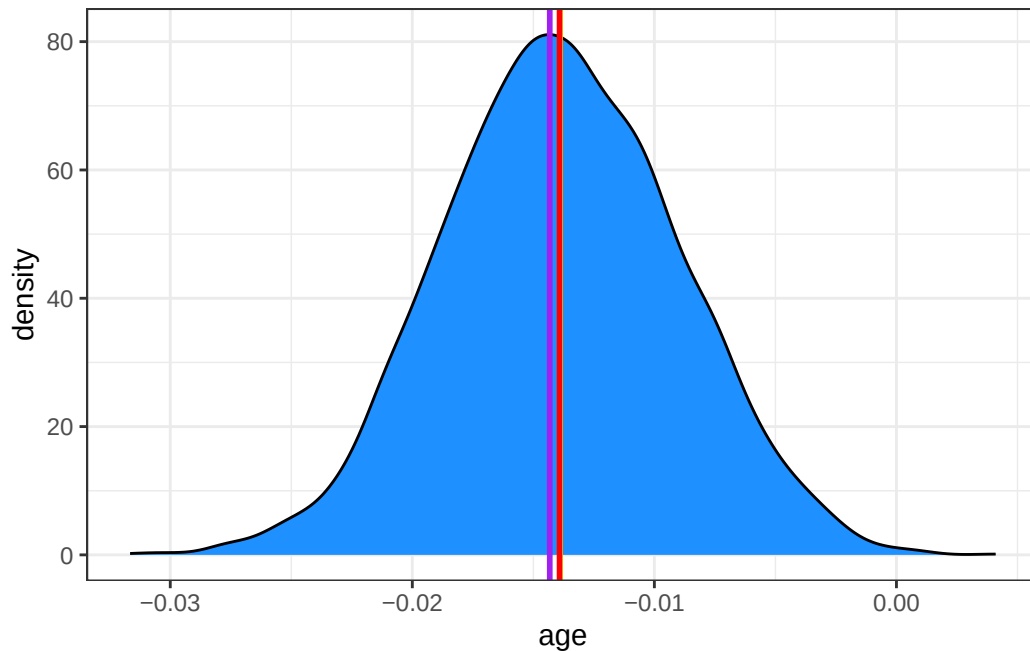
```
map_estimate(exp(posterior$age)) # on odds ratio scale
```

MAP Estimate

Parameter	MAP_Estimate
x	0.99

Adding these estimates to our plot, we can see that they are quite close:

```
ggplot(posterior, aes(x = age)) +  
  geom_density(fill = "dodgerblue") +  
  # The mean in yellow  
  geom_vline(xintercept = mean(posterior$age), color = "yellow", linewidth = 1) +  
  # The median in red  
  geom_vline(xintercept = median(posterior$age), color = "red", linewidth = 1) +  
  # The MAP in purple  
  geom_vline(xintercept = as.numeric(map_estimate(posterior$age)), color = "purple", linewidth = 1)
```



34.5.2 Describing Uncertainty

We might describe the range of estimates for the `age` effect.

```
range(posterior$age)
```

```
[1] -0.03166005  0.00409895
```

Instead of showing the whole range, we usually compute the highest density interval at some percentage level, for instance a 95% credible interval which shows the range containing the 95% most probable effect values.

```
hdi(posterior$age, ci = 0.95)
```

```
95% HDI: [-0.02,  0.00]
```

So we conclude that the `age` effect has a 95% chance of falling within the $[-0.02, 0.00]$ range. We can also exponentiate here, so as to provide the result in terms of an odds ratio.

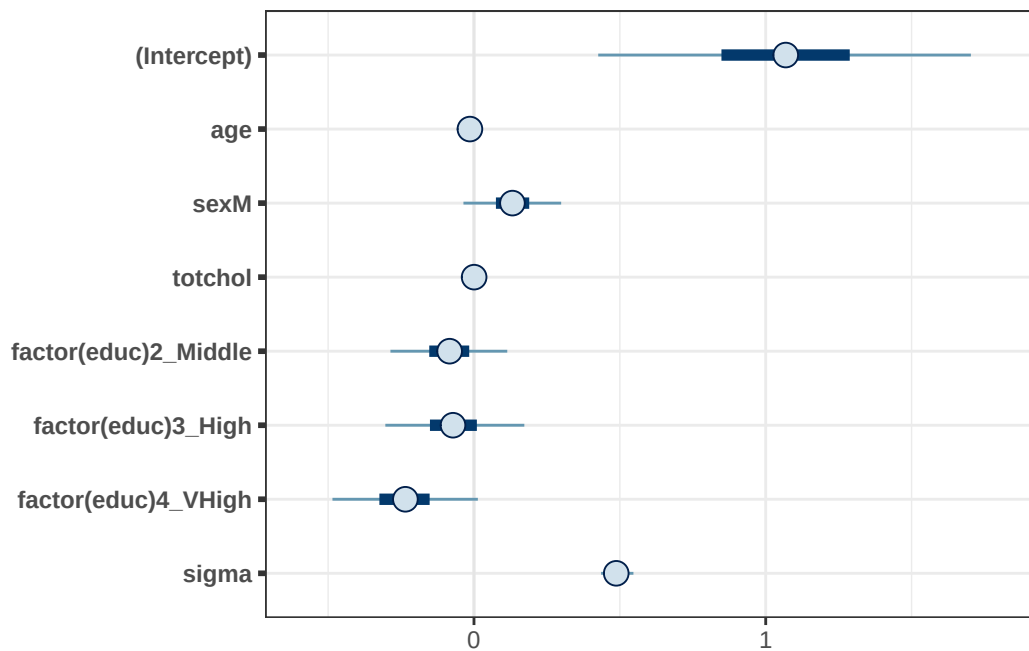
```
hdi(exp(posterior$age), ci = 0.95)
```

95% HDI: [0.98, 1.00]

34.5.3 Visualizing the Coefficients and Credible Intervals

Here is a plot of the coefficients and parameters estimated in `m2`, along with credible intervals for their values. The inner interval (shaded region) here uses the default choice of 50%, and the outer interval (lines) uses a non-default choice of 95% (90% is the default choice here, as it turns out.) The point estimate shown here is the median of the posterior distribution, which is the default.

```
plot(m2, prob = 0.5, prob_outer = 0.95)
```



34.6 Summarizing the Posterior Distribution

A more detailed set of summaries for the posterior distribution can be obtained from the `describe_posterior()` function from the `bayestestR` package.

A brief tutorial on what is shown here is available at <https://easystats.github.io/bayestestR/articles/bayestestR.html> and <https://easystats.github.io/bayestestR/articles/example1.html> and this is the source for much of what I’ve built in this little chapter.

```
describe_posterior(posterior, test = c("pd", "ROPE")) |> print_md(decimals = 3)
```

Table 34.2: Summary of Posterior Distribution

Parameter	Median	95% CI	pd	ROPE	% in ROPE
(Intercept)	1.07	[0.43, 1.70]	100%	[-0.10, 0.10]	0%
age	-0.01	[-0.02, 0.00]	99.85%	[-0.10, 0.10]	100%
sexM	0.13	[-0.04, 0.30]	94.15%	[-0.10, 0.10]	35.08%
totchol	5.36e-04	[0.00, 0.00]	69.92%	[-0.10, 0.10]	100%
factor(educ)2_Middle	-0.08	[-0.29, 0.11]	79.77%	[-0.10, 0.10]	54.87%
factor(educ)3_High	-0.07	[-0.30, 0.17]	72.32%	[-0.10, 0.10]	53.47%
factor(educ)4_VHigh	-0.24	[-0.49, 0.01]	96.88%	[-0.10, 0.10]	11.82%

Let’s walk through all of this output.

34.6.1 Summarizing the Parameter values

For each parameter, we have:

- its estimated median across the posterior distribution
- its 95% credible interval (highest density interval of values within the posterior distribution)

as we’ve previously discussed.

34.6.2 Probability of Direction (pd) estimates.

The `pd` estimate helps us understand whether each effect is positive or negative. For instance, regarding `age`, we see the proportion of the posterior that is in the direction of the median effect (negative), no matter what the “size” of the effect is, will be as follows.

```
n_negative <- posterior |> filter(age < 0) |> nrow()
100 * n_negative / nrow(posterior)
```

```
[1] 99.85
```

So we see that the effect of **age** is negative with a probability of 99.85%, and this is called the **probability of direction** and abbreviated **pd**.

We can also calculate this with

```
p_direction(posterior$age)
```

Probability of Direction

Parameter	pd
Posterior	99.85%

34.6.3 The ROPE estimates

Testing whether this distribution is different from 0 doesn't make sense, as 0 is a single value (and the probability that any distribution is different from a single value is infinite). However, one way to assess significance could be to define an area around 0, which will consider as practically equivalent to zero (i.e., absence of, or a negligible, effect). This is called the Region of Practical Equivalence (ROPE).

The default (semi-objective) way of defining the ROPE is to use (-0.1, 0.1) in this context. This is sometimes considered a “negligible” effect size.

```
rope_range(posterior)
```

```
[1] -0.1  0.1
```

So we then compute the **percentage in ROPE** as the percentage of the posterior distribution that falls within this ROPE range. When most of the posterior distribution does not overlap with ROPE, we might conclude that the effect is important enough to be noted.

In our case, 100% of the **age** effects are in the ROPE, so that's not really evidence of an important effect.

For **sex**, though, only 35% of the effects are in the ROPE, so that's indicative of a somewhat more substantial effect, but we'd only get really excited if a much smaller fraction, say 1%, 5% or maybe 10% were in the ROPE.

34.6.4 Summarizing the Coefficients as Odds Ratios

term	estimate	std.error	conf.low	conf.high
(Intercept)	2.912	0.949	1.531	5.492
age	0.986	0.005	0.977	0.996
sexM	1.141	0.096	0.965	1.348
totchol	1.001	0.001	0.998	1.003
factor(educ)2_Middle	0.920	0.093	0.751	1.121
factor(educ)3_High	0.931	0.111	0.738	1.188
factor(educ)4_VHigh	0.790	0.101	0.616	1.013

```
broom.mixed::tidy(m2, exponentiate = TRUE,
                  conf.int = TRUE, conf.level = 0.95) |>
  gt() |> fmt_number(decimals = 3)
```

34.7 Summarizing the Priors Used

From the `bayestestR` package, we also have the `describe_prior()` and `print_md()` functions to get the following summary of the priors we have assumed. Since we didn't specify anything about the priors in fitting model `m2`, we are looking at the default choices, which are weakly informative priors following Normal distributions. Details on the default prior choices can be found at <https://mc-stan.org/rstanarm/articles/priors.html>.

```
describe_prior(m2) |> print_md(decimals = 3)
```

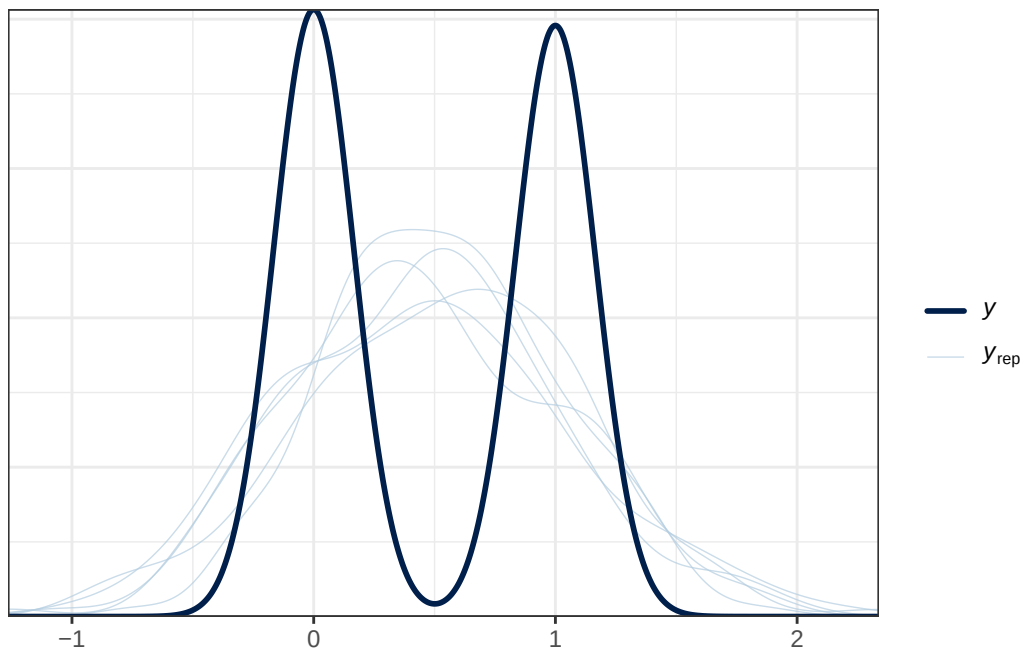
Parameter	Prior_Distribution	Prior_Location	Prior_Scale
(Intercept)	normal	0.49	1.25
age	normal	0.00	0.14
sexM	normal	0.00	2.53
totchol	normal	0.00	0.03
factor(educ)2_Middle	normal	0.00	2.73
factor(educ)3_High	normal	0.00	3.53
factor(educ)4_VHigh	normal	0.00	3.68

34.8 Graphical Posterior Predictive Checks

For more on these checks, visit <https://mc-stan.org/rstanarm/articles/continuous.html#the-posterior-predictive-distribution-1>, for example.

Here's the first plot which compares the density function of the observed outcome y (totchol) to several of the simulated data sets y_{rep} from the posterior predictive distribution using the same predictor values as were used to fit the model.

```
pp_check(m2, nreps = 5)
```

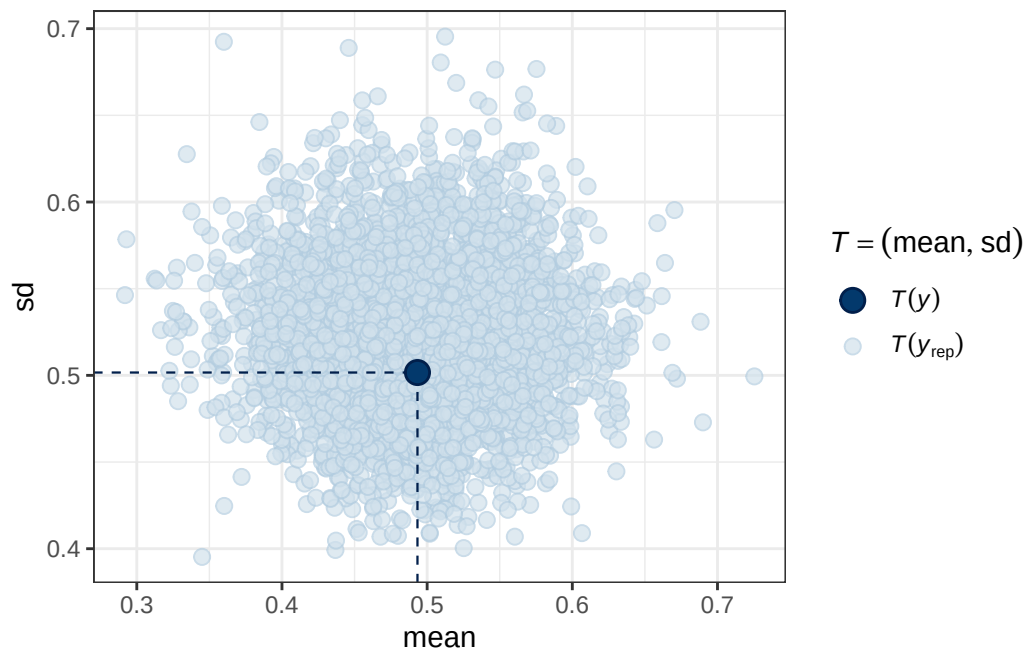


The idea is that if the model is a good fit to the data we should be able to generate data y_{rep} from the posterior predictive distribution that looks a lot like the observed data y . That is, given y , the y_{rep} we generate should look plausible. We'd worry a bit if this plot showed histograms that were wildly different from one another.

Another useful plot (shown below) made using `pp_check` shows the distribution of a test quantity $T(y_{rep})$ compared to $T(y)$, the value of the quantity in the observed data. I like this scatterplot version which allows us to look at where the simulations' mean and standard deviation fall compared to what the observed totchol values show us.

```
pp_check(m2, plotfun = "stat_2d", stat = c("mean", "sd"))
```

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inter



We can see that the cloud of simulated means and standard deviations has the observed statistics near its center, although perhaps the standard deviations are a bit higher than we might like to see, typically. Ideally, the dot would be right in the center of this cloud of simulated results.

References